AccDSL

Company Histories, Accounting Interpretations, and Scenario Simulation

Product and method proposal

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This document specifies a system that translates company disclosures into traceable evidence, represents several compatible economic and accounting histories, and simulates their future journals and financial statements under explicit economic scenarios.

The proposed data contracts, methods and acceptance cases specify the intended implementation. Current source and domain limitations remain explicit; this candidate is not a completed product or a claim of full proposal readiness. Proposed libraries can be replaced while preserving those contracts. Sector rule packages require their own accounting review before they can be described as compliant with a reporting standard.

Research materials, source qualifications and review records accompany this document in `docs/design/research` and `docs/design/review`.

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Chapter 1

The product decision

A company's accounts describe the financial consequences of its activities, but they leave much of the activity itself unobserved. Revenue combines many sales, a borrowing balance combines contracts with different terms, and a provision combines estimates about uncertain future payments. Aggregation is necessary for reporting. It also makes the inverse task—recovering economic history from published reports—fundamentally different from entering known transactions into a ledger.

AccDSL will preserve this distinction. Its first product is an evidence-backed representation of a company's historical record. Its second is a set of compatible histories and current economic states. Its third is a collection of conditional simulations, each explaining how a particular history, accounting treatment and business response produces future journals and reports. These products share identifiers and provenance, so an analyst can move from a forecast balance to the assumption or disclosure responsible for it.

The recommended design is a small, typed accounting language over a shared intermediate representation, combined with an evidence database, contract calculators and replaceable inference engines. A language earns its place by making recurring accounting decisions explicit and checkable: which entity and period an amount belongs to, whether it is known or assumed, which rule applies, and how an event changes a balanced ledger. The language does not supply missing facts or determine a company's future behaviour.

1.1 Why journals are necessary but insufficient

Consider a company with a borrowing of 100 currency units and annual interest expense of 5. Assume that the entire principal was outstanding throughout the year, interest was paid at year-end, and there were no fees, defaults, hedges or foreign-exchange effects. Both a fixed-rate loan at 5% and a floating-rate loan whose rate happened to be 5% generate the same entries:

Account	Debit	Credit
Cash, on initial borrowing	100	
Borrowing liability		100
Interest expense, at year-end	5	
Cash		5

Even the full historical journal would not distinguish these two contracts. Now impose a scenario in which the floating rate resets immediately to 7% for the whole next year. The fixed

loan produces interest of 5; the floating loan produces interest of 7. If the disclosure is also compatible with any fixed-rate fraction $\alpha \in [0, 1]$, future interest is

$$I(\alpha) = 100[0.05\alpha + 0.07(1 - \alpha)] = 7 - 2\alpha. \tag{1.1}$$

The historical accounts are compatible with the whole interval $[5, 7]$ of future interest. This is a range of feasible amounts. It becomes a probability distribution only when a distribution for α is supplied. Neither the number of generated interpretations nor a language model's confidence provides that distribution.

Journals remain essential because simulated economic events must have coherent accounting consequences. However, the simulated state must also contain contract terms, maturities, customer or borrower cohorts, collateral, unrecognised commitments, and management decisions. These determine what the journals will be under a changed economy. Reconstructing every historical transaction is unnecessary when a smaller state preserves the responses needed for the intended scenarios. Establishing that smaller state is an explicit modelling task, examined in Chapter 6.

The analogy with competing legal interpretations is useful at the level of representation. Several readings may survive the evidence and require separate reasoning. In accounting, some alternatives concern the meaning of a disclosure; others concern undisclosed facts or future behaviour. The system must identify which kind of uncertainty is involved. A known contractual term must not become an arbitrary interpretive choice merely because another choice is convenient for a simulation.

1.2 Users, questions and deliverables

An accounting analyst needs to inspect a number in its original table, verify its entity, currency and period, and see how it reconciles with the notes. A financial modeller needs to distinguish a reported balance from assumptions about its composition. A language engineer needs deterministic semantics and tests. A researcher needs to compare explanations under the same scenario, including explanations that fit history equally well.

The first release must answer the following complete workflow. An analyst loads a set of reports and specifies an information cutoff. AccDSL preserves the original documents and extracts source-linked observations. It identifies reconciliation failures and constructs an explicitly bounded family of compatible histories. The analyst declares additional assumptions where simulation requires them. The system executes each requested scenario and returns statements, journal traces, explanation-specific distributions or bounds, and unresolved limitations. When a later filing arrives, the analyst can compare the revised interpretation with the earlier information set.

Requirement	Required behaviour
R1 Evidence	Every observed amount or narrative assertion retains its document version, location, extraction method and original representation.
R2 History	Queries distinguish economic time, reporting period, publication time and database ingestion time; restatements remain recoverable.
R3 Meaning	Entity, perimeter, dimensions, currency, measurement basis, standard version and accounting concept accompany values.

Requirement	Required behaviour
R4 Ambiguity	Discrete alternatives and continuous unknowns retain their joint constraints across accounts and periods. Missing values never mean zero.
R5 Accounting	All committed journals balance in the book's functional currency, and balances reconcile to explicitly classified movements.
R6 Economics	Contracts, exposures and behavioural assumptions drive events. Accounting identities alone never determine scenario responses.
R7 Results	Conditional results survive by interpretation and scenario. Weighted results require declared, normalised probability measures.
R8 Explanation	A report cell can be traced to journals, events, rules, assumptions and evidence, including consolidation adjustments.
R9 Extensibility	Standard, industry and company packages extend shared types through versioned interfaces, with explicit unsupported cases.
R10 Reproduction	A saved run records source versions, model versions, parameters, calendars, seeds and numerical conventions.

These are acceptance requirements. Python, SQL storage, a particular parser, and a particular constraint solver are implementation choices. Initial support will cover a non-financial company with cash, receivables, payables, inventory, fixed assets, ordinary debt and ordinary equity. Banks and insurers must be representable in the core data model from the outset, but their executable measurement and behavioural packages follow dedicated milestones. Until a package supports an instrument, the result is explicitly incomplete.

1.3 What constitutes a useful first implementation

The first demonstrator should reconstruct three reporting dates and the intervening annual and interim flows for one company, including at least one restatement or revised comparative. It must preserve a borrowing-composition ambiguity and execute the fixed/floating example through both journals and all affected statement lines. A synthetic company supplies known hidden history; a public company supplies realistic disclosure and extraction problems. Their roles differ: the synthetic case tests preservation and rejection of known alternatives, while the public case tests traceability and practical coverage.

Success is a reproducible chain of evidence and computation, with identified gaps. A convincing-looking complete ledger reconstructed from incomplete disclosures would fail this product contract if its assumptions were hidden. Forecast accuracy requires a later evaluation with historical information cutoffs and suitable baselines; it is not established by the first demonstrator.

Chapter 2

A complete company and a real disclosure

2.1 What a balance must remember

Northstar Components is a synthetic seller of one homogeneous good. Its disclosures, transactions and contract terms are specified together, so we can check whether reconstruction preserves a known possible history. All amounts in this example are currency units, not thousands. The company has no foreign currency, subsidiaries, leases, returns or other comprehensive income. These are declared features of the example, not assumptions to impose on an imported company. Its source package and expected results are supplied in `fixtures/northstar.json`.

At the beginning of 2024, Northstar holds cash of 60, receivables of 20, inventory of 30 and plant costing 100 with accumulated depreciation of 20. It owes suppliers 20 and lenders 100. Share capital is 50 and retained earnings are 20. Thus assets of 190 equal liabilities of 120 plus equity of 70. Inventory consists of six units costing 5 each. The existing plant depreciates by 10 a year, with no residual value. The loan remains outstanding throughout 2024; both possible contract types charge 5% during that year.

The six months ending June and December provide two successive observation periods. Here the chosen accrual convention assigns one half-year to each. Cash-flow dates within a half-year are not recovered from a period aggregate; the synthetic history supplies a feasible ordering in which inventory is purchased before delivery and customer receipts precede the payments they fund. More detailed liquidity questions require more detailed dates.

Movement	First half	Second half
Delivered sales on credit	110	150
Customer collections	110	135
Inventory purchases on credit	70	85
Supplier payments	65	80
Cost of goods delivered	75	90
Payroll paid	20	22
Plant purchased at period end	10	12
Depreciation	5	6
Interest accrued and paid	2.5	2.5
Tax expense	1.5	5.9
Tax paid	3.5	3.9
Dividends paid	4	5

Each row has an economic meaning before it has a posting. Delivery of the good transfers control and creates an unconditional receivable; invoicing is assumed simultaneous in this example. Collection settles that receivable. Purchases increase inventory and trade payables; paying the supplier settles the payable. Delivery separately removes inventory at cost. Plant purchases are capital expenditure, while depreciation consumes its service potential. The addition of 10 enters service after June and has a five-year life, producing one unit of second-half depreciation. The addition of 12 enters service after December and has a twelve-year life.

Tax is deliberately simple: 20% of positive profit before tax, with all these expenses deductible and no deferred tax. The first-half payment exceeds its tax expense by 2, leaving prepaid tax; that prepayment is used in the second half. A genuine corporate tax package would need jurisdiction, tax basis, losses, timing differences and payment schedules. The example's tax rule governs its own numbers only.

2.2 Recovering movements without inventing invoices

Suppose the annual report discloses sales of 260, cost of sales of 165 and the opening and closing working-capital balances, while the collection and purchase totals are hidden. Under Northstar's explicit no-write-off, no-acquisition and no-currency-effect scope, the three bridges are

$$35 = 20 + 260 - C, \tag{2.1}$$

$$20 = 30 + P - 165, \tag{2.2}$$

$$30 = 20 + P - S. \tag{2.3}$$

Here C denotes collections, P purchases and S supplier payments, all in currency units over the same year. Substitution gives $C = 245$, $P = 155$ and $S = 145$. These are identifiable aggregate movements within this model. They do not identify which invoice was paid on which day. If acquisitions or write-offs are admitted, their movement variables enter the same bridges and these totals may cease to be unique.

The loan remains different. Write F and V for fixed and floating principal. The source imposes $F + V = 100$, $F, V \geq 0$ and $0.05F + 0.05V = 5$. The last equality adds no information about the split. Both endpoints and every intermediate allocation survive. Northstar therefore has identified working-capital totals and an unidentified financing exposure in the same joint historical family. A solver must not turn its convenient witness $F = 0$ into the company's recovered contract.

The model's historical slots are one delivery, collection, purchase, supplier settlement, payroll, plant addition, depreciation, interest, tax payment and dividend per half-year. Each slot represents a classified aggregate event. The interim filing supplies the information for allocating the annual movements between the two half-years; the annual bridges alone identify only their totals. The template for collection contributes $+C$ to cash and $-C$ to receivables; its state effect reduces the unpaid receivable pool by C. The purchase template adds inventory quantity $P/5$, inventory cost P and a payable P. A cash-balance identity by itself cannot substitute for either template.

2.3 The complete historical statements

The corrected statements follow from those movements. Temporary income and expense accounts close to retained earnings at each half-year boundary; annual income is the sum of the two periods' classified movements, excluding closing entries.

Balance	Opening	June	December
Cash	60	65	74.6
Receivables	20	20	35
Inventory	30	25	20
Plant, net	80	85	91
Prepaid tax	0	2	0
Total assets	190	197	220.6
Trade payables	20	25	30
Debt	100	100	100
Share capital	50	50	50
Retained earnings	20	22	40.6
Liabilities and equity	190	197	220.6

Annual revenue of 260 less cost of sales 165, payroll 42, depreciation 11 and interest 5 gives profit before tax of 37. Tax expense is 7.4 and profit is 29.6. Other comprehensive income is zero under the declared scope, so total comprehensive income is also 29.6. Equity moves from 70 to $70 + 29.6 - 9 = 90.6$ after dividends.

The cash-flow statement supplies an independent reconciliation. Collections 245 less supplier payments 145, payroll 42, interest 5 and tax payments 7.4 produce operating cash of 45.6. Investing cash is -22 and financing cash is -9, giving a cash increase of 14.6. The indirect route gives the same operating cash: profit 29.6 plus depreciation 11, minus the receivables increase 15, plus the inventory decrease 10 and the payables increase 10. The prepaid-tax balance is zero at both annual endpoints. The two cash routes use different information and therefore provide a useful check on classification.

2.4 A correction changes what was knowable

The original interim filing reported revenue of 120 and receivables of 30. Ten units of revenue had been recorded before delivery. Its tax expense was 3.5, profit 14 and equity 80. The annual filing subsequently corrected the interim revenue and receivable by 10 and recognised the excess tax payment as prepaid tax of 2. Corrected interim profit is 6 and equity is 72, as above. The original statement remains an accurate record of what was published, even though its revenue recognition was wrong under the stated delivery rule.

For the temporal fixture, the original report becomes available on 1 August 2024 and is ingested on 3 August. The correction and annual report become available on 2 March 2025 and are ingested on 5 March. An investor-information query at 1 September 2024 00:00 UTC returns 120. For this synthetic example, take UTC as the reporting timezone and use an instant cutoff of 3 March 2025 at 00:00 UTC, after the correction's entire publication-date interval has ended. That query returns 110 when evaluated using the subsequently acquired archive. A query that additionally asks what the database contained on 4 March still returns 120. Publication and ingestion answer different questions.

The annual total 260 and corrected interim total 110 share revision group `v2`, so second-half revenue is $260 - 110 = 150$. Subtracting the original 120 would yield 140, but that result mixes incompatible revisions. The query must reject the subtraction unless a reviewed compatibility relationship supports it. A later date alone does not establish such a relationship.

2.5 One economic scenario, several complete futures

For 2025, assume demand for 30 units at price 10, unit purchase cost 5 and a target closing inventory of eight units. Opening inventory is four units, so the purchase decision is $30+8-4 = 34$ units, costing 170. Deliveries cost 150, leaving inventory of 40. Collect 90% of current sales and 10 of old receivables: $270 + 10 = 280$, leaving receivables of 55. Pay old suppliers 30 and 130 of the new purchases, leaving payables of 40. Payroll is 50, capital expenditure of 20 occurs at year-end and dividends are 8.

These decisions are inputs to a conditional forecast. They are not estimates of demand or optimal management responses. No new borrowing is available. Under the existing asset schedules, depreciation is $10+10/5+12/12 = 13$. The new year-end asset begins depreciation in the next period. Let $\alpha = F/100$. With the floating rate resetting to 7%, interest is $I = 7 - 2\alpha$. Profit before tax is $300 - 150 - 50 - 13 - I = 80 + 2\alpha$, giving tax $16 + 0.4\alpha$ and profit $64 + 1.6\alpha$.

2025 statement amount	All fixed	All floating
Revenue	300	300
Interest expense	5	7
Tax expense	16.4	16
Profit and comprehensive income	65.6	64
Operating cash flow	48.6	47
Investing cash flow	−20	−20
Financing cash flow	−8	−8
Closing cash	95.2	93.6
Receivables / inventory / net plant	55 / 40 / 98	55 / 40 / 98
Assets	288.2	286.6
Payables / debt	40 / 100	40 / 100
Equity	148.2	146.6

The joint symbolic result is cash $93.6 + 1.6\alpha$, profit $64 + 1.6\alpha$ and equity $146.6 + 1.6\alpha$, with the same $\alpha \in [0, 1]$ throughout. Selecting the smallest cash and largest equity independently would describe no feasible company. A requested distribution requires a measure over that same historical family; the accounting equations do not supply one.

The later method schedules use the floating share x rather than the fixed share α used here: $x = 1 - \alpha$. Both parameterisations describe the same joint family. Its 2025 opening cash is 74.6 and equity is 90.6; the forecast spends 20 on period-end capital expenditure, distributes 8 and settles its stipulated tax in the same year. These teaching assumptions are retained when the later lease example introduces a different tax-payment schedule.

2.6 Keeping the same company through every calculation

The first inventory purchase shows what a reproducible company calculation must retain. Before the purchase, Northstar has six units costing 30 and owes suppliers 20. It buys fourteen more units for 70 on credit. The journal debits inventory and credits trade payables by 70; the resulting inventory is twenty units costing 100, payables are 90, and cash remains 60. A balanced journal accompanied by twenty-one units of inventory is wrong under the stipulated unit cost of 5. The quantity change and the monetary posting must therefore be checked against the same purchase and opening state.

Delivery follows purchase. Its four journal lines recognise the receivable and revenue together with cost of sales and the reduction in inventory. For the first half-year, delivery of fifteen units leaves five units costing 25. Separating the revenue and inventory effects into independently committable events could expose a state in which the customer has the goods but the seller still reports them. Treating delivery as one economic event prevents that intermediate inconsistency. Collections then precede the payments they fund. This order establishes a feasible aggregate history; it does not recover the dates of individual invoices.

The portable company example retains twenty-two successive states and journal steps: ten economic events and a closing entry in each half-year. Each step identifies its prior state, economic inputs, accounting template and resulting state. The corrected June statements and December statements refer to this same chain. The December closing state, with cash 74.6 and equity 90.6, is also the explicit opening of the 2025 projection. Consequently, changing a historical purchase or tax payment cannot leave the future opening silently unchanged. These are proposed histories; publishing them as candidates does not post them to an accounting book.

The rate comparison checks a different connection. Recall that x is the floating share of the unchanged principal of 100. For an imposed floating rate r, the fixed part earns interest at 5%, so annual interest is

$$I(x; r) = 100(1 - x)0.05 + 100xr = 5 + (100r - 5)x. \tag{2.4}$$

At 7%, increasing x from zero to one adds 2 to interest; at 9%, it adds 4. Under the stipulated 20% tax rule, profit and cash therefore fall by 1.6 or 3.2 respectively. The second rate supplies a useful control: two authoring frontends might agree because they share an erroneous constant. Independent arithmetic must recover the changed slope from the changed rate. The reference exchanges retain the exact expression for every $x \in [0, 1]$, including its dependence across cash, profit, assets and equity. They attach no probabilities to the endpoints or to intermediate debt allocations.

The opening also determines what kind of forecast this is. The corrected December and interim information becomes public in March 2025, after the January origin of the scenario. Using it is an archival conditional exercise: given that corrected opening, what follows from these imposed decisions and rates? It cannot establish what an investor could have forecast in January. An investor-information request at that earlier cutoff rejects the later disclosure dependency. Separately, a management policy cannot read a future realised collection merely because the simulation scheduler already stores it. An imposed scenario assumption and a later observed outcome have different information meanings.

Partial information must survive the same chain. If the fraction of current sales collected is missing, delivery can still determine revenue and cost of sales. Cash, receivables and the ability to fund the proposed settlements remain unresolved. The partial report identifies its affected lines and retains the full debt family. If the tax method is unavailable, profit after tax, retained earnings and the final closing entry are unresolved as well. A zero in a temporary account would incorrectly suggest that closing had been completed. The portable values therefore carry an explicit reason and its outstanding obligation wherever the calculation is unavailable.

There is a similar distinction in reading a note. The separate authored phrase “substantially all customers pay promptly” does not imply a 90% threshold. An analyst may impose an all-payers assumption for a conditional calculation, but its decision must preserve the unquantified source reading. Likewise, interpreting a displayed 100 as nearest-unit rounding requires an explicit rounding assumption. With ties rounded upward, that assumption permits values in $[99.5, 100.5)$; absent the assumption, the display alone supplies no such interval. Neither decision grants permission to post a journal.

Finally, a correct purchase must survive an uncertain acknowledgement. The request identifies Northstar, its book and branch, and the particular purchase; it also fixes the opening state, expected revision and all input versions. Suppose the client sends the request but receives no response. If the store committed nothing, an exact retry may commit the purchase. If the store already committed it, the same retry returns the original receipt without another purchase. The client cannot distinguish these possibilities from a timeout.

Receipt lookup therefore precedes comparison with today's head, after the original request has passed its input and authorization checks. A later, separately identified purchase may have advanced the book, but it does not erase the first receipt. Replaying the first request still returns that receipt. Changing its request identifier or budget changes its full fingerprint and conflicts with the already recorded business event. A new event presented against an obsolete opening is instead rejected as stale. Every rejection leaves the committed states, journals, quantity changes and receipt index unchanged.

Review decisions, saved search progress and imports have their own owners and receipts. Correcting the reported revenue invalidates dependent current uses, including a selection that depended on the old set of eligible readings; it does not rewrite committed purchases. Saving one searched region of $[0, 1]$ preserves the other region and the next position to visit. Resuming with a larger budget updates that saved work, while a changed model requires a different calculation. These distinctions make the complete company example reconstructible from its inputs and record references. The operation-specific fields and exact documentary exchanges are described in Section A.1; they still require implementation and actual fault testing before any claim of database durability.

2.7 One company close across several accounting methods

Separate examples can each balance while disagreeing about the cash they share. A return settlement may draw on the same bank account that must redeem a bond; an internal sale may increase a parent's earnings without increasing the group's earnings. To see these interactions, consider a second synthetic company: a parent, P, and its subsidiary, S. P owns 80% of S at the beginning of 2025 and controls it throughout the year. Both use USD, the same calendar and the same accounting policies. All amounts are currency units. The opening balances are supplied facts; the example begins after acquisition and debt issuance.

P opens with cash 500, inventory 60, plant 100, a return-recovery asset 16, an investment in S of 80 and a deferred tax asset of 10. It owes a refund liability of 40, matured debt of 100 and a deferred tax liability of 4. Its capital 600, original compound-instrument equity 10 and retained earnings 12 give equity 622. S opens with cash and capital of 100. Eliminating P's investment against its share of S's opening capital leaves non-controlling interests (NCI) of 20. The group's opening assets are 786, liabilities 144 and equity 642.

This case has a deliberately narrow tax law: a supplied enacted rate of 25%, deductible wages, matching book and tax depreciation, and the return timing specified below. Recovery of the stated deferred tax assets is supported by a supplied finding. P controls the timing of reversal of taxable outside-basis investment differences, with no reversal expected in the foreseeable future, so the IAS 12.39 exception applies conditionally. These are teaching premises, not a statement of any jurisdiction's current law. The recognition, consolidation and cash-flow rules use the retained historical standard text (European Commission, 2023a, IAS 12.24,39,46–51; IFRS 10.B86–B94).

2.7.1 Cash paid and expense incurred answer different questions

The bond's original liability proceeds were 90 and its original equity component was 10. Accrued interest of 10 has already increased the opening liability to 100. Redemption on 2 January therefore debits debt 100 and credits cash 100. It creates no new 2025 interest expense. Under this company's historical IAS 7 operating election, the payment is financing principal 90 and operating interest 10. Interest paid must still be disclosed separately even when the associated expense belongs to another period (European Commission, 2023a, IAS 7.31–33).

A component adapter must consequently preserve the original debt cash basis and the accumulated liability. Carrying amount alone cannot reconstruct the cash-flow split. Nor can an event labelled "accrue" simply be moved across a year boundary: the opening states whether its expense has already been recorded. Here P's cash falls to 400, the debt becomes zero and original equity remains 10.

The return liability comes from a previous ten-unit sale. Each unit originally cost 12 and can be refunded for 20. Expected returns were two units; estimated handling cost 3 and deterioration 1 per unit gave a recovery asset $2(12-3-1)=16$ and refund liability $2(20)=40$. Two units actually return. Handling is 4 per unit, deterioration is 2 and the returned stock's supplied net realisable value is 20. All return rights are now closed.

The stock recovered is $\min\{2(12-2),20\}=20$. P pays refunds 40 and handling 8, releases the refund liability 40 and recovery asset 16, and records inventory 20. Because the actual handling is explicitly a selling expense, the remaining entries are a cost-of-sales credit of 4 and selling expense 8. Their net expense is 4; there is no revenue adjustment when actual refunds equal the liability. This separates the recovery estimate from the cash actually spent and avoids charging handling twice (European Commission, 2023a, IFRS 15.B21–B27; IAS 2.16).

Tax timing supplies the next link. The original sale was already taxable, and its original inventory cost was deducted. On return, tax deducts the refund 40, restores recovered inventory 20 to tax stock and deducts handling 8. Thus the return reduces taxable income by 28, compared with book expense 4. The difference is 24. Opening deferred tax asset 10 and liability 4 reverse, giving deferred tax expense 6. A zero-tax return example would miss this interaction.

2.7.2 The subsidiary's inventory contains the parent's margin

During the first half P also pays wages 10 and records depreciation 10. It then sells its other inventory, which cost 60, to S for 140 on credit. P recognises revenue 140 and cost of sales 60; S records inventory and a payable of 140. The legal entries are necessary: P and S have different owners' claims and separate obligations. But the group has not yet sold that inventory to anyone outside the group.

S sells two-fifths of the lot externally for 84, recording legal cost of sales 56, and pays P the full internal invoice of 140. The diagram distinguishes the two sales. Only the second crosses the group boundary.

After the external sale, S carries three-fifths of its purchase price, or 84. The same physical stock cost the group only $3(60)/5=36$. Its unrealised internal profit is therefore $84-36=48$. IFRS 10.B86(c) requires full elimination of intragroup amounts and of profit still contained in assets; it also directs the resulting temporary differences to IAS 12.

For clarity, let T be the internal price, C the group's original cost and s the fraction sold externally. The remaining internal profit is

$$U=(T-C)(1-s)=(140-60)(1-2/5)=48. \tag{2.5}$$

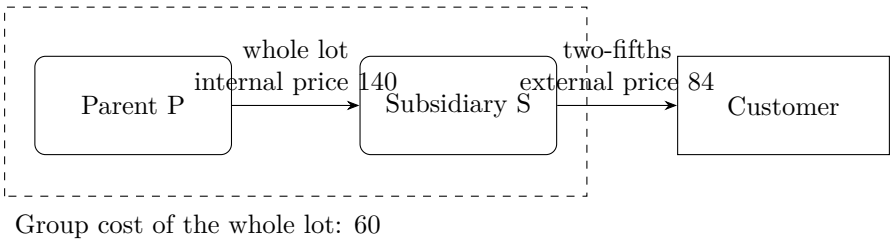


Figure 2.1: The internal price changes the legal books; the external sale determines group revenue.

The group removes internal revenue 140 and reduces the combined legal cost of sales by $T - U = 92$. S's tax base remains 84, while group carrying amount is 36. The deductible difference 48 gives a deferred tax asset 12 at the supplied 25% rate. The after-tax elimination is consequently 36. If all the stock were sold outside the group, $s = 1$, both the remaining profit and this deferred tax asset would be zero.

P's first-half pretax profit is $80 - 4 - 10 - 10 = 56$. The return timing difference reduces taxable income to 32, giving current tax 8; deferred tax expense 6 leaves P profit 42. S earns 28 before tax and 21 after current tax 7. Both current-tax amounts are paid. Group profit is therefore

$$\Pi_{H1} = 42 + 21 - 48 + 12 = 27. \tag{2.6}$$

The downstream margin originated in P's sale. Its after-tax elimination belongs to P's shareholders; NCI receives 20% of S's legal profit 21, or 4.2. At June the group has cash 511, inventory 56, plant 90 and deferred tax asset 12, totalling 669. It has no remaining external liability in this case. Equity is parent 644.8 plus NCI 24.2. These totals reconcile the same shared legal books used by the return and debt methods.

2.7.3 Buying more shares changes who owns the equity

On 1 July P pays 20 to the outside shareholders for another 10% of S. P now owns 90%, and control is unchanged. In P's separate books, its cost-policy investment increases by 20 and cash decreases by 20. S records nothing: the payment was between its shareholders. The group's accounting must instead describe a change in the relative claims of P's owners and NCI.

Immediately before purchase, S's equity is 121 and NCI owns 20%, carrying 24.2. P buys half of that interest. The NCI reduction is 12.1, not the cash price 20. IFRS 10.23 and B96 put the difference directly in parent equity:

$$\Delta NCI = -(0.90 - 0.80)121 = -12.1, \tag{2.7}$$

$$\Delta E_P = -(20 - 12.1) = -7.9. \tag{2.8}$$

No gain, goodwill or remeasurement is introduced. IAS 7.42A–42B classifies the group cash payment as financing, while the parent's separate purchase of an investment is investing under its own cash-flow presentation (European Commission, 2023a, IFRS 10.23,B96; IAS 7.16,42A–42B; IAS 27.10).

The purchase date matters for later earnings. First-half NCI profit remains 4.2; it is not recomputed at 10%. The new fraction applies to subsequent profit. Recording only an ending ownership percentage would lose that history and allocate the year's earnings incorrectly.

2.7.4 The external sale releases both margin and tax

In the second half S sells the remaining lot for 126. Its legal cost is 84, so legal pretax profit is 42 and tax is 10.5. The group's cost is only 36. The earlier elimination of 48 now reverses, and the associated deferred tax asset 12 is consumed. P separately sells its recovered inventory for 40 at cost 20 and records a further depreciation charge 10. Its second-half pretax profit is 10 and tax is 2.5. All current taxes are paid.

The resulting group profit is

$$\Pi_{H2} = (10 - 2.5) + (42 - 10.5) + 48 - 12 = 75. \tag{2.9}$$

NCI receives 10% of S's after-tax profit 31.5, or 3.15. The group release again belongs to P's shareholders because it reverses P's downstream margin. Annual NCI profit is $4.2 + 3.15 = 7.35$, and annual parent profit is $102 - 7.35 = 94.65$. Closing equity can now be reconstructed without using an asset total as a balancing plug:

$$E_P^{close} = 622 + 94.65 - 7.9 = 708.75, \tag{2.10}$$

$$NCI^{close} = 20 + 7.35 - 12.1 = 15.25. \tag{2.11}$$

Their sum is 724, equal to closing cash 644 and plant 80. The inventory has been sold, the return and debt obligations settled, and the temporary differences in this case reversed.

The comparison below follows the derivation rather than replacing it. The balance-sheet columns are closing amounts; income and cash columns are flows over the stated interval.

	First half	Second half	Full year
External revenue	84	166	250
Cost of sales	20	56	76
Profit before tax	36	100	136
Current tax expense	15	13	28
Deferred tax expense/(benefit)	-6	12	6
Profit after tax	27	75	102
NCI share of profit	4.2	3.15	7.35
Operating cash flow	1	153	154
Financing cash flow	-90	-20	-110
Closing cash	511	644	644
Closing group equity	669	724	724

Cash provides an independent check on the joins. External receipts are $84 + 126 + 40 = 250$. Operating payments are refunds 40, handling 8, wages 10, interest 10 and tax 28, totalling 96. Financing payments are debt principal 90 and the NCI purchase 20. Thus

$$C^{close} = 600 + (250 - 96) - (90 + 20) = 644. \tag{2.12}$$

The internal receipt 140 and payment 140 are removed from their respective gross totals. Merely observing that they net to zero would overstate both group receipts and payments. Group adjustments never alter either entity's legal cash ledger.

2.7.5 The implementation consequence of a joined close

Each method receives the current shared state, not its own duplicate cash balance. The internal sale and settlement each update both legal books as one transaction; an unmatched invoice or

one-sided publication is refused. A report must link an uninterrupted receipt sequence from its opening state to its closing state. Consolidation recomputes the required worksheet adjustment from legal balances and lot history at each date. It does not append the June elimination again in December.

The companion specification, `CompanyComposition/1`, fixes these inputs, events and outputs in closed records. An independent calculation reconstructs the legal and group reports, including the adverse cases of balanced but wrong journals, missing tax findings, changed ownership identities and partial publication. Its two narrow migration examples preserve the original source owners and dates; they do not rename a previous synthetic company into P. Section A.2 gives the exchange details.

This joined case establishes the stated calculation and record design for the supplied opening. It does not reconstruct the prior acquisition or issuance, nor does a corporate group with no bank or insurance business specify those sectors' combined statements. Foreign currency, wider ownership rights, actual tax authority and the remaining accounting methods retain their own required evidence. Those limits identify what the next case must add without obscuring what the shared company close already demonstrates.

2.8 Apple: source labels do not identify exposure

The real reference package is Apple's 2024 Form 10-K and first-quarter 2025 Form 10-Q, for fiscal dates 28 September and 28 December 2024 respectively (Apple Inc., 2024a, 2025c). The annual balance sheet also supplies 30 September 2023. The annual filing is archived as the company's PDF from AnnualReports; the interim filing is archived from SEC EDGAR with its inline XBRL. Apple's Note 1 describes a 52/53-week year ending on the last Saturday of September. Calendar-year assumptions would therefore misstate the periods.

The package inventory preserves every annual PDF page and every interim top-level section, all thirteen annual notes and all ten interim notes. It records internal cross-references and externally incorporated material, including the proxy statement, as distinct obligations. Statement and debt facts below are structured; policies, other notes and exhibits retain their source locations and specific modelling obligations. This is complete inventory coverage of the acquired files, with partial semantic formalisation. It does not claim to reconstruct every Apple contract or to have acquired every incorporated external document.

Apple's annual balance sheet reports assets 364,980, liabilities 308,030 and equity 56,950, in millions of US dollars. Note 9 reports term-debt principal 97,341, unamortised discount/premium and issuance costs of −321, and hedge accounting adjustments of −358. Thus its carrying amount is $97{,}341 - 321 - 358 = 96{,}662$, split into current 10,912 and non-current 85,750. Fair value is separately disclosed as 88.4 billion. None of these three measurements may replace another merely because a table calls it debt.

Note 9 identifies the notes as fixed-rate, then states that interest-rate swaps effectively convert a portion to floating rates. Note 4 reports interest-rate contract notional of 14,575 at the annual date. A notional is neither a liability carrying amount nor sufficient information for an exact interest sensitivity: reset timing, remaining periods, curve conventions and hedge coverage still matter. The cross-reference connects the derivative to the borrowing without erasing either contract.

At the interim date, total term-debt carrying amount is $10{,}848 + 83{,}956 = 94{,}804$. Cash is 30,299, total assets 344,085, liabilities 277,327 and equity 66,758. The interim debt note's rounded "94.8 billion" and the statement's 94,804 million can coexist; their precision differs. Treating them as conflicting exact numbers would create an artificial infeasibility.

These balances leave two practical financing questions open: how much cash can fund a payment, and when the payment falls due. The annual cash and cash equivalents balance of 29,943 includes 2.6 billion held in escrow and restricted from general use for the State Aid tax obligation. Note 4 states the inclusion explicitly, and Note 7 connects the escrow to that obligation (Apple Inc., 2024a, Notes 4 and 7, printed pp. 37 and 39). Subtracting the reported restricted component gives $29{,}943 - 2{,}600 = 27{,}343$ million as an indicative remainder at the displayed amounts. The restricted component is rounded to tenths of a billion, so this subtraction does not establish an exact freely available balance. The earlier case annotation that described the reported cash balance as excluding restricted cash was wrong relative to this footnote. Correcting that interpretation changes funding calculations without changing the issuer's reported number. It also requires revisiting any liquidity result that relied on the exclusion.

The debt movement supplies a second reason to preserve a calculation's missing terms. The first-quarter cash flow statement reports repayments of term debt of 1,009 million, while carrying amount falls by $96{,}662 - 94{,}804 = 1{,}858$ million. At the displayed amounts, the reconciliation is

$$94{,}804 = 96{,}662 - 1{,}009 - 849.$$

The 849 million is the residual after the reported cash repayment; it does not identify an exchange loss, an income-statement charge or a particular journal. Explaining it requires the missing movement information, including applicable currency, discount/premium and hedge-accounting effects. The quarterly cash flow statement still reconciles independently: operating inflow 29,935 plus investing inflow 9,792 less financing outflow 39,371 gives the reported cash increase of 356, from 29,943 to 30,299 million (Apple Inc., 2025c, statement of cash flows and Note 6). A balanced cash bridge therefore does not identify the debt history.

The annual filing's securities exhibit adds terms that a debt-total query cannot supply. Exhibit 4.1 describes two euro-denominated series bearing the year 2025 in their names. Its reported outstanding principal is dated 18 October 2024, after the annual balance-sheet date. The description concerns Apple Inc. as issuer, rather than every subsidiary in the consolidated group (Apple Inc., 2024a, Exhibit 4.1, pp. 3–4).

Series	Principal (EUR m)	Fiscal year	Scheduled maturity
0.875% 2025 Notes	1,250	2025	24 May 2025
0.000% 2025 Notes	1,000	2026	15 November 2025

Apple's disclosed 52-week fiscal 2025 ends on 27 September 2025 under its last-Saturday convention. The November maturity therefore belongs to fiscal 2026 even though the bond's name says 2025. The annual Note 9 principal-payment schedule uses fiscal years, whereas the instrument's stated maturity is a calendar date. Joining those fields on their year number would put a billion euros of scheduled principal into the wrong fiscal bucket. Neither the later holding date nor the euro principal may be silently substituted for a September US-dollar consolidated balance.

Suppose the disclosed faces remain outstanding until scheduled maturity, with no early redemption, default or additional withholding-related payment, and euro remains the payment currency. The annual coupon on the 0.875% series is then the reported face multiplied by the stated rate:

$$1{,}250 \times 0.00875 = 10.9375 \quad \text{million EUR}.$$

Its final scheduled principal plus annual coupon is consequently 1,260.9375 million EUR; the zero-coupon series has 1,000 million EUR of scheduled principal and zero stated coupon. Both scheduled maturity dates fall on a Saturday. Exact settlement requires the applicable payment-adjustment provision and business calendar. Exhibit 4.1 is expressly a summary qualified by

the 2013 and 2018 indentures; its payment-currency exceptions and redemption rights also matter. The retained 2021 indenture cannot replace those earlier contracts. The calculation establishes conditional scheduled amounts, while their exact settlement dates and subsequent holdings remain unresolved.

Those missing facts have different consequences for the financial statements. Principal and coupon payments reduce cash, but settling principal differs from recognising interest expense. The latter also depends on carrying amount, effective-interest accrual and the treatment of discounts and hedges; Note 9 expressly includes applicable amortisation and hedging adjustments in effective rates. Exchange rates, hedge designations and tax treatment can affect profit, other comprehensive income, retained earnings and closing debt differently. The cash flow statement additionally needs the payment classification and any associated hedge settlements. A usable answer thus carries the conditional euro schedule together with these unresolved dependencies. A dollar company forecast must supply consistent holdings, exchange, hedge and tax assumptions across all affected statements.

To demonstrate a source-to-forecast calculation, freeze the interim balance sheet and apply a *marginal* financing stress for a following 13-week period: a 200-basis-point rate increase, no principal change, no offsetting hedge beyond the explicitly selected exposure, immediate cash settlement and no tax effect. An assumed effective floating exposure $E = 14{,}575$ million produces $0.02 \times (13/52)E = 72.875$ million additional interest. This exposure uses the annual notional as a conditional input; it is not asserted to be the interim net sensitivity. Under that condition, debit interest expense and credit cash by 72.875. The resulting marginal statements have cash 30,226.125, assets 344,012.125, liabilities 277,327, equity 66,685.125, profit and comprehensive income of -72.875, and operating cash flow of -72.875 relative to the frozen baseline. The balance-sheet difference equals the equity difference exactly.

If exposure is instead only restricted to $0 \leq E \leq 14{,}575$, the conditional added interest lies in $[0, 72.875]$ before monetary rounding. That restriction is a stress assumption and is not claimed to bound every real Apple exposure. This real case verifies evidence, context, mappings and a coherent marginal propagation. Northstar supplies the complete-company forecast; a complete Apple forecast remains dependent on its unmodelled operations and undisclosed terms.

2.9 Apple: five statements constrain one year

The debt example leaves a larger question: can the reported profit, changes in equity and cash flows describe the same financial year? Apple's five consolidated statements supply a useful starting point. Their fiscal 2024 amounts refer to the year ending 28 September 2024; the opening balance sheet is dated 30 September 2023. All monetary amounts below are millions of US dollars. The operations statement's per-share amounts and weighted share counts keep their separate units (Apple Inc., 2024a, printed pp. 29–33).

Products revenue of 294,866 and services revenue of 96,169 sum to 391,035. Their costs, 185,233 and 25,119, leave gross margin of 180,683. Research and development of 31,370 and selling, general and administrative expense of 26,097 reduce that amount to operating income of 123,216. Other income of 269 and an income-tax provision of 29,749 give net income of 93,736 (Apple Inc., 2024a, printed p. 29). This same net income enters both the equity statement and the operating-cash reconciliation. It is one period result appearing in three statements, so it must not be added three times.

Other comprehensive income supplies the next connection. Foreign-currency translation contributes 395 after tax. Derivative fair-value changes of -832 and reclassification adjustments of $-1{,}337$ contribute $-2{,}169$; marketable-debt fair-value changes of 5,850 and reclassifications

of 204 contribute 6,054. Thus OCI is $395 - 2{,}169 + 6{,}054 = 4{,}280$, and total comprehensive income is $93{,}736 + 4{,}280 = 98{,}016$ (Apple Inc., 2024a, printed p. 30). Adding the period OCI to opening accumulated other comprehensive loss of $-11{,}452$ gives the closing $-7{,}172$. That accumulated balance, rather than the period flow, enters the balance sheet.

Despite positive comprehensive income, total equity falls. The equity statement explains the difference: common-stock issuance adds 1,423 and share-based compensation adds 12,034; withholding reduces the capital component by 3,993 and retained earnings by 1,612; declarations reduce retained earnings by 15,218 and repurchases by 95,846 (Apple Inc., 2024a, printed p. 32). These movements, apart from comprehensive income, total $-103{,}212$. The complete equity reconciliation is therefore

$$62{,}146 + 98{,}016 + 1{,}423 + 12{,}034 - 3{,}993 - 1{,}612 - 15{,}218 - 95{,}846 = 56{,}950.$$

The components reconcile separately: capital rises from 73,812 to 83,276, the accumulated deficit moves from -214 to $-19{,}154$, and accumulated other comprehensive loss moves from $-11{,}452$ to $-7{,}172$. These are the three closing equity components reported on the balance sheet. Their agreement prevents a model from explaining the correct total with the wrong allocation among capital, retained earnings and accumulated OCI.

The same change appears in net assets. Table 2.1 compares the reported endpoints; its changes are derived subtractions (Apple Inc., 2024a, printed p. 31). Assets rise by 12,397 while liabilities rise by 17,593, leaving a 5,196 decrease in equity, exactly the decrease from the equity reconciliation.

	30 September 2023	28 September 2024	Change
Assets	352,583	364,980	12,397
Liabilities	290,437	308,030	17,593
Equity	62,146	56,950	−5,196

Table 2.1: The balance-sheet change agrees with the complete equity movement. The agreement constrains a history without identifying its transactions.

Cash follows a different reconciliation. Starting from the same net income, the cash-flow statement adds depreciation and amortisation of 11,445, share-based compensation expense of 11,688 and other adjustments of $-2{,}266$. These three adjustments sum to 20,867. Its six operating-asset and liability adjustments are net receivables $-3{,}788$, vendor non-trade receivables $-1{,}356$, inventory $-1{,}046$, other assets $-11{,}731$, payables $+6{,}020$ and other liabilities $+15{,}552$. They sum to 3,651, giving operating cash of $93{,}736 + 20{,}867 + 3{,}651 = 118{,}254$ (Apple Inc., 2024a, printed p. 33). The full indirect bridge is thus present; none of its broad adjustment rows has been replaced by a presumed change in a similarly named balance-sheet account.

Investing cash is also positive: security purchases of $-48{,}656$, maturities of 51,211, sales of 11,135, plant-acquisition payments of $-9{,}447$ and other investing cash of $-1{,}308$ sum to 2,935. Financing uses 121,983: employee-award tax payments of 5,441, dividend payments of 15,234, share repurchases of 94,949, debt repayments of 9,958 and other outflows of 361 outweigh net commercial-paper proceeds of 3,960. The source shows a dash for net term-debt issuance; the displayed total treats that row as making no contribution, while the source reading retains the dash. The three activity totals give

$$118{,}254 + 2{,}935 - 121{,}983 = -794, \qquad 30{,}737 - 794 = 29{,}943.$$

The opening and closing amounts here are cash, cash equivalents and restricted cash and cash equivalents, as defined by the cash-flow statement. Opening balance-sheet cash is only 29,965,

so the cash-flow population exceeds it by 772; both closing amounts are 29,943. Consequently, the balance-sheet line falls by 22 while the cash-flow total falls by 794. The earlier 772 remains unallocated in the retained evidence. It cannot be relabelled as the closing 2.6 billion escrow, which Note 4 explicitly includes in reported cash and equivalents (Apple Inc., 2024a, printed p. 33 and Note 4, printed p. 37).

These joins expose the missing transaction links more precisely. Annual receivables rise by 3,902, leaving 114 after adding their −3,788 indirect adjustment. Inventory rises by 955, leaving −91 after its −1,046 adjustment. Payables rise by 6,349, exceeding their 6,020 adjustment by 329. Even the exact opposite vendor-receivable movements identify neither gross purchases nor collections. The notes examined below must supply the relevant recognition, valuation, currency and settlement information; the residuals themselves supply no such explanation.

Nor can the equity rows simply replace financing cash flows. The reported repurchase reduction of 95,846 exceeds cash repurchases of 94,949 by 897; withholding reductions total 5,605 against cash tax payments of 5,441; and equity compensation of 12,034 differs from the 11,688 expense adjustment. Cash dividends of 15,234 exceed declarations of 15,218 by 16. Those differences remain unassigned to particular settlements, fees, taxes or recognition periods (Apple Inc., 2024a, printed pp. 32–33). The later note examples retain their own evidence and conditional histories.

A reconstruction must now satisfy the statement relationships together. Correct total cash cannot excuse wrong equity components, and correct net assets cannot excuse omitted cash-flow adjustments. The five statements constrain a common history, but their agreement does not reveal all gross events, dates, counterparties or accounting treatments. Those unresolved links determine which historical alternatives can be supported and what additional facts a conditional future path still needs.

2.9.1 The following quarter must start from the same balances

The next filing makes the time connection testable. Its balance-sheet comparison is 28 September 2024, and all 27 monetary rows reproduce the annual closing column. Its income, comprehensive-income, equity and cash-flow comparisons instead cover the three months ending 30 December 2023. Those earlier-quarter flows cannot become the opening amounts for the quarter ending 28 December 2024. The interim statements are explicitly unaudited (Apple Inc., 2025c, printed pp. 1–5).

Quarterly products revenue of 97,960 and services revenue of 26,340 total 124,300. Cost of sales of 66,025 and operating expenses of 15,443 leave operating income of 42,832; other expense of 248 and tax expense of 6,254 give net income of 36,330. Translation OCI of −625, derivative OCI of 2,435 and marketable-debt OCI of −1,427 add 383, giving comprehensive income of 36,713 (Apple Inc., 2025c, printed pp. 1–2). The resulting equity movement begins at the same 56,950 that closed the annual record. Compensation adds 3,383, withholding removes 1,891 from capital and 1,102 from retained earnings, declarations remove 3,819 and repurchases remove 23,476. Together these other movements are −26,905, so $56{,}950 + 36{,}713 - 26{,}905 = 66{,}758$. Capital, accumulated deficit and accumulated OCI also join separately at both endpoints (Apple Inc., 2025c, printed pp. 3–4).

Assets decline by 20,895 to 344,085 and liabilities decline by 30,703 to 277,327. Their difference gives the same 9,808 equity increase. This check connects the period result to the balance sheet without confusing declining assets with a loss. The retained earnings deficit improves by 7,933, capital rises by 1,492 and accumulated OCI rises by 383; these three changes also sum to 9,808.

The cash reconciliation supplies a different account of the quarter. Depreciation and amortisation

of 3,080, compensation expense of 3,286 and other adjustments of −2,009 add 4,357 to net income. The six operating asset/liability adjustments are +3,597, +3,166, +215, +939, −6,671 and −11,998, in the same reported categories used for the annual reconciliation. They total −10,752, so operating cash is 36,330 + 4,357 − 10,752 = 29,935. Investing cash adds 9,792 and financing uses 39,371, leaving a 356 increase from 29,943 to 30,299 (Apple Inc., 2025c, printed p. 5). Both cash-flow endpoints now match the corresponding balance-sheet amounts. That numeric match still does not make every dollar unrestricted.

Adjacent flows may now be combined for a deliberately longer question. Over fiscal 2024 and the following quarter together, comprehensive income is 134,729 and the other equity movements total −130,117. Over that same interval, operating, investing and financing cash total 148,189, 12,727 and −161,354. The two cumulative reconciliations are

$$62{,}146 + 134{,}729 - 130{,}117 = 66{,}758, \qquad 30{,}737 + 148{,}189 + 12{,}727 - 161{,}354 = 30{,}299.$$

These are derived amounts for the interval from 1 October 2023 through 28 December 2024, not an issuer annual statement or a trailing-twelve-month result. Endpoint balances are carried forward, not added. EPS and weighted share averages also cannot be summed as if they were dollar flows.

The quarter preserves several distinctions that a common schema might otherwise erase. The annual equity table contains a common-stock issuance row and the annual cash-flow table contains a term-debt issuance row; the interim tables omit both. The interim reported totals reconcile over their own disclosed rows. A comparison record may describe those omissions, but must not manufacture two issuer-reported zeros. The prior-year quarter's term-debt repayment dash is likewise retained in its own dated column (Apple Inc., 2025c, printed pp. 4–5).

Transaction uncertainty remains across the joined interval. In the current quarter, the receivable change plus its indirect adjustment leaves −174; inventory leaves −160. The payable change is −7,050, differing from the −6,671 adjustment by −379. Cash dividends exceed declarations by 37, and cash repurchases exceed the equity reduction by 130. None identifies the missing settlement, valuation or timing components. The subsequent note inquiries must retain these relationships together when testing historical alternatives. Combining individually compatible note examples does not by itself establish that they share a compatible cash, income and equity history.

2.10 Apple: revenue recognition and cash collection

An iPhone sale can create revenue today and an obligation to provide services later. Apple's annual revenue note separates delivered hardware and bundled software from the right to receive product-related services such as iCloud, Siri and Maps. It allocates the price and related discounts by relative stand-alone selling prices. For the bundled services those prices are estimates, because observable separate prices are unavailable. The hardware portion is recognised when control transfers, generally on shipment; the service portion is deferred and recognised straight-line over the estimated period of provision (Apple Inc., 2024a, Note 2, printed p. 35). A historical model consequently needs the allocation and service cohort as well as the sale date and amount. Neither the product label nor the customer payment supplies those missing terms.

The note describes collection shortly after control transfer or the start of service delivery. It does not specify one collection lag for every contract. Products revenue also reflects expected returns, price protection and customer incentives. Revenue excludes taxes collected for remittance to governments (Apple Inc., 2024a, Notes 1–2, printed pp. 34–35). Together these qualifications prevent a simple conversion of reported revenue into gross customer receipts.

Presentation creates a second distinction. Where Apple obtains control of a third-party product before sale, it reports the gross amount billed. For third-party applications sold through the App Store, its stated treatment records only the retained commission in Services revenue. Services also already includes recognition of the deferred value of services bundled with products (Apple Inc., 2024a, Note 2, printed p. 35). Adding that recognition to reported Services a second time would double count it; treating Services as gross App Store customer spending would change the measured quantity.

The product and geographic tables describe different divisions of the same sales. The five product/service rows sum to 391,035 million dollars in fiscal 2024 and 124,300 million in the quarter ended 28 December 2024. The geographic segment sales sum to those same totals. They must not be added together. Apple says the product mix is generally similar across segments, except that iPhone represents a moderately higher share in Greater China. That qualitative statement does not identify a product-by-region matrix (Apple Inc., 2024a, Notes 2 and 13, printed pp. 35–36 and 46–47) (Apple Inc., 2025c, Notes 2 and 10, printed pp. 6 and 11).

2.10.1 The movement that the note leaves unidentified

At 28 September 2024 Apple reports total deferred revenue of 12.8 billion dollars. At 28 December it reports 13.2 billion, and says that quarter's net sales include 3.7 billion recognised from the deferred balance at the September date (Apple Inc., 2025c, Note 2, printed p. 6). The balance sheet's December current deferred-revenue line is 8,461 million, with 8,249 million at September (Apple Inc., 2025c, printed p. 3). Those are current liabilities; the note's larger amounts are totals. The precision differs too: the total is displayed to tenths of a billion, whereas the current line is displayed in millions.

Let D_0 and D_1 denote opening and closing total deferred revenue. During the quarter, let A be additions to deferred revenue, R_o revenue released from the opening balance, and R_n revenue released from amounts added during the quarter. Let M collect other signed movements, positive when they increase the liability. These symbols describe a balance reconciliation; M is not an assertion that a particular adjustment occurred. In billions of dollars,

$$D_1 = D_0 + A - R_o - R_n + M, \qquad A - R_n + M = D_1 - D_0 + R_o \simeq 13.2 - 12.8 + 3.7 = 4.1. \tag{2.13}$$

The source identifies R_o, not the whole release $R_o + R_n$. The 4.1 billion residual therefore measures net additions after new-origin releases and other movements at the displayed amounts. Calling it gross new deferrals, billings or cash collected would be wrong relative to what has been calculated.

Rounding also matters. If each of the three displayed inputs is rounded to the nearest 0.1 billion, a closed outer envelope for that residual is $[3.95, 4.25]$ billion: the lower endpoint uses $13.15 - 12.85 + 3.65$, and the upper uses $13.25 - 12.75 + 3.75$. This is a conditional precision calculation, not a verified issuer rounding rule or a bound on gross collections. Its endpoints leave the treatment of exact rounding ties unspecified.

Two constructed histories make the missing movement visible. For this illustration alone, use the displayed amounts as exact inputs and set $M = 0$. Keep the reported quarter's total revenue at 124.3 billion. The table supplies two different choices for A and R_n; other recognised revenue is the remainder needed to match that total.

Quarterly amount (USD billion)	History 1	History 2
New deferrals, A	4.1	5.1
Release from those new deferrals, R_n	0	1.0
Release from the opening balance, R_o	3.7	3.7
Other recognised revenue	120.6	119.6
Closing deferred revenue	13.2	13.2
Total recognised revenue	124.3	124.3

Both histories satisfy the selected note totals, yet their new deferrals differ by one billion. They are witnesses within this aggregate model, not reconstructed Apple contracts or a fit to every financial-statement constraint. Additional contract, cohort, return and movement information can rule out or refine them. No probability attaches to either merely because two examples were written down.

The corresponding recognition entry debits deferred revenue and credits revenue when service is provided. Receipt or billing of consideration that creates the deferral instead credits deferred revenue and debits cash or, where an unconditional right to payment exists, a receivable. Collection of that receivable changes cash and receivables without creating revenue a second time. For example, hold History 1's other revenue cash-settled and omit costs, taxes, refunds and other movements solely to isolate this subsystem. Receiving its 4.1 billion of new consideration gives revenue-side cash inflow 124.7 billion. If that new consideration is receivable at quarter end, cash inflow is 120.6 billion and receivables increase 4.1 billion. In both cases the asset increase is 124.7, matched by a 0.4 increase in deferred revenue and 124.3 of revenue before omitted expenses. These conditional entries explain the identification problem; neither inflow is an estimate of Apple's actual cash collections.

2.10.2 A dated recognition expectation is not a cash forecast

At both September and December, Apple expects 64% of the then-reported total deferred revenue to be realised in less than a year, 25% within one to two years, 9% within two to three years and 2% after three years (Apple Inc., 2024a, Note 2, printed p. 36) (Apple Inc., 2025c, Note 2, printed p. 6). These are two dated expectations about two balances. Multiplication at the displayed December values gives $13.2 \times 0.64 = 8.448$ billion in the first bucket. The corresponding September calculation gives 8.192 billion. Their difference does not measure revenue recognised during the intervening quarter: the dates, remaining horizons and contract populations have changed.

The percentages divide an expected amount by recognition horizon. They do not assign probabilities to alternative histories, fix collection dates or forecast all future sales. The 8.448 billion calculation also cannot replace the reported 8.461 billion current liability. Rounded amount and percentage disclosures do not justify an exact equality between those figures, and the query must retain the source's distinct description of each quantity. A future calculation that requires monthly service or receipts needs finer timing and payment evidence.

The note leaves another part of future service unquantified. For certain long-term arrangements, Apple has not yet delivered the service, has no right to bill for it and attributes the unbilled consideration entirely to that undelivered service. It reports neither revenue nor an amount for those obligations in this passage (Apple Inc., 2024a, Note 2, printed p. 35). A query must retain an unknown amount; setting it to zero or equating it with recorded deferred revenue would erase the distinction the company makes.

These disclosures therefore yield several different results. The total and current deferred balances are reported observations. The 4.1 residual is a derived net movement with unresolved

components. The recognition buckets are management's dated expectations, and the excluded consideration is unquantified. Contract-level allocations, service lives, new-origin releases and collection facts remain necessary for a particular future journal path. The annual and interim periods also stay separate: the December 2024 quarter follows the September 2024 fiscal year, so subtracting it from that annual revenue cannot produce the year's final nine months. The complete revenue-note reading makes these consequences explicit while the wider Apple disclosure package remains only partly formalised.

2.11 Apple: commitments and the cash needed to fulfil them

The same distinction between recognition and cash arises when Apple buys capacity and components. A lease liability measures one part of an existing obligation; a purchase commitment describes goods or services still to be acquired. A funding projection needs the payments arising from both, together with new orders and operating decisions. Adding every disclosed amount would mix measurements and could count the same requirement twice.

2.11.1 From a lease liability to future payments

At 28 September 2024, Apple's lease note reports the following amounts in millions of dollars (Apple Inc., 2024a, Note 8, printed p. 42). The maturity table first totals undiscounted payments and then deducts imputed interest to reach the recognised liability. The right-of-use assets are reported separately.

Amount (USD million)	Operating	Finance	Total
Undiscounted lease payments	13,518	1,272	14,790
Less: imputed interest	1,984	376	2,360
Recognised lease liabilities	11,534	896	12,430
Right-of-use assets	10,234	1,069	11,303

Thus $14{,}790 - 2{,}360 = 12{,}430$ reconciles the maturity and liability measurements. The recognised liability is already included in the balance sheet; it is not an extra cash requirement to add to 14,790. Its current components total $1{,}488 + 144 = 1{,}632$ million, whereas the fiscal-2025 maturity bucket is 1,991 million. A current carrying amount and a gross payment bucket answer different questions. The right-of-use asset likewise need not equal the remaining liability; these reported balances do not disclose the complete amortisation, prepayment and modification history.

Apple also reports 849 million of fixed payments under additional leases that had not commenced. It expects them to commence in 2025–2026, with terms from less than one year to twenty-one years. Adding this separately disclosed amount to the maturity total gives $14{,}790 + 849 = 15{,}639$ million. That sum is consistent, under a nearest-0.1-billion rounding assumption, with the 15.6 billion of fixed lease obligations in management's cash-requirements discussion (Apple Inc., 2024a, printed pp. 25 and 42). Numerical compatibility does not establish a contract-by-contract reconciliation. It neither adds 849 to the recognised liability nor identifies its commencement and payment dates.

The remaining lease information cannot supply those dates mechanically. The note describes original terms typically no longer than ten years, with renewal options, some reasonably certain to be exercised. Its weighted-average remaining term of 10.3 years describes a different quantity. The reported 3.1% discount rate, generally based on estimated incremental borrowing rates, does

not supply every contract's rate or the timing hidden in "thereafter". Applying that one rate to invented year-end payments would introduce new assumptions. The policy combining lease and nonlease components also has a stated scope: corporate, data-centre and retail facilities (Apple Inc., 2024a, Notes 1 and 8, printed pp. 34 and 42).

Fixed and variable costs add another distinction. For 2024, Apple reports 2.0 billion of fixed operating-lease cost, 1.9 billion of fixed operating-lease cash payments and 1.0 billion of noncash right-of-use assets obtained for lease liabilities. Variable costs across its leases were 13.8 billion; payments of that kind primarily depend on purchases of output from the leased assets (Apple Inc., 2024a, Note 8, printed p. 42). Historical expense is therefore not a fixed future cash schedule. Forecasting it requires output, price and settlement assumptions. Adding it to manufacturing commitments without identifying their relationship could double count procurement requirements.

2.11.2 Why a fall in commitments does not identify cash spent

The purchase note and management discussion disclose different views. Note 12 lists 11,226 million of future payments under unconditional purchase obligations whose remaining terms exceed one year. Its fiscal-2025 row is 3,206 million (Apple Inc., 2024a, Note 12, printed p. 45). The term restriction selects contracts; it does not place every payment beyond the coming year. The note describes these as off-balance-sheet commitments, primarily involving supplier arrangements, intellectual property, content and distribution rights.

Management's discussion provides additional near-term evidence. At the same September date it reports manufacturing purchase obligations of 53.0 billion, of which 52.9 billion is payable within twelve months. Other purchase obligations total 12.0 billion, with 4.1 billion within twelve months; that description includes capital assets as well as supplier, content and rights arrangements (Apple Inc., 2024a, printed p. 25). A model confined to Note 12 would miss those reported aggregates. Conversely, subtracting 11.226 from 12.0 and labelling the difference short-term contracts is unsupported: the category and term populations have not been reconciled, and the precision differs.

At 28 December 2024 manufacturing obligations are 39.8 billion, all payable within twelve months. The interim report says contractual cash requirements have not changed materially since the annual filing except for commercial paper, manufacturing obligations and the State Aid tax payable (Apple Inc., 2025c, printed p. 16). That qualitative statement is not an exact December lease measurement or evidence that every lease remained unchanged. The two manufacturing totals are dated observations over changing orders and payment horizons. Their 13.2 billion difference does not identify purchases or cash paid during the quarter.

To see what is missing, consider an explicitly simplified model in billions of dollars. Let O_0 and O_1 denote a comparable population of undelivered fixed-price orders at the two dates. Let N be new orders and F the value removed when goods are delivered. Assume no cancellation, price or currency change. Treat the displayed 53.0 and 39.8 as exact inputs for this illustration. These assumptions define the example; the disclosure does not establish this roll-forward convention for every reported obligation. The balance relation is

$$O_1 = O_0 + N - F, \qquad F - N = 53.0 - 39.8 = 13.2. \tag{2.14}$$

With no new orders, delivery is 13.2. With 5.0 of new orders, delivery is 18.2. Both leave the same closing undelivered amount of 39.8. The source balances alone cannot choose between these conditional movements; actual contract changes and delivery evidence could do so.

Payment is a further choice. Suppose the first history's 13.2 of delivered goods all remain unsold inventory, with no opening payables, prepayments or tax. Paying on delivery debits inventory

and credits cash by 13.2: total assets and profit do not change within this subsystem. Delivery on credit instead debits inventory and credits trade payables by 13.2: assets and liabilities rise together, with no immediate cash or profit effect. Settling that payable later debits the payable and credits cash; consumption, sale or impairment would require further entries. These examples isolate recognition and settlement. They are not estimates of Apple's actual purchases or payments, or a fit to all its financial statements. No probability follows from writing two histories.

Delivery assumptions also need the narrative that surrounds the purchase table. Apple describes single or limited sources for some components, industry-wide shortages and price fluctuations, and capacity or yield constraints for new technologies. Suppliers may favour common components over custom ones. Its named manufacturing locations describe outsourced production, not a product-by-revenue-region allocation or ownership map (Apple Inc., 2024a, Note 12, printed p. 46). A scenario in which input availability changes may alter output, variable lease payments, inventory and receipts together. The passage motivates those dependencies without identifying their size or probabilities.

The same note, and the interim contingency note, preserve management's dated assessment of unresolved legal claims. At each date, management states that there was not at least a reasonable possibility that Apple had incurred a material loss, or a material loss greater than a recorded accrual (Apple Inc., 2024a, Note 12, printed p. 46) (Apple Inc., 2025c, Note 9, printed p. 11). It does not make litigation exposure zero. Contract terms, supply constraints, commencement, delivery and settlement must therefore remain in the historical record alongside reported balances. They determine which future journal paths can be supported when a funding question is asked.

2.12 Apple: resources available for payment

The procurement and settlement choices just described determine when Apple needs money. Funding those payments requires a second account: what resources can be used, how securities become cash, and when customers pay. Annual and interim Note 4 distinguish these questions even though a single total might appear to answer them. The following table combines the notes' measurement and balance-sheet classifications. Amounts are in millions of US dollars at the displayed precision; the December date is the end of the first fiscal quarter, not another annual endpoint (Apple Inc., 2024a, Note 4, printed pp. 36–37) (Apple Inc., 2025c, Note 4, printed pp. 7–8).

Resource / measurement	30 Sep 2023	28 Sep 2024	28 Dec 2024
Cash and cash equivalents	29,965	29,943	30,299
Current marketable securities	31,590	35,228	23,476
Non-current marketable securities	100,544	91,479	87,593
Total fair value	162,099	156,650	141,368
Total adjusted cost	173,752	160,600	147,088
Net unrealised gains / losses	$-11,653$	$-3,950$	$-5,720$

Cash plus the two securities rows equals total fair value. Adjusted cost plus net unrealised gains and losses provides a second reconciliation of that same total; it is not an additional resource to add to it. The source calls the column *adjusted cost*. Relabelling every investment as amortised-cost debt would erase the different instruments represented in the table.

2.12.1 Restriction changes what can be spent

At September 2024, cash includes 2.6 billion dollars held in escrow, while marketable securities include a further 13.2 billion. Both are restricted from general use and designated for the State Aid tax obligation discussed earlier. Subtracting their displayed centres from the total gives $156{,}650 - 2{,}600 - 13{,}200 = 140{,}850$ million. MD&A separately reports unrestricted cash, equivalents and securities of 140.8 billion (Apple Inc., 2024a, printed pp. 25–26 and Note 4). The apparent difference does not establish conflicting disclosures: the restrictions and the MD&A total are rounded to tenths of a billion, whereas the table is in millions. If each amount is rounded to its nearest displayed quantum, the subtraction has possible bounds of 140,749.5 and 140,950.5 million. Those bounds overlap the 140,750 to 140,850 million rounding range for the MD&A total. This is a conditional compatibility check, not an issuer-stated rounding convention or a confidence interval.

The December footnote makes the distinction between maturity and restriction especially clear: *current* marketable securities include 3.3 billion dollars of restricted escrow funds, and cash includes 126 million. At the displayed centres, total resources less this disclosed escrow are $141{,}368 - 3{,}300 - 126 = 137{,}942$ million. Cash plus current securities is 53,775 million; removing the same escrow gives 50,349 million. These are indicative subtractions, not exact freely spendable balances. Under the same nearest-rounding hypothesis the first subtraction ranges from 137,891 to 137,993 million. Interim MD&A supplies no numerical unrestricted total (Apple Inc., 2025c, Note 4 and printed p. 16). A current classification therefore cannot serve as permission to spend an asset, and excluding escrow does not establish immediate liquidation at the reported price.

Maturity supplies a different kind of timing information. Note 1 tests the three-month cash-equivalent maturity at *purchase*, while annual Note 4 classifies marketable debt as available-for-sale and assigns current or non-current status using each instrument's underlying maturity (Apple Inc., 2024a, printed pp. 34 and 37). Among non-current debt excluding mortgage- and asset-backed securities, the September maturity shares are 86%, 10% and 4% for 1–5, 5–10 and more than 10 years. The separate mortgage- and asset-backed population has shares of 14%, 9% and 77%. At December these become 85/11/4% and 13/10/77%, respectively (Apple Inc., 2025c, Note 4, printed p. 8). Each triple divides its own population; the two triples cannot be added. Nor do contractual maturity bands establish a sale date, sale price or cash receipt schedule. Management's stated belief in sufficient resources also relies on ongoing operating cash and continued access to debt markets. It is a dated funding assessment, rather than a solvency guarantee derived from the table.

2.12.2 Security turnover differs from valuation

Securities can fund payments through sales or maturities, but a fall in their reported balance need not equal cash released. Fiscal 2024 cash proceeds from maturities and sales are 51,211 and 11,135 million, while purchases use 48,656. Net cash released is therefore $51{,}211 + 11{,}135 - 48{,}656 = 13{,}690$ million. The marketable-security balance falls by only $126{,}707 - 132{,}134 = -5{,}427$ million (Apple Inc., 2024a, cash-flow statement, printed p. 33, and Note 4). To expose the missing reconciliation, let M_0 and M_1 be total current and non-current marketable securities at the period's opening and closing. Let C be maturity and sale proceeds less purchases, positive when cash is released. Define R as the unallocated difference between the stock movement and

that cash measure, all in millions over the same period:

$$\begin{aligned} M_1 - M_0 &= -C + R, \\ \text{Fiscal 2024:} \quad -5{,}427 &= -13{,}690 + 8{,}263, \\ \text{First quarter 2025:} \quad -15{,}638 &= -13{,}335 - 2{,}303. \end{aligned}$$

For the quarter, the cash measure is $15{,}967 + 3{,}492 - 6{,}124 = 13{,}335$ million (Apple Inc., 2025c, cash-flow statement and Note 4). The relation defines a residual; it does not explain it. Cash sale proceeds differ from the carrying amount removed on disposal, and changes in measurement or classification may also enter the stock. Apple's specific-identification policy for securities sold requires the disposed lots to recover their cost (Apple Inc., 2024a, Note 1, printed p. 34). Neither the positive 8,263 nor the negative 2,303 identifies a fair-value, currency or reclassification journal.

The valuation columns help separate another tempting explanation. Across the annual dates, total adjusted cost falls by 13,152 million and net unrealised gains and losses rise by 7,703, producing a total fair-value decline of 5,449. Across the quarter, the corresponding changes are $-13{,}512$ and $-1{,}770$, producing $-15{,}282$. These totals include cash and equivalents, whereas M_0 and M_1 include only marketable securities. They also differ from the comprehensive-income statement's net-of-tax marketable-debt amounts: 6,054 million for the year and $-1{,}427$ for the quarter, including reclassification adjustments (Apple Inc., 2024a, comprehensive-income statement, printed p. 30) (Apple Inc., 2025c, comprehensive-income statement, printed p. 2). Endpoint unrealised balances and period OCI differ in population, tax and timing; their numerical differences cannot be assigned to a tax charge or realised gain without the missing movements.

Even reported cash needs matching populations. Annual cash-flow cash, cash equivalents and restricted cash opens at 30,737 million, exceeding opening balance-sheet cash and equivalents of 29,965 by 772. Both close at 29,943. Thus the cash-flow change is -794, while the balance-sheet line changes by -22; the cause and location of the earlier 772 remain unallocated. The quarter's endpoint populations agree, giving the same 356 million increase (Apple Inc., 2024a, 2025c, balance and cash-flow statements).

Valuation risk is likewise distinct from a payment. The notes describe Level 2 prices or models with significant observable inputs and counterparties that *generally* have high credit ratings. Annual Item 7A reports that a hypothetical 100-basis-point rate rise at all tenors would reduce investment fair value, including associated derivatives, by 2,755 million at September 2024 and 3,089 million at September 2023 (Apple Inc., 2024a, printed p. 27). Those are dated issuer scenarios, not cash outflows or loss probabilities; the issuer's valuation model has not been replicated. Interim Item 3's no-material-change referral does not issue a new December sensitivity.

2.12.3 Future receipts need customers and collection dates

Receivables may turn into cash later, but their movement does not identify collections. Annual net trade receivables rise by 3,902 million, while the indirect cash-flow adjustment is $-3{,}788$. Their sum leaves an unallocated 114 million difference. In the quarter, the balance change of $-3{,}771$ and adjustment of $+3{,}597$ leave -174. Vendor non-trade changes match the opposite indirect adjustments exactly: $+1{,}356/-1{,}356$ annually and $-3{,}166/+3{,}166$ quarterly. Even that match supplies neither gross sales nor total collections (Apple Inc., 2024a, 2025c, balance and cash-flow statements). Sales, allowances, currency and noncash movements still need their own evidence before receipts can be dated.

Concentration describes who might pay, but overlapping descriptions cannot be summed as separate exposures. Carriers account for 41%, 38% and 33% of total trade receivables at the three successive dates. At December the note also identifies two customers individually accounting for at least 10%, with shares of 18% and 11%. It does not say whether they are carriers (Apple Inc., 2024a, Note 4, printed p. 38) (Apple Inc., 2025c, Note 4, printed p. 9). Let A be those two customers, B the carrier group, and $s(S)$ the share of total trade receivables owed by customers in a set S. Each customer belongs wholly to a set or does not. Treat the displayed percentages as exact centres for this illustration. Inclusion-exclusion gives

$$s(A \cup B) = s(A) + s(B) - s(A \cap B) = 29\% + 33\% - s(A \cap B).$$

If both large customers are carriers, their 29% is already inside the 33% carrier share, so the union is 33%. If neither is a carrier, the union is 62%. For a concrete construction, allocate receivables across eleven hypothetical customers with shares $18, 11, 9, 9, 8, 7, 4, 9, 9, 8, 8$ percent. These sum to 100 and only two reach 10%. Carriers can be the 18, 11 and 4 customers, totalling 33, or the first 9, 9, 8 and 7 customers, also totalling 33. The same marginal disclosures therefore permit both endpoint unions. If only one large customer is a carrier, the union is 44% or 51%; exact-centre membership does not permit an arbitrary continuously divisible overlap. These are conditional percentage examples, not recovered customer identities or probabilities of loss. The note's trade denominator is not automatically the balance sheet's *net* receivable line, so multiplying it by that line would add an unsupported measurement assumption. Required credit support or collateral applies to *certain* customers; it does not establish universal recovery.

Vendor non-trade receivables have a separate economic origin and denominator. Apple buys components from suppliers and sells them to manufacturing vendors that make subassemblies or final products for Apple. Those component sales are excluded from product net sales; any gain reduces product cost of sales when Apple sells the related final products. The two largest reported vendor shares are 44% and 23% at September 2024, and 43% and 24% at December, each pair totalling 67% on its own dated vendor population (Apple Inc., 2024a, 2025c, annual and interim Note 4). Equal combined shares do not establish unchanged counterparties, component cohorts or collection risk. A funding forecast consequently needs settlements and restrictions on assets alongside payment dates on obligations. Keeping reported assertions distinct from those missing facts is the starting point for representing the company's historical record.

2.13 Sales geography is not a map of costs or assets

The liquidity question leaves a related operating question. If demand weakens in one place, which part of Apple's sales and profit is exposed? Apple reports five geographic segments, but a segment is a management reporting division, not a list of incorporated companies, factories or product lines. Reading its two segment notes as though those were the same partitions would make a regional scenario appear more precise than the filing permits.

2.13.1 What the regional labels measure

Apple's fiscal 2024 Note 13 names Americas, Europe, Greater China, Japan and Rest of Asia Pacific. Americas covers North and South America. Europe also includes India, the Middle East and Africa; Greater China includes China mainland, Hong Kong and Taiwan. Apple says regional net sales are *generally* based on customer location and sales through retail stores in those geographic locations. Its segments sell similar products and services, but are managed separately around customers, distribution partners and market conditions (Apple Inc., 2024a, Note 13, printed pp. 46–47). An India-customer sales shock therefore belongs with Apple's

Europe segment under this disclosure's general attribution rule, even though India is in Asia. Neither this mapping nor the label "Greater China" locates the legal seller, the manufacturing partner or the cash-receiving entity.

The prior revenue reading established that product and regional net sales are two classifications of the same 391,035 million dollars in fiscal 2024 and 124,300 million in the first quarter of fiscal 2025; the totals cannot be added. The full Note 13 and interim Note 10 tables confirm five reported regional sales and operating-income pairs in each dated column. For example, Greater China has fiscal 2024 sales of 66,952 million and segment operating income of 27,082 million; in the quarter ended 28 December 2024 the amounts are 18,513 and 8,160 million (Apple Inc., 2024a, Note 13, printed p. 47) (Apple Inc., 2025c, Note 10, printed p. 11). The quarter is a distinct three-month period, not a fourth of the fiscal-year observation. Apple's fiscal calendar can contain 53 weeks, as it did in fiscal 2023, and its interim MD&A describes historically higher first-quarter sales associated in part with the holidays (Apple Inc., 2024a, 2025c, Note 1 and MD&A).

Management attributes the annual Greater China sales decline mainly to lower iPhone and iPad sales and reports an unfavourable renminbi effect. For the first quarter of fiscal 2025 it attributes the decline mainly to iPhone, while describing a favourable currency effect (Apple Inc., 2024a, 2025c, annual and interim segment operating performance). These are dated issuer explanations, not measured contributions of each driver or an estimated response equation. A future region-specific demand or exchange-rate path needs product, currency and contract information that the regional totals alone do not supply.

2.13.2 Why segment income cannot be treated as fully allocated profit

Apple defines each segment's operating income as third-party sales less the related cost of sales and directly attributable operating expenses. The annual total of segment operating income is 162,044 million, but the consolidated operating statement reports 123,216 million. Note 13 supplies the bridge: 31,370 million of research and development and 7,458 million of other corporate expenses, net, are subtracted. The latter category includes general and administrative compensation, various nonrecurring charges and other separately managed costs. The reported relation is

$$162{,}044 - 31{,}370 - 7{,}458 = 123{,}216 \quad \text{million dollars, fiscal 2024.} \tag{2.15}$$

The first-quarter fiscal 2025 bridge has the same form: $53{,}174 - 8{,}268 - 2{,}074 = 42{,}832$ million. Both endpoints agree with the consolidated statements of operations (Apple Inc., 2024a, statements of operations and Note 13, printed pp. 29 and 47) (Apple Inc., 2025c, statement of operations and Note 10, printed pp. 1 and 11). The annual "other corporate" amount is not identical to the 26,097 million selling, general and administrative expense line in the consolidated statement. Equating the two would change the reported classification.

Equation (2.15) identifies the company-wide cost pool, not how it would be attributed to regions. To see the difference, take a hypothetical allocation of the 38,828 million total corporate bridge across Americas, Europe, Greater China, Japan and Rest of Asia Pacific of respectively 14,000, 10,000, 5,000, 4,000 and 5,828 million. Moving 1,000 from Americas to Greater China produces a second allocation of 13,000, 10,000, 6,000, 4,000 and 5,828 million. Both are nonnegative, sum to the same corporate total and reproduce the consolidated 123,216 million. Yet Greater China's hypothetical profit after these costs is either $27{,}082 - 5{,}000 = 22{,}082$ or $27{,}082 - 6{,}000 = 21{,}082$ million. Apple does not report either fully allocated figure, and the filing supplies no rule preferring one allocation. These constructions establish numerical nonidentification within the stated nonnegative illustration; they do not assert that all real cost allocations are nonnegative or that either construction describes Apple's internal books. Forecasting regional profit also

requires the behaviour of directly attributable and corporate costs under the scenario.

2.13.3 Country sales and long-lived assets answer different questions

The annual note separately presents country sales for fiscal 2024, 2023 and 2022 and country long-lived-asset stocks at the ends of fiscal 2024 and 2023. It separately reports countries individually accounting for at least 10% of the respective totals and groups the rest; that selection rule does not make "Other countries" a single location. The 2024 displayed amounts illustrate the different populations:

Country class	2024 net sales	Long-lived assets
	USD millions	
U.S.	142,196	35,664
China, including Hong Kong and Taiwan	66,952	4,797
Other countries	181,887	5,219
Total	391,035	45,680

The China country-sales row equals Greater China segment sales in all three annual columns. The note's China footnote includes Hong Kong and Taiwan, so the equality is a useful check on these displayed populations. It does not turn that row into a legal-entity or production-location ledger. The U.S. country-sales row is below Americas segment sales by 24,849 million in 2024; assigning that difference to particular countries requires the precise country-to-region attribution and customer facts. Europe's segment label cannot be substituted for a European-country total because it includes India, the Middle East and Africa (Apple Inc., 2024a, Note 13, printed pp. 46–47).

The 45,680 million long-lived-asset total equals net property, plant and equipment at the 2024 balance date, and the 2023 totals both equal 43,715 million at their displayed precision (Apple Inc., 2024a, Notes 5 and 13, printed pp. 39 and 47). The equality is a checked comparison, not a definition of every asset's nature, owner or productive capacity. Country asset stocks are measured at an instant, while sales cover a year. Further, Apple says the information supplied to its chief operating decision maker for segment assessment excludes asset information (Apple Inc., 2024a, Note 13, printed p. 46). Thus neither a segment return on assets nor an allocation of production capacity to the five sales regions is identified by the disclosed tables. The statement that hardware is made substantially by outsourcing partners located primarily in China mainland, India, Japan, South Korea, Taiwan and Vietnam concerns another population, and does not repair that missing asset/segment mapping (Apple Inc., 2024a, Note 12, printed p. 46).

2.13.4 The same totals permit different product exposure

Note 2 reports 201,183 million of fiscal 2024 iPhone sales. Its four other product and service rows sum to 189,852 million. Along with Greater China's 66,952 million regional total, these are *margins* of an unreported product-by-region table. Let z be iPhone sales assigned to Greater China, with all amounts in millions of dollars at the displayed centres. The four cells for Greater China versus the other four regions must be

	iPhone	Other products and services
Greater China	z	$66{,}952 - z$
Other four regions	$201{,}183 - z$	$122{,}900 + z$

(2.16)

The row and column sums in (2.16) hold for any nonnegative cells; they do not select z. For example, $z = 36{,}000$ and $z = 38{,}000$ both meet the displayed numeric margins. For an explicit extension, let S_r be each other region's reported sales; those four regions sum to $391{,}035 - 66{,}952 = 324{,}083$. Assign each $(201{,}183 - z)S_r/324{,}083$ of iPhone sales; these amounts sum to the remaining iPhone total and are below S_r. Then split every region's remaining sales across the four non-iPhone categories in their disclosed company-wide proportions. This produces all five product/service and five region margins exactly. These proportional splits are mathematical witness constructions, not issuer allocations. The source adds that the product mix was *generally* consistent across segments, with a *moderately* higher iPhone share in Greater China. Both numerical examples put Greater China's iPhone share above the combined other-region share. The qualitative words have no disclosed tolerance, so neither matrix is certified as a complete interpretation of that statement (Apple Inc., 2024a, Note 2, printed pp. 35–36). The interim note repeats the qualitative comparison for its own dated quarter (Apple Inc., 2025c, Note 2, printed p. 6); it does not supply the missing cross-classified cells.

If a scenario *assumes* a 10% decline in Greater China iPhone sales alone, the two illustrative fiscal 2024 matrices imply 3,600 and 3,800 million less sales, respectively. The scenario percentage is a teaching assumption, not an estimated shock; neither sales change determines profit, cash flow, currency translation or future behaviour. The gap between those two conditional answers comes from the unobserved joint allocation, even before cost responses are considered. A reproducible model should retain the reported marginals, the qualitative mix statement, any admitted cross-classification and its source, and the unresolved allocation when the evidence does not determine it.

Finally, the annual and first-quarter MD&A state Apple's then-planned fourth-quarter fiscal 2025 adoption of ASU 2023-07 with retrospective transition (Apple Inc., 2024a, 2025c, Recent Accounting Pronouncements). Those were dated plans, not the later disclosure itself; they do not establish the standard's operative provisions without inspection of that authority. The retained fiscal 2025 annual filing now supplies comparative 2024 segment expenses, examined in Section 2.16. A historical query keeps this later presentation separate from the older filing's content. The regional calculations here are checked source-bound comparisons and conditional witnesses, not a reconstructed Apple segment ledger or a forecast-ready regional model.

2.14 From geographic sales to legal entities

A regional sales forecast still leaves open which company sells the goods, owns the inventory or owes the supplier. Those identities matter when a scenario changes a particular entity's tax, funding or supply arrangements. Apple's geographic segments cannot answer that question: their sales are generally attributed by customer location and retail-store geography. The consolidation policy and subsidiary exhibit supply a different kind of information, about the companies represented in the group statements.

Annual Note 1 says that the consolidated financial statements include the accounts of Apple Inc. and its wholly owned subsidiaries. The filing also defines "Company" and "Apple" collectively in that way, unless otherwise stated. Interim Note 1 repeats the wholly-owned scope for its condensed statements and requires them to be read with the fiscal 2024 annual statements and notes (Apple Inc., 2024a, definition of Company and Note 1, printed pp. 1 and 34) (Apple Inc., 2025c, Note 1, printed p. 5). This is positive evidence about the reported consolidation perimeter. It is not a set of separate parent-company accounts, nor does it identify every legal entity or ownership link within that perimeter.

Exhibit 21.1 names the following subsidiaries and their jurisdictions of incorporation. The labels

below are transcribed from the exhibit, including the separate U.S. state jurisdictions (Apple Inc., 2024a, Exhibit 21.1, retained PDF p. 114).

Subsidiary	Jurisdiction of incorporation
Apple Asia Limited	Hong Kong
Apple Asia LLC	Delaware, U.S.
Apple Canada Inc.	Canada
Apple Computer Trading (Shanghai) Co., Ltd.	China
Apple Distribution International Limited	Ireland
Apple India Private Limited	India
Apple Insurance Company, Inc.	Arizona, U.S.
Apple Japan, Inc.	Japan
Apple Korea Limited	South Korea
Apple Operations International Limited	Ireland
Apple Operations Limited	Ireland
Apple Operations Mexico, S.A. de C.V.	Mexico
Apple Pty Limited	Australia
Apple Sales International Limited	Ireland
Apple South Asia (Thailand) Limited	Thailand
Apple Vietnam Limited Liability Company	Vietnam
Braeburn Capital, Inc.	Nevada, U.S.
iTunes K.K.	Japan

The exhibit expressly limits the list. Its footnote states: “Pursuant to Item 601(b)(21)(ii) of Regulation S-K, the names of other subsidiaries of Apple Inc. are omitted because, considered in the aggregate, they would not constitute a significant subsidiary as of the end of the year covered by this report.” This preserves Apple’s stated reason and date for the omission; it does not establish a numerical significance test for this model. The eighteen rows cannot be treated as an exhaustive group roster or as proof that an omitted subsidiary is irrelevant to every entity-specific question.

The same filing makes that incompleteness concrete. Its tax-risk discussion mentions Ireland and Singapore, “where a number of the Company’s subsidiaries are organized,” yet no Exhibit 21.1 row has Singapore as its jurisdiction (Apple Inc., 2024a, Risk Factors, printed p. 16). The evidence supports subsidiaries organized there while leaving their names and ownership paths unresolved. Creating a company named “Apple Singapore” would turn a geographic reference into an invented identity.

2.14.1 Membership does not determine the ownership path

Wholly-owned membership and immediate ownership answer different questions. For a simple hypothetical example, a subsidiary could be owned directly by the parent, or through another subsidiary that the parent wholly owns. If every link is wholly owned, both arrangements preserve ultimate wholly-owned membership while producing different immediate parents. This logical example does not select either arrangement for any company in the table. Neither the similar names of Apple Operations Limited and Apple Operations International Limited nor their common incorporation jurisdiction establishes an ownership link between them.

A model can therefore retain each named subsidiary, its reported jurisdiction and the general consolidation assertion without drawing unsupported edges. It should record an immediate parent, share class, ownership interest and effective interval only when the evidence identifies them. The September 2024 exhibit is not a control history for earlier years or an updated list at 28 December 2024. Repetition of the general policy in the interim note does not show that every subsidiary and ownership path remained unchanged. Determining the dated group

requires incorporation, acquisition, disposal, reorganisation and control evidence as relevant; it cannot be recovered from the fiscal calendar alone.

2.14.2 Selling, manufacturing and owning are separate relationships

Apple India Private Limited illustrates the distinction between incorporation and sales attribution. The exhibit places its incorporation in India, while Apple's segment definition places India in Europe. Under the general customer-location rule, an India-customer sale belongs in that sales region; neither statement identifies its legal seller or assigns all India sales to that subsidiary. Even a complete subsidiary list would not supply the missing seller-by-customer-region relationship.

The Business discussion also separates distribution channels: direct sales through Apple's stores and sales force accounted for 38% of fiscal 2024 net sales, and indirect channels for 62%. The latter include third-party cellular carriers, wholesalers, retailers and resellers (Apple Inc., 2024a, Markets and Distribution, printed p. 2). These are sales channel shares, not ownership shares or counts of companies. A trading relationship does not by itself establish that a counterparty belongs in the consolidated group.

Production adds a further distinction. Apple says substantially all its manufacturing is performed in whole or in part by outsourcing partners, primarily in China mainland, India, Japan, South Korea, Taiwan and Vietnam. It has also invested in manufacturing process equipment, much of which is held at certain outsourcing partners, and made prepayments under long-term supply agreements (Apple Inc., 2024a, Risk Factors, printed p. 8). Its MD&A describes partners manufacturing subassemblies and performing final assembly and testing, alongside purchases from component suppliers (Apple Inc., 2024a, Manufacturing Purchase Obligations, printed p. 25). Equipment custody, a supply contract and corporate control are thus separate relationships. A disruption at a partner may affect product supply, equipment recoverability and prepayments without supplying evidence that the partner should be consolidated. The affected assets, contracts and counterparties must be identified before amounts can be assigned to that scenario.

2.14.3 Tax and currency queries need facts beyond incorporation

The jurisdiction column also leaves entity tax residence and functional currency unresolved. Functional currency is the currency in which an entity measures its accounts; it is a different question from the dollars in which Apple presents its group statements or the currency of a particular invoice. The group foreign-exchange discussion describes Apple as a net receiver of currencies other than the U.S. dollar, with pricing and hedging responses to exchange-rate movements. It supplies no entity-by-entity functional-currency assignment (Apple Inc., 2024a, Foreign Exchange Rate Risk, printed p. 27). An Ireland incorporation row therefore does not justify entering euros as the entity's functional currency, just as the India row does not establish rupees.

Tax evidence has its own dates and units of analysis. Annual Note 7 describes the historic tax allocation of profits of Irish branches of two subsidiaries without naming those subsidiaries in that passage. The MD&A describes the resulting group payment obligation and restricted escrow (Apple Inc., 2024a, Note 7, printed pp. 39–40; State Aid Decision Tax Payable, printed p. 26). The four Ireland rows in Exhibit 21.1 do not identify which entities had those branches, their historic tax residence or the jurisdictional tax bases used in current accounts. A name match or country match would bypass the legal and temporal facts needed for the tax calculation.

These distinctions determine what a supported query can return. The retained sources answer

questions about the stated consolidated scope and the named subsidiaries' incorporation jurisdictions. Entity-specific tax, currency and intercompany questions require additional records: legal counterparties, dated ownership and control, compatible entity books, functional currencies, tax bases and matched transactions. Group eliminations then require the applicable accounting methods as well as those facts. The later IFRS group examples explain their own assumptions; they do not establish the operative US GAAP treatment for Apple. Keeping those dependencies explicit preserves useful source knowledge while leaving the unanswered entity questions open.

2.15 Legal proceedings beyond the booked liability

Knowing the companies within a group does not identify all the events that could change their earnings. A legal proceeding can require a payment, but it can also change how a company sells its products or earns licensing revenue. Apple's reported App Store injunction illustrates the second channel: it concerns developers' external purchase links on the U.S. storefront, rather than a disclosed amount of damages. A forecast that represents legal exposure only as a liability balance would leave that operating consequence outside the model.

The annual Item 3 and interim Part II Item 1 each discuss Digital Markets Act investigations, the Department of Justice lawsuit, Epic Games and other legal proceedings. These are dated issuer reports, with different kinds of statements inside each matter: allegations, reported decisions, pending motions, management's position and possible consequences. Keeping those distinctions allows a model to use what is reported without deciding the case itself (Apple Inc., 2024a, Item 3, printed p. 18) (Apple Inc., 2025c, Part II, Item 1, printed p. 18).

2.15.1 A loss assessment answers a narrower question than exposure

Annual Note 12 and interim Note 9 give the same qualified assessment: "In the opinion of management, there was not at least a reasonable possibility the Company may have incurred a material loss, or a material loss greater than a recorded accrual, concerning loss contingencies for asserted legal and other claims" (Apple Inc., 2024a, Note 12, printed p. 46) (Apple Inc., 2025c, Note 9, printed p. 11). The sentence identifies whose assessment it is, concerns an incurred material loss, and allows for a recorded accrual. It does not disclose a total legal accrual, allocate an accrual to one of the named matters, or assert that all exposure is zero. Nor does it supply a numerical probability or materiality threshold.

The annual MD&A separately reports Apple's recognition policy: the company records a liability when it is probable that a loss has been incurred and the amount is reasonably estimable, a determination requiring significant judgment (Apple Inc., 2024a, Legal and Other Contingencies, printed p. 26). That policy explains why a pending proceeding and a recognised liability are different facts. It is issuer-policy evidence; applying the operative US GAAP requirements to a particular case still needs the relevant authority and case facts. An allegation cannot supply those facts, and a possible change in future sales is not itself the incurred loss described by the policy.

Settlement adds a further distinction. Annual Item 3 says certain matters settled during the fourth quarter of fiscal 2024 without an individually or aggregate material impact on financial condition or operating results. The interim passage makes that statement about the first quarter of fiscal 2025. These are two period-specific assertions, not the same settlement observation copied forward. Neither passage identifies the matters, settlement amounts, payment dates or earlier accruals. Both retain the warning that adverse resolutions above management's

expectations could materially affect the reporting period. A payment schedule or an expense roll-forward therefore remains unresolved even though the settlement statements themselves are known.

2.15.2 The remedy, jurisdiction and procedural stage matter

The Digital Markets Act discussion reports two investigations opened by the European Commission on 25 March 2024. Article 5(4) concerns developers' communication and promotion of offers and conclusion of contracts with end users; Article 6(3) concerns default settings, app uninstallation and a browser choice screen on iOS. The Commission's 24 June announcement is described as preliminary findings alleging an Article 5(4) breach and the opening of a third investigation into new contractual requirements for third-party developers and app marketplaces. Apple states that it believes it complies and continues to change its compliance plan in response to Commission engagement. The preliminary allegation and Apple's position remain separate assertions.

Both filings say that, if the Commission makes a final violation determination, it can issue a cease and desist order and may impose fines up to 10% of Apple's annual worldwide net sales. They also say an order would apply immediately during an appeal unless a stay is granted. This preserves a conditional consequence, its stated sales base and the stay qualification. The percentage is not a fine already imposed, an expected loss or a booked liability. Even a conditional fine calculation would need the applicable operative law, relevant sales period and base, and assumptions about the determination and remedy. None is supplied by multiplying a convenient annual sales figure by the reported cap.

The Department of Justice discussion reports the civil antitrust lawsuit filed on 21 March 2024 in the District of New Jersey by the DOJ and state and district attorneys general. It alleges monopolization or attempted monopolization in the markets for "performance smartphones" and "smartphones." The DOJ seeks equitable relief; related state and federal civil litigation seeks monetary damages and other nonmonetary relief. Apple states that it has substantial defenses and intends to defend itself. Treating all these proceedings as one quantified damages claim would erase both the different requested relief and the absence of disclosed amounts.

Epic's case contains a reported court finding as well as unresolved procedure. Apple says the Northern District of California court found certain App Store Review Guidelines provisions to violate California's unfair competition law and issued an injunction against prohibiting developers' external purchase links. The stated scope is the U.S. storefront of the iOS and iPadOS App Store. Apple reports implementing a compliance plan and filing its statement on 16 January 2024. Epic's motion disputing that plan and seeking enforcement is reported as pending, opposed by Apple; Apple also reports filing a motion to narrow or vacate the injunction on 30 September 2024. These passages do not establish that the disputed plan was judicially accepted or quantify its sales effect. An EU-wide scenario cannot inherit the U.S. injunction merely because both matters concern the App Store.

The interim bodies repeat the substantive DMA, DOJ and Epic account, with acronym or trademark presentation differences. Repetition is evidence of what Apple reported in each filing, not proof that every intervening legal fact was unchanged. The September 30 motion also falls after the annual September 28 period-end. Event date, reporting period and the date on which the report became available must therefore remain distinct in a historical information query.

2.15.3 A partner's remedy can change Apple's revenue

The risk discussion connects these matters to operations. Both filings report changes to U.S. developer communications and EU changes to distribution, payment processing, fee structures and developer tools. They also describe licensing revenue from Google and other search providers, whose arrangements are subject to investigations and proceedings (Apple Inc., 2024a, Risk Factors, printed pp. 14–15) (Apple Inc., 2025c, Part II, Item 1A, printed pp. 19–20). The interim adds that Google was found to have violated U.S. antitrust laws on 5 August 2024 and that the DOJ proposed remedies ranging from restrictions on licensing agreements to prohibiting commercial search-distribution terms with Apple for up to ten years. Apple says that, if implemented, those remedies could materially affect its ability to earn that licensing revenue. The finding is an earlier event newly described in this retained comparison; the remedies are proposed, not reported here as imposed.

That channel differs from an Apple fine. The affected economic state would include the search agreement, its revenue terms, duration, geography and substitution possibilities. Likewise, an App Store remedy requires the affected storefront, developer terms, fees, customer purchases and behavioural response. The filings identify these channels without supplying the contract economics needed to calculate them. They do not justify applying one loss percentage to all Services revenue or treating every customer as affected.

A conditional calculation should consequently specify which matter and remedy it assumes, when that remedy takes effect, which contracts it changes and how the parties respond. A monetary-loss calculation additionally needs recognition facts, any opening accrual and the settlement and payment schedule; expense recognition and cash payment can occur in different periods. The State Aid tax charge and escrow settlement, developed in Section 9.8, retain their own tax records and must not be counted again as a DMA loss. Until the missing facts and applicable treatments are supplied, the model can retain the reported proceedings and affected queries as unresolved, without assigning probabilities or producing numerical journals. This preserves useful knowledge about future exposure while keeping the distinction between a known assertion and a supported financial consequence.

2.16 When later disclosures change the historical query

A later filing can reveal more about an earlier period without identifying a different economic history. For example, a new segment-expense disclosure may make a previously hidden cost visible, while the transactions that generated that cost remain the same. To use that information correctly, a model must retain the original report, the later presentation and the date at which the additional detail became available. Apple's announcement and its later segment tables show the difference between a planned disclosure and the comparative information eventually reported.

Annual and interim MD&A describe planned changes to tax and segment disclosures; the interim adds expense disaggregation and a subsequent effective-date clarification (Apple Inc., 2024a, Recent Accounting Pronouncements, printed p. 26) (Apple Inc., 2025c, Recent Accounting Pronouncements, printed p. 17). The following table preserves Apple's stated adoption plans. The quarters are fiscal, and the descriptions are issuer reports of the updates, rather than an independent determination of their operative requirements.

Update	Reported disclosure change	Apple's stated plan
ASU 2023-07	Segment expenses and information about the chief operating decision maker	Fiscal 2025 Q4, retrospective
ASU 2023-09	Income-tax reconciliation and taxes-paid detail	Fiscal 2026 Q4, prospective
ASU 2024-03, clarified by 2025-01	Expense components and selling-expense amount and definition	Fiscal 2028 Q4, prospective

The tax discussion says the rate reconciliation will include additional information and detail for items meeting a quantitative threshold. Taxes paid will be disaggregated into federal, state and foreign taxes, with further detail for significant individual jurisdictions. It gives neither the threshold value nor the future jurisdictional amounts. Those cash-tax populations also remain distinct from tax expense, payable and deferred balances; their reconciliation is developed in Section 9.8. An announced disclosure change cannot be used to allocate an old tax total among jurisdictions by assumption.

The earlier segment discussion concerns expenses that are significant and regularly provided to the chief operating decision maker, together with that person's title, position and use of segment profit or loss in assessing performance and allocating resources. That anticipated disclosure could answer questions about the information used by management. It did not yet identify the regional allocation of corporate expenses or supply segment assets. The adoption sentence was a dated plan at that information cutoff. The retained fiscal 2025 annual report now lets us check what followed.

Apple reports that it adopted ASU 2023-07 for its 2025 annual period using a retrospective transition method (Apple Inc., 2025a, Note 1). Its Note 13 presents 2024 and 2023 comparative segment expenses. Take the Americas in fiscal 2024. The earlier Note 13 reports net sales of 167,045 million and operating income of 67,656 million, but no separate regional cost of sales or selling-and-marketing amount (Apple Inc., 2024a, Note 13, printed p. 47). The later comparative gives 89,587 million of cost of sales and 9,802 million of selling and marketing for that same region and fiscal year (Apple Inc., 2025a, Note 13, printed p. 47). The displayed relationship is

$$167{,}045 - 89{,}587 - 9{,}802 = 67{,}656 \quad \text{(USD millions)}.$$

The 2025 table gives corresponding expense rows for the other four regions. Together, their disclosed selling-and-marketing expenses total 18,639 million. The five operating incomes still sum to 162,044 million; after the displayed corporate research and development of 31,370 million and general and administrative amount of 7,458 million, the company reports the same 123,216 million of operating income. The 2024 report called its 7,458 million line "other corporate expenses, net" and qualified its contents. The identical displayed amount is not, by itself, proof that the two labels have identical scope. Both original and later assertions retain their source identities.

This later presentation answers a prior-period disclosure question without identifying new fiscal-2024 transactions or creating a correction journal. Reading the named 2024 source leaves the regional expense detail unresolved; reading the later source supplies its 2025 comparative assertions. An investor-information view additionally needs evidence that the exact chosen source version was publicly available by its requested cutoff. The accepted corpus and filing-event metadata do not prove that for the retained 2024 PDF or the later issuer-hosted HTML. Nor do the new regional expense rows reveal the cross-classified product-by-region sales, entity cash allocation, or behavioural response needed by the earlier regional stress example.

The interim expense passage reports ASU 2024-03's issuance in November 2024 and ASU 2025-01's clarification in January 2025. It describes purchases of inventory, employee compensation, depreciation and intangible-asset amortization, as applicable, within certain expense captions,

a qualitative description of the remaining amounts, and both the amount and the company's definition of selling expenses. This describes disclosure populations; it does not provide their values or make selling expenses identical to the entire selling, general and administrative line. Purchases included in an expense disclosure likewise cannot be identified with cash paid to suppliers without the relevant timing and population facts.

The observed segment comparatives show why the two source-content views must remain separate: the original cannot be overwritten even when a later filing reports the same amounts with more detail. This comparison checks the affected periods, dimensions, displayed units and issuer labels; it does not use the phrase "retrospective transition" as an amount. The prospective tax and expense plans still supply no backward allocation of missing components. Their exact transition requirements need the operative updates and any applicable amendments.

The January clarification illustrates the timing problem directly. It occurs after the interim period ended on 28 December 2024 but is described in the interim report. An event can therefore belong to the information in a report without occurring inside its reporting period. A historical forecast cutoff needs evidence of availability and the exact retained version, not merely the period-end or the update's issuance month. Likewise, the earlier announced segment adoption did not become an observed adoption until the later filing was checked. The tax and expense plans remain future claims until their own later evidence is inspected.

2.16.1 Unchanged critical estimates do not mean constant amounts

Annual Critical Accounting Estimates says management makes judgments, assumptions and estimates affecting reported amounts, using historical experience and other assumptions it considers reasonable to judge carrying values. Its uncertain-tax discussion says reserves can be adjusted as facts and circumstances change, and that final outcomes may differ from those reserves. Its legal-contingency policy is the incurred-loss and estimability condition examined above (Apple Inc., 2024a, Critical Accounting Estimates, printed p. 26). These are descriptions of judgment and policy, not a disclosed numerical estimation model or an assurance that the estimates will prove accurate.

The interim references both annual and interim Note 1 and reports no material changes to critical accounting estimates since the annual filing (Apple Inc., 2025c, Critical Accounting Estimates, printed p. 17). That statement preserves the annual dependency and its materiality qualifier. It cannot make every carrying amount, reserve or underlying assumption exactly constant, and it supplies no numerical tolerance for a comparison. A model should retain the reported policy and qualified continuity assertion alongside the actual dated balances, then request the missing estimate or transition facts when the query needs them. These estimate passages supply policy and continuity assertions; they do not establish numerical stability, current compliance or the tax and expense details still absent from these sources.

2.17 Claims against another party can still affect Apple

The estimate policy leaves a further question: whose claim can give rise to a cost for Apple? A company can face exposure through an agreement even when the claim is against someone else. Apple's annual risk discussion says its agreements sometimes include indemnification provisions that can subject it to costs and damages in the event of a claim against an indemnified third party (Apple Inc., 2024a, Risk Factors, printed p. 13). The direction matters: the passage describes a possible obligation of Apple arising from another party's claim, rather than compensation owed to Apple. A model limited to proceedings naming Apple as defendant would miss that

contractual channel.

The statement supplies neither an indemnified party nor a particular agreement. It also leaves the covered conduct, trigger, duration and limit unknown. To connect a third-party claim to a dated obligation, the model would need those terms and the facts that activate them. The existence of indemnification language somewhere in Apple's agreements cannot establish that a named case is covered, that a payment is due, or that a liability has been recognised. Those facts remain distinct from the management loss assessment examined in Section 2.15.

The full annual passage also describes unresolved ordinary-course claims and investigations, an increase over time in their number and alleged magnitude, and patent claims concerning cellular-enabled products. Apple reports defending infringement actions in U.S. and international courts, whose plaintiffs frequently seek broad injunctions and substantial damages. These are issuer statements about claims and requested relief. They do not identify a complete matter population, decide the infringement allegations or quantify the payments and product restrictions that will result.

Defense is a separate source of cost. The filing states, "Regardless of the merit of particular claims, defending against litigation or responding to government investigations can be expensive, time-consuming and disruptive to the Company's operations." Thus a scenario in which Apple successfully resists a claim need not be a scenario with no defense effort or disruption. The model therefore needs the response services, their amounts and dates, and related payment obligations. The damages outcome cannot identify those expenses or payments.

Settlement can alter recurring operations as well as resolve a dispute. Apple says it may enter agreements or other arrangements to settle litigation, without assurance of acceptable terms or that litigation will not occur. Those agreements can increase cost of sales and operating expenses, change business practices and limit products and services. A conditional settlement therefore needs its payment terms and its continuing effects on royalties, costs or offerings. An adverse resolution can additionally involve monetary damages, disgorgement, corporate measures or an injunction. The annual passage explicitly includes a resolution against an indemnified third party above management's expectations. Neither that possibility nor the separately reported quarterly settlements supplies a contract, amount or probability for a forecast.

2.17.1 Insurance describes a possible recovery, not a net loss

Apple says it maintains insurance for certain types of claims, while warning that coverage may be insufficient for all losses or all claim types. The qualification leaves both the covered population and the amount recoverable unresolved. A model cannot deduct an assumed insurance payment from damages or an accrued liability merely because insurance exists. It needs the policy period, insured party and beneficiary, covered event, exclusions, deductible, limit, insurer and recovery terms, together with facts about the claim. The applicable recognition and presentation treatment must also be established before a recovery amount becomes an accounting entry. The passages identify none of those amounts or terms.

Other annual insurance statements concern different exposures. The business-interruption discussion warns of recovery expenditures and lost sales, especially when critical components have single or limited sources, while saying insurance for certain types of losses may be insufficient (Apple Inc., 2024a, Risk Factors, printed p. 6). The data-security discussion says coverage is intended for certain aspects of those risks and may not cover all losses or claim types (Apple Inc., 2024a, Risk Factors, printed p. 12). The trade receivable discussion instead says a substantial majority of outstanding trade receivables lack collateral, third-party bank support or financing arrangements, or credit insurance (Apple Inc., 2024a, Risk Factors, printed p. 16). These

statements cannot be combined into one insured share of Apple's exposures. The receivable qualifier is qualitative; it identifies neither every uncovered customer nor a percentage. Insurance bought by Apple also does not make these exposures an insurer's issued-contract measurement problem.

The interim risk preamble incorporates the annual risks and states that there have been no material changes except those set out below (Apple Inc., 2025c, Part II, Item 1A, printed p. 19). It preserves the annual dependency and an express exception; it does not assert identical policies, claims, coverage or loss amounts at the later date. Keeping the original risk, the later qualified statement and the separately reported updates allows a historical query to retain what was actually disclosed. The resulting account connects direct proceedings, contract indemnities, defense, settlement, operating restrictions and possible recovery without collapsing them into a single loss balance. Calculating their consequences still requires the dated contracts, claim facts and accounting treatment for each channel.

2.18 A regulation can change the feasible operating plan

A legal exposure need not begin with a claim or end with a payment. An import restriction can prevent a product from reaching a customer, leaving the company to change its supply route, offering or production plan. Apple's annual and interim general-regulation risk both say that its products could be banned, delayed or prohibited from importation despite procedures intended to satisfy regulatory requirements. They also warn that measures may arrive with little or no advance notice and that Apple may be unable to mitigate all adverse impacts (Apple Inc., 2024a, Risk Factors, printed pp. 13–14) (Apple Inc., 2025c, Part II, Item 1A, printed p. 19). This describes a possible constraint on operations, separate from the damages, defense and insurance channels examined above.

The two reports preserve a broad list of regulated subjects: among them antitrust, privacy, consumer protection, intellectual property, financial services, trade, foreign exchange controls, cash repatriation, tax and employment. The interim adds national security to that list. It also expands the discussion of new and changing laws and regulations to name executive orders, directives and enforcement priorities. These additions appear in the cost-and-business-change, interpretation and specialized-application passages. They identify the scope of Apple's later risk statement without identifying a particular new instrument, effective date, jurisdiction or violation. A model cannot create a dated legal event merely because a subject is newly named in this comparison.

Both reports describe how changing requirements can increase costs, restrict a product or feature, alter its design, affect customer demand and change the business or supply chain. They describe further risks as products enter specialized applications such as health and financial services or expand the use of machine learning and artificial intelligence. The annual combines these consequences and the compliance-policy discussion in one paragraph; the interim splits and expands it. Paragraph count therefore cannot measure additional economic exposure. Conversely, the notice-and-import paragraph repeats the same substantive wording. That repetition establishes what Apple disclosed in each report, rather than identical underlying laws, products or supply routes.

The policy qualification is essential to the mechanism. Apple says it has implemented policies and procedures designed to ensure compliance, while giving no assurance that employees, contractors or agents will not violate laws, regulations or company procedures. Its separate import passage gives no assurance that programs will prevent a violation or a claim of one. Policy existence, compliance in a particular case, a claim of violation and effective mitigation

are thus different facts. Treating the reported procedures as proof of compliance would remove precisely the uncertainty the source preserves.

A conditional operating scenario should start with the actual or assumed measure and its effective date, legal scope, affected entity, product and jurisdiction. It then needs the available supply routes, inventory, lead times, customer demand and management response. For an assumed import delay, those facts determine whether delivery is postponed, rerouted or lost, and which storage, production or sourcing costs change. Recognition, payment and any later claim must follow their own facts and treatments. The filings identify these possible channels but supply neither a tariff rate nor a universal sales reduction. Their reporting dates also do not establish when a particular rule arose or when these exact report versions became available. Keeping disclosed risk, operative authority, assumed restriction and response separate lets the model retain a useful operating question without inventing its answer.

2.19 Demand, disruption and recovery follow different paths

An open supply route is useful only if a customer still wants the product. Conversely, a customer may be ready to buy while a supplier cannot deliver a critical component. The operating scenario introduced above must therefore describe both demand and the ability to serve it. Apple's annual risk discussion connects global and regional economic conditions to consumer spending, the financial condition of its business partners and its own access to finance. Its separate interruption discussion describes obstacles to manufacture, delivery and service provision (Apple Inc., 2024a, Risk Factors, printed pp. 5–6). These passages give reasons to retain several possible operating histories behind the same reported financial totals.

Apple says that sales outside the United States account for a majority of net sales and that a majority of its supplier facilities are outside the United States. The two statements concern different populations. A sales region does not locate a factory, and a factory's location does not identify the customers whose orders depend on it. The segment definitions examined earlier generally attribute sales by customer and retail-store location, while manufacturing purchase obligations describe purchases from outsourcing partners and component suppliers (Apple Inc., 2024a, Note 13, printed p. 46; Liquidity and Capital Resources, printed p. 25). Linking the two requires the actual product, component, supplier and route relationships. Geographic labels alone cannot create them.

The macroeconomic passage describes several possible effects of the same economic conditions. Recession, unemployment, inflation, tighter credit, higher interest rates and currency fluctuations can weaken consumer confidence and spending. Fiscal and monetary policy, market volatility and changes in income or asset values can also affect demand. Suppliers, manufacturers, logistics providers, distributors, carriers, other channel partners and developers may instead face financial instability, difficulty obtaining credit or insolvency. Apple further identifies trade-receivable collectibility, derivative-counterparty and financial-institution failure, limited access to new debt, reduced liquidity and declining financial-instrument fair values. The disclosure permits a scenario to connect these channels through common conditions; it supplies neither a joint probability law nor a requirement that every channel deteriorate together.

An interruption raises a different operating question: which activity becomes unavailable, and for how long? Apple's discussion includes political and trade disputes, geopolitical conflict, terrorism, natural disasters, public-health issues and industrial accidents. It says substantially all manufacturing is performed wholly or partly by outsourcing partners located primarily in China mainland, India, Japan, South Korea, Taiwan and Vietnam. Trade restrictions can raise input costs, require different suppliers or business relationships, or lead the company to cease

offering affected products, services or third-party applications. Changing the supply chain may itself be costly and disruptive, and restrictions may arrive with little notice. Apple reports that past U.S.–China tensions led to tariffs and other restrictions affecting its business. This historical assertion is distinct from its conditional warning about further escalation; a future scenario still needs a specified measure, effective date and affected trade.

The same passage describes facilities in disaster-prone locations and asserts that climate change is increasing the frequency or intensity of certain natural disasters and extreme weather. It also identifies fire, power shortages, nuclear and other industrial accidents, hostile acts, ransomware, labor disputes and health emergencies. These are Apple's stated exposures, without facility-specific hazard estimates. Its requirement that suppliers maintain safe workplaces is separate from whether an accident occurs. The company reports that major public-health issues, including COVID-19, have adversely affected demand, operations and distribution, and warns of possible future effects. Travel restrictions, freight limitations and interruptions to production ramps supply concrete mechanisms; the paragraph does not quantify an historical loss or a future duration.

Consider two explicitly assumed histories for a particular product. In one, the supply route remains available but customers reduce orders. In another, orders remain but a critical component cannot arrive. The first requires a demand response; the second requires an inventory and delivery response. They can also occur together. To compare the histories, the model needs product-level orders, available components and finished goods, lead times, contractual commitments and feasible management actions. An assumed alternate supplier must have capacity and suitable terms before it becomes an available response. A sales delay or lost order follows from those facts and assumptions; it cannot be obtained by applying one percentage to every reported balance.

Recovery adds another date. Apple warns that an interruption can require substantial time and expenditure to resume operations, with lost sales and greater consequences where critical components have single or limited sources. Note 12 separately explains that even components available from several suppliers can face industry-wide shortages and commodity-price fluctuations (Apple Inc., 2024a, Note 12, printed p. 46). Ending an assumed interruption therefore need not restore production or sales immediately. Recovery spending can precede or follow resumed deliveries. The model must retain the dates of the interruption, restored capacity, deliveries, expenditure and settlement separately; the applicable accounting treatment then determines the journals. The insurance qualification already examined in the litigation discussion does not collapse those dates into a net loss or guarantee reimbursement.

The reported receivables, investments, debt and liquidity resources constrain particular dated histories, but they do not determine these operating responses. The investment sensitivity discussed above measures a specified interest-rate effect. Chapter 10 examines the distinct debt and currency measures; none identifies the broader disruption response. The interim's incorporation of annual risk factors, except for the changes it sets out, likewise does not establish unchanged facilities, orders or funding conditions (Apple Inc., 2025c, Part II, Item 1A, printed p. 19). For the historical record, the useful distinction is now precise: retain Apple's assertions, the relationships needed to interpret them, and the analyst's scenario assumptions as separate information. Alternative demand, supply and recovery histories can then remain visible until evidence narrows them.

2.20 Product transitions change the relevant history

Keeping a supply route open does not ensure that it supplies the product customers now want. A new offering can replace demand for an earlier one while the company still holds its components, finished goods or purchase commitments. Apple's product-transition risk explicitly says that new products, services and technologies may replace existing offerings and produce lower revenues and lower profit margins (Apple Inc., 2024a, Risk Factors, printed p. 7). A model that equates a successful release with an improvement in financial performance would remove a possibility the source asks it to preserve.

The preceding competitive-market discussion explains why a transition is not merely a production date. Apple describes price competition, short product life cycles, changing standards, rapid adoption of technology and price-sensitive customers. It says that developing almost the whole product solution requires substantial research and development, without assurance of the expected returns or successful marketing. Its reported intellectual-property portfolio also does not assure protection in every jurisdiction. The discussion of competitors imitating products or infringing rights remains an issuer assertion; it does not decide a particular infringement dispute. Apple's description of minority share in three global device markets and of competitors able to sell at little or no profit supplies competitive context, rather than a numerical market share or a forecast of Apple's margins. Services face a further set of alternatives, including free content. The company's expectation of more intense competition is a dated expectation, not an observed future result.

The transition passage identifies several separate conditions for a new offering: development must succeed on time, customers must accept it, production must reach suitable quantities and costs, and application software or other third-party support must be available. New technology can create ramp-up or quality problems. Purchase commitments and inventory must also be managed in line with anticipated demand. Note 12 makes the supply condition concrete: new products often use custom components available from only one source; capacity can be constrained while yields mature or manufacturing expands, and a supplier may instead concentrate on common components (Apple Inc., 2024a, Note 12, printed p. 46). A forecast of customer acceptance cannot establish that those components will be available.

Suppose, as an illustration, that customers switch from an earlier product to its replacement. If existing components can be used in the replacement and the supplier can expand output, the transition presents one set of feasible production choices. If the components are incompatible or the new component has constrained capacity, the same assumed customer response presents another. Neither history is identified by aggregate revenue or inventory. The comparison requires the old and new product definitions, component compatibility, inventory cohorts, committed purchases, cancellation terms, lead times and available capacity. Prices, quantities and costs would then be separate scenario inputs or model consequences. The risk language gives no substitution fraction, write-down amount or probability with which to combine the histories.

Even a delivered replacement has more than one financial date. Apple's revenue policy generally connects product revenue to transfer of control, while allocated revenue for bundled services is deferred over the estimated service period; collection follows its own terms (Apple Inc., 2024a, Note 2, printed p. 35). Development, announcement, production, shipment, service provision and cash receipt therefore cannot share a single "launch" timestamp. The reported policy must be applied to the relevant contract and performance obligations. Any inventory disposition, commitment change or development-cost treatment needs its own facts and applicable accounting method. Preserving these distinctions lets the model ask whether an assumed transition is feasible and how its financial consequences would arise, while leaving actual product histories unresolved.

2.21 A supply agreement does not establish available output

Once a transition identifies the components a product needs, the next question is whether they can arrive on the required date. An agreement to purchase them does not answer that operating question. Apple's component-availability risk says that supply agreements cover many components but gives no assurance that they can be extended or renewed on similar terms, or at all. It also identifies supplier financial difficulties, business failure and industry consolidation as possible limits on supply (Apple Inc., 2024a, Risk Factors, printed p. 8). The historical record must therefore distinguish the agreement, its effective term and the supplier's ability to perform. Renewal and delivery need their own evidence before a scenario can rely on them.

More suppliers do not eliminate every shortage. Apple describes significant supply and pricing risks for single or limited sources, while warning that even components available from several sources can face industry-wide shortages and commodity-price fluctuations. Its example of past semiconductor demand and supply shortages reports an adverse effect on obtaining components and products on commercially reasonable terms. The warning that such disruptions could recur preserves a future possibility; the passage supplies no duration, quantity shortfall or loss with which to calibrate it. A scenario may specify a common industry constraint, but it must state that assumption rather than derive independence or a diversification benefit from the number of suppliers.

Custom components present a different constraint. A sole source may initially have insufficient capacity while yields mature or manufacturing expands. Alternatively, the supplier may give priority to common components. A yield problem and a production-allocation decision can both delay the same product, yet the possible responses differ. Apple separately reports past adverse effects from delayed or constrained component supply and delayed shipments of completed products by outsourcing partners. Component production, final assembly and delivery remain separate activities; an improvement in one need not establish completion of the others. The Note 12 supply discussion and the outsourcing passage retain the same distinction between sources, capacity, partner performance and control (Apple Inc., 2024a, Note 12, printed p. 46; Risk Factors, printed p. 8).

For an explicitly assumed delay, compare restoring the current source with switching to another. The disclosure says that consequences can depend on the time needed to obtain sufficient quantities from either route. A named alternative is therefore only a proposed response until component compatibility, available quantities, lead times, price and contractual terms are established. Existing purchase obligations do not identify those facts or the release of the old commitment. The model should retain both responses with their separate prerequisites and delivery dates, leaving an infeasible or unresolved response visible. Only after the relevant events and accounting facts are specified can the resulting purchases, settlements or inventory movements generate journals. The filing does not select a route or assign probabilities to the alternatives.

2.22 Available output can still create losses and obligations

Obtaining components and completing delivery do not establish that a product can safely serve its intended use. Apple's defects discussion covers its hardware and software, third-party components and products, and service outages, slowdowns and errors. It reports that services have sometimes failed to perform as anticipated. It also warns that defects can create personal-injury, property or environmental risks, that specialized uses such as health can increase those risks, and that complex technologies such as artificial intelligence can expose users to harmful or inaccurate content. Detection and repair of every defect are not assured (Apple Inc., 2024a,

Risk Factors, printed p. 9). For the model, a delivered unit, an acceptable unit and a repaired unit are different states. The source identifies possible transitions between them, not a defect rate or a particular incident.

The financial consequences can extend well beyond inventory. Apple names product liability, recalls, replacement or modification, write-offs of inventory, property, plant and equipment or intangible assets, warranty and other expenses, litigation costs and regulatory fines. Quality can also affect reputation, acceptance, demand and the timing of new offerings. These effects must not all become one inventory expense. A customer obligation and a possible supplier recovery are also separate. The outsourcing passage says partner arrangements may include defect-expense reimbursement while Apple generally remains responsible to the consumer for warranty and out-of-warranty service (Apple Inc., 2024a, Risk Factors, printed p. 8). Identifying a supplier does not establish a reimbursement right, amount or receipt date.

Excess stock raises a different question even when its quality is adequate. The inventory-risk passage says Apple records write-downs for product and component inventories that are obsolete, exceed anticipated demand, or have cost above net realizable value. It separately describes necessary cancellation fee reserves for orders of excess products and components. Inventory already held and an order not yet received are different exposures; cancellation depends on the actual order terms. Note 1's first-in, first-out policy identifies the reported cost-flow method, rather than a lot's realizable value or a cancellation charge (Apple Inc., 2024a, Risk Factors, printed p. 9; Note 1, printed p. 34).

Apple also describes impairment reviews when circumstances indicate that long-lived assets, including capital assets held at suppliers and inventory prepayments, may not be recoverable. It reports a write-down for excess carrying value over fair value if impairment is determined. This is the issuer's description of its assessment; an executable calculation still needs the actual asset classification, assessment basis and applicable accounting method. The company's belief that the named assets and commitments are currently recoverable is accompanied by no assurance against future charges. It neither proves a future impairment nor permits the model to erase that possibility. An inventory cost-flow calculation, a cancellation obligation and an asset impairment assessment cannot be substituted for one another.

The link to product transitions is the forecast used before delivery. Apple orders components and builds inventory ahead of announcements and shipments; manufacturing purchase obligations cover forecast requirements, typically for periods up to 150 days. It warns of excess or insufficient output, or failure to use firm commitments fully, when forecasts are wrong. The qualifier "typically" prevents that period from becoming a universal contract limit. Nor does the description make the separately reported Note 12 obligations with remaining terms above one year the same population as manufacturing obligations (Apple Inc., 2024a, Risk Factors, printed p. 9; Note 12, printed p. 45).

Consider an explicitly assumed reduction in expected demand. Two companies could hold the same reported inventory amount while one holds saleable components for the new offering and the other holds incompatible earlier components, defective finished goods or commitments it cannot cancel cheaply. Their feasible responses and possible losses differ. Lot identity and cost, quality, compatibility, forecast dates, anticipated demand, realizable value, order terms and customer or supplier obligations distinguish the histories. Recognition and measurement then require their own evidence. The disclosure does not supply those facts or probabilities, so the model preserves the conditional histories without inventing a reserve or treating the forecast as observed customer demand.

2.22.1 Service provision requires continuing access

Adequate components and saleable inventory still leave a service question: can the company continue offering the software and content its customers want? Apple says its products and services require third-party intellectual property, while giving no assurance that the necessary licences will be available on commercially reasonable terms or at all. Its digital-content arrangements can be short and need not renew on similar terms. Content providers can also require new rights-management or security technology before access can continue (Apple Inc., 2024a, Risk Factors, printed pp. 9–10). The relevant resource is therefore partly a dated right to offer particular content or functionality. Physical capacity alone cannot establish that right.

Developer support has a separate economic cause. Apple believes customers' hardware purchases depend partly on available applications, whose continued development and maintenance are not assured. It describes developers comparing platforms by market position, anticipated revenue, expected growth and the cost of developing and updating software (Apple Inc., 2024a, Risk Factors, printed p. 10). An application can consequently become unavailable or unsuitable even while the hardware remains usable. A scenario needs the affected applications and developers' response; an estimate of component capacity cannot supply either.

Access also depends on how a service uses data and takes payment. Apple reports restrictions on collecting, using and transferring personal information, including transfers within its international group. It separately warns that failure to follow payment-card security standards could increase transaction costs or remove the ability to process cards (Apple Inc., 2024a, Risk Factors, printed p. 15). These issuer descriptions identify possible constraints. Which constraint applies depends on the data, entity, jurisdiction, agreement and effective rule.

Suppose a content agreement permits a specified offering only until its expiry date. Under one explicitly assumed history it renews; under another it expires without replacement. Identical physical resources then permit different offerings. The comparison needs the service, customer population, territory, contract term and allowed substitutions, together with any new cost and demand response. A financial calculation must apply the relevant revenue, payment and remedy terms to the service actually provided. Apple's risk statements supply neither those contract facts nor probabilities for the alternatives. The historical record must preserve the continuing relationships as well as the reported balances if it is to support these different futures.

2.23 A useful answer before the company's history is complete

The Apple examples now give us enough information to pose a more demanding question: what can one connected answer say about the reported company and its possible cash requirements? The answer must carry both the reported amounts and the limits of the histories used to explain them. Otherwise a complete-looking statement could conceal a missing liability, a hypothetical retirement or an unavailable settlement date.

For the retained annual source, the selected statements report net income of 93,736 and other comprehensive income of 4,280, giving comprehensive income of 98,016 million dollars. Assets of 364,980 less liabilities of 308,030 give closing equity of 56,950. The following quarter reports net income of 36,330, other comprehensive income of 383 and closing equity of 66,758 (Apple Inc., 2024a, 2025c, consolidated financial statements). These are source-located readings of displayed amounts. Their units, periods, group perimeter and statement populations accompany them. A share count or a per-share amount keeps its own unit; it cannot enter a monetary total merely because it sits next to one.

The connected result retains the 102 selected annual rows, 100 quarterly rows and their separate

opening balance-sheet comparisons. It also retains all 23 note dispositions, 70 known references, seven package obligations, 32 proposed source assertions and seven additional frontier entries from the frozen inventory. A note's remaining facts stay attached to it. These records preserve the declared package; they neither discover every possible reference nor establish that every retained sentence has a formal meaning.

A query for the selected displayed statement tokens can therefore finish even though the question about actual whole-company histories remains unresolved. The family question specifies the issuer, sources and target histories, but has no admitted member list or probability measure. The distinction matters: an unresolved feasible remainder does not mean that no history exists, and two illustrated histories do not mean that the real company has exactly two possible histories. The combined answer is partial because the financial explanation has obligations that the reading task does not.

Preserving a source token also requires restraint in arithmetic. The annual cash-flow row for debt issuance displays a dash. Its observation remains a reported dash. The retained financing reconciliation separately states the narrow convention under which that dash contributes zero to its displayed sum. This convention does not turn all dashes into zeros. Similarly, the annual cash-flow opening amount of 30,737 differs from the opening balance-sheet cash amount of 29,965 by 772. That population difference is an explanatory residual, not permission to create a receipt or payment for 772.

2.23.1 Matching annual rows leaves different liabilities to settle

Two conditional asset histories illustrate why the connected result must retain journals as well as totals. In H1, gross additions of 10,529 are paid with 9,447 of cash and a capital payable of 1,082. Retirements remove gross cost of 6,000 and accumulated depreciation of 5,636, leaving a loss of 364. H2 adds and retires another 1,000 of gross cost: additions become 11,529, retirements 7,000, loss 1,364 and capital payable 2,082. Depreciation is 8,200 in both branches. The extra acquisition and retirement cancel in net property, plant and equipment; cash is unchanged. The asset journals alone leave H2 with 1,000 more liability and 1,000 less profit and equity.

Now suppose a distinct non-tax expense liability exists: its amount is q in H1 and $q - 1{,}000$ in H2, with $q \geq 1{,}000$. Each branch debits expense and credits that liability. This is an independent economic premise, not an event inferred from a statement residual. Suppose also that the retirement loss and this expense share the product-cost row, both liabilities share the same other-current-liability bucket, the extra asset cost has the same gross class, and current and deferred tax, all other activities, shares and distributions are unchanged. Then the expense difference of $-1{,}000$ offsets the extra retirement loss, and the liability difference offsets the extra capital payable.

The indirect cash-flow presentation still needs a specified allocation. Let $\alpha \in [0, 1]$ allocate the expense-liability adjustment once between the noncash adjustment and the working-capital adjustment. If Δ means H2 minus H1, the two displayed-row differences become

$$\Delta\text{other adjustments} = 1{,}000 + \alpha(-1{,}000) = 1{,}000(1 - \alpha), \tag{2.17}$$

$$\Delta\text{other liabilities} = (1 - \alpha)(-1{,}000) = -1{,}000(1 - \alpha). \tag{2.18}$$

The two differences sum to zero for every allowed α. At $\alpha = 1$, each is zero and all 102 selected annual row differences vanish under these assumptions. At $\alpha = 0$ or $1/2$, the same total operating cash has a different presentation in those two rows. Here α is a reporting assumption, not a probability over histories. Matching differences also does not construct an absolute feasible remainder for the company. The illustrative journals remain conditional subsystem histories.

Their future consequences remain joint. With $q = 2{,}000$, carrying-value settlement of the two liabilities gives the following changes, in millions, if both payments occur in the first future period:

Branch	Operating cash	Investing cash	Total cash
H1	−2,000	−1,082	−3,082
H2	−1,000	−2,082	−3,082

Paying the capital liability is an investing movement; paying the separately accrued expense liability is an operating movement. Each payment debits its liability and credits cash. Under the explicit assumption of no incremental tax, currency or interest effects, it creates no new profit, other comprehensive income or equity movement. Sufficient unrestricted funding is assumed; no opening cash balance is manufactured.

Taking the lowest operating amount from H1 and the lowest investing amount from H2 would produce −4,082. Neither history produces that total. The result must return the two pairs, not independently selectable marginal endpoints. Moving capital payment to the second period changes first-period cash to −2,000 in H1 and −1,000 in H2; moving the expense payment instead changes it to −1,082 and −2,082. Thus equal eventual cash does not identify its timing. The six specified paths retain both branch identity and payment period.

When dates are missing, the response keeps the conditional eventual amounts and marks period timing unresolved. It distinguishes these calculated amounts from reported tokens. A request for future depreciation remains unsupported without the asset cohorts, tax state and other required facts. A probability request remains unsupported without an admitted law. Neither refusal erases the amounts that a narrower question can answer.

2.23.2 The information used belongs to a particular view

The later annual source provides a different kind of progress. Its FY2024 Americas comparative shows sales of 167,045, cost of sales of $-89,587$ and selling and marketing expense of $-9,802$, giving operating income of 67,656 (Apple Inc., 2025a, Note 13, FY2024 Americas comparative). The selected original source lacks those two separate expense rows. An original-source query therefore returns unresolved components, while a query explicitly naming the later source can return them. The result links the later view separately; it does not silently enlarge the original view or assert that measurement scope was unchanged. Later detail is not, by itself, an accounting-error correction.

Named-source access answers what retained bytes contain. Investor-information and system-as-of queries additionally require accepted clocks for those exact bytes. Where publication, ingestion or review time is unestablished, the historical query remains partial with a named timing obligation. A filing event alone cannot date a separately converted PDF byte for byte. This prevents the later comparative detail from becoming evidence that an earlier investor could already have used.

JPMorgan's reference conflict illustrates the same discipline in another company. In the retained 2023 report, Item 15 faithfully says page 156; Item 8 and the auditor heading support printed page 163, physical page 165, for navigation (JPMorgan Chase & Co., 2024, Items 8 and 15; printed pp. 156 and 163). The wrong target is printed page 156, physical page 158. Both use the same two-page offset, so changing the offset cannot reconcile the targets. A proposed navigation correction preserves the literal statement. If a counterfactual earlier view had accepted the wrong target, its selected page set, bounded search and coverage summary would

need reconsideration, while the literal and an independently sourced statement amount could remain usable. A one-page negative search could never establish absence from the whole filing.

That review proposal has its own issuer and source dependencies. It supplies no Apple financial constraint, no posting permission and no observed human decision. The connected financial result and the separate review proposal can therefore be inspected together without joining their companies' histories. Section A.3 gives the exact exchange locations and rejection boundaries that preserve these distinctions.

Chapter 3

Representing a company's historical record

Northstar's corrected receivable, Apple's debt measurements and its deferred revenue show why the source record must retain more than a label and a number. Section 2.10 distinguishes a reported balance, a derived movement, a dated expectation and an unquantified obligation. Section 2.11 adds the contract and delivery facts needed to turn a commitment into a future payment. The procedures below determine which assertions enter a model and which remain unresolved.

3.1 The record is a collection of assertions

The original record includes financial statements, notes, accounting policies, auditor opinions, management discussion, segment tables, security terms, regulatory returns and subsequent announcements. A complete ingestion process accounts for every page and every extracted block. A complete financial model is a stronger claim: it would require economically relevant information that public reports may never disclose. AccDSL therefore reports document coverage, semantic coverage and simulation coverage separately.

Each source span receives a disposition: structured observation, narrative assertion, relationship, policy evidence, contextual material, duplicate, or unresolved extraction. "Unresolved" remains visible. A commitment buried in a note is not discarded because it has no current balance-sheet line. An auditor's qualification is retained as evidence about the record; it does not become an automatic numerical haircut. Management's forecast is a dated assertion with an author and modality, not an observed future outcome.

An observation is an immutable record with the fields below. Values inferred from observations are separate records, linked by a derivation. This avoids overwriting a company's statement with an analyst's preferred reconstruction.

Field group	Minimum contents
Identity and source	Observation ID; document hash; filing identifier; PDF page and bounding box or HTML/XBRL locator; literal text; extraction version.

Field group	Minimum contents
Reporting context	Legal entity; reported group perimeter; statement or note; instant or interval; standard and taxonomy versions; dimensions such as segment, product, geography, seniority or measurement category.
Amount	Original string; parsed number; sign interpretation; unit; scale; currency; precision or rounding interval; missing-value category.
Meaning	Source label; taxonomy QName when present; one or more proposed concept mappings; whether the assertion is reported, estimated or conditional.
Timing	Economic validity when known; reporting period; publication and availability timestamps; ingestion timestamp; correction or supersession links.
Reliability and review	Extraction confidence, reviewer decision, source status and known contradictions, all distinct from a probability over histories.

A narrative assertion adds actor, predicate, object, scope, quantifier and modality. For example, "substantially all debt is fixed rate" has a source span and a qualitative quantifier. It does not assert $\alpha = 1$. A reviewed interpretation can introduce a numerical restriction such as $\alpha \geq a$, but the threshold a remains an analyst assumption unless the source defines it. Boilerplate policy language also does not prove that a company holds a particular instrument.

3.2 Different names, different measurements

An accounting concept is more than a row label. "Debt" can refer to legal principal, amortised carrying amount, undiscounted contractual cash flows or a fair-value disclosure. These amounts can all be correct at the same date. Likewise, "revenue" may be reported gross or net, a segment measure may differ from the consolidated measure, and an alternative performance measure may exclude selected expenses. A mapping must state its relationship: equivalent meaning, component of, broader than, presentation reclassification, or measurement transformation.

The common vocabulary will preserve source concepts as first-class identifiers. Company extensions may inherit a broad economic category without being declared identical to a standardised concept. For instance, an extension can be a component of borrowings while retaining an unresolved split between lease liabilities and other financing. Many-to-many mappings are allowed; each mapping has applicable dates, dimensions and evidence. An aggregation mapping does not silently become a measurement conversion.

Standards differ in recognition, measurement, presentation and disclosure. Consequently, conversion between IFRS, US GAAP and Chinese Accounting Standards for Business Enterprises (ASBE) is not generally a dictionary lookup. The original basis remains preserved, and any comparison basis is a separate derived book. The same principle applies to tax and prudential capital. A tax loss carryforward, accounting deferred tax asset and regulatory capital deduction can concern the same economic loss while belonging to different measurement systems.

3.3 Time, revisions and information availability

At least four times matter. A machine may enter service in November; its cost appears in the accounts for the year ending December; the annual report is published in March; and AccDSL ingests the report in April. A later report may revise the December comparative. These times answer different questions and must remain distinct.

The storage model uses validity intervals for economic assertions and transaction intervals for database versions, following the distinction in Snodgrass and Ahn (1986). Publication and availability timestamps are additional domain fields. Database transaction time records what AccDSL contained; it cannot substitute for what investors could have known. An "as known on" query applies an explicit availability rule and does not include a later restatement simply because its economic period is earlier.

AccDSL supports three named views: the original filing view; the latest restated view, with a stated information cutoff; and a harmonised research view, with explicit transformations. No view changes the stored source facts. Corrections have links to the replaced assertion and a reason. Where two sources conflict without a clear supersession relationship, both survive with a conflict record.

Instant values and interval flows have different types. A balance at 30 June is not a six-month flow. Calendar intervals use half-open boundaries $[t_0, t_1)$ internally; a filing period ending 30 June maps to the next boundary at 1 July in its reporting calendar. Quarter-only earnings may be derived from year-to-date values only if the periods, basis, perimeter, units and revision vintage are compatible. The derivation retains both source observations and their correlated rounding uncertainty. Fifty-two/fifty-three-week calendars, stub periods and reporting-date changes require explicit calendar objects.

Accounting-policy changes, estimate changes and error corrections also have different temporal effects. The applicable rule package decides whether a change is prospective, requires restatement, or adjusts opening balances. The engine never infers that effect from a generic "revision" flag.

3.4 The company boundary changes

Legal entity, reporting group and economic exposure are separate dimensions. Acquisitions, disposals, changes in control, joint arrangements, special-purpose vehicles and non-controlling interests can change the reporting perimeter. A group's increase in property need not be capital expenditure; it may reflect an acquisition, a currency translation or a revaluation. The bridge between opening and closing property must identify these movements before a scenario model can extrapolate investment.

The company graph therefore stores dated ownership, control and counterparty relationships. Control is an accounting conclusion with supporting evidence, not a rule that blindly consolidates every majority-owned entity. Group reporting combines entity books with a separate adjustment and elimination book. Intragroup receivables and payables require matching counterparties, currencies and dates. Unmatched eliminations remain unresolved; a balancing "plug" is not a recovered transaction.

Acquired opening balances, goodwill, purchase-price allocation, divested operations and non-controlling interests each have their own event types. Analyses can preserve reported scope or define a constant perimeter, but the latter is a derived modelling assumption, not the original record.

3.5 Further problems that the representation must handle

Several difficulties extend beyond differences in standards, industries and company labels. Recognition and settlement occur on different dates: a sale, invoice, cash receipt and impairment can belong to four periods. Net balances hide gross flows: repeated borrowing and repayment can leave the same closing debt. Netting can conceal offsetting exposures and liquidity needs. Different contractual maturities and repricing dates can share the same current carrying amount. Guarantees, undrawn credit facilities, litigation and covenants can be economically important before an asset or liability is recognised.

Foreign currency introduces transaction currency, functional currency and presentation currency, together with different translation conventions for stocks and flows. Currency translation effects must not be mistaken for cash receipts. Inventory and depreciation depend on vintages or lots; an aggregate stock and average age may be insufficient under a shock. Tax bases can evolve differently from carrying amounts. Deferred tax recognition requires assumptions about recoverability, rather than multiplication by a tax rate alone.

Public information is selectively disclosed and rounded. Duplicate XBRL facts may agree, overlap within rounding bounds or contradict one another. A dash can mean zero, immaterial, unavailable or inapplicable. Tables can mix units, span pages and embed signs in parentheses. Text and table references can be wrong in the source itself. AccDSL retains these distinctions and gives an analyst a local repair path without pretending that extraction confidence settles economic interpretation.

Finally, strategic behaviour matters. A firm may refinance, cut investment, raise equity, change prices, breach a covenant or fail. A bank may preserve a capital ratio by shrinking assets; an insurer may change reinsurance or asset allocation. Such responses cannot be deduced from the balance-sheet identity. They require explicitly calibrated or assumed policies, and alternative policies must remain separate when the available record does not distinguish them.

3.6 Account for a disclosure before translating it

The importer first freezes a filing package: documents, byte hashes, reporting dates, availability evidence and incorporated references. It compares the document's contents page and statement index with detected section and note headings. A missing heading, table continuation or referenced exhibit creates an obligation to resolve; successful text extraction cannot discharge it. Each PDF page and each HTML content unit then has a stable locator and a disposition. Child units such as a table or note belong to a parent page/section; the importer records overlap instead of counting the same content twice.

Three denominators remain separate. Acquired-document coverage counts source units with a disposition, including unresolved units. Semantic coverage counts identified assertions whose meaning and context have been reviewed. Simulation coverage counts the operations and exposures required by a specified query that have supported state and rules. The first can be complete while the other two are partial. An assertion inventory can itself be incomplete; until the page/section reconciliation is reviewed, semantic coverage is a lower-bound inventory rather than a percentage of all financial meaning.

Cross-reference traversal is a work-list algorithm. Start with the filing's statement and note references, resolve each target against the acquired manifest, add newly discovered targets, and stop when no unvisited target remains or a declared resource limit is reached. Document hash plus target locator prevents cycles from expanding forever. A target has one of four results: resolved, duplicate of a resolved source, unavailable, or ambiguous. The last two

remain obligations. Apple's proxy and prior-filed agreements show why closure over acquired files is weaker than closure over incorporated disclosures. A manifest may declare a bounded acquisition scope, but must show these outstanding references beside its completeness status.

A narrative translator records the actor, predicate, object, quantifier, modality, applicable entities/dates and qualifications before proposing any constraint. "All notes are fixed-rate" constrains contractual coupon type; it does not constrain the combined note-and-swap exposure. "May repurchase" creates an authorised action with limits, not an obligatory payment. "Substantially all" remains qualitative unless a separately reviewed assumption gives it a threshold. A conditional guarantee becomes an obligation with a trigger and exposure, not a present liability merely because its maximum amount is known. Unsupported predicates retain their literal text and links to affected balances. The reviewer accepts the translation separately from accepting that the text was extracted correctly.

3.7 Mappings must establish a financial relationship

A mapping record identifies source concepts, target concept, relationship, applicable entity/basis/period/dimensions, evidence, reviewer and any required transformation. Equivalence permits substitution only when all these contexts match. A component relationship permits inclusion in a sum only through a declared disjoint partition. A broader concept supplies containment, not an equality. Reclassification changes presentation while preserving the stated population and total; measurement transformation must supply an actual formula and inputs. These relationships cannot compose indiscriminately: component followed by equivalence remains component, while two measurement transforms compose only if their intermediate bases and information dates agree.

For example, Northstar's product revenue might split into 180 domestic and 80 overseas, while its customer channel split is 200 wholesale and 60 retail. Both classify the same 260. Adding all four numbers gives 520 and is invalid. The partition records therefore share a population ID but have different axes. To remove an axis, `sumOver` requires evidence that members are disjoint and exhaustive for that context. Where the source supplies only named segments, an explicit unallocated member carries the residual; it is not assumed zero. Geography-by-product cells cannot be recovered from the two marginals without further constraints or assumptions.

Apple supplies another counterexample: its vendor non-trade receivables arise from component sales to manufacturing vendors, whose gains reduce product cost of sales when the final products are sold (annual Note 4, p. 38). They cannot be relabelled as ordinary revenue-generating trade receivables without changing meaning. Likewise gross derivative assets and liabilities cannot be silently netted. Company extensions can define these distinctions and dated presentation changes; they cannot override mandatory recognition rules in a selected standards package.

3.8 Selecting original and revised assertions

The query procedure first filters assertions by the declared availability cutoff, then by an optional database-time cutoff. A day-only filing timestamp is conservatively available at the next local midnight in its named timezone; any more precise rule needs evidence. Within the eligible set, explicit supersession removes only the targeted assertions in matching contexts. The superseding assertion must itself be eligible. Follow the acyclic graph transitively; retain all maximal assertions if two eligible corrections conflict. File order and ingestion recency never decide which is true.

The original view selects the initial assertion chain in a named filing; the latest-available view selects eligible maximal versions; the harmonised view adds reviewed transformations to

those selections. A revision to revenue does not automatically revise every note or change the company's accounting basis. An estimate change takes effect in future measurements from its specified date. An error correction may change a comparative and an opening balance bridge, each as a new assertion with its own lineage. Northstar's explicit query fixture in Chapter 2 checks all these clocks without assuming that the most recently stored row is always the appropriate history.

3.9 What an accepted reading permits

Apple's inventory policy illustrates the distance between reading a sentence and reconstructing a business. Its reported use of FIFO supports a proposed cost-formula reading for the inventory and reporting contexts covered by that policy. It does not reveal the quantities and ages of the actual lots, establish that no write-down occurred, or supply the authority for every US accounting treatment. Those are separate inputs to a cost and impairment calculation. Similarly, a straight-line depreciation policy leaves useful lives, residual values, components and available-for-use dates to be established. Section 9.6 tests those limits against the reported asset balances, depreciation and cash payments, then constructs two different asset books compatible with the selected amounts.

The initial source preparation retains six located Apple policy passages: consolidation of Apple and wholly owned subsidiaries; comparative reclassification; FIFO; straight-line depreciation; the interim statement's estimates and annual-report reference; and its 52/53-week fiscal calendar. Each proposed reading carries the original document hash, extraction hash and exact character slice. These are company assertions. The consolidation sentence, for example, cannot exclude every partly owned entity without a control analysis. The calendar sentence does not make every future fiscal year 52 weeks or make a contractual interest year 365 days. The exact quotations and proposed predicates are retained in `../methods/source-translations-v1.json`; the meaning explained here is not an invented human acceptance of them.

A reading progresses through separate decisions. First, extraction review asks whether the located words and numbers were transcribed faithfully. Translation review asks whether the proposed predicate preserves their scope, quantifier, exceptions and reporting context. A model then admits that predicate, or an explicitly labelled assumption, for a particular purpose and information view. Accounting eligibility subsequently asks whether a method applies to the actual event. A faithful transcription of an erroneous company assertion is still faithful transcription. The four decisions therefore cannot share a single "approved" flag.

An explicit presumption in a standard also differs from filling an unknown with a convenient answer. Article 6 of the retained official ASBE 21 text requires both practical ability throughout use and economic benefit for a substantive substitution right. It specifies a presumption when that right is difficult to determine. Applying that conditional rule requires its stated circumstances; it does not transform missing lessor information into a reported fact. Nor may a translator assign arbitrary numerical thresholds to "substantive", "probable" or "highly probable".

3.10 When a faithful reading preserves a wrong reference

An analyst looking for the auditor's report in JPMorgan Chase's 2023 Form 10-K receives two incompatible directions. Item 15 says that the consolidated financial statements, notes and auditor's report listed in Item 8 are "set forth commencing on page 156". Item 8 identifies that same set as appearing on pages 163–309. Both statements are printed in the retained

issuer-hosted PDF (JPMorgan Chase & Co., 2024, Items 8 and 15, printed pp. 36 and 40). Correctly copying each sentence therefore preserves the disagreement.

The page numbering does not explain it. Printed page 156 is physical PDF page 158 and contains management's discussion and analysis. Printed page 163 is physical page 165 and begins the report of the independent registered public accounting firm. The two-page offset applies to both locations. Item 15's explicit reference to the material listed in Item 8 also rules out treating it as a direction to a broader discussion beginning in MD&A (JPMorgan Chase & Co., 2024, printed pp. 156 and 163). For this document and target, the page-156 direction is wrong. The supported navigation correction is page 163; it is a separate interpretation of the retained document, not a replacement for what Item 15 actually says.

The review must retain those two meanings of correctness. The source record keeps the PDF, both located sentences, their unchanged extractions and images of the referenced pages. A proposed extraction decision can accept 156 as the faithfully transcribed number. A translation can likewise record that Item 15 directs the reader there. Neither decision accepts the proposition that the auditor's report actually starts there. That target proposition is refuted by the inspected page. A separate navigation reading points to 163, supported by Item 8 and the report heading. This reading permits locating the report; accounting admission still requires the relevant financial assertions, contexts and methods.

Consider the resulting review sequence as a specification example. Propose both literal readings and retain their common target and document identity. Accept their transcription and attribution at the stated layers. Reject 156 as the verified start location, keeping the rejected proposal and its reason, and propose 163 for navigation admission. The source disagreement remains visible after that repair. It does not require keeping 156 and 163 as equally admissible start locations: an `AcceptAlternative` action for 156 at the target-location layer would fail the evidence check. Uncertainty is preserved when evidence leaves alternatives open; a refuted reading is preserved as review history.

Different queries now have different exact answers. Asking what Item 15 says still returns 156, with its locator and the conflict annotation. Asking for the supported start of the auditor's report returns printed 163, physical 165, with both supporting passages. Asking whether a particular accounting treatment is valid remains outside this navigation decision. Nor can a search confined to page 156 conclude that the entire filing contains no auditor's report. Its proper negative result is only that the report was not found on the selected page.

Suppose an earlier analysis had nevertheless accepted 156 as the correct start. This is a counterfactual mistaken decision, not an observed human review in this project or a permitted new admission after the target check. Repair creates a new reading and an explicit `Supersede` decision in a new evidence view. The old accepted decision, its reason and its results remain available under the old view. The corrected view cannot reuse a cached result whose justification still depends on that old navigation choice.

That dependency includes the pages selected for a search. A cached "not found on the selected page" answer must be recomputed when the admitted location changes, even if its output contains no extracted assertion from the new page. A coverage conclusion derived from that answer changes with it. By contrast, the literal Item 15 quotation remains unchanged. A separately located statement value whose provenance does not depend on the repaired reference may also be reused after its other dependencies are checked. An annotated display of the unchanged quotation still needs to refresh when its conflict or review status changes; preserving the text does not preserve every display derived from it. The correction therefore invalidates dependent navigation, search and coverage results; it does not mechanically invalidate every financial amount in the filing.

This case supplies an actual source-reference conflict and a proposed account of its review consequences. The source inspection is author work; the review actions above are expected behaviour, with human adjudication and software execution still pending. The finding establishes no financial misstatement, publisher erratum or historical availability of the retained bytes. Those questions require their own evidence.

3.11 Three meanings of disclosure coverage

The initial Apple annual/interim preparation inventories 125 quantities, 13 annual notes, 10 interim notes, 70 exhibit references and seven further package obligations. In that starting assessment, all 23 notes retain unresolved semantic content; none is recorded as fully formalised. The later worked readings now trace representative revenue, commitment, asset, tax and funding consequences, while retaining missing facts and wider package obligations. Downloading a note, reviewing its material passages and using it in a complete company model remain different achievements.

Three populations make that distinction inspectable. Retained-unit coverage partitions acquired documents into nonoverlapping pages or smaller leaf units, including duplicates and unreadable material. Assertion-review coverage counts the separately enumerated material assertions and their review decisions. Query-exposure coverage identifies the sources, legal facts, methods and authority that a particular result requires. A fully downloaded annual report can have incomplete assertion review, and a reviewed debt total can leave a refinancing query unresolved because an incorporated agreement is missing. No weighted average of these populations establishes completeness.

Preparation begins with declared annual and interim roots and a policy for following incorporated references. Each edge is recorded as satisfied, duplicate or cyclic, outside the declared scope with a reason, inaccessible, or still unexpanded. A bounded traversal saves its unexplored frontier and budget so it can resume. An unknown external extent remains unknown. Acquired proxy and registration documents improve the Apple record, but do not prove that every executed amendment, issuance term and exhibit dependency has been closed. The present real package remains incomplete for that reason.

Container choice can itself create a false missing-source diagnosis. Apple's retained annual PDF contains securities and award exhibits absent from the retained HTML container; other incorporated award forms are present in earlier reports in the retained corpus. Locating those bodies resolves a source-access question. It still leaves their qualifications and outgoing references to be interpreted. Conversely, the cash restriction in Section 2.8 shows why a located note can change the meaning of an already extracted figure: correct arithmetic alone would not detect an erroneous "unrestricted" label.

3.12 Displayed precision does not determine a rounding rule

Suppose a statement displays 100 in millions. The scale establishes how the literal is read; it does not establish how the issuer rounded its underlying amount. Under an explicitly admitted nearest-rounding assumption, an integer display m at scale s permits the conservative compatibility enclosure

$$s(m - \tfrac{1}{2}) \leq y \leq s(m + \tfrac{1}{2}).$$

The closed endpoints accommodate an unknown tie rule within that nearest-rounding model. A specified tie rule can exclude an endpoint. Truncation of a positive amount instead permits

$sm \le y < s(m+1)$. Thus the illustrative display 100 corresponds to 99.5–100.5 million under the first assumption and 100–101 million, excluding 101, under the second. These are different restrictions on a history.

The retained Apple arithmetic uses the first enclosure as a conditional compatibility assumption; the issuer's rounding convention has not been established. Without that assumption or source support, the record states known display precision and unknown rounding convention and supplies no hard half-unit constraint. A one-unit difference between a displayed total and the sum of displayed components is consequently not enough to establish a contradiction. The source value is scaled once, while its literal and precision metadata remain available for inspection.

3.13 Public availability and system possession are different views

Northstar's correction explains why a result needs a reporting vintage. A historical forecast also needs to establish when the exact document bytes were available. A report downloaded today may carry an old date on its current web page; that does not prove that the same bytes were public on that date. Public availability evidence and an ingestion receipt answer different questions.

An investor-information query supplies a cutoff K. Each admitted source must have supported public availability no later than K and evidence linking the selected bytes to that availability. Later ingestion is allowed. Later extraction review can read eligible old bytes, but an interpretation that uses later information fails the cutoff on its own dependencies. This view establishes public availability, not actual investor awareness or the ability to trade.

A system-as-of query instead supplies an instant S and requires an immutable receipt showing possession of those bytes by S. Publication cannot replace that receipt. An accepted-layer query also requires review decisions visible by S; a raw-layer query may return a then-unreviewed document. A query asking both what was public and what the system possessed must pass both filters. Unknown publication alone need not block a pure possession question.

Archival exploration permits inspection of retained sources with incomplete timing. With a cutoff, unknown publication yields an unresolved candidate, not a cutoff-qualified selection. Without a cutoff, the archive may display the retained lineage, explicitly without a historical-knowledge claim. A known later version is excluded from an earlier view and cannot suppress an eligible old assertion. An unresolved potentially relevant version keeps the affected answer unresolved even when an older assertion is displayed.

Date-only evidence requires particular care. For publication date D in a named timezone, let $[a_D, b_D)$ be the interval from local midnight to the next local midnight. A cutoff before a_D excludes the source; a cutoff inside the interval leaves availability unresolved; a cutoff at or after b_D supports admission on this date evidence. No arbitrary noon or fixed 24-hour duration is inserted. Missing timezone leaves the instant comparison unresolved.

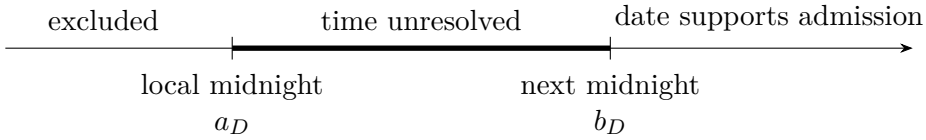


Figure 3.1: A dated publication supports historical admission only after its possible publication interval has ended. The line marks uncertainty about an instant, not a delay imposed on an already known timestamp.

Revision selection follows admission. Original selection returns eligible roots; latest-available

selection returns eligible assertions without an eligible superseding descendant for the same context. Unrelated roots and multiple live leaves remain competing assertions. A harmonised view adds explicit transformations with their own admitted sources, method and opening bridges. File order never selects a winner.

The synthetic interim example has original first-half revenue 120, revised first-half revenue 110 and revised annual revenue 260. Compatible subtraction gives second-half revenue $260 - 110 = 150$. The arithmetic $260 - 120 = 140$ is easy but mixes incompatible vintages. An original-view display may show both source amounts while refusing that derived flow. A change in calendar or perimeter likewise requires a specific bridge.

The financial-amount comparisons investigated so far have not supplied a qualifying unresolved same-context measurement contradiction. The reference conflict in Section 3.10 establishes the narrower source-navigation finding. Known revisions, different reporting bases and synthetic conflicts remain useful cases of their own kinds. A financial conflict case must first match entity, perimeter, concept, dimensions, unit, basis, period, versions and precision, and investigate extraction and supersession. The absence of a qualifying measurement conflict in a bounded search is not evidence that the corpus contains no conflicts.

3.14 A revised fact changes a conclusion without rewriting its history

A correction can change the amount that a model now explains while leaving both its earlier explanation and the company's committed journals intact. The bank and insurance examples later in this proposal make the distinction concrete. The bank needs the credit-risk baseline of its irrevocable commitment even after a loan is drawn. The insurer needs to know whether its reinsurance treaty preceded the recognition of an onerous group. Neither fact can be replaced by the time a document happens to arrive.

Consider a constructed bank record that supplies no original commitment baseline. The analyst can retain its cash flows and conditional loss scenarios, but cannot establish the required original-to-current credit comparison. A later record identifies the original commitment risk and supports the separately supplied finding of significantly increased credit risk. With the other assumptions in Section 11.6, the funded allowance is 4 and equity is 16. The additional record resolves the missing history; it does not reset the baseline to the draw date. The relevant AASB 9 provisions retain the irrevocable commitment as the initial comparison and treat the resulting asset as its continuation (Australian Accounting Standards Board, 2022g, paragraph 5.5.6 and B5.5.47). The probability tree remains an authored assumption, and the credit finding is not inferred merely because a probability doubled.

Suppose, for this constructed record, exact bytes were publicly available on 2 January, ingested into the system on 5 January, and accepted for the required reading on 6 January. An investor-information reconstruction with a 3 January cutoff may use those bytes if their public availability is verified, even though this particular system obtained them later. A system-as-of query for 3 January cannot use them. On 5 January a raw system query may retrieve the document, but a query for an accepted reading still lacks the 6 January decision. Later extraction is compatible with the earlier investor cutoff only if its interpretation introduces no later information. These dates are authored specification fixtures; they establish no actual historical availability.

A treaty correction separates knowledge from measurement

In Section 12.6, treaty R entered before group B's recognition produces an initial reinsurance asset of 85, initial equity of 85 and year-one equity of 83. If R instead enters immediately afterward, with all other stated facts unchanged, the initial asset is 70, initial equity is 70 and year-one equity is 74.75. The loss-recovery recognition condition causes the difference (Australian Accounting Standards Board, 2022e, paragraph 66A and B119C). Both histories eventually reach cash of 88.3 on the same supplied claim and recovery path. Agreement on that terminal amount therefore cannot establish correct recognition timing.

Now give the two order assertions different information dates. A constructed January disclosure says the treaty preceded recognition; a February correction says it followed recognition and explicitly replaces that order assertion. A latest-available query cut off in January retains the January assertion. The February record is known to be too late and cannot suppress it. A February query may select the correction and produce the later-entry measurement. The correction concerns the order assertion for B and R; it does not replace every other assertion in the company's report.

A competing record with unknown public availability has a different effect from a record known to be published after the cutoff. The known late record is excluded. The unknown record leaves an unresolved dependency if it might affect the requested conclusion, even while an older eligible assertion is displayed. Similarly, an asserted publication date without evidence linking the retained bytes to that date does not establish historical admission. If only the common calendar date of treaty entry and group recognition is known, their order remains unresolved; the model cannot choose 85 merely because that treatment is more favorable.

An extraction correction is different again. A source may consistently say that R followed recognition while its first accepted extraction mistakenly records the opposite. Correcting the reading creates a new reading and review decision linked to the same source bytes. The old result remains retrievable as a historical error; reproducing it does not make it correct. An issuer's correction instead changes the source assertion and usually its bytes. The system records which change occurred before deciding which derived results require reassessment.

Invalidate the selection, preserve the committed event

A result depends on the candidates considered by its selector as well as the candidate ultimately chosen. Adding a possible competing treaty record changes that candidate population. Reusing a cached selection solely because its chosen assertion identifier is unchanged would miss the new obligation. The proposed dependency graph therefore includes a version for candidate-set membership. The change propagates from membership through selection and interpretation to measurement and the resulting statement. A fresh selection may exclude a known late competitor and calculate the same amount, but it records that decision under a new view.

Newly created descendants must check their complete ancestry too. Otherwise a report created after an invalidation scan could inherit a stale measurement without appearing on the scan's earlier list. In contrast, a bank result whose complete dependency set contains no changed insurance inputs can remain reusable. Neither a shared company label nor an unchanged displayed value establishes independence. Missing ancestors leave reuse unestablished; a cycle or missing required reference rejects the affected dependency chain.

The retained explanation of a result uses its original evidence view. It can show a current warning that a dependency has changed, but it cannot replace the old source with today's correction and still purport to explain the old calculation. Requesting a different view requires a new calculation. This preserves the distinction between the company's historical record, what

the system then concluded, and what a later reconstruction supports.

The same rule protects committed accounting events. The connected insurance example commits financing, service, incurrence and settlement as four separate events. After settlement, retrying the exact original financing request returns its original receipt and reports no additional commit; today's evidence revision does not turn that retry into a new assessment. Changing the request while retaining the business-event identity is a conflict. The request's original bindings, historical admission and required bytes must still validate before receipt lookup.

A new conclusion that the treaty came later does not itself authorize an entry reducing the committed reinsurance asset. A correction proposal must identify its accounting treatment, effective period, affected accounts and opening-balance consequences under the applicable authority. Without that support, the proposed assessment returns `TREATMENT_UNSUPPORTED`, retains the revised conditional calculation and creates no journal. Versioning supplies the record of a correction; it cannot supply the accounting rule that makes the correction appropriate.

Chapter 4

What the literature contributes

The research problem crosses accounting, databases, programming languages, econometrics and document understanding. Each field resolves part of it. The review below distinguishes results that can be adopted directly, engineering ideas that need adaptation, and evidence that does not establish the intended product capability. Source versions and technical anchors are recorded in the accompanying research ledgers. The search includes work outside the original collection, with public-index searches and backward and forward citation searches through 2 October 2026. It is a scoped design survey, rather than a claim to enumerate every paper in these fields.

4.1 Accounting structure and company dynamics

Nehmer and Robinson (1997), especially Sections 2–5, supplies a useful algebraic starting point: balances occupy a structured state space, transactions change that state, and aggregation relates detailed accounts to reports. This supports a distinction between conservation constraints and permitted transactions. An entry can conserve the accounting total and still be an inadmissible transaction. AccDSL adopts that distinction in its posting validator. Its particular event vocabulary, evidence model and inverse-history semantics are new design decisions, not results established by the algebraic paper.

Doshi (2024) develops an accounting DSL with time, transaction logs and correction mechanisms. The relevant material is in Chapters 4 and 7–9 of the 2024 thesis; the original collection's filename misleadingly suggests 2026. It demonstrates why corrections and dates belong in the language rather than in informal comments. The thesis does not establish recovery of an entire company history from public reports or inference over economically distinct histories. Babcinețchi et al. (2022), pp. 342–345, offers a much smaller accounting language and grammar proposal. It is useful as a syntax precedent, but it provides insufficient semantics or evaluation to govern AccDSL by itself.

Shahnazarian (2004) connects accounting dynamics with behavioural and econometric equations. Sections 2.3–2.8 describe the accounting relationships; the optimisation and econometric discussion in Chapter 3 and the data and evaluation material in Chapters 4 and 8 show how the model is used with standardised Swedish corporate data. This is an important precedent for simulating companies under changed conditions. Its standardised tax-data starting point differs from heterogeneous public-report extraction, and its working-capital funds formulation should not be imported as a general modern cash-flow statement implementation. The transferable idea is the explicit separation of accounting evolution and behavioural assumptions.

Pierson (2020), Sections 3–4 and the reporting structures, makes that separation operational in system dynamics. Receivables, payables and wages accumulate differences between accrual and payment flows; reporting-period accumulators turn instantaneous flows into statements. AccDSL can use these structures as examples of coherent movement schedules. The model's payment delays, inventory approximations and simplified equity treatment remain modelling choices. They are not universal defaults for companies or evidence that an aggregate state is sufficient for all shocks.

Bank models demonstrate why journal generation needs economic state. Halaj (2016), Section 3, combines funding shocks, fire sales, loans, securities and risk constraints. The homogeneous portfolios and specified funding process make the model tractable, but limit instrument-level interpretation. Behn et al. (2019), Section 2, makes managers' choices over asset growth, debt, risk and payouts depend on capital and liquidity constraints. These models motivate interfaces for management policy and regulatory calculations. Their parameterisations must be calibrated and evaluated for the target application; passing their accounting identities would not validate an AccDSL forecast.

Two important neighbouring traditions remain explicit coverage qualifications. McCarthy's 1982 REA model is a foundational candidate for the relation between resources, events and agents; Caiani and co-authors' stock-flow-consistent agent-based benchmark is a candidate for simulation across interacting sectors. Verified primary PDFs could not be obtained in this retrieval campaign after institutional and publisher attempts. Their records are retained in the omission register, and no technical result from either is used to justify the design. AccDSL's event and counterparty records are specified and justified in Chapters 7 and 8. A later multi-company equilibrium engine would require expansion of the stock-flow-consistent literature before implementation.

4.2 Incomplete information and provenance

Financial disclosures are incomplete databases in a substantive sense. Multiple underlying records can yield the same published observations. Libkin (2014), Sections 2–6, explains the semantics of incomplete databases, labelled nulls, conditional tables and certain answers. Repeated occurrences of the same unknown must share a value. Conditional tables can represent dependence between alternatives that independent missing cells cannot express. This is directly relevant to a debt note and a cash-flow table that refer to the same unknown repayment.

AccDSL adopts shared symbolic unknowns and conditions. It extends the database idea with accounting constraints, continuous amounts, temporal transitions and economic event models. A "certain" answer means a statement that holds for every history in a nonempty declared feasible family. This is Boolean certainty; a table of individually certain rows can lose existential information and must not imply that omitted rows are absent. The payable example in Section 6.2 develops this distinction. The family itself may omit unknown mechanisms; certainty is therefore always relative to a model scope. Imieliński and Lipski's original 1984 paper is a foundational backward lead whose publisher copy was inaccessible in this campaign. The inspected Libkin paper supplies the technical discussion actually used here.

Green et al. (2007), Definitions 3.1–3.2 and Section 4, treats provenance as annotations that compose through relational operations. Combining sources and alternative derivations can therefore have formal semantics, rather than being a list of unrelated citations. This is useful for report aggregation and explanation queries. Positive relational operations covered by provenance semirings do not by themselves specify every operation AccDSL needs: negation, nonlinear valuation, optimisation and probabilistic conditioning require their own derivation

records. Accordingly, the implementation starts with a typed dependency graph and may use semiring annotations for supported relational subqueries. A dependency edge is not a posterior probability.

Snodgrass and Ahn (1986) distinguishes historical validity from the time facts are stored. The archived paper is *Temporal Databases*, by Snodgrass and Ahn, not the later Jensen–Snodgrass paper initially returned by a search. Its snapshot, rollback, historical and temporal examples support the need to recover both old business facts and old database beliefs. AccDSL adds public availability time because a database insertion date cannot establish whether a historical investor could have seen a filing.

4.3 Inverse problems, identification and probability

Stuart (2010), Section 2.2, starts from an observation equation $y = G(u)$ and distinguishes inversion, regularisation and Bayesian inference. The discussion makes explicit that a regularising norm or prior is an additional assumption. AccDSL has a similar observation problem, although its unknowns include discrete accounting interpretations and event structures as well as continuous parameters. The infinite-dimensional well-posedness theorems in that paper are not automatically applicable to this mixed, potentially discontinuous model.

The relevant issue is often identification rather than computation alone. Giacomini and Kitagawa (2021), Sections 2.1–2.6 and Theorem 1 of the archived 2020 working version, shows how a class of priors can preserve uncertainty within an identified set. A likelihood that is flat over the unresolved part does not learn the conditional prior over that part. Posterior lower and upper probabilities summarise a class of posterior beliefs. The paper's regularity conditions and application to structural vector autoregressions do not prove that an arbitrary accounting program admits the same algorithms or guarantees. The design borrows the distinction between learnable quantities and assumption-dependent distributions; the simple accounting bounds in Chapter 6 are derived directly.

Hoeting et al. (1999), equations (1)–(3), gives Bayesian model averaging as a weighted sum of model-conditional distributions, with weights determined by model priors and marginal likelihoods. Sections 3 and 5 discuss computation and prior specification. This supports an optional weighted mode, provided AccDSL specifies the likelihood, priors and candidate model space. It does not justify giving equal mass to every syntactically different journal reconstruction. Two encodings of the same economic history must not gain extra weight by duplication. Model averaging also remains conditional on the considered model family, rather than repairing missing mechanisms.

Scenario semantics need similar care. Pearl (2009), Section 3.2.1 and equations (5)–(6), distinguishes an intervention that replaces a structural equation from ordinary conditioning. AccDSL will distinguish a supplied exogenous path, an observational conditional forecast and an intervention. A rise in rates imposed by a stress test is not automatically distributed as the collection of historical periods in which rates happened to be high. Identifying a causal effect requires assumptions beyond reproducing accounts.

4.4 Languages for rules and contracts

van Deursen et al. (2000) surveys DSL benefits, risks and implementation options. Mernik et al. (2005), Sections 1.1 and 2.2–2.5 of the archived CWI report, treats application libraries as serious competitors and distinguishes deciding, analysing, designing and implementing a

DSL. Its implementation decision tree explicitly includes embedding in a host language. The literature therefore supports a conditional design decision: a DSL is attractive when domain constraints and repeated notation justify it, but an ordinary typed library may be cheaper and equally expressive. AccDSL will implement one shared representation accessible from both a Python API and a textual language, so syntax does not become a prerequisite for testing the accounting model.

Merigoux et al. (2021), Sections 3–4 and Figures 8–9, provides Catala's scopes, exceptions, default calculus and translation into a conventional functional target. Its explicit empty and conflict outcomes are valuable precedents for accounting rules with exceptions. They are different from probabilistic uncertainty. AccDSL will preserve this distinction: conflicting applicable rules produce a diagnostic, while an evidence-supported alternative reading produces a conditional model branch. The paper's compiler correctness results do not establish correctness of our accounting rules or of an automatic translation from disclosures.

The inspected LegalMath code offers reusable ideas in its shared typed interpretation, partial observations, source references and validity/knowledge times. Its ordinary frontend execution selects a reading. AccDSL needs an additional execution mode that retains a family of readings, continuous unknowns and economic state. A bounded candidate generator is useful for search, but its candidate limit does not prove completeness.

Peyton Jones et al. (2000), Sections 3–5, builds complex financial contracts from combinators and separates their compositional meaning from a chosen valuation model. The ACTUS technical specification similarly separates contract states, scheduled events, state transitions and payoffs, with risk factors supplied through an interface (ACTUS Financial Research Foundation, 2026). These are strong reasons to represent contractual obligations independently of accounting classifications. Neither contract cash flows nor a derivative price determine the recognition, measurement and presentation rules of a complete IFRS or US GAAP book.

4.5 Extraction and financial reasoning

The document literature supports a staged ingestion process. PubTables-1M (Smock et al., 2022), in its canonicalisation procedure and evaluation section, distinguishes table detection, structure recognition and functional analysis. Its GriTS evaluation compares reconstructed table structure. TableBank (Li et al., 2020), Sections 3–4, uses document structure to obtain weak supervision for table detection and recognition. GTE and FinTabNet (Zheng et al., 2021), Sections 3–5, provide a financial-document setting and company-level dataset splits. These sources help select extraction components and evaluation units. Their table scores do not establish that a debt note has been interpreted correctly across years or that every off-balance-sheet exposure has been found.

The Docling report (Auer et al., 2024), Section 3, describes PDF parsing, page layout analysis and table structure recognition within a document-conversion pipeline. The locally archived version lists Christoph Auer first; this version difference is recorded in the catalogue. Docling is an ingestion candidate whose output must retain source locations and undergo accounting validation. Its code and model licences must be considered separately when packaging a deployment.

TableFormer makes the division between structure and text concrete. Nassar et al. (2022a), Sections 4–5, describe a model that receives a cropped table image and predicts its HTML structure together with a bounding box for each cell. The cell's words and numbers come from separately supplied PDF text; they are not generated by the structure decoder. Matching the predicted cells to that text is itself consequential. The supplementary algorithm can remove duplicate columns, attach previously unmatched text to a cell, or create a new cell (Nassar et al.,

2022b, Section 2). A plausible grid therefore still needs its text assignments and structural corrections checked against the source.

The paper's evaluation preserves this distinction. Table 4 reports TEDS, a similarity measure based on tree edit distance, for structure *with content* on PubTabNet PDF documents containing cropped tables: 95.4 for simple tables, 90.1 for complex tables and 93.6 overall. The content comes from PDF cells; Table 2 separately reports structure-only results (Nassar et al., 2022a, Tables 2 and 4, p. 4620). These are results for that input population and text supply. They do not establish extraction accuracy for complete financial filings, or whether a cell has been assigned the correct entity, period, unit or accounting meaning. Those questions require the source-linked accounting checks developed here.

FinQA (Chen et al., 2021), Sections 3–6, provides questions with executable reasoning programs and supporting evidence. TAT-QA (Zhu et al., 2021), Sections 2–4, combines table cells and text spans with aggregation operations, and explicitly treats numerical signs in evaluation. DocFinQA (Reddy et al., 2024) extends financial questions to long documents and evaluates retrieval/context choices. FinBen (Xie et al., 2024) broadens the collection of financial tasks and model evaluations. These benchmarks can test retrieval and local numerical reasoning. Their question-answer format does not require a complete multi-period company representation, joint alternative histories or scenario-dependent journal generation.

A directly relevant recent competitor is the auditable RAG architecture of Cosentino et al. (2026). Sections 3.1–3.2 describe persistent extraction, enrichment, retrieval and provenance records; Section 4.6 checks citation resolvability and failure recovery. Section 5 limits the interpretation of those results to the evaluated document-intelligence tasks. AccDSL should reuse the idea of persistent evidence and recoverable processing, while adding executable accounting and economic semantics. No performance comparison with that architecture has been run here.

4.6 Solvers and probabilistic execution

de Moura and Bjørner (2008), Sections 2–3, describes Z3's interfaces and theory-solving architecture. It supports choosing an SMT solver for logical choices and arithmetic constraints, but does not make arbitrary nonlinear simulation tractable. The first solver fragment is deliberately bounded: finite choices, exact linear arithmetic and selected calendar relations. Unsupported formulas remain in the representation with a solver diagnostic.

van de Meent et al. (2018) develops probabilistic programs through sampling, observations and inference; the inspected material includes the first-order language and weighted sampling discussion. Stan (Carpenter et al., 2017), its model language and sampler-output discussion, and Pyro (Bingham et al., 2019), Sections 2–3, show different ways to separate a probabilistic model from inference machinery. They are candidates for optional inference over an already-defined economic model. Exact accounting identities and discrete rule selection should not be turned into nearly satisfied likelihood penalties merely to fit an inference library. Nor should successful sampler execution be described as evidence that all interpretations were explored.

4.7 The remaining design problem

The combined literature provides algebra for journals, semantics for missing information, temporal storage, provenance, contract languages, rule languages, inference methods and extraction tools. The unresolved integration problem is to retain their distinctions while connecting them: a disclosure constrains economic histories; a history supplies contract and behavioural state; a scenario drives events; accounting rules produce journals; reporting rules render statements; and every step remains explainable.

The design is therefore an integration proposal with several local mathematical constructions. Its originality and empirical advantage have not been established by this survey. Its practical merit will depend on the acceptance cases, accounting review and comparative implementation work in Chapter 14.

Chapter 5

Architecture and reuse

5.1 One representation, several authoring methods

The architectural centre is a typed intermediate representation (IR), not a text parser. A document extractor, a human editor, a Python library and the AccDSL compiler all create the same records. This allows a modeller to begin with a reviewed structured file and later add convenient notation. It also allows source extraction to improve without changing accounting semantics.

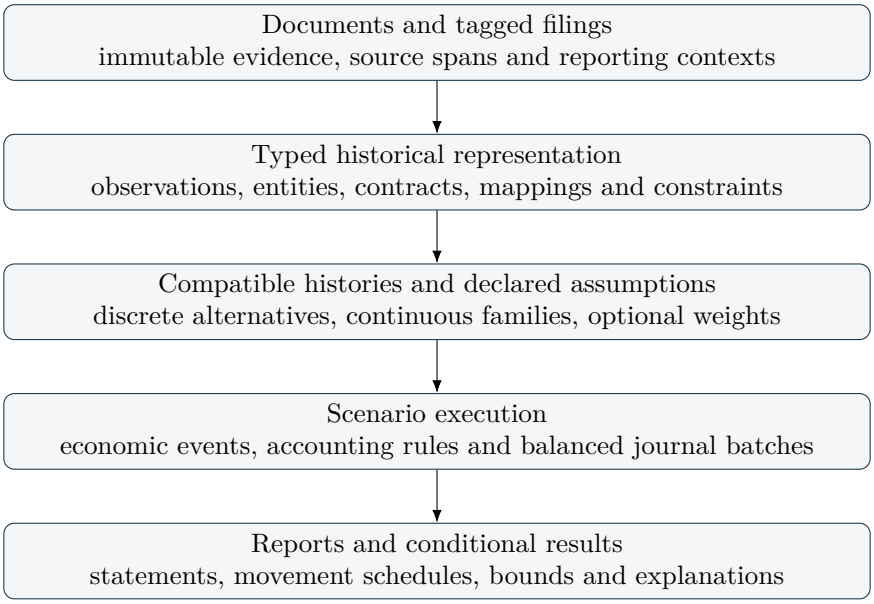


Figure 5.1: The simulation path. Every transformation records its inputs and assumptions; the source record remains available throughout.

Evidence storage and economic modelling must remain separate. If a company reports 100 of borrowing, a model can assume two loans of 60 and 40 without creating a new reported fact that the company disclosed those loans. The contract library consumes the assumed terms, the accounting module consumes the resulting events, and the explanation graph records the assumption at each affected result.

5.2 Why a DSL, and when another approach is better

The strongest baseline is a typed schema, SQL storage and ordinary Python code. That combination can implement the whole product. Its disadvantage is that repeated accounting requirements can become conventions scattered across functions: implicit currency conversion, missing period checks, hidden default values and duplicated rule selection. A bounded language makes these requirements part of typing and compilation. Its costs are language maintenance, diagnostics, migration and training.

Approach	Useful role	Limitation for this product
Typed schema and Python	Reference executor, flexible economic models, initial authoring API	Domain restrictions require disciplined APIs or a validator; arbitrary code complicates static analysis and provenance.
XBRL and OIM	Tagged evidence, contexts, dimensions, taxonomies and exchange	Reported facts and reporting relationships do not recover hidden economic histories or specify scenario behaviour.
Ledger languages	Recording known transactions, balance checks, export and differential testing	Alternatives, source interpretation and contract exposure need additional semantics.
Knowledge graph / SQL	Relationships, lineage, temporal queries and evidence navigation	A graph or schema alone does not define event execution or the probability meaning of alternatives.
Rules / Catala	Dated policy logic, defaults, exceptions, traceable rule evaluation	Ledger state, event scheduling and ensembles require an enclosing runtime.
Probabilistic program	Priors, likelihoods and inference over model parameters	Needs declared probability measures; cannot turn unspecified ambiguity into probability or supply accounting rules.
Spreadsheets	Familiar review and small transparent financial models	Versioned evidence, shared unknowns and branching histories are difficult to govern as the model grows.
AccDSL over shared IR	Common domain checks, reusable templates, explicit alternatives and deterministic posting semantics	Requires implementation and evaluation; broad notation without stable semantics would add cost.

The decision is to implement the IR, validators and Python reference API first, then the textual frontend. The same conformance cases must pass through both frontends. If the DSL fails to reduce ambiguity in representative authoring tasks, the IR and library remain useful. This is a reversible product decision, not a claim that a new language is inherently better than a library.

5.3 Open-source components

The following survey uses official repository descriptions, licence files and selected source interfaces archived on 2 October 2026. It is a source inspection, not an executed integration or performance evaluation. The machine-readable tool ledger records retrieval paths and available version metadata. Dependencies must be pinned when implementation begins; a branch name observed during this survey is not a release requirement.

Component	Proposed use and checked boundary	Licence / decision
Arelle	Parse and validate XBRL/iXBRL. Inspected `ModelFact`, context and unit accessors. Preserve taxonomy extensions and full contexts when importing.	Apache-2.0 core; first-choice adapter.
Docling	PDF layout, text and table conversion. Inspected converter interface; retain page coordinates and extraction settings.	MIT code; assess model licences separately. Candidate.
Table Transformer	Table extraction baseline associated with PubTables-1M; check finance-domain performance before selection.	MIT; repository marked archived in inspected metadata. Baseline, not a maintenance default.
Beancount	Posting and transaction data types; export selected concrete branches and compare balances.	GPL-2.0 project metadata; integration and redistribution choice needed. Optional adapter.
Ledger	Known-journal reference and export. Inspected transaction balancing code and manual date/assertion behaviour.	BSD-style licence; optional reference tool.
hledger	Alternative ledger import/export and accounting checks. Repository documentation inspected.	GPL-3.0 project metadata; optional process adapter.
Catala	Potential backend for deterministic policy modules; defaults and exceptions have useful semantics.	Apache-2.0; prototype only after core execution semantics stabilise.
ACTUS	Contract schedules, payoffs and transitions; external risk-factor interface. Verify event schedules against ACTUS tests.	Service Apache-2.0; technical specification CC-BY-SA-4.0. Candidate contract adapter.
QuantLib	Calendars, cash flows and valuation. Floating coupon source separates fixing dates and pricing. Accounting treatment remains external.	Modified BSD-style licence; candidate valuation adapter.
OpenFisca	Versioned tax-benefit formulas as a rule-system reference; inspected variable/formula metadata. Corporate-tax coverage must be supplied.	AGPL-3.0; optional, not the core runtime.

Component	Proposed use and checked boundary	Licence / decision
Z3	Finite alternatives, logical compatibility, linear constraints and conflict explanations.	MIT licence text; first solver candidate.
DuckDB	Local analytical queries over extracted facts and results.	MIT; candidate alongside SQLite transactional storage.
Stan	Continuous probabilistic models and diagnostics with an explicit target. Inspected model interface and paper.	BSD-3-Clause; optional inference backend.
Pyro / PyMC	Alternative probabilistic backends for model-specific inference. Pyro design paper and project descriptions inspected.	Apache-2.0 / Apache-2.0 licence texts; optional, no default selection.

The licence entries identify the inspected projects, not a legal conclusion about every combination of code, models and datasets. The default product architecture can communicate with optional tools through explicit adapters. This does not, by itself, settle redistribution obligations. That question is resolved when a concrete dependency and distribution method are selected.

XBRL's Open Information Model supplies standardised facts with concepts, entities, periods, units and dimensions (XBRL International, 2021). Those facts are an excellent ingestion source. AccDSL retains their original contexts and decimals rather than flattening them immediately into a company-by-year table. Calculation relationships can help validate reporting totals, but are not a general description of the transactions that generated those totals.

5.4 Responsibilities that AccDSL must implement

AccDSL owns the evidence and temporal model, typed accounting concepts, shared unknowns and alternatives, rule-package selection, journal validation, consolidation provenance and report explanations. It also owns the distinction between feasible sets and distributions. These responsibilities cannot be delegated implicitly to whichever parser or solver is convenient.

External components receive narrow requests and return typed responses. A contract adapter returns scheduled events, cash-flow quantities, state changes and source/model references; it does not directly mutate the ledger. A valuation adapter returns a measure, currency, date, method and numerical status; the accounting package decides whether and where to recognise the change. A solver returns feasibility or bounds with scope and completeness status. A language model proposes source-linked mappings or interpretations; it cannot commit an unsupported assertion as fact.

The architecture consequently permits deterministic reference execution, alternative solvers and later acceleration without changing the meaning of a historical observation or a journal entry. Performance work begins only after the same semantic cases pass through these interfaces.

Chapter 6

Historical compatibility and future simulation

6.1 Define what is observed before inverting it

Let D_k be the disclosures available under information cutoff k. Each numeric observation d_i has a context: entity or group perimeter, concept, measurement basis, dimensions, currency, period and precision. These contexts are part of the observation operator, not additional columns that can be discarded after extraction.

A candidate history h contains an opening economic and accounting state, events through the last historical date T, contract terms, accounting policies, consolidation relationships and latent quantities. Let M denote the declared model scope: its event templates, admissible rule versions and allowed unknowns. It excludes arbitrary events that the chosen model has no way to represent. The reporting operator $O_i(h)$ produces the quantity to which observation i refers, including its basis and perimeter.

Reported numbers are often rounded. If a number v_i is rounded to the nearest unit q_i, an illustrative admissible interval is $[v_i - q_i/2, v_i + q_i/2)$ under the corresponding tie convention. This half-open endpoint choice is used only when that tie rule is declared; Section 3.12 gives the conservative closed enclosure when nearest rounding is assumed but the tie rule is unknown. The actual rounding convention, scale and XBRL accuracy metadata govern the imported constraint. A disclosure with unspecified precision does not acquire a convenient tolerance silently. Write $\mathcal{I}_i$ for the justified observation interval, and $C_M(h)$ for the joint accounting, contractual and semantic constraints. Then the historical compatibility set is

$$\mathcal{H}_M(D_k) = \{h \in \mathcal{H}_M : C_M(h) \ \wedge \ O_i(h) \in \mathcal{I}_i \text{ for every applicable } i\}. \tag{6.1}$$

Qualitative assertions enter as reviewed logical conditions, rather than numeric intervals. Incompatible source contexts are never forced into the same equation.

Equation (6.1) distinguishes three outcomes. An empty set means that the observations and declared model are inconsistent. A singleton means that this model identifies a history at its chosen resolution. A larger set preserves unresolved alternatives. An empty set does not prove that the company's accounts are false: a wrong extraction, missing event type or incorrect perimeter can cause the inconsistency. A singleton does not prove that the model contains the real history.

6.2 A certain liability need not identify its creditor

Suppose the admitted records establish exactly one payable of 100 currency units, but leave its creditor unresolved between A and B. For this constructed example the declared family has two histories: in h_A the company owes A 100 and owes B nothing; in h_B it owes B 100 and owes A nothing. These are complete alternatives within the example, not two sampled points from a wider family. The question *does a payable exist?* has the answer yes in both histories. The question *is A owed 100?* does not.

Now ask for the rows of the payable schedule. If $Q(h)$ returns those rows, then $Q(h_A) = \{(A, 100)\}$ and $Q(h_B) = \{(B, 100)\}$. Their intersection is empty: no creditor–amount row occurs in both. Reading that empty table as the company's complete schedule would wrongly assert that it owes nothing. Reading the union as its complete schedule would instead assert that it owes 200. Evaluating the total within each history gives 100 in both. The uncertainty concerns the creditor, not the existence or total of the payable.

This distinction is central to Libkin's critique of intersection-based answers. His example retains a tuple with an unknown second component; discarding it loses knowledge that a tuple with the stated first component exists. An intersection read under a closed-world interpretation can also add an unsupported absence claim (Libkin, 2014, Section 3 and the opening of Section 6). A table of individually certain rows is therefore a restricted view of knowledge, not generally a replacement for the possible answer schedules.

For a nonempty feasible history family $H = \mathcal{H}_M(D_k)$ and a total, deterministic query Q on its histories, retain the family of answers

$$\mathcal{A}_Q(H) = \{Q(h) : h \in H\}.$$

Each answer remains linked to the histories that produce it. Equality of payable schedules alone does not license merging histories for another query. For a row-valued query, a separately requested certain-tuples view is

$$T_Q(H) = \bigcap_{h \in H} Q(h).$$

Membership in $T_Q(H)$ asserts that a row appears in every answer. Omission asserts only that the row is not established in every answer; it does not assert absence from every answer. A total computed after taking this intersection answers a different question from totals computed within histories. In the payable example, $T_Q(H)$ is empty while $\{\text{total}(Q(h)) : h \in H\} = \{100\}$.

For a proposition P about a concrete answer, Boolean certainty has the separate meaning

$$\text{Certain}_H(P \circ Q) \quad \Longleftrightarrow \quad \forall h \in H,\ P(Q(h)).$$

Taking P to mean that the schedule contains at least one payable proves existence without identifying A or B as the creditor. Allowing a third history with no payable would change that answer. The two-history result consequently depends on the admitted evidence and scope that require the liability to exist, not merely on the two rows chosen for display.

Libkin distinguishes a logical description of knowledge shared by all answers from an object representing their greatest lower bound in an information ordering. These representations need not exist in the chosen language; ordinary tuple intersection is not their general definition (Libkin, 2014, Section 5.3, equations (6)–(8)). AccDSL's answer family and Boolean propositions above are local definitions. They do not assert that every family has a finite relational representation. A symbolic answer must carry its conditions and preserve the stated family, or declare the restriction or approximation it makes.

Input completeness and output meaning must also be explicit. A model that fixes the available event templates and contract slots imposes a scope restriction; it does not establish that a public filing lists every economic event. Nor does that restriction entitle a display to interpret every omitted row as false.

Libkin's primary proof compares information by reverse inclusion of possible completions: an object is more informative when fewer complete objects remain compatible with it. For a query to be evaluated directly on an incomplete object, the chosen output interpretation must preserve that order as input information increases. The query must also respect the relevant renamings, and its output domain must have a representation system whose formulas describe each object's complete interpretations (Libkin, 2014, Sections 6.1–6.2, equations (9)–(10)). In Theorem 2, the proof first shows that evaluating the query preserves exactly the statements true in all completed answers; it then uses those statements to establish the greatest-lower-bound answer as an object (Libkin, 2016a, Definitions 4–7 and Theorem 2, pp. 11–18). Monotonicity alone is insufficient, and SQL null comparisons need not meet the genericity condition (Libkin, 2016a, Section 7.1, pp. 18–19).

The relational examples make the difference between input and output assumptions concrete. Positive existential queries have the stated result with open-world inputs and answers. A guarded positive fragment also has it with *closed-world inputs and open-world answers*; the latter is an interpretation of omitted result rows, not a claim that the input filing lists every economic event (Libkin, 2016a, Proposition 9 and Corollary 3, p. 20). These results do not authorize evaluating arbitrary financial negation, aggregates or nonlinear constraints by treating unknown amounts as ordinary constants. AccDSL has not established the theorem's abstract domain, information order and representation-system conditions for every financial output. The constructed payable example above instead follows directly from its two declared histories.

A related relational method illustrates what a sound but incomplete answer means in practice. For marked-null tables under closed-world set semantics, Libkin's modified three-valued evaluator returns a certainly positive answer when it yields 1 and a certainly negative answer when it yields 0; its value 1/2 carries no guarantee (Libkin, 2016b, Theorems 4.1 and 5.3, pp. 8–12). Let $T = \{(1, \bot)\}$ and ask whether there is a y for which $T(1, y)$ and either $y = 1$ or $y \neq 1$. Every completed table answers yes, yet the modified evaluation leaves the answer at 1/2: its null comparison and its negation are both unknown. The paper gives this case as a missed certain answer for ordinary SQL; the modified evaluator's outcome follows from its atomic rules. An inconclusive evaluation therefore does not show that two possible histories give opposite answers. AccDSL requires witnesses on both sides to establish that stronger claim; a timeout or unsupported operation remains undecided. The relational guarantee itself does not transfer to financial totals, nonlinear valuations, temporal histories or scenario conditioning.

The query contract records whether it returns a proposition, individually certain rows, or a family of answers, together with the family scope and completeness and the interpretation of omitted rows. Unsupported evaluation and unfinished search keep their existing failure or unresolved status. In particular, a sampled answer family cannot be relabelled as the full family merely because its displayed rows agree.

6.3 Why published statements cannot identify all transactions

The simplest derivation uses fixed account mappings and a fixed basis. Let $z \in \mathbb{R}^n$ contain unknown historical movement amounts. For m disclosed aggregates, let $A \in \mathbb{R}^{m \times n}$ be the aggregation matrix and $y = Az \in \mathbb{R}^m$. If there is a nonzero vector v with $Av = 0$, then

$$A(z + v) = Az + Av = y.$$

Both movement vectors give the same reports whenever both satisfy the remaining constraints. Thus the feasible family is the affine set $z + \ker A$ intersected with those constraints. Accounting requirements can shrink this family, but do not generally eliminate it. For example, cash collections in a receivables bridge are constrained by

$$AR_T = AR_0 + \text{credit sales} - \text{collections} - \text{write-offs} + \text{other classified movements}.$$

If the note does not distinguish acquisitions, currency effects and write-offs, several collections paths may fit the same closing balance.

There is a second ambiguity beyond aggregation. In Equation (1.1), identical historical movements arise from different contractual reset terms. Even recovering z exactly would leave these terms unidentified. AccDSL therefore infers economically relevant states and contract terms alongside journal movements. A linear inverse solver over statement lines alone cannot solve this second problem.

6.4 The accounting state and its invariants

For one entity book with n_a accounts in one functional currency, let $b_t \in \mathbb{Q}^{n_a}$ be signed account balances, with debit amounts positive and credit amounts negative. Exact decimal amounts can be represented as rationals with a power of ten denominator. A journal batch j_t contains the signed functional-currency amounts posted by one atomic event or explicitly grouped event batch. Its balance condition is

$$\mathbf{1}^{\mathsf{T}} j_t = 0, \qquad b_t = b_{t^-} + j_t. \tag{6.2}$$

If $\mathbf{1}^{\mathsf{T}} b_0 = 0$, substitution gives $\mathbf{1}^{\mathsf{T}} b_t = \mathbf{1}^{\mathsf{T}} b_{t^-} + \mathbf{1}^{\mathsf{T}} j_t = 0$. Induction establishes conservation through every committed batch. Opening balances must therefore be supplied as a balanced, explained entry or a balanced imported state with unresolved composition explicitly identified.

This proof establishes a bookkeeping invariant. It does not establish that an amount belongs in the chosen account, that a contract exists, or that an accounting standard permits a treatment. Posting validation additionally checks account roles, ownership, currency conversions, applicability of rules and event preconditions.

The balance vector includes temporary revenue and expense accounts during the period. Period closing transfers their net balance to equity through balanced entries. A reporting function can include current profit in equity before closing, or use the closed book afterwards, but must not include it twice. Income and cash-flow reports use classified movements and exclude mechanical closing transfers from operating flows. Consequently statements are functions of balances, journals, contract state and reporting rules:

$$Y_{[u,v)} = R(b_{v^-}, J_{[u,v)}, X_{v^-}, \rho, [u, v)). \tag{6.3}$$

Here ρ specifies the reporting basis and perimeter, and X contains economic state beyond ledger balances. Cash-flow classification is part of R, not recoverable from the closing cash balance alone.

Each book has its own functional currency. Quantities such as shares, tonnes and foreign-currency units are typed separately and never added to this balance equation. A foreign-currency posting stores its original units and the functional-currency valuation with its rate source. Translation into a group's presentation currency and consolidation adjustments occur in a separate book under explicit rules.

6.5 Simulate events from economic state

For a chosen history h, the state at T is $X_T(h)$. It contains balances, open contracts, lots or cohorts, unpaid accruals, commitments and other quantities needed for future evolution. Let s identify a scenario definition, U_t^s its supplied economic drivers, a_t management's actions, and ε_t future innovations. A behavioural policy π chooses a_t using the permitted information $\mathcal{F}_t$:

$$a_t = \pi(X_t, U_{\le t}^s, \mathcal{F}_t; \theta), \tag{6.4}$$

$$e_{t+1} = \mathcal{G}(X_t, a_t, U_{t:t+1}^s, \varepsilon_{t+1}; \theta), \tag{6.5}$$

$$(j_{t+1}, m_{t+1}) = \mathcal{A}(e_{t+1}, X_t, r_{t+1}), \tag{6.6}$$

$$X_{t+1} = \mathcal{T}(X_t, e_{t+1}, j_{t+1}, m_{t+1}). \tag{6.7}$$

The generator $\mathcal{G}$ emits events, such as a sale, rate reset, payment or default. The accounting function $\mathcal{A}$ applies rule package r and returns journal postings plus measurement annotations m. The transition $\mathcal{T}$ updates economic and ledger state atomically. A default can therefore change the loan's legal status, expected recoveries, recognised allowance and cash collections through related, but distinct, records.

This ordering is a reference abstraction. Within a date, event dependencies and phases specify the actual order. Expectations used to measure an allowance or provision must be calculated from the information permitted at that date. They cannot use the realised future path of the simulation. An inner expectation model may use quadrature, analytic formulas or nested simulation; its approximation and seed belong in the measurement annotation.

Some scenarios specify a deterministic macroeconomic path while leaving idiosyncratic defaults random. Others supply a joint stochastic driver model. A risk-neutral measure used to value a derivative is distinct from the real-world measure used to simulate business outcomes. Every stochastic adapter declares its measure and conditioning information. The accounting function receives the resulting valuation with that declaration.

6.6 Preserve sets and distributions separately

Let $K_{h,s}$ be the probability law of future reports produced by the forward model conditional on history h, scenario s and its declared behavioural assumptions. A deterministic simulation is the special case of a point mass. When no probabilities over historical explanations are justified, the natural answer is a family of laws,

$$\mathcal{K}_{M,s}(D_k) = \{K_{h,s} : h \in \mathcal{H}_M(D_k)\}. \tag{6.8}$$

The system can report each explanation, bounds on a deterministic outcome, or lower and upper event probabilities across this family. Those are different result types. A range of conditional means is not a confidence interval, and an envelope of conditional quantiles is not itself a single predictive distribution.

If a defensible probability measure $\mu(dh \mid D_k, M)$ supported on $\mathcal{H}_M(D_k)$ is supplied, histories and report outcomes must have specified measurable spaces. Assume that $K_{h,s}$ is a probability kernel: it is a probability measure on report outcomes for each h, and $h \mapsto K_{h,s}(B)$ is measurable for each measurable event B. The predictive law is then

$$\mathbb{P}(Y^s \in B \mid D_k, M) = \int K_{h,s}(B)\, \mu(dh \mid D_k, M). \tag{6.9}$$

For discrete explanations this is the usual weighted sum. For continuous unknowns, integration uses the declared measure over those unknowns, not a count of solver solutions. Bayesian weights

require a prior and a likelihood; expert weights remain labelled as elicited. A likelihood must not treat a statement total and all its components as independent repeated evidence. Deterministic equality constraints require a reduced parameterisation or proper conditional-measure treatment, rather than an undocumented Gaussian penalty with near-zero variance.

Suppose the fixed-loan explanation receives weight w and the floating-loan explanation receives $1 - w$, solely as an illustration. Under the rate shock,

$$\mathbb{P}(I = 5) = w, \qquad \mathbb{P}(I = 7) = 1 - w, \qquad \mathbb{E}[I] = 7 - 2w.$$

The expected expense is useful but hides the two different outcomes. For a general scalar output with finite second moment, put $m(h) = \mathbb{E}[Y \mid h]$. Write $Y - \mathbb{E}[Y] = (Y - m(h)) + (m(h) - \mathbb{E}[Y])$. The cross term vanishes because $\mathbb{E}[Y - m(h) \mid h] = 0$. Taking expectations of the squared expression gives

$$\mathrm{Var}(Y) = \mathbb{E}_h[\mathrm{Var}(Y \mid h)] + \mathrm{Var}_h(\mathbb{E}[Y \mid h]). \tag{6.10}$$

The first term is uncertainty within a chosen explanation; the second captures differences between explanations. This decomposition requires the measure μ. Without it, the system reports the family in Equation (6.8), rather than manufacturing a variance.

Weights themselves can be uncertain. For an admissible class $\mathcal{W}$ of measures, AccDSL can compute

$$\underline{p}(B) = \inf_{\mu \in \mathcal{W}} \int K_{h,s}(B)\, d\mu(h), \qquad \overline{p}(B) = \sup_{\mu \in \mathcal{W}} \int K_{h,s}(B)\, d\mu(h).$$

If every measure supported on the feasible histories is allowed and point masses are included, these bounds reduce to the infimum and supremum of $K_{h,s}(B)$ over histories: averages cannot lie outside the range of their integrands, and point masses attain, or approximate, the extremes. Narrower weight restrictions require an explicit justification and a separate optimisation problem.

6.7 Resolution and computational limits

For a requested horizon H and scenario family $\mathcal{S}$, histories may be grouped only when they have the same conditional law for all requested outputs:

$$h_1 \sim_{H,\mathcal{S}} h_2 \quad \Longleftrightarrow \quad K_{h_1,s} = K_{h_2,s} \text{ on those outputs for every } s \in \mathcal{S}.$$

This definition explains why individual sales can sometimes be replaced by cohorts, while fixed and floating loans usually cannot be merged for an interest-rate stress. Exact equivalence can be difficult to prove. A grouping justified only by approximate numerical agreement carries an error criterion and is not labelled exact. Source provenance remains available even when execution shares a common state.

Bounds require comparable honesty. A linear program can provide certified extrema for a supported linear model. A set of feasible sampled histories provides observed extrema: its minimum is no smaller than the true infimum, and its maximum is no larger than the true supremum. It therefore supplies an inner exploration of possibilities, not a certified outer bound. Interval methods may provide outer bounds that no single history attains. Every result records which of these meanings applies.

The initial implementation offers exact bounded choices and linear constraints, symbolic continuous families such as $\alpha \in [0, 1]$, and explicit scenario execution at selected feasible states. General nonlinear partial identification, automatic model discovery and exhaustive enumeration of all economic histories are research extensions, not promises made by the core language.

6.8 A solver formulation for ordinary rounded disclosures

The debt example used an exact principal. Real disclosures often do not. Suppose debt is shown as 100 and interest as 5, each rounded to the nearest unit with the declared half-open intervals $D \in [99.5, 100.5)$ and $I_0 \in [4.5, 5.5)$. Using an unknown fraction α and unknown total D would introduce the product αD. That expression is bilinear and is outside the promised linear fragment.

Instead use principal amounts F and V. The compatible family satisfies

$$F, V \geq 0, \quad 99.5 \leq F + V < 100.5, \quad 4.5 \leq 0.05(F + V) < 5.5. \tag{6.11}$$

Future interest is $I_1 = 0.05F + 0.07V$. Its infimum is $0.05 \times 99.5 = 4.975$, attained with $F = 99.5, V = 0$. Its supremum is $0.07 \times 100.5 = 7.035$, approached as $F = 0$ and V tends to 100.5 from below, but not attained. A closed interval is a valid enclosure, but its upper endpoint must not be reported as attainable. The exact solver check separately tests that no better value exists and whether each endpoint is attainable.

For $D > 0$, the amount representation is equivalent to the fraction representation through $D = F + V$ and $\alpha = F/D$. At zero principal the fraction is undefined and must be omitted or treated as irrelevant, not recovered. A known fraction restriction $\alpha \geq a$ becomes $F \geq aD$ when $D > 0$ and a is fixed. If both a and D are unknown, that product remains nonlinear. Reformulation has a domain of validity; it is not permission to send arbitrary formulas to a linear solver.

The supported exact fragment contains rational linear equalities, weak and strict inequalities, and finite Boolean or enumeration choices. Guard partitions become finite disjunctions. Posting in integer cents introduces a different integer problem if performed before bounding. Consequently the request specifies whether bounds apply before quantisation, after a monotone declared rounding map, or to a bounded integer model. Approximate numerical valuations carry error metadata outside the exact rational kernel. Solver timeout, unsupported theory and unbounded objective have distinct responses; none is relabelled as infeasibility.

6.9 Building the family with explicit templates

The reconstruction input is a model-scope record, not an invitation to invent unlimited journals. It lists event templates, slots per entity and period, allowed contract classes, source mappings, opening-state domains, calendars, resolution and excluded operations. A template gives an incidence vector for postings and a transition for economic state. Northstar's annual bridges arise by substituting its collection, purchase and payment templates into opening-plus-movement identities.

For each allowed slot, create an activity flag and nonnegative amount; an inactive slot has zero amount. Instantiate the template's posting and state constraints, unify variables referring to the same observation or economic contract, and match report expressions to source domains through validated context mappings. Add cross-note restrictions and choice-group dependencies to this same formula. No separate account-by-account reconstruction is allowed. Mark every constraint with its evidence, rule or assumption origin.

Enumerate the finite structural choices in a deterministic order. For each cell, ask for feasibility and retain the continuous formula, rather than enumerating real-valued solutions. An infeasible cell preserves a sufficient conflicting subset; an undecided cell preserves its formula and reason. Search is complete within the declared scope only when every finite cell has been decided, every applicable observation has been matched or explicitly qualified, and no resource truncation

remains. A witness establishes existence in one cell. It does not establish complete search, and complete search does not establish that the scope includes the company's real history.

In Northstar the declared templates identify annual collections, purchases and supplier payments, but retain every $F \in [0, 100]$. Adding a verified note that $F = 60$ narrows this family by conjunction. Adding a proposed "all floating" reading to that revised view produces a rejected cell with the conflicting note attached. Adding a rule that expenses a capital contribution would instead fail treatment admission: inconsistent evidence and unsupported accounting have different diagnoses.

6.10 Which details may safely be combined

A resolution record names the outputs, horizon, scenario family, grouping key, required state, transformation and exact or approximate status. Safe exact transformations are deliberately few. Cash balances in the same book and currency can be added. Homogeneous inventory lots may be combined under a weighted-average policy if ownership, cost basis, impairment state and future costing treatment coincide. Debt principal can be added only within a group sharing currency, coupon/reset function, fixing and payment calendars, maturity, options, guarantees and measurement method. The transition and reporting maps must be additive on that group.

The last condition supplies a short proof obligation. If a transition obeys $T(x_1 + x_2, u) = T(x_1, u) + T(x_2, u)$ and reporting is additive for every admitted driver u, induction preserves equality after grouping. This does not apply to covenant thresholds, counterparty netting, nonlinear guarantees or lot-specific impairment. Two fixed loans with the same dates and rate can share an execution state under this test. The fixed/floating pair cannot; neither can two policies with different guarantees merely because their current reserves match.

A changed horizon, scenario, rule version or requested output invalidates the resolution approval and reruns the sufficiency check. Extending beyond a debt maturity requires schedule detail that an earlier short-horizon view could omit. Approximate cohorts retain the detailed source references and a stated error criterion tied to outputs. Unless that error is bounded or evaluated, the grouping remains an approximation awaiting evidence, not an exact quotient of the history family.

6.11 A finite probability model that can actually be evaluated

Optional probability weighting begins with disjoint economic hypotheses and a declared joint measure. As an illustration, assign fixed and floating hypotheses prior probabilities 0.6 and 0.4. The stress itself leaves these weights unchanged and gives mean interest $0.6(5) + 0.4(7) = 5.8$. Suppose genuinely new observed evidence has likelihood 0.2 under fixed and 0.8 under floating. The unnormalised posterior masses are 0.12 and 0.32; normalisation by 0.44 gives 3/11 and 8/11. These likelihoods are illustrative inputs, not extraction-confidence scores.

The finite reference algorithm enumerates joint hypothesis and innovation states, evaluates their declared joint mass and any observation likelihood once, checks the nonzero normalising constant, and propagates the entire statement vector on each state. It sums masses only across disjoint events. Two overlapping labels such as "fixed share at least one half" and "all fixed" are query views and cannot become two independent mixture components. Totals and their components share an observation block so their evidence is not counted twice. Continuous inference needs an additional parameterisation, density or conditional measure and an assessed numerical method; the finite algorithm supplies no guarantee for those extensions.

Under the prior, before the new observation is admitted, if the floating branch has an unsupported valuation operation, its mass 0.4 remains unresolved. The run may report the fixed result with successful mass 0.6, but cannot report 5 as the unconditional mean by renormalising away the failure. A separately requested mean conditional on successful execution must show that conditioning event and its mass. For unweighted families, an unresolved branch blocks a complete envelope over the family. Economic default is retained as a model outcome with remaining obligations, rather than being treated as a computational error to remove.

6.12 A query keeps its domain, endpoints and memory

The same displayed amount can arise from different input domains, and the same ending cash can arise from different histories. A query therefore carries its domain, operation order, endpoint status and requested state. The examples below make those fields concrete before the implementation is asked to reuse a summary.

Real amounts, rounded outputs and a cent grid are different models

Return to the debt family in which nonnegative fixed-rate principal F and floating-rate principal V satisfy

$$99.5 \le F + V < 100.5, \qquad 4.5 \le .05(F + V) < 5.5.$$

The future interest query is $I = .05F + .07V$. The lower endpoint is attained at $F = 99.5, V = 0$, giving $I = 4.975$. The upper endpoint is the supremum $.07(100.5) = 7.035$, but it is excluded because the reported total is strictly below 100.5. The exact real image is $[4.975, 7.035)$, with an attained lower endpoint and an unattained upper endpoint. A solver result must carry both facts; writing a closed interval would change the set. To establish that there are no gaps, hold the total at 99.5 and vary V from zero to 99.5. Then $I = 4.975 + .02V$ covers $[4.975, 6.965]$. The all-floating histories with totals from 99.5 up to, but excluding, 100.5 cover $[6.965, 7.035)$. Their union is the stated interval. Conversely, $.05(F + V) \le I \le .07(F + V)$ excludes values outside those bounds.

Now ask for a display rounded to cents, using nearest rounding with exact halfway values away from zero. Every displayed value from 4.98 through 7.03 has a witness in the real family. The lower raw endpoint displays as 4.98. The excluded raw supremum would display as 7.04, but no admissible amount reaches it. Thus the rounded image has 206 values, 4.98–7.03. This is an image of the real family, not evidence that the underlying amounts are cents. For any cent amount x in the image with $x \le 6.965$, a witness is $V = (x - 4.975)/.02$ and $F = 99.5 - V$. Above 6.965 choose $F = 0, V = x/.07$. These formulas also show why enumerating cent displays has not discretised the unknown principal.

A different model restricts F and V to integer cents before computing interest. Its maximum is 7.0343 at $V = 100.49$, and its rounded display still runs from 4.98 to 7.03. Equal displays do not prove equal latent families: the real model admits values between 7.0343 and 7.035 that the cent-grid model excludes. A stored result must therefore retain the model identity and input quantum.

Signed and tie cases expose the same boundary. Negating the real image gives $(-7.035, -4.975]$; its rounded outputs are the cents from -7.03 through -4.98. The lower endpoint remains excluded. The interval $[0, 1)$ can round to 1.00 because 0.995 is an admitted value, even though 1 itself is not. A certified enclosure $[1.0049, 1.0051]$ crosses the 1.005 tie and cannot select one posting without more information. Unknown tie rule, zero denominator, and an unproved

nonzero denominator are separate failures. Rounding each of two 0.005 lines before adding gives 0.02, while adding first and rounding gives 0.01. Operation order belongs to the query method.

A projection is not a reusable transition state

For a concrete counterexample, take two separately retained histories in which the timing of a reinsurance treaty changes when income is recognised. Both end with cash 88.3. Their initial and first-year equities are 85 and 83 for the timely treaty, but 70 and 74.75 for the late treaty. These are the supplied comparison values here; Section 12.6 derives them from the separate issued and held contracts. The terminal-cash projection is therefore equal while the transition states and service results are different. A query requesting only terminal cash may project both histories to 88.3; a later query requesting first-year equity, loss recovery or treaty timing must revalidate the resolution and retain the full group states.

The bank commitment supplies the same warning. Funded gross loan 100 does not contain the original commitment date, credit baseline or observed draw. The funded allowance 4 cannot be recomputed from that scalar alone. If the original records were preserved, refinement can recover the state. If they were never known, the result is `DETAIL_UNAVAILABLE` or measurement unresolved, not a fabricated staging history. A source revision or changed query creates a new manifest; it does not silently resume an old result.

Conditional mass remains attached to unresolved outcomes

For a small authored law, assign prior masses $1/3, 2/3$ to immutable timely and late histories and likelihoods $3/4, 1$ to a newly admitted observation. The evidence normaliser is $11/12$, so posterior masses are $3/11$ and $8/11$. The invalid pooled treatment is not a third law atom: it was rejected as unsupported accounting treatment. If the late history's execution is unresolved, its mass remains $8/11$; reporting the timely result with probability one silently conditions on successful execution. If both first-year equities are available under that law, their mean is $(3/11)83 + (8/11)74.75 = 77$. With only the timely branch evaluated, the known weighted contribution is $(3/11)83$, not the complete mean. The same event IDs, measure provenance and unresolved mass must travel with the output.

These examples do not add a general solver or a cent-based accounting policy. They specify the distinctions that a compatible implementation must preserve: input domain, output operation, endpoint attainment, event order, retained economic memory, conditional mass and failure status. A changed output, horizon, grouping rule or scenario therefore returns `RESOLUTION_REVALIDATION_REQUIRED` until sufficiency is checked again.

6.13 Numerical precision is part of a financial method

The affine family is useful for Northstar because the scenario fixes prices, rates and operating coefficients while the debt share remains unknown. Other contracts require roots or discount factors. Their numerical treatment must state what is exact, what is enclosed and when a posted amount is determined. An accounting name on a library call does not make its result exact.

Consider positive payments C_i at positive times t_i and a positive price P. The effective annual rate solves

$$f(r) = \sum_i \frac{C_i}{(1+r)^{t_i}} - P = 0, \qquad r > -1.$$

Each term discounts a payment to the valuation date. In this positive-payment case,

$$f'(r) = -\sum_i \frac{t_i C_i}{(1+r)^{t_i+1}} < 0.$$

Consequently a certified sign-changing bracket contains exactly one root. The method uses stated bracket bounds and an expansion cap, evaluates signs with outward error bounds, and bisects until the requested rate or value enclosure is reached. An uncertified sign, unavailable bracket or exhausted accuracy budget yields an unresolved numerical result. Mixed-sign cash flows can have several internal rates of return; the preceding monotonicity argument does not apply and cannot justify returning the first Newton iterate that converges.

Net proceeds 100 and a payment 121 in two years give exactly 10%. An immediate fee of 2 changes proceeds to 98 and changes that root. Annual-effective 21% for half a year has the exact accumulation factor $\sqrt{1.21} = 1.1$, so interest on 100 is 10. Simple proration would give 10.5 and computes a different contract. At annual-effective 10%, the half-year factor $\sqrt{1.1}$ is irrational. For rational time p/q and positive $1 + r$, exact rational power comparisons can enclose the positive qth root of $(1 + r)^p$. A negative p first requires the reciprocal. General exponential discounting needs a separately specified error-controlled method.

A posting quantum of 0.01 does not eliminate this uncertainty. The whole certified monetary enclosure must lie in one rounding cell under the admitted tie rule. An enclosure from 1.0049 to 1.0051 crosses a half-cent boundary and does not establish one posted amount. Refine within a declared budget or retain rounding unresolved. This rule separates evidence for a numerical answer from the accounting authority to use that answer.

6.14 Updating a law without counting evidence twice

Two financial histories can be compatible with the same reports without being equally probable. Once an explicitly justified law is supplied, genuinely new evidence may change it. For finite histories h, prior mass $p(h)$ and an admitted observation likelihood $L(D \mid h)$, Bayes' rule gives

$$p(h \mid D) = \frac{p(h)L(D \mid h)}{\sum_g p(g)L(D \mid g)}.$$

The denominator is the probability of that evidence under the declared model. If it is zero, the condition is impossible in that model; dropping incompatible histories and renormalising an empty collection cannot repair it.

For two histories with assumed prior masses 1/3 and 2/3 and likelihoods 3/4 and 1, the unnormalised masses are 1/4 and 2/3. Their sum is 11/12, so the posterior masses are 3/11 and 8/11. If the second history's cash calculation fails, the first still has mass 3/11 and unresolved cash retains mass 8/11. A report giving the successful history probability one would answer a different, success-conditioned question.

An observation has a semantic information identity as well as a file locator. Reimporting the same observation through another file does not create a second measurement. Repeating an identical admitted condition is idempotent: it keeps the law and information lineage unchanged. Submitting an already consumed observation as a fresh likelihood block is rejected as duplicate information. Equal selected member sets alone do not prove that two conditions have the same meaning. An admitted conditional kernel may divide a region's existing mass when it is refined; the number of new representatives never determines that division.

Continuous inference adds a measure choice. One must specify coordinates z, domain, base measure, prior, observation mechanism and any transformation Jacobian. Stuart (2010, Section

2.2, equations (2.5)–(2.8)) supplies the inverse-problem formulation, but does not establish identification or easy sampling for a mixed accounting model. In particular, an exact accounting equality can confine a model to a lower-dimensional set. Multiplying an ordinary Lebesgue density by its zero-volume indicator is not a posterior construction. One must parameterise the constrained model with its declared measure or supply an actual measurement-error model. Adding convenient small noise changes the target.

6.15 A continuous example whose posterior can be checked

A linear Gaussian model supplies a tractable comparison for later numerical inference. Assume the unknown economic quantity has a Gaussian prior $z \sim \mathcal{N}(m_0, V_0)$ with positive definite covariance V_0. Let the observation be $y = Hz + \epsilon$, where ϵ is independent Gaussian noise with positive definite covariance R. The measurement-error assumption is substantive: it describes this example, not rounding in every financial statement.

Multiplying prior and likelihood adds their quadratic forms. Terms that depend on z are

$$\begin{aligned}&(z - m_0)^\top V_0^{-1}(z - m_0) + (y - Hz)^\top R^{-1}(y - Hz)\\ &\quad = z^\top (V_0^{-1} + H^\top R^{-1} H)z - 2z^\top (V_0^{-1} m_0 + H^\top R^{-1} y) + \text{constant}.\end{aligned}$$

Completing the square therefore gives posterior covariance V and mean m through

$$V^{-1} = V_0^{-1} + H^\top R^{-1} H, \qquad V^{-1} m = V_0^{-1} m_0 + H^\top R^{-1} y.$$

The prior and observation contribute precision and precision-weighted information. An implementation should solve factored linear systems rather than explicitly form these inverse matrices, and reject an invalid covariance or unsupported rank condition.

For $m_0 = 0$, $V_0 = 4$, $H = 1$, $R = 1$ and $y = 2$, posterior precision is $1/4 + 1 = 5/4$, variance is $4/5$ and mean is $(4/5)2 = 8/5$. A conditional cash quantity $100 + 10z$ has mean 116 and variance 80. A second *independent* measurement of 2 would give precision $9/4$, variance $4/9$ and mean $16/9$. Reusing the first measurement would produce those same numbers but would be wrong for the stated information. The analytic example can distinguish numerical error from information double counting.

For a bounded non-Gaussian scalar model, quadrature requires a stated domain or tail truncation and sufficient integrand error bounds to enclose the normalizer and requested moments. Without a positive lower normalizer bound, the ratio remains unresolved. Mixed models can enumerate finite choices and integrate each conditional continuous model, but combining branches requires their evidence factors, not draw counts. Optional MCMC needs target-specific parameterisation, tuning, independent chains, tractable correctness comparisons, convergence and Monte Carlo error checks. This proposal supplies no generally validated sampler or universal tolerance.

Nor can a more elaborate inference engine identify a quantity that the observation model never sees. If reports constrain only total debt 100, the likelihood may be constant in its floating share x. Its conditional prior then remains unchanged. A uniform prior is a chosen assumption, not a discovery from those reports. The inspected working version of Giacomini and Kitagawa (2021, Sections 2.1–2.3, equations (2.2)–(2.4)) motivates retaining posterior classes where a conditional prior cannot be revised. For Northstar there is a simple direct deduction: under all probability measures on $[0, 1]$, the expectation of $5 + 2x$ lies in $[5, 7]$, with endpoints attained by point masses. This deduction imports no asymptotic theorem from that paper's structural-vector-autoregression setting.

6.16 A stress is not an observed condition or an intervention

The interest-rate stress in Northstar fixes a future driver path. It supplies no new evidence about the historical debt share. An observational forecast instead conditions a joint law on information. A causal intervention needs structural equations and an exogenous disturbance law, because it replaces the mechanism determining a chosen variable while leaving other mechanisms in place (Pearl, 2009, Section 3.2, equations (5)–(9) and (15)–(17)).

An elementary example makes the difference numerical. Let U equal zero or one with probability $1/2$ each, and assume

$$X = U, \qquad Y = 2X + U.$$

Observing $X = 1$ establishes $U = 1$ and hence $Y = 3$. Intervening to set $X = 1$ replaces the first equation. It leaves the original law of U, so Y is 2 or 3 with equal mass and mean 2.5. Selection of a high-investment company and an imposed investment policy can therefore answer different questions even under one fully specified model. Financial filings alone do not establish these causal equations or identify their effects.

An intervention request must name its target, replacement equation, structural parents and disturbance law and satisfy acyclicity or a separately supported equilibrium rule. Correlated disturbances remain joint. The initial company profile supports imposed exogenous stress; the continuous and causal methods described here require separately admitted extension schemas and signatures. They do not silently expand the initial finite-law interface.

At every decision date, behavior and measurement receive the admitted historical view and permitted simulated past. The scheduler's knowledge of a complete future stress path is not permission for a manager to foresee it. A perfect- foresight policy needs an explicit additional assumption. Common random numbers can improve a paired numerical comparison while establishing no causal effect in actual company data.

Chapter 7

The accounting language

7.1 Scope of the first language

The historical AccDSL Core 0 proposal describes entities and books, accounts, observations, shared unknowns, constraints, alternative interpretations, contract instances, accounting rules and reporting mappings. It has pure expressions and a restricted posting effect. Economic forecasting algorithms can live in typed plugins; unrestricted numerical code does not need to be reimplemented in the DSL. The historical grammar in Appendix B fixes the preserved teaching examples. The proposed portable records and first company profile below define the new authoring boundary; the first matched text frontend supports the declared debt fragment.

The historical Core 0 compiler has six ordered stages: parsing with source locations; name and module resolution; type and temporal-context checking; lowering to the IR; constraint and rule-applicability analysis; and execution-plan construction. The compiler never silently supplies an economic assumption to make a program compile. A program can be valid as a historical representation while requiring additional assumptions before a particular simulation can run. The proposed portable compilation sequence is specified separately in Section 7.3.

The language's meaning is defined by its IR and evaluator, rather than by the surface spelling alone. Formatting changes cannot change a model's semantic hash. A changed source span or reviewer decision changes evidence provenance even if the numerical model remains the same. The system therefore maintains separate evidence, semantic-model and execution identifiers.

For the preserved Core 0 examples, the complete grammar is supplied as `spec/core-0.lark`; the original design directory retains `spec/core-ir.schema.json` and `spec/core-contract.md`. They define reference resolution, expression arity, type/context checks, operations, failures and identity. Structural schema validation alone does not establish those semantic properties. ASCII identifiers and UTF-8 strings are used in this revision; currency suffixes, dates and decimal literals have explicit lexical rules. The grammar is checked by parsing the complete demonstration file, not by claiming that its parser also implements a compiler or solver.

7.2 Types that protect meaning

The following table uses the preserved Core 0 type vocabulary. Its distinctions motivate the more precise portable types introduced below.

Type	Meaning and checks
`Money<C>`	Exact amount in currency C. Addition requires identical currency; conversion requires a dated rate and policy.
`Quantity<U>`	Units of shares, goods or other commodities. Quantities cannot be added to money or used as a functional-currency posting.
`Rate<B>` and `YearFraction`	Rate with basis B, plus a separately computed accrual fraction. Simple and compound accrual functions are distinct.
`Instant`, `Period`, `Calendar`	Instant observations and flows over intervals; interval operations preserve calendar conventions.
`Account<K,Book>`	Account role K and owning book. Cross-book effects require separate balanced batches linked by an economic event.
`Observation<T>`	Source assertion with contextual and precision metadata; accessing its value retains its constraint and provenance.
`Symbol<T,D>`	Shared unknown with domain D; repeated references denote the same value in a history.
`Choice<E>`	A finite structural choice over enumeration E. No probability is implied.
`Distribution<T>`	Explicit probability measure with parameter, conditioning and measure metadata. It is not a Symbol or a Choice.
`SourceRef`, `RuleRef`, `ModelRef`	Versioned evidence or implementation references; unresolved references prevent concrete execution.
`Measurement<T>`	Value plus valuation date, basis, method, assumptions and numerical status. Recognition remains a rule decision.

The portable profile spells the rate type `Rate<convention>` and makes both calendar and accrual convention explicit in `YearFraction<calendar, convention>`. `InstantValue<T>` and `Flow<T, [start,end)>` distinguish dated stocks from interval flows; their context carries the calendar. An approximate numerical component of a historical `Measurement<T>` uses `Approximation` with its method, error description and environment references. An exact measurement remains exact; valuation context and recognition authority are separate in either case.

Core expressions use exact integers and rationals, Boolean logic, enumerations, dates, records, lists, comparisons and registered pure functions. A decimal literal is parsed exactly, not first converted through binary floating point. Numerical plugins can return approximate measurements, but must declare their error or numerical status and the rounding used when posting money.

Missingness is a tagged type with distinct values: `notDisclosed`, `notApplicable`, `unreadable`, `withheld`, and `unresolvedMeaning`. Zero is an ordinary known number. A missing datum can be replaced by a declared Symbol or assumption in a model branch; the source observation stays missing. An unknown is not necessarily random, and an estimate is not necessarily missing. On portable import, `notDisclosed`, `unreadable`, `withheld` and `unresolvedMeaning` become `Unknown` with the corresponding reason and obligation references. `notApplicable` becomes the explicit `NotApplicable` variant, never an unknown or zero.

Addition of monetary observations also requires compatible entity, perimeter, time, dimensions and basis. These checks operate on contextual expressions in addition to the currency type. An analyst can deliberately aggregate dimensions or transform a basis using an explicit function.

For example, `sumOver(segment, ...)` records the dimension being removed and requires the segment partition to be declared; it does not assume every disclosed segment total is additive.

7.3 A portable record has an exact meaning

A constructor and a textual frontend can describe the same financial model only if they exchange values whose identity is independent of a Python object or an incidental JSON writer. The proposed portable record therefore has six fields: `kind`, `schema`, `key`, `revision`, `body` and `dependencies`. Kind and schema identify a registered version; key is its stable, schema-scoped logical name; revision is a nonnegative integer. The body obeys that schema's required, optional and nullable fields. Unknown fields reject.

A typed reference has exactly kind, schema and digest. The digest covers the complete target record. Every body reference is also listed in dependencies; additional dependencies need a schema-declared role. The submitted closure is finite and acyclic. A receipt can refer to its request and old state, but the old state cannot refer forward to that receipt. Stable business names can relate counterparties without creating a cycle of content references.

Financial quantities use a reduced rational with signed numerator text n and positive denominator text d, with $\gcd(|n|, d) = 1$ and zero represented by $0/1$. The decimal token 2.50 therefore denotes $5/2$ exactly. Revisions, ordinals and integer counts remain integers; a Boolean is never an integer. Approximate measurements instead retain decimal text, method, error description and environment. A rounded decimal cannot silently acquire exact status.

An absent required field is invalid. Unknown financial content is an explicit value with a reason and outstanding evidence references. It can be retained, but cannot satisfy a method that requires the missing fact. Null is permitted only where the particular schema gives it a meaning; not applicable, unknown and zero are distinct. A candidate also differs from a value admitted in a specific immutable view. Admission checks the actual decision and its ancestors; wrapping a candidate in an "admitted" type is insufficient.

Portable bytes use UTF-8 JSON with ASCII object keys sorted by their byte order, no insignificant whitespace, no floats, duplicate keys, nonfinite numbers or invalid Unicode scalar values. Decoded strings retain their code points. An escaped Unicode character and the same literal character encode equally; composed and decomposed spellings remain different unless an explicit named transformation relates them. Global NFC normalisation would silently change some source text and is therefore not part of this encoding.

The writer emits literal non-ASCII scalars, escapes quotation marks and reverse solidus, and escapes control characters U+0000–U+001F. It uses the five named JSON escapes where defined and lowercase `\u00xx` for the remaining controls; solidus is not escaped. Integers use minimal base-ten spelling without plus signs, leading zeroes or negative zero. Dependencies, module declarations by key, dimensions by axis and explicit all-of/any-of operand sets are sorted after rejecting duplicates. Other arrays, including journal lines, event schedules and diagnostic priority, retain order.

With these rules, the record digest is

$$\mathrm{SHA256}\big(\mathrm{UTF8}(\text{“accdsl.portable.v1/”} + \mathrm{schema}) \parallel \texttt{0x00} \parallel \mathrm{canonical}(\mathrm{record})\big).$$

This recognises representational identity, not economic equivalence or source truth. Original documents retain a separate hash of their raw bytes. A blob reference names that byte hash, algorithm and length; it is not a typed record reference. A submission record links those bytes to media type, transport and decoder, preserving a malformed request without pretending it was valid.

Historical Core 0 and replay examples keep their original encodings and IDs. A declared legacy import retains original bytes, decoder, old identity, portable output and field-by-field comparison. If one historical record maps to several portable records, an import mapping names every output and conversion. Import does not grant financial admission in a new view. This separation permits old examples to remain reproducible while the proposed portable representation becomes precise enough for two implementations to exchange.

7.4 The first company profile and its authoring boundary

The proposed `CorporateJointHistory/1` profile connects one legal entity and book per branch to annual/interim sources, revisions, working capital, payroll, tangible assets, simple financing, an explicitly declared tax model and complete statements. Its types preserve entity, perimeter and currency even where an operation is outside this initial scope. A real company cannot become eligible for the profile by dropping its subsidiaries, leases or contingencies.

Typed constructors must express every included declaration in the portable IR without opaque Python functions. The initially matched text frontend covers the bounded debt and constraint fragment. Whole-company textual authoring remains a subsequent frontend obligation; the whole-company typed model is part of the initial domain. Thus domain scope and authoring syntax are separate decisions. The following Core 0 listing teaches the shared debt family under its preserved historical grammar. It is not a promise that the first new parser supports every historical declaration. The matched portable debt example uses floating share x, while the following listing uses fixed share α. The substitution $x = 1 - \alpha$ preserves the entire family, including its endpoint results.

Backend admission examines expanded expressions. Multiplication of an unknown quantity by an unknown price is bilinear even when hidden inside a function called revenue. The initial exact reconstruction fragment admits rational affine constraints, finite choices and bounded integers under explicit templates. Unsupported functions return a capability diagnostic. A type-correct historical representation may therefore be retained even when a requested forecast still requires an unavailable method or an additional admitted assumption.

The portable `CompileRequest` supplies source kind, source bytes or a typed module, schema and profile references, the full package lock, evidence/time view and requested capability target. Its stages are strict decoding and source locations; name/import resolution; dependency and cycle checks; static type and context checks; admission-layer checks; expanded-operation capability analysis; and canonical record production. The earlier six-stage workflow describes the historical Core 0 frontend; it does not add execution planning to this portable compiler or permit it to call a solver.

The result contains candidate IR, source map, unresolved inputs, capability analysis and a validation receipt bound to every input. Invalid structure or types yield no executable IR. Valid structure with an unresolved financial input can yield a retained candidate whose affected target has execution admission false. Source maps retain decoded Unicode code-point half-open ranges for text or constructor paths for typed input. Formatting can change source-map identity without changing financial-model identity.

For a matched example, both authoring routes must describe debt 100, floating share $x \in [0, 1]$ and conditional interest $100((1 - x)/20 + 7x/100) = 5 + 2x$. Endpoints are 5 and 7; $x = 1/2$ gives 6. Changing the floating assumption from 7% to 9% changes the upper endpoint to 9. Independent arithmetic is needed because two frontends could share the same wrong constant. Agreement between them alone does not establish correctness.

7.5 A small model with a continuous interpretation family

The following source uses a synthetic observation whose full context is stored in the evidence registry. The example imports the demonstration debt package, which uses one-year simple accrual, no fees and immediate rate resets. It is not an IFRS rule package.

All dependencies are instantiated in `fixtures/demo-debt-package.json`: the source reading, balanced opening state, currency precision, contract state and dates, reset and interest operations, closing and report mappings, request and expected outputs. `fixtures/borrowing.acc` is the complete source, including the retained earnings account and an `opening` declaration. The shorter listings below explain its declarations in stages. They do not replace that full source.

```
module BorrowingExample version "1";
use DemoDebt version "1";
source note7 = external("synthetic-note-7");
entity Acme functional USD;
book Local entity Acme basis DemoDebt version "1";
account Cash : Asset book Local;
account Debt : Liability book Local;
account InterestExpense : Expense book Local;

observation debt : Money<USD> instant @2025-12-31
  = 100 USD from note7;
symbol alpha : Real in [0, 1]
  because "The fixed-rate proportion is not disclosed";
```

The remaining declarations use this same unknown fraction in both loan components. A change to the fraction therefore changes the whole debt mix consistently.

```
let fixedPrincipal = alpha * value(debt);
let floatingPrincipal = (1 - alpha) * value(debt);
constraint split:
  fixedPrincipal + floatingPrincipal == value(debt);

contract loans : DebtMix book Local {
  principal = value(debt);
  fixed_fraction = alpha;
  fixed_rate = 0.05 per year;
  initial_floating_rate = 0.05 per year;
}
alternative allFixed { assume alpha == 1; }
alternative allFloating { assume alpha == 0; }
scenario rateShock {
  set market_rate = constant(0.07 per year);
}
```

The parent model retains every $\alpha \in [0, 1]$. The two named alternatives are useful conditional views of that family, not a declaration that the endpoints exhaust it. Requesting the parent family's exact bounds produces $[5, 7]$ for the next year's expense under `rateShock`. Requesting a conditional run of `allFixed` supplies $\alpha = 1$. Requesting a weighted mixture requires a separate weight model; the two labels do not create equal weights.

An observation declaration is a reference to evidence plus a proposed typed reading. The compiler checks that the registry context agrees with the declared period and type. If the evidence is rounded, `value(debt)` denotes a shared constrained value rather than a silently exact 100. This synthetic registry explicitly declares an exact value, which keeps the illustration deterministic.

7.6 Posting rules and explicit exceptions

An event type declares the information needed by a rule. The example below assumes an event representing interest accrued and paid on the same date. Separate accrual and settlement events are needed when those dates differ.

```
event InterestPaid(principal: Money<USD>,
                   rate: Rate<annual_simple>,
                   fraction: YearFraction) book Local;

rule recogniseInterest for InterestPaid as e
when e.principal >= 0 USD {
  let charge = roundMoney(simpleInterest(e.principal, e.rate, e.fraction),
                          currencyScale(USD), "half_even");
  require charge >= 0 USD;
  post {
    debit InterestExpense charge;
    credit Cash charge;
  }
}
```

The rule cannot perform file I/O, mutate another entity or sample a random number. It returns a proposed batch. The runtime verifies balance and all preconditions before committing state. Negative interest requires an explicit supported rule; it is not mishandled by this intentionally restricted example. Rounding is a declared demonstration convention, not a universal accounting default.

Rules can declare `overrides ruleName`. The resulting override graph must be acyclic. For a concrete event, the evaluator determines all applicable rules and removes any that are overridden by another applicable rule. Exactly one maximal rule must remain for a required recognition operation. An empty set of applicable rules produces `UnsupportedTreatment`; more than one maximal rule produces `RuleConflict`. File order and a hidden numeric priority never break ties.

If a guard depends on a symbolic value, the solver partitions the current feasible family into the guard and its negation, subject to the existing constraints. This is a logical partition, not a random draw. An incomplete solver returns `UnknownApplicability`, not a guessed rule. Statically provable disjointness or total coverage can be cached; otherwise the runtime performs the same checks on every concrete event.

7.7 Alternative readings, assumptions and updates

An alternative adds constraints and possibly a supported semantic mapping to its parent model. Its record contains a rationale, supporting spans, opposing evidence, author, creation time and scope. Child alternatives inherit all parent constraints; they cannot quietly discard an inconvenient disclosure. A deliberate source rejection is a separate review decision with an explicit reason and a new evidence view.

Two independent alternatives may coexist, while two mutually exclusive readings of the same text require an explicit choice group. The representation stores these relationships in a constraint graph. It does not take a Cartesian product blindly. For example, interpreting an instrument as a liability affects both its balance-sheet account and the treatment of its periodic payments; those choices must be linked through one shared treatment variable.

New evidence narrows an existing family by conjunction when it concerns the same definitions and reporting basis. If it changes the basis or corrects an earlier assertion, the system creates a

revised evidence view and records the relationship. Posterior reweighting, when enabled, is a separate operation with likelihood provenance. A branch that becomes inconsistent is retained as historically considered and rejected, with the evidence that rejected it.

7.8 Modules and stable intermediate representation

A package identifies its namespace, semantic version, dependencies, effective dates, jurisdiction, supported entity classes, recognition operations and measurement policies. Standard packages define allowable treatments; industry packages add economic objects and specialised rules; company packages map source concepts and declare evidenced elections. A company extension cannot override a mandatory standard rule merely because its package is loaded last. An override is valid only at a declared extension point and within the selected standard's policy domain.

The IR has separate node classes for evidence, contextual observations, expressions, domains, constraints, alternatives, contracts, economic events, measurement results, rules, journal batches, report mappings and runs. Each node has a stable identifier and explicit input references. No expression stores executable host-language text. Plugins are referenced by versioned identifiers with typed signatures. This permits a second implementation to execute the same supported program without evaluating arbitrary embedded code.

The IR serialisation version is independent of the DSL version and of any accounting-standard edition. Migrations preserve the original payload and record changed meanings. A syntax migration can be automatic when semantics are unchanged. A changed recognition policy requires a new model version, even if the source file still parses.

7.9 Errors are part of the interface

The compiler and runtime return typed diagnostics with a source span, affected entity and period, dependency chain and suggested local action. The principal diagnostic categories have distinct recovery paths:

Diagnostic	Meaning and next action
`TypeMismatch`, `ContextMismatch`	Resolve incompatible units, periods, entities or measurement bases; do not coerce implicitly.
`MissingEvidence`	Supply a source or label a model input as an assumption.
`InfeasibleHistory`	Inspect the conflicting evidence and model constraints; preserve the rejected candidate.
`RuleConflict`, `UnknownApplicability`	Resolve rule scope or preserve a justified conditional partition.
`UnsupportedTreatment`	Add a reviewed capability or leave the affected output incomplete.
`UnbalancedBatch`	Reject the atomic update and diagnose the posting rule.
`SolverIncomplete`, `NumericalFailure`	Preserve the target and inputs; repair or change the computational method.
`InsufficientSimulationState`	Add declared conditional assumptions or request a family of results.

Warnings can describe low extraction confidence or an approximate economic model, but they do not replace these errors. A required unknown input prevents a concrete unconditional result. The analyst can add a declared conditional assumption, request bounds, or leave that output incomplete. Every path has a different, explicit result type; there is no generic fallback to zero.

Chapter 8

Reconstruction and execution procedures

8.1 Ingest, reconcile and construct histories

The historical procedure starts with a filing manifest. It records documents, reporting dates, publication dates, revisions and expected statements and notes. Tagged filings are parsed through an XBRL adapter; PDF and HTML sources are segmented into text, tables, captions and references. The original bytes and extraction output remain immutable. Re-running an improved extractor creates a new extraction version with links to the same source.

For each table, the importer first resolves headers, units, signs, periods and dimensions. It then proposes concept mappings. A deterministic validator checks subtotal arithmetic, duplicate facts, comparative consistency and cross-note references. Every proposed repair creates a new reading. The system may suggest that an OCR value is 10 rather than 70, but only a reviewed correction or an explicit alternative allows that reading into a validated model.

Reconciliation proceeds from local totals to movement schedules. A debt bridge may contain opening debt, issuance, repayment, acquired debt, interest accretion, currency effects, fair-value changes and closing debt. Every component receives a movement type. An unexplained residual is stored as a residual with a source and period; it is not automatically classified as repayment or issuance. A model can introduce an unknown decomposition of that residual, subject to evidence and economic constraints.

The reconstruction engine then performs the following finite procedure within a declared model scope:

1. Build shared variables for observations, opening states, contract terms and supported movement amounts. Attach original rounding domains.
2. Add identities, roll-forwards, account restrictions, contractual relationships and cross-period links. Add each semantic reading as a condition on the same joint model.
3. Propagate types, units, intervals and simple equalities. Reject context mismatches before invoking a numerical solver.
4. Ask the solver for feasibility. On infeasibility, return a sufficient conflicting subset of constraints, its source spans and model assumptions. A solver's conflict core need not be the smallest possible explanation.
5. Retain finite alternatives and continuous solution families symbolically. Materialise

example histories only when needed for review or execution.

6. Check whether each requested simulation has sufficient economic state. Missing maturities, reset dates or cohort composition become declared assumptions or explicit unresolved inputs.

Unrestricted journal enumeration would be unbounded: arbitrary offsetting transactions can be inserted without changing reports. The model therefore declares allowed event types, time resolution, relevant accounts, lot/cohort resolution and limits on latent contract instances. These limits describe the represented family; they are not evidence that all real possibilities were searched. The compiler canonicalises syntactic duplicates and equivalences established by its supported rewrite rules. It does not decide arbitrary economic equivalence. Probability measures attach to declared economic hypotheses or parameters, never to a count of encodings; adding a duplicate representation therefore cannot create extra probability mass or coverage.

The first implementation uses symbolic linear bridges and finite structural choices. Nonlinear economic models can be evaluated on feasible states through plugins. A solver timeout preserves the unresolved family and returns `SolverIncomplete`; it does not prune the family as impossible. Search budgets belong to run configuration and are recorded with the results.

8.2 Condition a run without rewriting history

A run selects an evidence view and cutoff, a historical family, a forecast starting date, scenario definitions, accounting packages, management policies, output queries and numerical settings. A run may use a concrete history, an exact symbolic family, a finite set of conditional histories, or weighted draws from a declared distribution. The output records which mode was used.

Suppose no debt maturity is disclosed. An analyst may run a conditional case that assumes all debt matures after the forecast horizon. The run is valid under that assumption even though the historical representation remains incomplete. The interface must display the assumption next to the result and provide sensitivity cases when the assumption is material. It must not label the assumed maturity as a discovered historical fact.

Management behaviour normally uses only information available at the decision date. A perfect-foresight stress case is a separate, explicitly labelled policy. Scenarios should share the same underlying innovations when estimating paired differences where appropriate, with stable stream identifiers so branch order does not change the draws. Common random numbers improve the comparability of paired simulations; they do not imply that every branch has the same economic event counts or shock distribution.

8.3 Schedule and commit events

Each event carries an economic occurrence time, accrual interval when relevant, settlement time, entity, contract or cohort identifier, currency, inputs and provenance. Its ordering key consists of date, declared phase, dependency edges and a deterministic tie identifier. The reference phases are: opening and scheduled contractual changes; market observations and rate resets; business events and decisions; accrual and measurement; settlement; period closing and reporting. A contract can require a more specific dependency. These phases are an engineering starting point, not a universal description of market timing.

Two events may share a tie order only if their order is either specified or irrelevant to the result. If payment before remeasurement and payment after remeasurement differ, an unresolved

ordering is a model ambiguity or an error. The engine does not conceal it behind alphabetical event IDs.

```
execute_event(operation_request):
    validate exact input closure, bindings and historical admission
    validate separate posting authorization, journal and delta
    lookup stable entity/book/branch/business-event key
    if exact committed fingerprint: return original receipt
    if different committed fingerprint: return event conflict
    compare expected opening state and head generation
    atomically recheck key/head and commit journal, delta, state, receipt
    return receipt, emitted dependent events and diagnostics
```

The atomic unit includes the economic state and its accounting effects. If posting fails, neither state changes. A repeated event ID with identical inputs returns the existing result; the same ID with different inputs is a conflict. Here the inputs include the request identifier, expected head and execution settings, not only the financial payload. Section 8.5 defines the stable key and full fingerprint. Consolidation runs after entity books reach the required date and stores its adjustments separately. A failure in one scenario branch cannot mutate another branch's state.

8.4 From an economic event to permission to post

A delivery can change revenue, inventory and receivables at once. Before any book changes, the method needs to know which contract was performed, which opening state it consumes, which evidence supports that reading and which accounting profile applies. Balance alone cannot supply those facts. AccDSL therefore separates the economic basis, permission to use a treatment, proposed entries and the transaction that makes them durable.

The event basis binds an immutable event to opening state, context, branch, time view, chart, authority, method, evidence, admissions, assumptions and view. Its business identity is the four-tuple of entity, book, branch and business event. The branch declares whether it is synthetic or real. A synthetic teaching profile can justify a teaching posting; it cannot authorize a real book merely because the resulting arithmetic is identical.

The method first evaluates reporting framework, jurisdiction and dates; then entity and transaction scope; contract and recognition facts; elections; numerical capability; and account/context bindings. Every material finding survives. A false required fact is ineligible, an unknown required fact is unresolved, and an event outside the method is outside scope. A known false finding must not disappear behind a later unknown one. Missing operative authority never becomes source-supported permission.

The native operation `accounting.assess@1` remains read-only assessment. Its `assess_only` request and false posting permission cannot be promoted by editing a flag. The separate `accounting.authorize@1` operation produces a `PostingAuthorization` from the complete basis and a supported posting profile. It permits posting only when the treatment is eligible, the method is fully specified for the event and the branch mode matches the profile's evidence scope. An unresolved or unsupported assessment is still a useful completed assessment, but it contains no permission to post.

An authorized proposal contains both a journal and an independent economic delta, each bound to the same basis and authorization. Journal lines name account role, debit or credit, positive exact quantity, currency/context, entity, book, event and method. Zero lines are omitted; a negative effect reverses sides. The journal balances within each currency and book. The delta separately checks quantities and contractual memory: inventory movements consume available

lots, cash follows settlement, share counts follow ownership events and debt face amount remains distinct from carrying value.

The commit arguments have exactly four references:

```
AccountingCommitArguments/v1 {
  basis_ref, authorization_ref, journal_ref, delta_ref
}
```

The expected opening state and branch generation appear once, in the outer operation request. A reference to a native assessment in the authorization field fails its reference type. False posting permission fails authorization. An unknown posting schema returns unsupported operation with no journal commit. All bindings are checked before looking up a prior receipt.

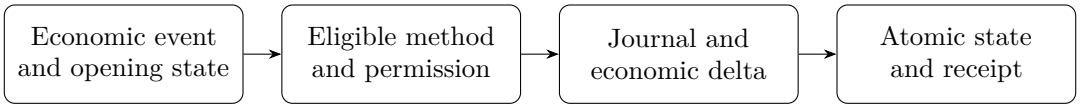


Figure 8.1: Each step answers a different question. Accounting assessment may inform the permission decision, but only the final transaction changes the book.

For a delivery, the basis identifies the exact performance obligation, quantity, allocation and control finding. The opening state contains cumulative revenue, billing, collection and inventory. The selected method derives the recognition and lot movements; the report later consumes that same committed path. Combining a journal from one method version with a delta or opening state from another is invalid even if the totals happen to balance.

8.5 A lost acknowledgement must not create a second sale

Suppose a sale commits but the response is lost. The client knows the request, but cannot tell whether the server committed it. Retrying under a new event ID could duplicate revenue. Retrying the original request must instead return the original immutable receipt.

The lookup key is

$$(\text{entity},\ \text{book},\ \text{branch},\ \text{business event}).$$

Payload, event version, opening state and request settings belong to the complete fingerprint, not this key. A renamed run or bumped event version cannot evade a conflict. A separately commissioned hypothetical projection uses a new branch namespace; reopening the same projection uses its existing one.

A correction of an already committed event is a new business event linked to its predecessor and an admitted correction method, such as reversal and replacement or a dated adjustment. It preserves the original receipt. The accounting treatment must apply to the actual correction; absent treatment authority leaves it unresolved rather than licensing a generic reversal.

Commit first validates input shape, closure, bindings and authorization in the requested historical view. It then looks up the stable event key. An equal fingerprint returns the old receipt with no additional commit; a different fingerprint returns an event-identity conflict. Only an absent key proceeds to the expected-state and generation comparison. This order is essential: after a successful sale the current head has changed, so checking freshness before replay would reject the very retry needed to resolve the lost response.

A new candidate is assessed and constructed without mutation. One transaction then rechecks key absence and the expected head, inserts journal, delta, closing state, dependencies and

receipt, and advances the head. It commits all or none. If another client has committed the identical event, return that receipt. A different event that advanced the head makes this request stale. A changed payload under the same key conflicts. Diagnostic or scratch records never count as economic transitions.

A timeout consequently has three possible observations: committed, not committed or unknown. Unknown retains the original key, fingerprint and expected head for receipt lookup or exact retry. Refreshing the head changes the request and cannot be presented as the same retry. Missing committed dependencies are an integrity failure, not permission to reconstruct an approximately similar receipt.

8.6 Funding, tax and other simultaneous relations

Some financial models have circular dependencies. Interest affects taxable profit; tax and investment affect funding needs; funding affects interest. The reference executor first tries to resolve these through explicit timing. For example, beginning-of-period debt can determine current interest, with new borrowing affecting the next accrual period. That is acceptable only when the model's timing assumption is stated and appropriate.

Where the economic model genuinely requires simultaneous decisions, the dependency graph identifies the strongly connected component and delegates it to a declared equation or optimisation solver. The solver must return a solution, residuals, admissibility checks and uniqueness or multiplicity status. A contraction iteration requires a justified convergence condition; an arbitrary fixed iteration count is not a mathematical solution. Multiple admissible equilibria become alternatives when material. No solution produces a model outcome such as financing failure, or an explicit numerical failure, depending on the cause.

Funding policy must include available facilities, limits, covenants, collateral and allowed capital actions. The program cannot issue unlimited debt to keep cash positive. An inability to meet payments may be a legitimate default outcome. Default ends or changes contractual and accounting processes under the relevant event rules; it is not quietly discarded as a bad simulation draw.

8.7 Worked example: identical history, different future journals

Return to the company in Chapter 1. Immediately after the historical interest payment and closing, cash is 95, debt is 100 and equity is −5. There are no other assets, liabilities or transactions. Next year's expense is 5 in the all-fixed branch and 7 in the all-floating branch under the rate shock. The generated entries and closing statements are:

Output	All fixed	All floating
Debit interest expense / credit cash	5	7
Closing cash	90	88
Closing debt	100	100
Current-year profit	−5	−7
Closing equity after profit transfer	−10	−12
Cash change	−5	−7

The balance-sheet identity holds in each branch: $90 = 100 - 10$ and $88 = 100 - 12$. The demonstration reporting profile classifies interest paid as operating cash flow. A different

permitted reporting profile changes that classification, while preserving the cash movement and source event. This example proves the intended distinction and its arithmetic; it does not estimate the probability that either debt structure describes a real company.

8.8 Worked example: a bond has several relevant amounts

A two-period zero-coupon bond raises cash of 100 and promises 110.25 at maturity. Assume a 5% effective rate per period, no transaction costs, no default, and amortised-cost treatment under a demonstration rule. The effective-interest schedule follows directly from $100(1.05)^2 = 110.25$:

Date	Opening liability	Interest accretion	Closing liability
Issue	0	0	100
End of period 1	100	5	105
End of period 2, before payment	105	5.25	110.25
Redemption	110.25	0	0

Issuance debits cash and credits the liability by 100. Each accretion debits interest expense and credits the liability. Redemption debits the liability and credits cash by 110.25. Legal face amount, carrying amount and market fair value remain separate fields. A market-rate shock may change fair value without changing this contractual payment schedule or this demonstration amortised-cost path.

Cash-flow reporting uses the actual payment event and its economic components; the annual accretion is non-cash. The applicable reporting package decides how to classify the maturity payment, rather than inferring it solely from the liability debit. Fees, modifications, early redemption, embedded options, hedges and foreign currency require additional event and measurement rules.

Ordinary equity issuance similarly separates cash, share quantity and legal capital. Ten new shares at 12, with par value 1, produce cash of 120, share capital of 10 and premium of 110 under a simple par-value profile. Shares outstanding increase by ten units. Direct issuance costs of 3 reduce the specified equity account and cash under the chosen illustrative treatment. The accounting rule and the cost's eligibility must be explicit; this example does not prescribe every jurisdiction's treatment.

8.9 Reporting and explanations

The reporting engine produces the balance sheet, income statement, comprehensive income, cash-flow statement, changes in equity and note schedules when the necessary modules are present. Each output carries its basis, perimeter, period, scenario and interpretation. A comparative table cannot combine an original prior-year figure and a restated current-year comparative without displaying that choice.

An explanation is a dependency graph with typed edges: extracted from, interpreted as, assumed by, constrained by, generated by, measured by, posted by, aggregated into and supersedes. A cash-flow line must lead back to payment events; a fair-value movement must lead back to valuation inputs; a consolidated amount must show eliminations. A provenance path establishes how a result was obtained. It does not by itself establish that the underlying economic model is correct.

8.10 The corporate reference package as operations

The Northstar example defines a closed, reproducible corporate package rather than a universal standard. Its events separate performance, invoicing and settlement even where the example makes the first two simultaneous. Every payload includes entity/book, currency, economic date, period, source or assumption IDs and an information-view ID. Amounts are exact before the declared posting quantisation. Quantities and money are checked separately.

Event	Preconditions and state change	Posting
Deliver/invoice	Goods owned and available; control transferred; quantity removed at selected cost; unconditional amount due added	Debit receivable, credit revenue; debit cost of sales, credit inventory
Collect	Receipt no greater than selected unpaid receivable; reduce that balance; excess must be separately classified	Debit cash, credit receivable
Receive goods	Accepted owned quantity at supported unit cost; create supplier obligation and inventory lot	Debit inventory, credit payable
Pay supplier	Identified obligation and sufficient available cash; reduce both	Debit payable, credit cash
Payroll	Service period and cash obligation evidenced or assumed	Debit payroll expense, credit cash; accrual variant uses a payable
Acquire plant	Control obtained, cost known, in-service date/life/residual recorded; add asset cohort	Debit plant cost, credit cash or payable
Depreciate	In-service asset with remaining depreciable amount; apply dated schedule, capped by that amount	Debit depreciation, credit accumulated depreciation
Interest	Contract, reset state and accrual fraction known; create interest obligation, then settle if cash available	Debit interest, credit interest payable; debit payable, credit cash
Current tax	Reference tax base positive; 20% teaching rule; apply prepaid tax before further payment	Debit tax expense; credit prepaid tax or payable; settlement separately affects cash
Dividend	Explicit authorised distribution, sufficient funds and permitted capital action	Debit retained earnings, credit dividend payable; then settle
Close	Required period operations completed; preserve original movements	Transfer temporary income/expense balances to retained earnings

The current example combines accrual and settlement when they share a date, but the separate obligation remains the general operation. Customer advances are debit cash/credit contract liability and become revenue only on the supported performance event. Inventory costing requires an admissible policy; negative inventory and collection beyond an identified claim are

rejected. Non-cash acquisitions, write-offs, impairment and currency movements require their own classified templates. An unknown residual never enters one of these operations merely to force a reconciliation.

Reporting maps asset and contra-asset balances into the balance sheet and liability/equity balances with their opposite sign. Before closing, current profit is included once in equity; after closing it is already in retained earnings. Income uses period movements in income/expense accounts, and OCI uses its own designated accounts. Cash flow sums cash legs by economic operation: Northstar's collections, suppliers, payroll, interest and taxes are operating, plant payments investing, and debt/equity proceeds or repayments and dividends financing. Non-cash depreciation is excluded from direct cash flow and added back only in the indirect reconciliation. Note schedules use the same contracts, lots and classified movements as the statements.

8.11 Scenario meaning and information access

An `ExogenousStress` supplies a path or joint driver law while holding the historical evidence model fixed. `ObservationalConditioning` additionally supplies actual observations and a joint probability model; the finite method in Chapter 6 evaluates the conditional distribution when its normalising mass is positive. `StructuralIntervention` requires a named structural model and replaces designated equations, retaining its exogenous noise. An absent structural model gives `UnsupportedScenario`, not an intervention inferred from an accounting identity (Pearl, 2009, Section 3.2.1). The textual Core 0 `scenario` declaration is an exogenous stress only; the other modes require the explicit typed API records.

At each decision date the runtime constructs an information view containing eligible disclosures, committed state through that date, currently observed drivers and the declared forecasting model. Future realised innovations and future outcomes are inaccessible. A stress path may be announced in advance, but this is an explicit visibility policy. Otherwise management sees only its current and past values. A bank's allowance at the opening date may use a forecast distribution of defaults; it must not inspect the outer simulation's future realised default. Inner valuation randomness has a separate stream domain from outer realised outcomes.

Funding follows a declared waterfall: available cash, committed eligible facilities up to their limits, and separately authorised capital actions. Each action checks collateral, covenants and settlement timing. If an amount remains due, record an economic funding failure and the unpaid obligation. If required valuation or accounting is unsupported, stop the affected dependencies as incomplete. The branch manifest preserves the last committed state, completed outputs, unavailable outputs and reason. These distinctions let an analyst see a meaningful bad outcome without confusing it with broken computation.

Chapter 9

Accounting methods for an operating company

9.1 A common core with distinct accounting books

The common core governs evidence, time, money, events, journals and provenance. Reporting standards govern recognition, measurement and presentation within that core. The design preserves three levels of meaning: economic terms such as a promise to pay cash; an accounting treatment under a particular standard; and a company's reported label or extension. These levels are connected by explicit rules and mappings, rather than collapsed into one universal chart of accounts.

A rule package includes a standards register with jurisdiction, issuer class, effective dates, transition provisions, permitted elections, paragraph references, implementation version and tests. The package selector uses the entity, reporting basis, period and adopted elections. It must distinguish a standard's mandatory effective date from permitted early adoption. For example, the IFRS Foundation's IFRS 18 overview states an effective date of 1 January 2027 with earlier application permitted (IFRS Foundation, 2026c). A 2026 historical report cannot be assigned that presentation regime solely because the latest installed package includes it.

IFRS, US GAAP and ASBE packages share economic contracts and events where appropriate, but produce separate accounting books. A reconciliation between books consists of explained adjustments with origins in differing treatments, not a renaming of accounts. The first implementation can preserve all three source bases even while only a selected subset has executable rules. Such a source is represented as observed information with unsupported simulation treatments, rather than being silently converted to the supported basis.

The official IFRS overviews for IFRS 9, IAS 32 and IFRS 17 establish the relevant scope of financial-instrument measurement, presentation and insurance contracts (IFRS Foundation, 2026d,a,b). They are orientation sources, not substitutes for reviewed paragraph-level implementations. The retained official Ministry of Finance ASBE 21 page now supports selected lease provisions. The official 2017 ASBE 22 text also supports the selected credit-loss, interest and write-off distinctions worked through in Chapter 11 (Ministry of Finance of the People's Republic of China, 2017). Entity-specific issuance and amendment chains and broader ASBE authority remain incomplete. Chapter 12 compares the dated ASBE 25 premium-allocation and incurred-claim provisions with the retained AASB 17 text (Ministry of Finance of the People's Republic of China, 2020). The Ministry's 2025 explanation supports the stated entity-class timetable, while Ping An's reports supply an actual adoption and comparative-restatement example (Section 12.23). Direct verification of the operative issuance and implementation

notices remains open, including the detailed conditions of the later simplifications. Current ASC access is also unresolved. Historical amendments and issuer policies inform proposed comparisons without substituting for operative authority. These source limits are addressed explicitly in the following dated-method discussion.

9.2 Non-financial companies

The initial industry package covers sales, collections, purchases, payments, inventory movements, payroll, capital expenditure, depreciation and basic financing. Revenue recognition is a separate rule from invoicing and cash collection. A service contract may generate a performance event before an invoice, creating a different asset from an unconditional receivable. Customer advances create obligations before revenue. Returns, rebates and variable consideration need separate assumptions and subsequent adjustments.

Inventory state includes quantity, cost, location, ownership and lot or cohort. A costing method is selected within the reporting package's permissible domain. An aggregate stock can support a simple model only if its approximation is declared. A commodity-price shock can expose differences between old and new inventory that an average value suppresses. Fixed-asset state likewise requires acquisition dates, components, useful lives, residual values, depreciation method, tax basis and impairment history when those affect the requested forecast.

Industry extensions add economic mechanisms without bypassing the accounting core. Construction requires performance and completion states; extractive industries require reserves, depletion and restoration obligations; utilities may require rate-regulation treatment; software businesses may require contract liabilities and development-cost decisions. The module manifest states what is represented, what is executable, and what remains aggregated.

9.3 A matched inventory case across reporting profiles

Standards packages need operation-level decisions. For interchangeable goods, the inspected AASB 102 text incorporating IAS 2 permits FIFO or weighted average in paragraph 25 and explains both in paragraph 27 (Australian Accounting Standards Board, 2015). The reproduced ASBE 1 text specifies FIFO, weighted average or specific identification in Article 14; its official Ministry version could not be retrieved in this review, so that reproduction has provisional source status (Ministry of Finance of the People's Republic of China, 2006a). Caterpillar's company-authored 2024 report states that it principally uses LIFO and supplies a FIFO comparison (Caterpillar Inc., 2024, Note 1.C). That is evidence of a disclosed US-GAAP company policy; full ASC 330 paragraph authority remains an acquisition task. It is not a rule that every US company must use LIFO.

Take the same purchases in three separate books: ten units at 5, followed by ten at 7, then delivery of twelve units. FIFO charges $10(5) + 2(7) = 64$ to cost of sales and leaves inventory 56. LIFO charges $10(7) + 2(5) = 80$ and leaves inventory 40. Periodic weighted average uses cost $120/20 = 6$, charges 72 and leaves 48. The physical deliveries and supplier cash flows agree; profit and carrying inventory differ by the selected costing rule. A concept mapping cannot remove that difference. The FIFO/average selector in an IAS 2 profile rejects LIFO as an unsupported election under the inspected paragraph. The US company profile admits its evidenced LIFO policy while retaining its narrower authority. An ASBE package remains source-provisional until the attributed text and applicable amendments are officially verified.

Inventory impairment illustrates another required memory. Under inspected AASB 102 para-

graphs 28–34, an item costing 56 written down to net realisable value 50 records expense 6. If the conditions reverse and net realisable value becomes 54, reversal is 4, capped by original cost; the previous write-down must therefore remain in state. The reproduced ASBE 1 Article 19 also addresses reversal within the earlier provision. This paragraph is not used to assert the US treatment, whose primary authority was unavailable. Source limitations constrain the treatment claim, while the separate-book method and exact costing calculations remain fully specified.

9.4 Operating events

The operating methods use the selected revenue, inventory and asset provisions of Australian Accounting Standards Board (2022d, 2015, 2022b). The lease and gross-tax case uses the dated lease, income-tax and cash-flow compilations (Australian Accounting Standards Board, 2023c,b,a). Each rule still needs the entity, period, transaction facts and elections described below.

Every event uses the event-basis, authorization and commit contracts described in Section 8.4. The event inputs described here are required in addition to identity and date. Each object identifier resolves to its contract, context and admitted source or assumption. A purported cash collection does not establish a sale; a purchase order does not establish receipt.

Revenue, billing and collection are separate transitions. A delivery event names the contract, performance obligation, quantity, allocated price and control evidence. Recognition first releases a related contract liability; earned consideration not yet unconditional creates a contract asset. Both credit revenue. A missing control finding or unresolved price constraint leaves the recognition unresolved.

An invoice or right-to-payment event names the contract, bill and unconditional amount. It debits the receivable and credits an existing contract asset, with any excess recorded as a contract liability. A collection names that receivable, settlement and amount and debits cash while crediting the receivable. An excess payment needs a separate customer-advance method; it cannot be credited to revenue merely because cash arrived. These transitions retain cumulative performance, billing and collection independently.

Purchasing starts when the company receives owned goods, not when it places an order. The event records lot, quantity, unit cost and ownership evidence, debits inventory and credits the payable. Consigned or unreceived goods cannot pass that ownership condition. Consumption or sale identifies an available lot allocation, quantity and admitted cost formula; it debits cost of goods sold or an eligible production asset and credits inventory. Overselling does not create stock. Supplier payment separately debits the payable and credits actual cash. Insufficient cash blocks that settlement and can trigger a separately modelled default; it never creates a hidden loan.

A tangible-asset event records component cost, residual value, useful life, depreciation method and the date it becomes available for use. Initial debit to PPE has a corresponding cash or payable credit; depreciation later debits expense or an eligible asset and credits accumulated depreciation. Employee service records earned pay, deductions and employer obligations for the service period, separately from settlement. Unknown labour or tax law cannot become a zero payable.

Closing derives revenue, expenses, OCI and owner movements from those events using admitted statement mappings. It checks cash, inventory, financing and tax roll-forwards rather than introducing a balancing entry. These event fields form the method's required inputs; each identifier resolves to the same contract, context and evidence used by its authorization.

AASB 15 paragraphs 9, 22, 31, 35–43, 47–59, 73–86 and 105–108 provide the selected revenue conditions; the retained compilation dated 31 December 2022 applies to annual periods beginning on or after 1 January 2023 and before 1 July 2026, covering the stated for-profit FY 2024 example. Variable consideration is estimated under a selected method and admitted only subject to the significant-reversal constraint, not recognized solely because its unconstrained expectation is positive. Principal versus agent, returns, warranties, contract modifications and distinct bundles require their own scoped decision records. If their facts are unresolved, retain alternative eligible methods; do not borrow the simple delivery rule by label.

The following schedules make the distinction between performance and billing explicit. For one performance obligation with price 100, cumulative revenue 60, unconditional bills 50 and collection 30, AR=20, CA=10, CL=0 and cash=30. Revenue 60 is represented by those three assets. Billing before performance changes intermediate balances but not the final result. For bills 80 instead, AR=50, CA=0, CL=20; net assets remain 60. Cumulative performed-service cost 40 then 70 posts 40 then 30, not 110. A downward revision follows the correction method and keeps the old result tied to its original view.

For this simple contract, let R be cumulative recognised revenue, B unconditional billings and C collections, with $0 \le C \le B$. The unpaid bills determine accounts receivable (AR); performance in excess of billing is a contract asset (CA), while billing in excess of performance is a contract liability (CL). After each event,

$$\text{AR} = B - C, \quad \text{CA} = \max(R - B, 0), \quad \text{CL} = \max(B - R, 0), \quad \text{cash received} = C.$$

This recurrence applies per contract, with no impairment, variable consideration or unbilled prepayment; it is not permission to offset unrelated contracts. For billing 80 before delivery, debit AR 80/credit CL 80. Recognizing service 60 later debits CL 60 and credits revenue 60. Collecting 30 debits cash 30/credits AR 30, leaving AR 50 and CL 20, without a second receivable or an artificial contract asset. If service comes first, recognize CA 60/revenue 60, then bill AR 80 against CA 60 and CL 20. Both orders end at the same state. Cash received before an unconditional bill needs an explicit advance event that debits cash and credits CL and records the advance applied to later billing; it cannot pass through collect(AR).

9.4.1 A payment before the bill must be applied only once

The preceding recurrence deliberately excludes an advance before billing. Suppose a customer pays 30 towards a service priced at 100. No service has yet been performed and no payment was already due. The company now has cash 30 and an obligation to provide the service, so it debits cash and credits a contract liability. Crediting revenue would balance the entry but recognise performance that has not occurred. Paragraphs 105–108 of the retained AASB 15 distinguish this payment from an unconditional receivable (Australian Accounting Standards Board, 2022d).

Now let the company earn 60 before issuing a bill. Recognition releases the liability of 30 and creates a contract asset of 30, with a credit to revenue of 60. The later invoice shows a gross amount of 80 and applies the customer's advance of 30, leaving a new unconditional right to 50. Its entry debits the receivable by 50, credits the contract asset by 30 and credits a contract liability by 20. Creating a receivable of 80 would count the prepaid 30 again. When the customer pays another 20, cash is 50, the receivable is 30 and the contract liability is 20. Their net contribution to assets is the earned 60.

The method therefore keeps the invoice's gross amount separate from the right still owed. Let G be cumulative gross billings, A cash advances, K the amount of those advances applied to bills, and C later collections of receivables. Each application names both its advance and its

invoice; the remaining unapplied credit is $A - K$. As before, R is cumulative earned revenue. The consideration already paid or made unconditionally due, counted once, is $T = G + A - K$. For this one fixed-price contract,

$$\text{AR} = G-K-C, \qquad \text{CA} = \max(R-T, 0), \qquad \text{CL} = \max(T-R, 0), \qquad \text{cash received} = A+C.$$

The minus K prevents the applied advance from appearing both in the cash payment and in the bill. Since $\text{CA} - \text{CL} = R - T$, the control

$$(A + C) + (G - K - C) + \text{CA} - \text{CL} = G + A - K + R - T = R$$

recovers the earned amount. This derivation uses a single contract, with $0 \le K \le \min(A, G)$, $0 \le C \le G - K$ and $0 \le R, T \le 100$. It provides no authority to offset different contracts or to ignore impairment, financing, tax, cancellation or refund terms.

Each event must carry its contract, identity, date, currency and admitted facts. In this restricted extension, advances precede all performance and billing; later payments settle identified receivables. A revenue increment first releases an existing contract liability and then creates a contract asset. For a new gross bill g that applies unused advance credit k, only $g - k$ creates a receivable. That amount releases the existing contract asset first; any remainder increases the contract liability. The invoice consumes its identified credit even if the advance has already been earned. A duplicate credit application, an overcollection or missing unconditional-right evidence cannot pass by adjusting a balancing account.

Billing before performance gives the same final balances in the example but a different intermediate history. The invoice creates a receivable of 50 and adds 50 to the existing contract liability of 30; earning 60 then releases 60 of that liability. After the later collection of 20, both paths have the balances above. Completing the service adds revenue 40, releasing the remaining liability of 20 and creating a contract asset of 20. A final bill for 20 and collection of the remaining 50 leave cash 100 and no contract or receivable balance. The actual sequence is retained even where endpoint amounts agree.

A wholly prepaid invoice exposes another implementation requirement. If its gross amount and applied advance are both 30, it creates no new monetary right and needs no accounting lines. It still consumes an identified credit. The proposed authored method therefore declares a state-only journal proposal: empty lines, zero account changes and explicit changes to the invoice and advance allocations. Authorization checks that the net new right is zero; the same atomic commit and replay rules protect the allocation from reuse. An empty journal cannot carry a nonzero monetary effect. The native acquisition schema requires nonempty lines and does not implement this case; an unavailable authored schema returns an unsupported-operation result without changing state.

Take the wholly prepaid invoice in a synthetic book containing only cash 30 and a contract liability of 30. No performance has occurred. Advance A1 has 30 unused; there is no invoice yet. The new invoice I1 records gross billings of 30 and applies all of A1. Cumulative G and K both rise from zero to 30, so $T = G + A - K$ remains 30. Cash, revenue, receivables and contract balances do not change. But A1 is now fully used and I1 is paid. Omitting this transition would let a later invoice claim the same credit again.

The authored advance-allocation profile therefore pins a closed journal variant and a separate economic delta. In addition to their common context, basis and authorization references, their distinguishing fields are:

```
JournalProposal {
  effect: "state_only", lines: []
}
EconomicDelta {
  account_changes: zero for every chart account,
  before_contract: advance A1 unused 30; no invoice,
  after_contract: A1 used 30; invoice I1 gross 30,
                  applied 30, unpaid 0; allocation A1->I1
}
```

This display abbreviates the required common references; the pinned schema bundle supplies the closed field lists and complete portable records. The independent calculation checks both levels. It derives zero account changes from the contract equations and verifies that the advance, invoice and allocation agree with the new cumulative values. Empty lines with a proposed receivable increase of 1 fail that reconciliation. Zero account changes with the allocation omitted also fail: the required economic effect is missing.

Authorization uses the identified unused advance, a fresh invoice identity, the zero unpaid amount and the admitted contract bounds. A caller cannot replace those findings with a true permission flag. The profile supports a synthetic book only; its historical source support and authored facts provide no authorization for a real company's posting. An assurance that the empty journal balances would establish none of these conditions.

For a valid first event E1, the atomic transaction publishes the state-only proposal, independent delta, new contract state, receipt and event-index entry, and advances the branch generation once. The receipt is evidence that the allocation was committed, even though every account balance stayed the same. The stable identity remains the entity, book, branch and business event. Its fingerprint covers the whole portable request, including the original expected head, generation, request identifier and execution settings.

Retrying the identical request after the commit returns the original receipt. The lookup must precede comparison with today's head; otherwise a successful retry would wrongly fail because its own earlier commit advanced the branch. Changing I1 to I2 under E1 instead conflicts, as does changing the request identifier while keeping the amounts. A genuinely new event E2 submitted against the old opening fails freshness. Rebuilding E2 against the current opening cannot rescue the reused credit: A1 has no unused amount, so authorization fails before lookup. These are different reasons for preserving the current book unchanged.

On replay, the response identifies the original committed event but reports no additional commit by this invocation. A lost acknowledgement remains unknown until lookup under the original identity resolves it; inventing a new event identity could double the allocation. Exports and recovery must also retain the economic transition. A format limited to monetary journal lines must preserve the state-only event separately or report that a lossless export is unsupported. Dropping it because revenue and cash did not change would make the recovered advance available twice. The complete documentary exchanges exercise these distinctions; transaction durability and concurrent execution remain implementation checks.

These are revenue and cash contributions before service costs and tax. Their missing measurements remain inputs to the complete company statements. The next example retains a different obligation: cash already received may have to be returned even after the goods have been delivered.

9.4.2 A return right survives the sale

Consider 100 goods sold for 10 each, with a former inventory cost of 6 each. The buyer obtains control, pays cash and has a contractual right to a refund for returned non-defective goods. Assume an eligible Australian for-profit contract in 2024, within the retained AASB 15 compilation's application period. The contract, control, price-estimation and significant-reversal conditions are separately satisfied; immediate cash receipt alone would not establish them. The numbers are synthetic measurements, not estimates of Apple's returns.

If the company initially expects 10 goods to be returned, it expects to retain 900 of the 1,000 received. AASB 15 B20–B25 requires revenue for the expected entitlement, a refund liability and a separate asset for the right to recover the goods (Australian Accounting Standards Board, 2022d). With no expected recovery costs or deterioration in this example, the refund liability is 100 and the recovery asset is 60. The company debits cash 1,000, credits revenue 900 and credits the refund liability 100. It removes inventory 600 by debiting cost of sales 540 and the recovery asset 60. Pretax profit is therefore 360. Recognising the full 1,000 as revenue would omit a known refund obligation, although the entry balances.

For a homogeneous cohort of N units, unit price p, former unit cost c and an admitted estimate q of returns, the initial no-recovery-cost case gives

$$\begin{aligned} \text{revenue} &= (N-q)p, & \text{refund liability} &= qp, \\ \text{cost of sales} &= (N-q)c, & \text{recovery asset} &= qc. \end{aligned}$$

The estimate and its measurement date are retained with the cohort. Paragraphs 53–59 require an appropriate estimation method and the constraint on a significant reversal; q is not a probability supplied by the accounting standard. Missing evidence for that measurement leaves the amount unresolved. After actual returns begin, the state separately records refunded units, remaining expected returns, remaining liability and recovery rights. A later outcome cannot be inserted into the original information view.

Suppose the sale occurs in 2024 and new information at year end raises expected returns to 12. The liability increases by 20, reducing revenue by 20. The recovery asset increases by 12, reducing cost of sales by 12. Closing revenue is 880, cost of sales 528 and pretax profit 352. The change in expectations does not itself establish an error in the original estimate. If the opening inventory of 600 was financed entirely by opening equity of 600, the resulting pretax balance sheet has cash 1,000 and recovery asset 72, against refund liability 120 and equity 952.

In 2025, assume the return period ends with eight accepted returns and no remaining legal right or unresolved claim. Before final settlement, remeasure the liability from 120 to 80 and the recovery asset from 72 to 48. The first change credits revenue by 40; the second debits cost of sales by 24. Refunding 80 then debits the liability and credits cash. Receiving the eight goods debits inventory by 48 and credits the recovery asset by 48. Settlement does not recognise the sale or its cost a second time. The 2025 pretax profit contribution is 16, closing cash is 920 and inventory is 48. With no other movements, equity is 968: opening 600 plus cumulative profit 368.

The cash and accrual statements can now be checked independently. Operating cash is 1,000 in 2024 and −80 in 2025. In the indirect method, the first year adds the inventory reduction and liability increase to profit while subtracting the recovery-asset increase. The second year reverses the relevant balance movements:

$$\begin{aligned} \mathrm{CFO}_{2024} &= 352 + 600 - 72 + 120 = 1{,}000, \\ \mathrm{CFO}_{2025} &= 16 - 48 + 72 - 120 = -80. \end{aligned}$$

There is no investing or financing cash flow, OCI or subsequent owner transaction in this slice. Tax has not been computed: these pretax contributions must enter the separate current/deferred-tax method before they can become complete after-tax company statements.

Recovery costs change the asset without necessarily changing the refund. For the 12-unit estimate, expected recovery costs of 1 per unit and deterioration of 0.5 give a recovery asset of $12(6 - 1 - 0.5) = 54$. Cost of sales becomes 546 and pretax profit 334, while the refund liability remains 120. The cost inputs are measurements, not automatic zeros; estimating them does not pay cash early. Paragraph B25 also requires separate presentation of the asset and liability. Showing only their net 48 in the original no-cost case would conceal both the gross right to recover goods and the gross obligation to refund cash.

Scope conditions matter as much as the arithmetic. Paragraph B26 treats an exchange for the same type, quality, condition and price differently from a return; B27 sends a defective-product replacement to warranty guidance. The refund method cannot accept those events simply because the goods move back to the seller. It also needs the contractual return window, accepted-unit identities, actual refund and recovery measurements, and evidence that any remaining rights have expired before clearing the balances. An unknown estimate yields unresolved, absent control makes recognition ineligible, and an event of a different kind remains outside this method. Inventory costing and the condition of recovered goods continue to have their own requirements.

9.4.3 A warranty can promise another year of service

A returned product and a warranty claim need different histories. The return example unwinds a sale when a customer exercises a refund right. A warranty may instead promise to keep a product working, or provide a service after the product has been delivered. The company must identify which promise it made before deciding how much of a customer's payment it has earned. The following cases use the same retained AASB 15 compilation and historical Australian for-profit scope as the return example.

Suppose a customer buys a product and a separately purchasable service warranty for a combined 108. Comparable standalone selling prices at contract inception are 100 for the product and 20 for the warranty. Both are distinct promises; the product works independently of the additional service. The contract has neither a discount attributable to just one promise nor variable consideration. Under paragraphs 73–86, the 12 bundle discount is therefore allocated in proportion to the standalone prices. The product receives $108 \times 100/120 = 90$ and the service receives $108 \times 20/120 = 18$. An invoice's labels would not by themselves establish those standalone prices or justify a different discount allocation (Australian Accounting Standards Board, 2022d).

On 1 July 2024 the company receives 108 and transfers control of the product, whose inventory cost is 60. First, receipt debits cash and credits the contract liability by 108. Delivery then debits that liability and credits product revenue by 90, while debiting cost of sales and crediting inventory by 60. Profit is 30 and a liability of 18 remains. Recognising all 108 on delivery would credit revenue for service the company still owes.

The service warranty runs for twelve monthly service periods to 30 June 2025. The company stands ready uniformly; the customer receives and consumes the benefits as service is supplied. The admitted measure gives each completed service month one twelfth of the allocation and faithfully depicts performance. These facts support over-time recognition under 35(a), 39–45 and B14–B19; the existence of a twelve-month date range alone would not suffice. At 31 December 2024, half the service has been provided, so half of 18, or 9, is earned. Debit the contract liability and credit service revenue by 9. Actual service costs of 3 are expensed and

paid. Opening inventory and equity were both 60, with no other opening balances. Year-one revenue is now 99, expenses 63 and pretax profit 36. Cash 105 equals the remaining liability 9 plus equity 96.

The second half-year releases the remaining liability of 9 and incurs another 3 of paid service costs. Its profit is 6 and closing cash and equity are both 102. Over the complete contract, revenue 108 less costs 66 gives profit 42. The cash statement follows the same history: operating cash is 105 in the first period and -3 in the second. The independent accrual bridges are $36 + 60 + 9 = 105$ and $6 - 9 = -3$, using the inventory reduction and changes in the contract liability. There is no OCI or subsequent owner transaction in this pretax slice; the complete company calculation still needs tax and any other obligations.

The contract record must therefore retain the two promises, the original allocation and its price evidence, delivery identity, coverage, progress measure, cumulative service revenue and actual costs. Each service update posts the difference between newly supported cumulative revenue and revenue already recognised. Replaying an accepted delivery cannot release the product allocation twice. Unknown progress is not zero performance: paragraphs 44–45 require a separate assessment of reasonable measurement and, where applicable, cost recovery. This percentage method cannot invent those findings.

The classification also has a substantive boundary. A separately purchasable warranty is a distinct service under B29. Without that option, B30 sends an assurance-only warranty to AASB 137 unless it provides additional service. Law, duration and promised tasks help distinguish the cases under B31; there is no universal number of months that makes a warranty a service. If assurance and additional service cannot reasonably be accounted for separately, B32 treats the two warranties together as one performance obligation. Harm or infringement compensation under B33 is not such a service. The historical AASB 137 population-measurement case in Section 9.14 supplies the separate assurance method. Actual coverage, claims, timing and reimbursement evidence remain required; missing facts cannot become a zero provision or convert assurance into this service example.

9.4.4 Changed promises can affect future units or past progress

A contract can change after some performance has already been recognised. The accounting consequence depends on what remains to be supplied. Consider an original prepaid order for ten distinct units at 10 each. Four have been delivered, earning 40. The parties approve an enforceable amendment adding six units for 48. Comparable standalone prices remain 10 each, and no appropriate contract-specific adjustment makes the 48 addition reflect those prices. The addition therefore fails the separate-contract condition in paragraph 20. Because the remaining units are distinct from those already delivered, paragraph 21(a) reallocates the consideration for the remaining performance (Australian Accounting Standards Board, 2022d).

The undelivered original units carry 60 of consideration. Adding the new 48 gives 108 for twelve remaining units, whose standalone prices are equal. Thus their new unit allocation is

$$\frac{\text{unrecognised original consideration} + \text{additional consideration}}{\text{remaining equal-price units}} = \frac{60 + 48}{6 + 6} = 9.$$

The first four units keep their revenue of 40. Spreading the entire revised price over all sixteen units would instead give $148/16 = 9.25$ and rewrite their revenue as 37. That calculation is wrong for the stated facts: this modification reallocates remaining performance, not completed deliveries.

Inventory and cash make the consequences concrete. Start with sixteen units costing 6 each, financed by opening equity of 96; six units were initially uncommitted. The first prepayment

debits cash and credits a contract liability by 100. The first four deliveries release 40 of revenue and remove inventory cost 24. Immediately before amendment, cash 100 plus inventory 72 equals the liability 60 plus equity 112. Receipt of the additional 48 increases cash and the liability. Approving the changed order itself earns no new revenue and creates no new stock.

Deliver six more units by the first year end. They release liability and revenue of 54, with inventory cost 36. Total year-one revenue is 94, cost of sales 60 and profit 34. Cash 148 and inventory 36 equal liability 54 and equity 130. Delivering the final six in the next year earns another 54 at cost 36, leaving cash and equity 148 with no inventory or contract liability. That period's profit is 18 but no cash arrives: the customer already paid. The operating cash bridges are $34 + 60 + 54 = 148$ and $18 + 36 - 54 = 0$. Lifetime revenue 148 less inventory cost 96 gives profit 52.

If the additional six units instead cost 60 and meet paragraph 20's standalone-price condition, the addition is a separate contract. The first four still earn 40; the six remaining original units earn 60 and the new six earn 60. Both contracts keep their own allocation records. The difference between these cases is the admitted price and scope assessment, not permission to choose whichever result produces the preferred margin.

An integrated promise can have the opposite effect on past cumulative revenue. Suppose a customer controls work in progress as the company creates it, satisfying paragraph 35(b). The company has one integrated performance obligation, and its remaining work is not distinct from work already supplied. Original price 100 and supported output progress of 40% yielded revenue 40. An approved enforceable amendment changes that same promise to price 130; the revised faithful output measure is 30% of the expanded work. Under 21(b), both the price and progress change the cumulative amount earned:

$$\text{revised cumulative revenue} = 130 \times 0.30 = 39, \qquad \text{catch-up} = 39 - 40 = -1.$$

The new work makes the completed work a smaller share of the promise. Revenue decreases by 1 at amendment even though the total contract price increases. The old 40 remains attached to its original information view; the modification does not establish that the earlier estimate was an error.

For a complete prepaid example, start with cash and equity 50. Receive 100 and pay service costs 30 before amendment. Recognition of 40 leaves cash 120, liability 60 and equity 60. At amendment, receive the additional 30 and debit revenue 1 while crediting the liability 1. Cash is now 150, the liability is 91 and equity is 59: cumulative revenue 39 less costs 30 has earned profit 9. Operating cash since opening is 100, also $9 + 91$. Completing the promise next period releases revenue 91 and incurs further paid service costs of 60. Period profit is 31, operating cash is $31 - 91 = -60$, and closing cash and equity are 90. Lifetime profit is $130 - 90 = 40$. The output progress percentages were independently supported; they were not inferred from these cash costs.

Both modifications need the approved rights, effective date, previous and revised promise, consideration, performance measurements and source versions. The distinct-goods method freezes the fulfilled deliveries and replaces the remaining allocation; the integrated method proposes only the difference between revised cumulative revenue and the amount already recognised. Economic state and journals change atomically. Different billing schedules can require a contract-asset adjustment instead of the prepaid liability entries used here. A request that has not been approved leaves the existing contract operative under paragraph 18. Unknown enforceability stays unresolved; approved scope with undetermined price requires paragraph 19's variable-consideration estimate and constraint, not a price of zero.

Paragraph 21(c) also admits mixed remaining obligations, requiring an explicit allocation consistent with the two principles above. The examples do not force such a contract into one

category. Price-estimate changes without a contract modification follow 87–89. A later price change after modification needs paragraph 90: variable consideration promised before a 21(a) modification can still be allocated to the earlier obligations under 90(a). It cannot automatically be spread over the twelve remaining units merely because their current allocation is 9 each.

9.4.5 The same cash can accompany different revenue

Suppose an entity receives 120 from a customer and sends 100 to a supplier. It keeps 20. Those cash movements cannot tell us whether it sold the supplier's good or arranged a sale by that supplier. Under B34–B38, the decision starts with the specified good or service and whether the entity controls it before transfer to the customer. Responsibility for delivery, inventory risk and price discretion inform that decision; none is a substitute for control, and momentary legal title does not by itself establish it (Australian Accounting Standards Board, 2022d).

In the principal case, the entity controls the identified good before customer transfer, can direct its use and obtain its benefits, and is responsible for delivering it. Acquisition on credit debits inventory 100 and credits the supplier payable 100. Customer transfer and cash receipt debit cash 120 and credit revenue 120; the inventory leaves through cost of sales 100. Settling the supplier debits the payable and credits cash 100. From zero opening balances, the result is cash and equity 20, revenue 120 and profit 20.

In the agent case, the supplier controls the good until transfer; the entity only arranges the sale and earns a fixed commission of 20 when that service is completed. Assume the entity controls the collected cash and owes the supplier 100 under its remittance agreement. Funds held outside its balance sheet in a fiduciary capacity are outside this example. Receipt therefore debits cash 120, credits the supplier payable 100 and credits commission revenue 20. Settlement clears the payable against cash 100. The agent has no inventory or supplier cost of sales. Its final cash, equity and profit are again 20, but revenue is 20.

Even the intermediate balance sheets agree: before remittance, both have cash 120, a payable 100 and equity 20. The different control facts explain the different revenue. The method must retain those facts and classify each specified promise separately, since the same entity can act as principal for one promise and agent for another. An unknown control finding leaves classification unresolved; the receipt cannot decide it. Both cash legs remain in the ledger. Whether the cash-flow statement presents receipts and remittances gross or net requires its own AASB 107 conditions, which do not follow merely from the revenue classification. These examples establish the cash change and pretax profit contribution while preserving that separate presentation question.

9.4.6 An idle factory does not make its remaining stock more valuable

Producing fewer goods leaves the factory's fixed costs spread over fewer units, but that arithmetic does not establish their inventory cost. Consider a factory that normally makes 100 good units each year after allowing for planned maintenance. Its annual fixed production overhead is 180. If it makes only 60 units, dividing all 180 by 60 would assign 3 to each unit, rather than the normal 1.8. The unsold goods would then carry part of the cost of idle capacity into the next year.

AASB 102.13 prevents that result. Fixed production overhead is allocated using normal capacity; low output does not increase the amount per unit, and unallocated overhead is expensed. Abnormally high output requires a lower allocation per unit so that inventory is not measured above cost (Australian Accounting Standards Board, 2021a). Normal capacity is an estimate of average production under normal conditions, including planned maintenance. It

requires evidence separate from the actual output or desired margin.

Use a synthetic manufacturer in calendar 2024 and the retained AASB 102 compilation dated 31 December 2021. Its documented application interval includes that year; subsequent amendment and actual-entity admission remain separate. The company makes one ordinary manufactured product, with no opening stock, work in progress, joint products or normal production loss. One material kit produces one good unit. A kit costs 2; direct labor costs 1 and variable production overhead costs 1 per good unit, all paid in cash. These goods are outside the agricultural, financial-instrument and broker-trader exceptions. Periodic weighted average is used for this homogeneous batch under paragraphs 25–27.

The fixed 180 consists of cash factory management and maintenance of 120 and depreciation of 60. Opening factory PPE is 600, with ten remaining years, zero residual value and admitted uniform time-based consumption. There is one component and recoverability has been separately established. The depreciation method therefore remains unchanged across the output scenarios. Paragraph 12 includes factory depreciation and production administration among production overheads; AASB 116.48–49 allows depreciation to enter another asset's carrying amount when appropriate (Australian Accounting Standards Board, 2022b). Recording it in inventory does not make it a second cash payment.

Keep excluded costs visible. A separately documented accident destroys six unused kits costing 12 before labor or conversion. They are additional to the kits used for good output and have no salvage value. Unrelated head-office administration of 10 and period selling services of 8 are also paid in cash. The abnormal loss and these unrelated expenses do not bring the good units to their present location and condition, so paragraphs 10–16 exclude them from inventory. Factory management and head-office administration cannot be classified solely from the word *administration*.

All completed goods have supported net realisable value of 10 per unit, with no further completion or sale costs. The example therefore still applies the lower-of-cost-and-NRV requirement rather than suspending it. Delivered units sell for cash of 10 each under an approved, enforceable single-product contract: rights and payment terms are identified, collection is probable, the contract has commercial substance, and delivery, acceptance, title and control transfer coincide. There are no remaining promises, returns or concessions. These facts admit the existing AASB 15.9 and 31–38 delivery method (Australian Accounting Standards Board, 2022d); cash receipt by itself would not do so.

Allocate the cost that production can carry

Let Q denote good completed units and S the units sold, with $0 \le S \le Q$. For this single cost pool, normal capacity is 100 and fixed overhead is 180. Each good unit first receives its direct and variable cost of $2 + 1 + 1 = 4$. When output is low, the fixed allocation is 1.8 per unit; when output is abnormally high, the total allocation is capped at the 180 actually incurred. For $Q > 0$, this gives

$$f(Q) = \frac{180}{\max(100, Q)}, \qquad u(Q) = 4 + f(Q), \qquad U(Q) = 180 - Qf(Q),$$

where f is fixed overhead per unit, u is admitted unit cost and U is unallocated overhead expensed now. This compact formula implements the stated homogeneous, constant-cost example. Multi-product pools, work in progress or changed production resources require their own allocation. With no output, there is no unit cost to divide by: stock is zero and all 180 is current expense.

At low output $Q = 60$, the good units carry fixed overhead of $60 \times 1.8 = 108$, leaving 72 as

current expense. Unit cost is 5.8. Selling 40 units removes inventory cost 232; the other 20 remain at 116. Abnormal waste 12 and the unrelated administration/selling cost 18 are also current expenses. Revenue 400 therefore leaves pretax profit $400 - 232 - 72 - 12 - 18 = 66$. Normal output of 100 absorbs all 180 at the same unit cost of 5.8. Abnormally high output of 150 instead allocates $180/150 = 1.2$ per unit, giving unit cost 5.2.

The comparison below changes output and sales explicitly. All arms retain the same fixed costs and the separately stated accident. They do not have equal total production spending: direct and variable costs change with good output. Sold-stock cost means the carrying amount removed on sale; AASB 102.38 also includes unallocated overhead and abnormal production costs in the broader inventory expense.

	Low	Normal	High	Idle
Good units / sold units	60 / 40	100 / 80	150 / 120	0 / 0
Unit cost	5.8	5.8	5.2	n/a
Unallocated fixed expense	72	0	0	180
Sold-stock cost	232	464	624	0
Closing inventory	116	116	156	0
Broader inventory expense	316	476	636	192
Pretax profit	66	306	546	−210
Closing cash	1010	1250	1450	850
Closing PPE	540	540	540	540
Pretax net assets	1666	1906	2146	1390

Opening cash is 1000 and PPE is 600, so opening net assets are 1600. There are no payables, borrowing or owner movements in this slice. Tax effects are uncomputed: the last row is a pretax net-assets reconciliation, not after-tax equity, distributable profit or a complete statutory set of statements. This boundary avoids manufacturing tax law to finish the example. Each arm has separately reconciled cash, stock and PPE; an after-tax extension must supply the relevant tax bases, rates and timing.

Preserve both the cost origin and the cash movement

The low-output journal path starts with 66 kits costing 132. Debit raw materials and credit cash 132. The accident debits abnormal-loss expense 12 and credits raw materials 12. Transferring the 60 good kits to production moves raw-material cost 120 into finished goods; direct labor 60 and variable overhead 60 enter the same goods account against cash. No raw materials remain, and the goods initially carry 240.

The fixed costs first enter a production clearing account: debit it 120 against cash and 60 against accumulated depreciation. Then debit finished goods 108 and unallocated-overhead expense 72, crediting the clearing account 180. Its balance is now zero, and finished goods cost 348. The separate administration and selling entry debits expense 18 against cash. On delivery, debit cash and credit revenue 400, then debit sold-stock expense and credit finished goods 232. The final 116 is the cost of 20 unsold goods, not a residual chosen to make the books balance.

Cash receipts of 400 less payments of $132 + 60 + 60 + 120 + 18 = 390$ increase cash by 10. The whole factory's depreciation is 60 even though part remains in inventory. To reconcile pretax profit with operating cash, add that noncash consumption and subtract the full inventory increase:

$$\Delta\text{cash} = 66 + 60 - 116 = 10.$$

The stock includes depreciation of $20 \times 60/100 = 12$. An equivalent reconciliation adds only the 48 of depreciation reaching current profit, but then subtracts only the cash-cost part of

stock, $116 - 12 = 104$: $66 + 48 - 104 = 10$. Adding back 48 while subtracting all 116 would wrongly report a cash decrease of 2. The two valid methods differ in presentation, not in the underlying cost population.

Selling all 60 good units in the low-output arm provides a second check. Production and its payments are unchanged. Revenue becomes 600, sold-stock expense 348, profit 150 and cash 1210; inventory becomes zero. The additional 200 cash receipt releases inventory cost 116 and increases profit by 84. Total pretax net assets become 1750. This is why a forecast needs both production and sell-through, rather than an inventory margin inferred from total factory spending.

Balanced books can still defer the wrong costs

If low output absorbed all fixed overhead using $180/60 = 3$ per unit, unit cost would be 7. Unsold stock would be 140 and profit 90, each 24 above the admitted result; cash would remain 1010. The books would balance, but they would postpone idle-capacity expense in the remaining goods. At high output, applying 1.8 without a cap would allocate 270 of fixed cost when only 180 was incurred. Crediting a variance of 90 to balance that over-allocation would leave stock 174 and profit 564, both overstated by 18. A balanced variance cannot create an incurred cost.

Capitalizing abnormal loss 12 and unrelated period costs 18 into the 60-unit batch would defer $30 \times 20/60 = 10$ of excluded expense in unsold stock. This is a separate classification error from the capacity error. Each resource therefore retains a purpose and identity, its membership in the cost pool, and its cash or noncash origin. The state also retains capacity evidence, good and lost units, batch cost and quantity, and the specific sale removals. A duplicate resource or negative stock is invalid; a revised capacity estimate creates a new supported allocation decision instead of overwriting an earlier result.

Unknown normal capacity leaves the fixed allocation unresolved. Unknown NRV prevents a complete closing measurement, and missing control-transfer facts prevent revenue recognition. Work in progress, joint products and heterogeneous cost populations require a different specified model rather than silent use of this unit formula. The proposed procedure and checked case define those boundaries; they do not establish production-engine conformance or an actual manufacturer's accounting treatment.

9.5 When a component reaches its next accounting event

A recognised balance also promises a subsequent method. A refund liability must eventually be settled; an impaired asset may recover; a modified lease must continue through payments and depreciation. The following methods specify those transitions under stated facts. They use the retained historical IFRS text, with current authority and actual-company measurement kept as separate admission questions (European Commission, 2023a).

9.5.1 Actual returns can differ from the estimate

Section 2.7 follows two expected and actual returns. More generally, let L be the opening refund liability, A the recovery asset, P the actual refund payment, K the actual selling-handling cost and V the returned inventory admitted after deterioration and its NRV ceiling. When all rights

have closed, the debit-positive entries are

$$\Delta\text{cash} = -P - K, \qquad \Delta\text{inventory} = V, \tag{9.1}$$

$$\Delta\text{refundliability} = L, \qquad \Delta\text{recoveryasset} = -A, \tag{9.2}$$

$$\Delta\text{revenue} = P - L, \qquad \Delta\text{costofsales} = A - V, \tag{9.3}$$

$$\Delta\text{sellingexpense} = K. \tag{9.4}$$

The debit-positive revenue amount is negative when releasing an excessive refund estimate increases revenue. These entries sum to zero. The separation of selling cost follows the explicit handling classification and IAS 2.16; IFRS 15.B25 governs the recovery asset. If handling instead creates eligible inventory cost, that needs its own admitted classification.

With $L = 40$ and $A = 16$, one actual return pays refund 20 and handling 4, and recovers inventory 10. Revenue increases 20, cost of sales is 6 and selling expense 4. Three actual returns instead pay refund 60 and handling 12, recover inventory 30 and reduce revenue by 20; cost of sales is credited 14. If two returns recover inventory worth only 14, the inventory ceiling raises cost of sales to 2. Unknown handling, inventory NRV or an open return window prevents final settlement rather than supplying a zero.

9.5.2 A reversal ceiling does not supply an allocation weight

IAS 36.122–123 allocates a cash-generating unit's reversal to assets other than goodwill, proportionately to their carrying amounts, subject to the individual ceilings. Suppose assets A and B carry 40 and 60 and goodwill is 10. The unit's recoverable amount rises to 170, allowing a total increase of 60. A's individual ceiling is 45 and B's is 150. The first proportionate allocation would assign 24 to A and 36 to B. A can accept only 5; reallocating the remaining 19 to B gives increases of 5 and 55. Goodwill remains unchanged.

Writing w_i for the original carrying amounts and c_i for the remaining headroom, the method distributes a remaining reversal R among eligible assets by $Rw_i/\sum_j w_j$, caps each amount at c_i, then redistributes any remainder among assets still below their ceilings. This terminates when the reversal is exhausted or no headroom remains. It requires a positive sum of eligible weights whenever a positive amount remains.

If all eligible carrying amounts are zero, the denominator is zero. Neither equal weighting nor weighting by headroom follows from the inspected source. The specification therefore retains an unresolved allocation with unchanged books. A different rule applies to an individual asset: a zero carrying amount, no-impairment carrying value of 80 and recoverable amount of 120 permits an individual reversal to 80 under IAS 36.117–119. That individual calculation does not resolve the allocation among several zero-carrying assets.

9.5.3 A lease reduction must identify the rights surrendered

A single physical percentage is insufficient when leased rights have different values. Consider right A with right-of-use asset 90 and liability 100, and right B with asset 30 and liability 50. The supplied modification terminates B, releasing asset 30 and liability 50 for a gain of 20. It leaves A with one payment of 121 in a year at a revised 10% rate. A's revised liability is $121/1.1 = 110$; the increase of 10 adjusts its asset from 90 to 100. These are the scope-reduction and remeasurement steps in IFRS 16.45–46.

A further same-day reduction surrenders an admitted portion of A carrying asset 10 and liability 11. The remaining liability before remeasurement is 99. Revised consideration of 99 due in one year has present value 90, so the remeasurement reduces both liability and asset by 9. The final

asset is $100 - 10 - 9 = 81$, liability 90 and additional gain 1. A year later, payment 99 contains interest 9 and principal 90; the supplied systematic depreciation amount of 81 exhausts the asset.

The event must therefore carry the actual surrendered asset and liability for each right, the revised payment schedule and the rate. The finite schedule uses annual-end payments and a checked one-year interval. Missing attribution, an unadmitted depreciation basis, a different calendar or a negative remaining asset requires a distinct finding or method. It cannot be repaired by applying the same physical fraction to every balance.

9.5.4 Contract costs survive more than one service period

IFRS 15.91–104 first distinguishes eligible contract costs from amounts governed by another standard. Acquisition costs must be incremental and recoverable. Fulfilment costs outside other standards must relate directly to the contract, create or enhance resources for future performance and be recoverable. Waste, past service and the specified administrative costs are expensed. The event records these findings before calculating the posting.

For example, expense waste 5 and capitalise eligible fulfilment cost 100 for five equal service units. Delivery of one unit amortises 20, leaving 80. After other applicable asset impairments have been recognised, credit-adjusted remaining consideration 70 less direct future cost 20 supports a contract asset of 50: recognise impairment 30. Delivery of another unit amortises $50/4 = 12.5$, leaving 37.5. Without the impairment, the unamortised balance would now be 60. If circumstances improve enough to support 60, reversal is limited to $60 - 37.5 = 22.5$; the final three units amortise 60.

That ceiling needs a separate no-impairment balance, not merely the present asset. Subsequent CGU assessment under IFRS 15.103 remains required after the component test. A successful component calculation does not establish that the wider CGU has no impairment.

The same history discipline applies to repeated revenue modifications. Under IFRS 15.21(a), eight remaining distinct units with consideration 80, followed by four additional units for 60 that do not form a separate contract, leave 12 units and consideration 140. Delivery of three recognises 35. Adding three more units for 30 then leaves 12 units and consideration 135, recognised on their delivery. Earlier revenue is not reallocated.

For a single integrated obligation, IFRS 15.21(b) instead requires the cumulative effect. A contract priced at 200 and 25% complete has revenue 50. Changing price to 260 and admitted progress to 20% changes cumulative revenue to 52, a catch-up of 2. Later 50% progress yields 130. A further modification to price 300 and progress 40% reduces cumulative revenue to 120, a catch-up of negative 10; completion ultimately brings it to 300. Arbitrary mixtures of distinct and integrated promises require their own allocation, not a selection of whichever result is convenient.

9.5.5 A tax benefit retains the origin of the underlying item

Acquisition-related recognition depends on information as well as dates. Under IAS 12.68 and IFRS 3.45–46, a newly recognised acquired deferred tax benefit from acquisition-date information within the open measurement period reduces goodwill, with any excess recognised in profit. A benefit of 20 against goodwill 5 therefore reduces goodwill by 5 and gives tax income 15. The measurement period cannot exceed one year; elapsed time alone does not establish that the new information concerns acquisition-date facts.

Other benefits retain their independently identified origin. In particular, IAS 12.67 excludes an

acquirer's own deferred tax benefit from acquisition accounting; it does not place every such benefit in profit. IAS 12.61A–63 still routes tax associated with OCI or direct-equity items to that origin. An acquirer-owned benefit of 20 with an OCI origin therefore adds DTA 20 and OCI 20, leaving goodwill and profit unchanged. Missing origin remains unresolved.

Mixed tax schedules retain one lot per entity, authority, temporary difference, recovery-period rate, recognition finding and origin. For a signed difference D_i, positive for deductible and negative for taxable, admitted rate r_i and supported deductible recognition fraction f_i, the signed deferred amount is $D_i r_i f_i$; taxable lots use $f_i = 1$. Changes go to the traced profit, OCI or equity account. A supplied schedule can thus move gross DTA from 30 to 19 and DTL from 20 to 30 while allocating expense 5 to profit, 10 to OCI and 6 to direct equity.

Gross asset and liability amounts stay available even when presentation is net. The finite offset route requires the same taxable entity and authority and an enforceable right to set off current tax under IAS 12.74. It does not implement the separate cross-entity simultaneous-settlement route. Neither a blended rate nor an assumed right of setoff can substitute for those findings.

9.5.6 Inspections, surrendered rights and changed employee benefits

A replacement inspection, a smaller leased area and a pension curtailment all remove something from an existing arrangement, but they remove different economic rights or obligations. The accounting must identify what survives before measuring the new arrangement. The following methods use the retained historical EU text and synthetic pretax amounts. Actual authority, contract and actuarial facts remain necessary for application.

Replacing the inspection component

IAS 16.14 (European Commission, 2023a, IAS 16.12–14,43–55) distinguishes a major inspection required for continued operation from the physical parts inspected. Capitalize its cost as a replacement only if the recognition criteria hold; derecognize the remaining previous inspection cost. That obligation persists even if the original purchase did not separately identify an inspection component. The estimated cost of a comparable future inspection may indicate the original component cost; it is not automatically today's replacement expenditure or an additional historical purchase.

The component record carries asset identity, inspection cohort, original cost allocation and evidence, availability date, residual value, consumption pattern, useful life until the next inspection, accumulated depreciation/impairment and the replacement event. Allocate the initial asset's existing total cost among its components; do not increase that total while recognizing the old inspection. Advance depreciation to the replacement date, remove the old carrying amount, add eligible new cost and begin the new component's depreciation when available. IAS 16.43–55 requires separate significant-component depreciation and retains depreciation during idle periods unless the applicable consumption method gives zero. Ordinary servicing is expense under IAS 16.12, not a new inspection asset. Other production-cost rules may place depreciation in inventory; these examples assume a non-production use, so charges enter profit.

Suppose original equipment cost 120 includes an inspection component 20. The remaining physical part costs 100, has ten-year straight-line life and zero residual; the inspection has four-year life and zero residual. After two full years the physical part carries 80 and the old inspection 10. A qualifying inspection performed early costs 24 and is available immediately for four years. Derecognize the old 10 as loss, capitalize 24, and retain physical carrying amount 80. The equipment now carries 104. One subsequent full year charges 10 physical and 6 inspection depreciation, leaving 88. Expensing all 24, retaining both inspection cohorts, or depreciating

104 uniformly over the physical life would compute different targets.

With opening cash 100 and equipment 90, equity 190, replacement leaves cash 76, equipment 104 and equity 180. Subsequent depreciation 16 leaves equity 164 and equipment 88. An otherwise separate routine service costing 3 reduces cash and profit by 3 and leaves equipment unchanged. An unavailable old-cost allocation does not justify writing off zero: reconstruct a supported component estimate or retain the unresolved derecognition amount. Revaluation/impairment histories must first update the component carrying amounts under their own methods.

Reducing a lease and paying usage-dependent rent

IFRS 16.44 (European Commission, 2023a, IFRS 16.38–46) first tests whether an added right and commensurate consideration make a separate lease. A scope reduction cannot meet that increase-in-scope test. For a modification that is not separate, paragraph 45 requires revised contract allocation, term and discount rate at the effective date. Paragraph 46(a) requires right-of-use derecognition and the gain/loss for a partial or full termination; the remaining liability remeasurement adjusts the surviving right-of-use asset. A mere decrease in price with unchanged scope belongs to paragraph 46(b), not automatic derecognition.

Represent the terminated rights explicitly, including the physical/use/time units and their carrying-cost allocation. In a homogeneous space lease with an unchanged remaining term and proportional rights, let fraction q of the rights be surrendered. The illustrative proportional allocation removes qA of the right-of-use carrying amount A and qL of the pre-modification liability L; the termination gain is $q(L - A)$. This allocation requires the homogeneous-rights facts. Do not use the payment reduction as q or apply floor-area proportions to heterogeneous equipment or a pure term change without a supported allocation.

For $A = 80$ and $L = 100$, surrender 25% of identical space with no termination fee. Remove right-of-use 20 and liability 25; recognize gain 5. The remaining $A = 60$ and $L = 75$ are then remeasured. Suppose the remaining contract requires one payment 70.4 after one year and the newly supported discount rate is 10%. Its present value is 64. Reduce the remaining liability by 11 and the asset by 11, giving asset 49 and liability 64. The complete journal debits liability 36, credits the asset 31 and credits gain 5. It is wrong to send the entire liability reduction 36 to income or to omit the termination gain by adjusting the asset 36.

With opening cash 120/ROU 80/liability 100/equity 100, the event leaves cash 120, ROU 49/liability 64/equity 105. Over the one remaining year, zero residual and straight-line use give depreciation 49, interest 6.4 and payment 70.4. Closing cash and equity are 49.6, with no lease balances. Present-value calculation and the right-of-use consumption pattern are independent assumptions.

A separate fee of 2 per actual unit of usage, not index/rate dependent, not in-substance fixed and not already in the liability, is recognized under IFRS 16.38(b) when the condition occurs. Three units create expense 6/payable 6; payment reduces cash/payable 6. It is not a lease-liability remeasurement. Qualifying production use can instead include that cost in another asset under the applicable standard. Missing usage evidence, uncertain payment character, heterogeneous surrendered rights or a termination penalty require their own measurement before the conditional example can be applied.

Curtailment, settlement and the asset ceiling

The benefits record separates obligation D, fair-valued plan assets P, asset- ceiling adjustment, recognized net liability/asset, actuarial assumptions, remaining-year service cost, discount rate,

legal discharge and actual payments. For a liability $D - P$ is positive. If there is a surplus, the recognized asset is limited to the admissible economic benefit C; the added ceiling adjustment is $\max(P-D-C, 0)$. This simple representation assumes no further minimum-funding adjustment; any applicable IFRIC 14 obligation remains a separate admission dependency. The example supplies C, not a method for inferring legal recoverability.

IAS 19.99 (European Commission, 2023a, IAS 19.99–112,120–126) requires current assumptions and asset fair values immediately before and after amendment, curtailment or settlement. First recognize pre-event remeasurements, then the change in benefits/covered employees, then any later settlement. Past service cost arises from an amendment or curtailment and is recognized at the earlier date specified by paragraph 103; a related restructuring or termination-benefit recognition can therefore precede the formal change. The examples have no such earlier trigger. A settlement discharges obligations; ordinary scheduled benefits included in assumptions are not a settlement gain event. Buying insurance while retaining an obligation is also not settlement under paragraphs 111–112.

Measure past service and settlement gain/loss without the asset ceiling, then remeasure the ceiling under paragraph 101A. Service cost and settlement enter profit unless eligible capitalization applies; actuarial and ceiling remeasurements enter OCI under paragraph 120 and are not later recycled under paragraph 122. For the rest of the year, use updated service assumptions and the post-event net position/discount rate, allowing for subsequent contributions and benefits, under paragraphs 122A–126. An unchanged beginning-of-year interest base would miss this required reset.

Consider opening $D = 1{,}000$, $P = 900$, net liability 100 and company cash 500/equity 400. At the event date, current measurement gives $D = 1{,}100$, $P = 920$: actuarial loss 100 less asset gain 20 creates OCI loss 80 and liability 180. A curtailment reduces the freshly measured obligation by 150 to 950. Recognize past-service gain 150, leaving net liability 30. A later transaction fully discharges a measured obligation 200 for plan assets 170 plus company cash 20. Settlement price 190 gives gain 10; D falls to 750, P to 750, liability to zero and cash to 480. Profit 160 and OCI−80 reconcile equity from 400 to 480.

For the remaining half-year, assume current service cost 30, annual discount rate 5%, simple half-year interest accrual and no contributions or benefit payments. Gross obligation and asset interest are both 18.75, so net interest is zero. Closing $D = 798.75$, $P = 768.75$, net liability 30, cash 480 and equity 450. Full period profit 130 and OCI−80 reconcile the change 50. Using the original net liability 100 for half-year interest would incorrectly add 2.5 expense. The simple accrual convention is an illustrative assumption, not a universal actuarial compounding rule.

To see the ceiling ordering, take D 100/P 130 and recoverable economic benefit 10. The surplus 30 yields recognized asset 10 and ceiling adjustment 20. An amendment reduces D by 20 to 80, while current evidence changes C to 15. Recognize past-service gain 20 without the ceiling. The new surplus 50 has ceiling adjustment 35, so its increase 15 is an OCI loss. The net recognized asset rises only 5 to 15. Recording just gain 5 would misclassify profit and OCI even though closing equity agrees.

Multi-employer information is a classification decision

IAS 19.32–36 (European Commission, 2023a, IAS 19.32–41) requires classification from actual plan terms, including constructive obligations. With adequate reliable allocation, a participant in a defined-benefit multi-employer plan recognizes its share of obligation, assets and cost. If the specific paragraph 34 information condition holds, account as if defined contribution and retain the paragraph 148 disclosures. Insufficient information is not proof that the plan economically has no defined-benefit exposure.

Paragraph 37 separately requires any contractual surplus/deficit-funding asset or liability and resulting profit effect. A synthetic agreed deficit payment 121 due in two years, with a supplied appropriate discount rate 10%, has liability 100 and expense 100 initially, followed by interest 10 and 11 and settlement 121. Ordinary current-service contributions are distinct. Do not omit the deficit contract because ongoing contributions use defined-contribution accounting or recognize the nominal 121 immediately while also accruing interest.

Withdrawal or wind-up liabilities follow IAS 37 under paragraph 39. A plan among entities under common control is not a multi-employer plan under paragraph 40; paragraph 41's plan-wide information and separate-entity charge/allocation rules must be applied instead. The record must therefore retain employer relationships, allocation reliability, contributions, deficit agreements, withdrawal terms and required disclosures. Unknown relationship or contractual terms block the classification, not the preservation of the underlying evidence.

9.5.7 Opening work carries both cost and completed effort

The earlier production example began with an empty factory account. An opening work-in-progress balance changes both the cost available and the work still needed. Materials may already be present even though conversion remains unfinished. Treating the opening units as entirely new would charge that effort twice; ignoring their costs would lose part of the inventory.

Consider homogeneous units with all materials added at the start and conversion effort incurred evenly. Ten opening units contain material cost 100 and conversion cost 50 and are halfway through conversion. The factory starts another 90 units, incurring material cost 900 and conversion cost 510. It completes 80 units and leaves 20 units halfway through conversion. Assume the opening units are completed first, no spoilage, no idle-capacity cost, and net realizable values above cost. Physical units reconcile as $10 + 90 = 80 + 20$; available cost is $100 + 50 + 900 + 510 = 1560$. These are supplied production facts, not estimates derived from a reported inventory total.

Under FIFO process costing, retain the opening 150 and allocate current cost only to current work. Completing the ten opening units requires five equivalent conversion units. Seventy units are started and completed, and the closing twenty require ten equivalent conversion units. Current conversion effort is therefore $5 + 70 + 10 = 85$, at $510/85 = 6$ per equivalent unit. Current material cost is $900/90 = 10$ per new unit. Completing opening work costs $150 + 5 \times 6 = 180$; the seventy new completed units cost $70 \times (10 + 6) = 1120$. Transfer 1300 from WIP to finished goods and retain $20 \times 10 + 10 \times 6 = 260$ in WIP. Completion is not a sale, so this transfer does not itself create cost of sales or a cash payment.

Weighted-average process costing pools the opening and current costs. Material cost remains 10 per equivalent unit, but conversion cost becomes $(50 + 510)/(80 + 10) = 56/9$. Completed goods then cost 11680/9 and closing WIP 2360/9, still totaling 1560. Posting at cents gives 1297.78 and 262.22 with a preserved rounding allocation. The methods distribute the same total cost differently because the opening conversion cost differs from the current-period rate. Neither method is selected merely to obtain a preferred closing inventory. IAS 2.23–27 requires specific identification where applicable and consistent FIFO or weighted-average treatment for the relevant interchangeable inventories; the equivalent-unit calculations are the local implementation of those cost policies (European Commission, 2023a, IAS 2.10–16, 23–27, 34).

For several stages, transfer the preceding stage's reconciled cost into the next stage as a separate cost component. Record when that component enters the process; it need not follow the completion percentage of later materials or conversion. Each stage retains opening units and costs, transfers in, completed transfers out, closing units, progress by component, inspection points, losses and recoveries. An impossible completion fraction, unexplained physical loss,

missing opening cost or inconsistent transfer amount blocks the cost allocation rather than becoming an unexplained unit-cost adjustment.

Abnormal loss cannot become a higher cost of the good units

Normal process loss and abnormal waste need different treatment. IAS 2.16(a) excludes abnormal wasted materials, labour and other production costs from inventory. Its normal-capacity rule also prevents idle fixed overhead from being hidden in a higher unit cost (European Commission, 2023a, IAS 2.12–16, 21). The normal-loss estimate must therefore be supported before observing the actual loss, with its inspection stage and recoveries specified.

In a separate fully processed batch, 100 input units cost 1000. Assume normal yield is 95 units, all lost units have consumed the same full effort at inspection, there is no salvage, and this cost pool contains no fixed overhead. The normal unit cost is $1000/95 = 200/19$. If actual good output is 90, five units of loss are normal and five abnormal. Good inventory receives $90 \times 200/19 = 18000/19$, while abnormal loss of $1000/19$ is expensed. At cents these are 947.37 and 52.63, which reconcile to 1000. Allocating all 1000 to the ninety good units would conceal the abnormal loss. Earlier inspection, partial conversion, recoverable scrap or a different supported normal yield changes the calculation; those facts cannot be supplied by the final good-unit count.

9.5.8 A pension surplus is useful only to the extent it is available

The preceding pension examples supplied a recoverable economic benefit. IFRIC 14 specifies how the plan's terms, statutory requirements and funding basis constrain that benefit. Separate an unconditional right to a refund from the ability to reduce future contributions, and separate funding for past service from funding for future service. An IAS 19 surplus can coexist with a shortfall measured on the funding basis (European Commission, 2023a, IAS 19.64; IFRIC 14.7–24).

Start with plan assets 130 and defined-benefit obligation 100, giving surplus 30. Suppose the employer has an unconditional right to the surplus but obtaining it costs the plan 5. The available refund is 25 and the recognized asset is 25, with a ceiling adjustment of 5. Costs depend on the permitted refund route, including settlement costs if winding up the plan is necessary. A right conditional on an uncertain event outside the employer's complete control does not establish this refund benefit. For a refund defined as the whole surplus or a proportion of it, IFRIC 14.15 makes no additional time-value adjustment to that refund (European Commission, 2023a, IFRIC 14.11–15).

Contribution reductions require a different calculation. Use employer service costs, excluding employee-funded amounts, over the shorter of expected plan life and entity life. Assumptions must be consistent with the benefit obligation and the situation at the reporting date. In a two-period example with zero discounting solely for illustration, future employer service costs are 20 and 20. Without a prepayment, required future-service funding would be 12 and 25. Their net difference is $8 - 5 = 3$. Suppose a prior prepayment separately reduces future required contributions by 7. The available reduction benefit is then

$$7 + \max\{0, (20 - 12) + (20 - 25)\} = 10.$$

The floor applies to the combined paragraph 20(b) amount, not separately to each period: ignoring the second period's excess would incorrectly produce 15. With no minimum funding requirement, use the available future employer service costs under paragraph 16 instead (European Commission, 2023a, IFRIC 14.16–22).

The refund and contribution routes may be mutually exclusive. In that case the available benefit is the greater of 25 and 10, not their sum 35; the recognized asset is also capped by the actual surplus. If the refund right fails but the contribution route remains supported, the asset is 10 and the ceiling adjustment is 20. Compatible combinations require their actual feasible terms and cannot consume the same surplus twice.

Past-service funding can require a liability before any payment. Suppose plan assets are 120 and the IAS 19 obligation is 100, but none of the surplus can ever be refunded or reduce contributions. The recognized pension asset is zero. A separate funding-basis assessment now requires a past-service contribution of 10 that will also be wholly unavailable after payment. IFRIC 14.23–24 requires the resulting liability when that obligation arises. With no intervening interest or other pension changes, recognize an OCI ceiling loss of 10 and a net pension liability of 10. On payment, debit that liability and credit cash 10. Plan assets become 130, but the recognized asset remains zero and payment causes no second loss or gain. Starting company cash and equity of 100 become cash 90 and equity 90. The obligation's recognition and the payment are distinct events (European Commission, 2023a, IFRIC 14.23–24; IAS 19.120(c), 126–127).

9.5.9 Restoration cash follows the activity being paid for

Return to the exhausted asset whose restoration liability is settled for 36.3. To complete its cash-flow example, specify the payment facts: the company pays wages, materials and contractors to remove the exhausted asset and restore the site; it acquires no new long-term asset and repays no borrowing. The proposed classification for this conditional case is an operating outflow of 36.3. This follows the activity definitions and supplier/employee examples in IAS 7.6 and 14. The earlier noncash creation of an asset and provision is not a cash outflow, and unwinding the provision does not by itself establish that part of the eventual contractor payment is interest paid (European Commission, 2023a, IAS 7.6, 12–16, 31–33, 43–44; IFRIC 1.8).

The competing interpretation classifies the payment as investing because the provision's initial measurement entered the asset's cost. The July 2012 IFRS Interpretations Committee staff paper examines comparable rehabilitation facts, presents both approaches and favors operating classification. That is support for the proposed reasoning, not a binding Interpretation or proof of universal treatment (IFRS Foundation staff, 2012, Appendix A, A35–A43; status notice on p. 1). The design records the chosen interpretation and its facts so that a different admitted rule can change the category without rewriting the legal journal or total cash.

Under the proposed branch, opening cash 100 plus operating cash flow minus 36.3, with no investing or financing flow, gives closing cash 63.7. In year 2 the indirect reconciliation is loss 74.3, plus noncash depreciation 71 and unwinding 3.3, less provision settlement 36.3, giving operating outflow 36.3. Under the competing investing branch, operating cash flow is zero and investing outflow 36.3; closing cash is identical. The classification difference is economically visible even though both cash bridges balance.

A new productive asset, actual borrowing repayment, separately evidenced interest payment, or restoration caused by current inventory production requires its own assessment. The proposed operating branch completes the specified exhausted-asset example; it is not a mandatory default for every restoration liability or evidence of an actual issuer's compliance. All three additions require explicit successor records and exchanges. They preserve the closed `CorporateJointHistory/1` profile rather than silently enlarging its accounts or event vocabulary.

9.5.10 Impairment across a group and the origin of tax

The following methods use the retained historical EU 2023/1803 text and the separately retained November 2023 IAS 12 amendment. Each example is conditional on its declared authority, perimeter, allocation and tax assumptions. These texts do not establish current issuer compliance or cross-GAAP equivalence.

Testing assets that serve more than one business

IAS 36.100–102 (European Commission, 2023a, IAS 36.100–108) identifies corporate assets by their contribution to multiple cash-generating units without independent cash inflows. Do not invent a separate recoverable amount for a headquarters just to make every row independently testable. When a reasonable consistent allocation to a unit exists, include that share with the unit. When it does not, first test the individual unit excluding the corporate asset, recognize that loss, then test the smallest larger group to which an allocation is reasonable and consistent. The second test uses the already adjusted carrying amounts.

The impairment record retains the hierarchy of units, legal/group/perimeter identities, corporate asset links, allocation basis, carrying and recoverable amount dates, goodwill allocation, individual floors and prior loss origins. The calculation is postorder: individual tests precede wider group tests. IAS 36.104 applies loss first to goodwill, then pro rata to other asset carrying amounts. Under paragraph 105, no asset falls below its measurable fair value less costs of disposal, determinable value in use, or zero, whichever is highest. Allocate a capped asset's unabsorbed loss again over the eligible remainder. Do not create a balancing liability for an unallocated remainder unless another standard actually requires it. Revised depreciation follows the adjusted amounts.

For a conditional example, unit A has two inseparable assets of 40 each, unit B two of 60 each, and headquarters 40. Headquarters cannot be reasonably allocated to either unit alone, but it serves their combined group. Unit A's recoverable amount 70 first causes loss 10, allocated 5 to each A asset. B's recoverable amount 130 causes no loss. The group's adjusted carrying amount is then $70 + 120 + 40 = 230$. Given group recoverable amount 207, additional loss 23 allocates 3.5 to each A asset, 6 to each B asset and 4 to headquarters. Closing amounts 31.5, 31.5, 54, 54, 36 sum to 207. No individual asset's value in use or fair value less disposal costs is determinable in this example, so only the zero floor binds. The unit-level recoverable amount 70 is not an individual A-asset floor.

Using the original group carrying amount 240 in the second test would repeat the first loss 10. Treating headquarters as costless would omit its exposure. These are different errors even if both versions produce balanced journals. The calculation supplies an allocation method, not a valuation method for the assumed recoverable amounts.

Partial goodwill and the non-controlling interest

IAS 36 Appendix C (European Commission, 2023a, IAS 36.C1–C9) distinguishes the consolidated carrying amount from the amount used for an impairment test. When acquisition-date NCI was measured as its proportionate share of identifiable net assets, the reported goodwill omits NCI goodwill, while the unit's recoverable amount includes its contribution. C4 therefore requires a notional gross-up before the test. C8 prevents recognizing an impairment of goodwill that was never recognized. Neither the notional goodwill nor its notional impairment is posted as an asset or expense.

Retain the acquisition's NCI measurement election, ownership and profit-allocation history, unit

allocation and evidence for the notional amount. Division of parent goodwill by ownership is a conditional illustration when proportional goodwill is justified; it must not silently assume away a control premium or differing synergies. For a subsidiary that is itself a unit, C6 allocates loss on the same basis as profit. C7 separately allocates a larger unit's goodwill losses by the parts' relative goodwill and identifiable-asset losses by their relative net identifiable carrying amounts, then allocates within each part. Preserve the NCI-bearing and parent-only parts rather than applying one ownership fraction to the entire group.

Assume the parent owns 80%, acquired proportionate-share NCI, has recognized goodwill 40 and supports proportional notional goodwill 50. Identifiable unit assets carry 200. If recoverable amount 220, test 250 against 220 and allocate loss 30 entirely to goodwill. The recognized goodwill loss is 24; the NCI notional loss 6 is not recorded. Reported goodwill falls from 40 to 16 and reported assets total 216; adding the remaining notional NCI goodwill 4 gives test carrying amount 220. Demanding the reported amount equal 220 would improperly capitalize that 4.

If recoverable amount instead falls to 180 in a separate branch, test loss 70 removes notional goodwill 50 then identifiable assets 20. Recognized loss is 60: parent goodwill 40 plus asset loss 20. Of the recognized loss, 56 belongs to the parent and 4 to NCI. The unrecognized NCI-goodwill loss 10 never enters either profit or NCI equity. Closing identifiable assets 180 and goodwill 0 reconcile the reported decrease from 240 to 180. Goodwill impairment cannot subsequently reverse under IAS 36.124; a rising valuation is not a new recognized goodwill asset.

A larger-unit branch adds a wholly owned part with goodwill 20 and identifiable assets 100. Total test carrying amount 370 includes subsidiary notional goodwill 50 and parent-only goodwill 20. At recoverable amount 270, loss 100 first removes all goodwill 70, then allocates 30 to identifiable assets in their $200 : 100$ ratio. Recognized loss 90 comprises goodwill 60 and assets 30; attribution is parent 86 and NCI 4. The notional 10 remains unrecognized. Asset floors and allocation of remaining goodwill must be respected when a branch does not exhaust goodwill.

Revalued impairment and its related deferred tax

IAS 36.60–64 (European Commission, 2023a, IAS 36.60–64,117–124; IAS 12.61A–65) treats a revalued asset's impairment as a revaluation decrease: charge the same asset's existing surplus through OCI before charging profit. Afterward, determine deferred tax from the revised carrying amount and tax base. IAS 12.61A attaches current and deferred tax to the related item even when the item was recognized in a different period. Preserve the origin of each temporary difference so a later tax-rate change does not move all tax to profit by default. Where exact attribution is exceptionally impracticable, paragraph 63 permits a reasonable allocation or a more appropriate method; document that circumstance.

Take an asset carrying 140, tax base 100, gross same-asset revaluation surplus 40 and related deferred tax liability 10 at a supplied 25% rate. There is no passage of time or tax-base movement. Recoverable amount 90 causes loss 50: OCI 40 and profit 10. The old DTL 10 reverses in OCI. A deductible difference 10 gives DTA 2.5 in profit only if recovery is probable; that condition is supplied here. Net OCI loss 30 plus net profit loss 7.5 equals the net-asset decrease 37.5, from $140 - 10 = 130$ to $90 + 2.5 = 92.5$. Offset presentation is not assumed: DTA and DTL eligibility must follow the applicable jurisdictional and legal-offset rules.

In the same no-time-passage illustration, recoverable amount later recovers to 130. IAS 36.117–120 limits reversal to the no-impairment amount and reverses the prior profit loss 10 first, with the remaining 30 through OCI. Reversing DTA 2.5 charges profit; establishing DTL 7.5 charges OCI. Net profit gain 7.5 and OCI gain 22.5 sum to 30, the increase from 92.5 to 122.5. A revaluation increase beyond the no-impairment amount is a separate IAS 16 event, not

unlimited reversal; future depreciation must update both the reversal cap and carrying amount.

Separately, if the original carrying 140/tax base 100 asset's admitted future tax rate changes from 25% to 30% before any impairment, DTL rises from 10 to 12. The additional tax 2 relates to the earlier revaluation and is charged to OCI, not profit. The example assumes enacted/substantively enacted rates and the same manner of recovery; it does not infer tax law or settlement method from the accounting carrying value.

Acquisition and investment tax are separate questions

IAS 12.66 (European Commission, 2023a, IAS 12.39–51,66–68) requires acquisition-date temporary differences on identifiable assets and liabilities to enter acquisition accounting, subject to recognition rules. For a simple full acquisition paying 150, identifiable fair-valued assets 150 with tax base 100 and an admitted 25% rate create DTL 12.5. Net identifiable assets 137.5 give goodwill 12.5. There is no recursive DTL on the initial recognition of that goodwill under paragraph 15(a). This exception is not permission to omit the DTL on the identifiable asset. The treatment of an acquirer's own previously unrecognized DTA, and acquiree benefits recognized later, follows paragraphs 67–68; the former does not automatically change acquisition goodwill.

Outside-basis differences on investments are tracked separately from temporary differences inside the subsidiary's asset book. Under IAS 12.39, a taxable difference avoids DTL only to the extent the investor controls reversal timing and reversal is probably absent in the foreseeable future. Both conditions are required. For a taxable difference 60 and supplied 20% tax on the expected reversal, DTL is 12 when the exception does not hold; it is not recognized where both conditions demonstrably hold. Unknown control or distribution intent is unresolved, not evidence for the exception. An associate investor usually cannot infer control from a minority shareholding. Under paragraph 44, a deductible difference requires probable foreseeable reversal and available taxable profits to recognize a DTA. The source gives asymmetric tests; copying the liability exception to assets is wrong. Rates and recovery paths follow paragraphs 46–51.

Pillar Two is not another deferred-tax rate

The original September 2023 compilation predates the specific amendment adopted by Commission Regulation (EU) 2023/2468 (European Commission, 2023b, arts. 2–3; annex 4A,88A–88D,98M) of 8 November 2023, published 9 November 2023. Its Article 3 enters into force the day after publication. Article 2 and the annex state the application and retrospective rules. The retained amendment adds IAS 12.4A, 88A–88D and 98M. For legislation within paragraph 4A, an entity neither recognizes nor discloses information about related deferred-tax assets/liabilities, and discloses that it applied this exception. It separately discloses current Pillar Two tax expense/income. Enacted or substantively enacted legislation not yet effective requires known or reasonably estimable exposure information; unavailable information requires the specified explanation and assessment progress.

The record therefore separates ordinary current/deferred tax from Pillar Two current tax, exception status, applicability authority and exposure disclosures. A supplied in-scope current top-up tax 7 creates current expense/payable 7 and later payment 7. A hypothetical Pillar Two temporary difference 40 does not create a DTL 6 by multiplying by 15%. That is the wrong target under the exception. Ordinary IAS 12 temporary differences continue under their own rules. The current top-up amount 7 is an illustrative input, not a calculation of an actual GloBE liability. Jurisdictional enactment, group thresholds, covered taxes, allocation, safe harbours and current-law measurement remain separate required facts/methods before actual application.

The inspected amendment supplies the accounting boundary; it does not itself calculate tax due for a company.

9.5.11 Inventory and assets retain their cost histories

Inventory uses identified cost for non-interchangeable items or the admitted cost formula for interchangeable items. AASB 102.9, 23, 25 and 28–34 support lower of cost/net realizable value, relevant cost formulas and write-down/reversal conditions. A method never chooses LIFO merely to fit a reported balance. In a second lot example, purchases are 10 units at 2 and 10 at 3, followed by sale of 12 units. FIFO gives cost 26 and stock 8 units/cost 24. Weighted average gives cost 30 and stock 20. Under the same given selling price 48, margin differs 22 versus 18. Both alternatives require their own permitted policy, consistent application and evidence; matching total cash does not remove the difference. NRV 16 for the remaining FIFO lot records expense 8, inventory allowance 8; recovery to NRV 22 reverses 6, leaving allowance 2. Recovery to 30 is capped at original cost 24. US LIFO remains a separately sourced question.

The retained AASB 116.7/15, 29–30 and 48–62 support cost and depreciation. Asset 100, residual 10, three-year even use gives 30 per full year. A year-two estimate change after first-year depreciation to residual 10 and three remaining years gives $(70-10)/3=20$ prospectively; an error in original cost is a correction, not an estimate change. Component replacements derecognize the replaced component's remaining amount before recognizing its replacement; no perpetual double asset. Borrowing-cost capitalization, impairment, provisions, employee benefits and share-based payment are specified as additional scoped methods below, not silently assigned zero in real-company cases.

9.5.12 Preparing equipment and selling its first samples

Preparing a machine can produce two things of value: equipment capable of operating as intended, and goods that can already be sold. Their costs and receipts answer different accounting questions. Necessary delivery, installation and functional testing contribute to the machine's cost under AASB 116.16–17. Paragraph 20A requires the proceeds from selling items produced during preparation, and those items' costs, to enter profit under the applicable standards. It measures the goods' costs using AASB 102 (Australian Accounting Standards Board, 2022b, 2021a). The samples' margin therefore does not reduce the cost of the machine.

Consider a synthetic Australian manufacturer in calendar 2024, starting with cash and contributed capital of 300. It pays 100 for equipment, 5 for its initial delivery, 12 for necessary installation and 3 for an engineering service that tests and calibrates its physical performance. Probable benefits and reliable cost measurement are admitted. These four amounts give PPE cost of 120. The engineering service uses different resources from those recorded as sample production below. Each receipt or resource has an identity and an assigned purpose; sharing a resource would require an evidenced allocation, not two recognitions of its full cost.

The sole sample batch uses normal materials costing 4 and directly attributable contracted conversion costing 2. The example has no other production overheads. Its goods cost is thus 6 under AASB 102.10–15. In a real factory, paragraph 13 requires fixed production overheads to be allocated using normal capacity; low output does not justify loading every unused resource onto a few units. Unknown overheads would leave the goods' cost unresolved here. Separate abnormal production waste of 1 is expensed under paragraph 16(a). Staff training of 3 and advertising of 2 are also period expenses, excluded from PPE by AASB 116.19. All these amounts are paid in cash.

On 30 June, the manufacturer sells the entire batch for cash of 8. These are ordinary manufac-

tured goods, outside the inventory standard's agricultural, financial-instrument and broker-trader exceptions. Net realisable value is 8, with no further completion or selling costs, so cost 6 needs no write-down. The approved, enforceable customer contract identifies the goods and price, has commercial substance and probable collection. Delivery, acceptance, title and control transfer together on that date. There are no remaining promises, returns or price concessions in this case. These facts admit the revenue treatment under AASB 15.9 and 31–38; cash receipt alone would not establish it (Australian Accounting Standards Board, 2022d).

The entries preserve both assets. Acquisition and preparation debit PPE 120 and credit cash 120. Sample production debits inventory 6 and credits cash 6; the excluded costs debit period expense 6 and credit cash 6. At sale, debit cash 8 and credit revenue 8, then debit cost of sales 6 and credit inventory 6. AASB 102.34 links the goods' expense to recognition of the related revenue. The batch's cost history remains available even though its closing stock is zero. Before depreciation, cash is 176, PPE is 120 and the loss is 4.

The machine becomes capable of its intended operation on 1 July but remains idle through 31 December. No further production occurs in this example. Management's admitted estimate is a five-year useful life, zero residual value and uniform consumption of benefits, including obsolescence while idle. The straight-line method uses equal monthly charges. Paragraph 55 starts depreciation when the asset is available for use and does not suspend this charge during idleness. It also allows a usage-based method to have zero depreciation when production is zero; that is a different admitted consumption pattern. Here the six available months give

$$\text{depreciation} = \frac{120 - 0}{60} \times 6 = 12, \qquad \text{closing PPE} = 120 - 12 = 108.$$

Debit depreciation expense 12 and credit accumulated depreciation 12. Profit is $8 - 6 - 6 - 12 = -16$, so equity closes at $300 - 16 = 284$. Cash 176 plus net PPE 108 gives the same total. The initial capital is already in the opening balance; it is not a financing receipt in this year.

Cash-flow presentation follows the purchases and sale. PPE payments of 120 are investing outflows under AASB 107.16(a). The sample receipt of 8, goods payments of 6 and period-cost payments of 6 give operating cash flow of $8 - 6 - 6 = -4$; profit of -16 plus noncash depreciation of 12 gives the same result under the indirect method. Paragraphs 14 and 18–20 support this classification and reconciliation (Australian Accounting Standards Board, 2023a). Closing cash is $300 - 120 - 4 = 176$.

Several wrong treatments would still balance. Netting the samples' cost and proceeds into PPE gives $120 + 6 - 8 = 118$, depreciation 11.8, net PPE 106.2 and a loss of 17.8; cash remains 176. It also shifts the sample margin from operating cash flow into investing cash flow. Waiting for productive use would omit depreciation 12 and overstate both PPE and equity by 12. Capitalising the excluded costs would instead produce cost 126 and closing PPE 113.4. A balanced trial balance cannot distinguish these errors from the required treatment. The method needs the cost purpose, goods population, sale conditions, consumption pattern and availability date.

The four retained compilations have an overlapping application period that includes this calendar 2024 example: AASB 102 compilation 3, AASB 116 compilation 6, AASB 15 compilation 8 and AASB 107 compilation 5. The example assumes no early adoption and isolates the pretax cost and sale mechanism; taxes, financing, grants, restoration obligations and later impairment require their own facts. With initial cost established, replacing part of the machine must next remove the cost history of the part that has gone.

9.5.13 Replacing a component removes an old carrying amount

An asset can remain in use while one of its components is replaced. The company then owns a new component and has given up the remaining benefits of the old one. Recording only the new purchase would preserve an asset that no longer exists. The register therefore needs component identity, original cost, accumulated depreciation, remaining life and the dates of replacement and use. Paragraphs 13 and 70 of the retained AASB 116 require the replaced part's carrying amount to be removed when an eligible replacement is recognised (Australian Accounting Standards Board, 2022b).

Consider a synthetic company with cash and contributed capital of 200 on 1 January 2024. It pays 100 for an administrative asset: a shell costing 80 with an eight-year useful life, and a significant component costing 20 with a two-year life. Both are available immediately, have zero residual values and consume benefits evenly. The company admits the cost model. These are authored facts, including recognition eligibility and the absence of any additional impairment; real assets require evidence for those assessments. Administrative use makes depreciation an expense. A production asset might instead contribute to inventory cost under paragraph 49.

Acquisition debits the two asset costs by 80 and 20 and credits cash by 100. Paragraphs 43–44 require separate depreciation of significant components; the full year's charge is $80/8+20/2 = 20$. Debit depreciation expense 20 and credit each component's accumulated depreciation by 10. At 31 December 2024, cash is 100, net PPE is 80 and equity is 180 after the loss of 20. No revenue or other operating activity is included in this asset-only example.

On 1 January 2025 the company replaces the old component. Its remaining carrying amount is $20 - 10 = 10$. Debit accumulated depreciation 10 and disposal loss 10, and credit the old component's cost 20. The replacement costs 30, is paid in cash and is available immediately for three years of even use, again with zero residual. Its entry debits new PPE 30 and credits cash 30. Cash is now 70 and PPE is $70 + 30 = 100$; equity is 170. Retaining the old component would instead report PPE of 110 and miss the loss of 10, although both acquisition entries would balance.

The requirement to remove the old part also applies when it was not depreciated separately. Paragraph 70 permits replacement cost as an indication of the old part's cost at acquisition if its carrying amount cannot practicably be determined. That does not make today's invoice the old part's remaining value. Acquisition-date cost and consumption would still need a supported estimate. Ordinary servicing has a different consequence: paragraph 12 expenses it as incurred. A separately assumed maintenance payment of 4 would debit maintenance expense and credit cash by 4. Capitalising it would balance but overstate both PPE and profit by 4 under those facts.

During 2025 the shell and new component each depreciate by 10. Immediately before sale on 31 December, gross active cost is $80 + 30 = 110$, accumulated depreciation is $20 + 10 = 30$, and net PPE is 80. Cash remains 70 and equity is 150. The buyer then obtains control and pays 90, with no selling costs or sale-and-leaseback arrangement. Remove gross cost 110 and accumulated depreciation 30, debit cash 90, and credit disposal gain 10. Paragraphs 67–72 locate the disposal at control transfer and measure the gain as net proceeds less carrying amount. The gain is not ordinary revenue; routine sales of formerly rented assets have a separate rule at 68A.

Pretax amount	2024	2025 including sale
Depreciation expense	20	20
Replacement loss	0	10
Disposal gain	0	10
Profit	−20	−20
Closing cash / net PPE	100 / 80	160 / 0
Closing equity	180	160
Operating cash	0	0
Investing cash	−100	60

The cash-flow statement separates the 2025 purchase of 30 from the sale proceeds of 90. Under the retained AASB 107.16(a)–(b), both are investing flows. The indirect operating reconciliation removes depreciation and the non-operating loss and gain from profit:

$$\mathrm{CFO}_{2024} = -20 + 20 = 0, \qquad \mathrm{CFO}_{2025} = -20 + 20 + 10 - 10 = 0.$$

Lifetime investing cash is $-100 - 30 + 90 = -40$, equal to the equity reduction once no assets remain. An ordinary servicing payment would instead be operating cash; paragraph 16 does not permit investing classification merely because an expense relates to equipment (Australian Accounting Standards Board, 2023a).

These cash-flow comparisons concern calendar 2024 and 2025. The retained AASB 107 compilation is dated 31 December 2023 and its official application record covers annual starts from 1 January 2024 to before 1 January 2026. The retained AASB 116 compilation is dated 31 December 2022 and covers annual starts from 1 January 2023 to before 1 January 2027. Their different intervals must remain attached to the respective rules. Tax has not been computed in these pretax statements; no tax-law or complete reporting claim follows.

9.5.14 Revaluation needs each asset's past gains and losses

Revaluation changes the amounts to be depreciated and can change profit and other comprehensive income (OCI) differently. A class total alone cannot determine the split. For the Australian for-profit case here, paragraphs 39–40 refer to the earlier decreases and surplus of the same asset. The Australian not-for-profit exceptions permit different class-level treatment; they cannot be imported into this case (Australian Accounting Standards Board, 2022b).

A second synthetic company owns two administrative machines, A and B, which form the entire class for which it elects the revaluation model. They became available on 1 January 2023 at costs 100 and 40, with useful lives of five and four years, zero residuals and even use. Cash after their acquisition was 60, funded by historical capital of 200. Every fair value used below is an explicitly assumed reliable measurement. Paragraphs 29, 31 and 36–38 require the class policy, sufficiently regular measurements and coverage of the whole class; they do not infer missing fair values from the ledger. No additional impairment is assumed after the separate assessment required by paragraph 63.

The 2023 history establishes the opening for the 2024 statement comparison. After depreciation of 20 and 10, A and B carry 80 and 30. At 31 December 2023 their assumed fair values are 120 and 24. A increases by 40, recorded in OCI and accumulated in its revaluation surplus. B decreases by 6; with no surplus of its own, that decrease is expense in profit or loss. The net class increase of 34 therefore cannot all be OCI. Such an entry would omit B's expense of 6 and understate OCI by the same amount while leaving total equity unchanged.

For clarity, this company chooses paragraph 35(b)'s treatment of accumulated depreciation. At each revaluation it first debits accumulated depreciation and credits gross PPE by the

accumulated amount, leaving the net amount as the adjusted gross balance. It then posts the revaluation difference. The prior gross cost, accumulated charge and original-cost history remain in the asset register. This choice changes neither the net revaluation nor its allocation between profit and OCI.

The opening 2024 balances are consequently cash 60, PPE $120 + 24 = 144$, capital 200, retained earnings -36 from depreciation and B's loss, and surplus 40. Assets and equity both equal 204. The register also remembers B's unreversed profit-or-loss decrease of 6. This prior-year derivation uses the retained AASB 116 scope; it does not assert a 2023 cash-flow presentation under the later AASB 107 compilation.

Let L_i denote an asset's outstanding earlier revaluation decrease in profit or loss, and S_i its remaining surplus credit. For an increase $u_i > 0$, the part reversing that earlier loss goes to profit; the rest goes to OCI. For a decrease $d_i > 0$, existing surplus absorbs the reduction through OCI before any excess reaches profit. The operative amounts are

$$\text{profit gain}_i = \min(u_i, L_i), \quad \text{OCI increase}_i = u_i - \min(u_i, L_i),$$
$$\text{OCI decrease}_i = \min(d_i, S_i), \qquad \text{profit loss}_i = d_i - \min(d_i, S_i).$$

An increase reduces L_i by its profit gain and adds its OCI increase to S_i. A decrease reduces S_i by its OCI decrease and adds its profit loss to L_i. These are revaluation-history rules; the separate impairment reversal ceiling is considered in the impairment method. No surplus is transferred during use in this example. If that permitted transfer were elected, the later surplus available to absorb a decrease would be smaller.

For 2024, the revalued amounts are depreciated over the remaining four and three years. The charges are $120/4 = 30$ and $24/3 = 8$, leaving carrying amounts 90 and 16. The company then revalues the whole class to 45 and 22. A falls by 45: its surplus absorbs 40 through OCI and the remaining 5 is a profit-or-loss expense. B rises by 6, reversing its own earlier loss through profit. After eliminating accumulated depreciation, gross PPE is 45 and 22, with surplus zero. Profit is $-38 - 5 + 6 = -37$ and OCI is -40. Cash 60 plus PPE 67 equals closing equity 127. A now retains a prior revaluation loss of 5; B retains none.

In 2025, the remaining three-year life of A and two-year life of B give depreciation $45/3 = 15$ and $22/2 = 11$. Carrying amounts become 30 and 11. At 31 December their assumed fair values are 50 and 11. A's increase of 20 first reverses its loss of 5 through profit, then adds 15 to OCI and surplus. B's valuation is included even though its monetary adjustment is zero. Before sale, cash 60 and PPE 61 equal equity 121.

Immediately after that measurement, the buyer obtains both machines and pays their same-date values, 50 for A and 11 for B. Debit cash 61 and remove PPE 61; the disposal gain is zero. The company elects the transfer permitted by paragraph 41: debit A's surplus 15 and credit retained earnings 15 directly. This transfer creates no income, new OCI or cash. Routing it through profit would falsely raise 2025 profit from -21 to -6.

Pretax amount	2024	2025 including sale
Depreciation	38	26
Revaluation loss / gain in profit	5 / 6	0 / 5
Disposal gain	0	0
Profit / OCI	-37 / -40	-21 / 15
Closing cash / net PPE	60 / 67	121 / 0
Closing retained earnings / surplus	-73 / 0	-79 / 0
Closing equity	127	121
Operating cash	0	0
Investing cash	0	61

The 2024 equity movement is $204 - 37 - 40 = 127$. In 2025, profit of -21 and OCI of 15 leave total equity at 121. Retained earnings move from -73 to -79: the year's loss reduces them by 21 and the direct transfer increases them by 15. That internal transfer does not change total equity. Operating cash follows independently from the year's profit and noncash effects:

$$\mathrm{CFO}_{2024} = -37 + 38 + 5 - 6 = 0, \qquad \mathrm{CFO}_{2025} = -21 + 26 - 5 = 0.$$

The entire 61 sale proceeds appear in investing cash. Tax consequences still need legal tax bases, applicable rates and the profit/OCI origin of each difference under AASB 112, as AASB 116.42 requires.

AccDSL must therefore preserve the class and its members, each valuation's date and evidence, original cost and accumulated charges, individual surplus and loss history, component exits and the chosen surplus-transfer policy. An absent valuation leaves that class remeasurement unresolved; it is not permission to select only appreciating assets. A missing disposal date or old component value likewise prevents the affected calculation. The journals and these economic changes belong to the same authorised transition. The zero adjustment for B illustrates why a valuation record may change even when no monetary line is needed. Together with the state-only advance-allocation rule above, this preserves the information needed by the next depreciation, revaluation or disposal.

9.5.15 A shorter useful life can change two different histories

A carrying amount does not reveal whether the estimate behind it was sound. Suppose an administrative asset costs 120, becomes available for use on 1 January 2023 and has no residual value. The company records straight-line depreciation of 10 in each of 2023 and 2024, leaving PPE of 100. On 1 April 2025, management concludes that the asset will be retired on 31 December 2028. That conclusion alone cannot tell us whether to change future expense or correct the earlier accounts. We need the information that supported the old estimate, including what was available when those accounts were authorized.

Consider two explicitly synthetic histories. In the first, twelve years was a reasonable useful life at acquisition and at both year-end reviews. A newly enacted operating restriction on 1 April 2025 now limits use to the end of 2028. That restriction neither existed nor was anticipated at the earlier dates. In the second history, the reliable, approved schedule at acquisition already specified six years, but a transcription error entered twelve. The original schedule was available and could reasonably have been obtained and considered at each authorization. Six years remains appropriate; the new discovery exposes the old mistake. These histories differ in facts, not in an accounting election available on the same known evidence.

AASB 108.5 makes that availability condition part of the definition of a prior-period error. Paragraphs 32–34A distinguish new information and developments from correction of an error; paragraphs 36–38 apply estimate changes to the current and relevant future periods (Australian Accounting Standards Board, 2022a). AASB 116.50–51 and 60–62 require systematic allocation, review of useful life and a method reflecting consumption (Australian Accounting Standards Board, 2022b). Here there is one component, even consumption, no capitalization into another asset, and an independently satisfied recoverability assessment. The shorter life does not replace that assessment.

Dates determine which accounts are being corrected. The 2024 statements were authorized on 31 March 2025; the discovery occurs the following day. The 2023 statements were authorized on 31 March 2024. No 2025 interim statements have been issued. Thus the error affects two authorized annual reports and an unissued current quarter. Moving discovery before the 2024 authorization would change the treatment of that report under paragraph 41. The example concerns newly

arising April conditions in the estimate history; it makes no separate determination about events revealing conditions that existed at the earlier reporting date.

The historical method comparison uses the retained December 2022 AASB 108 and 116 compilations and September 2023 AASB 112 compilation. It does not establish a later amendment chain or actual-company applicability. Assume a private, for-profit Tier 1 entity with no publicly traded ordinary or potential ordinary shares, no filing for public issuance and no voluntary EPS presentation. Those facts place this illustration outside the inspected AASB 133.2–3 scope (Australian Accounting Standards Board, 2021b). A public or voluntarily presenting entity needs the EPS restatement as well; the financing example's share schedule cannot simply be borrowed for this company.

New information changes the remaining allocation

Use equal monthly depreciation. In the estimate history the first three months of 2025 still consume $10 \times 3/12 = 2.5$, leaving 97.5 on 1 April. There are 45 months from April 2025 through December 2028. The remaining carrying amount, rather than original cost or a reconstructed earlier estimate, is allocated across those months:

$$d_{\text{Apr–Dec 2025}} = \frac{97.5}{45} \times 9 = 19.5, \qquad d_{2025} = 2.5 + 19.5 = 22.$$

Year-end PPE is $100 - 22 = 78$. Each full year from 2026 through 2028 has depreciation $97.5 \times 12/45 = 26$; the future allocations $19.5 + 26 + 26 + 26 = 97.5$ exhaust the April balance. The reported 2023 and 2024 depreciation remains 10 in each year. Revising those amounts to 20 would turn later information into a fact supposedly known earlier.

The estimate disclosure states the new restriction, the revised remaining life and the affected amounts under AASB 108.39–40. Relative to continuing the old schedule, 2025 depreciation increases by 12. Depreciation in each of 2026–2028 increases from 10 to 26; after retirement it is zero instead of the old schedule's 10 in each of 2029–2034. These changes redistribute the same total depreciable cost. They are consequences of the supplied retirement date, not forecasts inferred from the single reported balance.

An old error changes comparative profit and opening equity

In the error history, correct annual depreciation has always been $120/6 = 20$. Assume the error is material and all period-specific and cumulative effects are practicably determinable. The 2025 report presents 2024 as its comparative year. AASB 108.42 requires correction of that comparative and of the opening balances for errors before its start; paragraph 46 excludes the prior-period correction from 2025 profit. We therefore need two linked amounts: the error accumulated before 1 January 2024 and the additional error during 2024.

Tax makes the equity effect smaller than the PPE correction. Supply a conditional tax law that permits deductions of 20 annually in 2023–2028, with an enacted rate of 25% throughout the relevant recovery periods. The asset is recovered through use, and initial book and tax bases are both 120. There are no other differences or uncertain positions. Each year the company earns cash operating income of 50 before depreciation; current tax is paid on the following 31 March. These are authored tax assumptions, not statements of Australian tax law. AASB 112.7, 15–17, 46–47 and 51 support the tax-base and rate method; paragraph 53 prohibits discounting the deferred tax (Australian Accounting Standards Board, 2023b).

The originally reported 2023 PPE of 110 exceeded tax base 100 by 10, giving a deferred tax liability (DTL) of 2.5. In 2024, reported PPE 100 exceeded tax base 80 by 20, giving DTL 5.

Correct PPE is instead 100 and 80 respectively, equal to the tax bases, so those DTL balances disappear. Original pretax profit of 40 carried current tax 7.5 and deferred expense 2.5, leaving profit 30. Correct pretax profit of 30 carries current tax 7.5 and no deferred expense, leaving profit 22.5 in each year. Cash tax does not change: the tax deductions were correct throughout.

At the earliest comparative opening, 1 January 2024, reduce PPE by 10, DTL by 2.5 and retained earnings by 7.5. During comparative 2024, the extra depreciation of 10 and removal of deferred expense 2.5 reduce that year's profit by another 7.5. Thus the cumulative 1 January 2025 adjustment is PPE minus 20, DTL minus 5 and retained earnings minus 15. The opening and comparative movements sum to the same cumulative correction; they are not two separate postings of it.

To see the balance-sheet consequence, start with capital and cash of 200, then pay 120 for the asset at acquisition. The following figures retain the original publication alongside its corrected comparative. Initial capital remains 200; all later equity changes here are retained profit.

	2024 original	2024 restated	2025 error	2025 estimate
Depreciation	10	20	20	22
Profit after tax	30	22.5	22.5	21
Cash	172.5	172.5	215	215
Net PPE	100	80	60	78
Current tax payable	7.5	7.5	7.5	7.5
DTL	5	0	0	4.5
Total equity	260	245	267.5	281

The estimate column starts from the valid original equity of 260. Its 2025 pretax profit is $50 - 22 = 28$. Tax base is 60, so PPE 78 gives DTL 4.5, a decrease of 0.5 from the opening DTL. Current tax 7.5 less that deferred benefit gives tax expense 7 and profit 21. Compared with the old 2025 schedule, the estimate reduces pretax profit by 12 and after-tax profit by 9. The error column instead starts from corrected equity of 245 and adds current profit 22.5. Equal cash of 215 therefore does not establish equal earnings or equal equity.

The discovery entry joins the books to the restatement

The system discovers the mistake in April; it does not rewrite the time at which it learned the fact. Its correction records 1 April as the posting and knowledge date while attributing the prior-period amount to opening equity. Debit retained earnings 15, debit DTL 5 and credit accumulated depreciation 20. AASB 112.61A(b) and 62A(a) place the associated tax correction directly in equity with the prior error, rather than crediting current tax expense (Australian Accounting Standards Board, 2023b).

There is also a current-quarter error. Before discovery the books have already charged depreciation 2.5 and deferred tax expense 0.625 for January–March. Correct depreciation for those three months is 5. Debit current depreciation expense 2.5 and credit accumulated depreciation 2.5; debit DTL 0.625 and credit current deferred tax expense 0.625. That second tax entry reverses the current quarter's erroneous DTL increase. It has a different origin from the prior-period DTL correction of 5. After both corrections, April PPE is 75 and DTL is zero. Depreciation 15 over the remaining nine months leaves PPE 60; full-year current depreciation is $2.5 + 2.5 + 15 = 20$, not 40.

The comparative is a presentation of the corrected earlier periods, not an instruction to post the correction a second time. A correction identity binds the entity, affected periods, original close, accepted treatment and target restatement. An exact retry returns the original receipt

before testing later staleness. A changed payload under that identity is a conflict; a new request identifier cannot post the same target correction again. Original publications remain retrievable and explicitly identified as containing the error. These are proposed semantic compositions of the existing time, dependency and commit rules, not new portable request encodings or evidence of a running accounting engine.

The cash bridge provides a useful independent check. Cash after acquisition is 80; 2023 income increases it to 130. Subsequent annual income of 50, less payment of the previous year's tax 7.5, gives 172.5 and 215. Under the case's operating-cash convention, corrected 2025 profit 22.5 plus noncash depreciation 20 gives 42.5. The estimate history gives the same cash increase from profit 21 plus depreciation 22 less the deferred-tax benefit 0.5. Neither the opening-equity correction nor the comparative restatement creates cash.

When the classification cannot yet be made

Missing original-life documentation is not proof that management made an error, and finding a later shorter-life estimate is not proof that the old estimate was reasonable. The unresolved case retains both questions until evidence distinguishes them. A change of measurement basis requires the policy-change procedure. Paragraph 35's instruction for difficulty distinguishing a policy change from an estimate change does not resolve an error-versus-estimate ambiguity.

Nor does a missing local file establish impracticability. Paragraphs 43–45 distinguish inability to determine period-specific effects from inability to determine the cumulative effect; paragraphs 50–53 require the relevant effort and prohibit hindsight. Such a case needs its own supported starting date and correction route. The fully practicable example above cannot supply them. Unknown materiality also leaves the material-restatement route unestablished; it does not authorize ignoring a known error.

The error disclosure identifies the transcription mistake, comparative depreciation and tax/profit effects, the affected closing balance-sheet items and the earliest opening adjustment under paragraph 49. Cash and current tax are unchanged. If EPS is required, its effect must be added using that entity's actual numerator and share history. Missing tax law, original availability or accepted correction treatment blocks the related net-equity or posting claim. Balanced entries alone cannot supply those facts.

9.6 Apple: asset balances do not identify asset movements

The preceding methods need receipts, lots, useful lives and settlement dates. Apple's disclosures supply some of the balances those events must explain, but they do not supply the events themselves. Its FIFO and straight-line policies narrow the permitted cost calculations without identifying the actual inventory lots or asset cohorts (Apple Inc., 2024a, Note 1, printed p. 34). Comparing the notes with the cash-flow statement shows why filling those missing inputs by subtracting balances can produce a wrong history.

9.6.1 An inventory adjustment is not purchases or payments

Between 28 September and 28 December 2024, components fall from 3,627 to 2,792 million dollars while finished goods rise from 3,659 to 4,119 million. Total inventory falls from 7,286 to 6,911: the component changes, −835 and +460, sum to −375 million (Apple Inc., 2025c, Note 5, printed p. 9). Those two classes do not reveal quantities, unit costs or the transfers and sales that produced the balances.

The quarter's cash-flow statement reports +215 million for inventories within its reconciliation of net income to operating cash (Apple Inc., 2025c, printed p. 5). That is an indirect reconciliation adjustment, not total cash paid to suppliers. Even replacing it with minus the stock change would give +375, a different quantity. The difference $215 - 375 = -160$ million remains unexplained by these selected disclosures. In fiscal 2024, inventory rose from 6,331 to 7,286, while the cash-flow adjustment was −1,046; relative to the −955 stock-change shortcut the difference is −91 million (Apple Inc., 2024a, printed pp. 31 and 33). Neither residual is an identified currency effect, write-down or other journal entry. Production, consumption, ownership, valuation and payment histories must be established before they can explain the differences.

9.6.2 Testing a proposed fixed-asset history

The fixed-asset notes distinguish three measurements. The table shows gross property, plant and equipment (PP&E), its accumulated depreciation and the remaining carrying amount, all in millions of dollars (Apple Inc., 2024a, Note 5, printed p. 39) (Apple Inc., 2025c, Note 5, printed p. 9).

Amount	30 Sep 2023	28 Sep 2024	28 Dec 2024
Gross PP&E	114,599	119,128	120,615
Accumulated depreciation	70,884	73,448	74,546
Net PP&E	43,715	45,680	46,069

Accumulated depreciation is shown here as a positive deduction: gross less that deduction equals net. During fiscal 2024, gross cost increases 4,529, accumulated depreciation increases 2,564 and net PP&E increases 1,965. Yet the note reports PP&E depreciation expense of 8.2 billion, and the cash-flow statement reports capital payments of 9,447 million. Its broader depreciation-and-amortisation adjustment is 11,445 million (Apple Inc., 2024a, printed pp. 33 and 39). These amounts differ in meaning; the 8.2 billion also has coarser displayed precision. The change in accumulated depreciation cannot be substituted for the year's depreciation charge.

Consider a deliberately limited asset book to test a tempting reconstruction. Let G_0, G_1 be opening and closing gross cost and B_0, B_1 accumulated depreciation, recorded as positive amounts. Let A be gross additions, R the original cost of retired assets, d the depreciation charge and S the accumulated depreciation removed with those assets. Assume there are no other movements and all depreciation is expensed. For this illustration, use the displayed amounts as exact, including $d = 8{,}200$ million. The two balances must then satisfy

$$G_1 = G_0 + A - R, \qquad B_1 = B_0 + d - S, \qquad 0 \le S \le R. \tag{9.5}$$

The inequality requires a nonnegative carrying amount for the retired assets in this cost model. If additions are equated with the 9,447 of cash payments, the gross balance requires $R = 9{,}447 - 4{,}529 = 4{,}918$. The depreciation balance requires $S = 8{,}200 - 2{,}564 = 5{,}636$. The proposed book removes more accumulated depreciation than original cost, by 718 million. It is infeasible under these combined assumptions. That finding rejects the proposed history; it does not establish an error in Apple's filing or identify which assumption must change.

The result is not sensitive to small rounding differences under a specified rounding model. If the million-dollar balances and payment are rounded to the nearest million, and 8.2 billion to the nearest 0.1 billion, the excess $d - 9{,}447 + (45{,}680 - 43{,}715)$ lies in the closed outer envelope $[666.5, 769.5]$ million. Its lower endpoint is $8{,}150 - 9{,}447.5 + 45{,}679.5 - 43{,}715.5 = 666.5$. This

is a conditional precision check, not a verified issuer rounding rule; the exact tie convention remains unknown.

Allowing additions to be partly unpaid provides two simple repairs. Retain the same selected opening and closing balances, 8,200 depreciation and 9,447 cash payments. Assume retirement at the start of the year with zero proceeds, depreciation of surviving assets, and new acquisitions at year end. There are no opening capital payables, taxes or other asset movements. The figures below are constructed histories, not estimates of actual Apple transactions.

Annual movement (USD million)	History 1	History 2
Gross additions	10,529	11,529
Gross cost retired	6,000	7,000
Accumulated depreciation removed	5,636	5,636
Retired carrying amount, expensed	364	1,364
Cash paid for additions	9,447	9,447
Additions remaining payable	1,082	2,082

Both histories satisfy equation (9.5); their gross additions can also be allocated to match the three reported asset classes. For History 1, acquisition debits PP&E cost 10,529 and credits cash 9,447 and a capital payable 1,082. Depreciation debits expense and credits accumulated depreciation by 8,200. Retirement debits accumulated depreciation 5,636 and loss 364, while crediting gross PP&E cost 6,000. Net PP&E rises 1,965 and cash falls 9,447, so total assets fall 7,482. Liabilities rise 1,082 and equity falls 8,564 through the two expenses: $1{,}082 - 8{,}564 = -7{,}482$. History 2 has the same asset change but liabilities rise 2,082 and equity falls 9,564. These are coherent subsystem consequences, not a fit to all of Apple's statements or an exhaustive family of real asset histories. Neither history receives a probability or preference from this arithmetic.

The following quarter supplies a new snapshot, not the missing annual movements. Its gross, accumulated and net changes are 1,487, 1,098 and 389 million, whereas capital payments are 2,940 and broad depreciation/amortisation is 3,080 (Apple Inc., 2025c, printed pp. 5 and 9). The inspected interim note does not separately identify PP&E depreciation expense. The lease note adds a related distinction: finance-lease right-of-use assets are included in PP&E, while operating-lease right-of-use assets are in other noncurrent assets. The 1.0 billion of noncash right-of-use acquisitions covers the disclosed lease population (Apple Inc., 2024a, Note 8, printed p. 42); adding it all to PP&E or to cash capital expenditure would erase that distinction.

The method therefore preserves reported balances, cash payments, expense and unresolved movement components separately. A future cost or funding query still needs lot and asset ages, useful lives, residual values, impairment and settlement facts. The real filing has supplied constraints and ruled out one simple history, while leaving several different explanations open.

9.6.3 Local alternatives need different company histories

The two asset books expose a constraint on combining explanations. Both fit the selected PP&E movements and capital payments. Can they therefore be inserted interchangeably into one otherwise unchanged history of Apple? The journals above answer no. Relative to History 1, History 2 retires an additional 1,000 of gross cost without removing additional accumulated depreciation, so it recognises an additional 1,000 loss. It acquires an additional 1,000 of gross cost on credit. Those changes cancel in net PP&E and leave cash unchanged, but they add 1,000 to liabilities and subtract 1,000 from profit and equity. Amounts remain millions of dollars, under the same conditional asset-book assumptions, including exact illustrative depreciation of 8,200 and no tax effect within that subsystem.

Let P_i denote the vector of statement effects generated by asset History i, and let R_i denote the effects of the rest of the company's history. For this comparison, order each vector as asset change, liability change, profit, equity change and cash change. Both full histories must explain the same reported vector Y. If an admissible remainder exists for each branch, their necessary relation is

$$P_1 + R_1 = Y = P_2 + R_2, \qquad R_2 - R_1 = -(P_2 - P_1) = (0, -1{,}000, +1{,}000, +1{,}000, 0).$$

Thus the rest of History 2 must contribute 1,000 less liability growth and 1,000 more profit and equity growth, with no additional asset or cash change. Holding the remainder fixed is wrong relative to the common-statement target. The balance-sheet identity alone fails to detect the error: adding 1,000 to liabilities and subtracting 1,000 from equity still leaves their sum unchanged.

The reported annual totals make the required differences concrete. Apple's asset growth is 12,397, liability growth is 17,593, profit is 93,736 and equity change is −5,196 (Apple Inc., 2024a, printed pp. 29–32). Since History 1 contributes liability growth of 1,082, profit of −8,564 and equity change of −8,564, its hypothetical remainder must contribute 16,511, 102,300 and 3,368 respectively. History 2 instead requires 15,511, 103,300 and 4,368. In both cases the remaining asset change must be 19,879. These are necessary subtraction results. They do not supply the transactions, expense classifications, taxes or liability assignments that would make a remainder admissible.

The distinction matters for cash as well. In the isolated asset book, depreciation and the retirement loss are noncash expenses. Adding both back to its negative profit gives zero operating cash; acquisitions use 9,447 of investing cash. Switching branches therefore adds 1,000 to the noncash loss and 1,000 to the corresponding addback, leaving subsystem cash unchanged. If that loss were mapped into the annual cash-flow statement's broad Other adjustment of −2,266, the remaining contributions to that row would have to be −2,630 in History 1 and −3,630 in History 2 (Apple Inc., 2024a, printed p. 33). That mapping is itself conditional: the source does not identify either retirement loss or the composition needed to offset it. A matching operating-cash total cannot substitute for the reported adjustment rows.

Nor may a remainder erase the cash-population difference from Section 2.9. After the common asset payments, the required remaining balance-sheet cash change is 9,425, whereas the required remaining cash-flow change is 8,653. Their 772 difference is inherited from the disclosed opening populations. Calling it another cash receipt would manufacture a transaction rather than explain the record.

A proposed completion must now identify the liability class containing the capital payable, the expense class containing the loss, the tax consequences and the noncash cash-flow mapping, and satisfy the detailed statement rows under the same assignments. It must also retain the asset cohorts and payment terms needed at the next date. The December snapshot does not establish that either hypothetical annual payable remained outstanding or identify when it was settled. Failure of the unchanged-remainder construction therefore calls for a coupled historical model. It rejects that construction without deciding whether either local asset book has an admissible full-company completion.

9.6.4 An offset must explain the individual rows

The required offset can be tested without pretending that its transaction has been observed. Suppose a separate economic obligation would justify an expense and an unpaid liability of q in History 1 and $q - 1{,}000$ in History 2, with $q \geq 1{,}000$. These are alternative amounts in two histories; they are not a release booked within either history. For a numerical example,

take $q = 2{,}000$: each branch debits expense and credits its own liability, for 2,000 or 1,000 respectively. The difference supplies the 1,000 lower liability and 1,000 higher profit required of History 2's remainder. The economic obligation is an additional premise. Subtracting reported totals has not established that it exists.

Even conditional on that premise, a matched total is insufficient. Assume both the retirement loss and the new expense enter product cost, and both the capital payable and the new liability enter the same non-tax current liability category. History 2's extra loss then offsets its smaller expense within that expense row, and its larger capital payable offsets its smaller expense liability within that liability row. Other operations, distributions, share counts and OCI remain fixed in this comparison. Equal tax expense and tax balances are a further condition: equal pretax profit does not imply equal deductions, tax bases or deferred tax.

The indirect cash-flow statement exposes another restriction. If the expense liability is treated as an ordinary operating accrual, its 1,000 smaller increase reduces the working-liability adjustment by 1,000. The extra retirement loss increases the noncash addback by 1,000. Total operating cash flow is unchanged, but the separately reported Other adjustment and Other current and non-current liabilities adjustment each change. That construction fails to reproduce Apple's detailed cash-flow statement, even though it balances and matches total operating cash (Apple Inc., 2024a, printed p. 33).

A different conditional reporting map identifies the expense liability's recognition as a separate noncash adjustment. Its amount must then be excluded from the working-liability adjustment to avoid adding it back twice. Let α be the fraction of this hypothetical recognition allocated to the separate noncash adjustment; the remaining $1-\alpha$ enters the operating-liability adjustment. This parameter describes an assumed allocation, not a probability or a freely selectable accounting policy. With the retirement loss also assigned to Other, the differences between History 2 and History 1 are

$$\Delta\text{Other} = 1{,}000 - \alpha\,1{,}000, \qquad \Delta\text{WorkingLiabilities} = -(1-\alpha)\,1{,}000, \qquad \Delta\text{CFO} = 0.$$

For $\alpha = 0$, the two rows differ by $+1{,}000$ and $-1{,}000$. For $\alpha = \frac{1}{2}$, they still differ by $+500$ and -500. Both row differences vanish only at $\alpha = 1$ within this construction. Including the full liability movement as well as the separate addback would instead understate the History 2 minus History 1 operating-cash difference by 1,000.

Under the common expense and liability mappings, equal taxes and the disjoint noncash mapping with $\alpha = 1$, all five annual statements are unchanged by the combined transformation. Fixed shares also preserve EPS; fixed OCI and distributions carry the profit equality into comprehensive income and the equity statement. This is a conditional invariance result. It assumes a feasible full-company History 1 exists and that the transformed events remain admissible. It neither constructs that base nor proves that either complete history is compatible with all notes. Assigning the two expenses to different rows, the liabilities to different categories, or different tax consequences immediately breaks the relevant equality.

The notes make these remaining conditions specific. Note 5 separates gross PP&E by asset class: the extra acquisition and retirement must cancel within each affected class to leave that table unchanged. Note 6 separates 26,601 of current income taxes payable from 51,703 of other current liabilities, and 9,254 of non-current income taxes payable from 36,634 of other non-current liabilities. A total-liability offset cannot choose its own location inside those amounts. Note 8 identifies finance-lease right-of-use assets of 1,069 within PP&E and distinguishes lease liabilities and noncash lease acquisitions. Those disclosures do not identify the hypothetical capital purchases or expense liability. Nor do Note 12's purchase commitments and contingency assessment establish the assumed obligation or its amount (Apple Inc., 2024a, printed pp. 39, 42, 45–46).

The distinction therefore survives the repair: an algebraic offset is available; a row-preserving transformation is available under stated economic, tax and reporting assumptions; an actual Apple completion is unestablished. To admit one, the historical model needs the underlying obligation, gross asset classes, expense and liability assignments, tax bases and noncash treatment, together with a feasible rest of the company. The subsequent quarter additionally needs settlement dates and surviving asset cohorts. Identical annual statements do not make those future responses identical.

9.6.5 Equal liabilities can require different cash paths

The conditional repair leaves different obligations beneath the same annual liability total. In the $q = 2{,}000$ illustration, History 1 owes 1,082 for capital purchases and 2,000 for the separate expense; History 2 owes 2,082 and 1,000 respectively. Each owes 3,082. Consider what paying those balances would do, conditional on the economic and reporting assumptions needed for the preceding construction. Assume payment at carrying value, sufficient unrestricted cash, and no additional interest, foreign exchange, remeasurement or tax effect. Capital settlement is classified as investing cash and expense settlement as operating cash in this example. Neither the obligation nor its settlement terms have been identified in Apple's disclosures.

If both obligations are paid at the first subsequent date, each payment debits its liability and credits cash. No new expense arises: the expense or acquisition was recognised in the prior period. Both histories reduce cash and liabilities by 3,082, leave assets other than cash unchanged, and leave profit, OCI and equity unchanged. The indirect operating-cash calculation includes the expense liability's decrease once; it excludes capital settlement from the operating-liability adjustment. The cash-flow statement nevertheless distinguishes the histories. Writing each ordered pair as operating cash followed by investing cash, the joint alternatives are

$$(\mathrm{CFO}, \mathrm{CFI})_{H_1} = (-2{,}000, -1{,}082), \qquad (\mathrm{CFO}, \mathrm{CFI})_{H_2} = (-1{,}000, -2{,}082).$$

Both pairs sum to $-3{,}082$. Selecting the smaller outflow from each category would instead produce $(-1{,}000, -1{,}082)$, whose total is only $-2{,}082$. No branch produces that pair. Selecting both larger outflows would invent an extra 1,000 payment. The category intervals can describe marginal possibilities, but their Cartesian product contains paths excluded by the shared history.

Timing makes the distinction relevant even to total cash. Suppose capital obligations are paid at the first date and expense obligations at the second. History 1 then pays 1,082 followed by 2,000; History 2 pays 2,082 followed by 1,000. Relative to History 1, History 2 has 1,000 less cash after the first date and the same cumulative cash after the second. Paying the expense obligations first reverses the first-date difference. An aggregate liability of 3,082 suffices for the narrow query of total undiscounted cash after both payments, under these fixed assumptions. It does not suffice for cash by date or by statement category. Retaining liability kind, amount, settlement terms and joint branch identity is therefore necessary for those queries.

The actual following quarter constrains a proposed continuation without identifying it. Apple reports PP&E payments of 2,940, operating cash of 29,935, investing cash of 9,792 and total cash growth of 356 in the quarter ended 28 December 2024 (Apple Inc., 2025c, printed p. 5). If both hypothetical obligations settled in that quarter, the remaining PP&E payments would have to be 1,858 under History 1 or 858 under History 2. The rest of the company would have to contribute operating cash of 31,935 or 30,935, and investing cash of 10,874 or 11,874, respectively. After unchanged financing cash of $-39{,}371$, either required remainder has cash growth of 3,438; subtracting the hypothetical 3,082 settlement recovers the reported 356. These are necessary conditions on unconstructed remainders. Positive remaining payments and a matching cash identity do not establish the underlying transactions or prove that either continuation fits the complete quarterly record.

This calculation extends the conditional history into explicit journals and statement effects while keeping actual-history admission unresolved. If cash is insufficient, the model must retain the resulting funding or default question; it cannot invent a loan to preserve the path. A later depreciation or production query requires different information again: asset cohorts, remaining lives, residual values, capacity and tax bases. The settlement example supplies none of those quantities. A change in the query must therefore reopen the state-sufficiency check, even when the annual statements and the final cash total coincide.

9.7 Lease and tax integration

A lease connects asset use, a financing obligation and income tax. The following schedule extends the operating-company example by separating these three measurements. The retained lease and tax dossiers supply the dated provisions, modification case and assumptions. The official lease compilation is dated December 2023 despite the older filename. The right-of-use (ROU) asset represents the admitted use of the leased asset. Its cost, the lease liability, accumulated depreciation, remaining payments, rate version and option judgments are separate state fields. Tax retains carrying amounts, legal tax bases, reversal dates, enacted/substantively-enacted rate evidence, recovery support, and original profit/OCI/equity allocation. The deferred tax asset (DTA) and deferred tax liability (DTL) are retained gross before any permitted offset presentation. A liability schedule is not a tax-base schedule.

For the two annual 121 payments at 10%, initial liability and ROU are 210. Year-one interest is 21, liability after payment 110, depreciation 105 and ROU 105. Year two interest is 11, and both balances finish at zero. A same-asset modification just after year one creates two future payments 144 at revised 20%, PV 220, so liability increases 110 and ROU increases 110 to 215. A payment index reset has different rate rules; it cannot reuse that modification blindly.

To demonstrate connection to company statements, assume a synthetic entity with opening cash/equity 500, no other opening balances, and annual cash service revenue 400 less cash operating costs 100, all recognized in their years. There are no dividends, other assets, debt, losses or permanent differences. Add the base lease above. Assumed tax law deducts cash lease payments against taxable income, attributes those deductions to the lease liability, gives zero tax bases to ROU/liability, and taxes profits at enacted 25%. Sufficient usable profits and all recognition conditions for DTA are assumed. Tax is paid the following year. No current-period tax payment is invented from tax expense.

At commencement debit DTA 52.5 and credit DTL 52.5 in addition to the lease recognition. At year-one close DTA is 27.5 and DTL 26.25. The DTA reduction 25 is deferred expense; DTL reduction 26.25 is deferred benefit, net benefit 1.25. Taxable profit is 300−121=179; current tax/payable 44.75. Pretax book profit is 300−105−21=174, total tax 43.5 and profit 130.5. At year two, deferred expense 1.25 combines with current tax 44.75 to give total tax 46 on pretax profit 184.

Closing balance	2024	2025
Cash	679	813.25
ROU asset	105	0
Deferred tax asset	27.5	0
Lease liability	110	0
Deferred tax liability	26.25	0
Current tax payable	44.75	44.75
Equity	630.5	768.5

2024 assets 811.5 equal liabilities 181 plus equity 630.5; 2025 assets 813.25 equal liabilities 44.75 plus equity 768.5. For this case, elect financing classification for lease interest under the retained AASB 107 regime. The operating and financing cash totals below depend on that presentation election. Direct operating cash is 300 then 255.25 after the first tax payment; financing lease cash is −121 each year; no investing cash arises. Indirect operating cash in year one is profit 130.5 plus depreciation 105 plus interest 21 minus deferred benefit 1.25 plus unpaid current tax 44.75 = 300. Year two gives profit 138 +105 +11 +1.25 =255.25. The second tax payable settles next year. Zero OCI and no owner transactions mean the equity change equals cumulative profit. This case tests gross tax, cash timing and financial-statement coherence, not a real entity's tax law.

9.8 Apple: recognising tax and settling the obligation

The lease example separates a tax charge from the payment that follows it. Apple's fiscal 2024 disclosures show why that separation matters when several tax jurisdictions, deferred balances and a large disputed obligation enter the same accounts. Its annual income-tax expense was 29,749 million dollars; net cash paid for income taxes was 26,102 million (Apple Inc., 2024a, printed pp. 29 and 33). The 3,647 difference is not, by itself, an increase in tax payable. Deferred tax and other balance movements must first be distinguished from current tax and settlement.

9.8.1 The expense and balance reconciliations answer different questions

Note 7 divides the 2024 expense between current and deferred amounts for federal, state and foreign taxes. In millions of dollars, the reported components are as follows (Apple Inc., 2024a, Note 7, printed p. 40). Parentheses in the source represent negative expense, or a tax benefit.

Tax category	Current	Deferred	Total
Federal	5,571	−3,080	2,491
State	1,726	−298	1,428
Foreign	25,483	347	25,830
Sum	32,780	−3,031	29,749

The current charge is therefore 32,780 and the deferred benefit 3,031. The tax payable has its own scope. Note 6 reports current income taxes payable of 26,601 and noncurrent income taxes payable of 9,254 at 28 September 2024, against 8,819 and 15,457 a year earlier. Combining both maturity classes gives 35,855 against 24,276, an increase of 11,579 (Apple Inc., 2024a, Note 6, printed p. 39). Current expense less net cash taxes is only 32,780 − 26,102 = 6,678. Reclassifying an amount between current and noncurrent cannot explain the difference in their combined total.

To expose the missing movements, let T denote that combined tax payable and N the net deferred tax asset reported in Note 7. Let C be current tax expense, D deferred tax expense (negative for a benefit), and P net cash taxes paid, all for the same annual period and in millions of dollars. Define U and V as the residuals needed to connect the reported amounts:

$$\Delta T = C - P + U, \qquad \Delta N = -D + V. \tag{9.6}$$

These are diagnostic definitions, not assertions that every tax balance has the same recognition or settlement scope. The first relation gives $U = 11{,}579 - 6{,}678 = 4{,}901$. Setting that residual

to zero would produce the wrong closing payable. Assigning it to currency translation, uncertain tax positions or another journal would require the missing transaction and classification evidence.

The deferred balance illustrates the second relation. At September 2024, gross deferred tax assets of 34,873 less a valuation allowance of 8,866 leave 26,007. Deducting deferred tax liabilities of 6,805 gives net deferred tax assets of 19,202, compared with 17,251 a year earlier (Apple Inc., 2024a, Note 7, printed p. 41). The increase is 1,951, so $V = 1{,}951 - 3{,}031 = -1{,}080$. The deferred benefit in profit does not explain the whole net-balance movement. Its allocation among currency, equity, other comprehensive income and other causes remains unresolved here.

Note 6's 19,499 deferred-tax-asset presentation is also different from Note 7's 19,202 net position. Their difference of 297 is an arithmetic consequence, not a separately disclosed liability line or authority to offset jurisdictions. The reported temporary differences—including research expenditure, deferred revenue, leases and depreciation—need their own tax bases and recovery conditions. The note reports indefinitely available Irish foreign-tax and California R&D credit carryforwards, but also a valuation allowance for those credits and some other differences. An indefinite term does not establish that an asset is recoverable (Apple Inc., 2024a, Notes 6–7, printed pp. 39–41).

9.8.2 A recognised charge, restricted assets and a later payment

Apple attributes its large 2024 State Aid charge to the September 2024 court decision confirming the European Commission's recovery decision. This is the issuer's account of the dispute and its accounting consequence. Note 7 describes a 15.8 billion dollar payable to Ireland, partly offset by a 4.8 billion US foreign tax credit and an 823 million decrease in unrecognised tax benefits (Apple Inc., 2024a, Note 7, printed pp. 39–40).

The narrative calls the net charge 10.2 billion; the statutory-rate reconciliation gives the more precise amount 10,246 million. Subtracting the displayed narrative centres gives $15{,}800 - 4{,}800 - 823 = 10{,}177$, which is 69 million lower. That is compatible with the coarser inputs: if the two billion-dollar amounts are rounded to the nearest 0.1 billion and 823 to the nearest million, the component calculation lies in the closed outer interval $[10{,}076.5, 10{,}277.5]$ million, containing 10,246. The 10.2 billion charge is compatible with it as well. This is a conditional rounding check, with the issuer's precise rule and tie convention unverified; the centre difference does not establish a conflicting source assertion.

The tax charge does not make the settlement funds disappear at recognition. At 28 September 2024 the escrow comprised 2.6 billion dollars of cash and cash equivalents and 13.2 billion of current marketable securities, restricted from general use (Apple Inc., 2024a, Notes 4 and 7, printed pp. 37 and 39). The annual liquidity discussion separately identifies the obligation and the funds held to settle it. The following quarter reports both a payment and the remaining obligation (Apple Inc., 2024a, printed p. 26) (Apple Inc., 2025c, printed p. 16).

State Aid obligation and settlement	EUR billion	USD billion
Obligation at 28 Sep 2024	14.2	15.8
Payment during the following quarter	11.0	11.9
Obligation at 28 Dec 2024	3.3	3.4

The three rows are different dated quantities. In dollars, a movement equation with opening obligation 15.8, payment 11.9 and closing obligation 3.4 requires an additional net movement of $3.4 + 11.9 - 15.8 = -0.5$ billion. Under nearest rounding to 0.1 billion, its outer interval is $[-650, -350]$ million, excluding zero. The same calculation in euros gives $3.3 + 11.0 - 14.2 = 0.1$

billion, with outer interval $[-50, 250]$ million, which includes zero. Neither calculation supplies a common exchange rate or an exact exchange-gain entry. The interim tax discussion mentions foreign-currency revaluation effects related to the decision, but does not allocate this 500 million dollar residual (Apple Inc., 2025c, printed p. 16).

The remaining funds also retain their asset classifications. At December 2024, restricted cash and cash equivalents were 126 million dollars and restricted current marketable securities 3.3 billion (Apple Inc., 2025c, Note 4, printed p. 8). Their displayed sum, 3,426 million, is compatible with the rounded 3.4 billion obligation. It is not all cash available for ordinary operations. The opening 2.6 billion of restricted cash alone could not have funded the reported 11.9 billion payment; securities sales, maturities, transfers and valuations need their own histories.

For an isolated settlement illustration, suppose 11,900 million of restricted cash is available and the amount paid equals the liability's carrying value at settlement. Debit tax payable and credit restricted cash by 11,900. Assets and liabilities each fall by 11,900; equity and current-period tax expense do not change in this posting. This assumes the required cash conversion has already occurred and omits any associated gain, loss or fee. It explains settlement of an existing obligation without pretending to reconstruct Apple's funding and currency entries. The quarter separately reports tax expense of 6,254 and net cash taxes of 18,651 million (Apple Inc., 2025c, printed pp. 1 and 5); neither should be replaced with the Ireland payment or treated as another copy of the preceding year's charge.

9.8.3 Uncertain tax positions do not supply a payment distribution

The last part of Note 7 preserves another population: gross unrecognised tax benefits. Its 2024 movement is $19{,}454 + 1{,}727 - 386 + 2{,}542 - 1{,}070 - 229 = 22{,}038$ million. The increases concern prior- and current-year positions; the decreases concern changed prior-year positions, settlements and expired limitation periods. Interest and penalties are excluded from this rollforward (Apple Inc., 2024a, Note 7, printed p. 41). The 1,070 settlement-related decrease is a movement in that gross balance, not an independently identified total of cash paid to tax authorities.

Of the rounded 22.0 billion gross amount, the issuer says that 10.8 billion, if recognised, would affect its effective tax rate. It also describes a reasonably possible decrease of approximately 5–13 billion over the next twelve months, primarily concerning transfer pricing and deemed repatriation tax, while noting uncertainty about examination timing. These are different conditional statements. The range is neither a probability law nor a promised cash payment or expense release. A forecast still needs jurisdictional outcomes, recognition, interest, penalties and settlement dates. Retaining those unknowns keeps the tax note connected to the company's funding needs without turning management's dated assessment into an invented future transaction.

9.9 Returning to the company forecast

The Northstar case remains the broader ordinary-company teaching baseline: collections, purchases/payments, inventory, payroll, PPE, financing, tax, statements, annual/interim and revision views are already specified. Its 2024 collection/purchase/payment solution is 245/155/145. The 2025 forecast in the Northstar reference fixture opens with cash 74.6 and equity 90.6, spends 20 on capital expenditure at period end, pays dividend 8 and uses a base floating rate 7%. Its simplified tax is settled in the year earned. The running case uses fixed-rate share α; here floating-rate share $x = 1 - \alpha$ describes the same uncertainty. For 2025 the same unknown

floating fraction $x \in [0,1]$ determines every statement amount:

$$\begin{aligned} I(x) &= 5 + 2x, & \text{pretax profit}(x) &= 82 - 2x, \\ \text{tax}(x) &= 16.4 - 0.4x, & \text{profit}(x) &= 65.6 - 1.6x, \\ \text{operating cash}(x) &= 48.6 - 1.6x, & \text{closing cash}(x) &= 95.2 - 1.6x, \\ \text{equity}(x) &= 148.2 - 1.6x, & \text{assets}(x) &= 288.2 - 1.6x. \end{aligned}$$

Receivables 55, inventory 40, net PPE 98, payables 40 and debt 100 complete the balance sheet. Liabilities are 140 for every x, so the common cash/equity slope preserves the accounting identity. There is no default distribution on x. At a 9% floating rate the all-floating history has interest 9, profit 62.4, cash 92, equity 145 and assets 285. The annual cash bounds still cannot establish daily payment solvency without event-date cash flows.

The lease/tax example extends the interaction evidence without changing the original Northstar fixture or pretending that its teaching 20% tax formula is an IAS 12 method. In a real-company package, each required method is admitted separately and unsupported operations remain visible. The extended obligations for pensions, impairment, provisions, valuation and share-based payment are in the additional corporate methods that follow; omission from a finite company case never proves the actual company has none.

9.10 Employee benefits

The additional methods below use the selected IAS 19, IAS 36, IAS 37, IFRS 2, IFRS 13 and IAS 23 provisions in the original 2023 EU incorporating regulation (European Commission, 2023a). This is historical mechanism support; later amendments and actual entity facts remain separate applicability inputs.

DefinedContributionState retains service periods, promised contribution formula, contributions due/paid and applicable legal terms. For earned service requiring contribution 10, debit employee expense 10/credit payable 10; payment settles it. Calling a plan defined contribution needs the limited-obligation facts; a guaranteed benefit or additional funding obligation cannot disappear by name.

The defined-benefit obligation (DBO) measures accrued employee benefits under the stated actuarial assumptions. DefinedBenefitState retains employee/cohort service, benefit formula, projected salary, retirement/turnover/mortality assumptions, discount-curve currency/term, obligation, plan-asset fair value, asset ceiling, contributions, benefits and remeasurements. IAS 19.67–74 requires projected unit credit and the stated service attribution; .83 supplies discount-rate requirements. Assumptions are separately admitted measurements, not constants provided by the accounting rule. A one-employee toy plan promises 100 for each completed service year, paid after year two, with certain completion, zero salary growth and supported 10% discount. At year-one end accrued benefit 100 due in one year has PV 1000/11. In year two that opening obligation accretes 100/11 and the second earned benefit adds 100; obligation 200 then settles 200. Merely using current salary times years without timing would give the wrong year-one obligation.

A separate funded-plan roll-forward supplies an independent statement example: opening DBO 100, plan assets 80 and net liability 20; current service 12; interest 5 on DBO, interest 4 on assets at 5%; contributions 10 at year end; benefits 8 paid from plan at year end; actuarial obligation loss 7; actual asset return 6. DBO closes 100+12+5−8+7=116, assets 80+6+10−8=88, net liability 28. Expense is 12+1=13, OCI loss is 7−(6−4)=5. Debit service expense 12/credit net pension liability 12; debit interest expense 1/credit liability 1; debit OCI 5/credit liability 5; debit liability 10/credit cash 10. End liability 20+13+5−10=28. Plan-paid benefits reduce

both DBO and plan assets and do not cause a second employer cash payment. IAS 19.113, .120–127 supports these components; remeasurement OCI is not recycled to profit under 122. A surplus requires the asset ceiling and applicable minimum funding analysis, not recognition of any calculated negative liability.

Unknown actuarial inputs give conditional obligations or measurement_unresolved. Curtailment, settlement, amendment and multi-employer/insured arrangements need their own fact conditions and measurement-date remeasurement. A buyout is not settlement if obligations remain. Sensitivities vary one assumption only when that conditional meaning is explicit; correlated risks require joint scenarios.

9.11 Impairment and provisions

ImpairmentState retains asset/CGU boundaries, carrying amounts, allocated goodwill, corporate assets, recoverable-amount inputs and previous impairments/depreciation. IAS 36.9–10 determines indicators and mandatory annual tests. Recoverable amount is the higher of value in use and fair value less disposal costs (.18); it is not the lower of two pessimistic scenarios. VIU needs the supported cash flows, forecast horizon/growth, discount and exclusion of inappropriate financing/tax or uncommitted restructuring effects under 33–57. Compute conditional values if those estimates are unknown; do not infer them from historical profitability.

For a cash-generating unit (CGU), assets excluding goodwill 100 plus goodwill 20, recoverable amount 90: loss 30 first removes goodwill 20 and then reduces other assets 10 subject to their floors under 104–105. Debit impairment 30/credit goodwill 20/credit asset allowance 10. If the asset floor is 95 the requested reduction to 90 is impermissible; apply the remaining allocation/recognition conditions, not a forced plug. For a standalone cost-model asset with carrying 80, recoverable 60, recognize 20 loss. At a later measurement before further depreciation, supported recovery to 90 permits reversal only 20 to the no-impairment 80 ceiling (.117–119). Goodwill impairment is not reversed (.124). Different dates require the hypothetical unimpaired depreciation path, not the old nominal cost as cap.

9.11.1 An asset floor changes the allocation of a unit's loss

A cash-generating unit can lose value while one of its assets retains a substantial disposal value. Allocating the unit's loss in proportion to asset balances just once may then reduce that asset too far. IAS 36.104–105 first allocates loss to goodwill, then to the other assets pro rata, while protecting each asset from a reduction below the highest of zero, its fair value less costs of disposal if measurable, and its value in use if determinable. A share that cannot be assigned to one asset is redistributed to the others (European Commission, 2023a).

Consider a synthetic cost-model unit containing equipment A with carrying amount 60, equipment B with carrying amount 40, and allocated acquired goodwill 20. Cash 80 lies outside the tested unit. With no liabilities, opening equity is 200. The unit boundary and goodwill allocation are assumed to have been admitted under paragraphs 66–80; there are no shared corporate assets, noncontrolling interests or liabilities requiring an adjustment to this comparison. In a real case those judgments precede the calculation. In particular, paragraph 78 requires consistent treatment of a liability that is included in the unit's recoverable value; subtracting it from only the carrying amount would compare different populations.

At measurement date T_0, admitted value in use is 70 and fair value less disposal costs is 65, so the unit's recoverable amount is 70. Individual values in use are supported as not determinable, while the individual disposal values are 55 for A and 5 for B. These are supplied valuation

assumptions, not values inferred by an allocation routine. They establish floors of 55 and 5. A missing valuation would be a different state: the program may omit a term because its non-determinability is admitted, but may not replace an unexamined input with zero.

The unit's loss is $60 + 40 + 20 - 70 = 50$. Goodwill takes the first 20. An unconstrained allocation of the remaining 30 would assign $30 \times 60/100 = 18$ to A and $30 \times 40/100 = 12$ to B. A, however, can absorb only $60 - 55 = 5$. Its excess share of 13 passes to B, whose total loss becomes $12 + 13 = 25$, within its capacity $40 - 5 = 35$. The closing balances are therefore

$$A = 60 - 5 = 55, \qquad B = 40 - 25 = 15, \qquad \text{goodwill} = 20 - 20 = 0.$$

Together they equal the unit's recoverable amount of 70. Debit impairment expense 50, credit goodwill 20, and credit the separate asset impairment allowances by 5 and 25. Cash 80 plus equipment 70 equals equity 150. The unconstrained 18/12 entry would also balance, but would leave A at 42, below its admitted floor of 55. Clipping A's loss to 5 without redistributing the excess would instead recognize only 17 of the required remaining 30.

For an implementer, each round has a remaining amount and an active set of assets. Compute shares from their pre-allocation carrying amounts. Fix each share that reaches an asset's available capacity, remove that asset from the active set, and redistribute the remaining amount. A round that needs further redistribution removes at least one asset, so the procedure terminates after finitely many such removals. For positive carrying weights b_i and capacities c_i, its allocation has the form

$$x_i = \min(c_i, \lambda b_i), \qquad \sum_i x_i = \min\left(q, \sum_i c_i\right),$$

where q is the requested allocation and $\lambda \geq 0$ gives the common proportional rate among assets that have not reached their limits. In the example A is fixed at 5 and B receives the remaining 25. A source floor above an asset's existing cost gives zero reduction capacity; it does not create permission to increase that asset during an impairment calculation.

The capacity limit also explains how to handle an unsuccessful allocation. A diagnostic input requesting 50 against capacities 5 and 35 can allocate only 40, leaving 10. That arithmetic does not establish that the supplied unit and individual valuations are jointly coherent. It calls for checking measurement assumptions, the unit boundary and any separate obligations. Paragraph 108 permits a liability for a remaining unit loss only if another IFRS requires one. AccDSL must retain the residual and the unresolved decision; it cannot invent a payable to make the unit reach a requested total. This diagnostic is not a second admitted company example.

9.11.2 A recovery is limited by the depreciation that would have occurred

Suppose both pieces of equipment have five remaining years of even use and zero residual values at T_0. Under paragraph 63, their revised carrying amounts give next-year depreciation $55/5 = 11$ and $15/5 = 3$. At T_1, one full year later, they carry 44 and 12. The company must also retain the path that would have occurred without impairment. On that path depreciation would have been $60/5 = 12$ and $40/5 = 8$, leaving 48 and 32. Restoring a past loss therefore cannot mean restoring the old nominal balances.

Assume a supported change in expected future cash flows now gives the unit a recoverable amount of 90. Individual recoverable amounts remain admitted as not determinable. Paragraphs 114–116 require the change in estimates; mere passage of time and discount unwinding would not support this reversal. Paragraphs 122–123 allocate a unit reversal over assets other than goodwill pro rata, subject to each asset's recoverable amount if determinable and its depreciated no-impairment amount. Goodwill itself cannot be restored under paragraph 124 (European Commission, 2023a).

The requested increase is $90 - (44 + 12) = 34$. A has room for only $48 - 44 = 4$; B has room for $32 - 12 = 20$. The same redistribution procedure therefore recognises $4 + 20 = 24$, leaving equipment at 48 and 32 and goodwill at zero. The recognized total of 80 is below the unit's value of 90. That difference does not identify an omitted asset, authorize goodwill recognition or imply cash has been received.

Debit the impairment allowances by 4 and 20 and credit reversal gain 24. The original allowances of 5 and 25 consequently retain balances of 1 and 5. Accumulated depreciation is 11 and 3. The register now reconciles as

$$\text{PPE} = 100 - \underbrace{(11 + 3)}_{\text{depreciation}} - \underbrace{(1 + 5)}_{\text{impairment allowances}} = 80.$$

The remaining allowances record the different past depreciation paths. Erasing all 30 of the original allowances would produce PPE of 86, exceeding the no-impairment ceiling by 6. Future annual depreciation is $48/4 = 12$ and $32/4 = 8$, consistent with the remaining lives.

Pretax amount	T_0 impairment	Following year to T_1
Impairment expense	50	0
Depreciation / reversal gain	0 / 0	14 / 24
Profit / OCI	−50 / 0	10 / 0
Closing cash / PPE	80 / 70	80 / 80
Closing goodwill / liabilities	0 / 0	0 / 0
Closing equity	150	160
Operating cash	0	0

The first column isolates the impairment measurement; the second follows the next full year. With no cash or owner movements in this slice, the operating cash reconciliations are $-50+50 = 0$ and $10 + 14 - 24 = 0$, removing the noncash loss, depreciation and reversal under IAS 7.20. Equity moves from 200 to 150, then to 160. Tax remains a separate calculation under IAS 36.64 and IAS 12. These technical rules use the retained original 2023 EU text, with its date corrigendum; the relative dates do not assert current reporting admission or a complete later amendment review.

The method thus needs actual and no-impairment depreciation histories, measurement dates, individual valuation-status judgments, goodwill identity and evidence of the changed estimate. There is also an explicit numerical boundary. If every eligible asset has zero carrying amount but a positive reversal is requested, the positive-weight procedure above has no denominator. It returns an unresolved allocation basis and preserves the history; it does not choose equal weights without an additional justified interpretation. That extension, along with shared corporate assets and other scope interactions, remains distinct from the fully specified positive-weight case.

9.11.3 Recognising a separate obligation

ProvisionState retains obligating past event, present legal/constructive duty, outflow assessment, estimate distribution, settlement dates, risk/discount basis, reimbursements and evidence. IAS 37.14 requires present obligation, probable outflow and reliable estimate; .36–47 governs best estimate/risk/discount. An intended future operating expenditure without that duty is not a provision. Assume an established obligation to pay 121 in two years, deterministic amount, appropriate 10% rate and no double-counted risk: recognize expense/provision 100, accrete

interest 10 then 11, settle 121. The passage-of-time increase is borrowing cost under 60. If the obligation is an eligible asset restoration cost, initial debit follows that asset's method rather than always expense; later changes need the applicable restoration interpretation. Contingent liabilities/disclosures remain visible when recognition conditions are not met; no generic 0.5 threshold may substitute for the relevant standard's required analysis.

9.12 Share-based compensation

AwardState distinguishes equity versus cash settlement, grant/vesting/service dates, granted units, grant-date fair value, service/nonmarket/market conditions, expected units vesting, cumulative expense and changes/cancellations. IFRS 2.7–11, 15–23 supports the employee equity-settled method. With 100 units, grant value 3, two-year service condition and all expected to vest, year-one cumulative expense 150 and equity 150. At vesting only 80 satisfy the service condition: cumulative 240, second-year expense 90. Debit employee expense and credit equity. Failure of a market condition is different under 21; if the employee satisfies other vesting conditions, it does not automatically reverse the grant-date measured service expense. Unknown conditions cannot be represented by a single forfeiture percentage without their type and evidence.

Cash-settled awards remeasure the liability under 30–33. Same 100 expected units, half service and current unit fair value 4 gives liability/expense 200 at year one; 80 actual units and value 5 at settlement gives total 400, second expense 200, then debit liability 400/credit cash 400. Reusing the equity grant value 3 would be wrong. Actual award valuation requires the specified model/market inputs, option rights, volatility, dividends and calibration; the fair values 3/4/5 are case inputs. Modifications and settlement-choice arrangements require their own conditions, with the original minimum expense and increment analyzed under the applicable provisions rather than overwriting the original grant.

9.13 Fair value and capitalization boundaries

A ValuationRequest names the unit of account, market/date, asset/liability characteristics, principal/accessible market, technique, input hierarchy, counterparty/own non-performance risk where relevant, calibration and sensitivity. IFRS 13.24, .42–43 and .67–80 guide the selected measurement. Quoted identical asset price 7 for 10 units gives 70 in a qualifying Level 1 case; a blockage discount based only on holding size cannot reduce it to 63 under 69/80. A present-value illustration 111 due next year at an independently supported 11% rate gives 100; that equality does not establish the rate, market comparability or accounting classification. Historical EIR, an impairment discount, risk-neutral pricing and a physical expected return answer different questions.

Borrowing-cost capitalization requires a qualifying asset, eligible costs, commencement/suspension/cessation dates and the specific/general-borrowing calculation, before moving interest into cost. IAS 23.8–25 supplies these rules in the retained historical EU text: for specific borrowing, eligible actual costs less temporary investment income; for general borrowing, the applicable weighted rate and cost ceiling. Commencement requires expenditures, borrowing costs and necessary preparation activities; extended inactive periods can suspend it, and substantial completion ends it. Assume specific borrowing 100 outstanding during a full eligible construction year at 5%, with an admitted expenditure schedule and temporary investment income 1 on temporarily unused funds during that period. Net capitalized cost is 4. Real admission still needs the dated standard, actual schedule and eligibility evidence. It is not a default 5% add-on

to all assets. Internally developed intangibles likewise need research/development separation and all applicable recognition criteria; a forecast of future sales does not establish an asset.

The extension interface returns the stated measured components, proposed journal, economic delta, disclosures and unresolved inputs under the event-basis, authorization and commit contracts. Calculation success does not permit another operation to fill a missing authority/fact with a default. Real-company acquisition/tax/employee/valuation evidence and the full exceptions remain explicit internal preparation, with proposed answers and competing treatments retained for any later authorized domain review.

9.14 When a component reaches its next accounting event

A recognised balance also promises a subsequent method. A refund liability must eventually be settled; an impaired asset may recover; a modified lease must continue through payments and depreciation. The following methods specify those transitions under stated facts. They use the retained historical IFRS text, with current authority and actual-company measurement kept as separate admission questions (European Commission, 2023a).

9.14.1 Actual returns can differ from the estimate

Section 2.7 follows two expected and actual returns. More generally, let L be the opening refund liability, A the recovery asset, P the actual refund payment, K the actual selling-handling cost and V the returned inventory admitted after deterioration and its NRV ceiling. When all rights have closed, the debit-positive entries are

$$\Delta\text{cash} = -P - K, \qquad \Delta\text{inventory} = V, \tag{9.7}$$

$$\Delta\text{refundliability} = L, \qquad \Delta\text{recoveryasset} = -A, \tag{9.8}$$

$$\Delta\text{revenue} = P - L, \qquad \Delta\text{costofsales} = A - V, \tag{9.9}$$

$$\Delta\text{sellingexpense} = K. \tag{9.10}$$

The debit-positive revenue amount is negative when releasing an excessive refund estimate increases revenue. These entries sum to zero. The separation of selling cost follows the explicit handling classification and IAS 2.16; IFRS 15.B25 governs the recovery asset. If handling instead creates eligible inventory cost, that needs its own admitted classification.

With $L = 40$ and $A = 16$, one actual return pays refund 20 and handling 4, and recovers inventory 10. Revenue increases 20, cost of sales is 6 and selling expense 4. Three actual returns instead pay refund 60 and handling 12, recover inventory 30 and reduce revenue by 20; cost of sales is credited 14. If two returns recover inventory worth only 14, the inventory ceiling raises cost of sales to 2. Unknown handling, inventory NRV or an open return window prevents final settlement rather than supplying a zero.

9.14.2 A reversal ceiling does not supply an allocation weight

IAS 36.122–123 allocates a cash-generating unit's reversal to assets other than goodwill, proportionately to their carrying amounts, subject to the individual ceilings. Suppose assets A and B carry 40 and 60 and goodwill is 10. The unit's recoverable amount rises to 170, allowing a total increase of 60. A's individual ceiling is 45 and B's is 150. The first proportionate allocation would assign 24 to A and 36 to B. A can accept only 5; reallocating the remaining 19 to B gives increases of 5 and 55. Goodwill remains unchanged.

Writing w_i for the original carrying amounts and c_i for the remaining headroom, the method distributes a remaining reversal R among eligible assets by $Rw_i/\sum_j w_j$, caps each amount at c_i, then redistributes any remainder among assets still below their ceilings. This terminates when the reversal is exhausted or no headroom remains. It requires a positive sum of eligible weights whenever a positive amount remains.

If all eligible carrying amounts are zero, the denominator is zero. Neither equal weighting nor weighting by headroom follows from the inspected source. The specification therefore retains an unresolved allocation with unchanged books. A different rule applies to an individual asset: a zero carrying amount, no-impairment carrying value of 80 and recoverable amount of 120 permits an individual reversal to 80 under IAS 36.117–119. That individual calculation does not resolve the allocation among several zero-carrying assets.

9.14.3 A lease reduction must identify the rights surrendered

A single physical percentage is insufficient when leased rights have different values. Consider right A with right-of-use asset 90 and liability 100, and right B with asset 30 and liability 50. The supplied modification terminates B, releasing asset 30 and liability 50 for a gain of 20. It leaves A with one payment of 121 in a year at a revised 10% rate. A's revised liability is $121/1.1 = 110$; the increase of 10 adjusts its asset from 90 to 100. These are the scope-reduction and remeasurement steps in IFRS 16.45–46.

A further same-day reduction surrenders an admitted portion of A carrying asset 10 and liability 11. The remaining liability before remeasurement is 99. Revised consideration of 99 due in one year has present value 90, so the remeasurement reduces both liability and asset by 9. The final asset is $100 - 10 - 9 = 81$, liability 90 and additional gain 1. A year later, payment 99 contains interest 9 and principal 90; the supplied systematic depreciation amount of 81 exhausts the asset.

The event must therefore carry the actual surrendered asset and liability for each right, the revised payment schedule and the rate. The finite schedule uses annual-end payments and a checked one-year interval. Missing attribution, an unadmitted depreciation basis, a different calendar or a negative remaining asset requires a distinct finding or method. It cannot be repaired by applying the same physical fraction to every balance.

9.14.4 Contract costs survive more than one service period

IFRS 15.91–104 first distinguishes eligible contract costs from amounts governed by another standard. Acquisition costs must be incremental and recoverable. Fulfilment costs outside other standards must relate directly to the contract, create or enhance resources for future performance and be recoverable. Waste, past service and the specified administrative costs are expensed. The event records these findings before calculating the posting.

For example, expense waste 5 and capitalise eligible fulfilment cost 100 for five equal service units. Delivery of one unit amortises 20, leaving 80. After other applicable asset impairments have been recognised, credit-adjusted remaining consideration 70 less direct future cost 20 supports a contract asset of 50: recognise impairment 30. Delivery of another unit amortises $50/4 = 12.5$, leaving 37.5. Without the impairment, the unamortised balance would now be 60. If circumstances improve enough to support 60, reversal is limited to $60 - 37.5 = 22.5$; the final three units amortise 60.

That ceiling needs a separate no-impairment balance, not merely the present asset. Subsequent CGU assessment under IFRS 15.103 remains required after the component test. A successful

component calculation does not establish that the wider CGU has no impairment.

The same history discipline applies to repeated revenue modifications. Under IFRS 15.21(a), eight remaining distinct units with consideration 80, followed by four additional units for 60 that do not form a separate contract, leave 12 units and consideration 140. Delivery of three recognises 35. Adding three more units for 30 then leaves 12 units and consideration 135, recognised on their delivery. Earlier revenue is not reallocated.

For a single integrated obligation, IFRS 15.21(b) instead requires the cumulative effect. A contract priced at 200 and 25% complete has revenue 50. Changing price to 260 and admitted progress to 20% changes cumulative revenue to 52, a catch-up of 2. Later 50% progress yields 130. A further modification to price 300 and progress 40% reduces cumulative revenue to 120, a catch-up of negative 10; completion ultimately brings it to 300. Arbitrary mixtures of distinct and integrated promises require their own allocation, not a selection of whichever result is convenient.

9.14.5 A tax benefit retains the origin of the underlying item

Acquisition-related recognition depends on information as well as dates. Under IAS 12.68 and IFRS 3.45–46, a newly recognised acquired deferred tax benefit from acquisition-date information within the open measurement period reduces goodwill, with any excess recognised in profit. A benefit of 20 against goodwill 5 therefore reduces goodwill by 5 and gives tax income 15. The measurement period cannot exceed one year; elapsed time alone does not establish that the new information concerns acquisition-date facts.

Other benefits retain their independently identified origin. In particular, IAS 12.67 excludes an acquirer's own deferred tax benefit from acquisition accounting; it does not place every such benefit in profit. IAS 12.61A–63 still routes tax associated with OCI or direct-equity items to that origin. An acquirer-owned benefit of 20 with an OCI origin therefore adds DTA 20 and OCI 20, leaving goodwill and profit unchanged. Missing origin remains unresolved.

Mixed tax schedules retain one lot per entity, authority, temporary difference, recovery-period rate, recognition finding and origin. For a signed difference D_i, positive for deductible and negative for taxable, admitted rate r_i and supported deductible recognition fraction f_i, the signed deferred amount is $D_i r_i f_i$; taxable lots use $f_i = 1$. Changes go to the traced profit, OCI or equity account. A supplied schedule can thus move gross DTA from 30 to 19 and DTL from 20 to 30 while allocating expense 5 to profit, 10 to OCI and 6 to direct equity.

Gross asset and liability amounts stay available even when presentation is net. The finite offset route requires the same taxable entity and authority and an enforceable right to set off current tax under IAS 12.74. It does not implement the separate cross-entity simultaneous-settlement route. Neither a blended rate nor an assumed right of setoff can substitute for those findings.

9.15 Rule selection

The proposed selection procedure distinguishes operative authority from historical source support. The retained sources include official AASB compilations, the original EU incorporating regulation and official ASBE lease and income-tax texts (Australian Accounting Standards Board, 2022g,d, 2023c,b; European Commission, 2023a; Ministry of Finance of the People's Republic of China, 2019, 2018, 2006b, 2022). Historical US lease additions are separately identified in the retained amendment source (Financial Accounting Standards Board, 2016b).

AuthorityRequirement contains issuer/framework, jurisdiction, entity class, annual-period/ap-

plication interval, paragraph and local source Ref, incorporating instrument, amendment chain, transition, elections and unresolved dependencies. RuleSelection binds those requirements to the accounting event and evidence view. The algorithm evaluates chronological applicability first, then scope and facts, then available elections/transition, then measurement/presentation. If two supported rules can apply, retain the explicit alternative and required election; if one is contradicted by facts, mark ineligible. Missing current authority gives unresolved, never eligibility through a familiar rule name.

An unchanged historical report remains an assertion under its own basis and vintage. A harmonized comparison creates an adjustment book and reconciliation, with changed recognition/measurement explained. Label mapping alone cannot convert US GAAP to IFRS or ASBE. The four clocks and source version remain part of each rule dependency; later knowledge cannot leak into an original as-of view. Transition entries reconcile opening assets, liabilities, tax and equity, rather than changing old source bytes or plugging an unexplained amount.

9.16 IFRS and incorporating jurisdictions

The retained official AASB texts provide explicit IAS/IFRS incorporation for their stated entity/version scope. AASB 9/15 compilation dates are 31 December 2022 with the source covers' application window; AASB 16 compilation 8 is 31 December 2023, AASB 112 compilation 7 is 29 September 2023. Their filenames are not date evidence. AASB 102's older retained edition is selected historical support for inventory, not current 2024 applicability certification. The retained EU 2023/1803 original provides official EU-adopted IFRS 3/10/IAS 19/21/36/37/IFRS 2/13 passages inspected in this method investigation; its historical adoption scope and later amendments remain separate. The retained amendment investigation supplies additional source evidence, not a blanket acceptance of all rules in these standards.

Initial real-method admission is therefore claim-specific: the selected revenue, lease/tax, financing, credit, group/FX and insurance cases must resolve each named applicability prerequisite. For synthetic demonstrations, declare the chosen dated provisions and assumptions and use the synthetic book mode. No assertion of whole-IFRS compliance follows from those entries.

The base lease in the saved dossiers (two annual 121 payments, 10% rate, no qualifying exemptions or extra costs) has liability 210 initially, 110 after year one, 0 after year two. IFRS-equivalent cost-model ROU is 210/105/0; expenses 126 then 116. Income tax needs its own legal deductions and gross temporary differences, as the connected corporate case demonstrates. Revenue timing, inventory cost formula, financial instrument classification, control and contract boundaries are required rule predicates, not accounting options chosen after seeing which outcome fits best.

9.17 US GAAP proposed methods and authority limitation

The saved official FASB ASU 2016-02 Section A and later 2020-05/2021-09 amendments contain inspected historical ASC 842 additions. Their covers explicitly say an ASU is not authoritative. Current Codification access returned 403 again in this source investigation; the proposed methods cannot be marked current-authority-admitted. ASU amendments, historical FAS 109, issuer accounting policy and a model's memory are all insufficient substitutes for the operative ASC text and full applicable amendment chain.

The same lease facts include ten-year remaining equipment life, fair value 1000, alternative use to lessor, no ownership transfer, no purchase option, two-year term and PV 210. Under

the inspected historical 842-10-25-2/25-3 provisions, the stated facts support the proposed operating classification. Historical 842-20-25-6 and 35-3 yield single lease expense 121 per year, liability 210/110/0 and ROU 210/ 110/0. First year debit lease expense 121/credit liability 21/credit ROU 100; payment debits liability 121/credits cash 121. Next expense 121 decomposes into liability interest 11 and ROU reduction 110. The finance comparison uses the separate depreciation 105 plus interest 21/11 path. Changing classification facts is not a free election. The 75%/90% heuristics in the historical guidance cannot become universal requirements. This complete numerical contrast is useful despite the remaining current-authority limitation.

Proposed ASC 740 temporary-difference treatment retains tax base, reversal dates, enacted rates, valuation allowance and source-specific allocation/presentation. The saved historical FAS 109 and ASU 2019-12 provide inspected enactment support, not a completed current ASC 740 implementation. Under the matched PPE case, book 800/tax 600 at assumed 25% yields DTL 50 and current tax 75 on taxable 300. A future 30% rate that is merely substantively enacted but not legally enacted cannot be imported from the AASB 112 rule into this proposed US treatment; the declared 2025 closing temporary difference is 300, giving DTL 75 at the enacted 25% rate versus DTL 90 at 30%. This deferred-tax comparison is separate from current tax 75 on taxable profit 300. Actual legal enactment remains a separate jurisdictional fact.

The proposed CECL method for in-scope amortised-cost assets estimates lifetime expected cash shortfalls from origination using past/current/forecast evidence, with a documented reversion method beyond a reasonable/supportable forecast horizon and applicable instrument exceptions. It retains allowance roll-forwards, write-offs, recoveries, purchased-credit-deteriorated handling and off-balance commitments separately. It must not copy IFRS 9's 12-month/SICR staging to the same record. A candidate contrast: lifetime discounted loss 6 with 2 attributable to default events in the next year gives IFRS general Stage 1 candidate 2 versus CECL lifetime candidate 6, conditional on scope and method support. No current ASC 326 text was retrieved, and the attempted official amendment path returned 403; this contrast remains an explicitly unadmitted proposed reading, not a source-verified US requirement or posting example. Full current ASC 326, insurance 944, consolidation 810, FX 830 and relevant later amendments are specific internal source tasks before corresponding real admission.

9.18 ASBE lease adoption and deferred tax

For an ASBE lease, selecting the lease standard and selecting the related income-tax treatment require different application dates. Article 68 of the revised ASBE 21 states a general date of 1 January 2019, but the Ministry of Finance's 2018 issuance notice specifies the enterprise groups. Paragraph I requires application from that date for enterprises listed both domestically and overseas, and for overseas-listed enterprises preparing financial statements under IFRS or ASBE. Other enterprises applying ASBE must apply the revision from 1 January 2021 (Ministry of Finance of the People's Republic of China, 2018, issuance notice, para. I).

Paragraph II permits earlier application for enterprises whose parent or subsidiary is listed overseas and prepares its overseas financial statements under IFRS or ASBE. Their early date cannot precede simultaneous application of the specified 2017 financial-instruments and revenue standards. Paragraph III discontinues the 2006 lease standard and its application guidance when the enterprise applies the revision. Thus a 2020 reporting date alone does not establish use of the revised lease standard for an enterprise in the second group: its qualifying early adoption would need evidence. The retained official Chinese texts include the 2018 notice and attachment and the Ministry's later ASBE 21 page, published on 28 October 2019 (Ministry of Finance of the People's Republic of China, 2018, 2019). They replace the earlier secondary reproduction as support for the inspected wording.

Articles 4–8 require control of identified use, economic benefits and direction; article 6 includes both conditions for substantive substitution and the express presumption when difficult to determine. Articles 14–18 specify commencement, ROU cost and PV, using implicit rate or incremental borrowing rate if the former cannot be determined. Articles 20–23 separate cost-model depreciation and periodic interest; articles 25–29 distinguish reassessment/index changes/modifications and their discount-rate rules. Articles 30–33 define the exemptions; a two-year term is not short-term. Articles 60–67 specify transition choices and disclosures; the chosen transition cannot be inferred from a present closing liability (Ministry of Finance of the People's Republic of China, 2018, attached standard, arts. 4–8, 14–33, 60–67).

For the declared base lease, these inspected recognition/measurement provisions support proposed ROU 210/105/0 and liability 210/110/0 with expense 126/116, under the same stated facts, depreciation assumption and fixed rule version. A same-asset modification is tested against 28–29, rather than labeled a separate lease without increased scope and commensurate consideration. A 2019 first-adopter using article 61(2) records cumulative opening changes without adjusting comparatives, while 61(1) is retrospective; comparison datasets must retain that distinction and the entity's actual election.

The tax question is why equal opening lease balances can still require separate deferred-tax assets and liabilities. The Ministry's original ASBE 18 defines tax bases and temporary differences in articles 4–9 and contains initial-recognition exceptions in articles 11 and 13. Article 13 also limits a deferred-tax asset to the probable taxable profit available to use the deduction. Articles 16–23 govern measurement, recovery or settlement assumptions, subsequent review, allocation and noncurrent classification (Ministry of Finance of the People's Republic of China, 2006b). The original text must be read with later interpretations.

Interpretation 16 removes the initial-recognition exemption for a single transaction that is not a business combination, initially affects neither accounting profit nor taxable income or loss, and creates equal taxable and deductible temporary differences. Section I(a) expressly includes qualifying initial lease liabilities and right-of-use assets; the corresponding deferred-tax liabilities and assets are recognized separately under ASBE 18. Equal carrying amounts alone do not establish equal temporary differences, and the asset's recoverability condition still applies. Section IV makes this treatment effective from 1 January 2023; Section I(b) permits early application from the year of issuance, 2022, with disclosure (Ministry of Finance of the People's Republic of China, 2022, secs. I(a)–(b), IV). This tax date is distinct from the enterprise's lease-standard adoption date. The operative text now supports the treatment previously illustrated by the retained official 2023 tax application case.

For the synthetic two-year lease, retain the assumed payments of 121, 10% discount rate and straight-line depreciation. All amounts use the same illustrative monetary unit. Assume that tax deductions are allocated to the liability, both tax bases are zero, the applicable tax rate is 25%, and sufficient taxable profit satisfies the deferred-tax asset recognition condition. Also assume that the transaction meets Section I(a)'s conditions and falls within its application period. At commencement the ROU asset and liability are each 210. The taxable ROU difference gives a deferred-tax liability (DTL) of 52.5; the deductible liability difference gives a deferred-tax asset (DTA) of 52.5. The entry debits DTA and credits DTL by those amounts. At the first annual close, liability 110 and ROU 105 give DTA 27.5 and DTL 26.25. Reducing the DTA by 25 creates deferred-tax expense; reducing the DTL by 26.25 creates a benefit, leaving net benefit 1.25. Both balances reverse by expiry. Applying the old exemption to omit both opening deferred balances is wrong for these qualifying facts and dates: it conceals the required gross balances and the later tax effect.

Transition also needs more than an opening journal entry. Section I(b) requires adjustments for qualifying transactions between the earliest comparative opening and application. For qualifying

existing balances at that opening, cumulative effects adjust opening retained earnings and the relevant financial-statement items. Consider a separate transition example with cash 500, ROU 105, lease liability 110 and equity 495, with no deferred tax yet recognized. Under the same assumed tax bases, rate and recoverability, recognition debits DTA 27.5, credits DTL 26.25 and credits opening retained earnings 1.25. The resulting balance sheet is

$$\underbrace{500 + 105 + 27.5}_{\text{assets } 632.5} = \underbrace{110 + 26.25}_{\text{liabilities } 136.25} + \underbrace{495 + 1.25}_{\text{equity } 496.25}.$$

The equity adjustment records the cumulative effect at the comparative opening. If the remaining lease year falls within the comparative year, the DTA of 27.5 and DTL of 26.25 reverse during that year, giving comparative deferred-tax expense 1.25. The opening adjustment alone therefore does not complete the transition. Neither ASBE 21's article 61(2) lease-adoption election nor Interpretation 16 Section III's no-comparative-adjustment rule for its separate share-payment topic may be substituted for Section I(b)'s tax transition.

These are gross recognition and measurement amounts. Separate recognition and noncurrent classification do not determine whether an actual balance sheet may offset DTA and DTL. That decision needs the applicable presentation provisions and entity/tax-authority facts. Actual tax law, tax bases and rates, recoverable profit, the enterprise's adoption and elections, and later amendments remain to be established before applying the example to an issuer. Full ASBE 22 credit, consolidated statements, group combinations and insurance revisions also retain specific source dependencies; similar amounts do not establish full IFRS equivalence.

9.19 Reading an actual lease transition

Ping An's 2019 interim report shows what happens when the dated lease rules meet an actual disclosure. The group reports adoption of revised ASBE 21 on 1 January 2019 and leaves its 2018 comparative financial statements unadjusted. Its stated listings in Shanghai and Hong Kong place it within the enterprise class covered by the 2018 notice's 2019 application date (Ping An, 2019, pp. 112–113); (Ministry of Finance of the People's Republic of China, 2018, notice, para. I). This supplies an actual adoption fact for the representation. The later deferred-tax treatment described above still requires its own applicable date and tax facts.

The immediate reconstruction problem is to keep three observations apart: the original December presentation, the January adoption balances and the June closing balances. A right-of-use asset records the group's recognised right to use leased property; the lease liability measures the associated recognised payment obligation. Table 9.1 retains the amounts and their dates. All amounts in this example are displayed millions of renminbi; arithmetic on them does not establish exact underlying amounts or the issuer's rounding rule.

Table 9.1: Adoption balances and later carrying balances answer different questions.

Observation	Right-of-use assets	Lease liabilities
31 December 2018, unadjusted comparative	—	—
1 January 2019, reported at adoption	18,500	17,667
30 June 2019, carrying balance	17,390	17,212

The comparative dashes reproduce the presentation under the previous basis; they do not say that the group had no lease contracts in December. The January amounts come from the policy-change note, whereas the June amounts come from the consolidated balance sheet (Ping An, 2019, pp. 99–100, 112). Subtracting a December dash treated as zero from the June liability would mix recognition at transition with six months of subsequent activity.

The opening asset exceeds the liability by $18{,}500 - 17{,}667 = 833$. That difference alone does not identify an increase in retained earnings. Article 61(2) permits specified alternative measurements of the right-of-use asset for former operating leases, including a liability-based amount adjusted for prepaid rent; its cumulative approach adjusts retained earnings and other relevant financial-statement items (Ministry of Finance of the People's Republic of China, 2018, attachment, art. 61(2)). To see why the other items matter, suppose solely for illustration that pre-existing prepaid rent of 833 is reclassified. Debiting the new right-of-use asset by 18,500 and crediting the new liability by 17,667 and prepaid rent by 833 balances the entry with no equity movement. Assigning the same residual entirely to retained earnings would instead require a different assumption about all other transition adjustments. The report does not identify the illustrative prepayment or supply a complete opening journal. Reconstruction therefore retains the unexplained components rather than creating an equity plug.

The policy note nevertheless narrows the permitted interpretation. It says the opening liability discounts remaining payments at the lessee's incremental borrowing rate on 1 January 2019. It names five simplifications: treating operating leases with a remaining term shorter than twelve months as short-term leases; excluding initial direct costs at transition; using one discount rate for leases with similar characteristics; using actual option exercise and other recent information to determine the lease term; and assessing pre-existing onerous contracts instead of a separate right-of-use impairment test. These correspond to the specified provisions of Article 63, with the rate and direct-cost choices appearing in the same numbered provision (Ping An, 2019, p. 113); (Ministry of Finance of the People's Republic of China, 2018, attachment, art. 63). The record should preserve the reported elections without assigning an unreported rate or deciding

which asset-measurement alternative was used for each lease. The note also calls the difference between discounted prior operating-lease commitments and the recognised opening liability "not material". That qualification supplies neither an equality nor a numeric tolerance.

The June maturity disclosure adds a different measurement of the obligation. Under its heading for undiscounted contractual and expected cash flows, the lease-liability row gives 1,515 within three months, 5,885 in three to twelve months, 11,048 in one to five years and 485 beyond five years. Those numeric cells sum to 18,933; the immediate and no-term columns contain dashes (Ping An, 2019, pp. 164–165). The first two dated buckets sum to 7,400, but disclose no payment dates within those buckets. The 18,933 total and the 17,212 carrying liability can coexist because they measure different things. Their difference, 1,721, is not the interest expense for the half-year and does not by itself determine a discount rate.

Even on the post-adoption basis, a change in a balance is not a complete account of the transactions that produced it. The right-of-use asset falls by $17{,}390 - 18{,}500 = -1{,}110$ and the liability by $17{,}212 - 17{,}667 = -455$. Additions, remeasurements, disposals and other movements can accompany depreciation or payments. The first difference therefore does not identify depreciation, and the second does not identify cash paid. The consolidated cash-flow statement's finance-leasing business borrowing and receivable lines concern the group's financing activities as a business; their labels do not make them payments on its own lessee obligations (Ping An, 2019, pp. 106–107).

The same care prevents a false narrative conflict. In its major-contract discussion, the company says that it has no reportable custody, contracting, lease or other major-contract matters for the period (Ping An, 2019, p. 86). Preserving the company reference and the reportable-major-contract qualification gives a different assertion from saying that the consolidated group has no leases. Removing those qualifiers would create the apparent contradiction with the balance sheet.

For the proposed representation, the useful result is a set of dated, source-supported balances, policy elections and maturity restrictions. A query for the reported June carrying liability can return 17,212 with its basis and precision. A query for future scheduled payments must keep the disclosed maturity buckets and leave their internal dates unresolved. A query for the complete transition journal or the half-year's lessee cash payments must retain the missing inputs. These distinct answers let actual disclosures constrain histories without pretending that every balanced completion is the company's history.

9.20 Matched-case outcomes and decision

Input changed or withheld	Required consequence
Framework changes with same lease facts	Preserve different expense/ROU paths and source strength, not just rename accounts
Method source present but target entity's adoption unknown	Unresolved applicability, no real posting authorization
ROU/liability initially equal	Still evaluate gross tax differences and later divergent balances
Historical rule supports method but a later reporting year is requested	Recheck operative amendments/transition; no automatic extension
Issuer policy discloses LIFO, regulator text unavailable	Retain issuer assertion; do not upgrade it into universal US authority
Foreign basis label replaced by local label	Reject as a standards conversion without recognition/measurement bridge

These decisions establish a common procedure and specific proposed treatments. The recovered ASBE lease notice and tax interpretation support the stated application schedules and conditional example. Broader authority coverage remains incomplete: operative US sources, ASBE sector methods, actual tax law and presentation conditions, and the applicable amendment and entity-election evidence still require internal investigation.

Chapter 10

Financing, ownership and the group boundary

A company can preserve the same operating assets while changing who supplies its funding and who shares its results. The following cases move from debt cash flows to ownership, then to the reporting boundary of a group. Each numerical example states its own contract; the coupon-bearing convertible below and the later zero-coupon compound example are different instruments.

10.1 Debt and hybrid financing

The contract layer for a debt instrument stores issuer and holder perspective, currency, issue and settlement dates, issue proceeds and legal face amount. Its payment terms include coupon/index/spread, reset and fixing dates, day-count convention, payment calendar, amortisation schedule and maturity. Its rights and protections include seniority, collateral, covenants, options, conversion terms, guarantees and linked hedges. Missing terms are explicit unknowns. A single carrying amount cannot stand in for this record.

The accounting layer stores the classification decision and its evidence, initial measurement, carrying amount, accrued interest, effective-rate method, unamortised fees or discounts, fair-value disclosures, impairment or credit adjustments where applicable, and presentation mappings. Contractual interest, effective-interest expense, realised cash payments and fair-value changes are separate quantities. The zero-coupon example in Chapter 8 provides a minimal conformance case for this separation.

The lifecycle must support issuance, recognition, accrual, reset, coupon, principal repayment, modification, derecognition, default, recovery and conversion. A modification event first determines which accounting operation the selected standard requires; it does not always mean issuing a new loan. An embedded option can affect valuation and classification without being exercised. A derivative linked as a hedge requires both the economic contract and any applicable accounting designation; economic risk reduction alone does not establish hedge-accounting eligibility.

Hybrid instruments make interpretation particularly consequential. IAS 32's overview discusses liability/equity classification and the separation of a compound instrument (IFRS Foundation, 2026a). The executable classification must consider the full terms and the selected standard version. US GAAP or ASBE treatment must be implemented under its own rules rather than inferred from an IFRS classification. Where the terms are not disclosed sufficiently, alternative

classifications can coexist only if each is actually compatible with the known terms and applicable rules.

10.1.1 Apple: debt balances and cash repayment

Apple's borrowing shows why the distinction between a contract and its carrying amount matters for a funding forecast. A fall in reported debt might suggest how much cash the company repaid. Its annual disclosure lets us test that interpretation by separating principal from the adjustments included in the carrying amount. The following amounts are in millions of US dollars at the displayed precision; changes run from 30 September 2023 to 28 September 2024 (Apple Inc., 2024a, Note 9, printed pp. 43–44).

Term-debt component	2023	2024	Change
Principal	106,572	97,341	$-9,231$
Net premium, discount and issuance costs	-356	-321	$+35$
Hedge accounting fair-value adjustment	$-1,113$	-358	$+755$
Carrying amount	105,103	96,662	$-8,441$

The cash-flow statement reports no net term-debt issuance proceeds and cash repayments of 9,958 million for fiscal 2024 (Apple Inc., 2024a, printed p. 33). To connect that cash movement to the table, first calculate the difference between the change in reported principal and net term-debt cash flow. It is $-9{,}231 - (-9{,}958) = 727$ million. Adding the reported measurement-component changes then gives the carrying-amount movement. With Δ denoting closing less opening balance, the two displayed-amount identities are

$$\Delta\text{principal} = -9{,}958 + 727 = -9{,}231,$$
$$\Delta\text{carrying amount} = -9{,}958 + 727 + 35 + 755 = -8{,}441.$$

The 727 is an unallocated balance-to-cash difference. It does not identify an exchange movement or a journal. Likewise, the 35 is the change in a net unamortised balance, rather than an identified issuance fee or interest charge; the 755 is a change in the debt's hedge adjustment, rather than a cash receipt. Together they explain 790 of the 1,517 million difference between the carrying-amount movement and net cash repayment. A forecast that used the 8,441 carrying decline as repayment would therefore compute the wrong cash amount for this observed year. The cash statement's proceeds measure is net; it must not automatically become legal face issued in another period.

The maturity schedule answers a different funding question. Its six future principal buckets sum to 97,341 million, with 10,930 in fiscal 2025. The current carrying amount is 10,912. Their 18 million difference does not allocate a discount or hedge adjustment to the current portion. The measurements differ, and their time boundaries must also be retained: Note 1 supplies the fiscal-year default, the issuance table explicitly labels its 2024–2062 range as calendar years, and MD&A separately reports amounts payable within twelve months. MD&A also reports future term-debt interest payments of 38.5 billion dollars, including 2.6 billion within twelve months (Apple Inc., 2024a, printed pp. 25, 34 and 43). Those cash requirements cannot replace period interest expense or a present value. The dated Level 2 fair-value disclosures likewise measure something different: the annual note reports 88.4 billion dollars at September 2024, while the interim note reports 82.7 billion at December 2024 alongside a rounded carrying amount of 94.8 billion (Apple Inc., 2025c, Note 6, printed p. 10).

10.1.2 Net borrowing can conceal refinancing

Commercial paper exposes a second limitation of aggregate financing data. Apple describes unsecured short-term promissory notes used for general corporate purposes, including dividends and repurchases. The annual note says their maturities are generally less than nine months; that description does not identify each note or trace individual borrowed dollars to a use. At the annual statement amounts, commercial paper increases from 5,985 to 9,967 million, a rise of 3,982, while net cash proceeds are 3,960. The resulting 22 million difference remains unallocated. During the following quarter the balance falls from 9,967 to 1,995, while net cash repayments are 7,944; the corresponding unallocated difference is −28 million (Apple Inc., 2024a, Note 9 and balance and cash-flow statements) (Apple Inc., 2025c, Note 6 and balance and cash-flow statements).

Even a complete explanation of those differences would leave refinancing activity unresolved. Annual Note 9 reports net cash for paper with maturities of at most 90 days, but separate proceeds and repayments for longer maturities. In fiscal 2022, for example, the longer-maturity proceeds of 5,948 and repayments of 7,257 give net repayments of 1,309 million. Adding short-maturity net proceeds of 5,264 gives total net proceeds of 3,955. For fiscal 2024 the longer-maturity cash rows are zero and the short-maturity net inflow is 3,960. Neither that net amount nor a closing balance reveals how many issues and redemptions occurred between reporting dates.

Consider a hypothetical incremental borrowing of 1,000 million US dollars, issued at par and repaid after 60 days. Stipulate simple interest of 5% using actual days over a 360-day year, with no fees, taxes, hedges or early settlement. The rate is an assumption of this example; Apple's disclosed 5.00% weighted rate at September 2024 does not establish the rate on such a transaction. Issuing and repaying the principal adds $1{,}000 - 1{,}000 = 0$ to net principal cash flow and leaves zero principal outstanding, yet supplies 1,000 million between those dates. The cost over the 60 days is

$$\text{interest} = 1{,}000 \times 0.05 \times \frac{60}{360} = \frac{25}{3} \approx 8.3333 \quad \text{million US dollars.}$$

Issue debits cash and credits principal 1,000. Accrual debits interest expense and credits interest payable 25/3. Settlement debits principal 1,000 and interest payable 25/3, and credits cash 3,025/3. Each entry balances; after settlement the incremental liability is zero, while cash and equity are each lower by 25/3 relative to the otherwise funded baseline.

This example isolates what a net principal-flow observation fails to reveal. It is not a reconstructed Apple history: interest expense, interest cash and other disclosures could distinguish or exclude the added borrowing. Within that scope, it shows why a financing model needs dated issuance, accrual and redemption events. Matching a net financing row does not establish either intraperiod liquidity or funding cost.

10.1.3 A rate scenario needs an exposure definition

The contract's coupon is also insufficient to describe exposure after hedging. Apple's annual debt note states that interest-rate swaps effectively convert some fixed-rate US-dollar notes to floating rates, while cross-currency swaps effectively convert some foreign-currency notes to US-dollar notes. The effective-rate ranges include interest, discount or premium amortisation and applicable hedge adjustments. These are relationships between separate contracts and measurements, not a single rate that can be applied to total principal (Apple Inc., 2024a, Note 9, printed p. 43).

The filing supplies a useful, more specific constraint. Annual Item 7A reports that a hypothetical increase of 100 basis points at all tenors, including associated derivatives, would increase annual term-debt interest expense by 139 million dollars at September 2024 and 194 million at September 2023 (Apple Inc., 2024a, printed p. 27). Multiplying Note 4's 2024 interest-rate contract notional of 14,575 million by 1% instead gives 145.75 million. It does not reproduce the reported sensitivity, even allowing for nearest rounding of the displayed amounts. The difference cannot be assigned to a particular reset date or swap feature from these aggregates. For 2023 the same shortcut gives $19{,}375 \times 0.01 = 193.75$, consistent with rounded 194; that numerical match does not validate the shortcut or recover the issuer's model.

The scenario's measure matters as much as its shock. In the same table, the 2024 investment-portfolio response is a 2,755 million decline in fair value. It cannot be added to an annual interest-expense increase as though both were the same loss measure. At December 2024 the reported interest-rate notional is still 14,575 million, while the carrying amount of term debt subject to fair-value hedges is a rounded 13.3 billion, compared with 13.5 billion at September. These aggregates describe different populations or measurements. The interim statement that market risk has not changed materially refers the reader to annual Item 7A; it does not publish an updated 139 million sensitivity (Apple Inc., 2025c, Note 4 and Part I, Item 3).

Hedge accounting and cash presentation also remain distinct. Annual Note 1 presents derivative assets and liabilities at gross fair values. Note 4 classifies derivative cash flows with the items hedged, which are *generally* operating activities; that qualifier does not classify every derivative settlement as operating. The maximum disclosed horizon for hedging debt-related foreign-currency cash-flow variability is 18 years, and Apple states that some exposures may be left unhedged and that offsets may be partial (Apple Inc., 2024a, Notes 1 and 4, printed pp. 34 and 37–38).

Finally, Item 7A's currency-risk statistic has its own population. Apple reports a Monte Carlo estimate, at 95% confidence over one day, of 538 million dollars at September 2024 and 669 million at September 2023 for adverse fair-value changes in its foreign-currency derivatives. The assessment excludes underlying currency exposures, whose changes generally offset derivative changes. This derivative VaR is neither a hard maximum loss nor a measure of the company's net currency risk; the issuer's simulation has not been replicated here. A financing simulation can retain these dated scenarios as disclosed constraints, but still needs actual holdings, currencies, payment calendars, reset terms and hedge relationships to calculate future cash and accounting consequences. The next question is how financing choices change the quantities and claims of the company's owners.

10.2 Equity and ownership

Equity has both monetary accounts and security quantities. The security register records share class, issued and outstanding quantities, treasury holdings, par or stated value where applicable, voting rights, dividend rights, redemption features, options and conversion relationships. Monetary equity accounts include contributed capital, reserves, retained earnings and other components required by the selected basis. These records evolve together through issuance, repurchase, cancellation, split, dividend and conversion events.

A stock split changes quantities and per-share measures without necessarily changing aggregate monetary equity. A cash dividend affects distributions and cash rather than current operating expense under the corresponding ordinary equity treatment. Repurchases and employee share awards require their own rules. Earnings per share uses a dated quantity schedule and rules for potential ordinary shares; simply dividing profit by the closing share count is insufficient when

shares change during the period.

Funding simulations must connect the quantity register to market assumptions. An equity issue at a scenario-dependent price changes cash raised, shares and dilution. The model must state whether it targets a fixed cash amount, a fixed share count or a capital ratio. These are different management decisions even when the historical statement would report the same aggregate proceeds.

10.2.1 A share history changes the denominator

A company can earn the same profit with different numbers of shares outstanding during the year. Dividing by the closing count would attribute the entire year's earnings to shares that may have existed for only a few months. Basic earnings per share instead relates earnings attributable to ordinary shareholders to the time those shares were outstanding. Diluted earnings per share asks a further, conditional question: how would that measure change if eligible potential shares had become ordinary shares? Neither calculation exercises an option or converts a bond in the company's actual accounts.

The following synthetic case uses IAS 33 in the retained original EU Regulation 2023/1803, including its integral Appendix A. It develops a method under that historical text; applying it to a real issuer requires the issuer's reporting basis, operative version and instrument terms. It is separate from the Apple disclosures that follow (European Commission, 2023a, IAS 33, paragraphs 2–4, 9–12, 30–36). Assume one fully paid ordinary-share class, one consolidated reporting population, and a 365-day year from 1 January to 31 December 2025. Monetary amounts are currency units; share counts are numbers of shares, so EPS is currency units per share.

The year opens with 1,000 ordinary shares and no treasury holdings. On 1 April, consideration becomes receivable for 200 new shares at 12 each and is paid that day: cash and contributed equity rise by 2,400. On 1 July, a two-for-one split doubles every outstanding ordinary share. On 1 October, the company buys 300 shares for 7 each and holds them in treasury. Cash and equity fall by 2,100; issued shares remain 2,400, treasury shares become 300, and outstanding shares become 2,100. The split contributes no additional resources. IAS 33 therefore restates earlier share counts onto the new share basis, whereas the resource-changing issue and repurchase enter the average from their respective dates (European Commission, 2023a, IAS 33, paragraphs 19–21, 26–29).

Interval, including first date	Days	Actual outstanding	Split-adjusted
1 January–1 April	90	1,000	2,000
1 April–1 July	91	1,200	2,400
1 July–1 October	92	2,400	2,400
1 October–1 January 2026	92	2,100	2,100

Each interval excludes its end date, so no day is counted twice. The basic denominator is consequently

$$D = \frac{90(2000) + 91(2400) + 92(2400) + 92(2100)}{365} = \frac{162480}{73} \approx 2225.7534.$$

The closing count of 2,100 answers a different question. Nor should the split-adjusted count overwrite the historical register: there really were 1,000 outstanding shares in January. If the comparative year had constant shares of 1,000 and ordinary earnings of 6,000, its previously calculated EPS of 6 would be presented as 3 after this split. The same retrospective principle

applies to a split occurring after year-end but before authorization of the statements. A later cash issue does not receive that treatment (European Commission, 2023a, IAS 33, paragraphs 64–65, 70(d)–71).

10.2.2 Attribute earnings before testing potential shares

Suppose consolidated profit is 15,000 from continuing operations and 18,000 in total. Non-controlling interests receive 1,000, all from continuing operations. A cumulative, nonparticipating preference class, classified here as equity, has a current-year entitlement of 2,000. Deduct that entitlement even though it has not been declared. A payment of 500 of prior-year arrears is not another deduction from this year's earnings. These are supplied attribution facts; the EPS calculation does not infer them from cash dividends (European Commission, 2023a, IAS 33, paragraphs 12–14 and A1).

Earnings attributable to the parent's ordinary shareholders are thus $15000 - 1000 - 2000 = 12000$ from continuing operations and $18000 - 1000 - 2000 = 15000$ in total. Discontinued operations contribute 3,000. Basic continuing EPS is $12000/D \approx 5.3914$; basic total EPS is $15000/D \approx 6.7393$. Their difference, approximately 1.3479, is basic discontinued EPS. Profit already includes the interest and tax effects described below; adding interest again at this stage would be wrong.

There are three potential-share series. First, cash options have been outstanding all year: 200 shares at an exercise price of 4 before the split, and 400 at 2 afterward. Use a supplied average market price of 5 on the post-split basis. Earlier prices must be converted to that same basis. The options require no future employee services and contain no performance condition, debt tender, redemption requirement or cash-settlement alternative.

Assumed exercise would provide $400(2) = 800$. At the average market price, 800 corresponds to 160 fairly priced shares. Only the remaining $400 - 160 = 240$ shares enter the diluted denominator; there is no earnings adjustment. Adding all 400 shares would ignore the assumed proceeds. IAS 33 describes this as the difference between assumed issue and the shares represented by proceeds at the average price (European Commission, 2023a, IAS 33, paragraphs 45–47 and A4–A5). The calculation records neither cash of 800 nor an actual share issue.

Second, a convertible debt series is issued on 1 July. Its original terms permit voluntary conversion into 600 post-split ordinary shares; conversion is neither mandatory nor subject to a cash alternative. For this case, supplied issue proceeds of 10,000 divide into a liability of 9,200 and conversion equity of 800. The recognized effective-interest expense for the remaining year is 800, with a related tax benefit of 200 and no other earnings consequences. The original terms pay a coupon of 400 on 1 October and principal of 9,200 plus a final coupon of 400 on 1 January 2026. The supplied effective rate is 1/23 per 92-day period: discounting those payments gives $400/(24/23) + 9600/(24/23)^2 = 9200$. Interest of 400 first raises the liability to 9,600; the October coupon returns it to 9,200, and another 400 of interest gives a year-end liability of 9,600 before January settlement. The intervening payment explains why the two interest amounts are equal. These supplied amounts include the relevant cost and discount effects; substituting a coupon alone need not do so. Instrument classification and component valuation are separate accounting inputs, illustrated by the compound-bond example below.

Assumed conversion from issue eliminates the net expense of $800 - 200 = 600$ and adds shares only for the 184 days that the instrument exists. Its contributions are

$$\Delta N_b = 600, \qquad \Delta D_b = 600\frac{184}{365} = \frac{22080}{73}, \qquad \frac{\Delta N_b}{\Delta D_b} = \frac{365}{184} \approx 1.9837.$$

IAS 33 requires the after-tax interest effect and any consequential earnings changes, together

with this issue-date weighting (European Commission, 2023a, IAS 33, paragraphs 33–39, 49–50). For example, a profit-sharing expense caused by removing interest would also change the numerator; it cannot disappear merely because a coupon and a tax rate are available.

Third, the preference class can convert into 400 post-split ordinary shares and has existed throughout the year. Its assumed conversion adds back the current entitlement of 2,000. Earnings per incremental share are therefore $2000/400 = 5$. The class has no participation in residual earnings; participating rights would require a different attribution calculation.

10.2.3 The order of dilution changes the answer

Continuing-operations earnings attributable to ordinary shareholders, 12,000 here, control admission. Each series is assessed separately in ascending order of earnings per incremental share: options at zero, debt at about 1.9837, then preference shares at 5. Recalculate after each admitted series; comparing every instrument only with basic EPS loses this dependency (European Commission, 2023a, IAS 33, paragraphs 41–44).

The options reduce continuing EPS from approximately 5.3914 to 4.8667. The debt then reduces it further:

$$D_{\text{dil}} = D + 240 + \frac{22080}{73} = \frac{202080}{73}, \qquad E_{\text{dil,cont}} = \frac{12000 + 600}{D_{\text{dil}}} = \frac{7665}{1684} \approx 4.5517.$$

The preference class would increase this result because its incremental ratio, 5, exceeds 4.5517. Exclude it. Yet $5 < 5.3914$, so it would have appeared dilutive if tested against basic EPS alone. That is why the order matters.

The admitted population also determines the other EPS amounts. Total earnings increase from 15,000 to 15,600 through the same debt addback; discontinued earnings remain 3,000. Thus

$$E_{\text{dil,total}} = \frac{15600}{202080/73} = \frac{4745}{842} \approx 5.6354, \qquad E_{\text{dil,disc}} = \frac{3000}{202080/73} = \frac{1825}{1684} \approx 1.0837.$$

Continuing and discontinued EPS sum to total EPS before presentation rounding. The report retains that reconciliation, each admitted numerator and denominator increment, and the excluded preference series. Basic and diluted EPS receive equal prominence; negative EPS is still presented (European Commission, 2023a, IAS 33, paragraphs 66–70).

Two sign changes test the control number. If continuing ordinary earnings are instead a loss of 5,000 while total ordinary earnings remain a profit of 15,000, none of these three series is admitted: each would reduce the loss per share from continuing operations. Total profit cannot reverse that decision. Conversely, retain continuing earnings of 12,000 but change total ordinary earnings to a loss of 9,000. Options and debt still enter, so diluted total EPS uses $-9000 + 600 = -8400$ over the same diluted denominator. It is less negative than basic total EPS. That is the consequence of the continuing-operations control, illustrated explicitly in the standard's Appendix A (European Commission, 2023a, IAS 33, paragraphs 41–43 and A3).

10.2.4 Actual conversion, expiry and incomplete terms

Actual events divide the year differently. Suppose the debt pays its coupon and then converts under its original terms on 1 October; interest ceases that day. Before conversion, recognized interest is 400 with a tax benefit of 100. Paying the coupon reduces the liability to 9,200, which transfers to ordinary equity; the original conversion-equity component remains in equity. After conversion, 600 actual shares enter basic EPS for 92 days. The diluted calculation adds only the preceding 92 days of potential shares and the preconversion net expense of 300. There are

therefore $600(92 + 92)/365$ additional share-years in the diluted comparison, with no overlap. Under the otherwise identical supplied earnings assumptions, basic ordinary earnings rise by the 300 of avoided later expense, and the remaining addback of 300 restores the same fully diluted numerator as before. Treating the potential shares as outstanding for all 184 days *in addition* to the actual shares would double count (European Commission, 2023a, IAS 33, paragraphs 21(c), 36–38).

Expiry has no subsequent actual-share leg. If the options instead lapse on 1 October, their contribution is $240(273/365)$, using a supplied average price of 5 for this variant's relevant period. They do not vanish from the earlier part of the year. An exercise price equal to or above the average price gives zero incremental shares in this simple option profile. More generally, a conversion that leaves continuing EPS exactly unchanged is not dilutive: paragraph 41 requires a decrease in profit per share or an increase in loss per share.

Employee awards require another input even when their other terms fit the option calculation. Under paragraph 47A, assumed proceeds include the IFRS 2 value of future goods or services. If that supplied value were 400 across these 400 options, the assumed price would be $2+400/400 = 3$, and the increment at market price 5 would be 160 shares. This conditional arithmetic neither values the services nor calculates award expense. Performance-based awards follow the contingent-share provisions instead; share-or-cash alternatives, written puts, rights issues and participating or partly paid shares also need their own methods (European Commission, 2023a, IAS 33, paragraphs 47A–63, A2 and A13–A16).

An EPS assessment therefore references a dated book, reporting population, share class, attributed earnings, share history, prices and complete instrument terms. It produces report amounts and their reconciliation, without changing the book. Missing average prices, future-service values, conversion consequences or settlement terms leave diluted EPS unresolved even when basic EPS is available. Missing attribution or ordinary-share timing can prevent both. Known complex terms select the relevant further method; they are not grounds for silently dropping the instrument. These distinctions let the following issuer disclosures supply facts without being mistaken for a complete method or a complete history.

10.2.5 Apple: shares through time

Apple's disclosures show why the ownership record needs dates as well as quantities. In fiscal 2024, its opening 15,550,061 thousand common shares, less 499,372 thousand repurchased and plus 66,097 thousand issued net of employee-tax withholding, give 15,116,786 thousand closing shares. That bridge accounts for the change in the reported stock. It does not say how long each share participated in the year's earnings. The reported basic weighted average is 15,343,783 thousand shares (Apple Inc., 2024a, Notes 3 and 10, printed pp. 36 and 44).

The unit conversion matters before any timing inference. Net income is 93,736 million dollars, whereas the denominator is in thousands of shares. Thus $93{,}736 \times 1{,}000/15{,}343{,}783 \approx 6.109$ dollars per share, consistent with reported basic EPS of \$6.11. Using closing shares instead gives about \$6.201: it computes a different ratio. Adding the reported 64,312 thousand incremental shares from dilutive awards gives the diluted denominator of 15,408,095 thousand and EPS of about \$6.084, reported as \$6.08. These are checks using displayed amounts. Conditional nearest-rounding intervals also overlap the reported two-decimal EPS; the issuer's tie convention and undisclosed exact amounts remain unverified.

The first quarter of fiscal 2025 makes the distinction again. Income of 36,330 million divided by 15,081,724 thousand basic shares, with the same factor of 1,000, gives about \$2.409. The diluted denominator adds 69,141 thousand shares and gives about \$2.398, consistent with the reported \$2.41 and \$2.40 (Apple Inc., 2025c, Note 3, printed p. 7). Those denominators describe three

months; adding them to other quarterly averages would not produce an annual share count. Period lengths and consistent measurement rules are needed to combine averages.

Even a weighted average leaves dates unresolved. To see exactly what it constrains, consider a deliberately simplified common-share model for the 364 days of fiscal 2024. Apple reports a 52-week year ending 28 September 2024 (Apple Inc., 2024a, Note 1, printed p. 34). Let S_0 be the opening number of ordinary shares, and let ΔS_j be a signed change after d_j whole days have elapsed since 1 October 2023. Positive changes issue shares; negative changes repurchase them. An event then affects the remaining $364 - d_j$ days. Assume no splits or other quantity changes, and aggregate all net issuance into one event. The average is the opening holding plus each change weighted by its remaining time:

$$\overline{S} = S_0 + \sum_j \Delta S_j \frac{364 - d_j}{364}, \qquad 0 \le d_j < 364. \tag{10.1}$$

All quantities in this equation are ordinary shares and event days are integers. An immediate change receives full weight; a change after 363 days receives only one day's weight. Closing shares depend on the sum of changes, while equation (10.1) also constrains their timing. Neither relation separately identifies every change and date.

The construction must also respect information outside the annual EPS note. Apple reports fourth-quarter purchases of 35,697, 42,910 and 33,653 thousand shares in the intervals 30 June–3 August, 4–31 August and 1–28 September, respectively (Apple Inc., 2024a, Item 5, printed p. 19). A history putting all repurchases in the middle of the year fails that disclosure even if its annual average is correct. The later interim EPS note also reports a first-quarter fiscal 2024 basic average of 15,509,763 thousand shares. Both constraints are included in the following two authored quantity histories.

Event	History 1	History 2
First-quarter purchase	−40,298 (45)	−40,298 (45)
First-quarter purchase	−40,298 (46)	−40,298 (46)
Net issuance	+66,097 (182)	+66,097 (91)
Further purchase	−131,136 (179)	−15,643 (159)
Further purchase	−175,380 (180)	−290,873 (160)
June–August interval purchase	−35,697 (290)	−35,697 (290)
August interval purchase	−42,910 (322)	−42,910 (322)
September interval purchase	−33,653 (350)	−33,653 (350)

The table expresses changes in thousands of shares; parentheses give elapsed days, not negative signs. Each quantity is multiplied by 1,000 when used in equation (10.1). Treat the displayed source centres as exact solely for this illustration. Both histories start at 15,550,061,000 shares, repurchase 499,372,000, issue a net 66,097,000 and end at 15,116,786,000. Their first 91 days average 15,509,763,000 shares: the two equal early purchases reduce the average by $40{,}298{,}000(46 + 45)/91 = 40{,}298{,}000$. For the full year, the signed time-weighted changes sum to −75,085,192,000 share-days in each history. Dividing by 364 reduces the opening count by 206,278,000, giving the reported 15,343,783,000 average. An independent sum of daily balances gives the same result.

After 100 elapsed days, however, the first history has 15,469,465,000 shares and the second 15,535,562,000. The earlier issuance makes the difference. The fourth-quarter purchase dates lie inside their reported intervals, but none of these illustrative dates is an estimate of Apple's actual transactions. This establishes non-uniqueness for the stated quantity constraints. Prices, award-level settlement and monetary journals impose further conditions; the two examples

are not a complete reconstruction of the filing or an exhaustive family, and no probability attaches to either. A query for shares on an intermediate date must therefore retain the timing alternatives or obtain additional evidence.

10.2.6 Awards, repurchases and cash measure different movements

The next quarter shows how another disclosure can narrow an unknown. Note 7 describes about 100 million repurchased shares, while Item 2's interval table gives 41,627, 32,784 and 25,379 thousand, totalling 99,790 thousand. The balance sheet reports 15,040,731 thousand closing shares against 15,116,786 thousand at the preceding year end. At the displayed centres, the quantity bridge requires a net other increase of $15{,}040{,}731 - 15{,}116{,}786 + 99{,}790 = 23{,}735$ thousand shares (Apple Inc., 2025c, printed pp. 3, 10 and 20). This narrows the total without identifying the events composing it. Repurchase authority is different again: both the note and programme footnote explicitly say that the programme does not oblige Apple to acquire a minimum amount of shares.

Restricted stock units have their own movement record. In thousands of RSUs, the fiscal 2024 bridge is $180{,}247 + 80{,}456 - 87{,}633 - 9{,}744 = 163{,}326$: opening awards plus grants, less vesting and cancellations. The next quarter gives $163{,}326 + 61{,}391 - 35{,}683 - 2{,}278 = 186{,}756$ (Apple Inc., 2024a, Note 11, printed pp. 44–45) (Apple Inc., 2025c, Note 8, printed p. 10). These gross award movements are different from net common shares issued or incremental shares in the diluted EPS calculation. Apple reports that most vested RSUs were net share settled and that approximately 31 million shares were withheld for employee taxes in 2024. Subtracting that rounded amount from 87.633 million vested RSUs does not explain the 66.097 million net issuance line in full.

The annual report describes several award types under the 2022 employee plan and the usual terms of its RSUs. In that reported practice, RSUs generally vest over four years subject to continued employment, settle one-for-one in common shares and carry dividend-equivalent rights. "Generally" cannot determine the vesting dates of every grant. The prior plan ceased authorising new grants in the third quarter of 2022; that does not say its outstanding awards vanished. Likewise, the approximately 24 million RSUs excluded as antidilutive in fiscal 2023 are a period-specific disclosure, not a universal exclusion rule (Apple Inc., 2024a, Notes 3 and 11, printed pp. 36 and 44–45). A future dilution calculation needs the applicable award terms and EPS rules as well as quantities; the arithmetic above does not validate that methodology.

Monetary statements impose another set of distinctions. The following amounts are millions of dollars, with reductions and payments shown as positive magnitudes (Apple Inc., 2024a, printed pp. 32–33 and 45) (Apple Inc., 2025c, printed pp. 4–5 and 10).

Reported measure	Fiscal 2024	Fiscal 2025 Q1
Repurchase reduction in equity	95,846	23,476
Cash-flow repurchases	94,949	23,606
Employee-tax withholding reduction in equity	5,605	2,993
Cash-flow employee-tax payments	5,441	2,921
Share-based compensation in equity	12,034	3,383
Share-based compensation expense	11,688	3,286

The withholding reductions combine the contributed-capital and retained-earnings rows: $3{,}993 + 1{,}612 = 5{,}605$ annually and $1{,}891 + 1{,}102 = 2{,}993$ in the quarter. These differences remain unallocated here; assigning them to excise tax, transaction fees or unsettled trades would require further evidence. The narrative's $95.0 billion and $23.3 billion purchase amounts also retain

their own precision and transaction scope. The separate employee-tax settlement narrative cannot silently replace the cash-flow line.

Finally, grant-date value, vesting-date value, compensation expense and future unrecognised cost answer different questions. Apple's stated policy recognises expense on a straight-line basis for its estimate of awards that ultimately vest (Apple Inc., 2024a, Note 1, printed p. 34). Its unrecognised RSU cost increases from $19.4 billion over a weighted-average 2.4 years to $29.2 billion over 2.9 years at the quarter end. Dividing either amount by its weighted period does not disclose a constant annual expense schedule or a cash liability: different cohorts have different remaining recognition periods, and future vesting estimates can matter even under the stated straight-line policy. The ownership model therefore needs linked award cohorts, recognition periods, share events and cash settlements. Keeping those dates and measurements distinct allows an actual later grant or settlement disclosure to narrow the histories without overwriting the reported EPS or inventing a missing event.

10.3 A plan's capacity is different from an employee's award

The share register has already separated awards, vesting and common shares issued. The documents governing an award explain why those quantities can move differently. Apple's 4 March 2022 Form 8-K supplies the 2022 Employee Stock Plan, a time-based restricted-stock-unit form and a performance-award form as Exhibits 10.1–10.3. These are the counterparts of three references in the annual report. The filing reports shareholder approval and effectiveness on 4 March 2022. The plan also records Board adoption on 9 November 2021 and uses that earlier date as its defined Effective Date when setting its term. The two dates perform different jobs; replacing them with one undifferentiated date would lose information needed to interpret the plan (Apple Inc., 2022, Items 5.02 and 5.07; Exhibit 10.1, Sections 1 and 15(a)).

Start with the capacity to make awards. Section 5(a) supplies 510 million shares plus specified amounts available or recycled from the 2014 plan, subject to a maximum of 1,274,374,682. That maximum is neither the current unused capacity nor a promise to issue that many common shares. Section 5(b) counts a full-value award at two reserve shares for each underlying share, whereas exercised options and stock appreciation rights use one reserve share for each share to which the exercise relates. Options and stock appreciation rights count their gross underlying amount even when fewer shares are delivered (Apple Inc., 2022, Exhibit 10.1, Section 5(a)–(b)).

A small hypothetical award makes the distinction concrete. Suppose 100 restricted stock units were initially charged against the reserve at two for one, all 100 vest and 20 shares are withheld for employee taxes. Assume no other adjustments. The employee receives 80 shares. Under the plan's full-value withholding rule, the 20 withheld shares return 40 reserve shares, because the original charge used that ratio. The two quantities are therefore

$$\underbrace{100 - 20 = 80}_{\text{common shares delivered}} \quad , \qquad \underbrace{2(100) - 2(20) = 160}_{\text{net reserve shares used}} .$$

The reserve records authority used, rather than shares in the employee's account. An option exercise covering 100 shares can also deliver 80 after withholding, yet uses 100 reserve shares under the gross-count rule; the option withholding does not restore capacity. The same observed delivery therefore cannot identify the reserve movement without the award type and counting rule. Cash settlement creates another distinction: the plan excludes the shares that would otherwise have been delivered from its share count. These are consequences of Section 5(b), not an assumed employee tax rate, cash payment or compensation expense (Apple Inc., 2022, Exhibit 10.1, Section 5(b)).

The 2025 proxy illustrates why the reporting population must travel with each number. At 28 September 2024 its equity-compensation table reports 125,973,077 shares subject to outstanding awards under the 2022 plan, 37,387,521 under the 2014 plan and 10,612 under the director plan, totalling 163,371,210. The 2014 amount includes 39,984 options and 37,347,537 restricted stock units. Removing those options leaves 163,331,226 shares subject to restricted stock units across the three listed populations. The annual note's 163,326 thousand restricted stock units is a different reported amount; these disclosures alone do not identify a reconciliation. The proxy separately reports 1,022,428,528 shares available under the 2022 plan, 66,924,428 under the employee purchase plan and 4,110,122 under the director plan, totalling 1,093,463,078. That total includes different plans from the outstanding-award total and is governed by their counting conventions (Apple Inc., 2025b, printed p. 97 and table footnotes). A reconstruction must retain these populations rather than subtracting unlike totals to infer grants, dilution or unused authority.

Capacity still does not tell the employee what will be delivered or when. Section 9 permits restricted-stock-unit agreements to differ: the agreement specifies vesting, settlement may be in shares, cash or a combination, and delivery may be deferred under its terms. Dividend-equivalent rights may be granted; the units themselves carry no voting rights. The filed time-based form then makes a more specific promise. Its Section 7 provides for shares equal to vested units less the applicable tax-related withholding, delivered promptly and no later than two and one-half months after vesting, subject to the stated conditions. The plan's wider permission does not replace that form's narrower promise (Apple Inc., 2022, Exhibit 10.1, Section 9; Exhibit 10.2, Sections 7 and 11).

The form's grant notice leaves the recipient, quantity, award date and vesting schedule blank. Those blanks matter: the annual report's general four-year practice cannot fill every individual notice. For example, the 2025 proxy describes named executives' 1 October 2023 time-based awards as vesting in three equal annual instalments beginning 1 April 2026, over approximately four and one-half years. It also describes termination terms under which earned or vested rights and delivery dates must be kept separate: Cook's retirement treatment does not accelerate the originally scheduled delivery of shares or dividend equivalents (Apple Inc., 2022, Exhibit 10.2, grant notice and Section 8) (Apple Inc., 2025b, printed pp. 51 and 53–54). These examples establish that a general vesting convention cannot substitute for the applicable award; they do not identify every employee's terms or a complete amendment history.

For the simulation, the consequence is concrete. Retain the plan version and reserve rule alongside the actual award terms, vesting conditions, units earned, settlement medium, delivery date and withholding choice. A permitted plan alternative becomes a branch for a particular award only when its terms or explicitly stated unresolved facts allow it. Otherwise cash-settlement paths would be invented for a share-only promise, or vesting would create shares before their contractual delivery. Tax amounts, accounting expense and diluted EPS require their own facts and methods. The performance-award form poses the next question: how are units earned when the contractual performance schedule itself has not been filled in?

10.4 Performance conditions determine units before settlement

The performance form turns the distinction between a plan and a grant into a calculation problem. Its notice leaves the target quantity, grant date, performance period and vesting schedule blank. Exhibit A likewise leaves the percentile thresholds and payout percentages blank, although it specifies interpolation between thresholds, a cap when Apple's absolute return is negative, and rounding to whole units. The form therefore supplies part of the method without supplying an individual award's inputs (Apple Inc., 2022, Exhibit 10.3, notice and

Exhibit A).

The 2025 proxy provides a concrete schedule for the named executives' 1 October 2023 performance awards. It describes a three-year performance period from the beginning of fiscal 2024 through the end of fiscal 2026, with vesting on 1 October 2026 subject to the agreements. This is a dated description of that grant population, not the result of the performance period. The proxy separately reports results for awards granted on 27 September 2020 and vested on 1 October 2023. Those older results must not be attached to the newer grants merely because the same proxy reports both (Apple Inc., 2025b, printed pp. 51–53).

For the October 2023 schedule, first suppose that the relative-return percentile has been supplied correctly. Write it as p on a scale from 0 to 100, so that $p = 70$ means the seventieth percentile. Let $m(p)$ be the number of units earned per target unit before applying the negative-return cap and employment conditions. The proxy gives zero below the twenty-fifth percentile, 25% of target at that percentile, 100% at the fifty-fifth and 200% at the eighty-fifth or above. Between those points it requires straight-line interpolation (Apple Inc., 2025b, printed p. 52). In these units,

$$m(p) = \begin{cases} 0, & p < 25, \\ \dfrac{1}{4} + \dfrac{p - 25}{40}, & 25 \le p < 55, \\ 1 + \dfrac{p - 55}{30}, & 55 \le p < 85, \\ 2, & 85 \le p \le 100. \end{cases}$$

The first slope is $(1 - 1/4)/(55 - 25) = 1/40$; the second is $(2 - 1)/(85 - 55) = 1/30$. Thus a supplied percentile of 40 earns 0.625 units per target unit, while 70 earns 1.5. The threshold at 25 is a jump, not a line running from zero at some unreported lower percentile. If two unresolved rank calculations produced only $p = 24.9$ or $p = 25$, a hypothetical target of 1,000 units would yield either zero or 250 units before other conditions. Replacing those two possibilities by every quantity from zero to 250 would create unsupported outcomes.

Relative performance is only one input. Let R be Apple's absolute total shareholder return over the same performance period, expressed as a fraction. The source caps vesting at 100% of target whenever $R < 0$, regardless of relative rank. For this part of the calculation the multiplier is

$$g(p, R) = \begin{cases} \min\{m(p), 1\}, & R < 0, \\ m(p), & R \ge 0. \end{cases}$$

At the supplied percentile 70, a hypothetical target of 1,000 units therefore gives 1,500 units when the return is nonnegative but only 1,000 when it is negative, assuming the other vesting conditions are met. A company can rank well in a falling market: the rank does not identify the sign of its return. The example avoids fractional units; Exhibit A's nearest-whole-unit rule still needs an explicit treatment of exact half-unit ties before implementing all inputs (Apple Inc., 2022, Exhibit 10.3, Exhibit A).

Computing R introduces its own source requirements. The form uses twenty-trading-day beginning and ending average prices, dividend reinvestment, and adjustments for specified capital changes. Consider a deliberately simple path with no dividends, distributions or corporate actions. Suppose each of the twenty beginning prices is 100; nineteen ending-window prices are 80 and the final price is 120. The beginning average is 100 and the ending average is $(19 \times 80 + 120)/20 = 82$, giving $R = (82 - 100)/100 = -18\%$. Substituting only the first and last prices would instead give +20%. With the supplied percentile held at 70, that substitution would wrongly remove the cap and increase the hypothetical award from 1,000 to 1,500 units. Even in this simple case, the observation window changes a contractual consequence (Apple Inc., 2022, Exhibit 10.3, Exhibit A, Beginning Price and Ending Price).

The general return and rank calculation requires more than those two averages. The form defines the comparison group using initial S&P 500 membership and continued listing, with specific bankruptcy, successor and merger provisions. It ranks Apple ahead of a comparison member with equal return. That tie rule does not by itself specify a unique numerical percentile formula; a library's default rank conversion cannot silently supply one. Dividends enter the return through reinvestment, noncash distributions need values, and corporate actions need the stated adjustments. A calculation must document how those provisions are implemented, including avoiding a second addition of dividends to a price series that already includes them. Unresolved ambiguities belong with the Committee's determination under the form; they are not permission to invent historical inputs (Apple Inc., 2022, Exhibit 10.3, Exhibit A).

Employment conditions then determine which otherwise earned units survive. Under Section 8 of the filed 2022 performance form, ordinary termination generally cancels unvested rights. Death or disability before vesting instead allows a fraction of the performance-qualified award to vest on the specified vesting date. The fraction is the total performance-period days less the days remaining at severance, divided by total days. For an expressly hypothetical eligible fraction of 3/5 and the uncapped 1,500-unit result, 900 units survive to that date; if the negative-return cap first reduces the result to 1,000, 600 survive. These calculations use the filed form's terms and assume no other modification, leave adjustment or recoupment event (Apple Inc., 2022, Exhibit 10.3, Sections 3–4 and 7–9).

The proxy's termination summary must remain alongside that versioned form, rather than overwrite it. It describes pro-rata *accelerated* vesting for named executives other than Cook upon death or disability, and different provisions for Cook. The 2022 form's Section 8 instead points to the vesting date. An applicable executed agreement, amendment or further disclosure is needed to establish which timing governs a particular executive. This is an unresolved applicability and timing distinction between the inspected sources, not an established contradiction for one identified award (Apple Inc., 2025b, printed p. 53).

Dividend-equivalent rights add a cash consequence to the same conditional award. The filed form credits ordinary cash dividends using the outstanding target units and pays the vested proportion in cash, subject to withholding and the other terms. Suppose one dividend of $0.25 occurs while all 1,000 target units are outstanding before severance, and there are no other dividends or adjustments. The initial credit is $250. A 1.5 performance multiplier with full eligibility pays $375 gross; under the assumed 3/5 eligible fraction, the vested proportion is 900/1,000, so $225 is payable. Applying the negative-return cap reduces that conditional payment to $150. This cash right is different from the hypothetical dividend reinvestment used to measure shareholder return; adding it to that return would count a different entitlement (Apple Inc., 2022, Exhibit 10.3, Section 5(b)).

The model consequently needs distinct stages: determine the applicable award and observations; calculate the return and rank under evidenced conventions; apply the schedule and cap; determine surviving rights; and apply settlement and withholding terms. The checked examples cover selected consequences with explicit inputs. They do not recover the October 2026 outcome, reproduce all individual awards, value compensation expense or establish diluted EPS. Those questions require further data and their own accounting methods.

10.5 The applicable agreement changes which rights survive

The timing question can be narrowed by reading the later forms already incorporated into Apple's filings. The annual exhibit list identifies a CEO performance form effective 25 September 2022, filed as Exhibit 10.2 with the quarter ended 31 December 2022. It also includes general

and CEO performance forms effective 29 September 2024 as Exhibits 10.20 and 10.22. These are different versions and participant categories, even though their notices use similar headings and leave the individual grant particulars blank (Apple Inc., 2024a, annual exhibit list; Exhibits 10.20 and 10.22) (Apple Inc., 2023, Exhibit 10.2).

The 2024 general form does not remove the earlier timing distinction. Section 8 still provides that qualifying death or disability before vesting allows the performance-qualified award, multiplied by the severance fraction, to vest on the Vesting Date. Section 7 requires share delivery as soon as administratively practical thereafter, within two and one-half months and subject to its conditions. Thus neither inspected general form supplies a rule to deliver the shares immediately on severance. The proxy's description of accelerated vesting remains a summary whose application to a particular award needs further evidence (Apple Inc., 2024a, Exhibit 10.20, Sections 7–8) (Apple Inc., 2025b, printed p. 53). Later filing alone is not evidence that an earlier grant was amended.

The CEO forms change the surviving entitlement. In both the September 2022 and September 2024 versions, Section 8(b) allows unvested units to remain eligible for the Vesting Date following death, disability, or qualifying retirement, subject to the Committee's stated authority. These clauses do not apply the general form's severance fraction. Retirement combines an age of at least 60 and at least ten years of service with the stated service scope; the retirement exception normally also requires termination on or after the first anniversary of the Award Date. Earlier retirement does not receive that exception unless the Committee determines otherwise. Continued eligibility still depends on performance and does not make the eventual number of shares known on departure (Apple Inc., 2023, Exhibit 10.2, Sections 3–4 and 8) (Apple Inc., 2024a, Exhibit 10.22, Sections 3–4 and 8).

To isolate the effect, reuse the hypothetical 1,000-unit target and a performance multiplier of 1.5, with no modification, recoupment or other condition changing the result. Suppose death or disability occurs at a point giving a general-form severance fraction of 3/5. Under the general form, $1{,}000 \times 1.5 \times 3/5 = 900$ units remain eligible to vest. Under the CEO form's stated continuation provision, all $1{,}000 \times 1.5 = 1{,}500$ units remain eligible. With a negative-return cap reducing the multiplier to one, the corresponding quantities are 600 and 1,000. These comparisons hold the performance result fixed to expose the contract difference; they are not facts about any executive's award.

The difference carries into cash rights without creating an earlier payment. With the same hypothetical $250 dividend-equivalent credit established before departure, the uncapped gross payments are $225 under the general-form fraction and $375 under the CEO continuation provision. The units and cash rights remain linked to the same applicable terms and performance conditions. The CEO forms' Section 7 anchors delivery to the Vesting Date, and Section 5(b) ties dividend equivalents to the award's vesting and payment terms. Moving those amounts to the severance date would change the specified cash and share path (Apple Inc., 2023, Exhibit 10.2, Sections 5(b), 7–8) (Apple Inc., 2024a, Exhibits 10.20 and 10.22, Sections 5(b), 7–8).

A second boundary is independent of the performance calculation. An employee aged 60 with ten years of service has not necessarily satisfied the CEO form's first-award-anniversary condition. Conversely, passing that anniversary does not establish the age and service conditions. The record therefore needs the applicable agreement, Award Date, termination date and reason, service history and any relevant Committee decision. Unknown values remain unknown; they cannot be converted into automatic eligibility or forfeiture.

The later forms retain unfilled performance schedules and the same kinds of return, comparison-group and adjustment dependencies. They do not fill the rank convention or establish the actual October 2023 award terms by themselves (Apple Inc., 2024a, Exhibits 10.20 and 10.22, Exhibit A). The appropriate result of this comparison is a narrower conditional model: the

general and CEO forms support different survival rules, both preserve later delivery, and the proxy's timing description still needs an award-specific explanation. A source-reference match establishes which form was retained; applicability establishes which form can govern a financial path.

10.6 A compound bond exposes classification dependencies

Consider an issuer receiving 100 for a two-year bond paying annual coupons of 5 and principal 100 at maturity, with a holder option to convert that principal into ten ordinary shares. Assume its currency is the issuer's functional currency, no other options, fees or contingent settlements, and that the conversion feature qualifies as equity under the selected rule. A comparable liability without conversion yields 10%. The inspected IAS 32-equivalent text requires separate liability and equity components and measures equity as the residual (Australian Accounting Standards Board, 2022c, paragraphs 28–32, AG30–AG32).

The liability's present value is

$$L_0 = \frac{5}{1.1} + \frac{105}{1.1^2} = \frac{11050}{121} \approx 91.3223, \qquad E_0 = 100 - L_0 \approx 8.6777.$$

Issue debits cash 100 and credits these two components. Under the stated effective-interest schedule, the first coupon leaves liability $1.1L_0 - 5 = 1050/11 \approx 95.4545$; after the second coupon it is 100. Redemption debits liability and credits cash 100. Alternatively, conversion at maturity debits liability and credits equity 100 while issuing ten shares; the original equity component remains in equity. AG32 supports the absence of a gain or loss on that maturity conversion. Exact rational valuations are kept before the declared monetary rounding.

A variable-number share settlement, cash alternative or other clause can change classification. The contract record must therefore retain these terms; it cannot select liability/equity accounts independently of the coupon, conversion and settlement rules. Early repurchase and induced conversion invoke separate operations: the inspected AG33–AG35 text describes allocation and the induced-conversion consideration difference. They are not implemented by changing the maturity date of the simple example. US-GAAP and ASBE classification require their own dated authority; the IFRS-oriented split is never silently reused under another basis.

10.7 Schedules and initial classification

The following liability and ownership methods use selected dated AASB 9 and AASB 132 provisions (Australian Accounting Standards Board, 2022g,c). Issuer, holder, contractual cash flow and accounting measurement remain separate throughout the schedule.

DebtContract contains parties, currency, principal, issue/settlement dates, payment calendar, fixed/index/spread/reset terms, compounding/day-count, business-day adjustment, principal changes, fees, options, security, priority and legal default/remedy terms. Each term has evidence/admission provenance. The economic schedule emits due cash independently of the recognition method. A market-value adapter may value that schedule but cannot choose amortised cost, FVOCI, FVTPL or liability/equity classification.

An investor's classification evaluates business model and contractual cash-flow features; a debt label alone cannot establish the AASB 9.4.1.2/4.1.2A conditions. At initial recognition the selected 5.1.1 rule measures fair value and qualifying costs as appropriate. An unknown day-one observable-value condition yields unresolved under 5.1.1A/B5.1.2A. The retained acquisition cases distinguish FVTPL expensed fees, amortised-cost fees, original EIR and POCI

credit-adjusted EIR. No rate method inferred from market value can overwrite the contract rate.

For an issuer with a plain amortised-cost debt, net proceeds 100 and one legally due payment 121 at the end of two annual periods, the exact EIR solves 100=121/(1+r)^2, hence r=10%. Issue debit cash 100/credit debt 100; year-one debit interest 10/credit debt 10; year-two debit interest 11/credit debt 11; redemption debit debt 121/credit cash 121. Outstanding contractual payment stays 121 until settled even while carrying value is 100 or 110. Total interest 21 equals cash outflow 121 minus proceeds 100. An immediate transaction fee 2 changes net proceeds to 98 and the EIR root; retaining 10% is wrong. The numerical root method is in the numerical-method discussion; an irrational rate is not silently an exact rational.

A fractional period uses the declared convention. Annual-effective 21% over half a year produces factor 1.1 and interest 10 on 100; simple proration gives 10.5 and is wrong for that contract. A floating coupon resets from the admitted fixing available at the reset date, not a later observed rate. Actual day-count and holiday-calendar versions are immutable inputs. Unknown fixings preserve conditional paths, rather than defaulting to the latest value.

10.8 Modification, extinguishment and default

At a modification, retain original and replacement terms, counterparty, fees, effective date, legal-release evidence and original EIR. Determine derecognition before measurement. AASB 9.3.3.1–3.3.3 and B3.3.1–B3.3.6 govern the selected issuer-liability cases. The quantitative ten-percent test compares the PV of new terms, including specified borrower/lender fees, at original EIR against remaining original PV. It is not a universal asset test, nor proof that every change below ten percent is immaterial; changed currency, rights and other qualitative terms need the applicable analysis. Insufficient evidence remains unresolved. Third-party payment without legal release does not extinguish debt.

On the first anniversary above, old remaining 121 has PV 110. Replace with 133.1 due at the original maturity date, one year after the modification, with no fees and an explicit supported substantial-terms conclusion: PV 121 differs by exactly 10%, satisfying the inclusive test. Assume separately supported new fair value 118; debit old debt 110, debit extinguishment loss 8, credit new debt 118. The new EIR is 133.1/118−1, not automatically the old rate. If the replacement is instead a stated non-substantial same-currency reduction to 115.5, no fees or other changes, PV 105 leads to debit debt 5/credit modification gain 5 under the B5.4.6 recalculation mechanism, then interest 10.5 and settlement 115.5. The actual change below ten percent still needs the qualitative finding. Repurchase of carrying amount 110 for 108 extinguishes the repurchased obligation: debit debt 110/credit cash 108/credit gain 2. Unamortised fees belong in the carrying amount before that comparison, not a second gain adjustment.

If cash available is 80 against debt 121 due, record the due obligation and an unpaid/default economic state; payment 80 leaves contractual unpaid 41 only when that partial payment actually occurs. Do not cancel the remaining liability, recognize a waiver or issue a loan unless a separately admitted event authorizes it. Covenant acceleration, cross-default, cure periods, collateral realization and restructuring require the actual legal terms and priority waterfall. A scenario may explore specified alternatives with explicit assumptions; it may not silently select the cheapest settlement as an accounting fact.

10.9 Equity and compound instruments

EquityInstrument retains class, voting/distribution/liquidation rights, issued, treasury and outstanding counts, par/legal-capital requirements and conversion terms. Share counts are integer quantities separate from currency balances. Issue 100 ordinary shares at 2 with no fees: debit cash 200/credit equity 200, issued and outstanding 100. Repurchase 10 at 3 for treasury: debit treasury-equity 30/ credit cash 30; outstanding 90, issued 100, treasury 10. Reissue 5 at 4: debit cash 20/ credit treasury-equity 15/credit other equity 5; outstanding 95, treasury 5. There is no trading profit under the selected AASB 132.33 rule; a statutory par split requires that jurisdiction's facts. Dividend declaration creates a liability only at the admitted point at which the obligation exists; proposal of a dividend is not itself payment or automatically an expense.

For a qualifying fixed-for-fixed issuer conversion option in functional currency, AASB 132.28–32 separates liability fair value and residual equity. Assume total issue proceeds 100, standalone debt fair value 90, zero coupon, maturity two years and cash redemption 100 or fixed 10 ordinary shares; no extra derivative, tax or fee. Issue debit cash 100/credit debt 90/credit conversion-equity 10. Effective interest accretes debt 90 to 100. At maturity conversion under original terms debits debt 100 and credits issued-share equity 100; the original equity 10 stays in equity (a permitted internal transfer is not income). Shares increase 10 and no redemption cash leaves. Redemption instead debits debt 100/credit cash 100; unexercised equity is not remeasured into profit. AG 32 supports no conversion gain/loss in that stated case. Variable share count, cash alternatives, foreign currency and changed conversion terms require classification/modification analysis; they cannot be forced into this example.

Financial statement mappings separate interest, modification/extinguishment results, owner transactions and cash settlement. A financing-liability note reconciles opening carrying amount, cash proceeds/repayments, effective interest, fees, conversions, modifications and FX. Total shares and earnings-per-share denominators need dates and dilution conditions; outstanding shares at year end are not a substitute for a weighted period denominator. Full option valuation, hedge accounting, EPS and legal-capital regimes remain explicitly additional methods; the proposed records preserve their required inputs.

10.10 Changed conversion terms change the accounting question

A maturity conversion, an early repurchase and an inducement to convert all remove or alter debt, but they do not allocate the resulting consideration in the same way. The issuer must retain the old conversion rights, the revised rights, the date of amendment and the actual settlement. A record that says only *converted* loses the facts needed to distinguish these events. The following cases use the retained Australian standards for explicit calendar-2024/2025 for-profit Tier 1 illustrations, without optional later amendments. They are pretax: tax consequences remain uncomputed.

For an early repurchase with unchanged conversion privileges, the issuer allocates the consideration between the liability and equity components using the method of the original separation. The liability allocation uses its current fair value; the difference from its carrying amount enters profit or loss, while the equity allocation stays in equity (Australian Accounting Standards Board, 2022c, AG33–34 and IE39–46). The standard's example has a face amount of 1,000, half-yearly coupons of 50 and ten payments remaining. The original half-year discount rate is 5.5%; the contemporaneous comparable nonconvertible debt rate is 4%. For either rate r, the

remaining debt value is

$$V(r) = 50\frac{1-(1+r)^{-10}}{r} + 1{,}000(1+r)^{-10}. \tag{10.2}$$

The first term values the ten coupons; the second values principal at the last payment. The source rounds the carrying and current fair values to 962 and 1,081. Repurchase consideration of 1,700 then allocates 619 to equity and produces debt settlement expense of 119. The entries debit debt 962, expense 119 and equity 619, and credit cash 1,700. Expensing the whole $1{,}700 - 962 = 738$ would misclassify the equity purchase. The original equity component of 60 remains within equity; it does not cap the 619 deduction. Exact present-value fractions and the source's rounded presentation remain separate in the accompanying calculations.

An inducement asks a different valuation question. At the amendment date, compare the fair value of what the holder receives under the revised conversion terms with what the holder would have received under the original terms. Their difference is an inducement loss (Australian Accounting Standards Board, 2022c, AG35 and IE47–50). In the retained example, changing the conversion price from 25 to 20 on face amount 1,000 increases shares from 40 to 50. At the amendment-date share quotation of 40, the additional ten shares are worth 400. A later quotation of 50 would give 500 and would use the wrong measurement date.

For an explicitly immediate, unconditional share-only conversion, suppose the liability's admitted carrying amount is 960 and the original equity component is 60. This carrying amount is a separate illustrative input. The resulting entry debits debt 960 and inducement expense 400, and credits issued equity 1,360; 50 shares are issued. Original conversion equity remains in equity. The 400 expense offsets the corresponding part of the issued equity in total equity, while derecognizing the debt increases net equity by 960. There is no cash movement. Noncash financing disclosure preserves the event without inventing a financing receipt (Australian Accounting Standards Board, 2023a, 43–44).

This immediate case does not determine the recognition timing of a delayed, contingent or cash inducement. Those terms must first establish the obligation. Nor does an ability to settle in shares establish equity classification. A derivative with a net-cash alternative fails the all-equity-alternatives condition, and an obligation to deliver a variable number of shares worth a fixed sum does not represent a residual ownership interest (Australian Accounting Standards Board, 2022c, 21 and 26–27). The system must refuse the fixed-for-fixed method when these facts are missing or contradicted, while retaining the instrument and its unresolved classification.

10.10.1 An offer, an amendment and delivery can occur on different dates

The immediate conversion above combines several events that need not coincide. An issuer can announce an offer before it changes any enforceable right; the holder can accept a binding amendment before shares are delivered. Retain the announcement, approval or acceptance conditions, legally effective amendment, discharge of the old debt, and delivery separately. The following conditional examples use IAS 32 and IFRS 9 in the retained September 2023 EU compilation. Their dates and terms are synthetic. IAS 32.AG35 measures the inducement at the amendment date, while IFRS 9.3.3.1 and B3.3.1 remove the debt when it is discharged, cancelled or expires, including a qualifying legal release by the creditor. Administrative delay alone does not keep a discharged debt outstanding (European Commission, 2023a, IAS 32.13, AG35; IFRS 9.3.3.1–3, B3.3.1–6).

Suppose an offer is announced on 1 January 2024 but, under its supplied legal terms, changes no rights until approval and the holder's binding acceptance on 5 January. Acceptance releases

the issuer from the old cash obligation in return for 50 fixed ordinary shares; delivery occurs on 20 January. There is no cash alternative, further condition or variable-share obligation. The ordinary shares satisfy the equity definition. The old terms would have delivered 40 shares. Share quotations are 30, 40 and 50 on the three dates, respectively. The incremental ten shares give an amendment-date loss of $10(40) = 400$. Using the announcement quote would give 300; using the delivery quote would give 500. Both would measure the wrong date.

With debt carrying amount 960 and original conversion equity 60, debit debt 960 and inducement expense 400, and credit equity to be issued 1,360 on 5 January. Original conversion equity remains within equity. On 20 January, transfer 1,360 from equity to be issued to issued equity; the new quotation does not remeasure that equity-classified obligation (European Commission, 2023a, IAS 32.15–16, 21–22, AG32–35). Starting with cash 1,020, debt 960 and equity 60 gives closing cash and total equity 1,020: original equity 60 plus new equity 1,360 less loss 400. Neither the announcement nor administrative delivery creates another expense or cash flow. Recognition within equity does not settle the separate IAS 33 question of when shares enter the weighted denominator; retain the actual participation and issue terms for that assessment.

Change the accepted consideration to the original 40 fixed ordinary shares plus a fixed cash bonus of 100 payable later. Assume zero discount and no intervening interest in this short illustration, with the old debt legally released at acceptance. The incremental consideration and loss are 100. Debit debt 960 and expense 100; credit equity to be issued 960 and bonus payable 100. Equity is then $60 + 960 - 100 = 920$, with cash 1,020 and the payable 100. Later payment debits the payable and credits cash 100, leaving cash and equity 920 without a second loss. The payment buys the debt holder's early conversion and is a financing outflow under the stated facts; the share exchange is disclosed as noncash financing (European Commission, 2023a, IAS 32.AG35; IAS 7.6, 17, 43–44).

These entries require the supplied legal and classification findings. A binding instrument with a contingent cash clause can already be a liability under IAS 32.25; it is not equivalent to the expressly nonbinding offer above. An obligation to deliver shares worth a fixed cash amount uses a variable quantity and is also a liability. A derivative's cash settlement alternative fails equity classification unless all alternatives qualify as equity (European Commission, 2023a, IAS 32.19, 21, 25–27). The procedure first determines whether rights changed, then classifies each new obligation, measures the inducement at the effective amendment, and separately tests debt discharge. If the debt has not been released, retain and measure that debt and assess the actual modification under IFRS 9; do not copy the 960 debit from this case. Unknown approval effects, contingent consideration values or discharge terms remain unresolved inputs to that procedure, with the original history intact.

10.11 A hedge reserve is neither cash nor a second gain

Suppose a company hedges a highly probable inventory purchase. A derivative gain can arrive before the inventory affects earnings. Cash-flow hedge accounting preserves that timing through a separate equity reserve, but only after the relationship qualifies. The designation must identify an eligible instrument and item, the hedged risk, the risk-management objective and the assessment method at inception. An economic relationship, the absence of credit-risk dominance and an appropriate actual hedge ratio are separate requirements (Australian Accounting Standards Board, 2022g, 6.2–6.4 and B6.4.1–19). Favorable price correlation alone does not establish them. Unknown designation or valuation information remains unresolved; it cannot be repaired by assigning zero or placing a gain in other comprehensive income.

Take one documented, offsetting exposure and a derivative worth zero at designation. The

illustration includes the whole derivative, no excluded option or forward component and no intermediate reserve release. Let D_t be its cumulative gain, including settled gains, and H_t the cumulative change in the hedged item's present value attributable to the designated risk. Their signs refer to their economic values. With admitted offsetting changes, the cumulative effective gain E_t and ineffectiveness I_t are

$$E_t = \operatorname{sgn}(D_t)\min\{|D_t|, |H_t|\}, \qquad I_t = D_t - E_t. \tag{10.3}$$

The reserve follows the lower absolute cumulative amount; effective changes enter other comprehensive income and the remainder enters profit or loss (Australian Accounting Standards Board, 2022g, 6.5.11). This formula handles the stated offsetting case; a complex exposure outside it needs its own measurement method. In particular, one same-direction period is not itself proof that no economic relationship exists. Item valuation must include time value and only the item's relevant features; a hypothetical derivative cannot import features that exist solely in the actual hedging instrument (Australian Accounting Standards Board, 2022g, B6.4.5 and B6.5.4–6).

Initially, supplied values $D_1 = 12$ and $H_1 = -10$ produce effective gain 10 and profit 2. At the next measurement, $D_2 = 15$ and $H_2 = -14$ produce effective gain 14 and cumulative profit 1. The second period therefore records a derivative increase of 3, other comprehensive income of 4 and a loss of 1. Applying the minimum only to that period's changes would incorrectly give effective gain 3. The retained cumulative values are essential economic memory.

Now buy the inventory for 114 and settle the derivative for cash of 15. Settlement removes the derivative asset; it does not by itself release the 14 reserve. Because the forecast transaction has produced a nonfinancial asset, debit the reserve 14 and credit inventory 14. Inventory cost becomes 100. This direct basis adjustment is not an other-comprehensive-income reclassification (Australian Accounting Standards Board, 2022g, 6.5.11(d)(i)). Selling the inventory for 130 subsequently produces revenue 130 and cost of sales 100.

With opening cash and equity of 200 and no other movements, the full sequence has cumulative pretax profit 31, other comprehensive income 14 and a separate equity-to-inventory basis transfer of −14. Closing cash and pretax equity are both 231. The equity reconciliation is

$$200 + 31 + 14 - 14 = 231, \tag{10.4}$$

where omitting the direct basis transfer would overstate equity. Gross operating cash is $-114+15+130 = 31$: the identified hedge's cash flows follow the inventory position's classification (Australian Accounting Standards Board, 2023a, 16). A matching terminal cash total does not excuse a wrong interim reserve, profit or basis.

If the hedged payments are interest, the reserve instead enters earnings when the interest affects earnings (Australian Accounting Standards Board, 2022g, 6.5.11(d)(ii)). With an effective gain of 8, derivative settlement of 8 and cash interest of 18, release 8 from the reserve into earnings. Expense is then 10. The release has an OCI reclassification of −8, but no second cash receipt. The illustrative nonfinancial company consistently elects operating classification for interest under the retained rule; net operating cash is −10 (Australian Accounting Standards Board, 2023a, 31–33). A later presentation regime needs its own applicability decision.

Stopping hedge accounting also requires a reason. If the future cash flows remain expected, the accumulated reserve remains until the flows occur, subject to the unrecoverable-loss rule. If they are no longer expected, the reserve is reclassified immediately to profit or loss. No longer highly probable is not the same as no longer expected (Australian Accounting Standards Board, 2022g, 6.5.11(d)(iii) and 6.5.12). The worked alternatives keep a gain reserve of 10 in the first case and release it in the second. A loss reserve of 10 with an unrecoverable portion of 4 instead releases expense 4 and leaves a loss reserve of 6. Each decision changes a named exposure and reserve; it cannot be inferred from the derivative's cash settlement alone.

10.11.1 Fair-value adjustments and benchmark changes

An admitted fair-value hedge of amortised-cost debt uses a different route. Both the derivative change and the risk-attributable debt adjustment enter profit or loss; the latter changes the debt carrying amount (Australian Accounting Standards Board, 2022g, 6.5.8). For debt initially carried at 100, a derivative gain of 8 and debt-value increase of 10 give a derivative asset of 8, debt of 110 and net loss of 2. No cash-flow hedge reserve is created.

Once the debt ceases receiving hedge adjustments, the retained adjustment must be amortized using a recalculated effective interest rate (Australian Accounting Standards Board, 2022g, 6.5.10). Suppose contractual principal of 100 and coupon of 21 are payable together one year later. From carrying value 110, the recalculated rate is $121/110 - 1 = 10\%$, giving effective interest expense 11 before settlement. Cash interest remains 21 and principal remains 100; the difference between cash interest and effective expense amortizes the 10 adjustment. Reusing the old 21% rate would over-accrete the liability. Missing contractual dates or payments leave that calculation unresolved.

Benchmark reform is not a general exemption from this analysis. The retained standard's rate-update expedient applies to necessary reform changes whose new cash-flow basis is economically equivalent to the old basis. Additional changes then receive the ordinary modification or derecognition analysis (Australian Accounting Standards Board, 2022g, 5.4.5–9). Comparing one 4% coupon with 3.8% plus a 0.2% spread verifies one number, not equivalence of all future terms. The amendment record must retain the legal change, rates, spreads, reset terms, dates and the supporting equivalence analysis.

A qualifying change to formal hedge designation is neither automatic discontinuation nor a new relationship. Existing cash-flow reserves follow the alternative benchmark under the specified conditions rather than being reset to zero. Other qualifying requirements continue to apply; additional changes and mixed benchmark groups need their own assessment (Australian Accounting Standards Board, 2022g, 6.9.1–10). The rate-specific 24-month exception for a separately identifiable risk component likewise requires a supported expectation and ends on the stated adverse reassessment (Australian Accounting Standards Board, 2022g, 6.9.11–13).

The exact conditional journals and adverse alternatives are retained in `data/accounting-master/finite-method-decisions-20261015/financing/`; the field and event specification is `docs/design/proposal/methods/financing-changes-20261015.md`. These cases specify the financial consequences of admitted values and eligibility findings. The next section extends the treatment to excluded components and rebalancing. Actual valuation, delayed or contingent inducements and group hedges retain their named method and evidence obligations. Supplying a market value is useful input; it is not a completed valuation method.

10.12 Excluded hedge components and rebalancing

The following conditional methods use the retained AASB 9 compilation for calendar-2024/2025 Australian for-profit Tier 1 illustrations, without optional later amendments. Actual authority admission and component valuation remain separate requirements.

10.12.1 The option premium pays for protection

A purchased option can protect an inventory purchase even if it expires with no intrinsic value. Its time value is the price of that protection. When only intrinsic value is designated, applying the earlier whole-derivative method would mix the price of protection with the payoff that

offsets the purchase. AASB 9.6.5.15 (Australian Accounting Standards Board, 2022g, 6.5.15 and B6.5.29–33) separates them. The designated intrinsic-value movement follows the applicable hedge method; the related time-value movement accumulates in a different equity component. The nature of the hedged item determines when that cost affects earnings. A transaction such as an inventory purchase uses a cost attached to that transaction. Protection over a period uses systematic, rational amortisation over the period in which the intrinsic-value hedge adjustment can affect earnings. The life of the option alone does not decide which route applies (B6.5.29–30).

The event record must identify the instrument, designated component, underlying item and risk, critical terms, actual and aligned time values at designation and every measurement date, intrinsic-value changes, separate reserve balances, the transaction or time-period classification, amortisation policy, and each basis transfer or reclassification. Existing intrinsic-value effectiveness requirements remain necessary. Unknown component valuations cannot be replaced with a total derivative value and a zero time value.

Suppose a qualifying inventory hedge begins with cash and equity of 200. Buy an option for 6, entirely time value. Cash is 194 and the option asset is 6; there is no initial expense. The first supplied valuation is intrinsic value 12 and aligned time value 4. The option is worth 16. Its increase of 10 comprises an effective intrinsic gain of 12 and a time-value loss of 2, both in OCI but in separate reserves. At purchase and expiry, intrinsic value is 20 and time value zero. The option is worth 20, the intrinsic reserve is 20 and the cost reserve is negative 6. The last period's change of 4 comprises intrinsic gain 8 and time-value loss 4.

Buy the inventory for 120 and settle the option for 20. Settlement removes the option asset and leaves cash 94. The intrinsic reserve reduces inventory cost by 20. Removing the negative cost reserve adds 6 to inventory cost:

$$120 - 20 + 6 = 106.$$

Both direct transfers affect the asset and equity; neither is a second cash flow or an OCI reclassification. Cash 94 plus inventory 106 equals equity 200 after the transfers. Selling for 130 gives profit 24 and cash 224. The cash bridge is $200 - 6 - 120 + 20 + 130 = 224$. The equity bridge is opening 200, profit 24, cumulative OCI 14, and a direct basis transfer of negative 14. Omitting the premium would report cost 100 and overstate profit by 6. For a forecast sale instead, the accumulated cost enters earnings when the sale does. An unrecoverable portion enters earnings immediately under 6.5.15(b)(iii), rather than waiting for a transaction that cannot recover it.

10.12.2 Protection over time and imperfectly matching terms

Consider separately an option that protects an existing inventory holding for two equal periods. Admit aligned initial time value 6 and a rational even allocation over those periods. Intrinsic-value and inventory movements are zero in this illustration, isolating the protection cost. After one period, supplied time value 2 gives an OCI loss of 4. Amortisation of 3 is reclassified to expense, leaving the cost reserve at negative 1. After expiry, the final time-value loss of 2 and second amortisation of 3 leave no reserve; cumulative expense is 6. Immediately expensing each observed time-value fall would instead charge 4 and 2. That is a different timing policy. If the intrinsic-value hedging relationship is discontinued after the first period, 6.5.15(c) requires the remaining net reserve loss of 1 to enter earnings immediately. The remaining option asset of 2 is still an asset; discontinuation does not destroy it.

When critical terms differ, B6.5.32 requires an aligned option valuation using terms that perfectly

match the hedged item. The hypothetical option is a measurement input for the related cost, not a replacement of the actual contract. Under B6.5.33(a) (Australian Accounting Standards Board, 2022g, B6.5.32–33), if actual time value exceeds aligned time value at inception, base the cost reserve on aligned changes and put the difference between actual and aligned changes in profit or loss. For initial actual/aligned values 6/4 and later values 3/3, actual loss 3 separates into OCI loss 1 and profit loss 2. At expiry, cumulative losses 6/4 leave OCI loss 4 and profit loss 2, before the required transfer or reclassification.

If actual time value is initially lower, B6.5.33(b) limits the reserve by the lower cumulative change of actual and aligned time value; the remainder of actual changes enters earnings. In a same-direction decline example, initial values 2/3 become 1/2.5. Their cumulative losses are 1 and 0.5, so the cost reserve holds loss 0.5 and profit holds loss 0.5. At expiry, cumulative losses are 2 and 3: the reserve loss becomes 2 and cumulative profit from the mismatch returns to zero. The second period therefore records OCI loss 1.5 and a profit gain of 0.5, while actual value falls 1. Applying a fresh lower-of rule only to each period would lose this reversal. These numbers test declining positive time values; they do not define a signed-value selector for every collar or complex combination. Such cases must retain the source-defined alignment and valuation analysis rather than infer it from this small example.

A zero-cost collar can have zero initial time value and nonzero intermediate time value. B6.5.31 still requires the related changes in OCI. A supplied movement from zero to 1.2 and back to zero produces OCI gain 1.2 and loss 1.2; zero initial premium is not permission to omit both. The terminal transaction cost adjustment and initial-cost amortisation are zero in that example.

10.12.3 Forward elements and currency basis spreads

AASB 9.6.5.16 (Australian Accounting Standards Board, 2022g, 6.5.16 and B6.5.34–39) permits the corresponding treatment when the forward element is excluded from a spot-designated forward, or a foreign-currency basis spread is excluded. The accounting election is explicit; designation of spot alone does not automatically make the excluded component an OCI reserve. Apply the transaction/time-period distinction and aligned-value rules in B6.5.34–39. For the forward element, the inception comparison uses absolute actual and aligned amounts. Retain their signed fair values and the changes used for the reserve; do not compare signed inception amounts as though a negative forward element were necessarily smaller in economic magnitude. B6.5.39 carries the same guidance to currency basis spreads. The exact valuation of unmatched forward elements and collars remains a required method for those applications.

10.12.4 Rebalancing preserves the history of the hedge

A changed hedge ratio can preserve the original risk-management objective. It does not erase the accounting before the change. Under 6.5.5–6.5.6 and B6.5.7–21 (Australian Accounting Standards Board, 2022g, 6.5.5–6.5.6 and B6.5.7–21), first measure and recognise ineffectiveness under the existing designation, then adjust designated quantities prospectively. Distinguish ordinary fluctuations around a valid ratio from a changed relationship that requires a new ratio. A changed objective requires discontinuation of that relationship; a counterparty default is not repaired by changing quantities. Update the documented expected sources of ineffectiveness after rebalancing.

Before a change, cumulative derivative gain 12 and offsetting item loss 10 give reserve 10 and profit 2. Immediately before rebalancing, admitted values 18 and negative 14 instead give reserve 14 and cumulative profit 4. Record the period's derivative gain 6 as reserve gain 4 and profit gain 2 before adjusting the quantities. A reset to equal amounts at the new designation

date would hide the accumulated ineffectiveness.

Store each retained, added and removed volume with its designation date and valuation reference. An added item layer begins at the rebalancing date; the earlier layer keeps its old reference. The source's example has 100 tonnes designated at forward price 80 and adds 10 tonnes when the price is 90. In a zero-discount illustrative comparison at price 95, their changes are $100(95 - 80) + 10(95 - 90) = 1550$. Applying the old price to all 110 tonnes gives 1650, inventing a gain of 100 on the added layer. Actual measurement must use the admitted risk, dates, discounting and contractual terms.

Adding an instrument similarly starts its change at the new designation date without resetting the old instrument. Decreasing designated instrument volume leaves any retained undesignated derivative at fair value through profit or loss unless another qualifying designation applies. For example, removing 10% of a homogeneous instrument from the designation leaves its total carrying value unchanged; of a subsequent total fair-value gain 10, the undesignated 1 enters profit and the designated 9 follows the continuing hedge's effectiveness test. This split does not assert that all 9 is effective. Decreasing item volume invokes partial-discontinuation rules for the removed part. Whether its forecast cash flows remain expected determines the reserve's later release; removal from designation alone does not mean cancellation.

These procedures retain the history needed when a hedge changes. Combining exposures creates a further timing question: sales and expenses can offset economically while affecting different reporting periods. The following net-hedge case retains those gross transactions; the award methods then distinguish service conditions and settlement rights. Actual rights, complex component valuation and complete successor exchanges remain separate financing obligations.

10.12.5 A net hedge still needs its gross transactions

A company expecting dollar sales and dollar expenses may manage only their net currency exposure. Hedge accounting still needs both gross amounts and the periods in which they affect earnings. IFRS 9.6.6.1 and B6.6.1–8 require individually eligible items and actual group risk management. For offsetting cash-flow positions, the designated risk must be foreign currency risk, and the designation must specify nature, volume and the relevant reporting periods. An abstract label such as "net dollars 20" is insufficient (European Commission, 2023a, IFRS 9.6.6.1–3, B6.6.1–8).

Use a euro-functional company with highly probable external cash sales of USD 100 and payments of USD 80 for services consumed immediately. The designation identifies those transactions, their earnings periods and a forward to sell USD 20 for EUR 20. Assume actual net risk management, an eligible external derivative, contemporaneous documentation and all effectiveness conditions. Interest, discount, forward points and credit effects are zero in this illustration; the whole forward is designated. At inception the rate is EUR 1 per dollar and the derivative is worth zero. At execution the rate is EUR 1.10, so sales are EUR 110, service expense EUR 88 and the forward loses EUR 2. No inventory asset is created in this example.

The effectiveness comparison retains the gross changes: the expense's currency effect of minus 8 and the forward loss of 2 offset the sales effect of plus 10. IFRS 9.B6.6.5 and B6.6.9 require the offsetting item to enter that comparison alongside the instrument, without recognizing a forecast transaction before it occurs. When both transactions affect the same period, recognize the effective forward loss of 2 in OCI and subsequently release it as a separate hedge expense of 2. Revenue remains 110 and service expense 88; profit is $110 - 88 - 2 = 20$. The hedge result is presented separately because the offsetting risks affect different statement lines (European Commission, 2023a, IFRS 9.6.5.11, 6.6.4, B6.6.5, B6.6.9, B6.6.15). Showing revenue 108 would preserve profit but misstate the required presentation. Opening cash and equity 200 become

220 after gross operating receipts 110, service payments 88 and forward settlement 2. The reserve and derivative liability both close at zero. The designated forward follows the operating classification of the hedged flows (European Commission, 2023a, IAS 7.16).

Move only the service expense into the next period, retaining the same EUR 1.10 rate and a forward that settles with that expense. The designation already identifies these different earnings periods. At the first close the company has cash revenue 110 and a forward liability of 2. The revenue line must still use its transaction-date spot rate. To reflect the net hedge's timing, recognize a separate hedge expense of 10 when the sale occurs, with the corresponding credit to the cash-flow hedge reserve. Together with the forward's OCI loss of 2, the reserve is a credit of 8. First-period profit is 100 and OCI is 8; cash 310 less the derivative liability 2 equals equity 308. The 8 reserve carries the offset relating to the later expense; it is neither an additional derivative asset nor recognized service income (European Commission, 2023a, IFRS 9.B6.6.9, B6.6.15).

In period 2, record service expense and payment 88, settle the forward liability for 2, and release the reserve 8 as separate hedge income. Profit is minus 80 and OCI minus 8. Cash falls by 90 to 220, which also equals $308 - 80 - 8$. Across both periods profit remains 20 and OCI nets to zero. Releasing only the derivative loss when the sale occurs would give first-period profit 108 and miss this timing rule. A later rate change would require new measurements of all relevant components; the constant second-period rate isolates the recognition issue without supplying a general valuation model.

With sales and service payments both USD 100, the position is nil and no derivative is needed. Such a designation is allowed only under all four conditions in IFRS 9.6.6.6: an actual rolling net-risk strategy, changes in net size over that strategy's life, normal use of eligible instruments and hedge accounting when the position is nonnil, and otherwise inconsistent recognition of the offsetting risks. Conditional on those facts, the earlier-sale example has revenue 110, separate hedge expense 10 and reserve 10 in period 1. Expense 110 and separate hedge income 10 in period 2 leave lifetime profit and cash flow zero. Equal gross totals alone do not establish eligibility (European Commission, 2023a, IFRS 9.6.6.6, B6.6.3, B6.6.6, B6.6.10).

The group record therefore preserves each gross exposure's identity, reporting entity, currency, risk, amount, recognition period and admitted valuation, plus the strategy, designation, instrument, effectiveness calculation and reserve history. A layer also needs an identified and tracked overall pool and the conditions in 6.6.3. A group without offsetting risks apportions the instrument's net gain or loss systematically to the affected lines without grossing it up; a fair-value hedge adjusts each relevant asset or liability separately (European Commission, 2023a, IFRS 9.6.6.3–5, B6.6.11–16). A group of items is not permission to ignore consolidated-group external-party requirements: retain 6.2.3 and 6.3.5–6's distinct tests. These methods need explicit successor records and complete exchanges; the existing closed financing profiles are preserved until those additions are specified.

10.13 Conditions, settlement choices and the earnings that each share receives

The following conditional methods use IFRS 2 and IAS 33 in the retained September 2023 EU compilation. They are separate from the Apple contractual examples and do not determine their US GAAP treatment. Employee counts, fair values, service patterns and participation rights below are explicit synthetic inputs. Valuing the awards and admitting an entity's applicable rules remain distinct requirements.

10.13.1 A failed condition does not always reverse an expense

An employee's failure to complete required service differs from a share-price target that is not reached. For employee equity awards, IFRS 2.10–21 (European Commission, 2023a, IFRS 2.10–23) measures the services using grant-date fair value. Service and non-market performance conditions affect the estimated number of instruments expected to vest; market conditions enter fair value. Revise the former estimate as information changes and use the actual eligible number at vesting. If a market condition fails but the other vesting conditions are met, the service expense remains. Freezing the quantity at grant and subsequently marking every equity award to the current share price would get both sides of this distinction wrong.

The award record therefore separates grant and measurement dates, recipient and employing entity, each service/performance/non-vesting condition and who can choose to fulfil it, service already received, expected eligible quantity, grant-date valuation and its incorporated conditions, settlement rights, cumulative expense, equity and any liability. A contract's use of the word performance does not decide whether the condition is a market condition. Missing terms or fair value remain unresolved rather than selecting the most convenient expense pattern. Services are expensed in these examples; qualifying goods or services that create an asset require their own recognition treatment.

Give 100 employees ten equity instruments each, with grant-date fair value 3 per instrument and three years of equal service. At the first year-end, 90 employees are expected to complete service; at the second, 80; at vesting, 75 have completed it. The cumulative expenses are

$$90(10)(3)/3 = 900, \qquad 80(10)(3)(2/3) = 1600, \qquad 75(10)(3) = 2250.$$

The annual charges are therefore 900, 700 and 650, credited to award equity. If an additional non-market performance condition ultimately fails for all awards, cumulative expense becomes zero and previously recognised expense reverses. In a separate award whose supplied grant-date fair value of 3 already incorporates a market hurdle, failure of that hurdle does not reverse the 2250 for the 75 employees who satisfy the other conditions. Equal supplied fair values in these examples do not imply that different contracts have the same economic value. Once an equity award has vested, later non-exercise does not reverse total equity under paragraph 23; a transfer within equity remains possible.

Cancellation also differs from forfeiture. Paragraphs 27–28 (European Commission, 2023a, IFRS 2.27–28A) preserve the minimum original grant-date expense, subject to failure of the specified non-market vesting conditions, and accelerate unrecognised service expense on cancellation during vesting. For one grant with total grant-date value 300, one of three equal service years gives expense 100. An entity cancellation that is not a forfeiture accelerates the remaining 200. If it pays 330 when the repurchased award's fair value is 320, debit equity 320 and expense 10, and credit cash 330. Total service/cancellation expense is 310. Starting from cash and capital 1000 leaves cash and equity 670 after all entries. Calling the event a service forfeiture would wrongly reverse the first 100.

Non-vesting conditions enter fair value under paragraph 21A. However, when the entity or counterparty can choose whether to meet such a condition, failure to do so during vesting is a cancellation under paragraph 28A. Consequently the condition's classification and the party's control of it must be preserved together. Beneficial modifications require their incremental value analysis under paragraph 27 and Appendix B; this cancellation example does not replace that calculation.

10.13.2 Cash obligations change as the award's value changes

For a cash-settled award, IFRS 2.30–33D (European Commission, 2023a, IFRS 2.30–33D) requires remeasurement at each reporting date and settlement. Service and non-market vesting conditions affect quantity; market and non-vesting conditions enter the current valuation. Apply the service fraction to the current eligible quantity and fair value, then charge the difference from the previously recognised liability. The final cumulative expense equals cash paid, unlike the fixed grant-date measurement of an equity award.

Use the preceding employee quantities and ten rights per employee, but admit cash-settled fair values of 3, 6 and 8 at the three year-ends. Cumulative liabilities are 900, 3200 and 6000. The annual expenses are 900, 2300 and 2800. After vesting, a settlement-date value of 9 raises the liability to 6750 and adds expense 750. Paying 6750 removes the liability. A model that freezes the original value of 3 would leave expense 2250 and an unrecorded cash obligation of 4500. The changing values are supplied, not estimated from this example.

10.13.3 Who chooses settlement matters

If the employee can demand cash or shares, paragraphs 35–40 (European Commission, 2023a, IFRS 2.35–43) treat the award as a compound instrument. Measure the debt component first, then the equity component taking account of the cash right that must be surrendered for shares. Do not add the standalone values of both alternatives. Over the service period, remeasure the debt component under the cash-settled method and recognise the grant-date equity component under the equity-settled method.

For an immediately vested award, admit debt value 100 and incremental equity value 20. Recognise expense 120, liability 100 and equity 20. At settlement, remeasure the liability to 130 and charge expense 30. If the employee takes cash, pay 130 and retain the original 20 within equity. If the employee takes shares, transfer the liability 130 directly to equity, leaving total award equity 150. Both branches have cumulative expense 150; only the cash branch has a cash outflow. The values are an admitted compound valuation, not a claim that standalone alternative values uniquely determine the equity part.

If the entity chooses, paragraphs 41–43 first ask whether it nevertheless has a present obligation to settle in cash. Commercially insubstantial share settlement, a past practice or stated policy of paying cash, or normally paying cash whenever requested can create that obligation. A written share option therefore does not by itself establish an equity award. If no cash obligation exists, use equity-settled accounting. At settlement, cash is an equity repurchase, except that choosing the higher-valued alternative creates expense for the excess. For an immediately vested grant with admitted value 100, settlement cash 130 and alternative shares worth 120 require a debit to equity of 120, additional expense 10 and cash credit 130. A 130 expense would count the repurchase as remuneration again.

These cash choices are distinct from statutory withholding. Under paragraphs 33E–33H, an otherwise equity award can retain equity classification when shares are withheld to satisfy a legally imposed employee-tax withholding obligation. Preserve the legal obligation and required quantity. The exception does not cover an arrangement without that obligation or excess shares paid out in cash. It cannot justify classifying every net-settled award as equity.

10.13.4 A beneficial amendment adds a second service-cost clock

Cancellation accelerates an award's remaining cost. A beneficial amendment instead normally preserves the original grant-date cost and adds value measured at the modification date.

IFRS 2.B42–B44 requires the original minimum, subject to failure of the relevant original non-market vesting conditions, and recognizes the beneficial change. The comparison uses the award's value immediately before and after modification at the same date (European Commission, 2023a, IFRS 2.27, B42–B44). A higher current value is not a replacement grant-date value.

Grant 100 instruments with fair value 3 each for three years of equal service. All ultimately vest. After the first year, a beneficial repricing raises their supplied fair value from 2 to 4 per instrument, with the original vesting date unchanged. Original total cost is $100(3) = 300$. Incremental cost is $100(4 - 2) = 200$, spread over the two remaining service years. The first year's expense is 100; each later year records original cost 100 plus incremental cost 100. Total expense and award equity are 500. Replacing the grant with its current value of 400 would omit 100; subtracting the grant-date value 3 from the modification-date value 4 would also use the wrong comparison. Each annual entry debits service expense and credits award equity. With no cash or other movements, the increase in that equity component offsets the reduction in retained earnings, leaving total equity unchanged.

The two service periods need not end together. In a separate modification, leave the original 100 instruments unchanged and add 20 instruments worth 4 each that require only the second year's service. IFRS 2.B43(b) measures the additional 80 at the modification date over their own service period. Annual charges are now 100, 180 and 100, totaling 380. Spreading the additional 80 over both remaining original years would understate year 2 by 40. This example does not shorten the original award's service condition: a beneficial change to that condition must itself be reflected under B43(c). An adverse value-only change generally leaves the original 300 cost, while reducing the number of instruments invokes the cancellation rule for the affected portion (European Commission, 2023a, IFRS 2.B43(b)–(c), B44).

A change from cash settlement to equity settlement follows a distinct rule. Suppose two of three equal service years have passed. The cash award liability, remeasured at the modification date, is 180. Replacement equity has a total modification-date fair value of 300 and retains the third year's service requirement. IFRS 2.B44A recognizes $300(2/3) = 200$ in equity for services already received, removes the liability 180 and charges the difference 20 immediately. Debit liability 180 and expense 20; credit award equity 200. The remaining year charges 100 to expense and equity. Prior cash-award expense 180 plus the modification charge 20 plus later service 100 gives lifetime expense 300, with no cash payment and no remaining liability. If the correctly remeasured liability had instead been 220, the immediate difference would be an expense credit of 20. It is not automatically an extra positive cost (European Commission, 2023a, IFRS 2.B44A–B44C).

Keep the original grant, every modification's before-and-after valuation, eligible quantity, original and modified service conditions, settlement classification, cumulative expense and component balances together. Apply each event once and recognize the change from the previously recorded cumulative amounts. B44B requires the modified vesting period when applying the cash-to-equity rule; B44C extends it to an identified replacement for a cancelled cash award. Supplied fair values in these examples do not provide an option-pricing method. Missing valuation or legal terms prevent application without erasing the original award or assuming that it costs zero.

10.13.5 Participation allocates the numerator before division

IAS 33.A13–A14 (European Commission, 2023a, IAS 33.A13–A16) allocates earnings according to the actual dividend and undistributed-earnings rights of the classes. First deduct the declared and contractually required dividends from the period's allocable result. Allocate the remainder as if it were distributed, using the participation formula and any caps. Add each class's dividends

back to its allocation, then divide by the instruments to which that allocation belongs. A class's right to profits does not establish a right or obligation to share losses; preserve the actual terms. Unknown participation rights block the allocation.

Suppose parent-attributable profit is 120, 100 ordinary shares participate with weight one, and ten nonconvertible participating shares have weight two. Assume all are outstanding throughout the period and there are no senior claims, caps or other instruments. Declared dividends are 20 and 4. Remaining profit 96 allocates over total weight $100 + 2(10) = 120$: ordinary shares receive 80 and participating shares receive 16. Adding dividends gives class earnings 100 and 20, so per-share earnings are 1 and 2. Allocating all 120 to the ordinary shares would overstate their EPS as 1.2. Dividends are an allocation of this profit, not a second expense. A capped or senior participation formula needs its own piecewise allocation; the simple weights cannot replace it.

Diluted EPS then applies the relevant potential-share rules and dilution tests; it is not the same calculation as award expense. Under paragraphs 47A–48 (European Commission, 2023a, IAS 33.47A–60), future goods or services form part of the assumed proceeds for employee options. A separate case with profit 100, 100 ordinary shares, ten options, cash exercise price 5, period-average future service value 2 per option and average share price 10 has proceeds 70. Seven shares are assumed repurchased, so incremental shares are three and diluted EPS is 100/103. Omitting future services would instead add five. This comparison assumes the awards are outstanding for the whole period and pass the applicable dilution test; performance-conditioned awards also require paragraphs 52–57's period-end contingency assessment.

Under paragraphs 58–60, an entity cash/share choice presumes share settlement if dilutive, whereas a holder choice uses the more dilutive outcome. Required numerator adjustments preserve the profit effect of assumed equity treatment. For a separate nonparticipating contract, let reported profit be 90 on 100 shares after liability-remeasurement expense 10 that would be absent under wholly equity classification, and let assumed share settlement add 20 shares. With no tax effect or other changes, cash gives EPS 0.9 and share settlement gives $100/120 = 5/6$. The share outcome is more dilutive. Adding shares while leaving profit at 90 would produce 0.75 and compute the wrong assumed world. Continuing-operations control, instrument ordering, time weights and antidilution remain the earlier EPS method's requirements, not assumptions waived by this example.

10.14 Control and an elimination book

The consolidation procedure selects the control graph at each date, aligns supported accounting policies and presentation currencies, combines controlled entity books, and posts acquisition, elimination and non-controlling-interest adjustments in a separate book. A control conclusion needs evidence and a rule version. Ownership percentage alone is insufficient. Unknown purchase allocation, currency translation or unmatched intragroup differences remain named unresolved adjustments rather than unexplained balancing entries.

For a simplified acquisition of a different subsidiary from the goodwill case that follows, assume control is established, all net assets are already at the relevant acquisition measurements, no tax or goodwill arises, and non-controlling interest is measured proportionately. A parent has cash 60, investment 40 and equity 100; an 80%-owned subsidiary has cash 30, plant 20 and equity 50. Consolidation debits subsidiary equity 50, credits investment 40 and credits non-controlling interest 10. Group assets are cash 90 plus plant 20, and group equity is 100 attributable to the parent plus 10 to other owners.

A subsequent intragroup loan of 10 shifts cash from parent to subsidiary, creates a receivable and

payable, and is eliminated by debiting payable and crediting receivable 10. Group totals remain 110. An unmatched loan currency or settlement date would block that exact elimination. If the parent instead sells its entire interest for 44 immediately after the initial state and loses control, the group removes subsidiary cash 30, plant 20 and non-controlling interest 10, recognises cash consideration 44 and gain 4; remaining cash/equity is 104. More complex acquisitions and retained interests require additional measurement operations; this example fixes the structure and arithmetic without certifying a complete consolidation standard.

10.15 Perimeter and consolidation state

The acquisition, control and currency methods below use the inspected IFRS 3, IFRS 10 and IAS 21 provisions in the retained 2023 EU text (European Commission, 2023a). The acquisition with goodwill is a new example; it does not change the preceding zero-goodwill case.

ControlAssessment records investor/investee, relevant activities, rights and their substantive/protective character, variable returns, ability to use power, principal/agent analysis, effective interval and evidence. IFRS 10.5–9 requires power, returns and their connection, with reassessment on changed facts. An 80% share count is evidence to analyze, not an automatic substitute for control. Unknown control yields separate conditional perimeters and unresolved group results; no entity disappears merely to make reconciliation easy.

PerimeterVersion binds control intervals, ownership classes, report cutoffs, functional/presentation currency, policy-alignment adjustments and the exact source entity books. ConsolidationAdjustment is a distinct group book with matched counterparty/event references; legal-entity journals remain intact. The procedure is: select time-compatible entity books; apply supported policy alignment; translate them; add like balances/movements; eliminate investment against acquisition equity; eliminate intragroup balances, income/cost and unrealised gains; allocate profit/OCI to owner/NCI; reconcile all statements and movement notes. IFRS 10.19–23/B86–B96 support these steps. Reporting-date differences need B92–B93's conditions and intervening adjustments; a generic nearest-date join is invalid.

10.16 Acquisition and ownership case

Assume a business combination, no common-control transaction, parent obtaining control and 80% of ordinary residual interests for cash 96. Acquiree identifiable assets fair value 160 and liabilities 60 give net 100. There is no prior interest, contingent consideration, tax difference, separately unrecognized intangible, transaction cost or impairment; those absences are synthetic facts. Select proportionate-share NCI under IFRS 3.19: NCI 20 and goodwill $96+20-100=16$ under .32. In parent books debit investment 96/credit cash 96. In the consolidation overlay, debit acquiree acquisition equity 100 and goodwill 16, credit investment 96 and NCI 20. Only post-acquisition income enters group performance. Asset measurement exceptions, deferred tax and a bargain gain reassessment under .36 would change the calculation; goodwill cannot absorb an unexplained missing balance.

If post-acquisition net profit is 20, all accrued in retained earnings with no dividend/OCI, NCI increases 4 to 24 and parent's attributable profit is 16. Parent buys another 10% for 15 while retaining control: transfer 12 from NCI (10% of current net 120), credit cash 15 and debit parent equity 3. Remaining NCI 12; goodwill 16 unchanged; no new profit. IFRS 10.23/B96 makes this an equity transaction. This example has equal ordinary rights; preference rights need their actual allocation and cannot use a simple ownership fraction.

In a separate alternative starting before that additional purchase, sell all 80% for 130 when net assets 120, goodwill 16, NCI 24 and OCI zero. On loss of control, group gain=130+24−(120+16)=18. Derecognize individual assets/liabilities and goodwill, remove NCI, recognize consideration and gain. The separate parent book's investment gain is not added again. If 20% is retained and 60% sold for 100 with retained fair value 35, group gain=100+35+24−136=23; recognize the retained investment 35. Its subsequent associate/financial-asset method needs separately established significant-influence/classification facts. IFRS 10.B98–B99 also requires the appropriate prior-OCI treatment; this case assumes none.

10.17 Intercompany inventory and tax

Assume parent's external inventory cost 60 is sold to its controlled subsidiary for 100 on credit, all still held at report date, tax ignored in the base case. Parent records AR 100/revenue 100 and COGS 60/inventory 60; subsidiary inventory 100/ AP 100. Consolidation debits AP 100/credits AR 100 and debits revenue 100/credits COGS 60 and inventory 40. Group inventory 60, revenue and COGS zero, and no group profit arises. When subsidiary later sells externally for 120, group COGS is 60 and profit 60, not 20; release the previously eliminated 40 in the continuing adjustment roll-forward. Reversing a closing overlay and recreating a new one must not recognize the profit twice.

If the purchaser's independently supported tax base is 100, group carrying amount 60 creates deductible difference 40. At assumed applicable 25% and with all IAS 12 DTA conditions satisfied, recognize group DTA 10/deferred-tax benefit 10; reverse on outside sale. The original tax expense and cash obligations in the seller's tax jurisdiction remain. Different rates, recoverability or tax-law transfer rules change the answer; tax is not simply eliminated with group profit. For an upstream subsidiary sale, attribute elimination and related tax effects to the appropriate subsidiary/NCI performance; downstream parent's profit elimination does not mechanically reduce NCI by the same fraction.

10.18 Functional currency versus group presentation

EntityCurrencyState names the functional currency and its evidenced rationale, foreign-currency nominal balances, monetary/nonmonetary classification, historical or fair-value measurement dates, rates and sources. IAS 21.21–23 and .28 require transaction-date translation and closing-rate remeasurement of monetary items; historical-cost nonmonetary assets retain transaction-date rates. A functional currency change is prospective under .35–37. Hyperinflation, lack of exchangeability and hedge accounting require separate applicable methods; do not import the historical .26 text into a later amended period without checking it.

Example: buy goods for foreign 100 at functional 1.2 per foreign unit; debit inventory 120/credit payable 120. Closing rate 1.3 gives payable 130 and FX expense 10, while inventory remains 120 before any NRV test. Pay at 1.25: debit payable 130/ credit cash 125/credit FX gain 5. Cumulative expense 5 matches 125 cash less 120 asset cost. Translating the inventory again to 130 would be wrong for this declared basis.

Group presentation is a different operation under IAS 21.39–41. A subsidiary has opening net assets 100, equity 100, no debt or distributions, annual profit 20 entirely earned at rate 1.25, and closing assets 120. Opening historical rate 1.2 and closing 1.3 give translated assets 156, contributed/opening equity 120 and profit 25; translation OCI 11 reconciles the difference. It is explained as 100×(1.3−1.2)+20×(1.3−1.25)=11, not a free balancing plug. If ownership

is 80%, parent/NCI shares of profit are 20/5 and OCI 8.8/2.2 under the stated equal rights. Use transaction rates; an average is acceptable only as a justified approximation, not when significant fluctuation invalidates it. Goodwill and fair-value adjustments follow IAS 21.47's foreign-operation treatment, rather than freezing them automatically in the parent's currency.

Group reports retain the translation reserve by entity, ownership and event. Loss of control requires the appropriate reserve reclassification under IAS 21.48 and IFRS 10.B99; a no-control-loss change transfers the relevant attribution in equity under applicable conditions. Cash translation effects reconcile cash separately from operating/investing/financing movements. An intercompany monetary balance's exchange result cannot simply be eliminated as if no currency exposure existed; IAS 21.45 and net-investment .32 require the substantive classification.

10.19 A foreign subsidiary from acquisition to disposal

The preceding examples establish acquisition, translation and disposal separately. Their interaction matters because a foreign subsidiary carries more than translated net assets into a sale: goodwill and accumulated exchange differences also have owners and dates. A sale can leave total equity correct while putting the wrong amount in profit. We follow one subsidiary through these steps using the selected historical IFRS 3, IFRS 10, IAS 21 and IAS 7 provisions in the retained EU text (European Commission, 2023a).

A parent starts with USD 300 of cash and equity. USD is its functional and presentation currency; the subsidiary's functional currency is EUR. At acquisition the parent pays EUR 96 for 80% of equal ordinary interests, when each euro buys USD 1.20. It obtains substantive power over the relevant activities, exposure to variable returns and the ability to affect them. The other owners have no substantive rights that defeat that control. Neither investment-entity treatment nor a common-control transaction applies.

The acquired business already produces outputs and includes a skilled organised workforce performing a process critical to continuing them, as required by the assumed IFRS 3.B12C(a) facts. The optional concentration test is not elected. Identifiable fair values are EUR 20 of cash, inventory of 140 and a payable of 60. There is no other separately identifiable asset, tax difference, previous holding, contingent consideration or acquisition cost in this synthetic case. The workforce itself is subsumed into goodwill under B37; calling a residual goodwill would not justify omitting an identifiable customer relationship or other asset in a real acquisition.

Identifiable net assets are $20 + 140 - 60 = 100$ euros. The parent chooses proportionate-share measurement for the non-controlling interest (NCI) under IFRS 3.19. NCI is therefore EUR 20 and goodwill under paragraph 32 is $96 + 20 - 100 = 16$ euros. This recognised goodwill belongs to the acquired parent interest; no NCI goodwill has been recognised. The parent's separate entry debits investment and credits cash by USD 115.2. The consolidation adjustment debits acquisition equity 120 and goodwill 19.2, and credits the investment 115.2 and NCI 24. Immediately after acquisition, group cash is 208.8, inventory 168, goodwill 19.2 and the payable 72. Net assets of 324 equal parent equity 300 plus NCI 24.

The subsidiary then sells all the inventory for EUR 160, receives cash, and pays its EUR 60 payable. Both events occur at USD 1.25 per euro. The specified event interval has no further expense, impairment or distribution. In the subsidiary's books, revenue 160 less cost of sales 140 gives profit 20; cash closes at $20 + 160 - 60 = 120$ euros. Translated revenue is 200, cost of sales 175 and profit 25. Parent and NCI shares of profit are 20 and 5. The translation worksheet translates the opening balances to the event rate before recording these movements; it does not charge cost 175 against unadjusted acquisition inventory 168 and leave a temporary negative stock. These are presentation adjustments, not a remeasurement of historical-cost inventory in

the subsidiary's own functional currency.

At reporting, and at an immediate disposal considered below, the rate is USD 1.30 per euro. Subsidiary cash translates to 156. IAS 21.47 also treats the acquisition goodwill as an asset of the foreign operation: its EUR 16 becomes USD 20.8. To explain the currency reserve, translate the opening net assets and the subsequently earned profit from their respective rates:

$$\underbrace{100(1.30-1.20)}_{\text{opening net assets}}+\underbrace{20(1.30-1.25)}_{\text{later profit}}=11.$$

Goodwill adds $16(1.30-1.20)=1.6$. The total translation gain in other comprehensive income (OCI) is 12.6, but it is not divided mechanically 80/20. Only the net-asset gain is shared: parent OCI is $0.8(11)+1.6=10.4$, and NCI OCI is $0.2(11)=2.2$. This follows from the chosen proportionate NCI measurement and the separate attribution required by IAS 21.41 and IFRS 10.B94. NCI closes at $24+5+2.2=31.2$, equal to 20% of the subsidiary's identifiable net assets of 156. Parent equity is $300+20+10.4=330.4$. Group cash 340.8 plus goodwill 20.8 equals their combined equity of 361.6.

Cash has its own translation bridge. At acquisition, payment of 115.2 less cash acquired of 24 gives investing outflow 91.2. The subsequent operating receipt of 200 less payment of 75 gives operating inflow 125. Exchange changes increase the translated value of the acquired cash by $20(1.30-1.20)=2$ and of the later EUR 100 net receipt by $100(1.30-1.25)=5$. IAS 7.25–28 reports that total exchange effect of 7 separately from transaction cash flows. Thus $300-91.2+125+7=340.8$. The cash exchange effect is 7; the total translation OCI is 12.6. They measure different populations.

10.19.1 Losing control removes the whole subsidiary

Suppose an unanticipated transaction immediately after reporting transfers control to a buyer. No earlier held-for-sale condition is assumed in this event slice; a real planned disposal needs that assessment. Compare two alternatives: sell the whole 80% interest for USD 169, or sell 60% for 126.75 and retain 20% with independently admitted fair value 42.25. Both transactions remove the seller's substantive power. The equal price per share is an illustrative valuation assumption, not an accounting requirement. It makes proceeds plus retained value equal to 169 in each alternative.

IFRS 10.B98 removes all subsidiary assets, liabilities and NCI when control is lost, and measures any retained investment at fair value. If C is cash consideration and V the retained fair value, the gain before the currency reserve is reclassified is

$$C+V+\underbrace{31.2}_{\text{NCI removed}}-\underbrace{156}_{\text{subsidiary cash removed}}-\underbrace{20.8}_{\text{goodwill removed}}=23.4.$$

For the full sale, debit cash 169 and NCI 31.2, credit subsidiary cash 156 and goodwill 20.8, and credit gain 23.4. In the retained-interest alternative, replace the debit to cash 169 with cash 126.75 and retained investment 42.25. The subsidiary still holds its own bank account; removal of cash 156 is a change in the consolidated reporting population, not a payment by the parent.

IAS 21.48–48B and IFRS 10.B99 then reclassify the parent's entire accumulated exchange reserve of 10.4 into profit: debit that reserve and credit disposal gain. Total disposal gain is $23.4+10.4=33.8$. NCI's reserve of 2.2 is already within the NCI balance of 31.2 removed above; it is derecognised without being recycled to profit. Retaining 20% does not justify recycling only the fraction corresponding to the shares sold. Paragraph 48A treats a loss of control as a disposal even when a minority interest remains.

The two alternatives therefore have equal total ending equity but different cash and investment positions:

USD amount	Sell all 80%	Sell 60%, retain 20%
Cash consideration	169	126.75
Retained investment	0	42.25
Disposal gain including recycling	33.8	33.8
Net disposal cash flow	13	−29.25
Closing group cash	353.8	311.55
Closing equity	353.8	353.8

IAS 7.39–42 reports acquisition and loss-of-control cash separately, each net of the whole subsidiary's cash. Net disposal cash is $169-156=13$ for the full sale, but $126.75-156=-29.25$ when 20% is retained. A positive parent receipt can therefore accompany a negative consolidated disposal cash flow. The retained investment's 42.25 is not cash. The two cash bridges are $300-91.2+125+7+13=353.8$ and $300-91.2+125+7-29.25=311.55$.

For the whole interval, operating profit 25 plus disposal gain 33.8 gives profit 58.8, attributed 53.8 to the parent and 5 to NCI. Total OCI is $12.6-10.4=2.2$, all attributed to NCI; period OCI does not vanish merely because closing NCI is zero. The equity roll-forward is

$$300+\underbrace{24}_{\text{NCI at acquisition}}+58.8+2.2-\underbrace{31.2}_{\text{NCI at disposal}}=353.8.$$

Parent equity also follows directly from $300+53.8=353.8$. In the full sale, the parent's separate cost-based investment gain is $169-115.2=53.8$, already equal to its cumulative consolidated profit in this example. Adding that separate gain to the group result would count the investment return twice.

Intermediate statements are essential checks. Freezing goodwill at USD 19.2 understates both goodwill and parent OCI by 1.6 before disposal. Yet its larger pre-recycling disposal gain and smaller recycled reserve cancel, leaving the same final gain of 33.8. Dividing goodwill translation 80/20 would misattribute 0.32 to NCI. Recycling all 12.6 while still removing NCI 31.2 would overstate profit by 2.2 and leave a spurious negative parent reserve; recycling only the sold fraction in the retained-interest case would leave 2.6 after control is lost. Final total equity alone misses these different failures.

The case uses relative event dates and historical technical support. Actual valuation, tax, impairment, acquisition exceptions and applicable amendments remain required inputs. A retained 20% interest also needs a separate significant-influence and classification assessment before subsequent measurement; its percentage does not supply that conclusion. The same separation of legal-entity balances and group adjustments explains why eliminating an intragroup loan can leave a currency exposure.

10.20 Keeping the group calculation attached to its books

The disposal example gives the parent a cash receipt while removing cash from the consolidated balance sheet. A software implementation must know whose cash each entry describes. If it posts the removal of the subsidiary's 156 to the parent's legal book, it records a payment the parent never made. If it leaves the subsidiary's cash in the group after control ends, it overstates the group's assets. The consolidation calculation therefore needs its own book and exact references to the legal books it reads.

Call the legal entities Parent-P and Subsidiary-S, and the reporting entity Group-P. Parent-P's functional currency is USD; Subsidiary-S's is EUR. Their identifiers remain distinct even if two imported records have the same display name. Each legal-book snapshot identifies its entity, book, currency, relative date and signed balances. At acquisition the parent's investment 115.2 and cash payment stay in its own USD book. The subsidiary's opening cash 20, inventory 140 and payable 60 stay in its EUR book. Group-P's consolidation entries refer to both snapshots without altering either. This implements the separation between combining balances and eliminating the parent's investment against subsidiary equity (European Commission, 2023a, IFRS 10.B86–88).

The selected perimeter has equally precise dependencies. Its control assessment records the investor, investee, substantive rights, variable returns and ability to use power, together with the interval for which those findings hold. Ordinary ownership of 80% and proportionate NCI measurement are separate inputs. The currency record distinguishes the subsidiary's functional currency from the group's presentation currency. A rate of 1.30 is usable here only when its units are USD per EUR and its date is the closing date R. An inverse quote or a later rate is a different input, even when its label looks familiar. The relative A, O, R and D calendar defines this synthetic history; it supplies no evidence of when an actual investor could have known its facts.

The group needs more state than the initial corporate example. Goodwill, the parent's translation reserve and NCI have distinct balances, while profit, OCI and cash movements retain their period histories. The proposed `GroupForeignOperation/1` profile supplies those fields under its own schema lock. It preserves the common request and result fields rather than silently adding accounts to the closed corporate profile. Loading this extension would still require an implementation to declare support for it; there is no automatic conversion of a corporate state into a group state.

Follow the existing history through that representation. At A the group records the acquisition. At O it translates the opening balances before the subsidiary's operating entries. At R it finishes cash and goodwill translation and allocates operating profit. A and O are intermediate worksheet states; they are not presented as fully closed interim reports. The R state retains cash 340.8, goodwill 20.8, parent translation reserve 10.4 and NCI 31.2, together with profit 25, OCI 12.6 and the separate cash exchange effect 7. Closing the income accounts does not erase those movements.

Full disposal and disposal with a retained 20% interest then start from separate copies of that same immutable R state. Their branch identities prevent the second alternative from being posted after the first. Each disposal proposal identifies its opening state, control change, source books, cash consideration, retained fair value, rates and accounting method. Its permission to post is conditional on this synthetic profile and exact proposal. It is not permission to post to an actual company's books.

In the retained-interest alternative, the parent's receipt is 126.75. The group removes subsidiary cash 156, producing investing cash $126.75 - 156 = -29.25$. The retained investment 42.25 is a separate noncash asset. Both disposal alternatives remove the entire NCI balance 31.2, including its translation reserve 2.2. Only the parent's reserve 10.4 is recycled into the disposal gain. These are checks on the meaning of the journal, beyond the fact that its debits equal its credits (European Commission, 2023a, IFRS 10.B98–99; IAS 21.48A–B; IAS 7.39–42).

The report reads one branch and one dependency view. In both alternatives, period profit is 58.8, of which 53.8 belongs to the parent and 5 to NCI. Period OCI is 2.2, attributable to NCI. Ending NCI is nevertheless zero, because the disposal removes its accumulated balance. The

equity movement must therefore retain both its introduction and removal:

$$300 + 24 + 58.8 + 2.2 - 31.2 = 353.8.$$

Reading only the closing NCI account would lose the subsidiary's contribution while it was controlled. Reading only final equity would also miss the difference between full-sale cash 353.8 and retained-interest cash 311.55 plus investment 42.25.

Publication follows the common accounting transaction rule. The request identifies the group owner and expected opening revision; its business-event identity includes the branch. Before publishing, an implementation must validate the source and currency bindings, posting authorization, ordered journals and independently specified state changes. It then publishes the closing state, journal, economic delta, receipt and new group head together. The parent and subsidiary legal-book owners remain untouched.

An acknowledgement can be lost after that publication. Retrying the exact original request then finds the existing receipt and has no second effect, even though its original opening revision is now old. If publication never occurred, the same request can commit once. Changing the request identifier, budget or expected head changes the fingerprint and cannot masquerade as that retry. A newly proposed event with an outdated opening is rejected. The documentary packet preserves both complete acknowledgement histories and an invalid partial-publication example; later runtime tests must exercise those failures in the chosen storage system.

A corrected exchange-rate source creates a new evidence view. It makes the old group's calculation unsuitable for a request using the corrected view, but does not overwrite the old books or remove the old explanation. Unknown control or an unknown rate instead returns the missing-input obligation, with no fabricated zero amount. The companion group-binding specification supplies exact records and exchanges for these cases. Actual control, valuation, tax, policy alignment and subsequent accounting for the retained investment retain their separate evidence requirements.

10.21 Eliminating the loan, retaining the currency exposure

An intragroup loan disappears from the consolidated balance sheet, yet its exchange gain can remain in group profit. The reason becomes visible when we follow both the loan and the money it financed. IAS 21.45 requires the exchange result to survive elimination; paragraph 32 determines when it belongs in other comprehensive income instead (European Commission, 2023a, IAS 21.32–33, .45).

Consider a parent whose functional and presentation currency is USD and a wholly owned subsidiary whose functional currency is EUR. Control and both functional currencies are established assumptions. At an opening rate of USD1.20 per EUR, the parent's balances are cash 56, an investment in the subsidiary carried at historical cost 24, and a EUR100 loan receivable worth 120. Its equity is therefore 200. The subsidiary holds EUR120 in cash, owes EUR100 to the parent and has EUR20 of paid-in equity. At the opening rate, its equity matches the parent's investment. Eliminating that investment and the loan leaves group cash and equity of 200.

We isolate an exchange-rate remeasurement of these established balances, holding every other measurement input fixed. This is neither the initial valuation of a new loan nor a complete reporting period. Interest, tax, impairment, hedges and new transactions are outside this calculation; IAS 21.24 leaves the underlying carrying amount to the relevant measurement standards. Thus EUR100 is a given monetary carrying amount, not an assumption that an interest-free long-term loan must be recognised at its face value (European Commission, 2023a, IAS 21.23–24, .50). All following statement amounts are in USD unless explicitly marked EUR.

Now let the rate rise to USD1.30 per EUR. The parent remeasures its receivable from 120 to 130: debit the receivable 10 and credit exchange gain 10. The subsidiary records no exchange gain on its EUR loan in its own EUR books. For group presentation, its cash becomes $120 \times 1.30 = 156$ and its payable becomes $100 \times 1.30 = 130$. Its historical capital remains 24. The remaining 2 is translation OCI, explained by its EUR20 net assets and the rate change: $20 \times (1.30 - 1.20) = 2$ (European Commission, 2023a, IAS 21.28, .33, .39–41).

Consolidation now debits the translated loan payable 130 and credits the parent's receivable 130. It also debits the subsidiary's historical capital 24 and credits the parent's investment 24. These entries eliminate the internal balances; neither entry removes the parent's exchange gain. Group cash is $56 + 156 = 212$, with no external liability in this example. Opening equity 200, exchange profit 10 and translation OCI 2 reconcile to the same 212. Deleting the profit merely because the receivable was internal would leave equity short by 10. The group still owns the euros whose dollar value has risen (European Commission, 2023a, IFRS 10.B86; IAS 21.45).

The loan's settlement facts can change the presentation of that 10. In the ordinary-loan case, settlement is planned in the foreseeable future, so the gain stays in consolidated profit. In a separate conditional case, assume settlement is neither planned nor likely in the foreseeable future and the loan forms part of the net investment under IAS 21.15. This is a substantive condition, not an election created by calling a loan long term; trade receivables and payables are explicitly excluded. The parent's separate gain is still 10. In the consolidated statements, debit exchange gain 10 and credit OCI 10, giving total OCI of 12 and no exchange profit. The two cases in Table 10.1 have identical group cash and total equity but different profit and OCI (European Commission, 2023a, IAS 21.15, .32–33).

Table 10.1: Settlement facts change where the exchange result appears.

USD amount	Ordinary loan	Qualifying net-investment loan
Parent's separate exchange gain	10	10
Consolidated exchange profit	10	0
Consolidated OCI	2	12
Consolidated closing cash and equity	212	212

Neither case includes a cash receipt. The same EUR120 remains in the subsidiary, so its translated value rises by $120 \times (1.30 - 1.20) = 12$. IAS 7.28 presents this exchange effect separately from operating, investing and financing cash flows. Group cash reconciles as opening 200, transaction cash flows zero and exchange effect 12, giving closing 212 (European Commission, 2023a, IAS 7.25–28). Adding a further gain of 12 would count the same currency movement twice.

The representation must therefore retain both entities' functional currencies, the monetary balance, the settlement assessment and the origin of each OCI component. If the settlement facts are unknown, these conditional presentations remain separate even when they agree on total equity. The group adjustment does not overwrite the parent's separate journal, and the OCI history remains available for the applicable later disposal treatment. An equal consolidated total cannot establish that the histories answer the same profit, cash-flow or disposal question.

10.22 An elimination belongs to an identified transaction

Equal balances need not describe the same debt. Suppose the parent records a receivable of 100 from subsidiary S, while another subsidiary T reports a payable of 100. Cancelling those numbers would balance the worksheet while misstating both relationships. The group must first establish the seller, buyer and economic event that connect the two entries. IFRS 10.B86 requires the elimination of intragroup transactions; it does not make equality of amounts evidence that two entries are counterparts (European Commission, 2023a, IFRS 10.B86–88, B92–93).

For each proposed elimination, retain the reporting group and perimeter version, both legal entities and book revisions, the common economic-event reference, invoice or settlement allocations, transaction currency, reporting cutoff and history branch. The two books' local event identifiers can differ; an evidenced mapping connects them to the common transaction. For balance elimination, use the perimeter at the reporting date; for income and cash-flow elimination, use the intervals during which control exists. A receivable created before control began can be intragroup at the reporting date without making the original sale an intragroup revenue transaction for that period. Acquisition measurement and subsequent movements must remain distinct. Align accounting policies, reporting dates and currency measurements before comparing balances. A reporting-date exception needs the conditions and intervening

adjustments in IFRS 10.B92–93.

One invoice may correspond to several ledger lines. An explicit allocation of 40 and 60 can exhaust a receivable of 100; reusing the 40 creates an invalid allocation of 140. Each source amount is consumed at most once in a given consolidation version. A receivable of 100 against a payable of 90 leaves an unresolved difference of 10 until a settlement, cutoff correction or other supported adjustment explains it. Equal labels, a balancing entry or a stale source revision cannot supply that explanation. Unknown matches remain visible and prevent a complete-consolidation result for that view. These identity rules are implementation choices for preserving the source meaning; they are not a claim that IFRS prescribes a database key.

Now follow one correctly matched inventory sale. The wholly owned subsidiary begins with cash and capital of 50. The parent holds cash 90, inventory 60 and the investment 50, with equity 200. All amounts use one currency; taxes, impairment and other transactions are absent by assumption. The parent sells the inventory to the subsidiary for 100 on credit. The subsidiary then sells 40 percent of those goods to an external customer for cash 60, recording cost 40 and retaining inventory 60. The parent's profit is 40 and the subsidiary's profit 20, but the group has earned only 36: external revenue 60 less the original cost 24 of the goods sold.

Eliminate the investment 50 against subsidiary opening equity 50 and the payable 100 against receivable 100. Of the internal markup 40, the fraction still in inventory is 60 percent. The performance adjustment therefore debits internal revenue 100 and credits cost of sales 76 and inventory 24. Combined cost of sales 100 becomes 24, inventory 60 becomes 36, and profit 60 becomes 36. Closing group cash 200 plus inventory 36 equals opening equity 200 plus profit 36. The same result follows directly from the external history: opening group cash 140 plus customer receipt 60, with inventory cost 60 less 24. The adjustment retains the lot and external-sale references needed to release the remaining markup when those goods leave the group. Actual tax bases, upstream NCI allocation and evidence that an intragroup loss reflects impairment require their own consequences under IFRS 10.B86(c). The earlier foreign-loan case also remains: eliminating a monetary balance does not automatically erase its exchange gain (European Commission, 2023a, IFRS 10.B86(c); IAS 21.45).

10.23 Restating purchasing power before translating currency

An inventory cost and a bank balance of the same nominal amount do not retain the same purchasing power. IAS 29 addresses this problem when the functional currency belongs to a hyperinflationary economy. Its paragraph 3 requires judgment across economic indicators; three-year inflation approaching or exceeding 100 percent is one indicator, not an automatic test. The selected assessment, effective period and general price index must therefore accompany the entity book (European Commission, 2023a, IAS 29.1–4, 8–19, 24–28, 37–39).

For historical-cost nonmonetary amounts, retain each acquisition or measurement date and restate from that date's index to the closing index. Restate income and expense from recognition dates, and contributed equity from the relevant contribution dates. Monetary balances remain at their closing nominal amounts; contractually index-linked amounts first follow their contractual adjustment. Amounts already measured at closing current values are not indexed again. After restatement, apply the relevant recoverability or net-realizable-value limit. Missing dates or indices require the specific permitted estimation procedure and its support; an unexplained common multiplier is insufficient.

Consider a period beginning with cash 100, inventory 60, fixed nominal debt 80 and contributed capital 80. All arose at the opening date, and there are no transactions, interest, tax or impairment during the period. Assume IAS 29 already applies and the general index rises from

100 to 150. Inventory becomes 90 and capital 120; cash 100 and debt 80 remain unchanged. Holding cash loses 50 units of closing purchasing power, while owing the fixed debt gains 40. The net monetary loss is therefore 10. If M denotes the constant excess of monetary assets over monetary liabilities, and I_0, I_1 the opening and closing indices, this special case gives

$$L_{\text{mon}} = M\left(\frac{I_1}{I_0} - 1\right) = (100 - 80)(1.5 - 1) = 10.$$

Positive L_{mon} is a loss. A net monetary liability reverses the sign; an unchanged price index gives zero. The restated balance sheet reconciles as $100 + 90 - 80 = 110 = 120 - 10$. The restatement entry debits inventory 30 and monetary loss 10 and credits restated capital 40. If inventory's supported closing net realizable value is 84, a separate write-down 6 reduces equity to 104. Indexing both monetary balances would incorrectly increase net assets by 10 and conceal the purchasing-power loss.

The constant-exposure formula does not apply to arbitrary turnover. For a piecewise constant nominal exposure, split the history at every monetary change. If M_j is the net monetary amount held between index observations I_{j-1} and I_j, its loss expressed at closing index I_T is $M_j(I_j/I_{j-1} - 1)I_T/I_j$; sum those interval losses and separately handle contractually indexed balances. Recognition times and index observations determine those intervals. IAS 29.27 also permits a weighted-average estimate; its approximation must be identified. Reconcile the result with restated nonmonetary amounts, equity and income rather than inventing a residual profit. For example, external cash income 30 at index 125 changes the preceding net monetary exposure from 20 to 50. The two losses are 6 and 10, totaling 16; restated income 36 and monetary loss 16 reconcile closing net assets 140 to restated opening capital 120. Cash-flow items likewise use closing purchasing-power units. In the no-flow case, opening cash expressed in closing units is 150 and ending cash 100; the purchasing-power reconciliation of minus 50 is not a payment (European Commission, 2023a, IAS 29.26–34).

First application requires earlier acquisition history. IFRIC 7.3–5 requires restatement as though the economy had always been hyperinflationary and a particular ordering for opening deferred tax. In a separate inventory example, cost 40 at index 80 becomes 50 at the opening index 100 and 75 at the closing index 150. Assume a fixed nominal tax base 40, tax rate 20 percent, profit-origin differences and no recognition exception. Opening deferred tax is $(50 - 40) \times 0.2 = 2$, then restated to 3. Closing deferred tax is $(75 - 40) \times 0.2 = 7$; the period expense is 4. Indexing a nominal opening deferred-tax balance of zero would miss this sequence (European Commission, 2023a, IFRIC 7.3–5; IAS 29.32).

Only then translate the restated statements. At a supplied closing rate of USD 0.20 per local unit, the first example gives cash 20, inventory 18, debt 16, capital 24 and monetary loss 2: net assets 22 equal 24 minus 2. Translating unrestated inventory 60 would report 12 and understate the asset by 6. IAS 21.42–43 uses the closing rate for these restated current-period amounts. When the presentation currency is not hyperinflationary, prior comparative amounts remain those presented in the relevant prior-year statements. Thus prior cash 100 translated at the prior rate 0.25 remains USD 25, even though the current local comparative is restated. Keep these two comparative representations distinct (European Commission, 2023a, IAS 21.42–47; IAS 29.34–36).

10.24 A quoted rate may not be available for the required purpose

A subsidiary may obtain foreign currency to import goods while being unable to remit its net investment. Exchangeability therefore belongs to a currency direction, measurement date and

purpose. The inspected EU amendment 2024/2862 requires application no later than financial years beginning on or after 1 January 2025; IAS 21.60L permits earlier application with disclosure. The following method uses that amendment, rather than extending the older paragraph 26 rule about temporary lack of exchangeability to later periods (European Commission, 2024, Article 2; IAS 21.8–8B, 19A, 26, 60L–M).

Record the entity, both currencies, measurement date, purpose, amount needed, amount obtainable, permitted mechanisms and their enforceability, normal administrative delay and evidence for the finding. The assessment concerns ability, including indirect exchange, rather than management's intention. For individual transactions, the purpose is realization or settlement of those transactions; for presentation-currency translation it is realization or settlement of net assets or liabilities; for a foreign operation it is realization or settlement of the net investment. Assess the relevant direction and purpose separately. Obtaining only an insignificant amount fails the exchangeability test. The ratio 5 available to 1000 needed is evidence for a judgment, not a threshold supplied by this design (European Commission, 2024, IAS 21.A2–A10).

When exchangeability fails, IAS 21.19A requires an estimate reflecting an orderly exchange between market participants at the measurement date under prevailing economic conditions. A rate observable for another purpose, or the first subsequent exchange rate, can be used without adjustment only if it meets that objective. Examine incentives or penalties, restricted purposes, interventions, update frequency, elapsed time and inflation. Another estimation technique may adjust observable rates, including those from mechanisms that would not establish exchangeability. It must preserve the inputs, adjustments and reasoning supporting the measurement-date objective. Absence of that support leaves the rate unresolved; the design does not manufacture a universal estimator (European Commission, 2024, IAS 21.A11–A17).

For a conditional arithmetic illustration, assume an admitted estimate of USD 0.08 per local unit for a foreign operation's net investment. Local net assets 1000 translate to 80. A preferential import rate 0.10 would produce 100; a later rate 0.09 following changed conditions would produce 90. Neither alternative is interchangeable with the admitted estimate merely because it is observable. This calculation exposes the consequences of the choice; the numerical differences do not prove that 0.08 meets paragraph 19A.

Initial application has its own journal origin. Suppose an affected monetary payable of 100 foreign units moves from 10 to 12 functional units per foreign unit at initial application. The liability increase 200 adjusts opening retained earnings under paragraph 60M(a). In a separate nonhyperinflationary foreign-operation translation example, changing the rate for net assets 1000 from 0.10 to 0.08 reduces translated equity 20; paragraph 60M(b) assigns the initial effect to accumulated translation differences. Hyperinflationary functional-currency equity items also use the estimated initial rate. Comparatives are not restated for this amendment. Keep these transition adjustments distinct from subsequent-period profit or OCI (European Commission, 2024, IAS 21.60M).

The accompanying disclosures identify affected currencies and restrictions, transactions and carrying amounts, rates, estimation methods and assumptions, and exposure to risk. A foreign operation also needs the additional information in paragraph A20, including relevant support arrangements (European Commission, 2024, IAS 21.57A–B, A18–A20). These methods extend the documentary group design. They do not silently add fields or supported events to the closed `GroupForeignOperation/1` exchange profile; its successor must explicitly bind the additional histories and assessments before an implementation can claim these operations.

Chapter 11

Credit losses and the bank balance sheet

The funding and default distinctions become central in a bank. A loan creates an asset, customer payments affect settlement liquidity, and a forecast of credit loss affects an allowance before cash is lost. The first case connects funding to a simple allowance; the later dated loan schedule separates credit stages, interest recognition and eventual recovery.

11.1 Banks: accounting capital is not regulatory capital

A bank package needs instrument and cohort states for loans, securities, deposits, wholesale funding, repos, derivatives, collateral and commitments. For interest-rate analysis, currency and repricing buckets matter alongside remaining maturity. For credit analysis, exposure, default state, loss severity, recoveries, guarantees and their dependencies matter. For liquidity analysis, deposit behaviour, collateral eligibility, encumbrance and payment timing matter. These states can be represented at an explicitly chosen aggregation level; the public record rarely identifies every contract.

11.1.1 A reported group boundary

JPMorgan Chase's filings make the boundary between a bank, its parent and its securities issuers concrete. In the 2022–2025 reports, Item 1 identifies JPMorgan Chase Bank, N.A. as the principal bank subsidiary and describes an intermediate holding company, JPMorgan Chase Holdings LLC (the IHC). The reports also describe J.P. Morgan Securities LLC as a principal non-bank subsidiary and place J.P. Morgan Securities plc and J.P. Morgan SE beneath the bank. These are reported relationships for the stated dates, not a complete legal-entity graph (JPMorgan Chase & Co., 2023, 2024, 2025, 2026).

The attached Exhibit 21 tables qualify their scope: they list significant legal-entity subsidiaries, some parents of significant subsidiaries and certain entities designated for resolution planning, while excluding subsidiaries that would not be significant in aggregate under the stated SEC definition. Their indentation shows how the issuer presents the rows, but it does not supply ownership percentages, tax residence, functional currency or unrestricted transfer rights. A model therefore keeps a reported display relationship separate from a confirmed control edge.

Exhibit 22.2 adds a different edge. It names JPMorgan Chase Financial Company LLC as issuer of Global Medium-Term Notes, Series A, and JPMorgan Chase & Co. as guarantor. Financial

Company LLC does not appear in those qualified Exhibit 21 lists. The 2022–2023 text describes Series A notes as currently offered and sold under one cited registration; the 2024–2025 text also says that the issuer may issue debt securities and warrants under a later cited registration. The descriptions say that the relevant securities are fully and unconditionally guaranteed, but they give no outstanding principal, guarantee valuation or accounting recognition. They also do not establish that a warrant was issued or that an earlier note was redeemed.

The same reports state that the parent depends on subsidiary cash flows to pay its securities. The bank's dividends are subject to regulatory restrictions, and the IHC cannot pay dividends or extend credit to the parent when stated capital or liquidity thresholds, or management and board limits, apply. This means that consolidated cash, a subsidiary's assets and the parent's distributable funding are different state variables. The example gives the simulator a place to record the legal entity, guarantee and conditional funding path before it assigns a cash movement; it does not infer a balance or a probability of default.

Note 33 makes the funding distinction measurable for 2025. Its parent-only income statement reports net income of US$57,048m. The parent-only cash-flow statement subtracts US$71,163m of subsidiary and affiliate net income, reports a US$14,115m parent company net loss in that reconciliation, then adds US$50,000m of cash dividends from subsidiaries and US$8,583m of other operating adjustments to reach US$44,468m of operating cash. Neither the income figure nor the dividend alone is the period's ending cash.

At 31 December 2025, the parent reports US$39m of cash and due from banks and US$9,751m of deposits with banking subsidiaries, together US$9,790m. The cash-flow footnote says this measure includes US$8.0bn of intercompany time deposits. The same parent-only balance sheet reports US$669,449m of equity-method investments in bank and bank-holding subsidiaries; that asset is not parent cash. Operating cash of US$44,468m and investing cash of US$34m, less US$44,512m used in financing, leave closing cash and deposits US$10m below their opening balance. These dated amounts distinguish accounting income, existing cash and conditional subsidiary transfers. A future debt-service calculation still needs payment dates, permissible transfers, deposit terms and funding actions (JPMorgan Chase & Co., 2026, pp. 19, 310–311).

Note 26 supplies a dated transfer capacity, which differs from both cash held and cash already received. It reports that, at 1 January 2026, JPMorgan Chase Bank, N.A. could pay approximately US$30.1bn in dividends to the parent without prior approval from its relevant banking regulators. That is a capacity of the named bank at that date; the US$50,000m above is cash received from subsidiaries during 2025. The note also describes regulators' power to prohibit unsafe or unsound dividends and the IHC's separate capital, liquidity and management restrictions. It says the bank's 2026 earnings will supplement dividend capacity, without estimating those earnings. Turning the reported capacity into a future cash receipt still requires a dividend decision, payment date and the applicable conditions (JPMorgan Chase & Co., 2026, pp. 4, 292).

The same note reports US$29.0bn of consolidated restricted cash at 31 December 2025: US$19.4bn segregated for securities and cleared-derivative customers, plus US$9.6bn of non-US central-bank reserves and cash held for other general purposes. Its balance-sheet footnote allocates the total to US$27.8bn of deposits with banks and US$1.2bn of cash and due from banks. Those classifications describe the same restricted-cash total. Subtracting it from the parent's US$9,790m would mix consolidated and parent-only perimeters and cannot determine the parent's unrestricted cash. Other pledged cash and securities are disclosed separately; their category overlap must be established before adding them to a restricted-resource total (JPMorgan Chase & Co., 2026, p. 292).

For a future cash dividend, the model needs the bank's declaration as well as the eventual payment. The OCC's national-bank dividend rules, retrieved for 1 January 2026, use the

declaration date to determine compliance. They define retained net income as net income less all dividends declared in the period. Absent OCC approval under the earnings provision, current-year declarations, including the proposal, are compared with current-year net income to date plus the previous two years' retained net income, less specified required transfers. The current year is the calendar year of declaration. A cash-flow statement of dividends paid therefore cannot supply the declaration history by itself (Office of the Comptroller of the Currency, 2026f, 12 CFR 5.61–5.64).

The preceding two years are not always enough. Section 5.64(c)(2) provides conditional offsets for retained losses caused by dividends exceeding a single year's net income. Those offsets can require the third and fourth preceding years and follow a specified order; other earnings deficits do not qualify for that adjustment. An implementer must preserve the cause of the loss and the allocation of offsets before computing the earnings restriction. Capital, undivided-profit and supervisory conditions remain separate: an earnings calculation alone cannot establish a permissible transfer (Office of the Comptroller of the Currency, 2026f, 12 CFR 5.63–5.65).

Selecting the applicable capital rule also requires more than a calendar lookup. The final leverage rule published on 1 December 2025 is effective 1 April 2026 but permits early adoption from 1 January (OCC, Federal Reserve Board, and FDIC, 2025, p. 55248). JPMorgan's 2025 Form 10-K reports that the Firm elected early adoption effective 1 January 2026 (JPMorgan Chase & Co., 2026, p. 91). The rule's publication, general effective date, permitted election date and the issuer's reported election answer different questions. The later filing provides evidence about the election; it does not establish that the election was publicly known on 1 January. A historical information query must retain that distinction even when reconstructing an earlier regulatory state.

The amendment removes the additional 6% supplementary-leverage condition from the affected national banks' well-capitalized classification and adds separate leverage-buffer payout provisions. That classification change does not remove the distinct undercapitalization test or establish unrestricted distributions (OCC, Federal Reserve Board, and FDIC, 2025, pp. 55287–55288). The reported Firm election also does not supply every bank-specific application fact. A permissible dividend calculation still needs the relevant capital definitions, distribution limits, supervisory conditions and bank-level inputs. The US$30.1bn remains the issuer's dated capacity disclosure, rather than a recomputed legal limit.

One of those inputs, eligible retained income, differs from the retained net income used in the national-bank earnings rule. Part 3 takes the greater of two amounts: Call Report net income for the four preceding calendar quarters, net of distributions and associated tax effects not already reflected in that income; and the average quarterly net income over those four quarters. Its maximum payout amount for the current quarter multiplies this eligible income by the applicable payout ratio. The relevant capital buffers use the previous quarter's closing ratios, so the declaration, income window and capital-measurement date must remain distinct (Office of the Comptroller of the Currency, 2026d, 12 CFR 3.11(a)(2)–(4)).

The transactions being limited also extend beyond a cash dividend. Part 3's distribution definition includes dividend declarations or payments on Tier 1 capital instruments and specified capital repurchases or redemptions, with conditional exceptions for qualifying replacements (Office of the Comptroller of the Currency, 2026b, 12 CFR 3.2). The quarterly limit also covers discretionary bonus payments and the creation of obligations to make the specified payments. A bank's event history must therefore identify the instrument, declaration, obligation and settlement, then reconcile their regulatory treatment; a cash-dividend total cannot substitute for that history (Office of the Comptroller of the Currency, 2026d, 12 CFR 3.11(a)(4)).

A hypothetical bank in the first quarter of a year illustrates the income distinction. Suppose each of the preceding four quarters has Call Report net income of US$10m. Assume its only

distributions in that preceding calendar year were US$38m of declared dividends, with no associated tax adjustment. Part 5 retained net income for that year is then US$2m: US$40m less US$38m. Part 3 eligible retained income is US$10m, the greater of that US$2m remainder and the US$10m quarterly average. These are two calculations for the same hypothetical history, not interchangeable names for one balance.

Now suppose the bank falls within the amended rule's covered population and applies that rule. Take its previous-quarter capital conservation buffer as 1.5%, its applicable countercyclical amount as zero, its leverage buffer standard as 1%, and its measured leverage buffer as 0.2%. These are supplied example inputs, rather than estimates from a public balance sheet. The rule's Table 1 gives a 40% payout ratio, which alone would produce US$4m from the US$10m eligible income. Table 2 gives zero because the leverage buffer is at most one quarter of its standard. The required lower ratio therefore gives a zero table-based maximum, absent the specified OCC permission. Conversely, a table entry saying that no payout-ratio limitation applies does not mean a 100% ratio or permission under every other rule (Office of the Comptroller of the Currency, 2026d, 12 CFR 3.11(a)(4), (c), Tables 1–2).

Replacing the example's supplied leverage-buffer standard with an observed requirement needs a specific surcharge. For a bank within the amended rule's covered population, the standard is the lesser of 1% and, if applicable, half of the controlling global systemically important holding company's most recent required Method 1 surcharge, subject to the specified effective-date provisions. The holding company's overall surcharge instead takes the greater of its Method 1 and Method 2 surcharges. The overall figure alone does not identify the input this bank rule requires. Annual determination and the date an adjusted surcharge takes effect also remain separate (Office of the Comptroller of the Currency, 2026d, 12 CFR 3.11(a)(1)(v)) (Board of Governors of the Federal Reserve System, 2026c, 12 CFR 217.403(a)–(d)).

Method 1 compares twelve reported systemic indicators with published global amounts, applies their specified weights and caps the substitutability category's contribution at 100 basis points (Board of Governors of the Federal Reserve System, 2026b, 12 CFR 217.404). The denominators have their own measurement and publication dates. The Federal Reserve's notice published on 16 December 2025 supplies the 2025 denominators using year-end 2024 global data, converted from euros into dollars at the stated 31 December 2024 rate. The preceding notice, published on 18 December 2024, uses year-end 2023 data (Board of Governors of the Federal Reserve System, 2025, pp. 58245–58246) (Board of Governors of the Federal Reserve System, 2024, pp. 102908–102909). The release-year label therefore does not identify the underlying observation year or the date a bank's capital requirement changes. The published amounts are in dollars; reported indicator values must be brought to the same unit before division.

Reporting precision adds another constraint. The retained FR Y-15 general instructions require dollar amounts in thousands and permit rounding to the nearest million, with adjusted detail so the components add to the totals. Its payment-activity instructions instead say payment totals are not rounded, specify accuracy according to their magnitude and allow known overestimates when precise totals are unavailable. A reconstruction cannot treat every reported input as an exact value or give all components independent rounding intervals (Board of Governors of the Federal Reserve System, 2026d, GEN-3–4 and C-1; physical pp. 7–8, 25).

Rounding the resulting score is a separate operation. In the background to its March 2026 proposal, the Federal Reserve describes scores as rounded to the nearest basis point: fractions below one-half round down and fractions at or above one-half round up (Board of Governors of the Federal Reserve System, 2026a, p. 14909, n. 9). The earlier description appears in the 2015 final rule's preamble, within its account of the preceding proposal (Board of Governors of the Federal Reserve System, 2015, p. 49084, n. 22). These passages provide an official explanation of the convention; they do not complete the amendment history needed to establish its applicability

on a chosen date. The 2026 proposal also contains a separate ceiling formula for Method 2 surcharge bands and changes to indicator averaging and denominator exchange rates. Those proposed changes cannot be substituted for the rule applicable to the bank and date being reconstructed (Board of Governors of the Federal Reserve System, 2026a, pp. 14917–14922).

The report's quarter-end date differs from its submission deadline and public availability; amendments, later resubmissions and confidentiality can change what a historical reader could know. The retained instruction file itself combines a September 2021 cover with a January 2026 reporting-scope page. Applying it consequently requires the relevant page and reporting vintage, alongside the applicable rule and institution facts. These sources explain the input requirements; they do not yet supply a verified bank-specific score or a complete reporting-rule history (Board of Governors of the Federal Reserve System, 2026d, cover and GEN-1–5; physical pp. 1, 5–9).

Debt amounts add a second distinction. Note 20 puts US$253,039m of long-term-debt carrying value under Parent company for 2025, while Note 33's parent-only balance sheet shows US$254,633m. The US$1,594m difference equals Note 20's separately displayed junior-subordinated-debt subtotal. The same comparison gives US$1,603m in 2024. This exact two-year arithmetic match does not establish how that row relates to the parent-only balance, or which legal entity owes each instrument. Its accounting and contractual relation still needs source support.

Timing requires a different measure. Note 20 places US$14,298m of parent carrying value in its under-one-year bucket; Note 33 reports US$14.3bn of parent long-term debt contractually maturing in 2026. They align at the footnote's displayed precision, but carrying value includes premiums, discounts, costs and valuation adjustments rather than simply principal payable. For the separately reported consolidated zero-coupon population, Note 20 gives US$18.1bn of carrying value against US$57.6bn of aggregate principal at the notes' respective maturities. The firm's US$435,206m consolidated long-term-debt total and US$47.6bn of subsidiary debt and structured notes guaranteed by the parent are also separate from the parent's own debt balance; the latter guarantee amount supplies neither a liability valuation nor a payment forecast. A dated parent debt-service path still needs instrument terms, cash dates and permissible funding transfers (JPMorgan Chase & Co., 2026, pp. 281–282, 310–311).

The complete Form 10-Ks describe Series A as the issuer's direct, unsecured general obligations, ranking equally with its other unsecured and unsubordinated debt. Claims under the parent guarantee generally rank behind those of subsidiary creditors. Their summary of the 2016 Indenture requires the parent, on demand after issuer nonpayment, to make full and punctual payment of principal, interest and other amounts due: a guarantee of payment, not collection (Sections 14.01–14.02). It excludes amounts the issuer is excused from paying or beyond those actually due. Issuer defenses and counterclaims carry over, except those arising solely from insolvency or bankruptcy status. These descriptions refer to the Indenture and applicable prospectus supplements; they specify reported terms but do not supply a current balance, valuation or accounting recognition.

The complete filings also describe one particular Series A population: the Callable Fixed Rate Notes due 10 June 2032. They report US$5,000,000 of aggregate principal outstanding at 31 December 2022 in the 2022 filing, at 31 December 2023 in the 2023 filing, and at 31 December 2024 in both the 2024 and 2025 filings. The last report repeats a 2024 as-of date; this passage supplies no 2025 year-end balance. The notes were issued on 10 June 2021 and bear 2.60% annual interest payable semiannually on 10 June and 10 December. They mature on 10 June 2032 unless an earlier contractual outcome applies. The issuer may redeem all, but not part, on 10 June 2031 at principal plus accrued and unpaid interest. Upon acceleration after a continuing event of default, the stated amount is US$1,000 per US$1,000 principal plus accrued and unpaid interest. The descriptions report NYSE listing under JPM/32, DTC global book-entry issuance, and the possibility that a reopening increases aggregate principal outstanding (JPMorgan Chase & Co., 2023, 2024, 2025, 2026).

These dated amounts are reported principal, not carrying or fair value. Nor do they establish the accounting classification or value of the parent guarantee, the entire Series A population, or an unchanged balance between the stated dates. Each filing's Exhibit 4.4 identifies the 19 February 2016 Indenture and incorporates it by reference to Exhibit 4(a)(7) of Form S-3 file 333-209682, filed 24 February 2016. The complete filings summarize that instrument; its text and applicable supplements or amendments require their own inspection before being used as complete legal terms.

The filings then describe general Series A terms that may apply to securities including the Callable Notes. In their summary, unpaid interest must continue for 30 days to become an event of default; principal and other amounts default when due. An issuer covenant breach has a 90-day period after the specified written notice. For listed continuing payment, issuer-covenant or guarantee events, the trustee or holders of at least 25% of the aggregate principal then outstanding for the relevant series may, by written notice to the issuer and guarantor, declare principal and accrued interest due. Specified continuing issuer insolvency events accelerate the debt automatically without a declaration. Guarantor insolvency alone is excluded and does not authorize acceleration under this description, while a payment default by the guarantor remains a separate possibility (JPMorgan Chase & Co., 2023, 2024, 2025, 2026). A history must therefore keep the failure, notice, elapsed time, actor, declaration and changed due schedule distinct, rather than record one group-wide default flag.

Collective remedy control also differs from an individual's right to payment. The summaries condition an ordinary holder remedy action on prior trustee notice, a request by holders meeting the stated 25% threshold, 60 days of trustee inaction, indemnity and no inconsistent majority direction. They separately protect each holder's right to receive and sue for payment when due against impairment without that holder's consent. The summary permits certain amendments without holder consent under Section 8.01. For consent-based modifications under Section 8.02, it specifies a majority in aggregate principal amount of each affected outstanding series, whereas listed changes to maturity, amounts, currency, payment method, redemption, payment suit or specified holder repurchase rights, voting threshold or adverse guarantee terms require each affected holder's consent. These thresholds attach to different actions and voting populations, not a generic creditor vote. The 2022 default-and-waiver subsection omits the trustee-conflict paragraph that the 2023–2025 versions place there. The 2022 filing discusses a potential default, trustee conflict and possible resignation later under its trustee heading. This change in placement does not establish a contract amendment. The original Indenture and series-specific supplements remain uninspected (JPMorgan Chase & Co., 2023, 2024, 2025, 2026).

Reported succession terms give another reason to keep the obligor separate from the debt claim. Under the general Series A summary, a successor to the issuer in a covered merger, consolidation or transfer of all or substantially all assets must be a U.S. corporation or limited liability company that expressly assumes due payment and the issuer's covenants. A successor to the guarantor must instead be a U.S. corporation that assumes the full guarantee and the guarantor's covenants. The continuing issuer or guarantor is the other stated route. Each party's route also requires that no default or event that would become a default with notice or time has occurred or would be continuing immediately after the transaction. The descriptions deem certain resolution-related transfers of the guarantor's material assets to meet the all-or-substantially-all category; they do not define a numeric percentage (JPMorgan Chase & Co., 2023, 2024, 2025, 2026).

The same summaries say a recapitalization, change of control or highly leveraged transaction alone receives no additional holder protection from these merger provisions; a particular supplement may provide a put or higher interest. The model therefore needs a dated successor, assumed payment and covenant duties, separate issuer/guarantor conditions and any evidenced series-specific protection. Succession itself does not prove repayment, accounting derecognition or unrestricted access to the group's cash. These are reported terms that may apply, pending inspection of the original Indenture and applicable supplements.

Loan origination can debit loans and credit a customer deposit in the bank's book. Subsequent transfer to another bank produces settlement flows. Treating all loan growth as an immediate cash outflow would therefore encode a specific funding assumption, not a universal banking identity. A system-wide model must also link counterparties and settlement assets; single-company simulation records external counterparties without claiming to close the whole economy.

An allowance calculation is a dated measurement based on forecast information. For a single horizon, let EAD be a fixed exposure at default, PD the probability of default, and LGD the mean loss fraction conditional on default. Ignoring discounting, the illustrative expected loss in currency units is

$$L = EAD \times PD \times LGD.$$

With exposure 100, default probability 0.02 and loss given default 0.4, it produces 0.8. Changing those inputs to 0.04 and 0.5 produces 2, so the allowance increase is 1.2 if there are no other movements. This calculation is a demonstration model. A standard-specific expected-credit-loss implementation must determine horizon, staging or other classification, discounting, forward-looking information, write-offs and purchased-credit-impaired or other special cases. The simple formula is not an IFRS 9 or US GAAP compliance implementation.

The exposure entering an allowance calculation also needs a defined population. JPMorgan's 2025 report gives four different amounts for lending-related commitments at 31 December 2025:

Reported quantity	US$ millions
Contractual lending-related amount	1,817,307
Commitments in the allowance-methodology table	603,947
Allowance for lending-related commitments	5,071
Lending-related carrying value	6,278

Notes 13 and 28 define these populations and measurements separately (JPMorgan Chase & Co., 2026, pp. 258–262, 295–297). The contractual total contains consumer and wholesale commitments; the allowance table excludes specified populations. The commitment allowance is reported in accounts payable and other liabilities. Carrying value can also contain guarantee liabilities, fair-value measurements and deferred fees or purchase adjustments, as applicable. For standby letters of credit and other financial guarantees, the report explicitly reconciles a US$175m allowance and a US$441m guarantee liability to US$616m of carrying value. Treating either the contractual total or carrying value as the allowance would compute the wrong quantity.

The report also distinguishes contractual capacity from expected funding. Customers may leave facilities unused, refinance them or let them expire; some commitments can be reduced or cancelled under their contracts or law. The reported policy records no allowance for future draws on unconditionally cancellable commitments. That exclusion concerns the loss measurement; a liquidity scenario still needs the conditions under which a customer can draw, the bank can cancel, and a payment requires settlement funds. Those conditions and their dependence on borrower distress cannot be recovered from a commitment total. Comparisons also need the same reporting basis: specified wholesale amounts are net of risk participations here but gross in Federal Reserve filings. Parent guarantees of subsidiary debt are expressly outside the Note 28 table (JPMorgan Chase & Co., 2026, pp. 258, 295–296, 300).

The historical FASB amendment ASU 2016-13 helps explain these distinctions. Its paragraph 326-20-30-11 measures off-balance expected credit losses over the contractual period of a present obligation to extend credit, except for an obligation the issuer may cancel unconditionally. For a covered commitment, the estimate considers both whether funding will occur and the expected loss on commitments that do fund. Example 10 keeps funded credit-card loans within loss measurement while excluding an allowance on unconditionally cancellable undrawn credit, even though the bank has sometimes advanced funds before detecting default. Paragraph 460-10-30-5 separately requires a contingent expected-loss liability and a noncontingent fair-value guarantee liability for covered financial guarantees. These are historical amendment provisions, not verified current Codification; applying them to the 2025 report requires the actual contracts and later rule history (Financial Accounting Standards Board, 2016a, PDF pp. 2, 108, 118, 144, 173–176).

A commitment to acquire loans makes the time distinction especially clear. The report dates the Apple Card forward purchase commitment to 30 December 2025, its announcement to 7 January 2026, and expected closing to approximately 24 months later. The commitment disclosure includes about US$104bn of estimated credit exposure at closing, including about US$23bn of estimated drawn loans, while the reported transaction allowance is US$2.2bn (JPMorgan Chase & Co., 2026, pp. 47, 260, 296). The expected drawn loans at that future closing are not loans already owned at the 2025 reporting date. A reconstruction therefore retains the forward commitment, the future portfolio estimate and the dated allowance as different facts. A forecast must supply closing and subsequent draw assumptions before generating loan and settlement events. These reported policies and amounts support the distinctions; their application to a new calculation still requires the operative accounting rules and actual contract facts.

Regulatory capital is computed in a separate prudential layer with its own perimeter, eligibility, deductions, risk weights and transition provisions. Accounting equity feeds a reconciliation to capital; it is not automatically Common Equity Tier 1. Liquidity and capital constraints influence the management policy, which in turn generates asset sales, funding or capital events. A breached ratio can be an economically meaningful scenario outcome. No hidden capital injection is permitted to repair it.

The United States national-bank rules show why that reconciliation needs records for each instrument. Under the common-stock criteria in the retained January and April 2026 texts, classification as equity under GAAP is only one condition. The instrument must also satisfy conditions concerning payment and direct issuance by the bank, priority in insolvency, loss absorption, permanence, dividend discretion, purchase financing and guarantees. The rules distinguish common equity Tier 1, additional Tier 1 and Tier 2, each with its own eligible elements and deductions. A security's name and its balance-sheet category therefore cannot supply its regulatory treatment (Office of the Comptroller of the Currency, 2026c, 12 CFR 3.20(a)–(d)).

Regulatory recognition can also change while a debt remains outstanding. The general Tier 2 instrument criteria require an original maturity of at least five years. At the beginning of each of its last five years, the amount eligible for Tier 2 falls by 20% of the original amount, net of redemptions; the instrument is excluded when less than one year remains. Calls and early redemptions have further conditions, including OCC approval. This reduction in eligible capital does not itself reduce the accounting debt. The contract, issuance and redemption history must support both amounts separately (Office of the Comptroller of the Currency, 2026c, 12 CFR 3.20(d)(1), including footnote 7).

Deductions require similarly specific records. Section 3.22 deducts goodwill and specified other assets from common equity Tier 1, with separate rules for their tax effects. Its general rule permits an associated deferred tax liability to reduce an asset deduction only if that liability would be extinguished when the asset is impaired or derecognized. The same liability cannot offset two assets; separate conditions govern offsets against deferred tax assets, and the chosen netting treatment must be consistent across reporting periods unless the OCC approves a change. Accumulated other comprehensive income has its own adjustments and election conditions, including limits on which institutions may opt out and when they must elect. Reconstructing capital consequently needs asset-specific tax links, instrument facts, institution classification and the actual election history. The reported equity and aggregate deferred-tax balances do not determine that reconstruction (Office of the Comptroller of the Currency, 2026a, 12 CFR 3.22(a), (b), (e)).

11.2 A bank's cash, allowance and capital in one path

The bank reference state contains settlement cash 30, gross loans 100, allowance 0.8, deposits 110 and equity 19.2. Originating a loan of 20 first debits loans and credits the borrower's deposit. A subsequent transfer of 15 to another bank debits deposits and credits settlement cash. Loan origination and external cash use are consequently distinct events. Receive loan interest 6 in cash and credit deposit interest 2 to customer balances. Remeasure the allowance to 2.4, posting an additional loss of 1.6. The resulting state is cash 21, gross loans 120, allowance 2.4, deposits 117 and equity 21.6. Net assets are $21 + 120 - 2.4 = 138.6 = 117 + 21.6$.

Now 25 of deposits runs off. The declared funding policy uses cash first, then a committed facility with limit 10. It draws 4 and settles 25, leaving cash zero, deposits 92 and wholesale debt 4. A runoff of 40 instead leaves 9 unpaid even after the maximum facility draw. The simulator records default and retains the unpaid deposit obligation. It does not assume unlimited central-bank or shareholder funding.

The prudential calculation is a separate, illustrative policy. Deduct 2 from accounting equity to obtain capital 19.6; apply a 75% risk weight to gross loans for risk-weighted assets 90. The resulting ratio is 19.6/90. Neither the deduction nor the weight is a Basel default. Actual prudential packages require jurisdiction, exposure class, eligibility, buffers and dated rules, with a reconciliation from the accounting book. The structure follows the need to model capital and liquidity jointly in Behn et al. (2019, Section 2), without transferring that paper's estimated preferences or shock parameters to this bank.

11.2.1 An allowance affects capital and exposure in different ways

The illustrative ratio above shows where a prudential calculation belongs. Applying a particular rule requires a further step: reconcile its numerator and denominator to the same bank's accounting records. An allowance already reduces accounting equity through expense, yet a specified portion may enter Tier 2 capital. The regulatory loan exposure can meanwhile remain gross of that allowance. Using the net loan balance for every purpose would erase these distinctions.

Consider a separate synthetic US national bank, with amounts in millions of US dollars. Its conditional reporting window is January–March 2026. For this calculation, freeze the retained 1 January 2026 national-bank rules; this assumption does not establish the rules actually applicable to a 31 March filing. The bank has no subsidiaries, minority interests, AOCI, deferred taxes, other capital instruments or deductions apart from goodwill. It is outside the advanced-approaches, Category III, market-risk and community-bank leverage-ratio regimes, and has no transition or special supervisory adjustment. Its common stock is supplied as satisfying all thirteen conditions in section 3.20(b)(1). These are explicit case facts, not conclusions drawn from a security's name (Office of the Comptroller of the Currency, 2026c, 12 CFR 3.20(b)(1)–(2)).

Settlement cash is a direct Federal Reserve balance. The loans are ordinary, unsecured, performing corporate exposures measured at amortized cost: no purchased credit-deteriorated loans, nonaccrual, ninety-day delinquency, HVCRE, securities, derivatives or credit mitigation. The bank has adopted CECL. Its allowance amounts are supplied accounting measurements, not estimates made here, and all qualify as adjusted allowances for credit losses, or AACL. There is no allocated transfer risk reserve or off-balance credit exposure. These restrictions are consequential: a deposit at another commercial bank and a nonaccrual corporate loan need different exposure analysis (Office of the Comptroller of the Currency, 2026b, 12 CFR 3.2, definitions of AACL and carrying value).

The opening book contains cash 30, gross loans 100, goodwill 2 and allowance 0.8 against deposits

110, common stock 20 and retained earnings 1.2. Assets net of allowance are $30+100+2-0.8 = 131.2$, so equity is 21.2. Originate a loan of 20 by crediting the borrower's deposit, transfer 15 externally, receive earned loan interest 6, and credit deposit interest 2. Remeasure the supplied allowance to 2.4, recording credit-loss expense 1.6. The income statement reports $6 - 2 - 1.6 = 2.4$, which a separate closing entry transfers to retained earnings. Cash becomes 21, loans 120, goodwill 2, allowance 2.4 and deposits 117; net assets 140.6 equal deposits 117 plus equity 23.6. The cash bridge is $30 - 15 + 6 = 21$. The loan origination, deposit-interest credit and allowance change do not move settlement cash.

Construct the exposure base before applying the allowance cap

For these ordinary loans, the regulatory carrying-value definition does not subtract the associated allowance. The exposure amount is therefore the gross loan balance. Section 3.31 multiplies that amount by the applicable weight; section 3.32 assigns 100% to the stated corporate exposures and zero to the direct central-bank balance. Goodwill is deducted from CET1 and excluded from risk-weighted assets. With no associated deferred tax liability, its deduction is the full 2 (Office of the Comptroller of the Currency, 2026h, 12 CFR 3.31(a)–(b)) (Office of the Comptroller of the Currency, 2026g, 12 CFR 3.32(a)(1)(i)(A), (f)(1), (k)) (Office of the Comptroller of the Currency, 2026a, 12 CFR 3.22(a)(1), (e), (g)).

Let G be gross loans, A eligible AACL and B the risk-weighted base used to limit the allowance admitted to Tier 2. For this restricted book, $B = G$. The capital rule caps that admission at 1.25% of the relevant base. The official reporting instructions make the order concrete: Part I item 42 uses Part II item 26 for the cap; Part II subsequently subtracts the excess allowance in arriving at its total. Define admitted allowance T, excess X and final standardized risk-weighted assets D accordingly:

$$T = \min\{A, 0.0125B\}, \qquad X = A - T, \qquad D = B - X.$$

The last equality uses this example's exclusions. In a broader book, the cap base and pre-excess standardized denominator need not coincide. The instructions exclude section 22(a) deductions such as goodwill from the cap base but include specified section 22(c)–(f) deductions, which remain excluded from the final standardized denominator. The case has none of those other deductions or market-risk amounts (Office of the Comptroller of the Currency, 2026c, 12 CFR 3.20(d)(3)) (Federal Financial Institutions Examination Council, 2024a,b, Part I, item 42, pp. RC-R-27–28; Part II, items 26 and 28–31, pp. RC-R-109–110).

These retained instructions are the December 2024 files, with March 2024 revision footers on the cap pages. They corroborate the ordering in the specified historical rule calculation; they do not supply a complete amendment history for an actual later filing. The source date and the conditional reporting date remain separate.

Initially $B = 100$, the cap is 1.25 and all the allowance 0.8 enters Tier 2, leaving $D = 100$. At the final pre-runoff state, $B = 120$, the cap is 1.5, and excess allowance is $2.4 - 1.5 = 0.9$. Thus $D = 120 - 0.9 = 119.1$. CET1 is equity less goodwill, $23.6 - 2 = 21.6$. There is no additional Tier 1 capital, so Tier 1 is also 21.6; total capital is $21.6 + 1.5 = 23.1$. The following reconciliation retains the different measures rather than giving them one label.

Measure	Opening	Before runoff
Accounting equity	21.2	23.6
Goodwill deduction	2	2
CET1 and Tier 1	19.2	21.6
Eligible allowance in Tier 2	0.8	1.5
Total capital	20	23.1
Allowance-cap base	100	120
Excess allowance	0	0.9
Standardized risk-weighted assets	100	119.1

CET1 and Tier 1 ratios are $21.6/119.1 = 72/397$, about 18.14%; the total-capital ratio is $23.1/119.1 = 77/397$, about 19.40%. These are conditional ratios, not a determination of payout permission or all applicable capital requirements (Office of the Comptroller of the Currency, 2026e, 12 CFR 3.10(b)(1)–(3)).

Several plausible substitutions give the wrong quantity. Net loans 117.6 are the accounting amount, not this regulatory exposure. Deducting the full allowance from risk-weighted assets loses the rule's distinction between admitted and excess allowance. Deducting it again from CET1 counts an expense twice. Including goodwill in the exposure base reverses its required exclusion. Finally, using D to calculate its own allowance cap gives the circular equation $T = (117.6 + T)/80$, hence $T = 117.6/79$, rather than the required 1.5. Solving that equation accurately cannot make it the specified calculation.

A capital ratio does not settle the next cash payment

Now suppose 25 of deposits runs off. The supplied funding policy draws 4 within an available facility limit of 10, then pays 25. Cash becomes zero, deposits 92 and borrowing 4. Net assets 119.6 still equal liabilities 96 plus equity 23.6. Capital and risk-weighted assets are unchanged: the disappearing central-bank cash has zero weight, and no additional income or expense is assumed. The facility is borrowing by this bank, not an off-balance extension of credit by it to a customer.

If instead 40 is demanded, the same facility supplies at most 10. The bank pays 31 and leaves 9 unpaid, retained within its deposit liability of 86. Borrowing is 10 and cash zero. The capital ratios remain the same, although the payment obligation has failed. No capital or cash adjustment may be invented to remove that default. The two runoff outcomes are alternative histories of this bank, not balances to be combined.

The ordinary leverage ratio asks for another denominator: average total consolidated assets as reported for the relevant window, less specified Tier 1 deductions. Neither closing assets 140.6 before runoff nor 119.6 after it determines that average. If a separate, properly matched input supplies average assets 145, the denominator is $145 - 2 = 143$ and the ratio is $21.6/143 = 108/715$, about 15.10%. Without that input, the leverage ratio remains unresolved while the supported risk-based ratios remain available (Office of the Comptroller of the Currency, 2026e, 12 CFR 3.10(b)(4)).

The proposed `RegulatoryAssessment` therefore references the exact bank, book, perimeter, accounting state, information view, rule snapshot, capital eligibility, exposure classifications, deductions, allowance admission and reporting window. Average-assets evidence is a separate dependency. Its result names each numerator and denominator and records which outputs remain unresolved. Unknown stock eligibility blocks the dependent capital amounts; an incorrect perimeter rejects the input; a missing average blocks leverage alone. The assessment creates no accounting entry. A changed allowance, instrument, exposure class, average or rule

requires reassessing the affected outputs while retaining the historical result under its original assumptions.

For a richer credit adapter, partition borrower cohorts by currency, vintage, credit state, remaining maturity, collateral and measurement category. At date t, supply scenario probabilities, conditional default-time probabilities, exposure paths, conditional loss severity and discount factors. Its expected loss calculation has the declared form

$$L_t = \sum_s p_s \sum_{k \in H} d_{t,k} \Pr(\tau = k \mid s, \mathcal{F}_t) \mathbb{E}[\mathrm{EAD}_k \mathrm{LGD}_k \mid \tau = k, s, \mathcal{F}_t].$$

The conditional product expectation retains dependence between exposure and severity. The horizon H concerns eligible default times; associated cash shortfalls may continue later. The rule package decides twelve-month versus lifetime default scope, credit-state transfers, interest basis and write-off recognition. Changes in allowance must be attributed to origination, derecognition, transfers, revised expectations, currency and write-offs. Contract and prudential support can be specified now, while full IFRS 9/CECL/ ASBE classification and paragraph-level validation remain later obligations.

11.3 Classification comes before the loan calculation

A loan's contractual principal is not enough to determine its carrying amount. The dated amortised-cost examples that follow require both an eligible contractual cash-flow pattern and a business model for collecting those flows. These are findings about the instrument and the way assets are managed, not labels chosen to obtain a preferred profit figure. The following procedure uses the retained historical IFRS 9 clauses in EU Regulation 2023/1803; the surrounding Australian and Chinese examples retain their own stated jurisdiction and date boundaries.

First establish the financial asset's terms and the level at which the bank actually manages it. Evidence includes how performance is reported to key management, how risks are managed and how managers are compensated. An intention to retain one asset is insufficient. Sales must be explained by their reasons, timing, volume and place in that management objective; the source supplies no universal sale-count threshold (European Commission, 2023a, IFRS 9.4.1.1–3, B4.1.1–4B).

Next ask whether contractual payments are solely principal and interest on the outstanding principal. Principal here begins with fair value at initial recognition. Interest compensates for time value, credit risk, other basic lending risks and costs, and a profit margin. Equity-linked returns and leverage introduce different exposures. For a hybrid whose host is a financial asset, assess the entire contract under IFRS 9.4.3.2; the issuer's separate liability and equity components do not automatically become the holder's measurement units (European Commission, 2023a, IFRS 9.4.1.3, 4.3.2, B4.1.7A–9, B4.1.14).

If that cash-flow test passes, a hold-to-collect business model leads to amortised cost, while a model achieved by both collection and sales leads to fair value through other comprehensive income (FVOCI) for debt. Other financial assets generally use fair value through profit or loss (FVTPL), subject to the separate equity election below. At initial recognition, an otherwise eligible asset may be irrevocably designated FVTPL when the specified accounting-mismatch condition is met. Preserve that designation and its reason. A later change in one asset's intended sale date is not by itself a business-model change authorising reclassification (European Commission, 2023a, IFRS 9.4.1.2–5, 4.4.1, B4.1.2A–3).

11.3.1 The same price decline can reach profit and OCI differently

Consider a debt asset purchased for 100 and now worth 98. Supply cumulative ECL of 3, with no interest accrual, foreign-exchange movement, tax or fees in this measurement comparison. Opening cash and equity are each 200; purchase leaves cash 100. The three cases below have different admitted business models or an eligible FVTPL designation. They compare reporting consequences, rather than three freely selectable policies for one factual case.

At amortised cost, record impairment expense and allowance 3. Gross asset 100 less allowance 3 gives net asset 97, so cash 100 plus net asset 97 equals equity 197. A subsequent derecognising sale for 98 debits cash 98 and allowance 3, credits gross asset 100 and recognises gain 1. Cumulative profit is $-3 + 1 = -2$, and closing cash and equity are 198.

For debt FVOCI, the balance-sheet asset is 98. Record the fair-value decline 2 in OCI. Impairment expense 3 enters profit, with its allowance counterpart credited to OCI rather than deducted again from the asset's fair value. Net OCI is therefore $-2 + 3 = 1$; profit is -3, and total comprehensive income is -2. Cash 100 plus asset 98 equals equity 198. Reporting asset 95 by subtracting the allowance once more would count the credit effect twice. On a derecognising sale for 98, remove the asset and reclassify cumulative OCI gain 1 to profit. Cumulative profit then equals -2, with OCI cleared (European Commission, 2023a, IFRS 9.5.5.1–2, 5.7.10–11).

Under FVTPL, the decline from 100 to 98 is expense 2. There is no separate IFRS 9 loss allowance added to that fair-value loss. The sale at the same price creates no further gain. All three examples ultimately close with cash and equity 198, but their interim carrying amounts and placement of income differ. Reconciliation must preserve the chosen route rather than infer it from the equal final cash.

An equity investment is a distinct instrument. Where the irrevocable FVOCI election is available and made at initial recognition, a fall from 100 to 98 gives OCI loss 2 and no fair-value expense in profit. It excludes an investment held for trading and specified contingent consideration of an acquirer in a business combination. On sale, the cumulative loss is not recycled to profit; a transfer within equity is permitted. Dividends follow their separate recognition conditions, including whether they clearly represent recovery of part of the investment's cost. With no dividends in this example, cash and equity again end at 198 while cumulative profit is zero (European Commission, 2023a, IFRS 9.5.7.5–6, B5.7.1–1A). Copying the debt recycling entry into this case would wrongly create expense 2 on sale.

11.3.2 A familiar interest label can conceal a different exposure

A floating rate need not faithfully represent the passage of time. Suppose principal 100 pays monthly interest but resets each month to a one-year rate. A comparison instrument would reset monthly to a one-month rate, with the other relevant terms and credit quality matched. In one supplied six-month scenario, hold the contractual annualised rate at 12% and the benchmark rate at 6%. Each month's interest is respectively 1 and 0.5; six payments differ by 3. With principal repaid at month six, total undiscounted receipts are 106 and 103.

This calculation identifies the difference to investigate. IFRS 9's modified-time-value test compares undiscounted contractual and benchmark cash flows both by reporting period and cumulatively over the instrument's life, considering reasonably possible scenarios. A current yield curve, discounted present-value match or an invented percentage cutoff cannot settle whether the difference could be significant. Retain the benchmark construction, scenarios, period totals and significance reasoning before making that finding (European Commission, 2023a, IFRS 9.B4.1.9A–9E).

Nonrecourse terms likewise do not automatically fail the cash-flow test. The holder examines

the underlying assets or cash flows to determine whether the contractual restriction introduces payments inconsistent with basic lending. Contractually linked tranches require distinct tests of the tranche's terms, underlying pool and relative credit-risk concentration, including the permitted instruments in the pool. If the holder cannot perform the assessment required by B4.1.26, that clause requires FVTPL. A missing document in this system, however, is initially an unresolved evidence requirement; it is not proof that the holder made, or could not make, that assessment (European Commission, 2023a, IFRS 9.B4.1.15–26).

The method therefore records the governing rule, management evidence, contractual findings, designation or election, and remaining unknowns before selecting an interest, impairment and disposal procedure. Unsupported classification leaves those dependent calculations conditional. A supplied fair value or loss allowance can illustrate the reporting arithmetic while its valuation or credit model still requires separate specification and validation.

11.4 State and event order

The dated credit methods use the selected amortised-cost, interest and impairment provisions of AASB 9 (Australian Accounting Standards Board, 2022g). The following loan has different inputs from the earlier liquidity example, so its allowance and final equity are calculated separately.

BankState contains cash/reserves by settlement venue and currency; deposits by customer, access rights and maturity; assets by contract, gross carrying amount, allowance, original EIR, accrued cash flows, collateral and credit state; undrawn commitments; funding lines and covenants; accounting equity; regulatory mappings; and evidence/information views. LoanState separately retains origination credit risk and current risk, SICR finding, credit-impaired/POCI status, forbearance, modification, write-off and recovery history. Aggregating by today's loan balance alone loses both staging evidence and later response to shocks.

Each time step first admits information known then, forms permitted credit and liquidity forecasts, chooses behavior under that information, and schedules due events. Accrual, actual settlement, impairment measurement and regulatory queries have separate business identities. A deposit received from outside debits cash/reserves and credits deposit liability. A loan credited to the borrower's deposit debits loans and credits deposits without initially transferring reserves; a transfer to another bank then debits deposits and credits reserves. A loan paid directly outside instead debits loan/credits reserves. These two funding paths cannot be treated as identical cash events. Deposit interest debits interest expense/credits deposit or accrued payable; withdrawal requires settlement funds.

11.5 Expected credit loss

For each admitted scenario j and date t, calculate contractual due flows minus the cash expected to be received, including supported collateral/enforcement cash and timing; discount the shortfall under the applicable original EIR or specified exception. With mutually exclusive exhaustive scenario weights w_j, ECL is the sum of w_j times those present-value shortfalls. Default-time hazards must be converted to disjoint default-time probabilities; summing cumulative PDs over dates counts the same default repeatedly. Collateral recoveries cannot be credited both in LGD and as a second recovery cash flow. A shorthand PD×LGD×EAD is admissible only if its definitions and discounting reproduce the specified cash-shortfall calculation; it is not a universal replacement.

Expected credit loss (ECL) measures the discounted expected cash shortfall. A significant increase in credit risk (SICR) changes the general method's default horizon. Purchased or originated credit-impaired (POCI) assets follow a separate method. For the general approach, twelve-month ECL considers lifetime shortfalls arising from possible defaults during the next twelve months; it is not just unpaid cash in those twelve months. SICR uses the change in lifetime default risk since initial recognition, reasonable forward-looking information and the applicable rebuttable presumptions. Do not install a two-times-PD or 30-day rule as an unconditional universal trigger. Lifetime ECL follows a supported SICR finding; cure needs a supported reversal of that finding. Subsequently credit-impaired assets use original EIR on amortised cost for interest in subsequent periods; POCI has a separate credit-adjusted-rate and cumulative-change method.

The 2017 ASBE 22 text makes the two horizons explicit. Article 47 defines credit loss as the present value of all contractual cash flows less expected receipts; a late receipt can therefore create a loss even when its nominal amount is paid in full. Article 48 limits twelve-month ECL by the date of the possible default event, while retaining the resulting cash shortfalls over the asset's life. Article 52 assesses changes in default risk, rather than changes in the allowance, and Article 58 requires probability weighting, time value and reasonable, supportable information (Ministry of Finance of the People's Republic of China, 2017, Articles 47–48, 52, 58). These provisions support the following dated comparison. Article 84 states 1 January 2018, but applying the standard to a particular company also requires its issuance schedule, subsequent amendments and adoption facts.

The general measurement interface requires scenario_cashflows, law_ref, information_view, horizon_admission, credit_assessment and discount_rule. An unweighted history set can produce conditional losses or loss bounds, but cannot produce an unbiased probability-weighted ECL by silently using equal weights. Return measurement_unresolved for the ECL claim. Trade receivables and certain other assets use the separately applicable simplified approach; the general loan staging rule must not overwrite AASB 9.5.5.15–16.

11.6 A commitment carries its credit history into the loan

A promise to lend can create a credit-loss provision before it creates a loan or consumes cash. Once the customer draws, the uncertainty about whether funding will occur has been resolved, but the customer's earlier credit history has not disappeared. The distinction connects the commitment populations reported above to the funded-loan method that follows. AASB 9.5.5.6 dates the initial impairment comparison from the irrevocable commitment; B5.5.47 treats the resulting financial asset as a continuation of that commitment (Australian Accounting Standards Board, 2022g).

Consider a synthetic bank with cash 200, deposits 180 and equity 20. It grants an irrevocable commitment allowing one immediate draw of 100 or lapse. The contract is fixed rather than revolving, cannot be net settled, is not designated at fair value through profit or loss, and does not promise a below-market loan. No fee or separate noncontingent commitment liability is assumed. The bank will hold the resulting loan to collect principal and interest. Its admitted initial fair value is 100, equal to its cash price, and its payment of 121 after two years gives an effective interest rate of 10%. These are explicit measurement and classification facts. The credit probabilities below do not themselves establish market value; a different initial value or a below-market commitment would require the separate analysis under AASB 9.2.3, 4.2.1(d) and 5.1.1–5.1.1A.

All commitment and draw snapshots are ordered on 1 January 2024, before any material interest

accrues. Initially, the probability of a full draw is one half. Conditional on drawing, default occurs in the first year with probability 0.1, in the second with probability 0.1, or does not occur with probability 0.8. These are mutually exclusive outcomes, not cumulative default probabilities. Either default produces collection of 108.9 at the contractual two-year maturity; no default produces 121. The present value of the cash shortfall from either default is

$$\frac{121 - 108.9}{1.1^2} = 10.$$

The first-year default's loss is paid out at a later date, yet belongs in twelve-month expected credit loss (ECL). The horizon selects possible default events; it does not truncate their later cash shortfalls. Including expected draws as required by B5.5.30–31 gives an initial provision of $0.5 \times 0.1 \times 10 = 0.5$. Debit impairment expense 0.5 and credit the commitment provision 0.5. The nominal limit 100 is neither a funded asset nor an impairment expense.

Before the draw, a new forecast and a separately supported significant increase in credit risk (SICR) change the conditional default probabilities to 0.2 in each year and 0.6 for no default. The borrower is not credit-impaired. Doubling a probability is not a universal SICR rule; the credit finding is a distinct input in this illustration. With draw probability still one half, lifetime ECL is now $0.5(0.2 + 0.2)10 = 2$. Debit a further impairment expense of 1.5 and credit the provision by 1.5.

The customer now actually draws the whole 100. To see what changes, the joint probabilities immediately before funding are 0.5 for no draw, 0.1 for draw and first-year default, 0.1 for draw and second-year default, and 0.3 for draw without default. Observing the draw removes the first outcome and conditions the other three on their total probability of 0.5. The funded loan therefore has lifetime ECL

$$\left(\frac{0.1}{0.5} + \frac{0.1}{0.5}\right) 10 = 4.$$

Keeping the old draw probability would still report 2, which measures the earlier uncertain commitment. Resetting the funded loan to twelve-month ECL would also give 2, now by omitting the second default window. Equal wrong amounts can conceal different errors. The original commitment date, credit baseline, current finding and actual draw must all survive the transition.

For a direct payment outside the bank, debit gross loan 100 and credit cash 100. In the same draw event, debit the existing provision 2, debit additional impairment expense 2 and credit the loan allowance 4. The first 2 moves an existing loss estimate from a liability into a deduction from the funded asset; the other 2 reflects the information that funding has occurred. AASB 7.B8E distinguishes commitment provisions from financial-asset allowances, while 35H–35I and B8D require an explanation of their movements (Australian Accounting Standards Board, 2022f). Funding and its impairment transition are committed together under the existing event procedure, even though the entries are explained separately here.

There is no remaining undrawn capacity. Cash 100 plus gross loan 100 less allowance 4 gives net assets 196, equal to deposits 180 plus equity 16. Cumulative impairment expense is $0.5 + 1.5 + 2 = 4$. Keeping the old provision of 2 as well as the full funded allowance would count the loss twice and leave equity 14. Using allowance 2 would instead leave equity 18. All three sets of accounts can balance; only one measures the admitted funded exposure.

The route through the borrower's bank account has a different intermediate state. Debit gross loan 100 and credit the borrower's deposit 100, with the same provision-to-allowance transition. Cash stays 200, deposits become 280, and equity is 16 after allowance 4. If the borrower then transfers 100 to another bank, debit deposits and credit settlement cash by 100. The final state matches direct disbursement. Creating the deposit did not itself consume settlement funds; the later transfer did.

Alternatively, the customer lets the binding window expire without drawing. Debit the provision 2 and credit impairment expense 2. Cash 200, deposits 180 and equity 20 remain, with no loan. This is an admitted lapse, not an invented cancellation right. A partial or revolving draw needs separate funded and remaining-exposure measurements. B8E allows combined allowance presentation when their losses cannot be separately identified; any excess over the gross funded asset is a provision. Neither that condition nor the longer revolving exposure period in AASB 9.5.5.20 follows merely from calling something a credit facility.

The probability tree is an authored assumption, not a calibrated forecast or an observed default sample. Unknown probabilities can support conditional losses but cannot produce a probability-weighted ECL without a measure. Missing original credit history leaves staging unresolved, and missing initial valuation leaves the funded measurement unresolved. The retained AASB 9 compilation 5 and AASB 7 compilation 7 cover the stated historical 2024 case. Accounting equity of 16 does not determine regulatory capital: the applicable jurisdiction, perimeter, deductions and exposure rules must be supplied independently. The next example follows a funded loan through interest, deterioration, collection and write-off using different stated facts.

11.7 Full loan and allowance roll-forward

Assume a fixed-rate non-POCI amortised-cost loan held to collect its contractual cash flows, originated for 100, with one terminal payment 121 two years later, original EIR 10%, no fees, collateral, modifications, FX or taxes. Since no payment falls due before maturity, an earlier default in this example is a separately established qualitative credit event, such as financial distress under the admitted default definition; it is not a missed instalment. Article 52 of ASBE 22 requires a definition consistent with credit-risk management and consideration of qualitative indicators (Ministry of Finance of the People's Republic of China, 2017). At origination the assumed lifetime scenarios are: default in the first twelve months with probability 0.1, default during the following year with probability 0.1, and no default with probability 0.8. Either default gives terminal collection 60.5. The first twelve-month ECL includes lifetime loss from the first scenario only: $0.1\times(121-60.5)/1.1^2=5$; initial lifetime ECL would be 10. Debit loan 100/ credit cash 100; debit impairment 5/credit allowance 5. Gross 100, allowance 5, net 95.

At year-one accrual the loan has survived the first window and is not yet credit-impaired: debit gross loan 10/credit interest 10. Survival alone would give remaining default probability $0.1/0.9=1/9$. A subsequent, separately admitted credit forecast revises that conditional probability to 0.2; it is an authored forecast update, not a Bayes calculation from survival alone. Its lifetime weights are 0.8/0.2, with a separately established SICR finding; $\text{ECL}=0.2\times60.5/1.1=11$. Debit impairment 6/ credit allowance 6. Gross 110, allowance 11, net 99. The ratio change alone is not the defined SICR test. Only twelve months remain, so twelve-month and lifetime ECL both equal 11 under this revised forecast. This amount cannot distinguish the two classifications.

Immediately afterward, a distinct, ordered credit-impairment event changes expected maturity collections to 60.5. This is a forecast mean, not a guaranteed or maximum recovery. Required allowance is $110-60.5/1.1=55$; debit impairment 44/credit allowance 44. Net 55. ASBE 22 Article 59(5) measures this non-POCI credit-impaired asset by gross balance less discounted expected receipts. Articles 38 and 39(2) then make amortised cost, after the allowance, the basis for subsequent-period interest (Ministry of Finance of the People's Republic of China, 2017).

The bank is assumed still to have reasonable prospects of fuller recovery before maturity. The separate finding that some contractual cash will not reasonably be recovered first arises at settlement on this branch. If that finding arose earlier, ASBE 22 Article 43 would require

the corresponding earlier write-off. Expected collections alone do not justify postponing an already-required write-off.

In the next full year interest revenue is 55×10%=5.5. The gross contractual accretion is 11. One consistent decomposed posting debits gross loan 11, credits interest revenue 5.5 and allowance 5.5: gross 121, allowance 60.5, net 60.5. The allowance increase of 5.5 is attributed to discount unwind in its roll-forward, with no additional impairment expense in this posting. This reconciles gross and net-interest bases without counting the same expected loss again. Collect 60.5: debit cash/credit gross 60.5. When no reasonable expectation of remaining recovery is established, debit allowance 60.5/credit gross 60.5. Both balances reach zero. A mere delinquency is not that write-off finding. Subsequent unexpected cash recovery is recorded with its own recovery-income method and evidence; it does not restore an old live receivable by default.

Moment	Cash, opening 200	Gross loan	Allowance	Deposit liability	Equity, opening 20
Initial loan and allowance	100	100	5	180	15
Year-one accrual and SICR	100	110	11	180	19
Credit impairment	100	110	55	180	−25
Maturity accrual	100	121	60.5	180	−19.5
Collection and write-off	160.5	0	0	180	−19.5

Total interest 15.5 less impairment 55 is loss 39.5, equal to 100 advanced minus 60.5 received. The negative equity state survives; no missing branch or capital plug restores solvency. These five dates jointly test allowance, recognition and bank balance-sheet arithmetic. The same loss cannot also be deducted directly from gross before the stated write-off.

For a non-derecognized loan modification, AASB 9.5.4.3 requires gross PV at original EIR and a modification result, then allowance and SICR reassessment under 5.5.12. For example gross 110 changed to a single future 110 at original 10% gives revised gross 100 and modification loss 10 before any allowance change. ASBE 22 Articles 42 and 56 similarly separate remeasurement of a non-derecognised asset from its credit-risk reassessment (Ministry of Finance of the People's Republic of China, 2017). This is separate from the financing method's issuer-liability ten-percent test. A derecognition decision requires its own asset conditions and cannot reuse that threshold.

11.8 Changed terms and recovered credit

A borrower can receive easier terms without recovering its ability to pay. A bank can also recover its expectation of payment without recovering all the credit quality present at origination. These are different changes to the loan. The following historical Australian examples use the retained AASB 9 compilation for synthetic 2024–2025 reporting periods. Probabilities, legal findings and credit assessments are supplied assumptions; taxes and fees are excluded.

Start from cash 100, a gross loan 110, allowance 55 and deposits 180. Net assets 155 leave equity −25. At the 2024 close, the bank replaces a payment 121 due one year later with 110 at the same date. Assume the change preserves the existing asset's contractual rights and does not qualify for derecognition. The original annual effective interest rate is 10%. Discounting the amended contractual payment gives

$$G_{\text{modified}} = \frac{110}{1.1} = 100, \qquad \text{modification loss} = 110 - 100 = 10.$$

Debit modification expense 10 and credit gross loan 10. This calculation changes the contractual carrying amount before considering expected defaults. Paragraph 5.4.3 requires that separation. Fees would require an additional carrying-value adjustment and amortisation method (Australian Accounting Standards Board, 2022g, paragraph 5.4.3).

The bank separately expects full 110 with probability 4/5 and zero with probability 1/5. Expected cash 88 has present value 80, so the required allowance is $100 - 80 = 20$. Debit the old allowance 35 and credit impairment reversal 35. Cash 100 and net loan 80 now equal deposits 180 and equity 0. Neither this reversal nor the new payment terms establish recovery from credit impairment. For a loan that remains recognised, the comparison for a significant increase in credit risk retains the original, unmodified origination terms (Australian Accounting Standards Board, 2022g, paragraph 5.5.12 and B5.5.27).

A distinct, ordered credit review at that same close changes the supplied full-payment probability to 9/10. Expected 99 has present value 90; allowance falls 20 to 10, adding an impairment reversal 10 and leaving equity 10. Compare two histories at this point. In one, reinstatement of the borrower's operating licence and verified repayment capacity support a finding that the asset is no longer credit-impaired. The improvement is objectively related to an event after credit impairment. A significant increase in risk relative to origination nevertheless remains. In the other history, the supplied credit assessment still finds impairment despite the same improved cash expectation. A forecast mean alone does not decide between these findings.

The distinction changes the next year's income presentation. For the cured asset, paragraph 5.4.2 returns subsequent interest to gross 100: debit gross loan 10 and credit interest 10. At maturity, the expected shortfall 11 is no longer discounted, so debit impairment expense 1 and credit allowance 1. For the still-impaired asset, paragraph 5.4.1(b) applies the original rate to net 90: debit gross loan 10, credit interest 9 and credit allowance 1. Both histories reach gross 110, allowance 11 and net 99. Interest is 10 in one history and 9 in the other, with the compensating difference in impairment expense (Australian Accounting Standards Board, 2022g, paragraphs 5.4.1–2).

The borrower then pays 110, an outcome admitted by both supplied laws. Debit cash 110 and allowance 11, credit gross 110 and impairment reversal 11. Cash 210 equals deposits 180 plus equity 30. Profit since the opening is 55, equal to 110 received less the opening net loan 55. Equal closing wealth therefore cannot detect the wrong interest basis. Returning from lifetime to twelve-month expected credit losses would require a separate reversal of the significant-risk finding under paragraphs 5.5.3–12.

Ending contractual rights requires another decision. Suppose a different loan has gross 100 and allowance 20, and an enforceable agreement releases all remaining rights for an actual payment 85. There is no retained interest or replacement instrument. Debit cash 85 and allowance 20, credit gross 100 and gain 5: consideration 85 exceeds net carrying amount 80 by 5. Without the release, the same payment cannot by itself establish full derecognition (Australian Accounting Standards Board, 2022g, paragraphs 3.2.3 and 3.2.12).

For a transfer, the bank first identifies whether the unit is the whole asset or an eligible part, then assesses transfer conditions, risks and rewards, and control. A fixed-price repurchase can retain

the risks rather than establish a sale. The issuer-liability ten-percent comparison does not answer these asset questions (Australian Accounting Standards Board, 2022g, paragraphs 3.2.2–7). A replacement asset needs fresh classification, initial measurement and a purchased-or-originated credit-impairment assessment; a distressed replacement is not automatically an ordinary new loan with twelve-month losses (Australian Accounting Standards Board, 2022g, B5.5.25–26). Partial transfers and continuing involvement use paragraphs 3.2.13–23 (Australian Accounting Standards Board, 2022g, 3.2.13–23); the later revolving-capacity and transfer section develops those conditional methods.

11.9 Collateral received after maturity

Collateral changes expected receipts through its amount, timing and enforcement costs. Consider a separate loan advancing 100 on 1 January 2024 and requiring 121 on 1 January 2025. Its contractual half-year accumulation factor is 1.1, equivalent to 21% effective annual interest. This rate convention differs from the preceding annual 10% loan.

The supplied initial law assigns probability 3/5 to full payment at maturity and 2/5 to a default first observed within twelve months. In the default branch, collateral receipts arrive on 1 July 2025, one half-year after maturity. Possible net receipts 121, 66.55 and 12.10 have conditional probabilities $1/4, 1/2, 1/4$, respectively. Their conditional mean is 66.55. The bank has supported reasonable prospects of fuller recovery before the sale; 66.55 is not a known upper limit. These probabilities are illustrative assumptions.

The expected loss compares the present value of contractual cash with that of all expected receipts. The full-payment value is $121/1.1^2 = 100$; the default-branch mean has value $66.55/1.1^3 = 50$. Thus

$$\mathrm{ECL}_0 = 100 - \left(\frac{3}{5}\,\frac{121}{1.1^2} + \frac{2}{5}\,\frac{66.55}{1.1^3}\right) = 20.$$

The extra half-year belongs in the discounting. Omitting receipts after maturity produces the wrong loss 40. B5.5.55 expressly includes collateral cash beyond maturity and deducts obtaining and selling costs; B5.5.43 limits twelve-month losses by the default date, not by the recovery date. Here all possible defaults occur in the first year, so twelve-month and lifetime losses coincide (Australian Accounting Standards Board, 2022g, B5.5.43–44 and B5.5.55).

Start with cash 200, deposits 180 and equity 20. Advance 100 and recognise allowance 20. On 1 July 2024, first accrue gross interest 10. Rolling the original expectation forward gives net 88 against gross 110, so allowance rises 20 to 22 through impairment expense 2. A separately observed qualitative credit-impairment event then establishes the collateral-only expectation: $66.55/1.1^2 = 55$. Increase allowance 22 to 55 through impairment expense 33. The event is an admitted financial-distress finding, not an invented missed instalment before any instalment is due.

During the next half-year, original-rate interest on net 55 is 5.5. Debit gross loan 11, credit interest 5.5 and allowance 5.5. At contractual maturity gross 121 less allowance 60.5 leaves net 60.5. There is no contractual post-maturity interest, unamortised fee or premium. The initial 100-to 121 difference has been fully amortised. Over the final half-year the expected receipt value rises 60.5 to 66.55, giving net interest 6.05. Under these terms, gross remains 121 and allowance falls 60.5 to 54.45: debit allowance 6.05 and credit interest 6.05. This follows from the Appendix A definitions and B5.5.33's gross-minus-discounted-receipts measurement. It is a local derivation for these terms, not a rule that every overdue contract stops accruing contractual interest (Australian Accounting Standards Board, 2022g, Appendix A, paragraph 5.4.1(b), B5.5.33).

The middle recovery occurs: an enforcement agent obtains 79.86, deducts costs 13.31 already included in the forecast, and remits 66.55. Debit cash 66.55 and credit gross loan 66.55. Gross and allowance now both equal 54.45. At this sale, a separate supported exhaustion finding establishes that no further recovery is reasonably expected. Paragraph 5.4.4 requires write-off: debit allowance 54.45 and credit gross 54.45. An earlier such finding would require an earlier write-off; settlement is not a general waiting rule (Australian Accounting Standards Board, 2022g, paragraph 5.4.4).

Final cash 166.55 equals deposits 180 plus equity −13.45. Total interest 21.55 less net impairment expense 55 gives loss 33.45, also equal to 100 advanced less 66.55 received. The unpaid legal claim 54.45 remains in the contract history because no release has been supplied. Accounting write-off does not by itself waive that claim. The bank never recognised the agent's collateral as a separate owned asset, and never counted its proceeds a second time.

11.10 Revolving capacity, loan fees and the rights retained after transfer

These conditional methods extend the preceding full-draw example using the retained historical AASB9 and AASB7 compilations for explicit calendar-2024/2025 Australian for-profit Tier 1 cases. Legal rights, initial values and credit loss estimates are supplied assumptions. None is inferred from a facility label or from the fact that a receipt reached the bank's account.

11.10.1 A repayment can restore capacity without erasing credit history

A facility has both a funded balance and a right to request more funding. Record the legal limit, draw requests and actual settlements separately, together with the replenishment rule, expiry or cancellation conditions, currency, original credit-risk comparison date and current credit assessment. If the facility permits repeated drawings, paying down one loan may restore capacity. It does not establish a new low-risk borrower or reset the original credit history. Determine the resulting loan's classification and initial measurement from its business model, cash-flow terms and value, rather than assuming every funded amount qualifies for amortised cost.

For a partial draw, remeasure funded and remaining undrawn losses using the information available after the draw. Do not allocate the old provision by the percentage drawn unless the admitted loss model actually implies that result. AASB 9.5.5.6 and B5.5.47 (Australian Accounting Standards Board, 2022g, 5.5.6 and B5.5.47) preserve the commitment's impairment history. Where the two losses can be identified separately, AASB 7.B8E (Australian Accounting Standards Board, 2022f, B8E) presents the loan allowance and commitment provision separately. Where they cannot, the combined loss is recognised against the funded asset, with any excess over its gross carrying amount presented as a provision. Inseparability is a measurement fact, not an election to conceal a missing exposure model.

Start with cash 200, deposits 180 and equity 20. A revolving limit of 100 has an admitted undrawn provision of 2. No interest accrues between the following ordered events. The customer draws 40 externally; cash becomes 160 and gross loan 40. Conditional on the draw, supplied funded ECL is 3 and remaining undrawn ECL is 1.5. Debit impairment expense 2.5 and the old provision 0.5, and credit allowance 3. The result is allowance 3, provision 1.5 and equity 15.5. Cash 160 plus net loan 37 equals deposits 180, provision 1.5 and equity 15.5. The undrawn capacity is 60; it is not an asset of 60.

The borrower repays 10 externally. The contract restores that capacity, so cash is 170, loan 30

and available capacity 70. New supplied ECL values are funded 2.4 and undrawn 1.8. Reduce allowance by 0.6, increase provision by 0.3 and recognise a net impairment reversal of 0.3. Then a new draw of 20 gives cash 150, loan 50 and capacity 50. Supplied funded and undrawn ECL of 4 and 1.2 require allowance increase 1.6, provision reduction 0.6 and expense 1. Total loss balances are 5.2, cumulative impairment is 5.2 and equity is 14.8. Cash 150 plus net loan 46 equals deposits 180 plus provision 1.2 plus equity 14.8. Reusing the first funded estimate or treating the second draw as a new pristine commitment would compute a different exposure.

For a separate inseparable-loss example, gross funded loan 4 and combined ECL 6 require allowance 4 and provision 2. The rule does not create a negative net loan of 2, nor does it write off the contractual claim of 4. The reporting record preserves the combined estimate and the reason a separate attribution cannot be made.

A revolving facility does not necessarily use its short cancellation notice as its loss horizon. Under AASB 9.5.5.20 and B5.5.39–40 (Australian Accounting Standards Board, 2022g, 5.5.20 and B5.5.39–40), some jointly managed drawn/undrawn facilities expose the lender beyond that notice period because normal risk-management actions do not prevent those losses. Admit the exposure period from the facility's features, comparable historical exposure/default experience and expected risk actions. A one-day cancellation clause and a three-year assumed horizon are both insufficient alone. Missing evidence leaves the horizon unresolved; the mechanical capacity rollforward does not calibrate it.

11.10.2 Fee purpose determines whether it enters yield

Under AASB 9.B5.4.1–4 (Australian Accounting Standards Board, 2022g, B5.4.1–4), fees integral to origination or a probable specific lending arrangement enter effective interest for instruments measured at amortised cost. A fee's name does not settle its purpose. Servicing fees, commitment fees when a specific loan is unlikely, and qualifying syndication fees instead follow AASB15. Preserve the services promised, probability of a specific loan, instrument classification, timing, recipient and any allocation between distinct purposes. For instruments at fair value through profit or loss, the source's treatment of integral fees differs; do not copy the amortised-cost deferral.

Suppose a probable specific loan attracts commitment fee 2. Receive cash 2 and defer the fee; receipt alone is not service revenue. On origination, advance contractual principal 100 and incorporate the deferred fee in the loan's carrying amount, giving 98. A single contractual receipt of 107.8 one year later gives effective interest

$$r = 107.8/98 - 1 = 10\%, \qquad 98r = 9.8.$$

Accrete 9.8 and collect 107.8. Net cash from the fee and loan is $2 - 100 + 107.8 = 9.8$, equal to interest income. A supplied zero loss allowance isolates fee timing in this example; fee receipt does not imply zero credit risk. If the commitment instead expires without a loan, B5.4.2(b) recognises the fee as revenue on expiry. This branch contains no loan carrying amount.

If a fee is outside effective interest, determine revenue under the separate customer-service method. Calling it a service fee is not permission to recognise it immediately. For a separately admitted two-period stand-ready service with equal service each period and total consideration 2, the same cash receipt releases revenue 1 per period from a contract liability. A different obligation or satisfaction pattern changes that timing. The exact services must be supplied before this branch can be applied.

11.10.3 Selling a part requires identifying what was sold

AASB 9.3.2.2 (Australian Accounting Standards Board, 2022g, 3.2.2–3.2.23) first decides whether the derecognition test applies to a part or the entire asset. Eligible parts include specifically identified cash flows, fully proportionate shares of all cash flows, and fully proportionate shares of specified cash flows. The first or last collections from a pool do not become a qualifying pro rata part merely because their nominal amount is 40%. Then establish a transfer or qualifying pass-through arrangement and evaluate risks and rewards. If neither substantially all risks and rewards are retained nor transferred, evaluate control through the transferee's practical ability to sell. The retained legal and cash-flow evidence must support each branch.

When an eligible part is derecognised, 3.2.13–14 allocates the old carrying amount by the relative fair values of the transferred and retained parts at the transfer date. Retained servicing rights enter that analysis; inadequate servicing compensation can require a servicing liability. A nominal principal ratio is not a substitute for relative fair value.

Assume an eligible identified cash-flow part, completed transfer and admitted derecognition. The larger asset carries 100; transferred and retained parts have supplied fair values 45 and 55, and cash consideration is 46. Allocate carrying amounts 45 and 55. Debit cash 46, credit the asset 45 and recognise gain 1. The retained asset carries 55; net assets rise from 100 to 101. If the sold principal fraction happened to be 40%, using that fraction would report retained carrying value 60 and gain 6, despite the supplied relative values. This example excludes servicing, fees, tax and prior OCI and assumes no separate allowance at the measurement instant; those features need their own retained allocations, not silent deletion from a real transfer.

If substantially all risks and rewards are retained, 3.2.15 instead keeps the entire asset and recognises a liability for the proceeds. Receiving 46 then gives cash 46, unchanged loan 100 and liability 46, with no sale gain. The same cash receipt can therefore have a different balance sheet. Subsequent asset income and liability expense are recognised separately, not offset.

11.10.4 Continuing involvement preserves the guarantee exposure

Where neither substantially all risks nor rewards are transferred or retained and control remains, 3.2.16–22 requires continued recognition to the extent of involvement and an associated liability. A guarantee case follows B3.2.13(a) (Australian Accounting Standards Board, 2022g, B3.2.13(a)): the retained asset is the lower of its carrying amount and the maximum consideration that could be repaid; the initial associated liability is that guarantee amount plus the fair value of the guarantee. Subsequent loss allowance reduces the retained asset, while the initial guarantee fair value enters profit as the obligation is satisfied under AASB15 principles. Options and partial retained interests require their own measurement branches; the guarantee formula is not a general transfer formula.

Admit a transferred asset carrying 100, cash received 103 including fair value 3 for a guarantee, maximum repayment 20, and the required risk/reward/control findings. Retain an asset of 20 and recognise associated liability 23. The entry debits cash 103 and the retained asset 20, credits the old asset 100 and liability 23, and has no initial gain. Cash 103 plus asset 20 less liability 23 preserves net assets of 100. Derecognising everything would wrongly report gain 3 and omit the exposure.

At a later date, admit satisfaction of one third of the guarantee service and loss allowance 2. Release guarantee income 1 by reducing the associated liability to 22; recognise impairment expense 2 and allowance 2. Net assets are $103 + (20 - 2) - 22 = 99$, matching net expense 1. If the guarantee then expires with no claim and no remaining obligation, reverse that allowance 2 and remove retained asset 20 against liability 22, recognising the remaining guarantee income 2.

Cash and net assets are 103 and cumulative net income 3. An actual claim, changed valuation or unsatisfied obligation would require its actual settlement analysis; expiry cannot be presumed from a convenient reporting date. The continuing asset and liability are presented separately.

11.11 A transferred asset can still contain an option exposure

The guarantee example retains a possible credit loss. A call instead retains the right to buy an asset back; a written put obliges the bank to buy it if the counterparty exercises. Neither name determines whether a sale has occurred for accounting purposes. The transfer assessment first identifies the cash flows being tested and whether substantially all risks and rewards were retained or transferred. Only the intermediate case proceeds to control and, where control remains, continuing involvement. A transferee's practical ability to sell the whole asset unilaterally, without additional restrictions, matters. In particular, a call over an asset readily obtainable in the market need not prevent derecognition. Deeply in-the-money and out-of-the-money options can also lead to different risk-transfer findings (European Commission, 2023a, IFRS 9.3.2.2–16, B3.2.6–9, B3.2.16).

The following fair-value cases assume that this assessment requires continuing involvement and that the underlying asset is measured through profit or loss. They use the historical IFRS 9 text retained in EU Regulation 2023/1803. Amounts are in one currency, with no tax, coupons or transaction costs. Fair values and option time values are supplied measurement inputs. Applying the cases to an actual transfer requires its rights, market access, applicable rules and valuation evidence.

11.11.1 The asset less the liability represents the retained call

Suppose an asset worth 80 is transferred for 75 while the bank retains a call with exercise price 95 and value 5. The call is out of the money: exercise would currently cost more than buying the asset. Its entire value of 5 is therefore time value. Under the admitted continuing-involvement branch, the bank keeps an asset of 80 and recognises an associated liability of 75. Their difference is precisely the call's value. Opening cash 200, asset 80 and equity 280 become cash 275, asset 80, liability 75 and unchanged equity. The receipt debits cash and credits the associated liability; it creates no sale gain in this constructed example.

Let S denote the asset's current fair value, K the call exercise price and T_c its supplied time value, all in the same currency. The call's intrinsic value is $\max(S-K,0)$. The source's two measurement branches can consequently be written together as

$$A = S, \qquad L = \min(S,K) - T_c, \qquad A - L = \max(S-K,0) + T_c.$$

Here A and L are the recognised asset and associated liability. When the call moves into the money, the liability uses the exercise price instead of the higher asset value. This preserves the same net call exposure (European Commission, 2023a, IFRS 9.B3.2.13(c)).

At a later measurement, take $S = 100$ and $T_c = 4$. The asset becomes 100, the liability 91 and the net exposure 9. Asset income 20 and liability expense 16 produce net profit 4. At expiry, take $S = 110$ and zero time value. The asset becomes 110 and the liability 95, producing further net profit $10 - 4 = 6$. Physical exercise pays 95 and removes the liability; the already recognised asset remains. Closing cash 180 plus asset 110 equals equity 290, the original 280 plus profit 10. Recognising a separate call of 5 at transfer would have counted the same right twice. IFRS 9.B3.2.14 prohibits that duplication, while 3.2.22 preserves separate asset/liability and income/expense presentation (European Commission, 2023a, IFRS 9.3.2.19, 3.2.22, B3.2.14).

11.11.2 A written put caps the retained asset; a collar combines rights

For a written put, the bank bears the downside below the exercise price but does not retain the upside above it. With exercise price K and put time value T_p, the source therefore limits the recognised asset and measures the associated liability as

$$A = \min(S, K), \qquad L = K + T_p, \qquad A - L = -\big(\max(K - S, 0) + T_p\big).$$

The negative net amount is the full put obligation (European Commission, 2023a, IFRS 9.B3.2.13(d)). For $S = 120$, $K = 100$ and $T_p = 5$, asset 100 and liability 105 represent a net obligation of 5. Choose transfer proceeds 125, equal to the underlying value plus compensation for that obligation. From cash 200, asset 120 and equity 320, the entry debits cash 125, credits asset 20 and credits liability 105. Equity stays 320. Keeping the full asset value 120 would overstate the retained asset by 20.

If the asset falls to 90 and time value is 4, the recognised asset is 90 and liability 104. Net loss is $10 - 1 = 9$. At expiry with asset value 80, the liability becomes 100 and the further loss is $10 - 4 = 6$. Exercise pays 100 and removes that liability. Cash 225 plus asset 80 equals equity 305, reconciling the total loss 15. This is a physical-settlement case; different cash-settlement rights require their own derecognition analysis.

A collar combines a retained call at an upper exercise price K_c with a written put at a lower price K_p. Let P be the *full* put value, $\max(K_p - S, 0) + T_p$. The asset continues at fair value, and the associated liability is

$$A = S, \qquad L = \min(S, K_c) + P - T_c, \qquad A - L = \big(\max(S - K_c, 0) + T_c\big) - P.$$

Thus the recognised net position is the call less the put (European Commission, 2023a, IFRS 9.B3.2.13(e)). With $S = 100$, call strike 120, put strike 80 and time values 5 and 1, the liability is 96 and net asset 4. From cash 200, asset 100 and equity 300, proceeds 96 produce cash 296, asset 100 and liability 96 without an initial gain.

Two subsequent paths show why the full put value matters. At $S = 130$, call time value 2 and put time value 1 give liability 119 and net asset 11. At $S = 70$, call time value 1 and put time value 2 give full put value 12, liability 81 and net liability 11. Using only the put's time value in the latter calculation would understate the associated liability by 10. If these respective asset values persist to expiry, both time values vanish. The upper path pays 120 to exercise the call, closing with cash 176, asset 130 and equity 306. The lower path pays 80 on the exercised put, closing with cash 216, asset 70 and equity 286. Cumulative income is respectively 6 and -14, measured from the initial net option value 4.

11.11.3 Amortised cost uses two interest schedules

The fair-value formulas cannot be copied into an amortised-cost case. IFRS 9.B3.2.13(b) instead accretes the associated liability from proceeds to the transferred asset's gross carrying amount at option expiry, using effective interest. Any difference between that liability and the exercise price enters profit when the option is exercised (European Commission, 2023a, IFRS 9.B3.2.13(b)).

Take the source's gross asset 98, proceeds 95 and gross asset 100 at expiry, here one year later. Assume a zero loss allowance, no intervening coupons or cash flows, and collection of the asset after option expiry. The asset's effective rate is $100/98 - 1 = 1/49$, producing income 2. The liability's rate is $100/95 - 1 = 1/19$, producing expense 5. They cannot share a single rate merely because they concern the same transfer. If a written put is then exercised at 102, debit liability 100 and loss 2, and credit cash 102. Opening cash 200, asset 98 and equity 298 become cash 193, asset 100 and equity 293. Interest expense net of income is 3 and exercise loss is 2.

The zero allowance isolates the financing calculation; it does not infer an absence of credit risk from the transfer.

11.12 Funding, capital and decision policies

Liquidity simulation settles the dated net payment queue by venue/currency, respecting encumbrance, available cash, committed line terms and admissible actions. Haircut/eligibility and transferability come from a versioned policy; market value alone is not available liquidity. With cash 20, due withdrawal 30 and an admitted usable facility 8, even a full draw leaves shortfall 2. Record the default/unresolved settlement according to the event's legal model. Raising exactly 2 of new equity without an investor event is an invented financing source. Interbank contagion requires explicit counterparties and settlement priorities; it cannot be inferred from aggregate deposits.

11.12.1 A draw request and three cash histories

Make the facility example precise. Cash 20, securities 50 and gross loans 80 less allowance 10 give assets 140; deposits 120 leave equity 20. A synthetic agreement limits borrowing to 10 and accepts one unencumbered security lot with market value 10 and a 20% haircut. The other securities 40 are ineligible. With no existing drawing, usable capacity is $\min(10, 10(1-0.2)) = 8$. The agreement is an authored assumption, not evidence of a central-bank entitlement. The collateral remains the bank's asset; pledging it does not create another asset or sell the existing one.

Requesting 8 creates no cash journal. Confirmed receipt 8 debits cash 8 and credits borrowing 8. Confirmed receipt 6 instead posts 6, leaving the unfilled request visible. A third history retains a pending request and independent evidence that no receipt reached the bank before the withdrawal deadline. Its cash stays 20. An acknowledgement timeout alone cannot establish that third history: the receipt may have occurred even though the acknowledgement was lost.

Assume the deposit contract permits partial payment of the due withdrawal 30 and classifies the unpaid portion as a liquidity default without waiving it. Pay the cash actually available. The three outcomes are:

Confirmed borrowing	8	6	0
Cash before withdrawal	28	26	20
Withdrawal paid	28	26	20
Closing deposits	92	94	100
Unpaid due, within deposits	2	4	10

Each payment debits deposits and credits cash. In every column final cash is 0 and securities 50 plus net loans 70 give assets 120. Deposits plus borrowing total 100, leaving equity 20. Profit and OCI are 0. The unpaid amount is part of the deposit liability, so adding it again would overstate liabilities. Under the specified activities, deposit settlement is operating cash and the separate borrowing receipt is financing cash (Australian Accounting Standards Board, 2023a, paragraphs 14–17).

The corresponding records must retain facility version, borrower, currency, settlement venue, eligibility, encumbrance, valuation date, haircut and existing drawings. Capacity, request and settlement have distinct identities. A proposed request exceeding capacity is rejected. An independently evidenced unexpected actual receipt must nevertheless be recorded, together

with the control violation; admission rules cannot justify omitting money that arrived. The existing publication and replay rules apply to the cash and borrowing as one transaction. A repeated identifier with identical content has no second effect; a changed payload conflicts. Lost acknowledgement preserves the possible committed and uncommitted histories until reconciled.

Credit and collateral records likewise retain the original loan, current and original terms, dated risk assessments, information cutoff, receipt law, discount convention and source dependencies. Missing legal rights or credit findings produce unresolved or explicitly conditional measurements. Actual credit calibration, fees, continuing involvement and revolving collateral remain separate method obligations. The historical Australian cases do not automatically satisfy the distinct US prudential profile considered next.

A RegulatoryAssessment reads an accounting view and an independent jurisdiction, date, perimeter, eligibility, deductions and exposure-weight policy. It outputs capital components, RWA/exposure measures, liquidity measures and reconciliations with their distinct denominators. Illustratively, equity 20 minus an assumed regulatory deduction 2 yields capital 18; assumed RWA 125 gives 14.4%. Those inputs are synthetic, not a Basel rule, national capital approval or default threshold. Selected dated US national capital rules are inspected in the surrounding bank inquiry. A connected accounting-to-capital reconciliation for the same bank still requires its applicable perimeter, eligibility, deductions and exposure facts; the illustrative inputs above do not supply those facts.

Behavior under capital/liquidity constraints is a separate economic model. Specify feasible actions, objectives, information set and ties before solving; failure to find an action returns policy_failure or default under the stated contract. Halaj/Behn sources can motivate candidate behavior but cannot calibrate it for an individual bank. A forecast experiment must compare simple fixed-book, contractual runoff and explicitly estimated policy baselines with time-safe data, before claiming a predictive improvement. CECL and other bases retain separate authority/method branches in the dated-authority discussion.

Chapter 12

Insurance service, finance and claims

An insurer receives cash before much of its service is delivered. Cash premium, service revenue and profit therefore follow different schedules. A short coverage example first connects gross insurance to reinsurance. The subsequent two-year cases introduce unearned profit, loss allocation and finance one change at a time.

12.1 Insurance: cohorts and future service

An insurer requires policy cohorts or groups with issue dates, coverage, premiums, benefits, expenses, lapses, claims development, guarantees and reinsurance relationships. Asset portfolios need their own contracts and valuations. The liability state must distinguish expected future cash flows, discounting, risk adjustments or margins, unearned profit measures where applicable, incurred claims and movements between opening and closing states. The IFRS 17 overview's distinction between measuring future cash flows and recognising profit as service is provided motivates this separation (IFRS Foundation, 2026b).

The module API accepts policy/cohort state, economic assumptions, non-economic assumptions and information date. It returns a measurement vector, movement attribution and proposed recognition events. Experience differences, changes in assumptions, finance effects and service effects remain separately identifiable. Reinsurance held has its own record and measurement; it is not an automatic percentage offset to every insurance liability.

Scenarios can alter discount curves, mortality, morbidity, lapse rates, inflation, catastrophe losses and policyholder behaviour. Their joint dependence and cohort exposure must be declared. A portfolio total can match history while hiding very different guarantee and duration profiles. This makes alternative economic histories especially valuable, but also limits what public-data reconstruction can identify. A full insurance package needs actuarial and accounting review beyond the core release.

12.2 Insurance service and reinsurance in a complete cohort

The example uses one-year policies and a one-year proportionate treaty starting when those policies are recognised, with uniform service for each. Both groups qualify for premium allocation through the one-year conditions in paragraphs 53(b) and 69(b); paragraphs 62–62A determine treaty recognition. Assume no acquisition cash flows, investment components, significant financing or onerous remaining coverage. Claims and recoveries are expected within a year of the claims being incurred. The inspected AASB 17 text incorporating IFRS 17 specifies the movements in paragraphs 53–59, allocation in B126 and separate reinsurance requirements in 63–70A (Australian Accounting Standards Board, 2022e). Gross and ceded groups retain distinct identities.

Begin with capital and cash of 20. Receive premium 120 and recognise a remaining-coverage liability of 120. Pay reinsurance premium 24 and recognise a coverage asset of 24. After half the uniform service, recognise revenue 60 by releasing gross coverage and reinsurance expense 12 by releasing its coverage asset. Estimated incurred claims are 40, of which 30 is paid; the remaining-claim risk adjustment is assumed to be 2. Gross claims expense is therefore 42 and incurred-claims liability is 12. Assume the treaty reimburses half the gross expected claims, recovery cash of 15 has arrived, the risk transferred is valued at 1, and reinsurer non-performance has zero expected effect in this example. The reinsurance claims asset is $20 + 1 - 15 = 6$, and recovery income is 21.

Cash is $20 + 120 - 24 - 30 + 15 = 101$. Assets are cash 101, remaining reinsurance coverage 12 and claims recovery 6, totalling 119. Liabilities are remaining gross coverage 60 plus incurred claims 12, totalling 72. Profit is $60 - 42 - 12 + 21 = 27$ and equity is $20 + 27 = 47$. These balances reconcile without netting reinsurance against the insurance liability. If counterparty credit risk changes, the recovery measurement changes through its own inputs; it does not mechanically halve every gross liability or risk adjustment.

Longer-duration business adds a fulfilment-cash-flow vector, discount curve, risk adjustment, contractual service margin, loss component and coverage-unit schedule. Select portfolio and annual cohort, assess onerous grouping, project cash flows within contract boundaries, discount with the declared financial assumptions, measure non-financial risk separately, and attribute subsequent movements to new business, experience, assumptions, finance and service. Future-service changes adjust the supported service-margin operation subject to the loss rules; current/past-service effects enter their specified result components. Unsupported variable-fee, transition or loss-recovery operations must stop those dependencies explicitly. A PAA example does not implement these longer-duration methods by implication.

An actuarial reserve adapter also needs a concrete estimation contract. For cumulative claims $C_{i,k}$ by origin year i and development year k, the chain-ladder point method estimates each factor from matched observed cells, $\widehat{f}_k = \sum_i C_{i,k+1} / \sum_i C_{i,k}$, then projects the latest cumulative claim along the remaining factors. A triangular example with rows $(100, 150, 180)$, $(120, 180)$ and (140) gives factors 1.5 and 1.2, outstanding amounts 36 and 112 for the unfinished rows, and total 148. Mack (1993, Sections 2–3, equations (1)–(3)) makes explicit the conditional-mean and variance assumptions and independence across origin years; its aggregate-error discussion retains dependence from shared estimated factors. A single final-development observation cannot supply an empirical tail variance without an additional assumption. The adapter therefore returns the point projection and marks uncertainty estimation unavailable in this tiny example. It supplies neither a full predictive distribution nor an IFRS risk adjustment. Calendar effects, inflation, changing claims handling and tail selection require separate sensitivity and validation before real use.

12.3 Group state and general measurement

The following cases apply selected general-measurement and service-allocation provisions from the retained AASB 17 compilation (Australian Accounting Standards Board, 2022e). They extend the premium-allocation example without presuming actuarial calibration or a complete insurer model.

The general method distinguishes fulfilment cash flows (FCF), including an explicit risk adjustment (RA), from the contractual service margin (CSM), which holds unearned profit. The liability for remaining coverage (LRC) concerns future service; the liability for incurred claims (LIC) concerns service and insured events already experienced. Keeping these components separate makes a change in future estimates different from an unpaid claim.

Contract records retain insured risks, premiums/benefits, service and settlement dates, options/guarantees, cancellation/repricing rights and acquisition costs. Group admission evaluates portfolio/similar-risk management, onerous status and required issuance cohorts under AASB 17.14–24, recognition under 25–28 and boundary under 34. Cash outside the boundary belongs to future contracts, not this group's profit. GroupState retains expected dated cash flows, current discount curves, inception rates, explicit risk adjustment, CSM, loss component, LRC/LIC, coverage units, service experience and assumption vintages. Same premium totals are not sufficient state for merging different guarantees or claim tails.

General measurement follows 32–38: fulfilment cash flows are probability-weighted future net outflows, adjusted for timing/financial risk without double counting, plus explicit non-financial risk adjustment. CSM removes initial unearned gain under its conditions; an initial net loss is recognized under 47, not negative CSM. Paragraphs 40–44 separate remaining coverage and incurred claims. For a nonparticipating group, accrete CSM at the specified locked-in rate, apply future service changes at their required rate to CSM up to its available amount, record any excess loss and allocate current service over current plus future coverage units under B119. B96–B97 distinguish future service from incurred-claim changes, financial changes and current experience; not every revision is a CSM adjustment.

A MeasurementInput must resolve cash-flow/law, discount, risk-adjustment method, service grouping and coverage-unit evidence. Missing probabilities or RA support permit conditional sensitivity cases but block a source-supported measurement. Risk adjustment is the compensation the entity requires under 37; a standard deviation, confidence quantile or arbitrary percentage is only a candidate method until justified for that entity. Disclosures and the confidence-level equivalent, where applicable, are separate obligations. Market-consistent financial inputs and entity risk compensation must not be conflated.

12.4 Complete two-year general-measurement case

Take a two-year nonparticipating group with one premium 120 received at inception, claims expected/actually incurred and paid 40 each year, zero discount rate, RA 10 initially released 5 with each equal service year, no acquisition costs, investment component, lapses, taxes, finance changes or reinsurance. Coverage units are 1 in each year because equal benefits and coverage are assumed. Before premium payment, expected net cash outflow is $80-120=-40$, RA 10, FCF -30 and CSM 30, with net initial carrying amount zero. After the premium debit cash 120/ credit LRC 120. Its components are remaining expected claims 80, RA 10 and CSM 30.

Each year releases expected service claims 40, RA 5 and CSM 15: debit LRC 60/ credit insurance revenue 60. Incur claims 40: debit service expense 40/credit LIC 40; payment debits LIC 40/credits cash 40. End of year one: cash 80, LRC 60 (claims 40, RA 5, CSM 15), LIC 0

and equity/profit 20. End year two: cash 40, liabilities 0, cumulative equity/profit 40. Revenue 120 and claims 80 reconcile to premium 120 less claims cash 80. No investment component has been mistaken for revenue.

At year-one close, a separate pre-next-service change increases the remaining claim estimate from 40 to 60, RA unchanged. FCF rises 20; consume remaining CSM 15 and recognize service loss 5. LRC rises from 60 to 65, comprising FCF 65, CSM 0 and loss component 5 as a tracked portion, not an added liability on top. Debit service loss 5/credit LRC 5 with component transfer CSM→FCF 15. A decrease of 8 in the revised future claim estimate, before any additional service/finance movement, first reverses loss 5 and restores CSM 3. FCF 57 plus CSM 3 gives LRC 60. Credit loss reversal 5/debit LRC 5. This tests 48–52 ordering. Later systematic allocation of loss component to releases/finance must reduce it to zero by coverage end; it cannot be amortized as revenue a second time.

In a separate case, an already-incurred unpaid claim estimate rises 40 to 48. Debit service expense 8/credit LIC 8; do not consume future-service CSM. A discount-rate change is financial under B97; its profit/OCI route follows the selected permitted finance policy. The zero-rate base case does not remove the need for separate current and locked-in rate calculations in a real group.

For the unreversed onerous alternative, carry the loss component through the remaining service year. Opening LRC 65 comprises future claims 60 and RA 5, with CSM 0 and loss component 5 within that LRC. A declared proportional allocation assigns 5/65 of each expected claim/RA release to the loss component, consistent with paragraphs 49–52 and B124's revenue exclusions. The expected claim release allocates 60/13 to loss reversal and the RA release 5/13; the total is 5, leaving loss component 0. Debit LRC 65, credit revenue 60 and credit loss reversal 5. Separately incur and pay claims 60 through LIC. Period profit is 60−60+5=5; combined with first-year profit 20 and the earlier onerous loss 5, lifetime profit is 20. Closing cash 20 equals equity 20 and all liabilities are zero. Treating all 65 as revenue while also reversing loss 5 overstates profit by 5.

12.5 Nonzero discount and service allocation

A second complete case keeps premium 120, two equal coverage units and RA 10, but expected and realized claims are 44 after one year and 48.4 after two years. Both the inception and current discount curves remain a flat 10% throughout; there are no new estimates, options, fees, taxes, investments or OCI election. This RA schedule is an authored measurement input, not an actuarial calibration: it declines by 5 per year, with its entire change in the service result under 81. All other insurance finance effects go to profit or loss. The retained source supports the mechanics in 44, 80–88, B119 and B123–B124.

At inception the present value of the two expected claims is

$$\frac{44}{1.1} + \frac{48.4}{1.1^2} = 40 + 40 = 80.$$

FCF before premium is 80+10−120=−30, CSM 30, and after premium LRC 120. In year one the claim PV accretes 8; CSM accretes 3 at the locked-in rate. Allocate the resulting CSM 33 across the two current/future units, releasing 16.5. Revenue is expected claims 44 plus RA 5 plus CSM 16.5=65.5. Service expense 44 and finance expense 11 leave profit 10.5. After paying 44, cash 76 and LRC 65.5 (claim PV 44, RA 5, CSM 16.5) reconcile to equity 10.5.

In year two claim PV accretes 4.4 and CSM accretes 1.65. Release all remaining CSM 18.15, RA 5 and expected claims 48.4, giving revenue 71.55. Service expense 48.4 and finance expense

6.05 leave profit 17.1. Final cash/equity 27.6 equals premium 120 minus claims 92.4; all group liabilities are zero. Total revenue 137.05 less finance expense 17.05 equals premium 120. Setting revenue equal to cash premium each year or releasing 15 of CSM each year would ignore the specified financing. These are separate component calculations followed by debit finance expense/ credit LRC for accretion, debit LRC/credit revenue for service and the stated LIC recognition/settlement entries. Cash is deliberately assumed uninvested.

12.6 Two issued groups and their reinsurance share one balance sheet

An insurer's accounts contain several groups at once. A profitable group cannot lend its unearned margin to an onerous group merely because both belong to the same company. Reinsurance may recover part of the loss, but does so through a separately measured contract. The following extension puts those three measurements on one balance sheet and follows their cash through the two service years. It uses the retained AASB 17 rules for an Australian private-sector insurer's calendar 2024 and 2025 periods.

Start with cash and equity of 100. Group A is the preceding nonzero-rate example: premium 120, claims of 44 and 48.4 at the two year ends, initial claim present value 80, risk adjustment 10 and CSM 30. Group B receives premium 100 and expects claims of 66 and 72.6 at the same dates. At the same 10% rate, B's claim present value is $66/1.1 + 72.6/1.1^2 = 120$. Its separately supplied risk adjustment is 10, so it recognises a loss of 30 and has no CSM. After premium receipt its remaining-coverage liability is 130, including a loss component of 30. Both groups contain contracts issued at the beginning of 2024 and provide two equal annual coverage units. Their different profitability requires separate groups under paragraphs 16 and 47; assume no pricing-law exception under paragraph 20. Estimating cash flows together does not authorise pooling their CSMs (Australian Accounting Standards Board, 2022e, paragraphs 14–28, 32–38 and 47).

The current and inception discount curves remain at the assumed 10%. Each issued group's risk adjustment releases 5 per year, entirely in the service result under paragraph 81. All insurance finance enters profit or loss under paragraph 88(a). Claims are incurred and settled at each year end, with realised claims equal to the admitted expectations on this path. Cash is uninvested. There are no acquisition costs, investment components, taxes, currency effects, new contracts or estimate changes. The rates, expected claims, service units and risk adjustments are supplied conditional measurements; these calculations do not estimate them from insurer data.

Now add treaty R, entered into before B is recognised. It covers all B claims at 50%, with no exclusions, limits, settlement delay or expected non-performance. Its premium of 70 is paid immediately. Expected recoveries are 33 and 36.3, with present value 60. A separately supplied transferred-risk adjustment of 5 releases 2.5 each year. Its equality to half B's adjustment is an assumption of this example, not a general quota-share valuation rule. Without the initial loss-recovery adjustment, the net cost held in R's CSM would be $70 - 60 - 5 = 5$, and the asset after payment would be 70.

Because R was entered into in time and covers all B's contracts, paragraphs 66A and B119C–B119D recognise recovery income of $30 \times 0.5 = 15$. The adjustment increases R's cost CSM to $5 + 15 = 20$, so its complete remaining-coverage asset after payment is $60 + 5 + 20 = 85$. Its loss-recovery component of 15 is already within that asset. It is not another receivable. Debit B's initial service loss 30 and credit its liability 30; debit R's asset 15 and credit recovery income 15. Receive gross premiums 220 and pay the reinsurance premium 70. Cash is 250, issued liabilities are 120 and 130, and the reinsurance asset is 85, leaving equity 85 (Australian

Accounting Standards Board, 2022e, paragraphs 60–68, 78 and B119C–B119F).

Financing moves the loss component as well

B's loss component must be allocated across claim releases, risk release and finance under paragraphs 50–52. For this unchanged group, use the initial proportion $\alpha = 30/130 = 3/13$ for all three. If F is the period's financing of B's future claim value and S is its expected claims plus risk-adjustment release, its loss component K follows

$$K_{\text{close}} = K_{\text{open}} + \alpha F - \alpha S.$$

This is a declared systematic allocation for these facts. It preserves $K = \alpha$ times B's remaining-coverage liability and reaches zero at coverage end. New estimates, coverage or risk patterns would require their own reassessment of the allocation; the example does not prescribe this ratio for every insurer (Australian Accounting Standards Board, 2022e, paragraphs 49–52 and B124).

In year one, B's claim value accrues finance of 12. Its loss component therefore increases by 36/13, before service releases claims 66 and risk adjustment 5. Of that release of 71, 213/13 reverses the loss and 710/13 is insurance revenue. The closing liability is $130+12-71 = 71$, comprising future claim value 66 and risk adjustment 5. The loss component is $30 + 36/13 - 213/13 = 213/13$, within that liability. Debit finance expense 12 and credit B's liability; then debit the liability 71, credit revenue 710/13 and credit loss reversal 213/13. Separately incur and pay claims 66 through the incurred-claims liability.

R's claim value accrues 6 and its CSM accrues 2. The asset gains finance income 8. Its CSM of 22 is then allocated across two equal current and remaining service units, releasing 11. Recovery 33, risk release 2.5 and CSM release 11 reduce remaining coverage by 46.5; the closing asset is $85 + 8 - 46.5 = 46.5$, comprising 33, 2.5 and 11. Its loss-recovery component tracks half B's loss component: finance adds 18/13, service reverses 213/26, and the closing component is 213/26. No component is added on top of the measured asset. In this unchanged, fully covered case, the component reversal is represented by the scheduled cash-flow and risk release; paragraph 66(bb) must not produce a second CSM reversal for it.

The service entry debits reinsurance premium-allocation expense $46.5 - 213/26 = 498/13$ and loss-recovery reversal 213/26, and credits R's remaining-coverage asset 46.5. Separately recognise recovery income 33 and an incurred-recovery asset 33, then collect the cash. Thus R's service result is $33 - 46.5 = -13.5$, with finance income 8 shown separately. The component determines presentation within that service result; it does not change its total. Paragraph 86 also prevents the reinsurance premium allocation from reducing issued insurance revenue (Australian Accounting Standards Board, 2022e, paragraphs 66–67, 80–88, B72, B119 and B119F).

A retains its preceding result: revenue 65.5, claims expense 44 and finance expense 11 give profit 10.5, with remaining coverage 65.5. B has service profit 5 and finance expense 12, giving a further loss of 7. Adding R's loss of 5.5 makes the year-one movement in equity -2, after the initial net loss of 15. Cash is $250 - 44 - 66 + 33 = 173$, and $173 + 46.5 - 65.5 - 71 = 83$ confirms closing equity.

Settlement and timing expose different errors

In year two A produces profit 17.1 as before. B incurs finance expense 6.6, then releases expected claims 72.6 and risk adjustment 5. Its loss component first increases by 99/65, then reverses 1164/65, reaching zero. Of the total release 77.6, revenue is 776/13 and the

remainder is loss reversal. Paying claims 72.6 leaves B with no liability and period loss 1.6. R accrues finance income $3.3 + 1.1 = 4.4$, then releases recovery 36.3, risk adjustment 2.5 and all CSM 12.1. Its asset release is 50.9; its loss-recovery component also reaches zero. Collecting 36.3 leaves reinsurance service expense 14.6 and net expense 10.2. Total year-two profit is $17.1 - 1.6 - 10.2 = 5.3$.

The following balances are after each date's premium or claim settlements. All incurred claims and incurred recoveries have been settled; the exact event sequence preserves those intermediate balances before the cash entries.

	Initial receipt/payment	Year one	Year two
Cash	250	173	88.3
A: remaining-coverage liability	120	65.5	0
B: remaining-coverage liability	130	71	0
R: remaining-coverage asset	85	46.5	0
Equity	85	83	88.3

The final cash of 88.3 equals initial capital 100 plus gross premiums 220, less reinsurance premium 70 and gross claims 231, plus recoveries 69.3. Cumulative profit is $-15 - 2 + 5.3 = -11.7$. The two issued groups and the held treaty finish with every measurement component at zero.

Suppose instead that R is entered into immediately after B's recognition, with no intervening service, insured event or financing and otherwise the same prospective coverage. It is a valid different history. It cannot recover the original recognised loss under paragraph 66A because it fails B119C's ordering condition. R then starts with cost CSM 5 and asset 70, with no loss-recovery component. Opening equity is 70. Its year-one closing asset is $33 + 2.5 + 2.75 = 38.25$, giving total equity 74.75. Year-two cash and equity still finish at 88.3. The difference is the timing of recognition, not a difference in cash. Paragraph C20A's transition exception does not apply to this ordinary new-business example.

Incorrectly pooling A and B produces a third history of reported amounts. Their combined premiums 220 equal claim present value 200 plus the two supplied risk adjustments 20. A pooled calculation would therefore show neither CSM nor initial loss, conceal B's loss of 30, and suppress the timely treaty's recovery adjustment of 15. It reports equity 100 initially and 91.25 after year one, although final cash is again 88.3. Those accounts balance but violate the separate-group recognition requirements. Likewise, adding the recovery component to the full asset overstates opening equity by 15, while omitting financing from loss allocation makes the component run out before coverage ends. Intermediate group identity, component movements and recognition time are essential checks; terminal cash alone cannot distinguish these mistakes.

Actual use also requires the treaty's boundary, collectability, measurement inputs and eligible contract mapping. Partial coverage needs the systematic loss allocation in B119E; a nominal quota cannot supply it. Changes in those facts or in the estimates require new measurement events under paragraphs 66–67, rather than extending this fixed schedule unchanged. The shared event procedure commits journal and component movements for each dated event together, while preserving separate issued and held group identities. (Australian Accounting Standards Board, 2022e, paragraphs 24, 63–67, 78–86 and B119C–B119F)

12.7 Commit each insurance event as a complete book transition

Separate group measurements still have to produce one dated company balance sheet. In the preceding example, R's loss-recovery component depends on B's loss component. Publishing

B's financing while retaining R's opening measurement would combine amounts from different stages of the calculation. The proposed close operation therefore declares A, B and R as the required group set for each event in this example. Each group retains its own identity and measurement; the event publishes their consequences together in one entity, book and branch. This is a declared dependency for these facts, not a requirement to update every group in every insurance transaction.

The year end contains four events, in order: finance, service, claim and recovery incurrence, then settlement. Each has its own business-event identifier and immutable opening state. Keeping those boundaries preserves the incurred balances and the distinction between earning income and moving cash. The following intermediate states follow from the preceding journals. Coverage columns show liabilities for A and B and an asset for R.

After event	Cash	A coverage	B coverage	R coverage	Equity
Opening	250	120	130	85	85
Finance	250	131	142	93	70
Service	250	65.5	71	46.5	160
Incurrence	250	65.5	71	46.5	83
Settlement	173	65.5	71	46.5	83

Service releases the coverage amounts before the separate claim expenses and recovery income are recognised. Incurrence creates liabilities of 44 and 66 and a recovery asset of 33. Those balances explain why equity is 83 at that stage even though cash remains 250. Settlement clears them and moves cash by $-44 - 66 + 33 = -77$, leaving equity unchanged. Collapsing the four events into a single final balance would conceal this distinction.

A valid component calculation is part of permission to post

The proposed operation uses the common `OperationRequest` for `accounting.commit@1`. Its four arguments remain references to the accounting basis, posting authorization, journal proposal and economic delta. The expected company head remains in the common request envelope. Each group measurement identifies the group, issued or held role, covered group where applicable, event phase, opening and proposed closing components, book, branch, time, information view, method and source. A complete measurement has actual component values; an unresolved one has no closing value and gives its reason.

Before looking up a previous receipt, the operation validates the complete group set, common ownership and information view, admitted evidence, measurement calculation and posting authorization. If R is unresolved, the whole declared event lacks permission. A and B's completed proposals remain available for diagnosis, but they do not form a closing company state. A missing group, duplicate group, wrong branch or different method also prevents lookup. The existing receipt index cannot turn malformed historical input into an eligible request. This proposed profile admits only the authored synthetic case; submitting it to a real book fails that permission check.

For an eligible new event, the journal and economic delta must then agree with the measurements. During finance, R's asset rises by 8 because its claim present value gains 6 and its cost CSM gains 2. Its loss-recovery component separately rises by 18/13, as a component within the asset. An extra debit to R's asset and credit to finance income for 18/13 would balance, yet overstate both the asset and income. The operation rejects that proposal. It also rejects a proposal whose journal is correct but whose economic delta leaves the recovery component at its old value. The monetary accounts and component memory must both be correct.

Publication inserts the journal, complete component and account changes, closing company state, their dependency records and edges, and the immutable receipt, then advances the company head in the same transaction. The finite exchange examples list those published records separately from the global registry of proposed records. Merely retaining a rejected measurement for inspection therefore does not make it part of the book. A rejected request publishes none of its proposed transition. This requirement has one legal book as its owner; it supplies no transaction guarantee across different books or entities.

Retry the event without repeating its accounting

The business-event lookup key remains the entity, book, branch and business-event identifier. After input validation, an equal request fingerprint returns the original receipt before testing the freshness of today's head. Suppose all four events have committed, advancing the head from generation zero to generation four. Retrying the original finance request returns its original receipt and closing state at generation one. The response separately reports the observed generation-four head and that this invocation made no additional commit. It does not apply finance again or replace today's state with the old closing state.

Equality covers the complete request, including its request identifier, budget, original expected head and dependency identities. Changing any of those bytes while retaining the business-event key is a conflict, not an exact retry. A previously unseen event with an old head is stale. The transaction repeats both the key and head checks before publication: an identical request that has just won returns its receipt; different request bytes for that event conflict; a different event that advances the head makes this request stale. In each case the competing transaction's changes are distinguished from any changes made by the requester.

If the acknowledgement disappears, the client cannot infer whether the transaction committed. The exchange therefore records a transport observation with unknown commit status, the original submission and its exact request bytes. It does not invent a received operation result. One possible storage state still has the opening head and no receipt; the other has the complete closing state and receipt. Resubmitting the exact request commits once in the first case and returns the old receipt in the second. A timeout supplies no reason to change the event identifier, opening head or economic facts.

These finite examples check the equations, complete record bindings and required decisions for both storage states. They specify observable behavior; actual transaction isolation, crash recovery and durability need later implementation tests. Their completed coverage is a decision on one request, with no interpretation as probability mass over financial histories. The underlying source applicability and supplied actuarial assumptions of the connected-group example continue to govern its accounting meaning.

12.8 PAA, reinsurance and valuation methods

The preceding premium-allocation case states separate eligibility and recognition timing for gross insurance and reinsurance. PAA under 53–59 requires the one-year coverage or nonmaterial-difference condition; a short payment lag is not the eligibility test. LRC starts with received premium and applicable acquisition adjustments, then releases insurance revenue under the prescribed service pattern. LIC still measures incurred obligations using the applicable fulfilment-cash-flow rules; the allowed short-settlement discount simplification needs its own conditions. Emerging onerous facts require a loss assessment; a premium balance cannot be presumed sufficient.

A one-year policy can leave claims unpaid for several years. The distinction between the period of insurance service and the period until settlement is therefore essential even when PAA is available. The official ASBE 25 text separates three timing tests. Article 50 permits PAA when each contract's coverage is no longer than one year, or when the stated no-material-difference condition holds. Article 54 concerns a different interval: the time between each portion of service and the related premium due date. If that interval does not exceed a year, the remaining-coverage liability need not be adjusted for financing. Article 56 instead measures from claim incurrence to expected payment or receipt. Its option to omit the time-value and financial-risk adjustment requires settlement within a year of incurrence (Ministry of Finance of the People's Republic of China, 2020, Articles 50, 54, 56). The retained AASB 17 compilation makes the corresponding distinctions in paragraphs 53, 56 and 59(b).

Consider a separate group of issued policies with exactly one year of coverage. All amounts are illustrative currency units. Premium 120 is due and received when coverage starts, with no earlier onerous trigger; group recognition is therefore at that date under Article 16. Assume valid insurance grouping, uniform risk release, no acquisition costs, investment component, reinsurance, tax, other expenses or other movements, and no facts indicating onerous remaining coverage. Elect the permitted remaining-coverage financing exception. After half a year, recognise revenue 60 and retain remaining coverage of 60. Articles 53 and 57 support this roll-forward and time allocation; Article 57 requires a different allocation when risk release is not predominantly time-based (Ministry of Finance of the People's Republic of China, 2020). These assumptions isolate the claim measurement from other changes in the group's result.

At that reporting instant a claim is first incurred, with an expected payment of 60.5 exactly two years later. This is a mean estimate, not a certain settlement. Assume an appropriate flat annual effective discount rate of 10% and a separately estimated non-financial risk adjustment of 5 in current measurement units. The inputs are authored for the example; applying them to an insurer would require market and actuarial evidence. Articles 21, 23, 25 and 26 require current probability-weighted cash flows, appropriate discounting and a separate non-financial risk adjustment (Ministry of Finance of the People's Republic of China, 2020). The claim's cash-flow present value is $60.5/1.1^2 = 50$, so its liability is $50 + 5 = 55$. The risk adjustment is not discounted a second time. The claim has only just been incurred, so no intervening finance accrual enters this measurement.

Record debit cash 120/credit remaining coverage 120 on premium receipt, then debit remaining coverage 60/credit insurance revenue 60 for the service provided. Recognising the claim gives debit insurance service expense 55/credit incurred claims 55, consistent with Article 31. The group then has cash 120, remaining coverage 60, incurred claims 55 and a cumulative result of 5: $120 - 60 - 55 = 60 - 55 = 5$. These are balances attributable to the group at mid-year, not a complete insurer forecast. Using the undiscounted $60.5 + 5 = 65.5$ for this two-year claim would overstate the required liability by 10.5 and understate the result by the same amount. The shortcut is wrong under the stated rules and assumptions. A zero rate would remove that discounting difference while leaving the separate risk adjustment intact. Articles 86(3) and 97 also require disclosure of the distinct claim components and the PAA/financing choices (Ministry of Finance of the People's Republic of China, 2020).

This comparison uses identified editions. The Ministry page reproduces the 2020 ASBE 25 text; Article 120 states 1 January 2023, but that sentence does not settle every entity group's adoption schedule. Section 12.23 distinguishes the Ministry's later explanation of that timetable from a company's reported adoption. The retained December 2022 AASB 17 compilation applies to annual periods beginning on or after 1 January 2023 but before 1 July 2026. The shared result in this example establishes a scoped measurement comparison. Applying it to a particular entity still requires its applicable rules and contract facts; later amendments and

broader equivalence require separate evidence.

12.8.1 Acquisition costs and claim finance follow different clocks

The one-year coverage example establishes PAA eligibility without shortening the claim's settlement period. Acquisition costs add another choice. Under the historical IFRS 17 text, costs incorporated into the PAA liability reduce its opening balance and are subsequently amortised as an expense. Immediate expense is permitted under 59(a) when each contract's coverage period at initial recognition is no more than one year (European Commission, 2023a, IFRS 17.53–59).

Consider a separate eligible one-year group with premium 120 and acquisition cash paid 12 at recognition. Opening cash and equity are each 200. Assume uniform service and risk release, no investment component, tax or FX, and the permitted LRC financing exception because the service-to-premium-due interval is at most a year. No claims are incurred before coverage ends, and the base case has no onerous indication. Cash after receipt and costs is 308. Deferring the acquisition cost gives opening LRC $120 - 12 = 108$ and equity 200. Expensing it gives LRC 120 and equity 188.

At mid-year, both cases recognise revenue 60. In the deferred case, debit acquisition expense 6 and credit LRC 6 for amortisation, then debit LRC 60 and credit revenue 60. LRC is $108 + 6 - 60 = 54$, and cumulative profit is 54. In the expensed case, LRC is 60 and cumulative profit is $60 - 12 = 48$. After the second half's service, both LRC balances are zero and both cases have recognised revenue 120 and total acquisition expense 12. Equal final costs do not justify erasing the interim choice. Time allocation also requires the supplied uniform risk pattern; B126–127 instead uses expected incurred-service-expense timing when the risk-release pattern differs significantly (European Commission, 2023a, IFRS 17.55,59(a),B126–127).

An adverse mid-year branch requires a fresh onerous calculation. If current FCF for the remaining coverage is 70 while deferred-cost LRC is 54, recognise loss 16 and increase LRC to 70. Adding the entire FCF 70 to the existing 54 would overstate the liability by 54. The measurement must use the discount treatment required by paragraph 57, including its connection to the LIC simplification (European Commission, 2023a, IFRS 17.57–58).

Return to the base case. At coverage end a claim is incurred, with expected payment 48.4 two years later, current discount rate 10% and separately supplied risk adjustment 2. Initial LIC is $48.4/1.1^2 + 2 = 42$; cash 308 less LIC 42 leaves equity 266. The claim-incurrence rate of 10% differs from the group's assumed inception rate of 5%. If the portfolio elects to split LIC finance between profit and OCI, B133 uses the claim-incurrence rate specified in B72(e)(iii), not the inception rate (European Commission, 2023a, IFRS 17.59(b),B72(e)(iii),B133).

One year after incurrence, suppose the current rate falls to zero and stays there. The claim's current present value becomes 48.4, while a separate schedule using the claim-incurrence rate reaches 44. Finance expense is 8.4 in total: 4 in profit and 4.4 in OCI. Assume the risk adjustment releases 1 in each of the two years, wholly in the service result under paragraph 81. First-year LIC is then 49.4 and equity 258.6. In the second year, current claim value remains 48.4, so total finance is zero; the claim-incurrence schedule reaches 48.4 through profit expense 4.4 and OCI income 4.4. Release the remaining risk adjustment and pay the claim. Final cash and equity are 259.6, equal to opening 200 plus premium 120 less acquisition costs 12 and settlement 48.4. Cumulative OCI is zero. Using the inception rate would give first-year profit finance of only 2 and the wrong allocation, even if the total liability were correct.

12.8.2 Reinsurance and direct participation

Reinsurance has its own group, boundary, recognition time, counterparty risk, cash-flow, CSM/cost and possibly loss-recovery component under AASB 17 paragraphs 60–70A. Do not multiply every gross insurance liability by a quota share: excluded claims, limits, deductibles, timing, reinstatements and counterparty non-performance matter. Claim 100 with an eligible 50% recovery but expected 90% collectability gives conditional expected recovery 45, not 50; altering transferred insurance risk and reinsurer non-performance are different inputs. This simple calculation does not establish treaty recognition or RA by itself. Gross liability 100 remains; reinsurance recovery is a separate asset/measurement with its own authority.

For direct participation, test B101's conditions using the actual underlying items, substantial-share and variation judgments. Retain underlying fair values, entity fee share and guarantee/risk-mitigation choices; paragraphs 45 and B101–B118 specify the variable-fee adjustments to CSM, with onerous and mitigation exceptions. A local component illustration of a qualifying share: underlying-value increase 20 and entity's fixed 10% share gives candidate variable-fee change 2; eligibility, guarantees and other FCF changes must be evaluated before posting. The full option/guarantee valuation and transition calculations are further supported profiles, with source and numerical obligations, not an assumed linear fee model.

Eligibility depends on the contract and the insurer's expectations at inception. The policyholder must participate in a clearly identified pool, with the required substantial share of fair-value returns and substantial variability in payments. The variability assessment covers the contract's duration on a probability-weighted present-value basis. A minimum guarantee can bind in some scenarios without automatically excluding direct participation. Later market movements do not reopen the classification; contract modification follows its own rules. The identified reference pool also need not be held by the insurer, so the contractual link and the actual asset holdings require separate records (Australian Accounting Standards Board, 2022e, paragraphs B101–B108 and 72).

A small settlement example shows why the guarantee matters even when the stated share is fixed. Consider only a maturity-benefit component, in one currency at one date. Assume a held pool allocated to this component has amount U, and the policyholder is owed the larger of 90 and $0.9U$. These are illustrative contractual choices. The amount left from the pool after payment is $U - \max(90, 0.9U)$. Writing the payment as $0.9U + \max(90 - 0.9U, 0)$ shows that the amount left is $0.1U - \max(90 - 0.9U, 0)$: the nominal retained share less the extra payment needed to meet the minimum.

Pool amount	Policyholder payment	Amount left
80	90	−10
100	90	10
120	108	12

Moving from 80 to 100 leaves 20 more after payment: the ordinary share rises by 2 and the extra floor payment falls by 18. Moving from 100 to 120 leaves only 2 more because the minimum no longer binds. The negative amount at 80 is a funding shortfall for this component. These unweighted settlement outcomes explain the exposure; they do not establish insurance-contract eligibility, a present value or the insurer's profit. Other benefits and costs remain outside this component example, and its held-pool assumption does not replace the standard's separate accounting for investment assets (Australian Accounting Standards Board, 2022e, paragraphs B66(a), B107–B108).

The accounting calculation therefore needs more than this payoff schedule. Paragraph B104 defines the variable fee as the entity's share of underlying fair value less fulfilment cash flows that do not vary based on underlying returns. Changes in the obligation to pass through the underlying fair value do not adjust CSM, whereas the entity-share changes relate to future service. Financial-guarantee changes have their own future-service treatment under B113(b). Paragraph 45 retains the loss-component and risk-mitigation exceptions, as well as new-contract, currency and service-allocation movements. Applying those rules requires the group's measured cash flows, discount and risk adjustments, remaining CSM and loss component, service information and any eligible mitigation designation. The original increase of 2 remains a valid share-component calculation; the complete CSM movement needs these additional inputs (Australian Accounting Standards Board, 2022e, paragraphs 32–38, 45, 48–50 and B104–B118).

12.9 Pricing the floor before releasing the margin

The settlement schedule identifies a liability that increases when the underlying assets fall. To measure that liability before settlement, we need a financial model. A small, explicitly specified market makes the distinction between a price and a forecast visible. The retained standard permits a replicating portfolio to measure cash flows that it matches in amount, timing and uncertainty; other techniques must remain consistent with observable market prices (Australian Accounting Standards Board, 2022e, paragraphs B44–B48).

Suppose a non-dividend-paying traded asset costs $S_0 = 100$. Its next observed value is either $S_u = 120$ or $S_d = 80$, after which it remains at that value until payment. Cash earns zero interest, so its accumulation factor is $R = 1$. There are exactly two financial states, unrestricted borrowing and shorting, no transaction costs and no counterparty failure. These are assumptions of the illustration. The financial observation precedes all supplied coverage units; the subsequent service interval contains the mortality risk. An actual period with service already provided would require its own dated allocation.

For payoffs X_u, X_d, let δ denote the units of the traded asset and b the amount deposited in cash. Matching the two payments requires

$$\delta S_u + bR = X_u, \qquad \delta S_d + bR = X_d.$$

Subtracting the second equation from the first gives $\delta = (X_u - X_d)/(S_u - S_d)$. Substitution then determines b and the portfolio's current cost:

$$b = \frac{S_u X_d - S_d X_u}{(S_u - S_d)R}, \qquad V_0 = \delta S_0 + b = \frac{qX_u + (1-q)X_d}{R}, \qquad q = \frac{RS_0 - S_d}{S_u - S_d}.$$

Thus the weight q follows from today's asset and cash prices. Positive state prices require $S_d < RS_0 < S_u$, with $R > 0$. Here $q = 1/2$. Additional financial states or restrictions that prevent the matching trades would invalidate this particular valuation method.

The policyholder's investment payment is $\max(0.9S, 90)$. Its top-up above the ordinary share is $(90 - 0.9S)^+$, where $x^+ = \max(x, 0)$. That top-up pays zero in the up state and 18 in the down state. The matching portfolio has $\delta = -0.45$ and $b = 54$: $-0.45(120) + 54 = 0$ and $-0.45(80) + 54 = 18$. Its cost is $-0.45(100) + 54 = 9$. Adding the ordinary share's value of 90 gives a financial-benefit value of 99. This derives the option value; no option price has been supplied as an unexplained input.

For a separate physical forecast, suppose the up state has probability 3/4. The expected top-up is then $18/4 = 4.5$, although its market-consistent price remains 9. Suppose also that a covered death triggers an additional payment of 20, with physical probability 1/10, independently of the market state. Its expected cost is 2. Nonfinancial risk adjustment of 1 is a separately supplied measurement assumption, with its entire release in the service result. The financial state prices, physical joint law and compensation for nonfinancial risk therefore occupy different records (Australian Accounting Standards Board, 2022e, paragraphs 33, 37, 81 and B42–B49).

12.9.1 The same measurement through service and payment

Use the retained historical Australian provisions for a conditional calendar-2024/25 example. Assume the contract has significant insurance risk, an enforceable link to the identified pool and the required direct-participation characteristics. The component rights terminate together; neither can be retained separately after the other matures. This supplies the specified interrelationship under B32(b), together with an assessment that no embedded derivative must be separated. These admission findings need actual terms before real use; neither q nor the 90% share proves them (Australian Accounting Standards Board, 2022e, paragraphs 10–13, B17–B23, B31–B32 and B101–B108).

The investment payment is due whether the insured lives or dies. There are no surrender, lapse, acquisition or tax cash flows in this example, and all figures are pretax. The insurer holds one unit of the underlying asset at fair value through profit or loss and makes no risk-mitigation election. The portfolio used to calculate the guarantee price is not its actual hedge. Insurance finance also enters profit or loss under paragraph 89(a). The investment component remains within the insurance measurement but is excluded from service revenue and expense under paragraph 85 (Australian Accounting Standards Board, 2022e, paragraphs 85, 89 and B115–B118).

Start with owner capital and cash of 50, and receive premium 110. Before receipt, fulfilment cash flows are $99 + 2 + 1 - 110 = -8$, giving CSM 8. After receipt, the remaining-coverage liability is 110: financial benefits 99, mortality estimate 2, risk adjustment 1 and CSM 8. Buying the underlying for 100 leaves cash 60 and investment 100 against liability 110 and equity 50.

At the financial observation, the future-service fee change is the change in the insurer's 10% share less the change in the top-up value (Australian Accounting Standards Board, 2022e, paragraph 45 and B111–B114). In the up state this is $2-(0-9)=11$, increasing CSM from 8 to 19. Fulfilment cash flows become $108+2+1=111$ and the liability becomes 130. Investment gain 20 and insurance finance expense 20 offset.

In the down state the fee change is $-2-(18-9)=-11$. The existing CSM absorbs 8; the excess 3 becomes a service loss. Fulfilment cash flows and the liability are $90+2+1=93$, with CSM zero and a loss component of 3 *within* that liability. Investment loss 20 and insurance finance income 20 offset, while the service loss reduces equity to 47. The loss component does not create a second liability, and the binding floor does not reopen the inception VFA classification (Australian Accounting Standards Board, 2022e, paragraphs 45, 48–52, 87(c) and B102).

Completing the service interval releases the expected mortality cost, risk adjustment and remaining CSM. Up-state revenue is $2+1+19=22$. In the down state the service-related release of 3 reverses the loss component, leaving revenue zero. Actual mortality expense is either zero or 20, first recognised through the incurred-claim liability and then paid. Selling the underlying and paying the investment component of 108 or 90 completes all balances.

Joint outcome	Physical weight	Profit	Final cash/equity	OCI
Up, survival	27/40	22	72	0
Up, death	3/40	2	52	0
Down, survival	9/40	0	50	0
Down, death	1/40	−20	30	0

Each row is a complete conditional history. Operating cash is premium 110 less the investment benefit and mortality payment; the investment purchase and sale give investing cash of +20 or −20. This classification assumes nontrading investments under the retained cash-flow provisions; fair-value measurement alone does not determine trading purpose (Australian Accounting Standards Board, 2023a, paragraphs 14–16). The physical expected profit is 14.5 under the supplied law, but the four rows preserve the separate downside exposures. An unknown mortality law, risk adjustment, dependence assumption or admission finding leaves a named unresolved measurement rather than a default probability or zero cost.

The exact state prices, journals, component changes and adverse alternatives are retained with the acquisition/treaty cases in `data/accounting-master/finite-method-decisions-20261015/insurance/`. The companion method, *Pricing the guarantee and retaining separate insurance groups*, binds valuation, probability and risk-adjustment records to the same group, date and evidence view, while keeping their meanings distinct. General market models and actuarial calibration require additional methods and validation.

12.10 The variable fee separates investment returns from service

The preceding guarantee valuation measures a contractual payoff. Direct participation adds a service question: which changes belong to the insurer's unearned fee? Admission requires all three B101 conditions, assessed using expectations at inception: an enforceable share of a clearly identified pool, a substantial share of fair-value returns payable to policyholders, and substantial variability of their payments with those returns. The assessment considers the contract's duration and probability-weighted present values, including scenarios in which a guarantee binds. It is reassessed only upon a relevant modification. Reinsurance issued or held cannot qualify as direct participation under B109 (European Commission, 2023a, IFRS 17.B101–109).

For an admitted group, the source separates the obligation equal to the underlying items' fair value from the variable fee deducted for future service. Changes in the former do not adjust CSM. Changes in the insurer's share and in eligible future-service costs do, subject to loss and risk mitigation conditions. The latter adjustments use current rates; a GMM locked-rate calculation cannot be copied into this model (European Commission, 2023a, IFRS 17.45,B104,B110–114).

Suppose a qualifying insurance group receives premium 100, invested in its identified pool. Policyholders are entitled to 80% of the pool's value regardless of whether an insured event occurs, together with separate insured benefits. Supply remaining nonfinancial claims of 10 and risk adjustment 2, two equal current/future service units, and zero discount, tax and FX effects. The 80% amount is an investment component under the stated terms. All eligibility and service-quantity findings are explicit case premises. With pool value 100, FCF is $80+10+2 = 92$, leaving CSM 8 and LRC 100. Opening cash and equity 200 become cash 200, pool asset 100 and liability 100 after receiving and investing the premium.

Before the first service allocation, the pool rises to 120. The policyholder's share rises by 16 to 96. The insurer's 20% share rises by 4, increasing CSM to 12. FCF becomes $96 + 10 + 2 = 108$, and total LRC becomes 120. The financial increase in the liability is therefore 20, not merely the policyholder's increase of 16. With all pool gains and insurance finance in profit, asset gain 20 offsets insurance finance expense 20; the extra fee 4 remains unearned within CSM.

Now supply expected and actual first-period insured claims 5 and risk adjustment release 1. Allocate the CSM 12 equally across the current and remaining service units, releasing 6. Revenue is $5 + 1 + 6 = 12$, claims expense 5 and service profit 7. Paying the claims leaves cash 195, pool asset 120 and LRC $96 + 5 + 1 + 6 = 108$, giving equity 207. In the second period, with unchanged pool value and the same claims and release, profit is another 7. Sell the pool for 120 and repay the investment component 96. Final cash and equity are 214. The repayment reduces the liability and cash; it is excluded from both insurance revenue and service expense (European Commission, 2023a, IFRS 17.81,83–85,B119).

12.10.1 Losses and risk mitigation change the CSM adjustment

Consider a separate adverse branch from the initial FCF 92 and CSM 8, before any service. Future nonfinancial costs rise by 15. Use the CSM 8 first, then recognise service loss 7 and a tracked loss component 7 within LRC 107. Equity falls from 200 to 193. A subsequent cost reduction of 6 reverses loss 6 and leaves loss component 1, CSM zero and LRC 101. A further reduction of 4 reverses the last loss 1 and restores CSM 3; FCF 97 plus CSM 3 brings LRC back to 100. Those loss reversals remain service results under the specified rules even when a qualifying source of the original loss was a decline in the insurer's underlying-item share (European Commission, 2023a, IFRS 17.45,48–50,87(c)).

A financial guarantee change needs a different cause record. From the same initial position, suppose its supplied future-service value increases by 6 while a derivative gains 6. Without applying the risk-mitigation choice, FCF rises to 98 and CSM falls to 2, leaving LRC 100. The derivative gain enters profit, increasing equity to 206. If B115–116's conditions are met and the insurer applies the choice to this change, CSM remains 8 and LRC rises to 106. Insurance finance expense 6 then offsets the derivative gain, leaving equity 200. The choice changes recognition timing, not the derivative's value or the obligation.

This treatment requires a previously documented risk-management objective and strategy, an economic offset and credit risk that does not dominate that offset. B115 specifies which derivatives, FVTPL non-derivative instruments or held reinsurance can mitigate each exposure. It cannot be elected retrospectively because a mismatch became inconvenient. B117A takes the relevant finance to profit for derivative/FVTPL mitigation and specifies the matched policy

for held reinsurance. When B116's conditions cease, B118 stops the treatment prospectively without undoing previous recognition (European Commission, 2023a, IFRS 17.B115–118).

12.11 Partial reinsurance and changes in insurance estimates

The following methods use IFRS 17 in the retained September 2023 EU compilation. They extend the preceding nonparticipating general-measurement examples. Contract rights, group assignments, recovery percentages, risk adjustments and discount rates are explicit conditional inputs. A change in one of these inputs must retain its cause: changed service expectations, financial risk, counterparty performance and a new legal agreement do not have interchangeable accounting consequences.

12.11.1 Only the covered contracts support loss recovery

IFRS 17.62–66B (European Commission, 2023a, IFRS 17.62–68 and B119C–F) keeps a held treaty separate from the issued contracts, with its own entry, coverage and recognition dates. Proportionate coverage can delay recognition until an underlying contract is recognised. B119C permits the initial onerous-loss recovery only if the treaty was entered into before or at the same time as the underlying onerous contracts were recognised. The record must distinguish these dates; the treaty's later cash payment does not identify when it was entered into.

When a treaty covers only some contracts within an onerous issued group, B119E requires a systematic and rational allocation of the group's loss to the covered contracts. B119D then multiplies that covered loss by the percentage of their claims expected to be recovered. A treaty's headline quota percentage must not be applied to uncovered losses. Preserve the contract-to-issued-group and contract-to-treaty joins, covered risks and periods, attachment points, limits and exclusions, the loss-allocation method, expected claims/recoveries and the separate transferred-risk adjustment. For excess-of-loss or capped cover, expected recoveries require the admitted claim distribution and treaty payoff; a constant quota share cannot be used without support.

Take an issued group with premium 100, expected claims 118 and risk adjustment 2, all measured with zero discounting at recognition. Initial loss is 20 and CSM is zero. A source-supported allocation in this synthetic contract set identifies a covered portion with premium 41, claims 48 and risk adjustment 1, giving loss 8; the uncovered portion has premium 59, claims 70 and risk adjustment 1, giving loss 12. These amounts sum to the group; they are not separate groups created to improve the result.

A treaty entered into on time covers half of the covered contracts' claims. Expected recovery is 24, transferred-risk adjustment is a separately supplied 0.5, and premium paid is 26.5. After payment, the ordinary fulfilment asset is 24.5 and cost CSM is 2. Loss recovery is

$$8 \times 50\% = 4,$$

so cost CSM rises to 6 and the measured held asset to 30.5. The loss-recovery component of 4 sits inside that asset; it is not another receivable. Starting from cash and equity 50, receive issued premiums 100 and pay treaty premium 26.5. Cash 123.5 plus held asset 30.5 equals issued liability 120 plus equity 34. Applying 50% to the entire loss would recognise 10 and overstate equity by 6. If this same treaty had been entered into only after the underlying recognition, the initial loss-recovery adjustment would not apply.

12.11.2 A changed loss estimate affects both groups without netting them

Before any service or time passes, reduce expected claims on the covered contracts from 48 to 44, leaving other assumptions unchanged. Issued fulfilment liability falls from 120 to 116. Paragraph 50(b) first reduces the loss component, from 20 to 16, and recognises a loss reversal of 4; there is no new issued CSM while that loss component remains.

Expected held recovery falls from 24 to 22. Under paragraph 66(c)(i) (European Commission, 2023a, IFRS 17.66(c),66(bb),B119F), this change follows the underlying change that did not adjust the issued CSM, so it reduces the held asset and creates expense 2 rather than being hidden in held CSM. Held fulfilment asset becomes 22.5, cost CSM stays 6 and total held asset becomes 28.5. The covered loss is now 4, so B119F's recoverable loss-component limit is 2 and the loss-recovery component falls from 4 to 2. That component change tracks the same recognised recovery loss; posting a second expense for it would double count the event. Equity becomes $123.5 + 28.5 - 116 = 36$, an increase of $4 - 2 = 2$.

If all remaining services are then provided and claims equal these revised estimates, issued liability 116 is released through insurance revenue 100 and loss reversal 16; claims expense and payment are 114. Issued service-period profit is 2. Held coverage asset 28.5 is released as premium allocation 26.5 and reversal of loss recovery 2, with incurred recovery 22 and its collection recorded separately. Held service-period loss is 6.5. The entity ends with cash and equity $123.5 - 114 + 22 = 31.5$ and no group balances. Initial combined loss 16, estimate-change gain 2 and service-period loss 4.5 sum to lifetime loss 18.5, equal to the change from opening equity 50. Reinsurance has not removed or netted away the issued obligation.

B119F requires later loss-recovery components to reflect changes in the underlying loss component and never exceed the portion expected to be recovered. Track separately adjustments already included in fulfilment-cash-flow changes and reversals under 66(bb) that are not. The cap does not authorise inventing recovery income. Under paragraph 67, a change in reinsurer non-performance risk does not adjust CSM as future service; preserve collateral and dispute assumptions required by paragraph 63. Transferred-risk adjustment must be measured under paragraph 64, not mechanically copied as a fraction of the issued adjustment.

12.11.3 Changed terms and changed estimates are different events

For a treaty change, retain the old and new legal terms, effective date, covered-contract links and consideration. An exercise of a right already contained in the contract is not a modification under paragraph 72. A new treaty requires its own recognition and grouping decision and cannot be backdated to create initial-loss recovery. For a modification, apply paragraph 72's tests (European Commission, 2023a, IFRS 17.72–77) to the actual changed contract: changed scope or separated components, a substantially different boundary, a different initial group, changed direct-participation status, or loss of eligibility for a previously used premium-allocation approach can require derecognition and a new contract. If none applies, paragraph 73 treats the resulting cash-flow changes as estimate changes, with the held-contract modifications in paragraphs 63–68 where applicable.

For required derecognition, paragraphs 76–77 remove the affected cash flows and risk adjustment, adjust the remaining group's CSM under the applicable model and revise remaining coverage units. A transfer or qualifying modification uses paragraph 77's actual transfer consideration or equivalent new-contract premium adjustment rather than simply deleting the old balance. The equivalent premium is a required valuation input, not the old carrying amount chosen for convenience. Missing terms or that valuation leaves the particular modification unresolved. The generic tests identify the required procedure; they do not price an unspecified changed treaty.

12.11.4 The replacement premium connects the old and new groups

A qualifying modification changes both the removed obligation and the new contract. Paragraph 77 prevents the insurer from choosing the new opening balance simply by copying the old one. For a contract derecognised under paragraph 72, determine the premium that would be charged for an equivalent new contract at the modification date, then subtract any additional premium actually charged for the modification. This difference adjusts the old group's CSM together with the removed fulfilment cash flows, and is treated as a premium received when measuring the new contract (European Commission, 2023a, IFRS 17.72–77).

Take a general-measurement group with FCF 100, comprising expected claims 90 and risk adjustment 10, and CSM 30. Cash is 200, so liability 130 leaves equity 70. A legally effective modification substantially changes one contract's boundary and requires a new contract under paragraph 72. Its old allocated FCF is 40, comprising claims 36 and risk adjustment 4. Assume zero discount rates and no intervening service, tax, FX or OCI. The independently supplied equivalent new premium is 60, of which 10 is actually charged and received as an additional premium. The new obligation has FCF 52, comprising claims 48 and risk adjustment 4. These values are case inputs; they are not inferred from the old carrying amount.

The deemed premium is $60 - 10 = 50$. Removing FCF 40 while transferring value 50 reduces the old group's CSM by 10. Its remaining FCF is 60 and CSM 20, giving liability 80. The new contract's total consideration of 60 less FCF 52 gives CSM 8. One accounting sequence debits the old liability 50 and credits the new liability 50, then debits cash 10 and credits the new liability 10 for the actual receipt. Cash 210 equals old liability 80, new liability 60 and unchanged equity 70. Recording a second cash receipt for the deemed 50 would invent liquidity.

If the old group previously had three equal remaining coverage units and the removed contract contributed one, it now has two; the new group has its own service assessment. A proportional removal of 40% of the old CSM would leave 18 instead of 20 and would disregard the equivalent-premium rule. Over the remaining service, assume actual claims equal the supplied estimates and all remaining risk adjustment and CSM are released. The old group produces profit $80 - 54 = 26$; the new group produces $60 - 48 = 12$. Final cash $210 - 54 - 48 = 108$ equals opening equity 70 plus profit 38.

The same removed FCF with an actual third-party transfer premium of 50 instead reduces the old liability by 50 and pays cash 50; there is no new issued contract in this separate case. Cash 150, liability 80 and equity 70 reconcile. If the original CSM were only 6, either adjustment requiring 10 would consume it and recognise service loss 4. For the modification, old liability 60 plus new liability 60 would then leave equity 90 from cash 210, compared with opening equity $200 - 106 = 94$. The resulting loss component 4 is tracked within the old liability, not added again. Paragraphs 44(c), 48–50 and 77 govern that limit (European Commission, 2023a, IFRS 17.44,48–50,77).

Where the removed contract carries allocated accumulated OCI, paragraph 91 adds a separate presentation decision. Amounts arising under the systematic allocation in 88(b) are reclassified to profit on the specified transfer or derecognition; amounts arising under the held-underlying-item policy in 89(b) are not. Preserve that origin and the allocated amount before closing the old contract. The no-OCI example above does not eliminate this step (European Commission, 2023a, IFRS 17.91).

12.11.5 Future service consumes CSM before creating a loss

For issued contracts without direct participation, paragraph 44 and B96–B97 (European Commission, 2023a, IFRS 17.44,48–52,B72,B96–97) distinguish future nonfinancial service

changes from financial changes and incurred-claim experience. Use initial-recognition discount rates for the specified CSM adjustments under B72(c), while fulfilment cash flows use current rates. Changes in current rates, financial risk and incurred-claim estimates do not enter CSM by simply labelling them future cash flows. Preserve the current and initial-rate valuations and the reason for each change; their difference is part of the finance analysis, not an unrecorded balancing amount.

To isolate the nonfinancial rule, assume no passage of time, zero rates, no new contracts and no service release. A group has fulfilment liability 80, CSM 20 and total liability 100. An adverse future-service estimate of 15 raises fulfilment liability to 95 and reduces CSM to 5, with no immediate profit effect. A further adverse change of 10 raises fulfilment liability to 105, exhausts CSM and creates loss 5 and a loss component of 5. A subsequent favourable change of 8 first reverses that loss 5; only the remaining 3 restores CSM. Closing fulfilment liability 97 plus CSM 3 is 100. Crediting all 8 to profit would overstate the reversal by 3. The loss component is an internal allocation of the liability, never an additional liability of 5.

12.11.6 Finance presentation preserves total income and expense

For the nonparticipating case in paragraphs 87–90 (European Commission, 2023a, IFRS 17.87–91 and B128–132), a portfolio-level policy can present insurance finance entirely in profit or disaggregate it using the required systematic allocation. Under B129–B131, the choice and the contract characteristics determine the method. If financial assumptions do not substantially affect the nominal amounts paid, the systematic allocation uses the initial-recognition rates specified by B72(e)(i). OCI is the difference between that allocation and total finance, not a discretionary smoothing reserve. Over the group's duration it totals zero under B130.

Consider a GMM group with a now-incurred, fixed claim of 121 payable in two years, no remaining coverage or CSM, and zero risk adjustment because no nonfinancial uncertainty remains in this illustrative obligation. Initial and current discount rates are 10%, so liability is $121/1.1^2 = 100$. The next year's current rate falls to zero; one year remains, and current liability becomes 121. Total first-year finance expense is 21. Under the admitted OCI policy, the initial-rate allocation is 10 in profit and the difference 11 in OCI. In the second year, with rate zero unchanged, current liability stays 121, so total finance is zero. The initial-rate allocation accretes the alternative carrying amount 110 to 121, giving profit expense 11 and OCI income 11. Paying 121 extinguishes the liability. Cumulative profit finance is 21 and OCI is zero.

With opening cash 150, liability 100 and equity 50, first-year equity falls to 29 and stays 29 in the second year because profit expense and OCI income offset. Final cash and equity are 29. Presenting all finance in profit gives expenses 21 and zero instead, with the same total equity and cash. The OCI choice changes presentation, not the contractual claim or current measurement. This is the B131 fixed-payment case; variable-payment B132, VFA/underlying-item B134–B136, PAA and actual curve/RA/process valuation retain their separate methods. A successful fixed-claim example does not establish those models.

12.12 Variable benefits require a different finance allocation

The fixed-claim example uses initial rates because financial assumptions do not substantially change the nominal payment. B132 covers a different case: financial assumptions substantially affect the amounts paid. For the specified OCI policy, the cash-flow finance allocation may use a rate that spreads revised expected finance over the remaining duration at a constant rate. Contracts using a crediting rate have the alternative based on amounts credited and expected

to be credited. Nonparticipating CSM finance retains its prescribed initial-rate calculation; any separately allocated risk-adjustment finance must follow the source's consistent allocation (European Commission, 2023a, IFRS 17.B130–132).

Take a group with no remaining service, CSM or nonfinancial uncertainty, but a payment that remains sensitive to financial assumptions. Its liability and the amount measured using the systematic allocation both start at 100. A new financial assumption raises the expected payment in two years from 121 to 144. Supply a market-consistent current valuation of $144/1.1^2 = 14400/121$, approximately 119.0083. Under the chosen constant-rate allocation, the remaining rate solves $100(1 + r)^2 = 144$, so $r = 20\%$. The remeasurement at that instant produces OCI expense approximately 19.0083; no time has yet elapsed for the revised allocation to accrue interest.

With the current discount rate staying at 10%, current liability rises to 144/1.1, approximately 130.9091, after one year. Total finance expense is approximately 11.9008. The separate allocation rises from 100 to 120, giving profit expense 20 and OCI income approximately 8.0992. In the final year, current liability rises to 144, giving total finance approximately 13.0909; the allocation rises from 120 to 144, giving profit expense 24 and OCI income approximately 10.9091. All amounts are checked as exact fractions before display rounding. Total profit finance is 44, and accumulated OCI is zero. Opening cash 200 and equity 100 close at cash and equity 56 after payment. Retaining the original 10% allocation would reach only 121 and leave an unexplained 23 at settlement.

For a general payment schedule, retain the revised dated cash flows, carrying amount under the allocation, remaining duration and solved rate, then reconcile the full expected allocation and OCI unwind. A numerical failure or non-unique admissible solution remains unresolved; it is not permission to use an arbitrary rate. Current measurement and systematic allocation are separate calculations even when their values happen to agree.

12.12.1 Held underlying items determine the matching presentation

When a direct-participation insurer holds the underlying items, paragraph 89(b) permits a different OCI policy. B134 places in profit insurance finance that exactly offsets the profit income or expense on those items, with both amounts presented separately. The residual insurance finance goes to OCI. Revisit the preceding pool increase of 20, but now supply a valid asset measurement that reports 8 in profit and 12 in OCI. The matching insurance finance is expense 8 in profit and expense 12 in OCI, eliminating both net financial movements. The insurer's fee increase still enters CSM under the service method; matching is not an additional fee adjustment (European Commission, 2023a, IFRS 17.87A–90,B134).

Actual holdings must be recorded separately from the pool named by the contract. If holdings change the available choice between 88(b) and 89(b), B135–136 preserves the previously accumulated OCI under a separate reclassification schedule based on the former approach and assumptions at the change date. Those assumptions are then frozen for that schedule. Prior comparatives are not restated and accumulated OCI is not recomputed as if the new policy had always applied. New-period finance follows the newly applicable policy. A method that simply replaces the old rate and reserve with a newly computed one would discard required history (European Commission, 2023a, IFRS 17.B135–136).

12.13 A claim process needs joint probabilities

The preceding calculations take cash-flow and risk-adjustment values as inputs. Obtaining them requires a law for the dated obligations, including their dependence. Means alone can support a linear expected-claim total while giving the wrong answer for a treaty with a deductible. IFRS 17 requires probability-weighted discounted cash flows, with supportable non-market assumptions and consistency with market variables. It does not make independent claim draws a default when their dependence is unknown (European Commission, 2023a, IFRS 17.B37–41,B49–55).

Let claims X_1 and X_2 at two specified dates each be either 0 or 40, with probability one half for either amount. Assume zero discount and no counterparty loss solely to isolate dependence. Total expected claims are always 40. A treaty pays $\max(X_1 + X_2 - 60, 0)$. If both claims always occur together, total claims are either 0 or 80 and expected recovery is 10. If exactly one claim occurs, total claims are always 40 and recovery is zero. The two models have identical single-date distributions.

All joint laws on this declared binary support can be described by $a = \Pr(X_1 = 40, X_2 = 40)$. The two marginal constraints force the following probabilities:

	$X_2 = 0$	$X_2 = 40$
$X_1 = 0$	a	$1/2 - a$
$X_1 = 40$	$1/2 - a$	a

$$0 \le a \le 1/2.$$

Nonnegativity gives the displayed interval. Only the final cell produces a recovery, of 20, so its expectation is $20a$, ranging exactly from 0 to 10 within this finite family. Independent claims correspond to $a = 1/4$ and recovery 5; independence is one member, not a conclusion from the marginals. Substituting mean claims into the treaty would give $\max(40 - 60, 0) = 0$, which is wrong for every $a > 0$.

The same law supplies a sequential event procedure. Draw the first claim with probability one half. After observing 40, the conditional probability that the second claim is 40 is $2a$; after observing zero it is $1 - 2a$. These transitions reproduce the joint table and provide a conditional recovery forecast after the first event. This is an exact finite-law specification, not a simulated result. If the evidence leaves a unresolved, retain its admissible interval and separate conditional results; do not assign a probability distribution over possible values of a without additional evidence.

A general valuation method must similarly specify exposure and contract boundaries, event times and transitions, observation conditioning, claim severity, lapse and expense dependence, payment delays, discount treatment and financial-variable dependence. Its version must distinguish changes in the law from new observations under the same law. Risk adjustment then requires an entity-supported compensation method and diversification assumptions; the mean, variance or stop-loss value alone does not identify that compensation (European Commission, 2023a, IFRS 17.B65,B74–78,B86–89). The exact family above supplies a small test of such a method's dependence handling. Selecting and calibrating a richer law still requires the target population, credible data and downstream validation specified in the empirical protocols.

12.14 Claims development estimates the remaining payments

The claims-development example above supplies an estimation input, not a recognition rule. Its point projection is 148, but the single observed final factor leaves its variance unresolved. In

the general method the factors are

$$\widehat{f}_k = \frac{\sum_{i \in O_k} C_{i,k+1}}{\sum_{i \in O_k} C_{i,k}},$$

where O_k contains origin years with both required observed cells. Each incomplete row is projected through the remaining factors. The same triangle therefore has ultimate claims 180, 216 and 252, with unpaid amounts 0, 36 and 112. Tail development, calendar inflation, zero or negative cells and dependence need explicit extensions. The conditional means, variance scaling and independent-origin-year assumptions behind the inspected Mack error formulas (Mack, 1993, equations (1)–(3), Theorems 1–3) cannot turn this tiny triangle into a predictive law or a justified insurance risk adjustment.

12.15 A reserve estimate is not a future payment

The reserve of 148 leaves an important question unanswered: how much might the insurer ultimately pay? Repeating an estimation procedure can show how its fitted answer varies. Even if that answer were known exactly, however, the future claims would still vary. England's retained 2001 working addendum makes this distinction operational: first resample and refit to obtain future conditional means, then simulate payments conditional on those means. The inspected text is Actuarial Research Paper 138; its exact equivalence to the separately published 2002 journal addendum has not been established (England, 2001, Sections 2–3 and Appendix A).

Consider a small probability example, separate from the triangle. Suppose a fitted future mean, denoted by M, is either 2 or 8 with equal probability. These are authored possibilities, not estimates from an insurer. If $M = 2$, let a payment Y be 0 or 4, equally likely; if $M = 8$, let it be 4 or 12, equally likely. Within each fitted case, the payment has mean M and variance $2M$. Across cases, the fitted mean has expectation 5 and variance 9. The total variation in payments includes both the variation left within a case and the variation between case means:

$$\mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y \mid M)] + \mathrm{Var}(\mathbb{E}[Y \mid M]) = \mathbb{E}[2M] + \mathrm{Var}(M) = 10 + 9 = 19.$$

One can check the result directly: Y is 0, 4 or 12 with probabilities 1/4, 1/2 and 1/4. Its mean is 5, and its variance is $(25/4) + (1/2) + (49/4) = 19$. Reporting the distribution of fitted means alone would instead give variance 9 and no payment above 8. The full authored payment law gives probability 1/4 to that event.

Even those two moments do not select a unique payment law. Keep the same two fitted means, but, conditional on $M = 2$, pay 1 or 6 with probabilities 4/5 and 1/5; conditional on $M = 8$, pay 6 or 16 with those probabilities. The conditional means remain 2 and 8. Their variances are $(4/5)1 + (1/5)16 = 4$ and $(4/5)4 + (1/5)64 = 16$, respectively. The overall mean and variance remain 5 and 19, but the chance of paying more than 8 is now 1/10. Matching a reserve and a prediction error therefore cannot determine a tail probability or the insurer's compensation for non-financial risk.

12.15.1 What the two stages require

The addendum starts from a particular model for incremental payments. For an observed incremental payment X_{ij} in origin period i and development period j, its fitted mean m_{ij} is strictly positive. The model assumes $\mathbb{E}[X_{ij}] = m_{ij}$, $\mathrm{Var}(X_{ij}) = \phi m_{ij}$ and $\log m_{ij} = c + \alpha_i + \beta_j$, with reference levels fixed to identify the parameters. This incremental model and Mack's

conditional cumulative model have separate assumptions, even where their fitted reserve amounts agree. A common point estimate does not make their uncertainty models interchangeable (England, 2001, equations 2.1–2.3).

The first stage measures each observed increment's departure from its fitted mean using the Pearson residual $r_{ij} = (X_{ij} - m_{ij})/\sqrt{m_{ij}}$. With N observed cells and p fitted parameters, the paper estimates $\widehat{\phi} = \sum r_{ij}^2/(N-p)$ and resamples residuals after multiplying them by $\sqrt{N/(N-p)}$. The adjustment must be applied once. It cannot be followed by another correction of the same estimation variance (England, 2001, equations 2.4–2.6 and 3.1).

Start from the observed latest cumulative amounts, work backwards through fitted development factors, and difference the fitted cumulative amounts into increments. Calculate the residuals and dispersion from those increments. For each bootstrap draw, resample adjusted residuals with replacement and invert the residual relation to form pseudo-increments. Accumulate them, refit the development factors, project future cumulative claims and difference them into future means. Retain the same fitted draw across all cells. This produces a joint array of conditional means, not yet a joint array of payments (England, 2001, Section 2 and Appendix A, Triangles 1–9).

The second stage chooses a process law. For a positive future mean m and positive dispersion ϕ, the paper proposes ϕZ, where Z has a Poisson law with mean m/ϕ. Its mean is m and variance is ϕm. A gamma law with shape m/ϕ and scale ϕ has the same two moments but different support and tail. That is a modeling choice, not a change of random-number syntax. Each future payment is drawn conditionally, and aligned payments are summed by origin, payment period and total (England, 2001, Section 3 and Appendix A, Triangles 9–10).

Dependence must survive aggregation. In the first small example, suppose two cells share the same M but are independent conditional on it. Their conditional total variance is $4M$, with expectation 20, while their conditional total mean is $2M$, with variance 36. Their total variance is therefore 56. Independently redrawing each cell from its marginal payment law would give $19 + 19 = 38$, losing the dependence created by the shared fit. Marginal distributions are insufficient instructions for a joint simulation.

12.15.2 Sparse data, failed draws and implementation choices

The six observed increments in the earlier triangle fit the incremental model exactly. It has $p = 2(3) - 1 = 5$ parameters and only $N - p = 1$ residual degree of freedom, giving zero fitted dispersion. This arithmetic does not establish that future payments are deterministic. It also does not identify Mack's final development variance. The proposed interface returns `DISPERSION_UNRESOLVED` for this illustrative input and retains the point reserve of 148 separately. More data or a justified additional assumption is required before a predictive distribution can be admitted.

Every draw records its model, source version, data vintage, residual pool, fitted parameters, process law and random stream. Nonpositive degrees of freedom, an unsupported negative mean, a zero projection denominator or non-finite arithmetic receives an explicit failure. Structurally absent triangle cells have a separate mask. A failed payment is not converted to zero or silently replaced by another draw. In the authored example, if the $M = 8$ case cannot be evaluated, its probability 1/2 remains unresolved: the known contribution to the full mean is $\frac{1}{2}(2) = 1$, and a complete mean is withheld. In a finite bootstrap, attempted failure frequency is reported as such; it does not identify population probability.

The retained source itself needs a qualification. Its negative-mean suggestion draws X with mean $|m|$ and then subtracts $2m$. Read literally for $m < 0$, the resulting mean is $|m| - 2m = -3m$,

not m; at $m = -4$ it is 12. That prescription is wrong relative to its stated mean target in the inspected working version. A translation or reflection can preserve a negative mean, but changes the support and requires a different variance model. AccDSL leaves that extension unsupported pending an explicit signed-claims model; this local finding does not reject the positive-mean two-stage procedure (England, 2001, Section 3, final paragraph).

Software names also conceal consequential choices. In the inspected ChainLadder 0.2.17 source snapshot, `od.pois` calls a negative-binomial generator for dispersion greater than one. The paper's scaled Poisson and that negative binomial can share a mean and variance while assigning different probabilities to possible payments. The same snapshot uses signed gamma draws and replaces missing process values by zero. An adapter must distinguish structural absence from failed computation, identify the exact law and its parameter domain, and preserve separate resampling and process streams before its output can satisfy this proposal. These are source-code observations, not completed runtime validation (Gesmann and contributors, 2023).

The paper compares 1,000 simulations with analytic and earlier bootstrap summaries. That provides a worked implementation comparison under its assumptions. It does not establish future tail coverage, select an insurer's risk adjustment or justify 1,000 draws as a universal budget. A later empirical protocol must address calendar effects, tail development, pooling, heldout diagonals, simulation uncertainty and consequences for measured insurance groups (England, 2001, Section 4, Tables 1–3 and Figure 1).

12.15.3 The measurement date changes the question

A lifetime payment distribution differs from a one-year change in estimated liabilities. The 2019 accepted manuscript by England, Verrall and Wüthrich defines the claims development result as opening reserves less payments during the year and less the revised closing reserve. The closing estimate uses the information that has emerged during that year (England et al., 2019, equations 5–7 and Section 3.2). If the opening reserve is 100, payments are 30 and the revised closing reserve is 80, the result is $100 - 30 - 80 = -10$. Subtracting cash alone would yield 70 and answer a different question. The scenario must carry both realized next-year payments and the prescribed re-estimation rule, with the correct information cutoff. A lifetime draw cannot be relabeled as a one-year result. Neither statistic by itself supplies the discounting, group allocation or risk compensation required for insurance measurement.

12.16 What a bootstrap must reproduce

Separating fitted means from future payments is necessary, but it still leaves the statistical target to be chosen. An insurer knows an observed claims triangle. A method assessment may instead hold only its latest diagonal fixed, allowing different earlier histories to lead to those same cumulative amounts. These are different conditioning questions. Let D denote the observed triangle, Q its latest diagonal, R the future reserve payment and $\widehat{R}(D)$ the estimate calculated from the triangle. A distribution of $R - \widehat{R}(D)$ conditional on Q allows the earlier entries of D, and therefore the estimate, to vary. Conditional on the whole D, that particular estimate is fixed.

A small authored example makes the distinction concrete. Suppose two earlier histories are equally likely given the same Q and produce estimates of 90 and 110. Let future payments be 80 or 120 with equal probability, independent of those histories conditional on Q. The prediction error given Q is then $-30, -10, 10, 30$, each with probability $1/4$: its mean is zero and variance 500. Fixing the history whose estimate is 90 instead gives errors $-10, 30$, mean

10 and variance 400. Both calculations are correct for their stated conditioning information. They cannot serve as interchangeable validation targets. Nor does holding an estimate fixed imply that all uncertainty about model parameters has disappeared; a Bayesian parameter distribution, for example, would be a further explicitly specified model.

Steinmetz and Jentsch analyze a Mack bootstrap against the prediction error conditional on the latest diagonal. Their asymptotic construction adds older accident years and additional development observations while retaining the current calendar boundary. It does not obtain more information merely by increasing the number of bootstrap draws. The framework assumes positive initial cumulative claims with finite moments, specified independence, development factors tending to one, sufficiently fast decay of development variances, and further moment and regularity conditions. It is distinct from the incremental overdispersed-Poisson procedure in the preceding section (Steinmetz and Jentsch, 2024, Sections 2–4, especially Assumptions 2.2–2.5).

Their analysis separates an unscaled process term from a smaller estimation term. Under the stated assumptions, the process term has a nondegenerate, generally non-Gaussian limit. Correctly reproducing its distribution requires the correct conditional family for future development factors; matching two moments within a wrongly chosen family is insufficient. The original bootstrap can reproduce that limit while missing the conditional mean and covariance of the estimation term at its own, square-root sample-size scale (Steinmetz and Jentsch, 2024, Assumptions 4.6–4.9, Theorems 4.8–4.11 and Remark 4.12).

There is no contradiction in these two findings. For illustration, let a process term be an equally likely -1 or 1, and an independent estimation term an equally likely $-1/n$ or $1/n$. Their sum has variance $1 + 1/n^2$. Dropping the estimation term produces the same limiting distribution for the unscaled sum, yet multiplying that omitted term by n would have left variance one. This example explains the distinction between scales; it is not a derivation of the paper's Mack limit or evidence that a finite insurer triangle is adequately approximated by it.

12.16.1 Reconstructing the past requires a backward law

The paper's proposed repair holds Q fixed, simulates earlier cumulative claims backwards, refits the resulting triangle, and centers simulated future reserve payments at that refitted estimate. Future development uses the originally estimated forward factors. The purpose of reconstructing the past is to retain the effect of conditioning on the observed diagonal when measuring estimation uncertainty. Theorem 6.3 requires Assumption 6.2, including conditional limit properties for the backward reconstruction. Appendix C invokes those properties; it does not establish them for every positive distribution chosen by a programmer (Steinmetz and Jentsch, 2024, Section 5, Assumption 6.2 and Appendix C).

Forward conditional moments do not determine backward conditional moments. Let X be an earlier cumulative claim and Y the next cumulative claim. Suppose $X = 1$ or 2 with equal probability. Given $X = 1$, let $Y = 1$ or 3, equally likely. Given $X = 2$, let $Y = 3$ or 6 with probabilities 2/3 and 1/3. These positive claims satisfy $\mathbb{E}[Y \mid X] = 2X$ and $\mathrm{Var}(Y \mid X) = X$. Thus the forward factor Y/X has conditional mean 2 and variance $1/X$, the form of a one-step Mack moment relation.

Now observe $Y = 3$. The joint masses for $X = 1$ and $X = 2$ are 1/4 and 1/3, respectively. Their sum is 7/12, so Bayes' rule gives $\Pr(X = 1 \mid Y = 3) = 3/7$ and $\Pr(X = 2 \mid Y = 3) = 4/7$. The backward conditional mean is 11/7, and its variance is 12/49. Dividing the observed cumulative amount by the forward mean factor would instead give 3/2. Choosing a backward factor with mean 1/2 and variance 1/3 would imply a variance of 3 for X, also wrong for this joint law. The example establishes the missing implication; it does not assert that this finite distribution satisfies every assumption of an infinite-triangle theorem.

In the published paper, Section 5.1 proposes reciprocal-factor backward moments, while Remark 5.1 and Assumption 6.2 separately require adequate backward conditional behavior. The proposal therefore keeps such a generator as a hypothesis requiring its own justification. Given a joint model, the backward distribution must be derived from that model or approximated with a checked error criterion. Choosing a gamma family and matching reciprocal moments alone cannot establish the required conditional limit properties (Steinmetz and Jentsch, 2024, Section 5.1, Remark 5.1 and Assumption 6.2).

12.16.2 The center and the comparison must use the same quantity

Reserve payments exclude claims already paid. Write q_i for the latest cumulative payment in origin period i, U_i^+ for its simulated ultimate payment, and $\widehat{U}_i^+$ for the ultimate payment predicted from a reconstructed past. The reserve error must subtract like quantities:

$$R^+ - \widehat{R}^+ = \sum_i (U_i^+ - q_i) - \sum_i (\widehat{U}_i^+ - q_i) = \sum_i (U_i^+ - \widehat{U}_i^+).$$

For $q = 100$, simulated ultimate payment 130 and fitted ultimate payment 120, the reserve is 30, its estimate is 20, and their difference is 10. Subtracting the ultimate estimate 120 directly from the reserve 30 gives -90, shifting the error by the already-paid amount.

This distinction also resolves a literal inconsistency in the retained published text. Equation (5.3), printed on page 96, writes the centering quantity as a sum of projected ultimates without subtracting the latest diagonal, although the preceding step defines reserves net of that diagonal. Read literally, those two formulas produce the wrong reserve error. The adjacent product-difference formula and decomposition (6.1) instead cancel the diagonal and agree with the derivation above. The proposed implementation uses that explicitly derived reserve center; it does not copy (5.3) literally. No claim about the authors' unavailable execution code follows (Steinmetz and Jentsch, 2024, Section 5.1, equation 5.3 and equation 6.1).

The final journal article also changes the preprint's simulation comparison. Section 7.1.1 generates whole triangles to assess unconditional prediction interval coverage. Section 7.1.2 assesses conditional coverage using an approximate backward generator in equation (7.4), based on the same reciprocal-moment idea. The article acknowledges that the exact backward conditional law is generally difficult to obtain. Its conditional simulation therefore evaluates that constructed experiment; it does not by itself verify coverage under the exact backward law implied by every forward model. The published numerical differences remain reported simulation observations, without a local replication or statistically supported ranking here (Steinmetz and Jentsch, 2024, Section 7.1, Table 3 and Figures 1–3).

For AccDSL, a reserving-method specification must identify the payment horizon, conditioning information, process family, parameter-estimation procedure, backward law when one is needed, and reserve-centering convention. Validation must distinguish whole-triangle repeated-sample coverage, coverage conditional on a diagonal, and prediction conditional on a full observed history. A simulator with a convenient reverse rule is a candidate model, not an independent reference for that same rule. Exact centering and support checks precede simulations; target-specific comparison, uncertainty estimation and downstream financial-statement checks precede any promotion. The small triangle remains an accounting and arithmetic illustration, with its unresolved dispersion preserved.

12.17 A zero payment and a failed draw need different records

The distinction between payments and fitted means must survive the software boundary. Suppose a reserving calculation requests two draws, each containing payments for three future cells. In an authored example, the joint payment vectors are $(0, 4, 8)$ and $(2, 6, 10)$. Their totals are 12 and 18, and the sample mean is 15. These are possible nonnegative integer payments chosen to illustrate the exchange; they are neither fitted to the preceding triangle nor generated by a locally executed model.

Now let the last cell of the first draw fail. That draw's total is unresolved, while the second total remains 18. Reporting 18 as the requested sample mean would discard the failed draw. Replacing the failed value by zero would instead produce totals 4 and 18, and mean 11. Neither number is the mean of the requested complete sample. A diagnostic may state that the one completed draw has mean 18, provided it retains the failed draw and withholds the full sample mean. The fraction 1/2 describes completed computation, not a known probability mass assigned to unsuccessful economic histories.

This failure is relevant to the retained `BootChainLadder` source. Before forming its returned process triangles and totals, it replaces every process `NA` by zero. The returned array has no corresponding mask distinguishing failed future draws from structural cells. A valid payment vector $(0, 4, 0)$ and a vector $(0, 4, \text{failed})$ therefore become the same numeric vector and total. Inspecting the final array cannot recover the lost distinction (Gesmann and contributors, 2023, `BootstrapReserve.R`, process-triangle coercion and returned fields).

The unmodified output is consequently retained as source evidence but is unsupported as a complete predictive result under this proposal. A future instrumented implementation could preserve statuses before coercion, but would need its own source identity, dependency lock and numerical checks. Merely wrapping the returned array is insufficient. The documentary fixtures specify that boundary; they do not establish an implemented or tested R adapter.

12.17.1 The exchange retains the law and the whole draw

An optional `reserving.evaluate@1` signature uses the common operation envelope. Its declared profile names the claims triangle, method and process law, requested draw identifiers, information view and producer evidence. The present profile is a specification example that consumes authored evidence. It makes no financial-book mutation and grants no probability-law admission. Its request and result references use the same typed identity and dependency rules as the other operations.

The triangle carries the currency, base payment unit, scale, origin and development labels, payment-period mapping, valuation date and explicit cell mask. Observed cumulative amounts, future cells and unavailable observed amounts stay distinct. The inspected routine rejects unequal row and column counts, so the proposed initial adapter subset is square even though the native error message describes a broader shape. Rectangular support requires a separately inspected implementation.

Each replicate carries one fit identifier and its complete joint vector of future cells. A cell retains both its fitted mean and its process payment, with a status and reason if the payment is unavailable. A valid zero remains a value. Missing cells, repeated draw identifiers, a different fit inside one draw, or a currency mismatch prevent acceptance. Any failed cell leaves the draw total unresolved. Native evidence lacking the original status mask cannot be relabeled as complete by an adapter.

The process-law identity includes its formula. With positive mean m and dispersion $\phi > 1$,

the inspected `od.pois` helper calls a negative binomial with mean m and size $m/(\phi - 1)$. At $\phi = 1$ it calls a Poisson distribution. For $0 < \phi < 1$, its negative-binomial size would be negative. A gamma method with positive shape and scale is a different possibility; the helper's limitation does not reject that separate law (Gesmann and contributors, 2023, `rpois.od` and the process-law branches).

Nor can the negative-binomial path silently satisfy a request for the scaled-Poisson law from England's working addendum. For $m = 2$ and $\phi = 3/2$, both have mean 2 and variance 3. The scaled-Poisson support is $0, 3/2, 3, \ldots$, whereas the negative binomial can pay 1. The base payment unit thus matters to the model, not just to presentation. The positive-law profile excludes the package's separate signed handling; a different signed-claims model would need its own support and validation.

12.17.2 Stopping preserves what was actually computed

A result reports requested, attempted, completed, failed and pending draws separately. Its coverage domain is the set of computational replicate IDs. It is not the company's family of economic histories. Complete arrays do not establish calibrated forecasts, and an incomplete set of arrays does not justify a probability distribution conditional on computational success.

If cancellation follows the first complete draw in the example, retain its total 12 and mark the second draw unattempted. The diagnostic completed-draw mean is 12; the requested sample mean remains unresolved. A stop request alone does not establish that the worker has stopped. Unknown termination is reported as partial with unresolved worker state. Resume requires a durable checkpoint that binds the semantic manifest, original draw IDs and random generator state; the native function's seed argument supplies no such checkpoint.

The inspected source resets its supplied seed in both residual sampling and process generation. That behavior must retain its own method identity and cannot be described as verified independent streams (Gesmann and contributors, 2023, `sampleResiduals` and the process-law branches). A proposed replacement must specify generator version, stream derivation and restart behavior, and then check them in execution. Changing a method, law or request identifier creates a different request fingerprint. A resumed scheduling request must link to the old checkpoint under the persistence rules; it cannot silently alter the old request.

The retained snapshot's GPL terms, transitive dependencies and any future instrumented source remain part of the proposed runtime decision. No package installation, distribution clearance, crash-durability result or empirical performance follows from these exchanges. What they establish is narrower and necessary: a successful payment, a missing value and a failed computation remain distinguishable when reserving results enter the financial model.

12.18 Acquisition costs before the renewal contracts exist

An insurer may pay acquisition costs that help obtain both current contracts and their expected renewals. Some of the related future groups do not yet exist as recognised insurance balances. The retained AASB 17 provisions therefore require a separate acquisition-asset record for each related group, unless the permitted immediate-expense election is used. Qualifying costs are those directly attributable to the insurance portfolio; this is not a general permission to capitalise selling expenses. A systematic and rational allocation distinguishes the original group from groups expected to contain its renewals (Australian Accounting Standards Board, 2022e, paragraphs 28A–28B, B35A and Appendix A).

Consider a separate, authored example in currency units. Follow only 60 of qualifying costs already paid and allocated to expected renewals: 40 from one origination cohort to future group G1, and 20 from another cohort to future group G2. Other acquisition allocations are outside this example. Assume that both allocations are supported, neither future group has yet been recognised, the immediate-expense election is not used, and facts now indicate possible impairment. Those facts trigger the reporting-date recoverability assessment under paragraph 28E. An expected renewal remains a forecast, not a newly recognised premium receivable (Australian Accounting Standards Board, 2022e, paragraphs 28B, 28E and 35).

Each future group contains the relevant renewals together with other new contracts. For G1, suppose the whole group's expected net inflow is 35, of which the renewal cohort contributes 25. For G2, the corresponding amounts are 25 and 10. These are conditional measurements under paragraph 32(a), incorporating the required cash-flow, time-value and risk adjustments. They are neither gross premiums nor CSM. The acquisition cash already paid does not become a second future outflow. In this additive example, the other contracts contribute 10 to G1 and 15 to G2.

Paragraph B35D(a) first tests the asset against its related group's expected net inflow. G1's asset of 40 exceeds 35, so recognise impairment of 5 and retain 35. G2's asset of 20 is below its whole-group inflow of 25, so this test produces no loss for G2. Passing that first test does not settle whether the renewals themselves can recover the costs allocated to them.

The second test addresses that narrower population. Under B35D(b), the allocated renewal costs exceed the expected renewal inflows by $40 - 25 = 15$ for G1 and $20 - 10 = 10$ for G2. But 5 of G1's shortfall has already been recognised. B35D(b)(ii) requires that overlap to be deducted, leaving additional losses of 10 for G1 and 10 for G2. Total impairment is therefore $5 + 10 + 10 = 25$, and the remaining assets are 25 and 10. Adding the first loss to the two unadjusted renewal shortfalls would instead record 30, counting the same loss of 5 twice (Australian Accounting Standards Board, 2022e, paragraph B35D).

For these disjoint allocations, write A_g for the original asset, N_g for the whole-group expected net inflow and U_g for the relevant renewal inflow. With no prior impairment, the two losses and the resulting balance are

$$I_g = \max(A_g - N_g, 0), \qquad J_g = \max(A_g - U_g - I_g, 0), \qquad B_g = A_g - I_g - J_g.$$

The subtraction of I_g inside the second test preserves the already recognised loss. This local calculation relies on the identified, nonoverlapping renewal cohorts and positive inputs. A pool with several overlapping origins would need an explicit allocation of the relevant costs, inflows and prior losses before the same reasoning could be applied.

The entries also distinguish impairment from a cash payment. Debit impairment expense 5 and credit the G1 acquisition asset 5 for the first test. Then debit impairment expense 20 and credit each group's asset by 10 for the second. Cash does not move: the acquisition costs were already paid. For a small balance-sheet slice starting with cash and equity of 100 and no liabilities, paying the allocated costs leaves cash 40 and acquisition assets 60. After impairment, cash remains 40, acquisition assets are 35 and equity is 75. Paragraph 79 includes these acquisition assets within the related issued-contract portfolio carrying amounts; the separate records used here do not prescribe a separate line on the published balance sheet (Australian Accounting Standards Board, 2022e, paragraphs 28E and 79).

Recoverability can improve. At a later reporting date, suppose the full-group net inflows are 45 and 30, with renewal contributions of 35 and 15. Assume no intervening cash movement or recognition of either group. The allocation assumptions have been reviewed under B35B and remain unchanged; only the measurement inputs improve. Against the original allocations of 40 and 20, the remaining shortfalls are now 5 for each group. Paragraph 28F permits reversal to the extent the impairment conditions have improved. Reverse 10 for G1 and 5 for G2: debit the acquisition assets 15 and credit the impairment reversal in profit or loss 15. Assets become 35 and 15, totalling 50; cash stays 40 and equity becomes 90. The improvement restores 15 of the earlier loss of 25 without creating cash or recognising the expected renewals early (Australian Accounting Standards Board, 2022e, paragraphs 28F and B35B).

If the impairment conditions later disappear altogether, with these other facts unchanged, only the remaining loss of 10 can be reversed; the original allocated cost of 60 is the ceiling in this example. When related contracts are actually recognised and the acquisition cash flows enter their group measurement, the corresponding acquisition asset must be derecognised. B35C provides for partial incorporation when contracts are added over several periods. Keeping both the old asset and its incorporated amount would double count the same cost. The record must therefore preserve its origination cohort, future group, allocation history, impairment and reversal history, and the amount already incorporated into recognised contracts (Australian Accounting Standards Board, 2022e, paragraphs 28C–28F and B35B–B35C).

These are ordinary reporting-date impairment and reversal entries. They do not use the separate transition rules for acquisition assets. The historical edition, actual cost eligibility and supported renewal estimates must be checked before applying the example to an insurer.

12.19 When the acquisition asset enters the insurance group

Continue from the preceding reversal: cash is 40, the two acquisition assets are 35 and 15, and equity is 90. The next event is recognition of the related insurance groups. Assume that coverage begins and the first premiums become due at this date, with no earlier recognition trigger. All contemplated contracts are admitted together to their correctly identified groups. No service has yet elapsed and no claims have been incurred. Paragraphs 25–28 determine recognition; the later cash receipt is a separate event (Australian Accounting Standards Board, 2022e, paragraphs 25–28).

Keep the measured group net inflows of 45 and 30 from the previous example. To expose their components, suppose premiums of 100 and 60 are due immediately, the present values of future claims and expenses are 50 and 25, and each group has a nonfinancial risk adjustment of 5. Thus fulfilment cash flows before premium receipt are $50 + 5 - 100 = -45$ and $25 + 5 - 60 = -30$, using positive amounts for net outflows. These are supplied conditional measurements; present values do not imply a zero discount rate. Previously paid acquisition costs are not another future payment. Assume general measurement and no other assets, liabilities, taxes, reinsurance or intervening measurement changes.

Paragraph 38 requires initial CSM to prevent income or expense from the FCF, any initial cash flows and the derecognition of the acquisition asset, subject to its stated exceptions. The paid cost therefore affects initial CSM even though it is no longer a future cash flow. Let A denote the acquisition asset's remaining carrying amount, F the FCF and C the CSM. Removing the old asset without a gain or loss requires the new group balance to replace it. With positive L denoting a liability, the relation in this case is

$$L = F + C = -A, \qquad C = -F - A.$$

G1 consequently has CSM $45 - 35 = 10$ and a net group asset of 35; G2 has CSM $30 - 15 = 15$ and a net group asset of 15. Both margins are positive, so the onerous-group exception does not arise. There are no other previously recognised cash-flow assets to which paragraph B123A's revenue-and-expense exception would apply (Australian Accounting Standards Board, 2022e, paragraphs 28C, 32, 38, 47 and B123A).

An illustrative recognition entry debits the two group assets by 35 and 15 and credits the corresponding acquisition assets by those amounts. Cash is still 40 and equity is still 90. The separate acquisition balances are now zero. The amounts transferred are the carrying values after impairment and reversal, rather than the original allocations of 40 and 20. The earlier loss history remains available even though those assets have entered the group measurements (Australian Accounting Standards Board, 2022e, paragraphs 28C–28D and B35C).

Now receive the premiums in full, immediately after that measurement. Debit cash 160 and credit the group carrying balances by 100 and 60. Because the premiums are no longer future inflows, FCF becomes 55 for G1 and 30 for G2. CSM stays 10 and 15: no service or time has passed. The groups therefore carry liabilities of $55 + 10 = 65$ and $30 + 15 = 45$. Cash has increased to 200, and $200 - 65 - 45 = 90$ confirms unchanged equity. Paragraph 83 ties revenue to service, while B123(a)(i) expressly distinguishes premium cash inflows from the liability reductions that generate revenue. Later acquisition-recovery revenue and the equal service expense are allocated over time under B125; neither is created by this instantaneous receipt (Australian Accounting Standards Board, 2022e, paragraphs 40–41, 83–85 and B123–B125).

The three balance-sheet states make the transfer visible. Amounts are currency units; group assets and liabilities below are their net carrying amounts, not their separate FCF and CSM components.

	Before recognition	After recognition	After receipt
Cash	40	40	200
Acquisition assets	50	0	0
Group assets	0	50	0
Group liabilities	0	0	110
Equity	90	90	90

Receiving 160 has changed liquidity and the insurance balances without creating profit. Omitting the acquisition assets from the initial CSM calculation would instead give margins of 45 and 30, overstating total CSM by 50. Retaining the separate acquisition assets after the correct transfer would overstate assets and equity by 50.

Even a balanced statement can conceal both mistakes together. If the implementation leaves acquisition assets of 50 and uses the overstated margins, its post-receipt liability becomes 160. Cash 200 plus the retained asset 50 less liability 160 still equals equity 90. That identity passes, but the asset, liability and CSM are wrong relative to the stated measurement. Checking the components and the derecognition event is therefore necessary in addition to checking that the statement balances.

As a boundary case, if no acquisition asset exists, $A = 0$ gives $C = -F$ and a zero group carrying amount before premium receipt, under the same positive-margin assumptions. A general implementation must also handle onerous groups, other initial cash-flow balances and contracts added over time. The present calculation establishes the two-event bridge for the specified groups, rather than those additional cases or an insurer's actual valuation.

12.20 Acquisition costs and the timing of reinsurance

An acquisition asset can enter a group that is onerous at recognition. Reinsurance then raises a second question: how much of that initial loss may be recognised as recoverable, and when? Following the two groups separately prevents a recoverable loss from becoming either a second asset or a reduction of insurance revenue.

Consider a separate general-measurement group with opening capital and cash of 50. Before recognition, the insurer pays acquisition costs of 10, allocated wholly to that future group. At the prior assessment, expected premium 100, claims 80 and risk adjustment 5 give expected net inflow 15. The asset of 10 is therefore recoverable under the stated assumptions, leaving cash 40, acquisition asset 10 and equity 50. No renewal allocation is involved (Australian Accounting Standards Board, 2022e, paragraphs 28A–28F and B35A–B35D).

The expected claims increase to 100 when the contracts are recognised; premium remains 100 and risk adjustment remains 5. The example has no intervening unrecorded onerous period, discounting, tax, investment component or OCI. Before premium receipt, fulfilment cash flows are $100 + 5 - 100 = 5$. Removing the acquisition asset adds 10 to the initial loss, making 15, with CSM zero. The entry debits loss 15 and credits acquisition asset 10 and remaining-coverage liability 5. Receiving the premium then gives cash 140, liability 105 and equity 35 (Australian Accounting Standards Board, 2022e, paragraphs 28C, 38(c)(i) and 47).

The loss component of 15 is part of the liability of 105. Acquisition cash has already left the company: putting it back into future cash outflows would count the cost twice. Its original asset also disappears when it enters the group measurement.

A separate held treaty covers half of every claim of this group and costs 55. Assume no commissions, limits, disputes or counterparty shortfall. Expected recovery is 50; risk transferred is separately valued at 2.5. That supplied risk adjustment need not equal half the gross adjustment in a different contract. Ordinary cost CSM is $55 - 50 - 2.5 = 2.5$, so the held asset after premium payment would be 55 (Australian Accounting Standards Board, 2022e, paragraphs 62–65).

If the treaty was entered into at or before recognition of the onerous group, the specified initial loss recovery is $15(0.5) = 7.5$. It increases held cost CSM to 10 and the measured asset to $50+2.5+10 = 62.5$. The standard multiplies the recognised loss by the expected claims-recovery percentage; this does not mean the treaty separately reimburses half the acquisition invoice. The loss-recovery component of 7.5 is contained in the asset of 62.5. After paying the treaty premium, cash 85 plus that asset equals liability 105 plus equity 42.5 (Australian Accounting Standards Board, 2022e, paragraphs 66A–66B and B119C–B119F).

At the end of the single service interval, release the issued liability of 105 as revenue 90 and loss reversal 15. The ratio $15/105 = 1/7$ allocates the expected-claim and risk-adjustment releases to the loss component. Acquisition-cost presentation adds revenue 10 and an equal service expense 10 under B125. It creates neither cash nor a new acquisition asset. Recognising and paying actual claims of 100 therefore gives issued-group service-period profit $100 - 100 - 10 + 15 = 5$. Including the initial loss of 15, its lifetime loss is 10 (Australian Accounting Standards Board, 2022e, paragraphs 49–52 and B120–B125).

The held asset releases as premium allocation 55 and loss-recovery reversal 7.5, with incurred recovery and collection of 50 recorded separately. Its service-period result is $50-55-7.5 = -12.5$; the initial recovery of 7.5 makes its lifetime result -5. The company's final cash and equity are both 35:

$$50 + 100 - 10 - 55 - 100 + 50 = 35.$$

All issued and held balances are zero. Total operating cash is -15, profit is -15, and OCI is zero.

Premium allocation remains separate from issued revenue (Australian Accounting Standards Board, 2022e, paragraph 86).

If the otherwise identical treaty is entered into only after the issued group is recognised, B119C prevents recovery of that original loss. Its initial held asset is 55 and its cost CSM is 2.5. Equity is initially 35 and remains 35 after the combined subsequent service result of zero. The timely treaty instead gives initial equity 42.5 and a subsequent loss of 7.5. Equal final cash does not make those recognition histories interchangeable.

The transfer therefore requires the acquisition allocation, recoverability history and unconsumed asset version. The held measurement additionally requires treaty entry time, covered-contract mapping, claims recoverability and its own risk adjustment. Exact replay does not transfer the asset again; an unresolved treaty date does not default to timely recognition. Partial coverage or changing estimates require their stated allocations before the same procedure can be applied.

12.21 Reading acquisition costs in an actual report

Section 12.19 supplied the acquisition assets and the net inflows for two identified groups. A published insurer's accounts reveal a different level of detail. Munich Re's 2023 report shows contracts recognised for the first time during the year, separating concluded from acquired contracts and nononerous from onerous business. Note 27 covers segments predominantly using general measurement or the variable fee approach; it omits two predominantly PAA segments. These are segment disclosures, not a list of individual groups (Munich Re, 2024, printed pp. 252–254).

Consider the ERGO Life and Health Germany segment, identified in the paragraph preceding its table. Its newly recognised contracts combine GMM and VFA measurements. The following two columns reproduce its concluded-contract amounts in millions of euros. Preserve the report's signs: inflows are positive, outflows and the margin are negative. Its onerous CSM cell is blank; a dash below preserves that blank instead of assigning a reported zero (Munich Re, 2024, printed pp. 252–253).

	Nononerous	Onerous
Present value of future inflows	2,842	17
Future claims, expenses and investment components	−2,438	−12
Future acquisition costs	−243	−25
Risk adjustment	−8	0
Contractual service margin	−154	—
Reported total	0	−21

The nononerous amounts illustrate how expected future profit is deferred in CSM. The onerous column retains a loss-bearing population separately. Munich Re describes recognising an expected initial loss in profit or loss and keeping onerous contracts separate from profitable ones. Combining these columns into one net profit would erase that distinction (Munich Re, 2024, printed pp. 165–166).

Adding the displayed nononerous numbers gives −1, although the table reports zero. The onerous cash-flow and risk-adjustment rows give $17 - 12 - 25 - 0 = -20$, against a reported total of −21. The report states that amounts are rounded to millions and that totals may consequently differ. A possible set of unreported nononerous values is $2{,}842.3 - 2{,}437.8 - 242.8 - 7.8 - 153.9 = 0$; each rounds to its displayed amount. Similarly, $16.6 - 12.1 - 25.2 - 0.3 = -21$ is consistent with the onerous column if no CSM is assigned. Together these hypothetical values also reproduce

the table's outflow subtotals and aggregate rows after rounding. They demonstrate consistency under nearest-million rounding; they do not recover the issuer's unreported measurements or establish its exact rounding convention. Retain the original figures and blank cells (Munich Re, 2024, printed pp. 155, 253).

The acquisition-cost row appears within *future cash outflows*. Its reported total of $243+25 = 268$ therefore cannot simply be entered as the previous example's asset A, which arose from costs already paid before the related groups were recognised. Nor do the table's acquired-contract columns describe how sales acquisition costs were paid. Those columns concern a different distinction between contract populations. Their displayed zeros retain the same rounding qualification. The selected disclosure does not identify separate prior acquisition assets, their impairment history or their allocation to individual groups (Munich Re, 2024, printed pp. 252–253).

A third quantity appears in the segment's service revenue. For 2023, Munich Re reports €202 million as the portion of premium relating to recovery of acquisition costs. Its policy explains that acquisition costs enter insurance service expense in the same amount as the recovery included in revenue, allocated over the coverage period. This is a period service amount; the €268 million above is an estimate of future acquisition outflows for newly recognised contracts. Subtracting one from the other would not produce a remaining acquisition asset. The relevant populations, dates and movements must first be linked (Munich Re, 2024, printed pp. 177, 253, 264).

The segment's liability reconciliation makes that caution concrete: it shows €637 million in the service-expense row labelled "Recovery of acquisition costs", and €978 million of acquisition costs within cash flows. Thus the €202 million recovery in revenue does not describe the entire expense row. The difference of €435 million is an arithmetic residual; assigning its cause requires the relevant populations and expense treatments. Neither that residual nor cash less expense identifies a separate acquisition asset (Munich Re, 2024, printed pp. 230–231).

The same timing distinction applies to profit. The report gives €819 million of CSM for services provided during 2023 in this segment. That amount concerns service during the year, whereas €154 million is the margin attached to the newly recognised contracts in the selected table. It is not a release from that new-business amount alone. A reconciliation needs the opening CSM and every intervening movement. The segment's total insurance revenue also combines measurement approaches: €7,966 million under GMM/VFA and €1,976 million under PAA make the reported €9,942 million (Munich Re, 2024, printed pp. 253, 264).

Even within one insurer, an acquisition-cost policy needs its applicability conditions. Munich Re reports using the PAA option to expense acquisition costs directly for a substantial part of its property-casualty primary insurance business. The earlier general-measurement asset example therefore requires the relevant treatment and group facts before it can be applied to a real portfolio (Munich Re, 2024, printed p. 166). These are readings of the retained 2023 English translation, whose German original is binding. They identify reported aggregates and necessary further facts; the underlying group histories and valuations remain unresolved.

12.22 Following the margin through the year

Why can the segment release €819 million of CSM during 2023 when newly recognised contracts add only €154 million? The opening margin carries profit from earlier business. Munich Re's measurement-component reconciliation for ERGO Life and Health Germany begins with CSM of −10,449 million euros and ends with −9,569 million. Its signs present the margin as a negative balance: new business makes it more negative, while service release moves it towards zero. Keep that convention while reading the movements (Munich Re, 2024, printed p. 246).

The nonzero CSM movements are service provided during the year, contracts initially recognised, changes in estimates that adjust CSM, and derecognition or changes in the consolidated group. In that order, the displayed amounts give the complete bridge

$$-10{,}449 + 819 - 154 + 214 + 1 = -9{,}569.$$

In positive margin amounts, opening CSM of 10,449 receives 154 from new business, releases 819 for service, and falls by another 214 and 1 through the other two movements. The net reduction is 880. The table's CSM service subtotal of 879 already combines $819 - 154 + 214$; adding that subtotal to the individual movements would count them twice. The estimate movement here is the CSM column's 214, not a claim that every change in the underlying estimates is absorbed by CSM (Munich Re, 2024, printed p. 246).

The numbers connect three distinct disclosures. New-business CSM of 154 matches the initial-recognition table; the 819 release matches the segment's service revenue; and closing CSM of 9,569 is the balance used in the issuer's schedule of expected future releases. The latter match verifies which balance that schedule starts from, rather than the accuracy of the future releases. The comparative-year column also reconciles: $-9{,}412 + 948 - 221 - 1{,}764 = -10{,}449$, which becomes the 2023 opening amount (Munich Re, 2024, printed pp. 247, 253–254 and 264).

The larger liability totals require a different comparison. The measurement reconciliation includes the full GMM/VFA balances and the incurred-claims liabilities of PAA contracts. It excludes PAA remaining coverage. It would therefore be wrong to describe the whole table as GMM/VFA only. The footnote continued after its comparative table explicitly gives the omitted PAA remaining-coverage balance: −62 million euros in the comparative period and −1 million at the end of 2023 (Munich Re, 2024, printed pp. 241 and 247).

This scope difference accounts for the displayed opening and closing differences. The complete liability reconciliation starts at −109,556, against −109,494 in the measurement table: the difference is −62. Their closing amounts are −112,808 and −112,807, a difference of −1. Thus an apparently small missing balance is a disclosed excluded population, rather than permission to omit it or absorb it into rounding (Munich Re, 2024, printed pp. 230–231 and 246–247).

The cash rows preserve the same distinction. Premium receipts change the reported liability balance by −10,406 in the full table and −8,471 in the measurement table, leaving −1,935 attributable to the omitted PAA remaining-coverage portion at displayed precision. Acquisition cash movements of 978 and 502 leave 476 for that portion. These signs describe the liability reconciliation; they are not the debits and credits of a cash account. Nor are these differences all PAA cash flows, because PAA incurred claims are included in both tables (Munich Re, 2024, printed pp. 231, 241 and 246).

The 476 cash difference still does not explain the earlier acquisition expense difference of $637 - 202 = 435$. The two quantities differ by 41, but calling that amount a deferred acquisition asset would again invent a missing link. A cash payment, an expense and recovery included in service revenue can concern different dates and contract populations. The disclosed expensing election makes the treatment of PAA costs a relevant question; the selected tables do not establish that it alone explains 435. Preserve that unresolved attribution alongside the now-reconciled CSM and the identified PAA scope difference (Munich Re, 2024, printed pp. 166, 230–231, 246 and 264).

All these comparisons use the report's rounded euro millions. Exact addition of the CSM column does not establish unrounded measurement equality or exact conservation of every other column. The public reconciliation supplies movements and their reporting populations, while the group assignments, valuation inputs and detailed acquisition-expense allocation remain separate facts to obtain.

12.23 Adoption dates and the restatement of 2022

A standard's effective date and the economic date of a reported balance need not coincide. An insurer adopting new rules in 2023 may report a revised balance for December 2022 while retaining the earlier report as a record of what it originally disclosed. The system must preserve both versions and explain which one a calculation uses.

The Ministry of Finance's explanation dated 12 June 2025 describes the original ASBE 25 timetable. It specifies application from 1 January 2023 for enterprises listed both domestically and overseas, and for overseas-listed enterprises preparing financial statements under IFRS or ASBE. Other enterprises applying ASBE were scheduled to apply the revised standard from 1 January 2026. Its account of the 2025 notice says existing adopters, including enterprises that had adopted early under the 2020 notice, must continue applying it. Insurers facing genuine difficulties and seeking deferral must submit written reasons to the Ministry's Accounting Department and the relevant institutional supervision departments of the National Financial Regulatory Administration. That statement does not establish an automatic deferral merely because preparation is difficult (Ministry of Finance of the People's Republic of China, Accounting Department, 2025, answers on background and implementation arrangements).

The same explanation limits the new simplification choices to unlisted enterprises; matters outside those choices remain subject to ASBE 25. It names grouping, measurement, presentation and transition as affected areas, including comparative information and the transition date. These are source-supported descriptions of the Ministry's explanation. The operative 2020 and 2025 notices have not yet been directly verified in this collection, so the explanation does not supply the detailed conditions needed to implement those choices (Ministry of Finance of the People's Republic of China, Accounting Department, 2025, answer on the relationship between the notice and the standard).

Ping An provides a concrete adoption history. Its 2022 ASBE annual report states that the group would adopt the revised insurance standard for accounting years beginning on 1 January 2023. Its 2023 report confirms adoption on that date and restatement of comparative information (Ping An Insurance (Group) Company of China, Ltd., 2023, Note XIV.2, pp. 315–316) (Ping An Insurance (Group) Company of China, Ltd., 2024, Note IV.34, p. 192). The following balances all refer to the consolidated group at 31 December 2022, in RMB millions. The first column comes from the 2022 report; the second is the restated comparative in the 2023 report. Changes are calculated from those columns; the asset, liability and parent-equity changes also match the three rows of Note IV.34's reported bridge (Ping An Insurance (Group) Company of China, Ltd., 2023, pp. 146–148) (Ping An Insurance (Group) Company of China, Ltd., 2024, pp. 141–143 and 192).

Balance	Original	Restated	Change
Assets	11,137,168	11,009,940	$-127,228$
Liabilities	9,961,870	9,823,944	$-137,926$
Total equity	1,175,298	1,185,996	$10,698$
Parent shareholders' equity	858,675	869,191	$10,516$
Non-controlling interests	316,623	316,805	182

The distinction between total equity and equity attributable to the parent matters. Assets less liabilities rise by $-127,228 - (-137,926) = 10,698$. Of that increase, 10,516 belongs to parent shareholders and 182 to non-controlling interests. Treating the parent-equity row as total equity would leave an unexplained difference of 182. These are changes between reported accounting bases for the same date, not cash received in a later period.

The transition also changes how later growth is calculated. Ping An reports assets of 11,583,417 at the end of 2023. Subtracting its restated 2022 comparative gives a reported-balance increase of $11,583,417 - 11,009,940 = 573,477$. Subtracting the original 2022 amount instead gives 446,249, mixing the old and new bases and understating that comparable movement by 127,228 (Ping An Insurance (Group) Company of China, Ltd., 2024, p. 141). The first difference is a movement in reported asset balances; it is not a cash-flow measure or an explanation of every economic cause of growth.

An original-source query for assets at 31 December 2022, explicitly selecting the 2022 report, therefore returns 11,137,168. A query selecting the 2023 report's restated comparison for that date returns 11,009,940. Each assertion retains its report identity, consolidated scope, RMB-million scale, accounting basis and economic date. A link records that the later assertion restates the comparison; it does not delete the original. A request for a comparable annual movement must reject the mixed-basis pair unless an explicit reconciliation has first aligned it. Because the basis and disclosure version differ, these two amounts are not a same-context source conflict.

The information date remains a separate question. The 2023 report says its board authorised the statements on 21 March 2024 (Ping An Insurance (Group) Company of China, Ltd., 2024, Note I, p. 154). Neither that authorisation nor the 31 December reporting date establishes when the exact retained document became public. The corpus has archive-date leads but no verified publication date or historical byte vintage for either selected report. A query for information available to investors at 31 December 2023 must therefore retain an unresolved availability result for this source selection; it cannot silently import the later restated balance. The corpus's October 2026 revalidation timestamps describe local evidence handling, not investor knowledge in 2023.

Finally, Note IV.34 reports financial-asset reclassification, including use of the classification overlay for relevant assets derecognised during 2022. It also states that some insurance groups used modified-retrospective or fair-value transition because retrospective application was impracticable. The headline bridge consequently cannot be reconstructed by applying one percentage to every insurance liability. The example establishes how to retain and compare disclosed histories; reconstructing the transition itself still requires the affected groups, asset classifications and underlying measurement inputs (Ping An Insurance (Group) Company of China, Ltd., 2024, Note IV.34, p. 192).

12.24 Reconstructing balances at transition

A disclosed restatement tells us what balances an insurer reported under the new basis. It does not reveal the historical contract groups, assumptions or services needed to reproduce their opening measurement. The three transition methods address different limits on that reconstruction. The retained December 2022 AASB 17 compilation provides a useful, explicitly scoped account of those methods: its stated applicability covers annual periods beginning from 1 January 2023 to before 1 July 2026. These provisions explain a design requirement; they do not establish the rules governing Ping An's ASBE groups (Australian Accounting Standards Board, 2022e, Appendix C).

First distinguish the dates. Initial application begins the annual period in which the entity first applies the standard. Transition normally begins the immediately preceding annual period. Thus a calendar-year entity first applying the standard in 2023, with only 2022 presented as an adjusted comparative, starts its transition calculation on 1 January 2022. If it elects to present earlier adjusted comparatives, paragraph C25 instead takes transition back to the beginning of the earliest such period. Neither date establishes when the resulting report became public (Australian Accounting Standards Board, 2022e, paragraphs C2 and C25–C27).

Full retrospective application identifies and measures each group as if the standard had always applied. Acquisition-cash-flow assets are considered separately, balances that would not exist under the new rules are removed, and the resulting net difference is recognised in equity. This route needs the historical information required to reconstruct grouping, cash flows, measurement assumptions and service already provided. It is required unless application is impracticable or the specific paragraph C5A exception applies (Australian Accounting Standards Board, 2022e, paragraphs C3–C5).

Impracticability requires more than a missing historical workbook. Under AASB 108, the entity must have made every reasonable effort to apply the requirement. Relevant difficulties include effects that cannot be determined, assumptions that would have to be made about management's past intentions, and significant estimates for which the necessary contemporaneous evidence cannot be objectively distinguished from other information. Those conditions need to be established for the reconstruction at issue (Australian Accounting Standards Board, 2022a, paragraph 5).

An estimate prepared later can nevertheless describe an earlier period. The question is whether it reflects the circumstances that existed then, using evidence that would have been available when the earlier financial statements were authorised for issue. A newly assembled estimate can meet that test; replacing an earlier expectation with knowledge of what subsequently happened uses hindsight. The date of preparation alone therefore does not settle the question. Nor does authorisation establish when an outside investor could obtain the information: that requires the separate public-availability record. The insurance standard's specific transitional provisions govern their stated cases, including permitted information under the modified and fair-value routes described below (Australian Accounting Standards Board, 2022a, paragraphs 7, 19(a) and 50–53).

When full retrospection is impracticable for a group, the alternatives are modified retrospection and fair-value transition. Modified retrospection aims for the closest retrospective result possible using reasonable and supportable information available without undue cost or effort. Each specified modification is allowed only to the extent that the necessary retrospective information is missing. If the information needed even for this modified method cannot be obtained, paragraph C6(a) requires the fair-value approach. Missing history therefore does not license an unrestricted choice of present assumptions (Australian Accounting Standards Board, 2022e, paragraphs C5–C8).

For example, the modified route can reconstruct earlier expected cash flows from estimates at transition, or an earlier determinable date, adjusted for known intervening cash flows, including those of contracts that have expired. Its discount-rate approximations require the specified yield-curve evidence; its risk adjustment and allocation of past service require their own records. The source also prescribes nil amounts for particular acquisition-cash-flow items when the necessary information is unavailable. That narrow instruction cannot be generalized into a rule that every missing amount is zero (Australian Accounting Standards Board, 2022e, paragraphs C12–C16).

Fair-value transition takes a different starting point. Paragraph C20 determines the CSM or loss component from the difference between the fair value of the insurance group and its fulfilment cash flows, both measured at transition. The insurance group's fair value is distinct from the value of its underlying investment pool. Replacing one with the other would compute a different quantity. The calculation needs a supported group valuation as well as the cash-flow, discount and risk-adjustment inputs; C20 expressly excludes the demand-feature rule in AASB 13 paragraph 47. Its permitted information-date, grouping and accumulated-OCI choices also have specified conditions (Australian Accounting Standards Board, 2022e, paragraphs C20–C24B).

To interpret that group valuation, consider what would be paid to transfer the liability in an orderly transaction between market participants at the measurement date. The obligation remains outstanding, and the transferee must fulfil it; the thought experiment is not an immediate settlement with the policyholder. AASB 13 therefore calls for assumptions that market participants would use, in the accessible principal market or, in its absence, the most advantageous market. The unit being valued still comes from C20: the insurance group (Australian Accounting Standards Board, 2021c, paragraphs 2–3, 9, 13–24 and 34).

A quoted price is not required in every case. The valuation should maximise relevant observable inputs, while supported unobservable inputs can supply the participant assumptions for which market observations are unavailable. The insurer's own data can be a starting point, but require adjustment when reasonably available evidence indicates different participant assumptions or entity-specific features unavailable to other participants. A technique must also fit the circumstances and available data. These requirements make the source, date and interpretation of each valuation input consequential (Australian Accounting Standards Board, 2021c, paragraphs 61–67 and 87–89).

There is a further reason to retain separate fair-value and FCF calculations. AASB 13 includes non-performance risk, including the issuer's own credit risk, in liability fair value and assumes that risk is unchanged by the hypothetical transfer. AASB 17 expressly excludes the issuer's own non-performance risk from fulfilment cash flows for issued insurance contracts. Even when the two measurements draw on some of the same cash-flow records, their treatment of this risk differs. Neither a single imported FCF balance nor the value of the underlying investments supplies both C20 inputs (Australian Accounting Standards Board, 2021c, paragraphs 42–44) (Australian Accounting Standards Board, 2022e, paragraph 31).[1]

[1]The comparison uses AASB 108's 31 December 2022 compilation, applying to periods beginning from 1 January 2023, and AASB 13's 30 June 2021 compilation, applying from 1 July 2021. These are retained historical editions; later amendments and the actual entity's applicable requirements need their own check.

There is one important qualification to the ordinary method choice. Under C5A, a group with direct participation features may elect fair-value transition even when retrospective application is possible, if the entity chooses the B115 risk-mitigation option prospectively from transition and used qualifying mitigation before that date. The applicable instrument, qualification and designation conditions still matter. Treating every fair-value election as proof of impracticability would therefore be wrong (Australian Accounting Standards Board, 2022e, paragraphs C3(b), C5A and B115–B118).

The reconstruction must consequently retain the method and its basis for each group, each modification used, the information date and evidence supporting it, the separately measured FCF, CSM or loss component, relevant OCI and acquisition balances, and the opening-equity reconciliation. Some transition rules expressly permit transition-date information for specified assessments. That permission concerns the accounting restatement; it does not make a later report admissible in a forecast made before its publication. Detailed impracticability judgments under AASB 108 and valuation under AASB 13, together with actual group facts, remain necessary before this method description can produce a supported opening measurement.

12.25 From transition measurements to opening equity

The fair-value approach produces a margin or a loss component, but those two outcomes enter the liability differently. Consider two distinct groups of issued contracts for which the grouping and eligibility for this transition method have already been established. Use the historical provisions just described and assume that both groups contain only remaining coverage. All amounts below are authored currency units at the same transition date. There are no incurred claims, reinsurance, acquisition balances, tax, accumulated OCI or other transition adjustments in this example. These simplifications isolate the effect of the two measurements; they do not supply actual insurer valuations or evidence of eligibility.

For group A, suppose the measured fulfilment cash flows are 90 and the separately measured fair value is 110. The FCF amount already includes the discounting and nonfinancial risk adjustment required by the measurement rules. Paragraph C20 makes the difference, $110 - 90 = 20$, the CSM. This is unearned profit associated with future services. Keeping it within the liability gives $90 + 20 = 110$; recording it as immediate profit would contradict that role (Australian Accounting Standards Board, 2022e, paragraphs 32–38, 40(a) and C20).

Now reverse the ordering of the inputs. For group B, let FCF be 120 and fair value 100. The difference identifies a loss component of 20, while CSM is zero. The liability is 120. Paragraphs 49–50 describe the loss component as part of the liability for remaining coverage, with its own subsequent allocation and reversal rules. Adding another 20 to the FCF balance would count that component twice. Replacing the liability by fair value, 100, would instead understate the required FCF balance (Australian Accounting Standards Board, 2022e, paragraphs 40, 47, 49–50 and C20).

Write F for measured future-service FCF, V for the group's fair value, C for CSM and K for the nonnegative loss component. Positive amounts denote liabilities. The two sign cases give the following local derivation:

$$C = \max(V - F, 0), \qquad K = \max(F - V, 0), \qquad L = F + C = \max(F, V).$$

Here L is the recognised remaining-coverage liability. The loss component K is tracked within L and is not added to it. Thus $C - K = V - F$, and at $V = F$ both components are zero. These relations apply to the stated transition case; later changes in FCF, service provision and finance effects follow the subsequent-measurement rules.

The numerical comparison makes the distinction visible:

Currency units at transition	Group A	Group B
Fulfilment cash flows	90	120
Fair value of the group	110	100
CSM	20	0
Loss component within the liability	0	20
Recognised liability	110	120

The recognised liability equals fair value for A and exceeds it for B. Calling the method fair-value transition therefore does not mean that every resulting liability is carried at fair value.

To see the opening-equity effect, suppose the old carrying liabilities are 105 for A and 115 for B. Assets of 500 remain unchanged, and these are the only liabilities. Before transition, equity is $500 - 105 - 115 = 280$. Afterwards, it is $500 - 110 - 120 = 270$. Each group's liability increases by 5, so total opening equity falls by 10. Paragraph C5 replaces the retrospective measurement requirement in C4(a); the removal of incompatible old balances and the net adjustment to equity in C4(b)–(c) still apply. The ordinary initial-recognition loss rule does not turn this transition adjustment into current-period operating expense (Australian Accounting Standards Board, 2022e, paragraphs C4–C5).

Illustrative component entries show the same reconciliation. For A, debit the old liability 105 and opening equity 5, then credit the FCF liability 90 and the CSM liability 20. For B, debit the old liability 115 and opening equity 5, then credit the FCF liability 120. Record B's loss component of 20 within that 120 for its subsequent allocation; it produces no additional credit. Neither entry changes cash. An actual transition would also need the acquisition, reinsurance, tax and OCI effects excluded here.

The order of calculation matters as much as the signs. If an implementation first combined the two already distinct groups, it would find total FCF of 210 and total fair value of 210. Applying the sign rule to those totals would produce neither CSM nor a loss component and would report a liability of 210. The correct group-by-group calculation gives 230. Pooling would therefore understate liability and overstate equity by 20. The programme must retain the admitted group boundary while calculating CSM and loss components, then aggregate the resulting balances. Whether different grouping is itself permitted requires the separate grouping rules; an aggregation shortcut cannot decide that accounting question.

The example supplies the arithmetic once the group's two measurements and old balances are known. A real transition disclosure may reveal only some of those quantities. The later Munich Re case shows both what can be read from an insurer's report and what still requires contract and valuation records.

12.26 The recoverable part of a transition loss

Group B's loss component of 20 describes the issued contracts. If the insurer also holds reinsurance, how much of that loss is expected to be recovered? Extend the preceding example by adding a separately recognised, qualifying group of reinsurance contracts held. Assume it covers all contracts in B and only B, with remaining coverage and no incurred claims, and uses general measurement. This changes the earlier no-reinsurance assumption. The issued obligation to policyholders remains 120, with its loss component of 20; purchasing reinsurance does not extinguish it (Australian Accounting Standards Board, 2022e, paragraphs 60–64 and 75).

The transition calculation has a specific timing exception. Ordinarily, paragraph B119C permits the recovery-income treatment in paragraph 66A only if the reinsurance contract was entered into before or when the onerous underlying contracts were recognised. Suppose the underlying contracts' recognition date, applying these rules, is 1 July 2020, the treaty was entered into on 1 October 2020 and both coverages remain in force at transition on 1 January 2022. The treaty does not meet that ordinary condition for the original underlying recognition. Paragraph C20A nevertheless expressly removes B119C's condition when determining the loss-recovery component at transition. The exception does not remove the treaty's other recognition, grouping, boundary or measurement requirements (Australian Accounting Standards Board, 2022e, paragraphs 61–66B, B119C and C20A).

Let q be the percentage of claims on the underlying contracts that the insurer expects to recover from this reinsurance group, expressed as a fraction. Assume $q = 0.60$ for this conditional illustration. A nominal treaty quota alone does not establish that expectation: coverage, limits, exclusions and collectability matter. The full reinsurance measurement also includes issuer non-performance risk and the nonfinancial risk transferred, under paragraphs 63–64. With $K = 20$ still denoting B's loss component, C20A gives the reinsurance loss-recovery component R:

$$R = Kq = 20 \times 0.60 = 12.$$

This 12 belongs to the reinsurance asset for remaining coverage. It is not the issued liability, the amount of expected claims, or a valuation of the whole reinsurance asset. If only some contracts in the underlying group were covered, the required allocation of the losses would also have to be established; the all-covered assumption avoids that extra calculation here (Australian Accounting Standards Board, 2022e, paragraphs 63–64, 66B, B119E and C20A).

The complete asset still needs the treaty cash flows, discounting, transferred risk and net cost or gain of purchasing reinsurance. The reinsurance CSM reflects that net cost or gain; held reinsurance is not classified as onerous under paragraph 68. Applying the issued-group CSM and loss formulas to the held contract would therefore be wrong. Presentation likewise keeps the issued and held-reinsurance portfolios separate (Australian Accounting Standards Board, 2022e, paragraphs 65–68 and 78–79).

To extend the opening balance sheet, supply the missing total explicitly. Assume a separate complete measurement gives a reinsurance asset of 50, already including the loss-recovery component of 12, against an old asset of 45. These are additional conditional inputs; the multiplication above does not derive them. Other assets remain 500 and the two issued liabilities still increase from 220 to 230. Total assets therefore rise from $500 + 45 = 545$ to $500 + 50 = 550$, while equity moves from $545 - 220 = 325$ to $550 - 230 = 320$. The asset increase adds 5 to equity against the issued-liability increase of 10, giving a net reduction of 5. An illustrative reinsurance entry debits the new asset 50, credits the old asset 45 and credits opening equity 5. Adding another asset of 12 would count a component of the supplied 50 twice and overstate equity by 12. The net transition change belongs in equity under C4(b)–(c), rather than being inferred as current recovery income from the ordinary recognition rule (Australian Accounting Standards Board, 2022e, paragraphs C4–C5 and C20A).

The component also imposes a useful later check. If B's loss component falls to 14 and the recoverable portion remains 60%, paragraph B119F limits the reinsurance loss-recovery component to $14 \times 0.60 = 8.4$. Leaving it at 12 would violate that bound. The bound alone neither selects a unique new component nor determines the full reinsurance asset or period's profit; the actual changes and allocation rules still matter. Similarly, $q = 0$ gives no loss-recovery component and $q = 1$ gives 20, without erasing the issued obligation or determining the entire asset (Australian Accounting Standards Board, 2022e, paragraphs 66–67 and B119F).

12.27 What an insurer's transition disclosure reveals

Munich Re's 2023 report shows how much a public disclosure can resolve. Its consolidated statements apply IFRS as adopted by the European Union, together with the stated German commercial-law requirements. The first application of IFRS 17 is 1 January 2023, with transition at 1 January 2022 and restated insurance comparatives. Financial-instrument comparatives generally remain under IAS 39, subject to the disclosed classification overlay. A single year label therefore does not imply a single measurement basis for every line in the comparative balance sheet (Munich Re, 2024, pp. 155 and 183–184).[2]

The report identifies full retrospection for certain property-casualty contracts and parts of life reinsurance. Where historical information is insufficient in life reinsurance, it mostly uses fair-value transition; it also uses that method for specified property-casualty reinsurance groups and German primary business. Modified retrospection and fair value together account for over 90% of net insurance liabilities at transition. That is a statement about a liability denominator at a particular date. It supplies neither a count of contracts nor the assignment and supporting history of every group (Munich Re, 2024, p. 183).

The opening-equity bridge makes the scale of the different effects visible. The reported margin component is much larger than the net equity reduction because the other measurement changes also enter the reconciliation. The following amounts reproduce the report's signs and billion-euro scale (Munich Re, 2024, p. 184).

Opening-equity reconciliation at 1 January 2022	EUR billions
Equity under IFRS 4	30.9
Modified measurement basis	25.3
Contractual service margin	−22.3
Risk adjustment for non-financial risk	−5.6
Loss component	−1.4
Deferred taxes	1.5
Reported change in equity	−2.5
Equity under IFRS 17	28.5

The five displayed change components sum to −2.5 billion, whereas the displayed endpoints differ by −2.4 billion. The report warns that rounding can cause such deviations; these figures are compatible under rounding to the displayed precision. The representation should retain both the reported endpoints and the bridge, with their precision, instead of changing one to force an exact identity (Munich Re, 2024, pp. 1, 155 and 184).

[2]The retained English report states that it is an informational translation and the German original is binding (p. 1). The discussion attributes the disclosed policies and amounts; the historical AASB texts above are comparison sources.

The separate margin tables also clarify which contracts are being followed. At 1 January 2022 the reported issued-contract CSM is −22,688 million and the held-reinsurance net cost/gain is 368 million. Adding those reported totals gives −22,320 million, consistent with the rounded −22.3 billion margin component above. The issued-contract transition table follows groups in force at transition, and its CSM amounts concern the general measurement model and VFA; the associated revenue also includes PAA contracts. Its later balances therefore cannot be substituted for the whole subsequent portfolio or used as if CSM and revenue had identical populations (Munich Re, 2024, pp. 210–211 and 254–255).

Finally, disclosure of a method does not supply all its inputs. Munich Re reports a 6% cost-of-capital rate for its non-financial risk adjustment, applied to projected capital requirements with diversification and discounting. Its equivalent one-year confidence level is about 90% at transition, compared with 88% for 2023 and 87% for the comparative period. These are dated properties of the insurer's calculation, not universal IFRS parameters. The rate also cannot be imported into the separate fair-value calculation merely because that discussion mentions cost of capital. Reconstructing an individual group still requires its cash flows, capital projections, curves, allocations and supported transition-method choice (Munich Re, 2024, pp. 165, 183 and 185).

12.28 Interfaces, exclusions and acceptance cases

Insurance events are group_recognize, premium_settle, service_release, claim_incur, claim_settle, assumption_change, onerous_change and treaty_change, each bound through the event-basis, authorization and commit contracts to the exact group/version, evidence and information view. Journals and component roll-forwards must reconcile independently; a component reallocation is not always a net liability change. Invalid grouping, missing rate/law, unsupported option valuation, ambiguous contract boundary or unknown service units produces the corresponding unresolved/outside-scope result. Failed numerical paths retain their coverage/mass, without successful-only normalization. Modified contracts need the applicable derecognition analysis; no new ID can erase unfavorable old-group obligations.

Initial transition requires an actual method choice and eligibility for full retrospective, modified-retrospective or fair-value transition under AASB 17 Appendix C, with acquisition, grouping, past service and historical-rate records. This collection does not claim an executed transition or a general VFA valuation. The nonzero-rate general case and onerous allocation through expiry above are finite specification examples. Required later acceptance still includes a priced guarantee, acquisition/reinsurance interaction, transition and actuarial backtesting. These retained obligations prevent the two-year illustration from being described as full insurance-domain readiness.

Chapter 13

Implementation contracts

13.1 Reference implementation and component boundaries

The proposed reference implementation uses Python for orchestration and a deterministic reference evaluator, exact decimal/rational arithmetic for booked amounts, SQLite for transactional metadata and versioned records, and columnar files for large scenario outputs. DuckDB is an optional analytical reader. These are convenience choices justified by inspectability and modest initial scale; they do not establish production performance. The IR and conformance tests allow a later compiled executor without changing accounting meaning. Committed journals, economic deltas, receipts and branch heads belong to the one transactional store, including for simulation branches. Columnar files hold derived analytical exports, never independently writable committed books.

```
accdsl/
  evidence/       documents, spans, filing contexts, extraction versions
  model/          typed IR, entities, books, symbols, contracts
  language/       grammar, parser, resolver, type checker, lowering
  constraints/    propagation, solver adapters, bounds and conflict cores
  accounting/     rule selection, posting validation, closing, consolidation
  economics/      contract and behavioural plugin interfaces
  execution/      scheduler, branch state, stochastic streams, run records
  reporting/      statements, notes, movement schedules, explanations
  storage/        versioned records, migrations, result readers
  interfaces/     Python API, CLI, structured import/export
  packages/       reviewed demonstration and accounting modules
```

The first service boundary is local and synchronous. A job abstraction can be added for extraction and simulation without requiring distributed execution. The model compiler returns an immutable execution plan; the executor receives that plan and run inputs. Neither should retrieve live market data implicitly. Input datasets and rule packages are selected and versioned before execution.

13.2 Persistent records

The table identifies logical storage relations and their obligations; exact versioned record schemas determine their payloads. Candidate calculations and diagnostics remain separate from the committed accounting relations.

Relation	Key information and constraints
`document`	Hash, original path/URI, filing ID, entity, publication and availability times, media type, revision links. Original bytes are retained.
`source_span`	Document hash, page/element locator, bounding box, text/table representation, extraction run ID.
`observation`	Context ID, value or missingness tag, precision domain, source-span IDs, semantic-reading ID, review status.
`context`	Entity/perimeter, concept, basis, currency/unit, instant or interval, dimensions and taxonomy version. Contexts are immutable.
`assertion_version`	Logical assertion ID, payload hash, economic validity, ingestion time, public availability, superseded version and reason.
`model_node`	IR node ID, type, serialised payload, dependency IDs, package/model version; no executable arbitrary text.
`constraint`	Expression, scope, source/assumption references, supported solver fragment, applicable alternative condition.
`alternative`	Parent model, added conditions, rationale, evidence, coverage declaration and rejection history.
`economic_event`	Stable entity/book/branch/business-event key; versioned event, contract/cohort, dates, quantities and dependencies.
`branch`	Branch key and immutable definition reference, including entity, book and required synthetic or real book mode.
`branch_head`	Branch owner, current state reference and generation; updated only by the expected-head transaction.
`posting_authorization`	Exact basis, posting profile, method, authority, findings, permission and historical view. A permission record alone does not establish a committed event.
`journal_batch`	Committed balanced postings with event key, book/basis, date, method and measurement references. Candidate journals are separate; membership here requires the complete accounting transaction.
`economic_delta`	Committed contractual and physical state change, bound to the same basis and authorization as its journal.
`committed_receipt`	Unique stable event key, full request fingerprint, original request, opening/closing state, journal and delta references. Published atomically with those records and the advanced head.
`review_view`	Immutable parent view, decisions and dependency selection revisions; preserves historical interpretation.
`dependency`	Typed source and derived-record edges, including candidate-set membership needed for transitive invalidation.
`search_checkpoint`	Semantic manifest, partition, explored/unresolved regions, frontier, used budget and result references, committed together.

Relation	Key information and constraints
`run`	Evidence and model hashes, cutoff, scenarios, branch selection, weight model, solver/numerical settings, seeds and environment.
`report_cell`	Run, branch/scenario, context, amount or result family, dependency references and completeness status.

Persistent identifiers are opaque; display labels can change without changing identity. References use foreign keys or validated content-addressed links. The full logical key of a reported fact includes its dimensions, not merely company, label and year. Duplicate assertions are stored as distinct source claims until a documented duplicate-resolution operation relates them.

Version records are append-only. A derived transaction-time interval can be closed by the next version event without deleting the original payload. Validity intervals can overlap when assertions conflict. Such overlap is diagnosed and resolved by the evidence view, not prohibited by a database constraint that would erase one assertion. Scenario states use structural sharing of immutable historical nodes and branch-local changes.

Large simulation output is indexed by run, explanation, parameter draw, scenario, innovation path, entity, book, date and account. Reducing this to means across all dimensions would destroy the product's central feature. Online summaries may compress draws within a declared distribution, while retaining conditional summaries and representative replay identifiers. A retention policy that removes individual traces must state which explanations can still be reconstructed.

13.3 Public API and result types

The following unversioned names describe the user workflow. They are conceptual shorthand, not the portable operation registry. The exact versioned requests are specified below; Python, command-line and any service binding must exchange the same typed values:

```
ingest(manifest, extractor_config) -> EvidenceVersion
compile(source_or_ir, evidence_view, package_lock) -> ModelPlan | Diagnostics
reconcile(plan, cutoff, solver_config) -> HistoryFamily | Diagnostics
condition(family, assumptions) -> ConditionalFamily | Infeasible
simulate(family, scenario_set, run_config) -> RunResult
render(run_id, report_request) -> ReportBundle
explain(node_id, evidence_view_or_run_id) -> ExplanationGraph
compare(run_a, run_b, alignment_spec) -> Comparison
```

`HistoryFamily` contains a model-scope declaration, shared variables, constraints, finite choices, feasibility status and solver-completeness status. It may contain witness histories without claiming that those witnesses enumerate the family. `RunResult` is a tagged union of concrete outputs, conditional samples, certified bounds, explored ranges and unresolved outputs. Probability-bearing results reference a probability-model record. Results without that reference cannot be presented as probabilities.

Comparison requires an alignment specification: entity/perimeter, dates, reporting basis, scenarios and shared innovations where applicable. Comparing IFRS and US GAAP books holds economic events fixed and changes accounting rules. Comparing management responses holds the specified scenario and initial information fixed and changes the behavioural model. These are different questions and must not be combined into an unexplained difference.

13.4 Versioned requests and independent results

The workflow names above become portable operations only when their arguments, input versions and effects are fixed. Each request names an exact operation, profile and signature; an immutable view; typed arguments; execution settings; and the expected head for a mutation. The body is closed:

```
OperationRequest {
  operation, request_id, workspace_key,
  profile_ref, signature_ref, view_ref, arguments,
  execution: {mode, budget, environment_ref},
  expected_head: null | {owner_key, state_ref, generation}
}
```

All fields are present. The mode distinguishes a specification example from a runtime request. Budget states operation-count and elapsed-time limits, or a reason either is inapplicable. The expected head appears only in this envelope. For accounting its owner is the entity/book/branch; for review it is the review namespace; for probability selection it is the family/evidence/time owner. Resolving a mutable "current" alias happens before constructing the request, and a mutation compares the resolved revision again.

The fingerprint covers the complete request, including its request identifier, expected generation and execution settings. In accounting the lookup namespace is still the business-event tuple in Section 8.5, not request ID. Changing the request ID for a committed event changes the fingerprint and conflicts. Source and review mutations use their own operation/workspace/request namespace. A transport correlation identifier and an economic event are not interchangeable identities.

The result preserves the submission even when decoding fails. Once exact-byte retention succeeds, it has the following fields:

```
OperationResult {
  submission_ref, request_ref, operation, input_view_ref,
  disposition, statuses, payload, diagnostics, coverage,
  unresolved, continuation_ref, commit_observation,
  receipt_ref, evidence_mode
}
```

The validated request reference remains null until all required input checks, including bindings and authorization, pass. Rejected input has null payload. If submission retention itself fails, the caller receives a transport failure with its original correlation, not a fabricated stored submission or a known commit outcome. A read can inspect an earlier committed receipt while its own commit observation says no new commit occurred.

Completed, partial, rejected, unsupported, cancelled, internal failure, unknown commit and replay are operation dispositions. They do not replace the separate structural, mathematical, computational, empirical, accounting, forecast, product and human statuses. Computing a zero normalizer can complete while conditioning on that event is undefined. A valid symbolic family can have only partly explored regions. A supported insolvency scenario can be fully evaluated while its company defaults. These distinctions prevent an infrastructure error from erasing an economically important path.

For example, let an admitted law assign mass 3/5 to a path with cash 120 and 2/5 to an unavailable cash calculation. The output retains both regions and their masses. If separate evidence bounds the unavailable cash by $[80, 110]$, the unconditional mean is bounded by

$$\tfrac{3}{5}(120) + \tfrac{2}{5}[80, 110] = [104, 116].$$

Without those bounds, even this mean interval is unsupported. Reporting a mean of 120 would silently condition on computational success.

Portable-only validation proceeds through decoding, field shape, reference integrity, cross-record binding and time context, capability and admission, calculation, transaction comparison and publication. Native interfaces preserve their own explicit stage order; the native accounting sequence checks recognizable dependencies before shape and source versions before bindings. At the earliest failed stage, return every independently decidable diagnostic in canonical field order. A malformed parent prevents its dependent checks, but does not hide unrelated sibling errors. Later stages are marked unevaluated.

Appendix A gives the exact operation names and argument meanings. Financial method names such as delivery or claim settlement are versioned members of a package; they are not invented new platform endpoints. In particular, continuous inference and causal methods need their own admitted extension signatures before the finite/exogenous initial profile can execute them.

13.5 Correction changes a view without rewriting its history

A correction can affect a result that never explicitly selected the corrected source. Suppose a query selected one available report because it was the only eligible candidate. Admitting a second report may create a conflict. The query therefore depends on candidate-set membership as well as on its selected record. Invalidation must follow both dependencies transitively.

Each correction creates an immutable new view with parent and changed decisions. Old frozen outputs retain their original views. A current query must check ancestor dependencies even for descendants created after an invalidation scan; otherwise delayed computation could revive a stale reading. Unrelated closures remain usable. Self-supersession, incompatible replacement scope and a missing required predecessor reject before mutation.

The mathematical correction and the audit of affected work are separate records. A `CorrectionBasis` depends on the unaffected base law, replaced reading, corrected source, target admission and replacement method. An `InvalidationReport` additionally depends on the inspected store snapshot and lists affected descendants. Computing another optional descendant changes the report, not the corrected law. This also keeps the content-reference graph acyclic.

Probability calculation, admission and selection are three effects. A selected law belongs to the owner consisting of family, target evidence view, target time and purpose. Correcting a source does not mutate the old owner's selected pointer or automatically select the new candidate posterior. In the new owner, an inherited stale selection returns selection superseded with an explicit readmission and reselection obligation. With no inherited selection it returns selection required. A historical query naming the old view remains valid in that frozen scope.

13.6 Durability and migration have observable obligations

One logical transactional store owns records, dependency edges, review views, branch heads, journals, economic deltas, receipts and search checkpoints. Original documents reside in immutable byte storage. Their exact bytes must be durable and verified before a metadata transaction refers to them. A crash between blob creation and metadata commit may leave an unreferenced blob; it must not leave an accepted reference to absent bytes. Analytical tables, caches and PDFs derive from a named snapshot and do not become separate writable versions of truth.

The proposed local store serializes commits even when several clients submit requests. A single

writer still needs expected-head comparisons. An event-log implementation is also possible if its commit marker, projection visibility and replay obey the same observable rules. Neither a document trace nor a choice of SQLite establishes physical-media durability or multiwriter throughput. Those require crash, reopen and fault-injection evidence in the selected environment.

A search checkpoint commits the semantic manifest, partition identity, explored and unresolved regions, frontier, used budget and result references together. A cell is not complete before its result and frontier transition are durable. Resume names that exact checkpoint and manifest. Increasing a scheduling budget can continue it; changing evidence, methods, scenario, family or traversal meaning requires a new manifest and an explicit reuse relation. Cancellation preserves the last committed frontier.

Export first stages a closed manifest, verifies every required record and blob, then publishes the manifest atomically. An incomplete export remains explicitly incomplete. Import verifies required bytes before committing metadata and grants no new financial admission. A failed report export cannot undo a journal or justify posting it again.

A representation-only migration retains old and new bytes and IDs plus a field/meaning mapping. Its input names the old and new schemas, migration method, source snapshot and declared semantic relation; the common request envelope supplies the expected head for its transaction. It must preserve typed quantities, context, uncertainty category, reference closure, admission, event identity and supported reference query results. Changing a rounding policy, calendar, recognition rule or probability law is a new model, not a format migration. A failed comparison leaves old heads and objects usable.

Exact reruns compare values on the same evaluated domain. An elapsed-time cap can visit different search cells on different machines, so equal partial coverage cannot be presumed. Reproducible prefix comparisons need a pinned traversal and operation-count budget; resume uses the saved frontier. Approximate methods use their declared financial error comparison, and random streams name economic owner, date, shock kind and draw index independently of branch order.

13.7 Plugin interfaces

Every plugin declares a version, supported types, input information dates, determinism, numerical measure and capabilities. A contract plugin implements `schedule`, `observe`, `transition` and `payoff`; it may return future conditional events. A behaviour plugin implements `decide` and declares which state and scenario information it uses. A measurement plugin implements `measure` and returns an amount or vector with a basis and numerical status. A reporting plugin maps classified journals and balances to output contexts.

The plugin response must include all inputs that materially affect results. An external calculator's global evaluation date, market-curve cache, calendar version or random seed cannot remain invisible. For example, QuantLib's floating coupon interface makes fixing-date conventions and pricing objects relevant inputs; the adapter records them rather than relying on process-global state. ACTUS risk-factor observations receive explicit dates and identifiers.

The constraint interface accepts a typed formula and a requested operation: feasibility, witness, enumeration of declared finite choices, or bound. It returns `sat`, `unsat` or `unknown`, plus evidence appropriate to the operation. Timeouts and unsupported theories return `unknown`. The history layer exposes that solver outcome as `SolverIncomplete`, retains the affected region and marks its coverage unresolved. These are layer-specific descriptions of incomplete computation, not infeasibility. A witness proves existence within the formula, not completeness of a recovered historical family.

13.8 Human review and model-assisted extraction

The review interface should put the source passage beside the proposed observation, context and reconciliation impact. An analyst can accept a mapping, add an alternative, reject an extraction, define a symbolic unknown or request more evidence. These actions create structured changes with reasons. The useful unit of review is a coherent disclosure and its links, not an isolated number detached from its note.

A language model can identify candidate tables, draft concept mappings, propose contract terms and suggest alternative readings. Its output must reference source spans and a schema. Deterministic checks reject fabricated source references, invalid units, inconsistent periods and unbalanced postings. A model-generated interpretation that passes those checks remains a proposal until its evidentiary status is established. Token probabilities and self-reported confidence are never automatically used as weights over economic histories.

For routine local research use, a lightweight interface and immutable source versions are sufficient. Multi-user approval hierarchies, cryptographic attestations and deployment security systems are outside this initial design. They can be added for a concrete operational requirement without changing the scientific representation.

13.9 Failure recovery and reproducibility

Extraction jobs and simulation jobs write explicit progress records and atomic results. Restarting a failed document conversion reuses its document hash and creates either the same idempotent extraction result or a new version under changed settings. Restarting a simulation resumes only from a committed state whose model and input hashes match the run. Numerical failures retain the inputs needed for a focused diagnostic.

A run manifest records code revision when available, package lock, source hashes, cutoff rule, model hash, scenario data, priors or weights, seed/stream scheme, solver settings, decimal and rounding conventions, calendars, environment and result locations. Absence of a usable Git revision is recorded as unavailable with content hashes, rather than inventing a commit.

13.10 Review decisions have consequences for later results

A proposed reading starts in `Proposed`. Acceptance records a reviewer, date, reason, verified locator and context, together with the reconciliation result. Acceptance means that the stated layer has been reviewed: accepting an extraction of a company's sentence does not accept its accounting treatment. Competing defensible readings enter `AcceptedAlternative` under a choice group. Rejection preserves the source and reason. A correction creates a new reading; replacing an accepted reading requires an explicit `Supersede` action. A prior reading therefore remains reproducible without entering the latest model.

Suppose OCR reads Northstar's original 120 as 720. The source span supports 120; correcting extraction creates a new reading linked to the erroneous one. The later revenue correction from 120 to 110 is different: it changes the company's reported assertion. The original extraction remains faithful to its document. Finally, a reviewer may accept that two notes conflict without accepting either as a unique model constraint. The view then contains a conflict set requiring alternatives or further evidence.

A changed reading or mapping invalidates descendant constraints, families, measurements, reports and cached runs for a new view. Unaffected dependencies can be reused. Old frozen

runs retain their original decisions and evidence IDs; they are marked stale for current use, not overwritten. A journal derived under the old view is never silently attached to a new source. The local workflow needs no multi-user approval hierarchy to enforce these rules.

13.11 Identity must survive harmless reordering

The portable serialisation in Section 7.3 preserves decoded Unicode code points without global NFC normalisation. It uses sorted ASCII keys and exact rational financial quantities, while integer metadata remains integer. Only schema-declared sets are sorted. Source identity hashes original bytes. The old Core 0 encoding survives only for its historical fixtures through an explicit legacy import; it does not define new portable digests. Neither encoding proves arbitrary economic equivalence.

SHA-256 with a domain/version prefix identifies three different closures. Evidence identity includes source bytes, locations, extraction and review versions. Model identity includes typed IR, assumptions, rules, calculator locks and resolution, while excluding formatting and source byte offsets. Run identity adds the evidence view, horizon, scenarios, outputs, calendars, numerical policy, random-stream scheme and environment. A corrected source can change evidence and run identity without changing the numerical model; a new rate assumption changes the model as well.

Each economic business event has a stable identity within its entity, book and branch. Its lookup key is the four-tuple in Section 8.5; event version, payload, expected head and request settings belong to the full fingerprint. Exact replay returns the original receipt before a new-event freshness check. Changed inputs under that key conflict. A new run label or version cannot make the event post again. Independent events may reorder only when read/write sets are disjoint or a package proves commutation, including funding constraints. Alphabetical ordering cannot settle an economic timing ambiguity.

Stochastic keys contain seed, coupling group, entity, contract/cohort, economic date, shock kind and draw index. Branch enumeration order is absent. Shared coupling groups may use the same uniform draw with different marginal transforms. The reference fixture maps the first 64 bits of a SHA-256 key digest, interpreted as unsigned big-endian integer v, to $(v + 1/2)/2^{64}$. This fixes example replay; it does not validate a production Monte Carlo generator. Research runs select an assessed counter-based generator and record transforms and numerical environment. Inner measurement draws have different shock domains from outer realised defaults.

Schema migration preserves old and new payloads and a field/meaning mapping. A representation-only migration must demonstrate equivalent supported semantics; a changed recognition rule creates a new model. Restart resumes only a committed state whose locks and identities match. SQLite owns source, review, accounting and run metadata transactions; content-addressed files hold immutable bytes and JSON, while DuckDB reads versioned analytical exports. Each mutable transition has one owner.

13.12 Adapters exchange typed records

The portable interfaces described in this chapter govern new exchanges. The original `spec/core-contract.md` and `spec/core-ir.schema.json` retain the historical Core 0 examples. A named decoder and comparison link those examples to the portable records; their older fields do not silently extend a new closed signature. Every method returns a success record or a diagnostic with code, source IDs, entity/period, dependency path and retry status. The principal sequence

is `ingest`, `review`, `compile`, `reconstruct`, `simulate`, `report`, with `bound`, `explain` and `replay` over frozen identities.

Arelle supplies candidate facts with QName, entity, context, period, dimensions, unit, accuracy, nil status and locator. Duplicates survive for review. Its archived `ModelInstanceObject` interface exposes these properties; validation success does not establish economic truth. Docling supplies page/table proposals, locations and conversion status. Missing headers and failed pages remain unresolved. Z3 receives named rational-linear assertions and finite choices; witnesses are checked against exact inputs and unknown results retain their reasons. The reference checks exercise Z3; the other selected adapters remain interface designs grounded in inspected official source snapshots.

ACTUS receives contract terms, calendars and dated risk factors and returns schedules, transitions and payoffs. AccDSL applies accounting separately. QuantLib receives an explicit evaluation date, instrument, curves, fixings, calendar/day-count and measure; it returns dated measurements with numerical status. No process-global valuation date may leak between branches. Beancount exports concrete journals for later differential checks; it does not own symbolic families. Import failures retain request ID and source hash so retries are idempotent. Licence/version evidence remains in the software ledger; integration acceptance requires pinned releases or source digests.

13.13 A successful extraction can still lose the fact

Consider an Inline XBRL amount whose literal is `1,234`, whose admitted numeric transform treats the comma as grouping, whose scale is -3, and whose context and unit identify a particular dollar observation. The resulting amount is $1234 \times 10^{-3} = 1.234$. Scaling that normalized value again gives 0.001234 and is wrong. An adapter must retain the literal, transform, scale, context, unit, nil status and original occurrence so that this error can be found.

The same principle applies to dates and dimensions. A lexical instant of 31 December remains distinct from the exclusive 1 January boundary used for an interval. Publication and ingestion are separate timestamps. Two equal numbers with different contexts remain separate occurrences. A nil fact is unknown, not zero. Dropping a typed dimension or context reference yields a preservation failure for that fact even if the resulting Markdown looks clear.

A PDF can preserve characters that the reader never sees. The retained HSBC 2023 annual report has a cover reading “Annual Report and Accounts 2023”, but both PyMuPDF and Poppler also return 2024 in the extracted cover text (HSBC Holdings plc, 2023, cover, physical PDF p. 1). Inspection of the drawing order explains the discrepancy: the earlier 2024 text is covered by an opaque dark rectangle, after which the 2023 text is painted. This is an observed difference between extracted characters and the visible title. It does not establish an accounting contradiction or explain why the file was constructed that way. Agreement between two text extractors would not resolve it.

The proposed exchange keeps the original PDF, the unchanged raw extraction, the page locator and a render from the same bytes. The annotation’s `literal` retains both years; separate `readings` record the raw extraction and the proposed visible title. A linked inspection record identifies the extractor versions and options, rendered image, relevant drawing order and reviewer. Thus a raw-text search still finds 2024, while the inspected cover supports a visible report-year reading of 2023. Rejecting 2024 as the visible report year must not delete its extracted occurrence. The current example has author inspection; independent adjudication remains pending.

If an earlier review had already accepted 2024 as the report year, the correction would require a new reading and an explicit `Supersede` decision. Results depending on that reading would need revalidation in the new evidence view; frozen earlier results would retain their original evidence. This follows the review rules already stated above. Even after the visible title is resolved, the cover year supplies no publication or availability timestamp for these particular bytes. An as-of query must therefore retain the unresolved availability condition. This single-page case motivates preservation and review behavior; it supplies no extraction-accuracy ranking or general visibility test for other pages.

The inspected Arelle interfaces support native XBRL contexts, dimensions, transformations and validation. Its internal object IDs do not replace stable source identity. The inspected Docling source offers useful layout and table proposals, but its examined XBRL graph path does not retain every required fact/context/unit/typed-dimension field. That path needs a preservation sidecar or must report unsupported conversion for affected records. These are source audit findings, not measured accuracy rankings of the tools.

A returned layout conversion can also conceal missing pages. At the pinned Docling implementation, the timeout branch assigns `PARTIAL_SUCCESS` when pages were expected, even if none finished. The public `convert_all` interface permits that status through `raises_on_error=True` if the remaining stages return normally. Thus absence of an exception cannot establish that a three-page request produced any extracted content. The assembly code can restore missing page numbers with empty page objects; counting those objects does not count successful processing (Docling contributors, 2026). These are source-derived failure cases, not observed failures on Apple's filing.

The proposed exchange therefore keeps the original page inventory, requested range, successfully processed pages, empty placeholders and structured errors separate. If only page two was requested and processed, that is success on the declared subset; pages one and three remain outside the conversion claim. If the three-page request times out before any page finishes, all three remain unprocessed despite the partial-success status. Retain the source and the unfinished pages for another attempt. A deadline is also distinct from stopped execution: the inspected shutdown code waits for worker threads for a bounded time and can report that a thread is still alive. Later execution must confirm termination within its isolated process before treating the attempt as stopped (Docling contributors, 2026).

A requested processing stage may also remain unperformed. In the same Docling source, automatic OCR selects among engines according to platform, installed packages, languages and runtime availability. If it selects no engine and has recorded no language-specific rejection, it logs a warning; the subsequent call returns the pages without OCR. Language-specific rejection can instead produce an exception. Thus `do_ocr=True` records a request, not evidence that text recognition occurred. The proposed adapter records the requested mode, selected engine and model identity, and processing outcome for each page. A page that requires OCR but did not receive it remains unprocessed, even if other stages produced layout regions (Docling contributors, 2026).

Identifying the engine still leaves its model to be fixed. The inspected default layout preset names Heron at revision `main`, while the TableFormer downloader names the model repository's `v2.3.0` tag. Neither name supplies the hashes of the weights and preprocessing files actually used. The proposed adapter binds those files, configurations and dependency versions before use. Disabling remote inference services also does not disable model acquisition: the layout helper downloads when no artifact directory is supplied. With a supplied directory it requires the model's subfolder, but finding that folder does not check its contents against the requested revision. Local file verification and later isolated execution must therefore establish the intended offline behavior; a source setting alone cannot do so (Docling contributors, 2026).

The text supplied to table recognition also needs an identity. The retained `docling-ibm-models` 4.0.2 consumer uses OTSL cell and span tokens, converts them to HTML, and merges text from separately supplied PDF cells (Docling IBM Models contributors, 2026). This is a later representation than the original paper's HTML sequence. The Docling report explicitly describes a subsequent token-language refinement (Auer et al., 2024, Section 3); neither that statement nor a model-card score identifies the later files with the paper's experimental checkpoints.

The proposed adapter therefore retains the source page and crop, the supplied text cells and their locations, the predicted table cells, and the assignments and structural changes made during matching. If a correction combines columns or adds a cell for unmatched text, a reviewer must be able to recover the earlier structure and the text that prompted the change. Text obtained from an OCR engine also retains that engine's identity and outcome. A resulting table remains a proposal until its cell content, context and accounting interpretation have been checked; a well-formed HTML table alone cannot settle those questions.

Token syntax and cell geometry require separate checks too. In the inspected author version of OTSL, Lysak et al. (2023, Sections 4.1–4.3) use five tokens: C starts a cell, L joins the cell to its left, U joins the cell above, X joins both, and NL ends a row. Neighbor restrictions and equal row lengths allow invalid tokens to be detected during generation. The authors explicitly distinguish a syntactically valid sequence from a correct prediction.

Even those local rules need a further condition to ensure rectangular merged cells. Consider two rows, `C L NL` followed by `U C NL`. The L has C to its left, the U has C above, and the first-row, first-column and equal-length rules are satisfied; no X occurs. Yet the first cell joins the top-left, top-right and bottom-left positions, leaving the bottom-right position to the second C. It is L-shaped. Representing it with both a two-row and a two-column HTML span would also occupy the second cell's position. This finite example shows why the six printed rules alone do not establish a nonoverlapping partition into rectangular cells. The proposed adapter checks that each merged cell occupies a complete rectangle, that cells do not overlap, and that every declared grid position belongs to a cell. Passing those checks still leaves correspondence with the source table to be established.

The later decoder needs its own verification. In the retained 4.0.2 source, a correction described as applying to the first row uses a row counter that is initialized to zero and never updated. It therefore also rewrites an affected token on later rows (Docling IBM Models contributors, 2026). This is a specific source-code defect relative to that stated scope; its frequency and effect on actual model predictions have not been measured. Neither the paper's syntax argument nor these partial corrections establish that a deployed converter enforces all the proposed checks.

The retained Apple annual filing makes the dependency problem concrete. Its issuer archive includes the instance, the issuer's schema and its calculation, definition, label and presentation linkbases. Those files also refer to US GAAP and SEC Reporting Taxonomy definitions, SEC schemas and XBRL definitions held elsewhere. The issuer archive alone therefore cannot supply every definition needed to interpret its tagged facts. For this particular archive, a static inspection now resolves the recorded imports, link references and typed-domain references to retained files, checks target namespaces and fragment identifiers, and checks that instance facts have declared concepts, contexts and units. This establishes a source dependency map. XML Schema and XBRL validation, the issuer's interpretation of each concept, and the completeness of the wider disclosure package remain separate questions.

An offline parser needs more than a map of intended filenames. At the inspected Arelle source version, the cache lookup can return a local pathname when the file is absent; subsequent loading records an `IOerror`. The proposed adapter must check each mapped file and its hash before loading, retain the missing dependency and affected records, and return an unresolved-dependency result. It must not infer successful retrieval from a nonempty pathname or silently

substitute a newer taxonomy. Original schemas remain unchanged; an external resolver or isolated cache associates their declared addresses with the retained bytes. Any built-in cache fallback must belong to the same checked source set (Arelle contributors, 2026c).

Nor is a successful API return necessarily a validation verdict. In the same source version, the controller's return value takes accumulated validation errors into account only when `validationExitCode` is enabled. The proposed adapter sets it explicitly, requests validation, disables automatic formula execution, and examines the returned status, diagnostic messages and model errors together. It keeps the model open while copying the necessary facts, contexts, units and original occurrence references, then closes the session. One session runs at a time in its owning thread within a process; the inspected API uses shared package and plugin state. These are source-grounded interface requirements for a later implementation, not results from an executed Arelle adapter (Arelle contributors, 2026c).

13.13.1 An inline value is already a transformed reading

Return to the source spelling `1,234`. At the inspected Arelle version, an inline fact's `value` property first obtains assembled text and applies the selected transformation. In the ordinary non-nil numeric path, it then converts that result to a decimal number, applies the declared scale and applies the sign. Both `textValue` and `stringValue` delegate to this transformed value. Reading any of those three properties and applying scale again would therefore change the reported amount (Arelle contributors, 2026b).

For a concrete constructed case, select `numdotdecimal` from the transformation registry dated 31 July 2011, scale -3 and a minus sign. The inspected function accepts `1,234` and removes the grouping comma, giving 1234. Scaling gives 1.234; applying the sign gives -1.234. If a consumer treats that returned value as the original literal and reapplies both scale and sign, it obtains $+0.001234$. The difference is a consumer error relative to the selected source path. The format's namespace and function name matter: the dispatcher selects a registry by namespace and can also invoke a custom transformation. The example establishes no equivalence among those choices (Arelle contributors, 2026b).

Nor does `rawValue` recover the original source spelling in every case. The inspected property assembles inner text, omits inline excluded content, can follow nonnumeric continuation links, and collapses XML whitespace when a format is present. An original spelling containing surrounding spaces and a line break can consequently yield the same `1,234` reading. Escaped content follows an additional markup-generation path with URI handling; one inspected branch can modify an object element's `codebase` attribute in the parsed tree. Original retained bytes and occurrence locators are therefore needed even when assembled text or a later tree serialization is available (Arelle contributors, 2026b).

Consider an unescaped nonnumeric fact containing text A, an excluded child containing X, and text B after that child, with a continuation containing C. Under the stated traversal conditions, its assembled text is ABC. The excluded X remains part of the source document, while B survives as the child's tail text. The helper follows continuation links already attached by the loader; it does not resolve the source's continuation identifier itself. If the link is absent, a nonempty AB can still be returned. The loader separately records missing-target, wrong-target and cycle diagnostics. Nonempty text alone cannot establish that the disclosure was completely assembled (Arelle contributors, 2026b).

The proposed adapter thus preserves the original bytes and nodes, the ordered text fragments and exclusions, the continuation identifiers and outcomes, and the assembled and transformed readings. It records transform identity, scale, sign and validation state so that each step can be reconstructed. Computing a value is distinct from validating it: the inspected property

initializes an unvalidated state on first evaluation, and failures can set an invalid value and raise an exception. Nil facts take a separate branch and remain distinct from numeric zero. These source-derived requirements and constructed examples do not replace schema checks, an executed adapter or review of an actual issuer's facts (Arelle contributors, 2026b).

13.13.2 Equal amounts still have separate source occurrences

Suppose an issuer repeats one tagged amount in a statement and a note. A query may need a single amount, but an auditor still needs both locations and the evidence that they agree. These are different requirements. The proposed exchange preserves each occurrence before constructing a view that groups facts for a particular question. The inspected Arelle methods make the reason for that separation precise: displaying a value, comparing two values, grouping facts with matching aspects and removing repeated facts perform different operations (Arelle contributors, 2026a).

A displayed value is particularly unsuitable as the retained amount. The `effectiveValue` property produces a view: it rounds and formats numbers, represents nil with a string, and can return an error description. Even a convenient numeric property needs scrutiny. The inspected `vEqValue` converts numeric text to a binary floating-point number. The distinct integers $2^{53} = 9{,}007{,}199{,}254{,}740{,}992$ and $2^{53} + 1$ then have the same floating-point representation. This constructed example does not describe an issuer error; it shows why that conversion cannot establish exact identity. The adapter therefore keeps the source literal, the typed value and the displayed reading as distinct fields, with the transformation between them recorded (Arelle contributors, 2026a).

The same caution applies to a method whose name contains *equal*. In the inspected `isVEqualTo`, the ordinary numeric comparison checks context and unit and follows the selected rounding mode, but the method does not itself compare concept names. Its nil branch returns whether the other fact is nil before comparing contexts. A caller needing facts about the same accounting quantity must establish that condition separately. Conversely, the item branch of `isDuplicateOf` checks concept, parent, context and numeric unit without comparing the item values. It can identify candidates for investigating a disagreement; its result does not resolve that disagreement. The tuple branch has additional child comparisons and must not be described as the same test (Arelle contributors, 2026a).

The dedicated duplicate-fact module supplies more explicit grouping. It checks concept, context and unit after forming hash buckets, and includes language for the relevant text-fact types. Its value tests nevertheless remain specific to the operation. Consider two hypothetical valid numeric occurrences with the same concept, context, unit and declared decimals of zero, whose value strings are `1` and `1.0`. In the inspected code, the ordinary value-equality key retains those strings, so the all-complete classification distinguishes them. The complete-subset reduction instead keys on declared decimals and the typed value; decimal one equals decimal one, so it retains the first occurrence. Thus even within this module, the classification and reduction do not share one universal equality key. The source strings and membership of a reduced view must remain recoverable (Arelle contributors, 2026a).

Reported accuracy creates a second distinction. For a finite declared decimal accuracy d, the inspected non-truncating interval helper assigns a numeric value v the inclusive bounds $v - \frac{1}{2}10^{-d}$ and $v + \frac{1}{2}10^{-d}$. These are the helper's comparison bounds, not a confidence interval for the true economic amount. Values zero and one, both declared to zero decimals, have bounds $[-0.5, 0.5]$ and $[0.5, 1.5]$. They touch. The interval option of `isVEqualTo` therefore admits overlap under its other required conditions. The duplicate-set consistency check rejects the pair because unequal typed values with the same declared decimal accuracy fail its additional condition. Substituting

one test for the other would change the question being answered (Arelle contributors, 2026a).

Pairwise agreement also differs from agreement of the whole set. Take three constructed observations of the same quantity: 100 at zero decimals, and 99.6 and 100.4 at one decimal. The first has bounds $[99.5, 100.5]$; the other two have bounds $[99.55, 99.65]$ and $[100.35, 100.45]$. Each precise observation overlaps the coarse one, but the precise observations disagree and the three intervals have no common point. The inspected consistent-pairs reduction can remove the coarse occurrence while leaving both precise occurrences. The consistent-set path instead reports the inconsistent set and falls back to complete-subset reduction. Neither operation establishes which precise observation the accounting analysis should accept (Arelle contributors, 2026a).

These paths require their own admission checks. The initial grouping considers validation state and the availability of a concept, type and context; a singleton bucket is not itself a validity verdict. The numeric consistency path also expects an inferred decimal accuracy from its helper, which returns no value when the fact's decimals attribute is absent. An adapter must not silently reinterpret an unsupported or unvalidated case as an ordinary consistent numeric fact. It retains the declared decimals and precision, validation state, nil state, concept, full context and unit, together with the particular comparison operation and its options (Arelle contributors, 2026a).

Finally, the inspected deduplicated-instance export removes fact nodes from the model's underlying tree before saving. The controller performs it at the end and warns that subsequent use of the modified model can be unreliable. The proposed exchange therefore copies every original occurrence and its diagnostics before constructing any optional reduced view. A view records its input members and selection rule; it does not replace the source record. This preserves disagreement for review while allowing a query to avoid counting an accepted repeated amount twice. These are requirements derived from pinned source inspection and independently checked constructed arithmetic; an executed adapter and actual-filing validation remain to be established (Arelle contributors, 2026a).

13.13.3 The same balance can survive different source identities

Apple's following-quarter filing provides an actual counterpart to the constructed examples. Its balance sheet repeats the 28 September 2024 comparative balances already examined in the company case. The retained annual HTML is a converted presentation without inline tags, so it cannot supply the proposed inline trace. Instead, the selected comparison follows the next-quarter inline comparative amounts to the retained annual XBRL instance. This compares reports of the same endpoint across two filings; it does not show that one file was extracted from the other (Apple Inc., 2025c, 2024c,b).

In the interim filing, the comparative total-assets occurrence has identifier `f-135`, value text `364,980`, scale 6 and declared decimals -6. Its format names `num-dot-decimal` in the 12 February 2020 transformation registry. The selected transform removes the grouping comma and canonicalizes the numeric string; scaling gives $364{,}980 \times 10^6 = 364{,}980{,}000{,}000$ US dollars. That reported amount agrees with annual-instance occurrence `f-169`. Scale converts the displayed millions to dollars. Declared decimals retain the source's million-dollar reporting accuracy; adding zeros to the string does not reveal an exact unrounded asset value (Apple Inc., 2025c, 2024c; Arelle contributors, 2026b).

The context identifiers differ: the interim uses `c-18`, while the annual instance uses `c-21`. Comparing the identifiers would reject a valid candidate match. Inspecting their contents instead shows the same CIK entity identifier and scheme, the same 28 September 2024 instant and no segment, scenario or dimensional additions in these selected contexts. The expanded

concept name, USD unit and declared accuracy also agree. This is a checked comparison of these simple contexts, not a general context-equivalence algorithm or schema-validation result (Apple Inc., 2025c, 2024c).

The liabilities and equity comparisons use the same conditions and give 308,030 and 56,950 million dollars. Their reported totals reconcile: $308{,}030 + 56{,}950 = 364{,}980$. Equity, however, appears twice in the interim: `f-173` is the comparative balance-sheet amount and `f-176` is the following period's beginning equity. Both carry 56,950 million dollars under the same selected concept, context and unit. The annual instance also retains two matching occurrences, `f-207` and `f-256`. Amount and context matching therefore identifies a group, not a unique pairing of source locations. Adding the two interim occurrences would produce 113,900 million dollars and would double the reported equity balance. A balance query uses one accepted amount while preserving both locations and their statement roles (Apple Inc., 2025c, printed pp. 3–4); the annual counterparts are retained in the instance (Apple Inc., 2024c).

The accumulated other comprehensive loss exposes a different failure. Interim occurrence `f-171` contains numeric text `7,172`, with scale 6 and an explicit minus-sign attribute. Its reported value is consequently $-7{,}172{,}000{,}000$ dollars, matching annual occurrence `f-205`. Reading the digits alone loses the sign; reading the surrounding parentheses and then applying the sign attribute again would reverse it twice. The source text, markup attribute and displayed accounting reading must remain distinguishable (Apple Inc., 2025c, printed p. 3); the signed annual value is retained in the instance (Apple Inc., 2024c).

The bounded inventory retains all 24 interim and 32 annual-instance occurrences of the four selected concepts, then selects five occurrences from each source at the unsegmented September endpoint. Other dates and dimensional populations remain visible as outside this comparison. All selected reported amounts agree under the stated structural and arithmetic checks. The agreement supplies an actual preservation example; it establishes neither historical availability of the retrieved bytes nor schema validity, complete filing coverage or an executed adapter. Those distinct questions remain part of the wider acceptance work (Apple Inc., 2025c, 2024c).

Each adapter therefore pins code, dependency, model and taxonomy versions, licenses, input and output schemas, transformations and field-loss behavior. Preparation preserves bytes and receipts before parsing, inventories the package, attaches occurrences and contexts, records normalization, checks required fields independently and submits candidates for review. Partial conversion retains both successful units and the exact failures or unexplored frontier. Conversion counts cannot accept a package or an accounting treatment.

The proposed extraction comparison uses the same retained files, roots, budgets and taxonomy resources. Manual located source units and transparent native extraction provide the reference; applicable pdfplumber/Arelle configurations, plain Docling and explicit context/footnote repairs form the candidate ladder. Transcription, context, table structure, mapping, source support, retention, historical admission and cost are separate outcomes. Lost hashes or locators, look-ahead, silent omissions and untracked downloads veto the affected exchange. The current 31 development annotations cannot establish a heldout accuracy ranking, and cell counts within one report are not independent observations.

Reuse also has distribution conditions. Arelle's Apache and Docling's MIT code licenses do not by themselves clear every model, taxonomy or dataset. ACTUS technical-specification reuse retains its share-alike obligations; QuantLib, Z3 and ledger tools retain their own code and dependency terms. The Beancount and hledger comparison needs an explicit distribution boundary for copyleft components. License admission records actual versions, notices, models and data, rather than treating a repository's headline license as universal.

EasyOCR makes the dependency question concrete. Version 1.7.2 requires `python-bidi` without a version constraint and imports its `get_display` function when the OCR module loads. Although

the call reorders Arabic recognition results, selecting English does not remove the import or package requirement. In the inspected `python-bidi` 0.6.11 source, this import selects a Rust-backed implementation; the older Python implementation has a different import path. Its own source states LGPLv3-or-later terms. The inspected release is a candidate for dependency analysis, and its source-level call signature alone does not establish compatibility with an installed OCR configuration (Jaided AI, 2024; Kriheli and contributors, 2026).

The same package shows why an accompanying notice still needs checking. Its third-party list names older PyO3 and `unicode-bidi` versions than its bundled Cargo lockfile. The exact locked sources retain MIT-or-Apache terms, so the version discrepancy does not demonstrate a licence conflict. Other locked packages add conditions: `target-lexicon` includes the LLVM exception to Apache 2.0, while generated `unicode-ident` tables require Unicode data-file terms alongside the chosen MIT or Apache licence (Kriheli and contributors, 2026; PyO3, Servo, Rust and dependency contributors, 2026). The adapter must therefore bind the selected build and retain each applicable notice. Verified source archives help prepare that decision; actual wheel contents, model rights and the intended distribution still need separate assessment.

The actual taxonomy notices show why that distinction changes the design. The inspected FASB notices permit specified financial reporting and unchanged incorporation in works that explain or assist implementation, subject to their notice and third-party conditions. They do not provide a general permission to edit taxonomy definitions. Keeping the original files and recording location mappings separately respects that distinction. The notices retrieved for this inspection are unversioned; their relationship to the terms attached to the 2024 files still needs to be established before a distribution decision. XBRL's public-document notice separately directs software and schemas to a Software Notice, so permission for an explanatory document cannot simply be applied to a schema bundle. Exact dependency licences, component-specific terms and the intended distribution remain explicit preparation tasks (Financial Accounting Foundation, 2026b,a; XBRL International, 2026).

13.13.4 Choosing the first import path

The first optional importer uses the retained Arelle source (Arelle contributors, 2026c) for native XBRL, with the Apple 2024 instance as its concrete preparation case. Native facts already carry contexts, units and occurrence relationships that a flattened table cannot reconstruct. The selection therefore follows the preservation requirements established above. It selects a source and an interface for the first implementation slice; the source commit is not an installed release or a tested environment. The complete commit and body hashes are retained in the source-selection manifest.

Use the previously inspected session lifecycle with no optional plugins or additional package groups: request validation and its error-sensitive return, disable formula execution, and copy facts and their contexts before closing the session. Each owning process admits one session at a time. For this particular Apple archive, the recorded map has 27 resources and 3,551 reference edges. Resolving those static references does not establish XBRL validation or closure of every financial note.

The deterministic PDF comparator is the retained pdfplumber source, used for character and page geometry on machine-generated documents. Its output remains a proposal to be checked against the located page. A page that needs unadmitted OCR remains unresolved, with its original bytes and manual inspection route preserved. Docling, TableFormer and OCR are separate candidate paths. They do not become an automatic fallback that silently changes the selected method, model terms or resource budget.

There are three distinct forms of reproducibility here. A source selection identifies the code

and technical evidence. A resolved environment identifies the actual platform, Python build, every direct and transitive package version and hash, enabled extras, model and taxonomy bytes, and applicable notices. An executed compatibility check establishes what those installed components actually do together. The present selection establishes the first. Arelle's 15 base requirement ranges and three build requirements remain an inventory; they are not an exact transitive lock. Before runtime, resolve that inventory for the selected platform and inspect the resulting compatibility and terms. Any changed dependency, transformation registry, taxonomy or configuration creates a new identity and invalidates dependent admission.

Offline access is exact URI-to-bytes resolution. An absent file, changed hash, wrong namespace or missing required fragment returns the affected dependency as unresolved before accepting its financial values. A cached pathname is insufficient. Original schemas remain unchanged, and the isolated cache or external resolver may provide only admitted bytes. Substituting a newer taxonomy would change the interpretation problem.

Terms also attach to different things: code, dependencies, issuer documents, taxonomies and model weights. The earlier source-specific dispositions remain binding evidence limits. In particular, an unversioned FASB page displaying 2027 does not establish the terms applicable to a retained 2024 package; the missing component-specific XBRL Software Notice remains a source dependency. Unresolved model-weight terms constrain the corresponding model path. They do not resolve or negate the separate native-XBRL source decision. No distribution or blanket legal clearance follows from this selection.

Eleven complete documentary exchanges test the proposed boundary without executing a converter. Every synthetic source body, context, unit and occurrence span is supplied. Raw "1,234" under an admitted grouping transform and scale -3 produces 1.234; precision metadata does not scale it again. A nil occurrence remains unknown. Equal values in different contexts stay separate. Unknown transforms, missing contexts or typed-dimension values, absent resources and changed hashes retain unresolved records instead of guessed facts.

A cancelled case retains its completed proposal and unprocessed frontier. Repeating an identical request preserves its proposal receipt; changing bytes under the same identity is a conflict. All results remain proposed or unresolved, and none claims package completeness, review acceptance or an accounting commit. These are authored expected exchanges with independently checked transformations. Their closed documentary envelope is a verification format for the proposal; it does not alter the portable product schemas or establish runtime translation into them. That translation and the exact environment remain prerequisites of the later implementation.

Chapter 14

Acceptance and implementation sequence

The reference cases now make several of the acceptance obligations executable as design checks. Northstar provides full statements and a known compatible history; Apple supplies real source contexts and a bounded marginal stress; bank, insurance, consolidation and inventory fixtures test the core's broader state requirements. They do not constitute a production implementation.

14.1 Three different questions require different evidence

Engineering correctness asks whether the implementation follows the specified types, transitions and arithmetic. Accounting validity asks whether the selected rules faithfully implement the applicable standards and disclosed policies. Scientific validity asks whether the economic assumptions and uncertainty representation support the intended forecasts or comparisons. A balanced journal answers only part of the first question.

The implementation must therefore maintain separate validation results. Compilation and deterministic examples cannot establish forecasting quality. Agreement with a public report cannot identify an undisclosed contract term. A lower extraction error on a document benchmark cannot establish that the engine preserved every materially different scenario response.

14.2 Core conformance cases

Test	Required case	Pass condition
T01	Exact debit/credit batches, including reversals and period closing	Every committed book conserves signed balances; imbalance rejected atomically.
T02	Money, quantities, currencies, instants and interval flows	Invalid additions and context combinations rejected; explicit conversions traced.
T03	Original filing, later correction and as-known cutoff	Earlier query reproduces the original evidence view; later view shows revision.

Test	Required case	Pass condition
T04	Annual and interim cumulative flows with revised comparatives	Quarter derivation only across compatible contexts and vintages.
T05	Fixed/floating borrowing with identical historical postings	Both explanations survive; next-year expense is 5 and 7 under the stated shock.
T06	Continuous fixed share $\alpha \in [0, 1]$	Exact model returns $I = 7 - 2\alpha$ and extrema 5, 7 without assigning probabilities.
T07	Duplicated encoding of one history	Canonical duplicates preserve results and declared weights; no extra probability mass from an additional encoding.
T08	Linked interpretation across debt classification and payment treatment	Impossible combinations rejected through shared constraints.
T09	Rounded totals, duplicate facts and contradictory notes	Correct interval compatibility; no automatic zero or balancing plug.
T10	Infeasible or unsupported constraint problem	Conflicting sources exposed; timeout/unsupported case reported as unknown.
T11	Zero-coupon bond and ordinary equity examples	Carrying amounts, payments, share quantities and report mappings match derivations.
T12	Foreign-currency asset, translation and cash movement	No mixed-unit balance test; translation effects separated from cash flows.
T13	Acquisition and intragroup balance elimination	Perimeter and adjustment book reproduce group totals without inventing capex.
T14	Conflicting recognition rules and unknown guards	Explicit conflict or symbolic partition; file order has no effect.
T15	Insufficient cash with constrained funding	Funding failure/default retained as an outcome; no hidden borrowing.
T16	Expected-loss measurement before future default is realised	Measurement uses only information allowed at that date.
T17	Repeated/reordered independent events and interrupted execution	Idempotent replay; branch isolation; identical committed outputs.

Test	Required case	Pass condition
T18	Conditional distributions and optional weighted mixture	Branch distributions retained; mixture agrees with declared weights and total probability; unweighted family never labelled probabilistic.

For synthetic cases, expected histories, excluded histories and scenario consequences are known by construction. Tests should include counterexamples that balance but have the wrong economic effect, such as treating a non-cash acquisition movement as operating cash. Such tests discriminate between arithmetic validity and the intended semantics.

Differential tests against an existing ledger check only the shared supported subset. ACTUS schedule comparisons check contractual events, and QuantLib comparisons check selected valuation calculations under matched conventions. Disagreements must be classified as different conventions, unsupported cases or implementation errors before one tool is treated as an oracle.

14.3 Extraction and reconstruction evaluation

A company corpus must include different standards, industries, languages, filing formats, extensions, interim calendars and restatements. Company and time splits are required so repeated report templates or revised versions do not leak into evaluation. Public benchmark tables are useful development material, while acceptance uses complete disclosure packages that include notes and their relationships.

Reviewers annotate source facts, contexts, mappings, required relationships, known ambiguities and unsupported assertions. Relevant measures include correct source-linked values and contexts, unresolved extraction volume, coverage of note obligations, reconciliation failures and invented facts. Report each separately. A single overall score can hide a serious error in a small but consequential financing note.

Historical reconstruction should test whether the known synthetic history remains feasible, whether known incompatible histories are rejected, and whether economically distinct alternatives are preserved. On real companies, public data generally cannot establish the true journal. Review instead assesses evidential compatibility, unsupported commitments and conditional economic consequences. Later disclosures may provide discriminating evidence for previously ambiguous terms.

No universal numerical threshold for semantic coverage or extraction accuracy is fixed in this document. Before a release evaluation, an accounting-reviewed test corpus and task-specific acceptance thresholds must be recorded in an evidence contract. Hard conformance failures, fabricated citations, silent discarding of valid alternatives and undocumented accounting adjustments veto acceptance regardless of an aggregate score.

14.4 Evaluate the DSL against the strongest practical baseline

The comparison uses the same IR, accounting examples and tasks. One group of models is authored through the typed Python API; another through the DSL. Both receive equivalent documentation and examples. Tasks include encoding a movement schedule, adding an alternative

interpretation, changing an effective date and tracing a simulated amount. Measure correctness, material omissions, time to a reviewed result, ease of explaining a change and maintenance effort.

This comparison tests whether the notation and compiler help users; it is not a comparison between careful DSL code and unstructured spreadsheets. If the DSL offers little advantage, retain the shared IR and library and narrow the textual language. Small pilots nominate improvements. A claim of superiority requires a predeclared comparison and uncertainty evidence, not a few favourable demonstrations.

14.5 Forecasting evaluation comes later

The proposed experiment asks whether a conditional company method adds out-of-time predictive information while preserving statement coherence and unresolved exposure. The comparator ladder uses the same available sources and targets: last-observation and seasonal naive forecasts; contractual runoff and accounting-identity forecasts; tuned classical statistical and conventional financial models; the structural model with one selected interpretation; the same structural model preserving alternatives; and declared extensions. This separates changes in economics from changes in the representation of ambiguity. Tuning stays within training windows. Test issuers or time blocks remain untouched, and unavailable predictions stay in the reported denominator.

For probabilities, predeclare proper scores, calibration, interval coverage, sharpness and dependence-sensitive joint checks. Unweighted feasible sets have coverage and width under a stated set interpretation, not probability intervals. Promotion needs a declared score difference with uncertainty evidence on matched eligible targets and no accounting or information-time veto. Issuer and time dependence require a justified paired resampling or equivalent inference design; overlapping reports are not independent replications. Runtime, a few losses and small-seed tail differences remain descriptive without that support. No such forecast experiment has been run for this proposal.

Look-ahead, broken conservation, a changed target or missing scope can invalidate an experiment. A valid comparison in which the proposed predictor performs poorly instead rejects that candidate and motivates its stated repair. The two outcomes must not be conflated when choosing the next experiment.

Forecast evaluation uses historical information cutoffs, fixed data vintages, declared economic scenarios and proper separation of training and testing periods. Every level of the comparator ladder above uses that same information boundary, so extra information cannot silently account for an apparent benefit.

Construct the inexpensive comparators before choosing a complex forecasting method. A no-growth operating forecast with scheduled contractual financing tests whether the model adds value beyond existing obligations. A constant margin and turnover model tests whether detailed operations improve on familiar financial ratios. A trailing-average growth model tests whether fitted dynamics improve on recent experience. A fixed-exposure contractual stress model tests whether estimated management responses help beyond repricing the existing book. Each comparator must generate coherent accounts through the same reporting rules and use the same historical information. In synthetic cases, the known hidden state also supplies an oracle conditional on the stipulated scenario.

Evaluate these comparisons separately under stable conditions, large rate resets, refinancing pressure, credit deterioration and changes in reporting perimeter. A pooled average can conceal failure in the situation the product is intended to explain. Supported underperformance against

a cheap comparator in a declared critical situation vetoes promotion of the complex forecasting method and must lead the result discussion. These checks are falsification tests, not training targets or evidence of superiority when merely passed.

Evaluate point forecasts where they are a stated target, distributional calibration when probabilities are justified, and decision-relevant robustness when only feasible sets are available. A wide feasible set is not automatically a good prediction, and an unweighted set cannot be assessed as if it were a 95% predictive interval. In multi-scenario stress tests, scenario severity and probability remain separate inputs.

Any serious numerical comparison must follow the repository's research plan, assumption audit, baseline, veto, uncertainty and result-record policies. The present document runs only deterministic checks of its own examples. It neither selects a scientific forecasting default nor reports comparative performance.

14.6 Fixing the target before evaluating a method

The following protocols specify the comparisons needed before selecting a forecasting method. The first question is what a successful prediction would mean. A correctly parsed filing, a retained feasible history and a useful forecast are different results. Each receives its own target and comparison.

Information available at the forecast date

The nine-issuer, 2018–2025 panel is a purposive development resource. Its 192 accepted reports do not form 192 independent observations. All existing issuer roles remain provisional: subsequent source work on JPMorgan and insurers has also exposed proposed test families. The 31 pilot annotations are development answers. The 136 unverified publication dates remain unresolved. Even a verified filing date is insufficient when it does not establish availability of the exact retained version at the intended cutoff. No date is imputed from a filename.

An eligible information record identifies original bytes and full hash, issuer and consolidated perimeter, report period, original/revised status, actual public availability evidence, clock precision, timezone and ingestion time. All language versions, mirrors and amended reports belong to the same issuer/version family. An as-of view excludes any document whose latest possible availability exceeds the cutoff. If only a day is known, the origin is after that day in the declared local clock; an intraday claim is unavailable. Later comparatives cannot repair training data retrospectively without defining a different retrospective task.

Forecast origin and target period must be distinguished. A December year-end report first available in March cannot support an entirely future prediction of the following January–December total: part of that total has already accrued. Such a task is a combined nowcast and forecast. The all-future annual task uses the first complete fiscal year whose start is strictly after the origin. A shorter future interval is admissible only if its realized quantity can be measured independently. Neither subtracting incompatible cumulative vintages nor silently calling an annual period a twelve-month forecast repairs this issue.

E1. Disclosure extraction and preservation

The unit is a complete declared disclosure package, with a source inventory covering statements, note spans, policies and incorporated references. Labels preserve raw tokens, entity/perimeter, concept, dimensions, unit, scale, sign, precision, period, availability and admissible readings.

Independent manual location and transcription supply the reference; a parser's output cannot label its own test. Disputed readings remain adjudication cases rather than convenient single-valued answers. Internal preparation precedes any separately authorized professional question.

Compare manual located transcription, transparent PDF/native-XBRL extraction, tuned pdfplumber or Arelle on their supported formats, plain Docling and the same converter with explicit context/footnote repair. Equal source bytes, referenced taxonomies, allowed network resources, annotations and tuning budgets are mandatory; PDF and native-XBRL strata are reported separately. A converter unable to accept a format is unsupported in that stratum, not assigned a fabricated accuracy. TableFormer/OTSL scores nominate layout methods only; they provide no accounting labels.

Prespecify separate denominators for required source units, resolved units, exact context/value tuples, unresolved units and materially wrong accepted assertions. Missing package content stays in coverage. Any fabricated locator, wrong admitted entity/unit/time, silent source loss or untracked download vetoes that candidate in the reviewed conformance set. It does not prove a zero population error rate when none occurs. For accuracy comparisons, aggregate paired differences first by report, then issuer; state whether the estimand weights issuers equally or weights deployment volume. Nine selected issuers cannot justify population-wide accuracy or a reliable rare-error bound.

E2. Reconstruction, query and conditional accounting

Use the same admitted evidence, finite template registry and unknown domains for an enumerated exact reference, a transparent typed-library model, the plain proposed reconstruction and any pruning/abstraction extension. Small rational polyhedra and finite history trees provide exact known feasible and infeasible cases. Test both soundness and completeness within that declared finite scope: each returned history satisfies every constraint, each reference history remains represented, and each query interval agrees with independently derived bounds. The continuous floating-share example has interest in [5,7]; duplicate encodings must leave that interval and any separately supplied probability measure unchanged.

Include cross-statement dependence, open endpoints, contradictory evidence, unknown solver results, interrupted enumeration, unsupported nonlinear terms, correction/replay and failed funding. The oracle for an interrupted case is an explicit incomplete result, not an invented full answer. Check journal and statement invariants plus economic timing and conditional response. A timeout is a resource outcome; representing it as infeasible is a correctness failure. On actual companies, evaluate evidential compatibility and retained alternatives; the hidden true journal is generally unavailable. Exact small cases do not prove completeness of unrestricted histories or the correctness of an entire solver.

E3. Conditional financial prediction

Corporate prediction begins with operating cash flow as the primary reported quantity and closing cash and financial debt as a prespecified joint secondary vector. Use the issuer's actual presentation, perimeter and accounting basis; cross-basis harmonization requires its own admitted mapping. The target record pins whether the outcome is the first released amount or a later corrected amount. Primary evaluation uses first release; restated values are a separately labelled sensitivity with their own release dates. Each unknown target remains in the eligibility/failure report, with a reason fixed independently of forecasts.

Every comparator receives the same as-of information and target. The ladder is last value/seasonal naive where frequency permits, scheduled contractual runoff with no-growth accounts,

tuned constant-margin/turnover and conventional statistical models, the structural model with one fixed interpretation, the same model preserving alternatives, then a named extension. Classical tuning uses training/validation periods and an equal stated resource cap. Model complexity, information and ambiguity representation are separate ablations. Neither a poorly tuned comparator nor selecting the best baseline after test results gives a fair primary comparison.

For an admitted probability law with finite first moment, use the negatively oriented continuous ranked probability score (smaller is better). If X and X' are independent draws from the predictive law and y is the realized amount,

$$L(F, y) = \mathbb{E}|X - y| - \tfrac{1}{2}\mathbb{E}|X - X'|.$$

Gneiting and Raftery (2007, sec. 4.2, eqs. 20–21) supply this representation and its conditions; the paper uses the opposite orientation before explicitly introducing this loss orientation. A point forecast reduces to absolute error. For probabilities one half on 5 and 7 and outcome 5, the first expectation is 1 and the second is 1, giving loss 0.5. A point forecast 6 has loss 1 on this observation. This one outcome is a calculation check, not evidence of superiority. For outcome 6 their losses reverse to 0.5 and 0.

For scale comparisons, divide all amounts for an issuer-origin by the same strictly positive, as-of known total-assets amount, fixed before the outcome. Missing or nonpositive scale prevents that scaled comparison; it is never replaced by a test-outcome-dependent scale. Report unscaled amounts as well. The joint secondary loss replaces absolute differences by Euclidean norms after fixed component scaling: the energy score with exponent 1 (section 4.3, equation 22). It requires a joint law and finite first moments; the vector of independently optimized marginal extrema is not a joint forecast. The source's kernel argument in section 5.1 supports propriety under its stated hypotheses; it supplies no assurance that this finite corporate panel can detect every dependence error. Prespecified cash shortfall and linked-statement event checks therefore accompany it.

For unweighted admissible sets, report inclusion of the realized quantity and width only under a declared observation/scenario interpretation. An outcome outside an assumed scenario does not refute conditional arithmetic; an always unbounded set is not useful just because it contains observations. Compare equal-scope sets and their decision implications without assigning 95% labels or manufacturing equal branch probabilities. Robustness across a prior class remains a range of scores or decisions, not one averaged forecast.

Fix the comparison population and weighting before scoring. A candidate missing eligible predictions cannot win on an easier surviving subset. Report paired scores on the prespecified common coverage and the full eligibility/failure denominator; a claim about a complete service requires a declared fallback with its own results. Never renormalize away unresolved probability mass. A probability forecast with unresolved mass is incomplete for the above proper-score target.

The primary effect is the paired mean loss difference, proposed minus the preselected strongest admissible comparator, with issuer and time dependence accounted for. Promotion requires a predeclared uncertainty interval to lie below minus a decision-justified improvement margin, while all validity vetoes pass. A zero margin can test statistical improvement but cannot establish material business value. The interval level, dependence model, time blocks, multiple-comparison rule and margin are locked using development evidence before test predictions. Nine purposively selected issuers with eight years and overlapping horizons do not automatically support a bootstrap approximation. Without a defensible uncertainty design, report differences descriptively and retain all screened viable candidates without ranking.

E4. Bank losses and insurance reserves

Bank prediction distinguishes contractual cash collections, realized defaults, recoveries and reported allowance measurements. An allowance is an accounting estimate, not an observed realization of latent credit loss. A bank protocol needs exposure definitions, vintage/cohort boundaries, default/cure policy, original rating history, censoring, collateral and recovery timing, and the information available at each measurement. Aggregate filings do not supply those fields merely because they disclose a total allowance. Until eligible cohort data exist, compare exact conditional transition examples and reported aggregate bridges; do not claim calibrated borrower-level PD/LGD/EAD.

Insurance validation separates ultimate reserve prediction conditional on an observed upper triangle from one-year paid claims plus re-estimated reserves. Use the retained Mack and bootstrap dossiers for the exact conditioning and net-reserve center. Deterministic chain ladder is a point baseline; moment-based Mack uncertainty, a fully specified process-law bootstrap, and any enhanced model are different comparators. Mack moments alone do not specify a simulation law. Preserve shared fitted-parameter dependence across accident years and the backward-law assumptions required by the inspected consistency argument. One-year validation refits after simulated/revealed development; it cannot substitute an ultimate prediction distribution. Missing increments, negative/zero bases, unverified development dates or absent process law veto the dependent comparison and trigger data/method repair. Financial-disclosure panels alone do not provide independent insurance triangles or actuarial calibration evidence.

E5. Numerical inference and scenario interpretation

Start with rational enumeration and the normal conditional model derived earlier: prior mean 0 and variance 4, observation 2 with variance 1 give posterior mean $8/5$ and variance $4/5$. Cash $100 + 10z$ then has mean 116 and variance 80. Check the measure, normalizer, transformations and downstream cash distribution before speed. Compare analytic/factored linear algebra, error-controlled quadrature where applicable, an ordinary target-specific sampler, and only then an enhanced sampler. Mixed branches require evidence weights, not chain counts. Exact-equality constraints require a declared constrained measure; adding noise changes the target and requires a separate comparison.

Numerical tolerances derive from the requested monetary enclosure or decision accuracy. Posterior agreement, unresolved mass, divergences, convergence and Monte Carlo standard error have declared veto roles before throughput is interpreted. Independent chain initials are required; seed repetition alone is not an independent target. Failure on an analytic target triggers implementation or target repair. Failure only on a complex target triggers the specified parameterization/tuning repair, not abandonment of all inference.

The earlier causal example supplies a separate semantic test. Under $U \sim \text{Bernoulli}(1/2)$, $X = U$ and $Y = 2X + U$, conditioning on $X = 1$ gives $Y = 3$; intervening to set $X = 1$ gives Y in $\{2,3\}$ with mean 2.5. The test checks the supplied structural model. A successful test does not identify a real corporate intervention from filings. Empirical causal promotion requires its own identification design, confounding assumptions, data and estimand; observational forecast performance cannot provide those premises.

E6. Language usability and reuse

Compare the DSL and typed-library authoring over exactly the same IR, accounting rules, error messages, documentation and tasks. Tasks include encoding a movement schedule,

retaining an alternative, changing an effective date and tracing an amount. Correctness and material omissions are primary; completion time applies to a correctly reviewed result, with censored/failed tasks retained. A later counterbalanced task order must separate practice effects and prior expertise. The same author trying both interfaces is a development pilot, not independent usability evidence. Participant number, target expertise and uncertainty design require a justified future study before a superiority claim. No participants are contacted by this proposal.

Reuse compatibility first uses documentary exchange fixtures and field-loss cases. Arelle, Docling and other selected adapters need exact dependency/model/ taxonomy locks and terms before runtime evaluation. A successful source review does not establish installation, compatibility, license compliance or speed.

Budget, seeds, repairs and future records

The future ladder is bounded by decisions, not by an invented universal sample size. B0 runs exact tractable/adverse cases only. B1 uses exposed development packets to estimate failure rates, resource costs and within-target variability; its results cannot promote a method. B2 permits a frozen, honestly labelled retrospective panel comparison after availability and exposure repair. B3 uses newly sequestered, task-appropriate data or participants and a justified sample/ uncertainty design for external claims. Failure of B0 invalidates the dependent run; a weak candidate in B1/B2 triggers its declared repair. A later rung begins only when its prerequisite artifacts exist and execution is authorized.

For stochastic B1 debugging, seeds 101, 202 and 303 are a reproducibility convenience, not a sufficient replication count. Paired arms share randomness only when the coupling preserves each arm's marginal law. Subsequent replication counts, independent data units, chains/draws, search trials, wall-time and memory caps are selected from target-specific development evidence and frozen before evaluation. Optional learned components need a separate objective/scaling, capacity, optimizer/search, budget, heldout and downstream protocol. No neural architecture or optimizer is selected by this general design. Insufficient budget produces under_budgeted, not a convenient inherited default.

Each future run saves eligibility and exclusions; data, source and exposure hashes; split and temporal embargo; target and outcome-vintage definitions; comparison/tuning budgets; random seeds; environment and CPU/GPU mode; exact command and commit; wall time; per-unit predictions and failures; independent validity checks; uncertainty calculations; decision and inference-status tables; and a repair/next-action record. Changes after outcomes are inspected receive a new exploratory identifier and cannot overwrite the frozen confirmatory result. Unknown critical source time, corrupted data, wrong target, missing essential diagnostics or exceeded resource limits stop that run. A valid poor candidate blocks its promotion and leaves independently ready repairs available.

14.7 Capability contracts for sector packages

Package	Required state and operations	Minimum acceptance evidence
Core corporate	Working-capital bridges, lots/assets, ordinary debt and equity, period closing and cash-flow mappings	Exact synthetic cases, reviewed public filing reconstruction, all statement reconciliations.

Package	Required state and operations	Minimum acceptance evidence
Financial instruments	Terms, calendars, resets, options, measurement categories, modifications and hedges	Contract schedule comparisons, valuation checks, paragraph-based posting cases and transition cases.
Banking	Credit cohorts, deposits, collateral, liquidity, regulatory capital, management actions	Allowance roll-forwards, repricing and runoff stresses, capital reconciliation and default/funding-failure cases.
Insurance	Policy groups, future cash flows, service and finance components, reinsurance and asset interaction	Cohort movement reconciliation, assumption change cases, service recognition and reviewed actuarial calculations.
Multi-basis reporting	Versioned IFRS/US GAAP/ASBE rules, entity elections and transition treatment	Matched economic-event cases with explained basis differences and independent accounting review.

Every package declares capabilities at operation level. “Supports bonds” is too vague; “supports fixed-rate, single-currency, non-callable amortised-cost issuance and redemption under this reviewed rule set” is implementable and testable. Unimplemented operations retain their source and contract representation while producing an explicit capability error when execution requires them.

14.8 Implementation milestones and dependency order

Stage	Concrete implementation deliverable	Exit evidence
P0	Portable records and contextual types; source/review/time selection; typed constructors and versioned IR; separate posting authorization and atomic journal/delta/state/receipt transactions for the running cases	T01–T04, T09 and T17; source inspection and exact receipt replay.
P1	Admitted history templates, finite choices, shared symbols, linear solver adapter and historical bridges; conditional simulation and report explanations	T05–T10, T14 and T18; synthetic family preserved.
P2	Matched debt text frontend with identical portable IR; connected corporate methods through the typed authoring API; document/XBRL import adapters	Parser/API equivalence; T11–T13; reviewed three-date public-company case.
P3	Contract and valuation adapters, dated standard packages, richer financing and tax/funding policies	Contract/valuation differential cases; accounting review; T15 and transition cases.
P4	Bank package, allowance models, regulatory reconciliation and management response alternatives	T16; banking-specific reviewed histories and stresses.

Stage	Concrete implementation deliverable	Exit evidence
P5	Insurance cohort package, reinsurance and actuarial measurement adapters	Reviewed cohort roll-forwards, information-date and service/finance tests.
P6	Optional probabilistic inference, model-weight sensitivity and broader scenario research	Explicit targets, sampler/numerical validity, prior sensitivity and downstream evaluation.

These stages specify dependency order, not dates or staffing estimates. Exit checks are cumulative: a case can recur when a later stage changes the inputs or execution path that it tests. Banking and insurance representation requirements influence P0 types, although their measurement engines arrive later. Probability weighting is optional; P1 already produces useful conditional cases and bounds. A failure of a chosen extractor or inference algorithm triggers a local replacement or repair, rather than abandoning the product's representation requirements. The parser/API comparison in P2 covers the matched debt fragment. Complete textual authoring of the corporate profile is a further frontend task; its absence does not remove corporate declarations from the typed API.

14.9 Decisions and replaceable assumptions

The requirements fixed by this design are preservation of evidence and alternatives, explicit time and basis, balanced and admissible posting, economic state beyond journals, and distinct set/probability semantics. Replaceable choices include the storage engines, parser generator, solver, numerical libraries, aggregation resolution and user interface. Every scientifically material aggregation, behavioural parameter, prior or truncation rule must record provenance, purpose, failure mode and a diagnostic before it becomes a default.

The immediate work is to finish and review the governing proposal, including its real-source, authority, domain and research obligations. Product implementation follows completion of that work and explicit implementation authorization. In the table above, P0 establishes portable records, review/time selection, durable receipts and typed authoring; P1 builds admitted histories; P2 connects the corporate methods and matched textual frontend. These stages are a design sequence, not an already running product campaign.

14.10 Concrete conformance and corpus protocol

The fixture directory contains independently entered inputs and expected outputs for Northstar, rounded borrowing, evidence views and sector cases. `check_repair_conformance.py` recomputes exact journals and statements, constructs the joint linear bridges in Z3, checks strict bounds, and exercises replay, atomic rejection, finite Bayes updating and funding failure. `check_core_contract.py` checks grammar/schema examples, canonical identities, stable streams, information restrictions and source inventory. These scripts are reference oracles for portions of the design. They do not test an unimplemented full DSL compiler, all adapters or standards compliance.

Every conformance case must specify input files, the operation requested, expected values or

typed errors, comparison convention and supported scope. Exact money, finite probabilities, linear witnesses and example journals compare as rationals. Approximate valuations require an error bound or a justified tolerance tied to the valuation scale. A missing expected result cannot become a passing test simply because the operation returned something. An implementation release must pass all applicable positive and negative fixtures and preserve unresolved operations; the arithmetic oracle alone does not release it.

The retained panel contains all 192 target annual/interim report slots for 2018–2025 across nine issuers and three evidenced accounting families, with eight consecutive reporting years per issuer. Here "complete" applies to the source-acquisition panel: every declared slot has a retained report version. It does not mean that each selected document contains every related filing section. For example, the selected JPMorgan shareholder annual reports for 2022–2025 explicitly omit Form 10-K pages (respectively 1–42 and 304, 1–44 and 322, 1–48 and 334, and 1–42 and 328). The complete issuer-hosted 10-K candidates and their bounded semantic annotations are preserved as a separate source layer; they do not retroactively change the selected panel or establish full package closure.

Apple, Caterpillar and JPMorgan Chase each contribute one annual and three quarterly reports per year; the other six issuers each contribute one annual and one half-year report. Thus $3 \times 8 \times 4 + 6 \times 8 \times 2 = 192$. This is source acquisition, not semantic closure or a representative random sample. The 244 retained original versions include historical alternatives to the active panel. Apple remains a detailed development package, with Caterpillar supplying a contrasting inventory policy; bank, insurer, IFRS and ASBE reports are also retained.

Publication dates remain unverified for 136 of the 192 accepted reports; 56 have filing-date metadata. A historical forecasting population needs the actual byte-availability evidence described in Section 3.13, not imputed publication dates. The 31 provisional annotations across four reports are exposed development material, not heldout answers. Policies, note links, incorporated documents, changed comparatives and adjudicated meanings still require task-specific preparation. Synthetic revisions provide controlled cases without substituting for a required real conflict. The corpus protocol and active panel preserve their distinct populations and versions.

Annotators first record literal/context readings without seeing the model's predictions, then mark source conflicts, mapping relationships, narrative qualifications and admissible alternatives. A second accounting-aware reviewer adjudicates material disagreements while preserving both initial annotations. Reports from the same issuer family and duplicated tables cannot be split between development and heldout evaluation. Forecast tests freeze the actual publication cutoff; later revisions are separately labelled outcomes and may not enter predictor inputs. Extractor upgrades rerun the same frozen corpus.

For the initial supported release, loss of a source span, wrong entity/unit/ period/basis, invented evidence, hidden financing, dropped feasible branch, incorrect cash reconciliation or unreported unsupported treatment blocks acceptance. All selected critical facts and operations must have exact reviewed answers or the expected unresolved response. These zero-tolerance conditions follow from the product's meaning-preservation requirements. Non-critical extraction quality is reported by stratum and error category; no arbitrary overall accuracy threshold certifies whole-report understanding. An extraction benchmark needs its own sample size and uncertainty protocol before making statistical accuracy claims.

The DSL and typed-library comparison uses identical packages, evidence, schemas and runtime. Users perform the same tasks: correct a context, add a shared uncertainty, represent a restatement, add a supported event and explain a forecast difference. Task order is counterbalanced and experience recorded. Both frontends receive the same validation and error messages. Measure successful completion, material semantic errors, time and maintenance effort with uncertainty

appropriate to the participants; report no winner from the current syntax checks. A textual language earns its maintenance cost only if that comparison supports its use.

14.11 Workload limits and release evidence

The first integration workload is one local company package, up to 10,000 observations, 100 declared aggregate event slots per period, 64 materialised finite-choice cells and a 30-second budget per solver operation. These are provisional engineering limits for bounded failure behaviour, not measured capacity or scientific defaults. They must be configurable and recorded. Reaching a limit yields partial coverage with the unvisited formula/cells, never an exhaustive-history claim. Peak memory and latency are measured only after an implementation exists; a slower but semantically correct result does not become a scientific failure.

Engineering release needs compiler/API equivalence, transactional replay and adapter subset checks. Accounting release additionally needs paragraph-level authority, dated applicability, exact fixtures and independent domain review for every promoted operation. Forecast promotion additionally needs preserved information sets, the stated baseline ladder, statistical uncertainty and critical-regime vetoes. Human comprehension review remains separate from all three. This version supplies a richer design candidate and reference checks while preserving these outstanding acceptance decisions.

Appendix A

Portable operation reference

These proposed versioned operations share the request and result envelope in Section 13.4. Required inputs are explicit even when a list is empty. An expected head mentioned below refers to the common envelope, never a second argument. The registry is a lookup reference after the financial and transaction meanings have been explained in the body. These are specified exchanges, not deployed endpoints.

A.1 The corporate exchange package

The Northstar example in Section 2.6 supplies a closed instance of `CorporateJointHistory/1`. Its package is retained at `data/accounting-master/corporate-portable-20261015/`. The files `record-schemas.json` and `portable-exchanges.json` contain the closed record bodies, signature-specific arguments, complete immutable records, required byte references, exact submitted request bytes, results and the before/after store worlds. The accompanying semantic definition is `docs/design/proposal/architecture/corporate-portable-20261015.md`. These files specify exchanges; generating or checking them invokes no product endpoint.

Each request uses the common envelope. Its arguments have exactly the two fields `tag` and `value`: the tag equals the versioned operation name, and the value obeys that signature's closed schema. The profile and signature references bind the interpretation of those fields. Every reference has a kind, schema and content digest. Its target must be present, with the correct type and every required dependency; the submitted raw bytes are preserved even when decoding or validation fails. A record's dependency list contains its body references in canonical order. Required source and schema blobs are checked separately by byte length and digest.

The eight source units retain the original inventory. Corrected interim and annual coverage points to the complete historical reconstruction; the superseded original statement remains a literal source with a separately typed revenue reading. The source-reference closure does not imply that every source line has been admitted as accounting truth. A separate package variant retains a missing incorporated debt agreement as an explicit obligation. The new qualitative and rounding examples have their own authored source bytes.

The principal record relationships are useful when locating an exchange:

Record	Required financial connection
`HistoryFamily`	Frozen module and view, shared symbol and constraints, opening state and complete historical reconstruction.
`Projection`	That family, the imposed scenario and corrected closing state used as its opening; conditional journals, final balances and obligations.
`Report`	One projection and statement map; the same period, view, family and scenario throughout all reported lines.
`StateTransitionReceipt`	Validated request, stable event identity, authorization, journal, independent quantity/money change and prior/new states.
`WorkReceipt`	Saved search owner, prior/new checkpoint state and unchanged semantic manifest and continuous partition.

The text frontend deliberately has a smaller authoring scope than typed construction. Its published grammar accepts the ordered `NorthstarDebt` declaration: principal, the rational interval for x, fixed and floating rates, and the stated interest formula. The frozen module supplies the context, year fraction, source and assumption dependencies. Both principal tokens, both interval endpoints and both rates must match those module records. The typed debt input identifies the same module; the full company input separately includes all operating templates and economic facts. An unsupported declaration has a located diagnostic. This is a specified new bounded frontend, not a claim that the earlier Core0 parser implements it.

Accounting commits change the accounting owner, event lookup, journals, independent deltas and receipts atomically. Review commits change their review owner and publish a new immutable view. Checkpoint commits preserve the completed and unresolved cells under their work owner. Import verifies all original record mappings and required bytes before changing its metadata owner; it admits no financial claim. Each exchange states every unaffected domain as well, permitting exact comparison of the two worlds.

Failure location matters. Malformed JSON, extra fields, missing required references, mismatched operation tags or corrupt supplied blobs leave the validated request and financial payload null. A transaction conflict or stale head after valid input retains the validated request and structural validity, while reporting no new commit. The result's coverage retains the continuous family for financial queries and the original partition on resume. Neither an unexplored cell nor an unsupported calculation becomes an empty family.

The independent checks reconstruct both half-years from preserved economic inputs, derive the annual bridges and 2025 affine coefficients, replay the proposed journals, and compare the permitted store changes. Deliberately altered rate sensitivity, inventory quantity, receipt history, publication date and unknown closing balance challenge those checks. Their scope is the finite documentary specification. Generic frontend and solver execution, admitted-law extensions, crash/reopen testing, accounting acceptance and human comprehension retain their separate evidence requirements.

A.2 Joined company and explicit component migration

The company in Section 2.7 is specified by `CompanyComposition/1`. Its state contains both legal trial balances, the common currency and dates, ownership/control history, return and matured compound-instrument memory, the internal inventory lot, tax-close history and dated gross cash movements. The schema admits nine event kinds: redemption, return settlement, payroll, depreciation, internal inventory transfer, external inventory sale, reciprocal settlement, tax close and a purchase from NCI without loss of control. Every kind has a closed parameter record.

A basis binds the opening state, event, profile, information view and economic event identity. Authorisation checks these bindings and supplied findings. Proposal contains legal journals for both entities, the common closing state and dated cash origins. Commit publishes the proposal, closing state, receipt and immutable request decision together. An internal sale cannot publish only P's receivable while omitting S's payable. Unrelated owners remain unchanged.

An exact retry returns its stored result before current-head or current-view checks. Reusing the identity with changed bytes conflicts. A new request needs the exact current head and view. If an acknowledgement is lost, the observable outcome is unknown: either no commit exists, or the complete commit is durable. The retry commits once in the first world and replays in the second. The documentary examples specify both worlds; they do not measure runtime durability.

The report operation is read-only. It requires published receipts that connect the requested opening and closing in the complete date interval. It recomputes the consolidation worksheet, tax effects, gross external cash and ownership allocation. Reports carry the explicit joined-finite-company scope. A request for an uncomputed tax close, incompatible calendar or unsupported sector is not satisfied by adding independent component totals.

The source and schema locks bind exact bytes. Canonical JSON excludes duplicate keys, floats, non-finite values and unreduced rational quantities. Record identity includes schema and body; dependency lists are the exact transitive typed references. The retained independently checked packet is under `data/accounting-master/company-composition-20261016/attempt-03/`. It has 13 transactions, three reports and two migration demonstrations. Those counts describe the finite evidence, not a complete-company support measure.

Migration is deliberately narrower than method instantiation. The return projection maps the named `CorporateResiduals/1` opening state into a closed target containing its complete return memory, source context/view, owner/book/branch, dates and cash/equity accumulators. The compound projection does the same for the named matured `FinancingComponents/2` state. Nonzero balances outside the explicit map, unknown fields and foreign profiles are refused. Zero-only omitted accounts are reconstructed from the locked source schema in the round trip.

For these two operations, every quantity used by the subsequent settlement or redemption is preserved. Their exact sums, products and minimum therefore commute with the mapping: deriving the target result equals mapping the source result under the same findings. The checks exercise both the round trip and this following transition. They do not migrate old requests, annual-report labels or arbitrary methods. In particular, changing the original synthetic owner to P would be a new instantiation, not a lossless migration.

The separate `CorporateResiduals/1` namespace fixes the 17 finite methods in Section 9.14; its prior `CorporateComponents/5` requests remain bound to their original lock. The all-zero CGU allocation, full retrospective acquisition comparisons, general mixed-contract methods and current-law admission remain explicit residuals. No profile name itself provides interoperability or extends an earlier operation's accepted fields.

A.3 Real-source connections and partial answers

`RealDisclosureConnection/1` is the documentary extension for Section 2.23. Its closed record schemas, canonical request bytes, complete record registry and expected responses are retained in `data/accounting-master/real-package-connection-20261015/`. The semantic specification is `docs/design/proposal/architecture/real-package-connection-20261015.md`. The examples specify behavior; no product endpoint or live store was executed.

The five supported operations are source preparation, historical querying, conditional scenario projection, report construction and review proposal. Each uses the I1 request/result envelope with a profile-bound signature and tagged arguments. Record dependencies and separate byte references follow L1. The schema lock includes both schema bytes and the semantic specification. The 31 exchanges include the combined partial answer, named-source statement readings, original and later segment queries, unknown historical clocks, three conditional difference arms, six settlement paths and company-specific review proposals.

The family record represents an unresolved question about actual issuer histories. It admits neither the illustrative branches nor a probability law. Reported observations, reported dashes, unknown values and conditional derived amounts have distinct tags. A conditional amount references the assumptions under which it was calculated. The complete reported-token subquery does not upgrade the connected result's partial status.

Coverage partitions the requested claims. For a missing-date settlement request, eventual conditional amounts are completed and period timing is unresolved. These regions are neither probabilities nor proportions of all company histories. Future periods are the relative indices 1 and 2; an unknown period is null. An out-of-range period or allocation parameter is invalid. An unsupported depreciation or probability request retains its unresolved obligations instead of returning zero.

Decoding precedes operation-tag, signature, profile, issuer, view and parameter binding. Truncated bytes, a mismatched tag, invalid parameters and a cross-company binding return rejected input with no validated request, payload or coverage. Valid named-source results may only use observations eligible in their own view. Linking a later view in a connected result does not authorize an original-view query to read it.

Every exchange has null expected head, no commit receipt and unchanged before and after worlds for accounting, review, work and import owners. These empty worlds are explicit documentary examples of candidate/read-only effects; they do not establish persistence in a deployed service. Independent checks derive journal balances, all annual difference rows and all settlement-period effects, verify source spans and dependency closure, and preserve the actual source-conflict limits. They establish the stated finite specification and arithmetic, not actual-company history feasibility or accounting admission.

A.4 source.prepare@1

package manifest, source snapshot, adapter/config lock, prior review, traversal budget and optional continuation → ExtractionPreparation

Acquires/preserves sources and constructs candidates under the source-preparation method; no semantic admission

A.5 review.propose@1

target Ref, layer, decision kind, evidence Refs, actor, rationale, replacement Ref or null → ReviewProposal

Candidate construction; accepts no layer

A.6 review.commit@1

ReviewProposal, expected review view/revision → immutable ReviewDecision and new view

Review transaction under the review and storage rules; admissible action/layer required

A.7 time.select@1

package/view, investor/system/exploratory mode, cutoffs, revision and unknown-time policy → selected assertions, conflicts and temporal obligations

Pure selection under the time-admission rules; no invented ingestion or publication time

A.8 model.compile@1

the versioned CompileRequest → candidate Module, source map, diagnostics and target admission/capability report

Candidate construction; no solver or posting

A.9 history.construct@1

admitted Module, opening, scope, template registry, query/horizon, resource budget → HistoryFamily, partition, witnesses and unresolved formulas

History-construction contract; witnesses are examples within the family, not its definition

A.10 history.refine@1

frozen family, additional admitted constraint or refinement, dependency view, budget → child family and inclusion relation

Candidate construction; record whether correction replaces constraints instead of intersecting them

A.11 history.query@1

family, typed joint observable, bound/feasibility request, capability manifest → symbolic/partial/exact answer, coverage and endpoint attainability

Pure; unsupported expressions remain explicit

A.12 uncertainty.evaluate@1

exact native uncertainty request, law/attachment/admission and operation-specific event or observation block → native UncertaintyResult and candidate derived bundle

Native admission/attachment/chaining rules; never selects its result automatically

A.13 uncertainty.admit@1

candidate bundle, target time/evidence/purpose, actor, rationale, requested assumption scope, expected selection revision → admitted attachment and decision

Admission transaction; author-stipulated measures keep that label

A.14 uncertainty.select@1

admitted attachment/receipt, rationale and expected selection revision → new selected-view revision

Selection transaction; no probability recomputation

A.15 scenario.project@1

family, exogenous scenario, policy/method lock, horizon/calendar, explicit information policy, resource budget → conditional event/journal/state proposals by history and scenario

Candidate construction; budgets preserve unresolved regions; no historical-book commit

A.16 accounting.assess@1

native assessment request, event, opening, context/time, authority/method, evidence/assumptions and admissions → TreatmentAssessment

Pure accounting assessment; inherited `assess_only` profile cannot post

A.17 accounting.authorize@1

AccountingEventBasis and posting-profile Ref → PostingAuthorization

Pure evaluation under a separately supported posting profile; an assess-only result cannot grant posting permission

A.18 accounting.propose@1

PostingAuthorization, AccountingEventBasis and posting/rounding method → JournalProposal and independent EconomicDelta

Candidate construction; revalidate bindings; unsupported method gives no journal

A.19 accounting.commit@1

AccountingCommitArguments/v1 and envelope expected branch head/generation → StateTransitionReceipt

Single atomic transaction under the event-commit contract; the separate commit schema, separate from native assess-only

A.20 report.build@1

frozen branch/family result, statement-map version, interval/view, typed requested statements and presentation policy → joint statement family, reconciliation and coverage

Pure; derive from retained state/journals, no balancing plug or hidden branch selection

A.21 record.explain@1

output Ref, its frozen view, requested lineage depth → source/method/assumption/decision paths and unresolved dependencies

Pure; depth truncation is explicit; cannot substitute today's view

A.22 work.resume@1

checkpoint Ref, original semantic manifest, scheduling budget → operation-specific resumed result

Resume the original operation; checkpoint ownership/atomicity under the durability contract; no model change hidden as resume

A.23 store.export@1

snapshot/root Refs, format/decoder, destination → closed manifest or explicit incomplete export

Publish a verified export manifest atomically under Section 13.6; no domain mutation

A.24 store.import@1

closed manifest, exact blobs/records, decoder and target namespace → import receipt and original-to-local mappings

Verify before metadata transaction; import alone grants no financial admission

A.25 store.migrate@1

migration request under Section 13.6 → preserved originals, new representation, comparison/mapping and receipt

Transaction only after required semantic comparisons; no changed-model-as-format claim

A.26 Closed accounting records

The following field lists are exact for the named versions. Every reference uses the portable kind/schema/digest type; plural references are schema-declared sets. Unknown fields reject. Source and admission sets can be empty only when the method's admitted assumption scope explicitly permits and explains that.

```
AccountingEventBasis/v1 {
  identity: {entity_id, book_id, branch_key, business_event_id},
  event_ref, opening_ref, context_ref, branch_ref, time_ref,
  chart_ref, authority_refs, method_ref, evidence_refs,
  admission_refs, assumptions_ref, view_ref
}
PostingAuthorization/v1 {
  basis_ref, posting_profile_ref, method_ref, authority_refs,
  status, findings, posting_permitted, evidence_level, view_ref
}
AccountingCommitArguments/v1 {
  basis_ref, authorization_ref, journal_ref, delta_ref
}
```

Identity fields are nonempty ASCII names. Each singular reference is nonnull. Authority references identify applicable support or a specifically typed synthetic authority for an explicitly synthetic example. Authorization status is eligible, unresolved, ineligible or outside scope; posting permission is a Boolean and can be true only for eligible status under a supporting profile. A branch's immutable book mode is synthetic or real, and a profile names its supported modes. Synthetic evidence cannot authorize a real-book posting. Journal and delta both identify the same basis and authorization. Their domain payloads obey the package's pinned journal and state schemas. No earlier record refers forward to the final receipt.

A.27 Method and extension definitions

A versioned financial method supplies the following nonempty definitions:

```
MethodDefinition {
  method_id, version, event_kind,
  state_schema, event_schema, journal_schema, delta_schema,
  output_schema, required_contexts, authority_requirements,
  preconditions, algorithm, account_roles, report_mappings,
  numerical_capability, examples, unsupported_cases
}
ExtensionPackage {
  package_key, version, record_schemas, operation_signatures,
  supported_contexts, state_requirements, authority_refs,
  method_refs, capability_requirements, migration_refs,
  example_refs
}
```

Each precondition has an identity, typed predicate, required input/admission references and separate false/unknown consequences. Domain state and event fields are specified with the financial methods in the body. Schema names and package members resolve through immutable locks. An empty authority list makes no accounting-authority claim. Registering a schema does not establish the method's entity, instrument, reporting-period or probabilistic applicability. The full machine-oriented contracts and finite exchanges accompany the proposal in the architecture and methods directories; their exact versions are pinned by the manuscript's dependency index.

Appendix B

Core grammar, records and execution details

B.1 Preserved Core 0 teaching grammar

The grammar governing the preserved Core 0 teaching examples is reproduced from `spec/core-0.lark`. It does not enlarge the first matched text frontend beyond the portable profile described in the body. Names are case-sensitive ASCII identifiers; strings are UTF-8 with JSON escaping. All literal words in the grammar are reserved. The companion contract specifies valid dates, exact rational literals, argument arity, type/context checks and package resolution. A syntax parse is not semantic validation.

```
// Normative Core 0 syntax, Lark EBNF. UTF-8 input; ASCII identifiers.
// Longest token wins; keywords cannot be declaration identifiers.
start: "module" NAME "version" STRING ";" declaration*
?declaration: use | source | entity | book | account | observation | symbol
            | choice | binding | constraint | alternative | contract | event
            | rule | scenario | report | opening
use: "use" NAME "version" STRING ";"
source: "source" NAME "=" expr ";"
entity: "entity" NAME "functional" NAME ";"
book: "book" NAME "entity" NAME "basis" NAME "version" STRING ";"
account: "account" NAME ":" NAME "book" NAME ";"
observation: "observation" NAME ":" type time_spec "=" expr "from" NAME ";"
time_spec: "instant" DATE | "period" DATE "to" DATE
symbol: "symbol" NAME ":" type "in" domain "because" STRING ";"
domain: "[" expr "," expr "]" | "{" expr ("," expr)* "}"
choice: "choice" NAME "in" "{" NAME ("," NAME)* "}" "because" STRING ";"
binding: "let" NAME [":" type] "=" expr ";"
constraint: "constraint" NAME ":" expr ";"
alternative: "alternative" NAME "{" assumption* "}"
assumption: "assume" expr ";"
contract: "contract" NAME ":" NAME "book" NAME "{" field* "}"
field: NAME "=" expr ";"
event: "event" NAME "(" [parameters] ")" "book" NAME ";"
parameters: NAME ":" type ("," NAME ":" type)*
rule: "rule" NAME "for" NAME "as" NAME ["overrides" NAME ("," NAME)*] "when" expr "{" rule_statement* "}"
?rule_statement: binding | require | post
require: "require" expr ";"
post: "post" "{" posting+ "}"
posting: SIDE NAME expr ";"
scenario: "scenario" NAME "{" setting* "}"
setting: "set" NAME "=" expr ";"
report: "report" NAME "for" NAME "{" field* "}"
opening: "opening" NAME "for" NAME "from" expr ";"
type: NAME ["<" type ("," type)* ">"]
?expr: disjunction
?disjunction: conjunction (OR conjunction)*
?conjunction: comparison (AND comparison)*
```

```
?comparison: sum [COMPARE sum]
?sum: product (ADD product)*
?product: unary (MUL unary)*
?unary: UNARY unary | postfix
?postfix: atom ("." NAME)*
?atom: number | STRING | DATE | BOOL | call | NAME | "(" expr ")" | array | record
call: NAME "(" [expr ("," expr)*] ")"
number: NUMBER [NAME | "per" "year"]
array: "[" [expr ("," expr)*] "]"
record: "{" [record_field ("," record_field)*] "}"
record_field: NAME ":" expr
SIDE: "debit" | "credit"
OR: "or"
AND: "and"
UNARY: "-" | "not"
COMPARE: "==" | "!=" | "<=" | ">=" | "<" | ">"
ADD: "+" | "-"
MUL: "*" | "/"
BOOL.2: "true" | "false"
DATE: /@[0-9]{4}-[0-9]{2}-[0-9]{2}/
NAME: /[A-Za-z_][A-Za-z0-9_]*/
NUMBER: /(?:0|[1-9][0-9]*)(?:\.[0-9]+)?/
STRING: /"(?:[^"\\\x00-\x1f]|\\(?:["\\/bfnrt]|u[0-9a-fA-F]{4}))*"/
%import common.WS
%ignore WS
%ignore /\/\/[^\n]*/
```

B.2 Required core record definitions

The historical closed machine-readable definitions remain in the original design directory at `spec/core-ir.schema.json`. They define contexts, source spans, observations, narratives, coverage units, mappings, partitions, entity/control records, books, accounts, symbols, choices, expressions, constraints, assumptions, alternatives, contracts, events, rules, packages, measurements, states, journals, scenarios, information views, review decisions, evidence views, model scopes, historical families, probability models, report mappings, result cells, branches, runs and diagnostics. For those historical records the schema fixes field spelling and structural shape, and its companion contract fixes reference resolution and operations. New portable records instead obey the exact versioned contracts introduced in Chapters 7, 8 and 13. No historical field list grants new posting authority.

For example, these selected records reproduce the schema's field names. Each field is required; a nullable link is represented by explicit null.

```
EvidenceView {
  id, cutoff, system_as_of, mode,
  documents[], observations[], decisions[],
  conflict_sets[][], transforms[]
}
Context {
  id, entity, perimeter, concept, basis, measurement,
  time, unit, currency, taxonomy_version, dimensions[]
}
Symbol {
  id, type, context, domain, uncertainty_kind,
  scope, dependencies[], admission, provenance[]
}
Journal {
  id, event, book, date, accrual_period,
  postings[], rule, measurements[], assumptions[],
  precondition_hash, explanation
}
ResultCell {
  id, context, history_condition, scenario, path, kind,
  value, expression, lower, upper,
  lower_attained, upper_attained, sample_ref,
  certificate, reason, dependencies[]
}
```

A value is a reduced rational, or a typed symbolic/domain reference. A posting is nonnegative with its side carrying the sign, in the book's functional currency. Original units and the valuation reference survive conversion. Monetary contexts require currency; a period flow requires a calendar and exclusive end boundary. Historical zero postings may preserve traceability but establish no economic event. New portable proposals omit zero lines and retain their explanation in the event and findings instead. The amount may have finer precision before the declared posting-rounding operation.

Schema validation rejects unknown fields but cannot prove that a treatment is permitted, a reference points to the correct kind of record, or two periods are comparable. Those semantic checks precede solver or runtime dispatch. The structural fixture and complete demonstration source/package are supplied separately so that a future implementation can check both layers explicitly. `fixtures/core-ir-demo.json` supplies an authored expected lowering; the reference checker evaluates its selected debt expressions. A general source-to-IR compiler and complete external-registry validation remain to be implemented. Domain records retain disclosed display text and rounding rules alongside their exact bounds.

B.3 Typing and partial-information rules

For expressions x and y in the same currency C, the money addition rule is

$$\frac{x : \mathrm{Money}\langle C\rangle \quad y : \mathrm{Money}\langle C\rangle}{x + y : \mathrm{Money}\langle C\rangle}.$$

Context compatibility is a separate premise for observations. There is no analogous implicit rule for different currencies. Explicit conversion uses a dated measurement and records its rate and purpose. A rate-times-principal calculation is typed through a declared accrual convention; the example's annual simple rate and year fraction produce money in the principal's currency.

Known expressions evaluate normally. Shared symbols evaluate to symbolic expressions subject to a joint constraint store. Missing source values cannot enter arithmetic until the model supplies a symbol or a conditional assumption. Boolean guards over symbols produce formulas, not the value false. This avoids the common error in which "unknown whether impaired" silently means "not impaired." Division requires a proven nonzero denominator or a partitioned domain that excludes zero; otherwise the expression remains unsupported for concrete execution.

Logical queries first require a nonempty feasible family. An empty family returns `Infeasible`; it must not produce vacuous answers to substantive questions. For a nonempty family, constraint evaluation has three results. `entailed` means every state in the declared feasible family satisfies the proposition. `refuted` means none does. `undetermined` means both remain feasible or the solver has not decided; the response distinguishes those causes. Probabilistic truth is a separate query requiring a measure. A language-level Boolean value must never be inferred from an arbitrary probability threshold.

For a decided answer, let $F(h)$ be the exact constraint representation of the whole family requested by the query, and let $P(h)$ be its proposition. Establish feasibility first. Then `entailed` requires that $F(h) \wedge \neg P(h)$ has no solution, while `refuted` requires that $F(h) \wedge P(h)$ has no solution. Feasible witnesses on both sides establish the mixed case. A timeout, unsupported operation, or unfinished search establishes neither universal conclusion. A proof for a selected cell applies to that cell; it applies to the parent family only with coverage of the remaining cells or a proof over the full family. The result records the scope, query kind and evidence for that status. Section 6.2 distinguishes this Boolean test from a certain-tuples view and a family of answer schedules.

B.4 Operational rule selection

Let $V(e, h)$ be the applicable rules for event e in concrete history h, after package/date selection and guard evaluation. Let $r \prec r'$ mean that r' validly overrides r through the declared acyclic graph. Define

$$\mathrm{Max}(V) = \{r \in V : \nexists r' \in V \text{ with } r \prec r'\}.$$

For one required recognition operation, a singleton selects that rule; the empty set yields `UnsupportedTreatment`; a larger set yields `RuleConflict`. Multiple distinct operations, such as measurement and recognition, are evaluated through separate declared interfaces. A rule that intentionally produces no postings must belong to a supported non-recognition event operation, whose economic state effect is still recorded.

Given a selected rule, the evaluator computes all local bindings and preconditions, emits proposed postings, verifies Equation (6.2), validates account/context restrictions and commits the event transaction. Failure returns the original state plus a diagnostic. This defines the atomicity requirement independently of the storage technology.

B.5 Example run request and response

The following request executes two named conditions under one scenario. It does not request an aggregate probability distribution.

```
{
  "model": "BorrowingExample@1",
  "evidence_view": "synthetic-2025-exact",
  "starting_state": "after-2025-close",
  "horizon": {"start": "2026-01-01", "end_exclusive": "2027-01-01"},
  "conditions": ["allFixed", "allFloating"],
  "scenarios": ["rateShock"],
  "mode": "conditional",
  "weight_model": null,
  "reports": ["balance_sheet", "income", "cash_flow", "equity", "journals"],
  "package_lock": "demonstration-packages-v1"
}
```

The response contains two separately labelled results with interest expense 5 and 7, cash 90 and 88, debt 100 in both, and equity -10 and -12. A request for the parent symbolic family returns the formula $7 - 2\alpha$ and, under exact linear bounds, $[5, 7]$ before monetary quantisation. The API rejects an unconditional mean request without a weight model. A user can instead request an arbitrary clearly labelled representative case, which does not stand in for the feasible family.

B.6 Requirement traceability

Requirement	Primary implementation area	Conformance cases
R1	Evidence registry and extraction adapters	T03, T09
R2	Context/version model and cutoff queries	T03, T04
R3	Types, mappings and package selection	T02, T12, T13
R4	Symbols, choices and constraint graph	T05–T10, T14
R5	Posting, closing and consolidation	T01, T11–T13
R6	Contract and behavioural interfaces	T05, T11, T15, T16
R7	Conditional output and probability types	T06, T07, T18
R8	Explanation graph and report dependencies	T09, T11–T13
R9	Capabilities and versioned modules	T02, T14–T16
R10	Run records, scheduler and replay	T03, T17, T18

Appendix C

Research basis and outstanding review

C.1 Source collection and reading scope

The companion library contains 32 verified paper PDFs: 25 retrieved from external sources and seven preserved copies from the original collection. Filenames use the title, first author's surname and year. Publication years are used where an archived working version underlies a later publication; the catalogue identifies the actual version read. The original collection has not been renamed or modified. Standards pages, contract specifications, software source snapshots and metadata are stored separately from papers.

The survey considered accounting algebra and DSLs, corporate and bank simulation, incomplete and temporal databases, provenance, inverse problems, partial identification, model averaging, causal scenarios, contract languages, probabilistic programming, document extraction and financial reasoning. OpenAlex and Crossref supplied dated metadata, while primary publisher, institutional and author copies supplied technical support. Public-index results were checked against the downloaded title pages; HTTP success alone was not accepted as a valid paper download.

Technical reading was targeted to the mechanisms used in this design. The full PDF is archived even where only specified sections support the argument. No claim is made that every theorem, appendix or empirical result in every archived paper was independently replicated. Representative support anchors are listed below; the source-support ledger contains the per-source scope and qualifications.

Source	Inspected technical support	Design use
Nehmer and Robinson	Sections 2–5: account states, transactions and aggregation	Conservation plus admissibility.
Doshi	Chapters 4, 7–9 of the 2024 thesis	Dates, logs, corrections and DSL scope.
Shahnazarian	Sections 2.3–2.8; relevant Chapter 3 equations; data and evaluation sections; Appendix B	Accounting/behaviour separation; limits of data transfer.
Pierson	Sections 3–4 and reporting structures; supplementary Vensim equations	Accrual/payment bridges and reporting accumulators.

Source	Inspected technical support	Design use
Libkin	Sections 2–5: incomplete states, conditions and certain answers	Shared unknowns and joint alternatives.
Green et al.	Definitions 3.1–3.2; Section 4	Composable relational provenance; limits outside positive operations.
Stuart	Section 2.2, observation equation and prior/regularisation discussion	Inverse-problem formulation, not imported convergence guarantees.
Giacomini and Kitagawa	Sections 2.1–2.6 and Theorem 1 in CWP12/20	Prior sensitivity and classes of posterior beliefs.
Hoeting et al.	Equations (1)–(3); prior and predictive-evaluation discussion	Optional model-weighted predictions.
Merigoux et al.	Sections 3–4, default calculus and translation figures	Scopes, exceptions, empty/conflict outcomes.
Peyton Jones et al.	Sections 3–5, combinators and valuation semantics	Contract composition separate from valuation assumptions.
FinQA / TAT-QA	Dataset operations, annotation, model and evaluation sections; FinQA evaluation source	Evidence-linked local arithmetic; benchmark boundaries.
DocFinQA / FinBen	Context/retrieval and financial-task evaluations	Long-document difficulty and task coverage.
PubTables / GTE	Canonicalisation, dataset construction, methods and evaluation	Table structure and financial-domain extraction.
Cosentino et al.	Sections 3.1–3.2, 4.6 and 5	Persistent RAG provenance; scope of auditability evidence.

C.2 Version and access qualifications

Several discrepancies mattered. Doshi's thesis is dated 2024 despite the original filename. The archived Mernik text is the 2003 CWI report underlying the 2005 article. Giacomini and Kitagawa's copy is the April 2020 working paper underlying the 2021 article. The Docling v5 copy lists Auer as first author. The probabilistic-programming tutorial is an arXiv version revised in 2021, first submitted in 2018; a merged index DOI was not used. The Libkin author PDF has a footer DOI that resolves to another paper; title/author metadata from the indexes resolves the cited work to DOI `10.1145/2594538.2594561`. These discrepancies remain documented rather than being hidden by a cleaned bibliography.

McCarthy (1982), Imieliński and Lipski (1984), Tamer (2010), and Caiani and co-authors (2016) are retained as important blocked candidates. Access failures and rejected wrong-content downloads are recorded. The design does not use their uninspected technical claims. More specialised work on accounting measurement, process mining, multi-company equilibrium, actuarial inference and learned forecasting is recorded as expansion work for the corresponding implementation stage. The survey supports the core architecture without pretending that those later modules have a complete literature or rule base.

OpenAlex retraction flags and available Crossref correction/version relations were inspected

alongside local versions. No reviewed source was excluded for a discovered retraction. Missing correction metadata is marked unavailable; absence of a discovered notice does not certify that no notice exists. Citation counts are dated index metadata, and venue rankings were not available in this campaign. Neither is used as evidence of correctness.

C.3 Companion records and status

The six research ledgers are `source-support.json`, `citation-venue.json`, `backward-snowball.json`, `forward-snowball.json`, `claim-support.json`, and `omissions.json`. They record technical anchors, metadata sources, considered leads, omitted or blocked candidates and the distinction between source claims and this document's derivations. Citation leads with checked seed references are distinguished from cross-survey additions and topic expansion leads. Forward searches retain the most-cited and most-recent results returned by the recorded bounded queries; they do not establish exhaustive citation coverage. The separate `tools.json` records software evidence. Raw retrievals and text extracts permit a later reviewer to revisit the decisions.

The research library includes Mack's reserving paper, with its conditional-mean/variance assumptions, proofs, aggregate-error dependence and numerical comparisons inspected. It also archives official AASB texts incorporating IAS 2, IAS 32 and IFRS 17, distinguishing their Australian additions and applicable editions. The corporate filings are evidence of reported amounts and policies, rather than substitutes for every underlying standard. The ASBE 1 reproduction has not been verified against an accessible official current text, and official FASB/ASC access remains incomplete. These limitations are recorded at the specific operations they affect.

Every one of the 179 unique candidates in the acquired forward-query files has an individual title/metadata relevance decision. This includes previously omitted discovery files associated with inaccessible seed papers. The generation script requires an explicit decision and stops on an unseen candidate. Screening is still distinct from technical reading: deferred provenance, temporal-query, bank-optimisation, extraction and actuarial follow-ups remain identified expansion risks. An omitted-paper reason does not make that paper technically reviewed or make the survey exhaustive.

This document specifies the product; it does not report a completed engine. The companion review records cover deterministic example arithmetic, bibliography/build checks, rendered-page inspection and author review. Human comprehension review, independent accounting review and implementation conformance remain separate future evidence. They are not replaced by a clean LaTeX build or by the quantity of literature collected.

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